



## **User's Manual Version 4.12**

**Last Updated: February 25, 2026**

**Contributors**

Steve Maas ([steve.maas@utah.edu](mailto:steve.maas@utah.edu))

Dr. Jeffrey Weiss ([jeff.weiss@utah.edu](mailto:jeff.weiss@utah.edu))

Dr. Gerard Ateshian ([ateshian@columbia.edu](mailto:ateshian@columbia.edu))

**Contact address**

Musculoskeletal Research Laboratories, University of Utah  
72 S. Central Campus Drive, Room 2646  
Salt Lake City, Utah

**Website**

Weiss Lab: <https://weisslabutah.org>

FEBio: <http://febio.org>

**Forum**

<https://forums.febio.org/>

**Acknowledgments**

Development of the FEBio project is supported in part by a grant from the U.S. National Institutes of Health (R01GM083925).



# Contents

|          |                                 |           |
|----------|---------------------------------|-----------|
| <b>1</b> | <b>Introduction</b>             | <b>17</b> |
| 1.1      | Overview of FEBio               | 17        |
| 1.2      | About this document             | 18        |
| 1.3      | FEBio Basics                    | 19        |
| 1.4      | Units in FEBio                  | 20        |
| <b>2</b> | <b>Running FEBio</b>            | <b>23</b> |
| 2.1      | Running FEBio on Windows        | 23        |
| 2.1.1    | Windows                         | 23        |
| 2.1.2    | Running FEBio from Explorer     | 24        |
| 2.2      | Running FEBio on Linux or MAC   | 24        |
| 2.3      | The Command Line                | 24        |
| 2.4      | The FEBio Prompt                | 28        |
| 2.5      | The Configuration File          | 31        |
| 2.6      | Using Multiple Processors       | 31        |
| 2.7      | FEBio Output                    | 32        |
| 2.7.1    | Screen output                   | 32        |
| 2.7.2    | Output files                    | 33        |
| 2.8      | Advanced Options                | 34        |
| 2.8.1    | Interrupting a Run              | 34        |
| 2.8.2    | Debugging a Run                 | 34        |
| 2.8.3    | Setting break points            | 35        |
| 2.8.4    | Restarting a Run                | 36        |
| 2.9      | FEBio File Conventions          | 36        |
| 2.10     | FEBio Tasks                     | 37        |
| <b>3</b> | <b>Free Format Input</b>        | <b>39</b> |
| 3.1      | Free Format Overview            | 40        |
| 3.1.1    | Format Specification Versions   | 41        |
| 3.1.2    | Notes on backward compatibility | 42        |
| 3.1.3    | Multiple Input Files            | 42        |
| 3.1.3.1  | Include Keyword                 | 43        |
| 3.1.3.2  | The 'from' Attribute            | 43        |
| 3.2      | Module Section                  | 45        |
| 3.3      | Control Section                 | 47        |
| 3.3.1    | Control Parameters              | 47        |
| 3.3.2    | Time Stepper parameters         | 50        |

|         |                                                       |    |
|---------|-------------------------------------------------------|----|
| 3.3.3   | Common Solver Parameters                              | 51 |
| 3.3.4   | Quasi-Newton Solver Parameters                        | 54 |
| 3.3.5   | Solver Parameters for a Structural Mechanics Analysis | 54 |
| 3.3.6   | Solver Parameters for Biphasic Analysis               | 56 |
| 3.3.7   | Solver Parameters for Solute and Multiphasic Analyses | 56 |
| 3.3.8   | Solver Parameters for Fluid and Fluid-FSI Analyses    | 56 |
| 3.3.9   | Explicit-Solid Solver                                 | 57 |
| 3.4     | Globals Section                                       | 59 |
| 3.4.1   | Constants                                             | 59 |
| 3.4.2   | Solutes                                               | 59 |
| 3.4.3   | Solid-Bound Molecules                                 | 60 |
| 3.4.4   | User Variables                                        | 60 |
| 3.5     | Material Section                                      | 62 |
| 3.6     | Mesh Section                                          | 63 |
| 3.6.1   | Nodes Section                                         | 63 |
| 3.6.2   | Elements Section                                      | 64 |
| 3.6.2.1 | Solid Elements                                        | 64 |
| 3.6.2.2 | Shell Elements                                        | 65 |
| 3.6.2.3 | Beam Elements                                         | 67 |
| 3.6.3   | NodeSet Section                                       | 67 |
| 3.6.4   | Edge Section                                          | 68 |
| 3.6.5   | Surface Section                                       | 68 |
| 3.6.6   | ElementSet Section                                    | 69 |
| 3.6.7   | DiscreteSet Section                                   | 69 |
| 3.6.8   | SurfacePair Section                                   | 69 |
| 3.6.9   | PartList                                              | 69 |
| 3.6.10  | Implicit Partitions                                   | 70 |
| 3.7     | MeshDomains Section                                   | 71 |
| 3.7.1   | SolidDomain Section                                   | 71 |
| 3.7.2   | ShellDomain Section                                   | 73 |
| 3.7.3   | BeamDomain Section                                    | 74 |
| 3.8     | MeshData Section                                      | 75 |
| 3.8.1   | Data Generators                                       | 75 |
| 3.8.2   | ElementData                                           | 76 |
| 3.8.3   | SurfaceData                                           | 77 |
| 3.8.4   | EdgeData                                              | 77 |
| 3.8.5   | NodeData                                              | 78 |
| 3.9     | Mesh Adaptor Section                                  | 79 |
| 3.9.1   | The MeshAdaptor Section                               | 79 |
| 3.9.1.1 | The erosion adaptor                                   | 80 |
| 3.9.1.2 | The mmg_remesh adaptor                                | 80 |
| 3.9.1.3 | hex_refined2d                                         | 81 |
| 3.9.1.4 | tet_refine                                            | 82 |
| 3.9.1.5 | hex_refine                                            | 82 |
| 3.9.2   | Mesh Adaptor Criteria                                 | 82 |
| 3.9.2.1 | max_variable                                          | 82 |
| 3.9.2.2 | element_selection                                     | 83 |
| 3.9.2.3 | math                                                  | 83 |

|          |                                              |     |
|----------|----------------------------------------------|-----|
| 3.9.2.4  | stress                                       | 83  |
| 3.9.2.5  | max_damage                                   | 83  |
| 3.9.2.6  | relative error                               | 83  |
| 3.9.2.7  | min-max filter                               | 83  |
| 3.9.2.8  | contact gap                                  | 84  |
| 3.9.3    | Limitations                                  | 84  |
| 3.10     | Initial Section                              | 85  |
| 3.10.1   | The init_dof Initial Condition               | 85  |
| 3.10.2   | The displacement Initial Condition           | 85  |
| 3.10.3   | The velocity Initial Condition               | 86  |
| 3.10.4   | The shell velocity Initial Condition         | 86  |
| 3.10.5   | The Initial Fluid Velocity Initial Condition | 86  |
| 3.10.6   | Initial Fluid Dilatation                     | 87  |
| 3.10.7   | The Prestrain Initial Condition              | 87  |
| 3.10.8   | Fluid Pressure Initial Condition             | 88  |
| 3.10.9   | Initial Shell Fluid Pressure                 | 88  |
| 3.10.10  | Initial Concentration                        | 88  |
| 3.10.11  | Initial Shell Concentration                  | 88  |
| 3.10.12  | Rigid Kinematics                             | 89  |
| 3.11     | Boundary Section                             | 90  |
| 3.11.1   | Zero Displacement                            | 90  |
| 3.11.2   | Zero Shell Displacement                      | 90  |
| 3.11.3   | Zero Fluid Pressure                          | 91  |
| 3.11.4   | Zero Concentration                           | 91  |
| 3.11.5   | Zero Fluid Dilatation                        | 91  |
| 3.11.6   | Prescribed Displacement                      | 91  |
| 3.11.7   | Prescribed Shell Displacement                | 92  |
| 3.11.8   | Prescribed Fluid Pressure                    | 93  |
| 3.11.9   | Prescribed Concentration                     | 93  |
| 3.11.10  | Prescribed Deformation                       | 94  |
| 3.11.11  | Prescribed Fluid Dilatation                  | 94  |
| 3.11.12  | Prescribed Fluid Velocity                    | 95  |
| 3.11.13  | Prescribed Shell Fluid Pressure              | 95  |
| 3.11.14  | Normal Displacement                          | 96  |
| 3.11.15  | Rigid Deformation                            | 96  |
| 3.11.16  | Rigid                                        | 97  |
| 3.11.17  | Multiphasic Fluid Pressure                   | 97  |
| 3.11.18  | Fluid Pressure                               | 97  |
| 3.11.19  | Fluid Resistance                             | 97  |
| 3.11.20  | Fluid RC                                     | 98  |
| 3.11.21  | Fluid RCR                                    | 98  |
| 3.11.22  | Linear Constraints                           | 99  |
| 3.11.23  | Matching OSM Coefficient                     | 99  |
| 3.12     | Rigid Section                                | 100 |
| 3.12.1   | Rigid Constraints                            | 100 |
| 3.12.1.1 | Rigid_fixed Constraint                       | 100 |
| 3.12.1.2 | Rigid_displacement Constraint                | 101 |
| 3.12.1.3 | Rigid_rotation Constraint                    | 101 |

|                                           |     |
|-------------------------------------------|-----|
| 3.12.1.4 Rigid_rotation_vector Constraint | 102 |
| 3.12.1.5 Rigid_euler_angles Constraint    | 102 |
| 3.12.2 Rigid Initial Conditions           | 103 |
| 3.12.2.1 Initial Rigid Velocity           | 103 |
| 3.12.2.2 Initial Rigid Angular Velocity   | 104 |
| 3.12.3 Rigid Loads                        | 104 |
| 3.12.3.1 Rigid_force Load                 | 104 |
| 3.12.3.2 Rigid_moment Load                | 105 |
| 3.12.3.3 Rigid_follower_force Load        | 105 |
| 3.12.3.4 Rigid_follower_moment Load       | 106 |
| 3.12.3.5 Rigid_cable Load                 | 106 |
| 3.12.4 Rigid Connectors                   | 107 |
| 3.12.4.1 Rigid Spherical Joint            | 109 |
| 3.12.4.2 Rigid Revolute Joint             | 110 |
| 3.12.4.3 Rigid Prismatic Joint            | 111 |
| 3.12.4.4 Rigid Cylindrical Joint          | 113 |
| 3.12.4.5 Rigid Planar Joint               | 114 |
| 3.12.4.6 Rigid Lock Joint                 | 116 |
| 3.12.4.7 Rigid Spring                     | 117 |
| 3.12.4.8 Rigid Damper                     | 117 |
| 3.12.4.9 Rigid Angular Damper             | 118 |
| 3.12.4.10 Rigid Contractile Force         | 118 |
| 3.13 Loads Section                        | 119 |
| 3.13.1 Nodal Loads                        | 119 |
| 3.13.1.1 nodal_load                       | 119 |
| 3.13.1.2 nodal_force                      | 120 |
| 3.13.1.3 nodal_target_force               | 120 |
| 3.13.1.4 Nodal Fluidflux                  | 120 |
| 3.13.2 Surface Loads                      | 120 |
| 3.13.2.1 Pressure Load                    | 121 |
| 3.13.2.2 Traction Load                    | 121 |
| 3.13.2.3 Surface Force                    | 122 |
| 3.13.2.4 Bearing Load                     | 122 |
| 3.13.2.5 Pipette Aspiration               | 123 |
| 3.13.2.6 Mixture Normal Traction          | 123 |
| 3.13.2.7 Fluid Flux                       | 125 |
| 3.13.2.8 Multiphasic Solute Flux          | 126 |
| 3.13.2.9 Solute Flux                      | 126 |
| 3.13.2.10 Multiphasic Solute Natural Flux | 127 |
| 3.13.2.11 Heat Flux                       | 127 |
| 3.13.2.12 Convective Heat Flux            | 128 |
| 3.13.2.13 Fluid Traction                  | 128 |
| 3.13.2.14 Fluid Pressure                  | 128 |
| 3.13.2.15 Fluid Normal Traction           | 128 |
| 3.13.2.16 Fluid Backflow Stabilization    | 129 |
| 3.13.2.17 Fluid Tangential Stabilization  | 129 |
| 3.13.2.18 Fluid Tangential Damping        | 129 |
| 3.13.2.19 Fluid Viscous Traction          | 129 |

|           |                                         |     |
|-----------|-----------------------------------------|-----|
| 3.13.2.20 | Fluid Normal Velocity                   | 130 |
| 3.13.2.21 | Fluid Velocity                          | 131 |
| 3.13.2.22 | Fluid Rotational Velocity               | 131 |
| 3.13.2.23 | Fluid-FSI Traction                      | 132 |
| 3.13.2.24 | Biphasic-FSI Traction                   | 132 |
| 3.13.2.25 | Multiphasic-FSI Traction                | 132 |
| 3.13.2.26 | Solute Backflow Stabilization           | 133 |
| 3.13.2.27 | Fluid Mixture Viscous Traction          | 133 |
| 3.13.2.28 | Pressure Stabilization                  | 133 |
| 3.13.3    | Body Loads                              | 133 |
| 3.13.3.1  | Constant Body Force                     | 134 |
| 3.13.3.2  | Non-Constant Body Force                 | 134 |
| 3.13.3.3  | Centrifugal Body Force                  | 134 |
| 3.13.3.4  | Heat Source                             | 134 |
| 3.13.3.5  | Surface Attraction                      | 135 |
| 3.13.3.6  | Moving frame                            | 135 |
| 3.13.3.7  | Mass Damping                            | 136 |
| 3.13.3.8  | Fluid Centrifugal Force                 | 136 |
| 3.13.3.9  | Fluid Moving Frame                      | 137 |
| 3.14      | Contact Section                         | 138 |
| 3.14.1    | Sliding Interfaces                      | 139 |
| 3.14.2    | Biphasic Contact                        | 145 |
| 3.14.3    | Biphasic-Solute and Multiphasic Contact | 145 |
| 3.14.4    | Rigid Wall Interfaces                   | 146 |
| 3.14.5    | Tied Interfaces                         | 147 |
| 3.14.6    | Tied Elastic Interfaces                 | 147 |
| 3.14.7    | Tied Biphasic Interfaces                | 147 |
| 3.14.8    | Tied Multiphasic Interfaces             | 148 |
| 3.14.9    | Sticky Interfaces                       | 148 |
| 3.14.10   | Contact Potential                       | 149 |
| 3.15      | Constraints Section                     | 151 |
| 3.15.1    | Symmetry Plane                          | 151 |
| 3.15.2    | Frictionless Fluid Wall                 | 151 |
| 3.15.3    | Fixed Normal Displacement               | 152 |
| 3.15.4    | Normal Fluid Velocity Constraint        | 152 |
| 3.15.5    | Uniform Fluid Velocity Constraint       | 153 |
| 3.15.6    | The Prestrain Update Rules              | 153 |
| 3.15.6.1  | Using Update rules                      | 153 |
| 3.15.6.2  | prestrain update rule                   | 154 |
| 3.15.6.3  | The in-situ stretch update rule         | 154 |
| 3.15.7    | Node Distance                           | 155 |
| 3.16      | Discrete Section                        | 156 |
| 3.16.1    | Discrete Materials                      | 156 |
| 3.16.1.1  | Linear Spring                           | 156 |
| 3.16.1.2  | Nonlinear spring                        | 156 |
| 3.16.1.3  | Hill                                    | 157 |
| 3.16.2    | Discrete Section                        | 158 |
| 3.16.3    | Rigid Cable                             | 159 |

|                                                         |            |
|---------------------------------------------------------|------------|
| 3.17 Step Section                                       | 160        |
| 3.18 LoadData Section                                   | 161        |
| 3.18.1 The loadcurve controller                         | 161        |
| 3.18.2 The math controller                              | 162        |
| 3.18.3 The math-interval controller                     | 162        |
| 3.18.4 The PID controller                               | 163        |
| 3.19 Output Section                                     | 164        |
| 3.19.1 Logfile                                          | 164        |
| 3.19.2 Plotfile                                         | 167        |
| <b>4 Materials</b>                                      | <b>169</b> |
| 4.1 Elastic Solids                                      | 169        |
| 4.1.1 Specifying Fiber Orientation or Material Axes     | 169        |
| 4.1.1.1 Transversely Isotropic Materials                | 169        |
| 4.1.1.2 Orthotropic Materials                           | 172        |
| 4.1.2 Uncoupled Materials                               | 174        |
| 4.1.2.1 Arruda-Boyce                                    | 176        |
| 4.1.2.2 Ellipsoidal Fiber Distribution Uncoupled        | 177        |
| 4.1.2.3 Ellipsoidal Fiber Distribution Mooney-Rivlin    | 178        |
| 4.1.2.4 Ellipsoidal Fiber Distribution Veronda-Westmann | 179        |
| 4.1.2.5 Fung Orthotropic                                | 180        |
| 4.1.2.6 Gent Material                                   | 182        |
| 4.1.2.7 Uncoupled Holmes-Mow                            | 182        |
| 4.1.2.8 Holzapfel-Gasser-Ogden                          | 183        |
| 4.1.2.9 Mooney-Rivlin                                   | 184        |
| 4.1.2.10 Muscle Material                                | 185        |
| 4.1.2.11 Ogden                                          | 187        |
| 4.1.2.12 Tendon Material                                | 188        |
| 4.1.2.13 Tension-Compression Nonlinear Orthotropic      | 189        |
| 4.1.2.14 Transversely Isotropic Mooney-Rivlin           | 190        |
| 4.1.2.15 Transversely Isotropic Veronda-Westmann        | 192        |
| 4.1.2.16 Uncoupled Solid Mixture                        | 193        |
| 4.1.2.17 Veronda-Westmann                               | 194        |
| 4.1.2.18 Mooney-Rivlin Von Mises Distributed Fibers     | 195        |
| 4.1.2.19 Isotropic Lee-Sacks uncoupled                  | 197        |
| 4.1.2.20 Yeoh Material                                  | 198        |
| 4.1.3 Fiber Active Contraction                          | 199        |
| 4.1.3.1 Active Contraction                              | 199        |
| 4.1.3.2 Force-Velocity Active Contraction               | 200        |
| 4.1.4 Unconstrained Materials                           | 202        |
| 4.1.4.1 Arruda-Boyce Unconstrained                      | 202        |
| 4.1.4.2 Carter-Hayes                                    | 203        |
| 4.1.4.3 Cell Growth                                     | 205        |
| 4.1.4.4 Kinematic Growth                                | 207        |
| 4.1.4.5 Cubic CLE                                       | 207        |
| 4.1.4.6 Donnan Equilibrium Swelling                     | 209        |
| 4.1.4.7 Ellipsoidal Fiber Distribution                  | 211        |
| 4.1.4.8 Ellipsoidal Fiber Distribution Neo-Hookean      | 212        |



|          |                                                                 |     |
|----------|-----------------------------------------------------------------|-----|
| 4.1.4.9  | Ellipsoidal Fiber Distribution with Donnan Equilibrium Swelling | 213 |
| 4.1.4.10 | Fung Orthotropic Compressible                                   | 214 |
| 4.1.4.11 | Gent Compressible                                               | 215 |
| 4.1.4.12 | Isotropic Hencky                                                | 215 |
| 4.1.4.13 | Holmes-Mow                                                      | 216 |
| 4.1.4.14 | Holzappel-Gasser-Ogden Unconstrained                            | 218 |
| 4.1.4.15 | Isotropic Elastic                                               | 219 |
| 4.1.4.16 | Orthotropic Elastic                                             | 220 |
| 4.1.4.17 | Orthotropic CLE                                                 | 221 |
| 4.1.4.18 | Osmotic Pressure from Virial Expansion                          | 222 |
| 4.1.4.19 | Natural Neo-Hookean                                             | 223 |
| 4.1.4.20 | Neo-Hookean                                                     | 224 |
| 4.1.4.21 | Coupled Mooney-Rivlin                                           | 225 |
| 4.1.4.22 | Coupled Veronda-Westmann                                        | 226 |
| 4.1.4.23 | Ogden Unconstrained                                             | 227 |
| 4.1.4.24 | Perfect Osmometer Equilibrium Osmotic Pressure                  | 228 |
| 4.1.4.25 | Porous Neo-Hookean                                              | 230 |
| 4.1.4.26 | Lung Material                                                   | 232 |
| 4.1.4.27 | Solid Mixture                                                   | 233 |
| 4.1.4.28 | Spherical Fiber Distribution                                    | 234 |
| 4.1.4.29 | Spherical Fiber Distribution from Solid-Bound Molecule          | 235 |
| 4.1.4.30 | Coupled Transversely Isotropic Mooney-Rivlin                    | 237 |
| 4.1.4.31 | Coupled Transversely Isotropic Veronda-Westmann                 | 238 |
| 4.1.4.32 | Large Poisson's Ratio Ligament                                  | 239 |
| 4.1.4.33 | Shenoy-Wang material                                            | 241 |
| 4.2      | Fibers                                                          | 242 |
| 4.2.1    | Unconstrained Fiber Models                                      | 243 |
| 4.2.1.1  | Fiber with Exponential-Power Law                                | 244 |
| 4.2.1.2  | Fiber with Neo-Hookean Law                                      | 246 |
| 4.2.1.3  | Fiber with Natural Neo-Hookean Law                              | 247 |
| 4.2.1.4  | Fiber with Toe-Linear Response                                  | 248 |
| 4.2.1.5  | Fiber with Exp-Pow-Linear Response                              | 249 |
| 4.2.1.6  | Fiber Exp-Linear                                                | 250 |
| 4.2.1.7  | Fiber as Entropy Chain                                          | 252 |
| 4.2.2    | Uncoupled Fiber Models                                          | 254 |
| 4.2.2.1  | Fiber with Exponential-Power Law, Uncoupled Formulation         | 255 |
| 4.2.2.2  | Fiber Kiouisis Uncoupled                                        | 257 |
| 4.2.2.3  | Fiber with Toe-Linear Response, Uncoupled Formulation           | 258 |
| 4.2.2.4  | Uncoupled Fiber Exp-Linear                                      | 258 |
| 4.2.2.5  | Uncoupled Fiber as Entropy Chain                                | 260 |
| 4.3      | Continuous Fiber Distribution                                   | 261 |
| 4.3.1    | Unconstrained Continuous Fiber Distribution                     | 262 |
| 4.3.2    | Uncoupled Continuous Fiber Distribution                         | 263 |
| 4.3.3    | Fiber ODF Constitutive Model                                    | 264 |
| 4.3.4    | Distribution                                                    | 266 |
| 4.3.4.1  | Spherical                                                       | 267 |
| 4.3.4.2  | Ellipsoidal                                                     | 268 |
| 4.3.4.3  | $\pi$ -Periodic von Mises Distribution                          | 269 |

|          |                                                             |     |
|----------|-------------------------------------------------------------|-----|
| 4.3.4.4  | Circular                                                    | 270 |
| 4.3.4.5  | Elliptical                                                  | 271 |
| 4.3.4.6  | von Mises Distribution                                      | 273 |
| 4.3.5    | Scheme                                                      | 274 |
| 4.3.5.1  | Gauss-Kronrod Trapezoidal Rule                              | 275 |
| 4.3.5.2  | Finite Element Integration Rule                             | 276 |
| 4.3.5.3  | Trapezoidal Rule                                            | 277 |
| 4.4      | Viscoelastic Solids                                         | 278 |
| 4.4.1    | Uncoupled Viscoelastic Materials                            | 278 |
| 4.4.2    | Unconstrained Viscoelastic Materials                        | 279 |
| 4.5      | Reactive Viscoelastic Solid                                 | 280 |
| 4.5.1    | Relaxation Functions                                        | 282 |
| 4.5.1.1  | Exponential                                                 | 282 |
| 4.5.1.2  | Exponential Distortional                                    | 283 |
| 4.5.1.3  | Exponential Distortional User-Specified                     | 283 |
| 4.5.1.4  | Exponential Continuous Spectrum                             | 284 |
| 4.5.1.5  | Exponential Continuous Spectrum Distortional User-Specified | 284 |
| 4.5.1.6  | Fung                                                        | 285 |
| 4.5.1.7  | Malkin                                                      | 285 |
| 4.5.1.8  | Malkin Distortional                                         | 286 |
| 4.5.1.9  | Malkin Distortional User-Specified                          | 286 |
| 4.5.1.10 | Park                                                        | 286 |
| 4.5.1.11 | Park Distortional                                           | 287 |
| 4.5.1.12 | Park Distortional User-Specified                            | 287 |
| 4.5.1.13 | Power                                                       | 287 |
| 4.5.1.14 | Power Distortional                                          | 288 |
| 4.5.1.15 | Power                                                       | 288 |
| 4.5.1.16 | Prony                                                       | 288 |
| 4.5.2    | Weak Bond Recruitment Functions                             | 290 |
| 4.5.2.1  | Power                                                       | 290 |
| 4.5.2.2  | Exponential                                                 | 290 |
| 4.5.2.3  | Polynomial                                                  | 291 |
| 4.5.2.4  | User                                                        | 291 |
| 4.5.2.5  | Log-Normal                                                  | 292 |
| 4.5.2.6  | Weibull                                                     | 292 |
| 4.5.2.7  | Quintic Polynomial                                          | 293 |
| 4.5.2.8  | Gamma                                                       | 293 |
| 4.6      | Reactive Damage Mechanics                                   | 294 |
| 4.6.1    | General Specification of Damage Materials                   | 295 |
| 4.6.2    | Cumulative Distribution Functions                           | 296 |
| 4.6.2.1  | Simo                                                        | 297 |
| 4.6.2.2  | Log-Normal                                                  | 298 |
| 4.6.2.3  | Weibull                                                     | 299 |
| 4.6.2.4  | Quintic Polynomial                                          | 300 |
| 4.6.2.5  | Gamma                                                       | 301 |
| 4.6.2.6  | Step                                                        | 302 |
| 4.6.2.7  | User                                                        | 302 |
| 4.6.3    | Damage or Yield Criterion                                   | 304 |

|          |                                                          |     |
|----------|----------------------------------------------------------|-----|
| 4.6.3.1  | Simo                                                     | 304 |
| 4.6.3.2  | Strain Energy Density                                    | 304 |
| 4.6.3.3  | Specific Strain Energy                                   | 304 |
| 4.6.3.4  | Von Mises Stress                                         | 304 |
| 4.6.3.5  | Maximum Shear Stress                                     | 305 |
| 4.6.3.6  | Maximum Normal Stress                                    | 305 |
| 4.6.3.7  | Drucker Shear Stress                                     | 305 |
| 4.6.3.8  | Drucker-Prager Criterion                                 | 306 |
| 4.6.3.9  | Deshpande-Fleck Criterion                                | 306 |
| 4.6.3.10 | Maximum Normal Lagrange Strain                           | 307 |
| 4.6.3.11 | Octahedral Shear Strain                                  | 307 |
| 4.7      | Reactive Fatigue                                         | 308 |
| 4.7.1    | General Specification of Reactive Fatigue Material       | 308 |
| 4.8      | Reactive Plasticity                                      | 310 |
| 4.8.1    | Kinematic “Hardening” Response                           | 310 |
| 4.8.2    | Plastic Yield Surface                                    | 310 |
| 4.8.3    | Plastic Flow Curve                                       | 311 |
| 4.8.3.1  | User-Specified Flow Curve                                | 311 |
| 4.8.3.2  | Mathematical Expression for Flow Curve                   | 311 |
| 4.8.3.3  | Custom Flow Curve                                        | 312 |
| 4.8.4    | Specification of Reactive Elasto-Plastic Solid           | 314 |
| 4.9      | Reactive Elasto-Plastic Damage Mechanics                 | 318 |
| 4.9.1    | Theoretical Formulation                                  | 318 |
| 4.9.1.1  | Stress and Damage                                        | 319 |
| 4.9.1.2  | Damage Measures                                          | 319 |
| 4.9.2    | Specification of Reactive Elasto-Plastic Damage Solid    | 320 |
| 4.10     | Reactive Viscoplasticity                                 | 322 |
| 4.11     | Reactive Viscoelasticity and Viscoplasticity with Damage | 323 |
| 4.12     | Viscoelastic Damage                                      | 325 |
| 4.13     | Multigeneration Solids                                   | 327 |
| 4.13.1   | General Specification of Multigeneration Solids          | 327 |
| 4.14     | Biphasic Materials                                       | 329 |
| 4.14.1   | General Specification of Biphasic Materials              | 330 |
| 4.14.2   | Permeability Materials                                   | 331 |
| 4.14.2.1 | Constant Isotropic Permeability                          | 332 |
| 4.14.2.2 | Exponential Isotropic Permeability                       | 333 |
| 4.14.2.3 | Holmes-Mow                                               | 334 |
| 4.14.2.4 | Referentially Isotropic Permeability                     | 335 |
| 4.14.2.5 | Referentially Orthotropic Permeability                   | 336 |
| 4.14.2.6 | Referentially Transversely Isotropic Permeability        | 338 |
| 4.14.3   | Fluid Supply Materials                                   | 340 |
| 4.14.3.1 | Starling Equation                                        | 341 |
| 4.15     | Biphasic-Solute Materials                                | 342 |
| 4.15.1   | Guidelines for Biphasic-Solute Analyses                  | 344 |
| 4.15.1.1 | Prescribed Boundary Conditions                           | 344 |
| 4.15.1.2 | Prescribed Initial Conditions                            | 344 |
| 4.15.2   | General Specification of Biphasic-Solute Materials       | 345 |
| 4.15.3   | Diffusivity Materials                                    | 347 |

|          |                                                                |     |
|----------|----------------------------------------------------------------|-----|
| 4.15.3.1 | Constant Isotropic Diffusivity                                 | 347 |
| 4.15.3.2 | Constant Orthotropic Diffusivity                               | 348 |
| 4.15.3.3 | Referentially Isotropic Diffusivity                            | 349 |
| 4.15.3.4 | Referentially Orthotropic Diffusivity                          | 350 |
| 4.15.3.5 | Albro Isotropic Diffusivity                                    | 352 |
| 4.15.4   | Solubility Materials                                           | 353 |
| 4.15.4.1 | Constant Solubility                                            | 353 |
| 4.15.5   | Osmotic Coefficient Materials                                  | 354 |
| 4.15.5.1 | Constant Osmotic Coefficient                                   | 354 |
| 4.16     | Triphasic and Multiphasic Materials                            | 355 |
| 4.16.1   | Guidelines for Multiphasic Analyses                            | 358 |
| 4.16.1.1 | Initial State of Swelling                                      | 358 |
| 4.16.1.2 | Prescribed Boundary Conditions                                 | 359 |
| 4.16.1.3 | Prescribed Initial Conditions                                  | 359 |
| 4.16.1.4 | Prescribed Effective Solute Flux                               | 359 |
| 4.16.1.5 | Prescribed Electric Current Density                            | 359 |
| 4.16.1.6 | Electrical Grounding                                           | 360 |
| 4.16.1.7 | Neglecting Osmotic Effects                                     | 360 |
| 4.16.1.8 | Solutes as 'Solid-Bound Molecules'                             | 360 |
| 4.16.2   | General Specification of Multiphasic Materials                 | 362 |
| 4.16.3   | Solvent Supply Materials                                       | 365 |
| 4.16.3.1 | Starling Equation                                              | 366 |
| 4.17     | Chemical Reactions                                             | 367 |
| 4.17.1   | Guidelines for Chemical Reaction Analyses                      | 367 |
| 4.17.2   | General Specification for Chemical Reactions                   | 370 |
| 4.17.3   | Chemical Reaction Materials                                    | 371 |
| 4.17.3.1 | Law of Mass Action for Forward Reactions                       | 371 |
| 4.17.3.2 | Law of Mass Action for Reversible Reactions                    | 372 |
| 4.17.3.3 | Michaelis-Menten Reaction                                      | 373 |
| 4.17.4   | Specific Reaction Rate Materials                               | 374 |
| 4.17.4.1 | Constant Reaction Rate                                         | 375 |
| 4.17.4.2 | Huiskes Reaction Rate                                          | 376 |
| 4.18     | Rigid Body                                                     | 378 |
| 4.19     | Active Contraction                                             | 379 |
| 4.19.1   | Contraction in Mixtures of Uncoupled Materials                 | 379 |
| 4.19.1.1 | Uncoupled Prescribed Uniaxial Active Contraction               | 380 |
| 4.19.1.2 | Uncoupled Prescribed Transversely Isotropic Active Contraction | 380 |
| 4.19.1.3 | Uncoupled Prescribed Isotropic Active Contraction              | 381 |
| 4.19.1.4 | Prescribed Fiber Stress                                        | 382 |
| 4.19.2   | Contraction in Mixtures of Unconstrained Materials             | 384 |
| 4.19.2.1 | Prescribed Uniaxial Active Contraction                         | 384 |
| 4.19.2.2 | Prescribed Transversely Isotropic Active Contraction           | 384 |
| 4.19.2.3 | Prescribed Isotropic Active Contraction                        | 385 |
| 4.19.2.4 | Prescribed Fiber Stress                                        | 386 |
| 4.20     | Viscous Fluids                                                 | 387 |
| 4.20.1   | General Specification of Fluid Materials                       | 390 |
| 4.20.2   | Viscous Fluid Materials                                        | 391 |

|          |                                                  |            |
|----------|--------------------------------------------------|------------|
| 4.20.2.1 | Newtonian Fluid                                  | 392        |
| 4.20.2.2 | Bingham Fluid                                    | 393        |
| 4.20.2.3 | Carreau Model                                    | 394        |
| 4.20.2.4 | Carreau-Yasuda Model                             | 395        |
| 4.20.2.5 | Powell-Eyring Model                              | 396        |
| 4.20.2.6 | Cross Model                                      | 397        |
| 4.20.2.7 | Quemada Model                                    | 398        |
| 4.20.3   | General Specification of Fluid-FSI Materials     | 399        |
| 4.20.4   | General Specification of Biphasic-FSI Materials  | 399        |
| 4.20.5   | General Specification of Fluid-Solutes Materials | 400        |
| 4.21     | Prestrain material                               | 401        |
| 4.21.1   | Introduction                                     | 401        |
| 4.21.2   | The Prestrain Material                           | 403        |
| 4.21.2.1 | prestrain gradient                               | 403        |
| 4.21.2.2 | in-situ stretch                                  | 404        |
| 4.22     | Continuous-Discontinuous Damage                  | 405        |
| 4.22.1   | Damage Fiber Power                               | 407        |
| 4.22.2   | Damage Fiber Exponential                         | 408        |
| 4.22.3   | Damage Fiber exp-linear                          | 409        |
| 4.23     | First-Order Homogenization                       | 410        |
| 4.23.1   | Macro model definition                           | 410        |
| 4.23.2   | RVE model definition                             | 411        |
| <b>5</b> | <b>Restart Input file</b>                        | <b>413</b> |
| 5.1      | Introduction                                     | 413        |
| 5.2      | The Archive Section                              | 413        |
| 5.3      | The Control Section                              | 414        |
| 5.4      | The LoadData Section                             | 414        |
| 5.5      | The Step Section                                 | 414        |
| 5.6      | Example                                          | 415        |
| 5.6.1    | Example 1                                        | 415        |
| 5.6.2    | Example 2                                        | 415        |
| <b>6</b> | <b>Multi-Step Analysis</b>                       | <b>417</b> |
| 6.1      | The Step Section                                 | 417        |
| 6.1.1    | Boundary Conditions                              | 418        |
| 6.1.2    | Relative Boundary Conditions                     | 418        |
| 6.2      | An Example                                       | 419        |
| <b>7</b> | <b>Parameter Optimization</b>                    | <b>421</b> |
| 7.1      | Optimization Input File                          | 421        |
| 7.1.1    | Task Section                                     | 422        |
| 7.1.2    | Options Section                                  | 422        |
| 7.1.3    | Parameters Section                               | 424        |
| 7.1.4    | Objective Section                                | 424        |
| 7.1.4.1  | The data-fit model                               | 424        |
| 7.1.4.2  | The target model                                 | 426        |
| 7.1.4.3  | The element-data model                           | 426        |

|          |                                                                    |            |
|----------|--------------------------------------------------------------------|------------|
| 7.1.4.4  | The node-data model                                                | 427        |
| 7.1.5    | Constraints Section                                                | 427        |
| 7.2      | Running a Parameter Optimization                                   | 427        |
| 7.3      | An Example Input File                                              | 427        |
| <b>8</b> | <b>Troubleshooting</b>                                             | <b>429</b> |
| 8.1      | Before You Run Your Model                                          | 429        |
| 8.1.1    | The Finite Element Mesh                                            | 429        |
| 8.1.2    | Materials                                                          | 430        |
| 8.1.3    | Boundary Conditions                                                | 430        |
| 8.2      | Debugging a Model                                                  | 431        |
| 8.3      | Common Issues                                                      | 431        |
| 8.3.1    | Inverted elements                                                  | 431        |
| 8.3.1.1  | Material instability                                               | 432        |
| 8.3.1.2  | Time step too large                                                | 432        |
| 8.3.1.3  | Elements too distorted                                             | 432        |
| 8.3.1.4  | Shells are too thick                                               | 432        |
| 8.3.1.5  | Rigid body modes                                                   | 432        |
| 8.3.2    | Failure to converge                                                | 432        |
| 8.3.2.1  | No loads applied                                                   | 433        |
| 8.3.2.2  | Convergence Tolerance Too Tight                                    | 433        |
| 8.3.2.3  | Forcing convergence                                                | 433        |
| 8.3.2.4  | Problems due to Contact                                            | 434        |
| 8.4      | Guidelines for Dynamic Analyses                                    | 434        |
| 8.4.1    | Implicit Dynamics Solver                                           | 434        |
| 8.4.2    | False Transient Analysis                                           | 434        |
| 8.5      | Guidelines for Contact Problems                                    | 435        |
| 8.5.1    | The penalty method                                                 | 435        |
| 8.5.2    | Augmented Lagrangian Method                                        | 435        |
| 8.5.3    | Initial Separation                                                 | 436        |
| 8.6      | Cautionary Note for Steady-State Biphasic and Multiphasic Analyses | 436        |
| 8.7      | Guidelines for Multiphasic Analyses                                | 436        |
| 8.7.1    | Initial State of Swelling                                          | 436        |
| 8.7.2    | Prescribed Boundary Conditions                                     | 437        |
| 8.7.3    | Prescribed Initial Conditions                                      | 438        |
| 8.7.4    | Prescribed Effective Solute Flux                                   | 438        |
| 8.7.5    | Prescribed Electric Current Density                                | 438        |
| 8.7.6    | Electrical Grounding                                               | 438        |
| 8.8      | Guidelines for Fluid Analyses                                      | 439        |
| 8.8.1    | Degrees of Freedom and Boundary Conditions                         | 439        |
| 8.8.2    | Biased Meshes for Boundary Layers                                  | 440        |
| 8.8.3    | Computational Efficiency: Broyden's Method                         | 441        |
| 8.8.4    | Dynamic versus Steady-State Analyses                               | 441        |
| 8.8.5    | Isothermal Compressible Flow versus Acoustics                      | 441        |
| 8.8.6    | Fluid-Structure Interactions                                       | 442        |
| 8.8.7    | Fluid-Solutes Analyses                                             | 443        |
| 8.9      | Understanding the Solution                                         | 446        |
| 8.9.1    | Mesh convergence                                                   | 446        |

|           |                                                |            |
|-----------|------------------------------------------------|------------|
| 8.9.2     | Constraint enforcement                         | 446        |
| 8.10      | Guidelines for Using Prestrain                 | 446        |
| 8.11      | Guidelines for using the CG-solid solver       | 447        |
| 8.12      | Limitations of FEBio                           | 448        |
| 8.12.1    | Geometrical instabilities                      | 449        |
| 8.12.2    | Material instabilities                         | 449        |
| 8.12.3    | Remeshing                                      | 449        |
| 8.12.4    | Force-driven Problems                          | 449        |
| 8.12.5    | Solutions obtained on Multi-processor Machines | 450        |
| 8.13      | Where to Get More Help                         | 450        |
| <b>9</b>  | <b>Configuration File</b>                      | <b>451</b> |
| 9.1       | Overview                                       | 451        |
| 9.2       | Configuring Linear Solvers                     | 452        |
| 9.2.1     | Pardiso                                        | 453        |
| 9.2.2     | Skyline                                        | 453        |
| 9.2.3     | MKL_DSS                                        | 453        |
| 9.2.4     | FGMRES                                         | 453        |
| 9.2.5     | CG                                             | 455        |
| 9.2.6     | BoomerAMG                                      | 455        |
| 9.2.7     | Schur                                          | 457        |
| 9.2.8     | Accelerate                                     | 458        |
| 9.2.9     | Pardiso_64                                     | 458        |
| <b>10</b> | <b>FEBio Plugins</b>                           | <b>459</b> |
| 10.1      | Using Plugins                                  | 459        |
| 10.2      | Error Messages                                 | 460        |
| <b>A</b>  | <b>Heterogeneous model parameters</b>          | <b>461</b> |
| A.1       | Math parameters                                | 461        |
| A.2       | Mapped parameters                              | 462        |
| <b>B</b>  | <b>Referencing Parameters</b>                  | <b>463</b> |
| <b>C</b>  | <b>Math Expression</b>                         | <b>465</b> |
| C.1       | Functions                                      | 465        |
| C.2       | Constants                                      | 466        |
| <b>D</b>  | <b>FEBio Binary Database Specification</b>     | <b>467</b> |
| D.1       | Introduction                                   | 467        |
| D.1.1     | Changes in this version                        | 467        |
| D.2       | Block Structure                                | 468        |
| D.2.1     | Overview                                       | 468        |
| D.2.2     | Parsing the FEBio plot file                    | 468        |
| D.3       | Root Section                                   | 469        |
| D.3.1     | Header Section                                 | 470        |
| D.3.2     | Dictionary Section                             | 470        |
| D.4       | Mesh Section                                   | 472        |
| D.4.1     | Node Section                                   | 472        |

|          |                                      |            |
|----------|--------------------------------------|------------|
| D.4.2    | Domain Section . . . . .             | 473        |
| D.4.3    | Surface Section . . . . .            | 475        |
| D.4.4    | Node Set Section . . . . .           | 475        |
| D.4.5    | Parts Section . . . . .              | 475        |
| D.5      | State Section . . . . .              | 477        |
| D.5.1    | State Header . . . . .               | 477        |
| D.5.2    | State Data Section . . . . .         | 477        |
| <b>E</b> | <b>FEBio Output</b>                  | <b>481</b> |
| E.1      | Plotfile Variables . . . . .         | 481        |
| E.2      | Logfile Variables . . . . .          | 489        |
| E.2.1    | Node_Data Class . . . . .            | 489        |
| E.2.2    | Face_Data Class . . . . .            | 491        |
| E.2.3    | Element_Data Class . . . . .         | 491        |
| E.2.4    | Domain Data Class . . . . .          | 497        |
| E.2.5    | Surface Data Class . . . . .         | 497        |
| E.2.6    | Rigid_Body_Data Class . . . . .      | 497        |
| E.2.7    | Rigid_Connector_Data Class . . . . . | 499        |
| E.2.8    | Model Data Class . . . . .           | 500        |
|          | <b>Bibliography</b>                  | <b>500</b> |



# Chapter 1

## Introduction

### 1.1 Overview of FEBio

FEBio is a nonlinear finite element solver that is specifically designed for biomechanical applications. It offers modeling scenarios, constitutive models and boundary conditions that are relevant to many research areas in biomechanics, thus offering a powerful tool for solving 3D problems in computational biomechanics. The software is open-source and the source code, as well as pre-compiled executables for Windows, OS-X, and Linux platforms are available for download at <http://febio.org>. This chapter presents a brief overview of the available features of FEBio.

FEBio is a multi-physics code and can solve problems in structural mechanics, biphasic and multiphasic physics, fluid mechanics, and fluid-solid interaction (FSI). Both (quasi-) static (or steady-state) and dynamic (or transient) analyses can be performed in each of the different physics modules. For instance, in the structural mechanics module, the (quasi-) static response of the system is sought in a quasi-static analysis and the effects of inertia are ignored. In a dynamic analysis, the inertial effects are included in the governing equations to calculate the time dependent response of the system. In the biphasic module, a coupled solid-fluid problem is solved. In a transient biphasic analysis the time dependent response of both the solid and the fluid phase is determined. For the steady-state analysis the final relaxed state is recovered. Similarly, for multiphasic problems, both the time dependent transient response as well as the steady-state response can be determined. For fluid analyses, dynamic and steady-state responses may be specified.

Many nonlinear constitutive models are available, allowing the user to model the often complicated biological tissue behavior. Several isotropic constitutive models are supported such as Neo-Hookean, Mooney-Rivlin, Ogden, Arruda-Boyce and Veronda-Westmann. All these models have a nonlinear stress-strain response and are objective for large deformations. In addition to the isotropic models there are several transversely isotropic and orthotropic constitutive models available. These models exhibit anisotropic behavior in a single or multiple preferred directions and are useful for representing biological tissues such as tendons, muscles, cartilage and other tissues that contain fibers. FEBio also contains a *rigid body* constitutive model. This model can be used to represent materials or structures whose deformation is negligible compared to that of other materials in the overall model. Several constitutive models are available for representing the solid phase of biphasic and multiphasic materials, which are materials that contain both a solid phase and a fluid phase. For incompressible materials FEBio employs special algorithms for enforcing the incompressibility constraint. A three-field formulation is used for tri-linear hexahedral and wedge elements. This algorithm allows the user to capture the accurate response of highly incompressible materials.

FEBio can now also solve first-order computational homogenization problems. In such problems, the response of the macro-model is determined by the averaged local response of a representative volume element (RVE). The deformation of the macro-model, and more specifically the local deformation gradient, is applied to a RVE model which in turns determines the stress (and tangent) of the macro-model.

FEBio supports a wide range of boundary conditions and loads to model interactions between materials that are relevant to problems in biomechanics. Deformable models can be connected to rigid bodies. With this feature, the user can model prescribed rotations and torques for rigid segments, thereby allowing the coupling of rigid body mechanics with deformable continuum mechanics. FEBio provides the ability to represent frictionless and frictional contact between rigid and/or deformable materials using sliding interfaces. A sliding surface is defined between two surfaces that are allowed to separate and slide across each other but are not allowed to penetrate. Variations of the sliding interface, such as tied interfaces, tied-sliding (tension-compression) and rigid walls, are available as well. As of version 1.2 it is also possible to model the fluid flow across two contacting biphasic materials. Finally, the user may specify a body force to model the effects such as, gravity, base acceleration or centripetal acceleration.

FEBio has a large library of element formulations. These include linear and quadratic tetrahedral, hexahedral and pentahedral (wedge) elements. FEBio also supports triangular quadrilateral shell elements, with linear and quadratic interpolations.

In order to facilitate the process of customizing FEBio, a plugin framework is available. Plugins allow researchers to add new materials, boundary conditions, loads, and more, without the need to edit and recompile the FEBio source code itself.

FEBio is a nonlinear implicit FE solver and does not have mesh generation capabilities. The finite element mesh, as well as all constitutive parameters and loading is defined in an input file, the format of which is described in detail in this document. This input file needs to be generated by preprocessing software. The preferred preprocessor for FEBio is called [FEBioStudio](#). FEBioStudio can convert some other formats to the FEBio input specification. For instance, NIKE3D [61] and Abaqus input files can be imported in FEBioStudio and can be exported as a FEBio input file.

## 1.2 About this document

This document is part of a set of three manuals that accompany FEBio: the User's Manual, describing how to use FEBio (this manual), a [Developer's Manual](#) for users who wish to modify or add features to the code, and a [FEBio Theory Manual](#), which describes the theory behind the FEBio algorithms.<sup>1</sup>

This document discusses how to use FEBio and describes the input file format in detail. Chapter 2 describes how to run FEBio and explains the various command line options. It also discusses the different files that are required and created by FEBio. Chapter 3 describes the format of the FEBio input file. An XML-based format is used, organizing the data in a convenient hierarchical structure. Chapter 4 gives a detailed overview of the available constitutive models. Chapter 5 discusses the restart capabilities of FEBio. The restart feature allows the user to interrupt a run and continue it at a later time, optionally making changes to the problem data. Chapter 6 describes the multi-step analysis feature, which allows the user to split up the entire analysis into several steps. Chapter 7 explains how to setup and run a parameter optimization problem using FEBio's optimization module. Chapter 8 provides helpful information for troubleshooting an FEBio model

---

<sup>1</sup>The User and Theory manuals are also available in pdf form, but the Developer's manual is only available online.

and offers guidelines that help users avoid common problems. Chapter 9 discusses the configuration file, which allow users to customize how FEBio runs on their system. For instance, users can configure the default linear solver that will be used for solving models. Chapter 10 explains how use plugins with FEBio.

## 1.3 FEBio Basics

This section provides a brief overview of how FEBio works. For more details regarding the algorithms in FEBio, please consult the [FEBio Theory Manual](#).

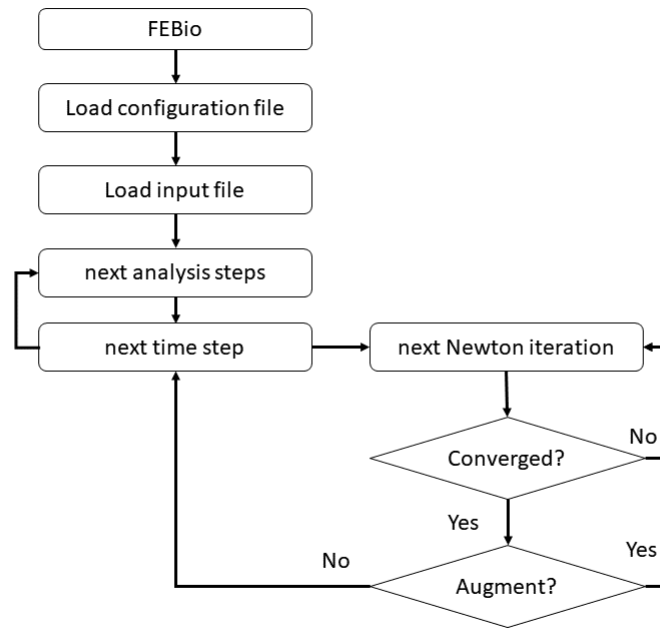
When FEBio starts, first it will load the configuration file where it will find instructions on what linear solver to use, what plugins to load, etc. Please see section 9 for more information regarding the configuration file.

Usually FEBio will read the input file that was specified on the command line next. The input file contains all the information that FEBio needs to build the FE Model and solve it. At the very least the input file will define the mesh, the materials, boundary conditions, time stepping, and analysis parameters. This document describes the structure of the FEBio input file in detail. See 3 and 4.

Next, FEBio will start solving the model defined by the input file and proceed as follows:

1. **Loop over all analysis steps.** A model can define multiple analysis steps. In each step the boundary conditions, time stepping, and analysis parameters can be modified (e.g. a two-step analysis where a model is loaded statically in the first step. In the second step the load is released and the dynamic response is sought.)
2. **For each analysis step, loop over all time steps:** In each analysis step, loads are usually applied incrementally for stability reasons over a period of time. The time parameter can represent the actual physical time (e.g. in dynamic simulations), or a pseudo-time (e.g. quasi-static loading of an elastic model.). For more on time stepping see 3.3.
3. **Solve each time step:** for each time step FEBio will solve the corresponding finite element equations. Usually they are nonlinear and thus are solved with a nonlinear solution strategy. In FEBio this is a variation of the Newton method. See section 3.3 for more information.
4. **Check convergence:** Since Newton's method is an iterative method convergence is determined by comparing the norms of the solution and residual to the user-defined values in the input file. What norms are used will depend on the physics of the problem, as well as on several user control parameters. See 3.3.
5. **Augmentation:** In models that define some type of constraint (e.g. contact, rigid joints, etc.), FEBio will do an additional calculation after a time step converges, called an augmentation. This is because FEBio does not calculate the exact Lagrange multiplier that enforce the constraints, but instead uses an iterative algorithm (called augmented Lagrangian method) to approximate the Lagrange multipliers. During the augmentation, FEBio updates the approximate Lagrange multipliers. If the updates to the multipliers are small, FEBio will terminate the time step, otherwise it will restart the time step with the updated multipliers.

See the figure below for an overview of the basic FEBio flow.



## 1.4 Units in FEBio

FEBio does not assume a specific unit system. It is up to the user to enter numbers that are defined in consistent units. For example, when entering material parameters in SI units, the user must enter all loads, contact parameters, and other boundary conditions in SI units as well. The units of all the parameters are given when they are defined in this manual. We use a generic designation of units for all the parameters using the following symbols.

| <b>Symbol</b> | <b>Name</b>         | <b>SI unit</b>                |
|---------------|---------------------|-------------------------------|
| <b>L</b>      | Length              | meter (m)                     |
| <b>M</b>      | Mass                | kilogram (kg)                 |
| <b>t</b>      | Time                | second (s)                    |
| <b>T</b>      | Temperature         | Kelvin (K)                    |
| <b>n</b>      | Amount of substance | mole (mol)                    |
| <b>F</b>      | Force               | Newton (kg·m/s <sup>2</sup> ) |
| <b>P</b>      | Pressure, stress    | Pascal (Pa=N/m <sup>2</sup> ) |
| <b>Q</b>      | Electric charge     | Coulomb (C=A·s)               |

Units are given using the bracket notation. For instance, the unit for density is **[M/L<sup>3</sup>]** and the unit for permeability is **[L<sup>4</sup>/F·t]**. When using SI units, this corresponds to units of kg/m<sup>3</sup> for density and m<sup>4</sup>/N.s for permeability, respectively. Unitless parameters are designated by empty brackets **[ ]**. The units for angles are either **[deg]** for degrees or **[rad]** for radians.

When adopting a consistent set of units, first choose a primary set of units, and then determine the remaining derived units. For example, in typical problems in solid mechanics, the primary set consists of three units. If you choose  $[M]=\text{kg}$ ,  $[L]=\text{m}$ , and  $[t]=\text{s}$ , then  $[F]=\text{N}$  and  $[P]=\text{Pa}$ . Alternatively, if you choose  $[L]=\text{mm}$ ,  $[F]=\text{N}$  and  $[T]=\text{s}$  as the primary set, then  $[P]=\text{MPa}$  (since  $1 \text{ N/mm}^2 = 10^6 \text{ N/m}^2 = 1 \text{ MPa}$ ) and  $[M]=\text{tonne}$  (tonne =  $\text{N} \cdot \text{s}^2/\text{mm}$ ). The primary set of units must be independent. For instance, in the last example, you cannot choose  $[P]$  as a primary unit as it can be expressed in terms of  $[F]$  and  $[L]$  (i.e.  $[P]=[F/L^2]$ ).

*Example:*

| Primary Units       |                                                                                |
|---------------------|--------------------------------------------------------------------------------|
| time                | s                                                                              |
| length              | mm                                                                             |
| force               | N                                                                              |
| amount of substance | nmol                                                                           |
| charge              | C                                                                              |
| temperature         | K                                                                              |
| Derived Units       |                                                                                |
| stress              | $\text{N/mm}^2$ , MPa                                                          |
| permeability        | $\text{mm}^4/\text{N} \cdot \text{s}$ , $\text{mm}^2/\text{Pa} \cdot \text{s}$ |
| diffusivity         | $\text{mm}^2/\text{s}$                                                         |
| concentration       | $\text{nmol/mm}^3$ , mM                                                        |
| charge density      | $\text{nEq/mm}^3$ , $\text{mEq/L}$                                             |
| voltage             | mV                                                                             |
| current density     | $\text{A/mm}^2$                                                                |
| current             | A                                                                              |



## Chapter 2

# Running FEBio

FEBio is a command line application which means it does not have its own Graphical User Interface (GUI) and must be run from a shell or command line. FEBio runs on several different computing platforms including Windows, Mac OSX, and many versions of Linux. The command line input and output options are described in this chapter.

### 2.1 Running FEBio on Windows

There are several ways to run FEBio on Windows. The easiest way is by simply selecting the FEBio program from the Programs menu or by double-clicking the FEBio icon in the installation folder. However, this runs FEBio with the installation folder as the working folder, and unless the FEBio input files are in this folder, you will need to know the relative or absolute path to your input files. A more practical approach is to run FEBio from a command prompt. Before you can do this, you need to know two things: how to open a command prompt and how to add the FEBio installation folder to your PATH environment variable so that you can run FEBio from any folder on your system.<sup>1</sup>

#### 2.1.1 Windows

In order to run FEBio from a command prompt, the path to the FEBio executable must be set in the PATH environment variable. The environment variables, including the PATH variable, are specified in the Environment Variables dialog box. There are a few different ways to get to this dialog box. The easiest way is to write “environment” in the Windows search bar, and select the “Edit the system’s environment variables” suggestion. This will open the “System Properties” dialog box. On the Advanced tab, click the Environment variables button. This will open the Environment Variables dialog box. Find the *path* variable and click the *Edit* button. On older versions of Windows a semicolon delimited text string appears. At the end of that string (don’t delete the current value) type a semi-colon and then the absolute path to the FEBio installation folder (e.g. C:/Program Files/FEBio/bin/). On Windows 10 or newer, the values are displayed in a list. Either click New or double-click on the first empty slot in the list and type the path the febio executable. Then click the *OK*-button on all open dialog boxes.

To open a command prompt, type *cmd* in the Windows search and press Enter. A command prompt window should appear. You can now use the *cd* (change directory) command to navigate

---

<sup>1</sup>The current FEBio installers should automatically setup the environment path so that FEBio can be run from anywhere on the file system.

to the folder that contains the FEBio input files. To run FEBio, simply type *febio4* (with or without additional arguments) and press *Enter*. Note that if you already had a command prompt open when changing the PATH environment variable, you'll need to close this window and reopen it after the changes to the environment variables are applied.

### 2.1.2 Running FEBio from Explorer

A third method of running FEBio, which often is very convenient, is to run FEBio from Windows Explorer. To do this, first open Explorer and browse to the folder that contains your FEBio input files. Next, right-click on the input file and select *Open With*. Now select *Choose default program*. A dialog box appears with a list of programs to open the input file. If FEBio is not on this list yet, click the *Browse* button. Locate the FEBio executable (e.g. in C:/Program Files/FEBio2/bin), select it and press the *Open* button. Now select FEBio in the *Open With* dialog box and press Ok.

After you have done this once, the process simplifies. After you right-click the input file, FEBio should now show up in the *Open With* menu item and can be selected immediately without having to go through all the previous steps.

## 2.2 Running FEBio on Linux or MAC

Running FEBio on Linux or Mac is as easy as opening up a shell window and typing FEBio on the command line. However, you may need to define an alias to the folder that contains the FEBio executable if you want to run FEBio from any folder on your system. Since this depends on your shell, you need to consult your Linux documentation on how to do this. E.g. if you are using c-shell, you can define an alias as follows:

```
alias febio '/path/to/febio/executable/'
```

If you don't want to define this alias every time you open a shell window, you can place it in your shell start up file (e.g. *.cshrc* for c-shell).

## 2.3 The Command Line

FEBio is started from a shell window (or the *command prompt* in Windows). The command line is the same for all platforms:

```
febio4 [-o1 [arg1] | -o2 [arg2] | ... ]
```

Where -o1, -o2 are options and *arg1*, *arg2*, ... are additional arguments. The different options (of which most are optional) are given by the following list:

A more detailed description of these options follows.

- i The -i option is used to specify the name of the input file. The input file is expected to follow the format specifications as described in Chapter 3.

*Example:*

```
> febio4 -i input.feb
```



| command line option                     | description                                                    | argument        |
|-----------------------------------------|----------------------------------------------------------------|-----------------|
| -i                                      | Specify the FEBio input file.                                  | feb file        |
| -r                                      | Specify the FEBio restart file.                                | restart file or |
| -g,-g1,-g2                              | Debug flags                                                    | -               |
| -p                                      | Specify name of the FEBio plot file.                           | plot file       |
| -o                                      | Specify the name of the FEBio log file.                        | log file        |
| -s                                      | Specify the optimization input file.                           | optimization    |
| -d                                      | Run an FEBio diagnostic                                        | diagnostic      |
| -break                                  | Set a breakpoint where FEBio will pause the run.               | breakp          |
| -dump[= <i>n</i> ]                      | Activate restart option and (optionally) set dump file name    | dump file       |
| -dump_stride[= <i>n</i> ]               | Set the dump interval.                                         | -               |
| -config                                 | Specify the FEBio configuration file to use.                   | configura       |
| -noconfig                               | Don't read configuration file.                                 | -               |
| -nosplash                               | Don't print the splash window to the screen on startup.        | -               |
| -import                                 | Import a plugin file.                                          | plugin file     |
| -silent                                 | Run FEBio in silent mode, which suppresses screen output.      | -               |
| -task[= <i>name</i> ]                   | Run an FEBio task                                              | task inp        |
| -info                                   | Print febio version information to screen or file              | (optional) outp |
| -noappend                               | Do not append the log and plot files on restart.               | -               |
| -norun                                  | Do not run FEBio. (Only process command line.)                 | -               |
| -output_negative_jacobians[= <i>n</i> ] | Specify the max nr of negative jacobians that will be printed. | -               |

Table 2.1: List of command line options.

This is the most common way to start an FEBio run. However, FEBio allows the omission of the `-i` when only a filename is given.

```
> febio4 input.feb
```

On Windows, this allows for starting FEBio by double-clicking on an input file (assuming you have chosen FEBio as the default program to open .feb files). Note that if additional options are specified on the command line the `-i` must be present.

It is also allowed to omit the .feb extension for FEBio input files. If no extension is given, FEBio will assume that .feb is the file extension. Thus, the following command will run the file *input.feb*:

```
> febio4 input
```

**-r** The `-r` option allows you to restart a previous analysis. The filename that must follow this option is an FEBio *restart input file* or a *dump file*. The restart input file and dump file are described in more detail in [2.8.4](#). The `-i` and `-r` options are mutually exclusive; only one of them may appear on the command line.

*Example:*

```
> febio4 -r file.feb
```

**-g,-g1,-g2** The `-g` options run FEBio in *debug mode*, in which case FEBio will print more information to the log and plot files. See [Section 2.8.2](#) for more information on running FEBio in debug mode.

*Example:*

```
> febio4 -i input.feb -g
```

- p** The `-p` option allows the user to specify the name of the *plot file*. The plot file is a binary file that contains the main results of the analysis. FEBio usually provides a default name for this file; however, the user can override the default name using this option. See Section 2.7 for more details on the output files generated by FEBio.

*Example:*

```
> febio4 -i input.feb -p out.xplt
```

- o** The `-o` option allows the user to set the name of the *log file*. The log file will contain a record of the screen output that was generated during a run. FEBio usually provides a default name for this file (see Section 2.7), but the user can override it with this command line option.

*Example:*

```
> febio4 -i input.feb -o out.log
```

- s** This option instructs FEBio to run a material parameter optimization on the specified input file. The optimization module is described in detail in Chapter 7. The `-s` option is followed by the optimization control file which contains among other things the parameters that need to be optimized. Note that the restart feature does not work with the optimization module.

*Example:*

```
> febio4 -i file.feb -s control.opt
```

- d** This option will run a FEBio diagnostics. A diagnostic is a special type of test that can be used to verify an implementation. For example, the *tangent diagnostic* allows users to check the consistency between the material's stress and tangent implementations.

*Example:*

```
> febio4 -d diagnostic.feb
```

- break** With this option a break point can be set which sets a time point or an event at which FEBio will interrupt the run and show the FEBio prompt. The following example sets a break point at time 1.0. FEBio will interrupt the run after the time step at time 1.0 is reached (i.e. has converged). (See section 2.8.3 for more information on setting break points.)

*Example:*

```
> febio4 -i file.feb -break 1.0
```

- dump** It is possible to restart a previous run using the restart capability in FEBio. This is useful when a run terminates unexpectedly. If that happens, the user can restart the analysis from the last converged timestep. Before this feature can be used, the user must request the creation of a *dump file*. This file will store all the information that FEBio will need to restart the analysis. FEBio will usually provide a default name for the dump file, but the `-dump` command line option allows the user to override the default name for the dump file. See Section 2.8.4 and Chapter 5 for more details on how to use the restart feature.

*Example:*

```
> febio4 -i input.feb -dump out.dmp
```

To control when FEBio will write the dumpfile, the `-dump` command can be appended by a number that defines the dump level. See section [2.8.4](#) for more information.

**-dump\_stride** Sets the interval between writing dump files. The default is 1 (although the dump level may affect this as well.) The following example will write a dump file after every 10 timesteps.

*Example:*

```
> febio4 -i input.feb -dump out.dmp -dump_stride=10
```

**-config** As of version 1.2, FEBio uses a *configuration file* to store platform specific settings. Usually FEBio assumes that the location for this configuration file is the same as the executable. However, the user can specify a different location and filename using the `-config` command line option. If the user does not have a configuration file or does not wish to use one, this can be specified using the `-noconfig` option. More details on the configuration file can be found in Section [2.5](#) and Chapter [9](#).

*Example:*

```
> febio4 -i input.feb -config C:\path\to\febio.xml
```

**-noconfig** Do not read and process the FEBio configuration file. FEBio will start with its default configuration.

**-nosplash** When the `-nosplash` command is entered on the command line, FEBio will not print the welcome message to the screen. This is useful when calling FEBio from another application and when the user wishes to suppress any screen output from FEBio. Other options for suppressing output can be set in the control section of the FEBio input file (see Section [3.3.1](#)).

*Example:*

```
> febio4 -i input.feb -nosplash
```

**-import** This command will load the plugin that is specified following this command line option. Although it is more convenient to list plugins in the FEBio configuration file, this option allows users to load a plugin from the command line, if such a need would arise.

*Example:*

```
> febio4 -import myplugin.dll
```

**-silent** When the `-silence` option is specified on the command line, FEBio will not generate any output to the screen. Unless explicitly instructed not to, FEBio will still create a log file which will have the convergence information.

*Example:*

```
> febio4 -i input.feb -silent
```

**-task** FEBio will always run a *task*. The default task is to solve a forward model defined by the input file specified using the `-i` command line option. However, other tasks are available, such as optimization and restart, as well as user-created tasks created in FEBio plugins. The name of the task is defined after an equal sign (=). An optional task control file can be specified as well. For instance, an alternative way for running an optimization problem is as follows.

*Example:*

```
> febio4 -i input.feb -task=optimize control.opt
```

**-info** Prints version information to the screen.

**-noappend** Do not append the log and plot file on restart. When restarting an analysis, FEBio will append the log and plot files with new data. However, when using this flag, new log and plot files will be generated for the run.

**-norun** Do not run FEBio. Only command line is processed. For instance, to get the version info from FEBio without running it, use the following command.

```
> febio4 -info -norun
```

**-output\_negative\_jacobians** Sets whether FEBio will output any negative Jacobians. A negative Jacobian is often an indication of a problem during the solution process and knowing where these negative Jacobians occurred may be helpful for debugging. By default, FEBio will only report that negative Jacobians happened, but won't print any other information. The command can also be appended by a number that indicates the max number of Jacobians will be printed.

```
> febio4 -i input.feb -output_negative_jacobians=100
```

## 2.4 The FEBio Prompt

The FEBio prompt is shown when you start FEBio without any command arguments (referred to as *interactive mode*), or when a run is interrupted either by the user (using `ctrl+c`<sup>2</sup>; See Section 2.8.1 for more details), or by reaching a breakpoint (referred to as *break mode*). The FEBio prompt will look something like this:

```
febio>
```

You can now enter one of the following commands. Note that some commands are only available in a specific mode (either *interactive* or *break* mode). Also see additional comments below.

*Comments:*

---

<sup>2</sup>This feature does not work on some Linux platforms and may abruptly terminate the run.

| command line option | description                                                     | argument                    |
|---------------------|-----------------------------------------------------------------|-----------------------------|
| break               | Add a breakpoint, which stops FEBio at a particular point.(1)   | breakpoint                  |
| breaks              | Print a list of breakpoints.                                    | -                           |
| clear               | Clears one or more breakpoints.                                 | breakpoint                  |
| config              | Load (or reload) configuration file.                            | configuration file          |
| cont                | Continue the current task                                       | -                           |
| conv                | Force conversion of the current time step. (2)                  | -                           |
| debug               | Toggle debug mode (3)                                           | [on/off]                    |
| events              | Print a list of events                                          | -                           |
| fail                | Force the current time step to fail. (4)                        | -                           |
| help                | Print a list of available commands                              | -                           |
| import              | Import a plugin file                                            | plugin file name            |
| out                 | Output to file the current stiffness matrix and right-hand side | [-txt]                      |
| plot                | Write the current model state to the plot file. (5)             | -                           |
| plugins             | Print a list of plugins.                                        | -                           |
| print               | Print the value of an internal variable. (6)                    | variable name               |
| quit                | Stop the current model or exit if no model is running           | -                           |
| restart             | Toggle the restart flag. (7)                                    | -                           |
| run                 | Run a febio input file. (8)                                     | (command line options)      |
| set                 | sets certain model and configuration variables. (9)             | variable name (space) value |
| svg                 | Write the sparse matrix profile to an svg file.                 | svg file name               |
| time                | Print progress time statistics.                                 | -                           |
| unload              | Unload a plugin from FEBio.                                     | plugin name                 |
| version             | Print version information.                                      | -                           |
| where               | Prints the current callback event                               | -                           |
| list                | Lists all the factory classes (10)                              | (command line options)      |

Table 2.2: List of options for the FEBio command prompt.

1. A breakpoint is a particular time point or an event at which FEBio will pause the run. See section 2.8.3 for more details on break points.
2. The *force* command is useful, for example, when a time step is having difficulty satisfying the convergence criteria. The user can then manually force the convergence of the time step. However, if the convergence difficulties are due to instabilities, forcing a time step to converge could cause the solution to become unstable or even incorrect. Also be aware that even if the solution recovers on later timesteps, the manually converged step might be incorrect.
3. The *debug* command toggles the debug flag. Adding *on* (*off*) will turn the debug mode on (resp. off). In debug mode, FEBio will store additional information to the log and plot file that could be useful in debugging the run. It is important to note that since FEBio will store all non-converged states to the plot file, this file may become very large in a short number of time steps. See Section 2.8.2 for more details on debugging.
4. If the current time step is not converging and if the auto-time stepper is enabled, the fail command will stop the current time step and retry it with a smaller time step size. If the auto-time stepper is not enabled, the fail command will simply exit the application.
5. The *plot* command is useful when you want to store the non-converged state at the current iteration. Note that this command only stores the state at the current iteration. If you turn on debug mode, all the iterations are stored to the plot file.

6. The following variables can be printed.

**nnz** Number of nonzeros in global stiffness matrix.

**time** The current simulation time

**neq** The number of equations

7. When the restart flag is set, FEBio will create a dump file at the end of each converged time step. This dump file can then later be used to restart the analysis from the last converged time step. See Section 2.8.4 and Chapter 5 for more details on FEBio's restart feature.
8. The *run* command is used to start an FEBio task. This command takes the same options that you can enter on the command line. For example, to run a file named *test.feb* from the FEBio prompt, type the following:

```
run -i test.feb
```

9. The following configuration file settings can be set from the febio command prompt.

**output\_negative\_jacobians** turn on or off the detailed output when negative jacobians are encountered.

**print\_model\_params** turn on or off the output of the parameter values of parameters that are load controlled.

**show\_warnings\_and\_errors** turn on or off the printing of warning and error messages.

To turn the option on (off), follow the set command by the variable name and then a value of 1 (0). For example,

```
set output_negative_jacobians 1
```

If you enter the set command without any additional options, a list will be printed of the current configuration settings.

10. The *list* command prints a list of all available FEBio features. Additional options can be specified to configure the output.

- m Specify the module name. (e.g. list -m solid)
- c Specify the category. (e.g. list -m FEMATERIAL\_ID)
- n Set the max nr of lines in the output (e.g. list -n 100)
- s Specify a general pattern that should match. (e.g. list -s neo)

## 2.5 The Configuration File

As of version 1.2, FEBio uses a *configuration file* to store platform-specific settings, such as the default linear solver and the list of plugins that need to be loaded at startup. The configuration file uses an xml format to store data and is detailed in Chapter 9. For backward compatibility, it is still possible to run FEBio without the configuration file. In that case, the default settings prior to version 1.2 are used.

*Example:*

```
> febio -i myfile.feb -noconfig
```

The default configuration file needs to be stored in the same location as the executable and named *febio.xml*. Alternatively, the location and the name of the file can also be specified on the command line using the `--config` option.

*Example:*

```
> febio -i myfile.feb -config /home/my/folder/FEBio/febio.xml
```

## 2.6 Using Multiple Processors

As of version 2.0, FEBio uses OpenMP to parallelize several of the finite element calculations, improving the performance considerably. Both the right-hand-side and the stiffness matrix evaluations for many types of problems have been parallelized. On a system with four processors, a speedup of 2-3 can be expected, depending on the size and type of model. Models with complex material behavior (such as EFD-type materials, biphasic, multiphasic materials, etc.) will benefit most from these parallelization efforts. In addition, FEBio implements the [MKL](#) version of the [PARDISO](#) linear solver, which is a parallel linear solver that uses OpenMP.

To use multiple processors, set the environment variable `OMP_NUM_THREADS` to the number of desired threads. You should set the number of threads to be equal or less than the number of processors on your system (Setting it higher may actually decrease performance). For example, on a system with four processors you can set the environment as follows. On Linux using the Bash shell, execute:

```
> export OMP_NUM_THREADS=4
```

Using the c-shell, execute:

```
> setenv OMP_NUM_THREADS 4
```

Or at a Windows command prompt:

```
> set OMP_NUM_THREADS=4
```

On Windows, you can add this environment variable as well from the System Properties dialog box.

**A note on repeatability** Ideally, when the same model is run repeatedly, either on the same machine or on different machines, it should produce the same convergence statistics and results. However, in practice this is not always the case and sometimes the convergence stats and even the results can differ. In most cases, the discrepancies should be small, however in some cases, and especially in models that are prone to ill-conditioning (e.g. contact), the discrepancies may be more significant. The underlying reason is that in different compute environments it can not be guaranteed that all calculations are executed in the exact same order and, due to numerical round-off, the results of these calculations will not always be the same. On a single machine, this can happen when the model is run using multiple processors. (However, running the same model on a machine with a single processor should always produce the exact same results.) This behavior can also be observed when running the same model on different machines, especially if these machines have different OS.

Similarly, although FEBio aims to be backward compatible, running the same model with different versions of FEBio, may also show discrepancies for similar reasons. In this case, the discrepancies are likely caused by algorithmic optimizations or changes in compiler settings.

## 2.7 FEBio Output

### 2.7.1 Screen output

By default, FEBio will print convergence and progress information to the screen that informs users how the analysis is progressing. The output depends on the particular module and solver that was chosen in the FEBio input file, as well as some user-defined settings, but in general provides the following information.

- **Time stepping information:** The current time step that FEBio is solving and a notification when the time step completed (or an error message why the time step failed to complete)
- **Convergence information:** for each time step, FEBio usually has to solve a nonlinear problem for which an iterative nonlinear solver is used. Several “norms” are printed to inform the user how FEBio is progressing toward the solution of the nonlinear problem.
- **Summary:** At the end of the analysis a summary is printed with heuristics that inform the user how well the model ran.

By default, all screen output will also be printed to the log file.

The summary printed at the end of a run also provides some useful timing information. Below follows an explanation of the different reported timings.

Comments:



| timing                | description                                                                          |
|-----------------------|--------------------------------------------------------------------------------------|
| Input time            | The time it took to process the FEBio input file.                                    |
| Initialization time   | The time to initialize and check the model                                           |
| Solve time            | The total time it took to solve the model. This is the sum of the following timings: |
| IO-time               | The time to write the output files (i.e. the plot, dump, and data files)             |
| reforming stiffness   | The time to reform the stiffness matrix (1)                                          |
| evaluating stiffness  | The time to evaluate the stiffness matrix (2)                                        |
| evaluating residual   | The time to evaluate the residual (3)                                                |
| model update          | The time to update the model (4)                                                     |
| QN updates            | The time to evaluate the Quasi-Newton updates (5)                                    |
| time in linear solver | The time spent in the linear solver (6)                                              |

Table 2.3: Table listing descriptions of the different timings reported at the end of an FEBio job.

1. Internally, FEBio uses a sparse-matrix storage format to store the global stiffness matrix efficiently. A matrix reformation refers to the process of calculating the structure of this sparse matrix. The structure depends mostly on the nodal connectivity and for many problems this only needs to be calculated once at the start of the run. However, for some problems the connectivity may change during an analysis and this may require a reformation of the global stiffness matrix. (For instance, changes in contacting facets.)
2. The global stiffness matrix needs to be evaluated often as it is needed by the Newton solver for finding the solution. The total time to evaluate this stiffness matrix may often be a significant portion of the total runtime, however, there are several parameters that affect this. For instance, the `max_ups` parameter sets the number of quasi-Newton updates. During these updates, the full stiffness is not re-evaluated. Instead, a rank-one update is performed, which often significantly reduces the time to evaluate the stiffness matrices.
3. The residual, which is the difference between “internal” forces (i.e. stresses) and external forces, is evaluated during each quasi-Newton iteration. Although the residual is calculated many times during an analysis, its evaluation rarely takes up a significant time of the total run time.
4. During the model update, all quantities that depend on the solution variables will be updated. This includes nodal positions, element stress, etc. This update is also called for each quasi-Newton iteration. Although the update is usually fast, the fact that it is required for every iteration may cause the total time spent in the model update to be significant.
5. The QN update refers to the process of calculating a rank-one update of the stiffness matrix. This is usually a fast process and does not take up a significant amount of time.
6. The time in the linear solver usually takes up most of the total runtime. Most of this time is spent calculating the factorization of the global stiffness matrix.

### 2.7.2 Output files

After running FEBio, two or three files are created: the *log file*, the *plot file*, and optionally the *dump file*. The log file is a text file that contains the same output (and usually more) that was written to the screen. The *plot file* contains the results of the analysis. Since this is a binary file, the results

must be analyzed using post processing software such as *FEBio Studio*. In some cases, the user may wish to request the creation of a *dump file*. This file contains temporary results of the run. If an analysis terminates unexpectedly or with an error, this file can be used to restart the analysis from the last converged time step. See Section 2.8.4 and Chapter 5 for more details. The names of these files can be specified with the command options `-p` (plot file), `-a` (dump file), `-o` (log file). If one or more of the file names following these flags are omitted, then the omitted file name(s) will be given a default name. The default file names are derived from the input file name. For example, if the input file name is *input.feb* the logfile will have the name *input.log*, the plot file is called *input.xplt* and the dump file is called *input.dmp*.

*Note 1.* The name of the log and plot file can also be specified in the FEBio input file. See Section 3.18 for more information.

*Note 2.* When running an optimization problem the name of the log file is derived from the optimization control file. See Chapter 7 for more information on running optimization problems with FEBio.

*Note 3.* FEBio may also generate additional *data files*. Although the log file is used as the default output file for data requested in the *logfile* section of the input file, users can specify the location of the output file. In that case, the data will be written to the output file instead of the logfile.

## 2.8 Advanced Options

### 2.8.1 Interrupting a Run

The user can pause the run by pressing `ctrl+c`. This will bring up the FEBio prompt, and the user can enter a command. The FEBio prompt will also be shown when FEBio reaches a break point. See Section 2.4 for a list of available commands. Note that it may take a while before the FEBio prompt is displayed after the user requests a `ctrl+c` interruption. This may be because the program is in the middle of a call to the linear solver or another time-consuming part of the analysis procedure that cannot be interrupted. NOTE: This feature may not work properly on all systems, although it should always work on Windows systems.

### 2.8.2 Debugging a Run

As stated in Section 2.3, FEBio can be run in debug-mode by specifying one of the `-g` options on the command line. When running in debug mode, FEBio performs additional checks and prints out more information to the screen and to the plot file. It will also store all non-converged states to the plot file. These non-converged states can be very useful for determining the cause of non-convergence or slow convergence. Because of this additional work, the problem may run slightly slower. Note that debug mode can be turned on/off while running an analysis by first interrupting the run with `ctrl+c` and then using the *debug* command to toggle the debug mode on or off. It is important to note that since FEBio will store all non-converged states to the plot file, this file may become very large in a short number of time steps. An alternative approach is to use the *plot* command to write out select non-converged states.

### 2.8.3 Setting break points

Although `ctrl+c` can be used to interrupt a run, it might sometimes be difficult to interrupt a run at a desired point. For this reason, you can set break points that will pause the execution and bring up the FEBio prompt. Breakpoints are set on the command line using the **-break** option, followed by the break condition. In addition, breakpoints can also be set via the FEBio prompt using the **break** option.

To break at a particular time point, simply add the value of the time after the **-break** option. This example will pause the run after time 0.45 completed (or the first time point past 0.45):

```
>febio -i input.feb -break 0.45
```

Instead of pausing at a particular time, you can also interrupt the run when a certain event happens. To break at an event, specify the event name at which to pause the run. The following list of events are defined.

| Event        | Description                                               |
|--------------|-----------------------------------------------------------|
| ALWAYS       | break on any event                                        |
| INIT         | break after model initialization                          |
| STEP_ACTIVE  | break after step activation                               |
| MAJOR_ITERS  | break after major iteration (i.e. time step) converged    |
| MINOR_ITERS  | break after minor iteration (i.e. Newton iteration)       |
| SOLVED       | break after the model is solved                           |
| UPDATE_TIME  | break before time is incremented                          |
| AUGMENT      | break before augmentation                                 |
| STEP_SOLVED  | break after step is solved                                |
| REFORM       | break after global matrix and right hand side is reformed |
| MATRIX_SOLVE | break right before the linear system solve                |

For example, to pause the execution after a matrix reformation (and right-hand-side evaluation), enter the following command line.

```
>febio -i input.feb -break REFORM
```

Once the code reaches a breakpoint, the febio prompt will be presented. There are several commands that relate to breakpoints.

**break** Add a breakpoint

**breaks** Prints list of current breakpoints.

**clear** Clear one or all breakpoints.

### 2.8.4 Restarting a Run

When the creation of a restart dump file is requested, the analysis can be restarted from the last converged timestep. This is useful when the run terminated unexpectedly or when the user wishes to modify some parameters during the analysis. To request the creation of a dump file, simply add the `-dump` flag on the command line. This will generate a dump file which then can be used to restart the analysis. You can specify an optional dump level, which sets how often the dump file is created, and an optional dump file name.

```
> febio -i input.feb -dump
```

With the optional arguments, the complete syntax would be:

```
> febio -i input.feb -dump=1 out.dmp
```

The dump level can be set to the following values:

1. write dump file after each converged time step
2. write dump file after each completed analysis step
3. write dump file after each converged *must-point*

To restart an analysis, use the `-r` command line option. This option requires a filename as a parameter, and this name can be either the name of a dump file or the name of a restart input file. The latter case is a text file that allows the user to redefine some parameters when restarting the run. The format of this file is described in [Chapter 5](#).

## 2.9 FEBio File Conventions

Although users are free to choose file names and file extensions, it might be beneficial to adhere to the following file naming conventions.

**feb** FEBio input file.

**xplt** FEBio plot file.

**log** FEBio log file.

**dmp** FEBio dump file (for restarts).

**opt** FEBio optimization control file.

**rst** FEBio restart input file.

| task               | description                                             |
|--------------------|---------------------------------------------------------|
| diagnose           | Run a diagnostic.                                       |
| jfnk tangent test  | Runs a tangent test on the JFNK solver                  |
| optimize           | Runs an optimization problem.                           |
| param_run          | Runs a parameter study                                  |
| parameter_sweep    | Runs a parameter study                                  |
| quick_restart_test | Runs a restart test                                     |
| rci_solve          | Runs a model using the RCI solver                       |
| reset_test         | Runs a test on the FEBio reset functionality            |
| restart            | Runs a cold restart problem.                            |
| restart_test       | Runs a test on the restart feature.                     |
| solve              | Runs the forward model (default task)                   |
| stiffness_test     | Allows users to export the stiffness matrix from FEBio. |
| test               | Runs the model as a test suite problem.                 |

Table 2.4: Table of available FEBio tasks.

## 2.10 FEBio Tasks

At its core, FEBio always executes a *task*. You can think of a task as the outer loop that instructs FEBio what to do. The default task will run the FEBio input file provided on the command line and calculate the solution. Other tasks perform a restart, solve an optimization problem, or run a diagnostic. Users can also create custom tasks via plugins and direct FEBio what exactly it should do. Many of these tasks perform tests on specific FEBio features and thus can be helpful for plugin developers to ensure that their plugin works correctly with these specific FEBio features.

The task to execute can be specified on the command line using the **-task** command line option. The name of the task is specified as follows.

```
>febio4 -i input.feb -task=task_name [task_input_file]
```

The name of the task follows after an equal sign ('=') (Notice that there are no spaces before or after the equal sign). Some tasks require an additional input file, which can be specified directly following the -task command line option. (There should be a space between the -task command line option and the optional task input file.)

In this section we'll document the tasks that are currently supported by FEBio. The following table lists the available tasks.

Some more details about these tasks follow.

- **diagnose:** The *diagnose* task runs a diagnostic.
- **jfnk\_tangent\_test:** The *"jfnk tangent test"* runs a test on the tangent produced by the JFNK linear solver. The JFNK linear solver is an iterative solver that aims to solve the nonlinear Newton problem without ever evaluating the stiffness matrix explicitly.
- **optimize:** Runs an optimization problem. (See [7](#) or more details on how to run optimization problems in FEBio.)
- **param\_run:** The *param\_run* task runs a parameter study on the input model. In other words, it runs the input file multiple times, each time using a different value for certain parameters.

- **parameter\_sweep:** This task runs a parameter study on the input file.
- **quick\_restart\_test:** The *quick\_restart\_test* runs a test on the restart feature after model initialization. This task initializes the model and then writes a dump file. After that, the model is cleared and the dump file is read to re-initialize the model. Any issues that occur during re-initialization will be reported. If no problems are encountered, the task finishes. (Note that the model is not solved.)
- **rci\_solve:** The *rci\_solve* task solves a model using the RCI solver. The RCI (Reverse Communication Interface) solver is a feature that allows users more control over how FEBio solves the model. This feature was added to give plugins that interface with third party solvers the ability to transfer some of the solution progress decisions (e.g. when to advance the time step) to the third party library. This task tests that interface and ensures that it produces the same result as the standard solver.
- **reset\_test:** FEBio's reset feature restores the original state of the model. This feature is used in several other tasks that need to solve the forward model repeatedly (e.g. optimization, parameter study). This task can be used to test the reset feature and ensure that it indeed properly resets the model. The test will run the forward model twice. After the first run, some stats are collected. The model is then reset and run again. The same stats are collected after the second run and compared with those from the first run. If the model reset was successful, the stats should be identical. If not, there is a problem with model reset. This task also produces a separate log file for each run that can be compared.
- **restart:** Runs a cold restart of a previously run model. See [5](#) for more details on how to run restart problems.
- **restart\_test:** This task tests the restart feature to ensure that it indeed correctly restarts a model. After each converged time step, the model's state is serialized (in memory) to a dump stream and then immediately read back from the dump stream. If the restart feature is working properly, the model should produce the exact same output to both log and plot file as the forward model (i.e. the model run with that default task). Users can compare outputs to see if there might be any problem with the restart feature.
- **solve:** This solves the forward model provided by the FEBio input file. This is the default task when the `-task` command line option is not provided.
- **stiffness\_test:** The *stiffness\_test* task allows users to export the stiffness matrix and right-hand-side vector out of FEBio.
- **test:** This task is used by the FEBio test suite and adds the solution norm as a log variable that can be queried once the model completes. (This is used in the test suite to see if the solution norm has changed compared to a golden standard.)

## Chapter 3

# Free Format Input

This chapter describes the XML-based input format used by FEBio. Since this format follows standard XML conventions, the files can be viewed with any file viewer that supports XML files. Since the free format input file is a text file, it can be edited with any text editor.

An XML file is composed of a hierarchical list of *elements*. The first element is called the *root element*. Elements can have multiple *child elements*. All elements are enclosed by two *tags*: a tag defining the element and an *end tag*. A simple example of an XML file might look like this:

```
<root>
  <child>
    <subchild> ... </subchild>
  </child>
</root>
```

The *value* of an element is enclosed between the name and the end tag.

```
<element> here is the value </element>
```

Note that the XML format is case-sensitive.

XML elements can also have *attributes* in name/value pairs. The attribute value must always be quoted using quotation marks (") or apostrophes (')<sup>1</sup>.

```
<element attr="value">...</element>
<element attr='value'>...</element>
```

If an XML element has no value, an abbreviated syntax can be used. The following two lines are identical.

```
<element [attribute list]></element>
```

or

```
<element [attribute list]/>
```

Comments can be added as follows.

---

<sup>1</sup>Support for apostrophes was not added until FEBio version 2.1.

```
<!-- This is a comment -->
```

The first line in the document – the XML declaration – defines the XML version and the character encoding used in the document. An example can be:

```
<?xml version="1.0" encoding="ISO-8859-1"?>
```

### 3.1 Free Format Overview

The free format organizes the FEBio input data into hierarchical XML elements. The root element is called *febio\_spec*. This root element also defines the format version number (Note that FEBio and the input format specification follow different version numberings). This document describes version 4.0 of the FEBio specification<sup>2</sup> (see Section 3.1.1 below for more details on the different input specification formats). The root element will therefore be defined as follows:

```
<febio_spec version="4.0">
  <!-- contents of file -->
</febio_spec>
```

The different sections introduced in this chapter are child elements of this root element. The following sections are currently defined:

**Module** defines the physics module for solving the model.

**Globals** Defines the global variables in the model

**Control** specifies control and solver parameters.

**Material** Specifies the materials used in the problem and the material parameters.

**Mesh** Defines the mesh of the problem, including nodal coordinates and element connectivity.

**MeshDomains** Assigns materials and other formulation attributes to element sets.

**MeshData** Defines element, facet, edge or nodal data that can be mapped to material parameters or certain boundary conditions and loads.

**MeshAdaptor** Defines mesh adaptors that modify the mesh.

**Initial** Defines initial conditions for dynamic problems, such as initial velocities, and for transient quasi-static problems.

**Boundary** Defines the boundary conditions that are applied on the geometry.

**Loads** Defines the loads applied to the model. This includes nodal loads, boundary loads and volume loads (or sources for heat transfer problems).

**Contact** This section defines all contact interfaces.

**Constraints** This section defines rigid and nonlinear constraints.

---

<sup>2</sup>FEBio continues to read some older formats, but they are considered to be obsolete.



**Rigid** This section defines rigid components, such as rigid constraints, rigid loads, and rigid connectors.

**Discrete** This section defines all the discrete elements (i.e. springs).

**LoadData** Defines the load controllers, which can control model parameters as a function of time.

**Step** Defines different analysis steps, where in each analysis the boundary, loads, contact and initial conditions can be redefined.

**Output** Defines additional data that needs to be stored to file.

Although there is some flexibility in the order in which these sections can be listed, it is recommended to list them in the same order as given above. Not all sections are required. Empty sections can be omitted. A minimal file must contain at least the Module, Control, Material, Mesh, and MeshDomains sections. The rest of this chapter describes each of these sections in more detail.

### 3.1.1 Format Specification Versions

This document describes version 4.0 of the FEBio input specification. This format differs in several aspects from the previous versions of the input specification. This section describes the major changes between the different versions.

- **Version 4.0:** The latest and recommended version of the FEBio input specification described in this document. The main motivation for this format was to improve consistency between the feb file structure and the way the FEBio model is constructed in FEBio Studio. This format is very similar to version 3.0 and only differs in a few sections: Some solver parameters have moved. The Rigid section was restructured. Node and element sets can be specified more concisely.
- **Version 3.0:** The major focus of this revision was adding support for FEBio 3.0's features for modeling heterogeneous parameters. Heterogeneous parameters can be created via mathematical expressions or via the MeshData section. It also introduces a more consistent way for referencing model parameters. The *Geometry* section was replaced with the *Mesh* and *MeshDomains* sections in order to further separate the definition of the mesh and its physical attributes (e.g. material assignments). This format is **still supported** but considered obsolete.
- **Version 2.5:** This format differs from its predecessor in some important aspects: all node-sets, surfaces, etc., that are used by boundary conditions, loads, contact, etc., must be defined in the Geometry section. Boundary conditions, loads, contact, etc., are now defined by referencing the sets in the Geometry section. This format also adds the *MeshData* section and re-formats the *Discrete* section. Rigid node sets and prescribed rigid degrees of freedom are moved to the *Boundary* section. This format is **still supported** but considered obsolete.
- **Version 2.0:** This is the first major revision of the input file format and redefines many of the file sections: The *Elements* section uses a different organization. Elements are now grouped by material and element type. Multiple *Elements* sections can now be defined to create multiple parts. Surfaces can now be defined in the *Geometry* section and referenced

by boundary conditions and contact definitions. A new *Contact* section contains all the contact definitions. A new *Discrete* section was defined that contains all the materials and definitions of the discrete elements (e.g. springs). The *Boundary* section is also redesigned. This format is **still supported** but considered obsolete.

- **Version 1.3:** This was an experimental version that redefined the *Geometry* section, but was later abandoned in favor of version 2.0. This version is **no longer supported**.
- **Version 1.2:** A *Loads* section was added and all surface and body loads are now defined in this section instead of the *Boundary* section. This version is **mostly supported** but considered obsolete.
- **Version 1.1:** Rigid body constraints are no longer defined in the rigid material definition but instead placed in a new *Constraints* section. This version is **no longer supported**.
- **Version 1.0:** The original input format specification. This version is **no longer supported**.

As of FEBio 4.0, only versions 1.2, 2.0, 2.5, and 3.0 are supported. Versions 1.2 and 2.0 are considered obsolete and it is highly recommended to convert older files to the newest specification for use with newer versions of FEBio. This can be done for instance using FEBioStudio.

### 3.1.2 Notes on backward compatibility

Some features that were available in versions prior to 4.0 are no longer supported or require a different syntax.

1. The *Parameters* section is no longer supported. This section allowed users to define file parameters that could be used as parameter values. This section was removed since it was difficult to support in FEBio Studio and a better mechanism was implemented for defining model-level parameters. (See the [3.4.4](#) section.)
2. Some fiber materials defined the *theta* and *phi* parameters for setting the fiber orientation. These parameters are no longer supported. Instead, the fiber direction should be specified via the *fiber* property, or if the *fiber* property is not available, the *mat\_axis* property.

```
<fiber type="angles">
  <theta>90</theta>
  <phi>0</phi>
</fiber>
```

### 3.1.3 Multiple Input Files

FEBio supports distributing the model definition across multiple input files. This can greatly facilitate defining large, complex models and allows the re-use of model input files without the need to create the entire model input file from scratch. When using multiple input files to define a model, you must create a control input file that may reference other input files. This control file will be used to run the model in FEBio. There are two ways of including a file into the control file: The *Include* section, and the *from* attribute.

Note that all input files must start with the *febio\_spec* tag and use the same format specification.

### 3.1.3.1 Include Keyword

The *Include* keyword<sup>3</sup> can be used to include the contents of another FEBio input file. The filename is entered as the value of the tag.

```
<Include>example.feb</Include>
```

The included file must be a valid FEBio file in that it must begin with the *febio\_spec* tag and contain sections defined in this document. However, the included file does not need to define a complete model definition. For instance, it can contain only the *Mesh* section.

Note that the contents of the entire file will be included. This is different from the *from* attribute discussed below, which can be used to include only certain sections from files.

### 3.1.3.2 The 'from' Attribute

The *from* attribute can be used to include sections from other files. All the main sections defined in Section 3.1 support the *from* attribute which can be used to load the section from another input file. For example, to load the *Material* section from the file *mat.feb*, defining the *Material* section in the control input file as follows.

```
<Material from="mat.feb"/>
```

FEBio will now read the *Material* section from this child file. The child file must be a valid FEBio input file, meaning it must begin with the *febio\_spec* root section, but does not have to be complete. For example, the file *mat.feb* only needs to define the *Material* section. However, the child file may contain other sections. In that case, only the section referenced in the control file will be read from the child file. For example, if the file *in.feb* contains both the *Material* and the *Mesh* section, the control file can read both these sections as follows.

```
<Material from="in.feb"/>
<Mesh from="in.feb"/>
```

To give a more concrete example, assume that the *Material*, *Mesh*, and *Boundary* sections are defined in the files *mat.feb*, *geom.feb*, and *bc.feb* respectively. The control input file could then look like the following.

```
<febio_spec version="4.0">
  <Module type="solid"/>
  <Control>
    <time_steps>10</time_steps>
    <step_size>0.1</step_size>
  </Control>
  <Material from="mat.feb"/>
  <Mesh from="geom.feb"/>
  <Boundary from="bc.feb"/>
</febio_spec>
```

---

<sup>3</sup>Supported from FEBio version 2.3 and up.

Notice that the *Control* section is still defined in the control file. The control file can contain a combination of explicit section definitions and referenced sections using the *from* attribute. As mentioned above, the control file is used to run the model in FEBio. So, if the control file is called *model.feb* then the model is run as follows.

```
>febio -i model.feb
```

When FEBio parses the control file it will automatically parse the referenced child files it encounters in the control input file.

## 3.2 Module Section

The module section defines the type of analysis to perform with FEBio. This section must be defined as the first section in the input file. It takes on the following format:

```
<Module type="[type]"/>
```

where *type* can be any of the following values:

| type            | Description                                                                                    |
|-----------------|------------------------------------------------------------------------------------------------|
| solid           | Structural mechanics analysis: quasi-static or dynamic, implicit and explicit solver available |
| biphasic        | Biphasic analysis: steady-state or transient                                                   |
| solute          | Biphasic analysis including solute transport: steady-state or transient                        |
| multiphasic     | Multiphasic analysis including solute transport and chemical reactions                         |
| fluid           | Fluid mechanics analysis: steady-state or dynamic                                              |
| fluid-FSI       | Fluid mechanics with fluid-structure interactions: steady-state or dynamic                     |
| fluid-solutes   | A fluid mechanics analysis with support for solute transport.                                  |
| multiphasic-FSI | The FSI module with support for multiphasic materials.                                         |

The module tag can also be used to specify the units of the model. Although FEBio doesn't care about units, if the user specifies the unit system, the proper units of variables are stored in the plot file and visible in FEBio Studio when opening the plot file.

To specify units, use the following syntax:

```
<Module type="[type]">
  <units>[unit system]</units>
</Module>
```

where *unit system* can be any of the following values.

| units   | Description                                                               |
|---------|---------------------------------------------------------------------------|
| SI      | Units follow the standard definition of the international SI unit system. |
| mm-N-s  | Millimeter-Newton-seconds unit system                                     |
| mm-kg-s | Millimeter-kilogram-seconds unit system                                   |
| um-nN-s | Micron-nanoNewton-seconds unit system                                     |
| CGS     | Centimeter-gram-seconds unit system                                       |

If the units are not defined, FEBio will not output any units for plot variables.

For example:

```
<febio_spec version="4.0">
  <Module type="solid"/>
  <!-- rest of file -->
</febio_spec>
```

The following example defines the model's units explicitly.

```
<febio_spec version="4.0">
  <Module type="solid">
    <units>SI</units>
  </Module>
  <!-- rest of file -->
</febio_spec>
```

*Notes:*

1. In version 1.2 the Module section was optional. If omitted it was assumed that the *solid* module was used. Since version 2.0 the Module section is required and must be the first section in the file.
2. Older versions of FEBio (format specification 1.2 and before) allowed you to run a poroelastic (now called biphasic) problem by simply defining a poroelastic material. This is no longer possible. You need to define the proper Module section to run a biphasic analysis. If you have a file that no longer works as of version 1.4 of FEBio, you'll need to insert the following Module section in the file as the first section of the file.

```
<febio_spec version="1.2">
  <Module type="biphasic"/>
  <!-- rest of the file unaltered -->
</febio_spec>
```

### 3.3 Control Section

The control section is defined by the *Control* element. This section defines all parameters that are used to control the evolution of the solution as well as parameters for the nonlinear solution procedure. These parameters are defined as child elements of the *Control* element and depend somewhat on the analysis as defined by the *Module* section. The control section defines the control parameters, the solver parameters, and optionally, the time\_stepper parameters.

Note that for multistep analyses (see chapter 6) the Control section is specified for each step separately and defined as a child element of each *step* section.

*Example:*

```
<Control>
  <analysis>STATIC</analysis>
  <time_steps>50</time_steps>
  <step_size>0.02</step_size>
  <solver>
    <max_refs>25</max_refs>
    <diverge_reform>1</diverge_reform>
    <reform_each_time_step>1</reform_each_time_step>
    <dtol>0.001</dtol>
    <etol>0.01</etol>
    <rtol>0</rtol>
    <lstol>0.9</lstol>
    <min_residual>1e-20</min_residual>
    <rhoi>-2</rhoi>
    <qn_method type="BFGS">
      <max_ups>10</max_ups>
    </qn_method>
  </solver>
  <time_stepper>
    <dtmin>0.0002</dtmin>
    <dtmax>0.02</dtmax>
    <max_retries>5</max_retries>
    <opt_iter>6</opt_iter>
  </time_stepper>
</Control>
```

#### 3.3.1 Control Parameters

The following parameters are common for all analysis types. If not specified they are assigned default values, which are found in the last column. An asterisk (\*) after the name indicates a required parameter. The numbers behind the description refer to the comments following the table.

| Parameter   | Description                                      | Default |
|-------------|--------------------------------------------------|---------|
| analysis    | Sets the analysis type (1)                       | static  |
| time_steps* | Total number of time steps. (= <i>ntime</i> )(2) | (none)  |

|                 |                                                                                                                          |                   |
|-----------------|--------------------------------------------------------------------------------------------------------------------------|-------------------|
| step_size*      | The initial time step size. (= $dt$ ) (2)                                                                                | (none)            |
| plot_level      | Sets the level of state dumps to the plot file (3)                                                                       | PLOT_MAJOR_ITRS   |
| plot_range      | Set the range of the states that will be stored to the plot file (4)                                                     | 0,-1              |
| plot_stride     | Set the stride of the states that will be stored to the plot file (4)                                                    | 1                 |
| plot_zero_state | Flag that controls whether the “zero” state will be written to the plot file, even if it is not defined in the range (4) | 0 (false)         |
| output_level    | Controls when to output data to file (5)                                                                                 | OUTPUT_MAJOR_ITRS |
| output_stride   | Set the output stride for writing out                                                                                    | 1                 |
| adaptor_resolve | re-solve time step after mesh adaptations                                                                                | 1 (yes)           |

*Comments:*

1. The *analysis* element sets the analysis type. The text value of this element is determined by the module, but the numeric value should be 0 for (quasi-)static or steady-state analysis and 1 for dynamic or transient analysis. In a quasi-static analysis, inertial effects are ignored and an equilibrium solution is sought. Note that in this analysis mode it is still possible to simulate time-dependent effects such as viscoelasticity. In a dynamic analysis the inertial effects are included.

| Value        | Description                                                                                                             |
|--------------|-------------------------------------------------------------------------------------------------------------------------|
| static       | (quasi-) static analysis                                                                                                |
| steady-state | steady-state response of a transient (quasi-static) biphasic, biphasic-solute, multiphasic, fluid or fluid-FSI analysis |
| dynamic      | dynamic analysis for solid, fluid and fluid-FSI analyses                                                                |

2. The total simulation time of the analysis is determined by  $n_{time} * dt$ . Note that when the auto-time stepper is enabled (see below), the actual number of time steps and time step size may be different than specified in the input file. However, the total simulation time will always be determined by  $n_{time} * dt$ .
3. The *plot\_level* allows the user to control exactly when the solution is to be saved to the plot file. The following values are allowed:

| Value              | Description                                                 |
|--------------------|-------------------------------------------------------------|
| PLOT_NEVER         | Don't save the solution to the plot file                    |
| PLOT_MAJOR_ITRS    | Save the solution after each converged timestep             |
| PLOT_MINOR_ITRS    | Save the solution for every quasi-Newton iteration          |
| PLOT_MUST_POINTS   | Only save the solution at the must points                   |
| PLOT_FINAL         | Only store the final converged state.                       |
| PLOT_STEP_FINAL    | Store only the final converged state of each analysis step. |
| PLOT_AUGMENTATIONS | Store states at the start of each augmentation.             |



The `PLOT_MUST_POINTS` option must be used in conjunction of a *must-point* curve. See the comments on the `dtmax` parameter for more information on must-point curves. When the `plot_level` option is set to `PLOT_MUST_POINTS`, only the time-points defined in the must-point curve are stored to the plotfile.

4. When using the fixed time stepper, several parameters control which time steps are stored to the plot file.

The `plot_range` parameter sets the range of the states that will be stored to the plot file. The range is defined by two values that specify the first and last time step that will be stored to the plot file. The value “0” refers to the initial time step, usually time zero, and negative values count backwards from the final time step (as defined by the `time_steps` parameter). For instance, the default values,

```
<plot_range>0,-1</plot_range>
```

store all time steps to the plot file, including the initial “zero” time step. As another example, to store only the last five time steps, set

```
<plot_range>-5,-1</plot_range>
```

By default, all time steps within the range will be stored to the plot file. Time steps can be skipped using the `plot_stride` parameter. For instance, to store only every 10 steps, set

```
<plot_stride>10</plot_stride>
```

Note that the first and last time step defined by the range will always be stored, regardless of the plot stride.

The “zero” time step refers to the initial state of the model, before any calculations are done. This state will only be stored to the plot file if the minimal value of the plot range is set to zero. To force storing this state to the plot file, set the `plot_zero_state` parameter to one.

```
<plot_zero_state>1</plot_zero_state>
```

This will store the zero time step to the plot file, even when it is not specified inside the plot range.

Again, the `plot_range`, `plot_stride`, and `plot_zero_state` parameters are only used by the fixed time stepper. Currently, these parameters are not used with the auto time stepper.

5. The `output_level` can be used to control when FEBio outputs the data files. The following values are supported.

| Value              | Description                                           |
|--------------------|-------------------------------------------------------|
| OUTPUT_NEVER       | Don't generate any output                             |
| OUTPUT_MUST_POINTS | Only output at must points                            |
| OUTPUT_MAJOR_ITRS  | Output at end of each time step                       |
| OUTPUT_MINOR_ITRS  | Output at each iteration                              |
| OUTPUT_FINAL       | Only output the data at the last converged time step. |

### 3.3.2 Time Stepper parameters

The optional *time\_stepper* element sets the parameters that control the FEBio auto-time stepper. This auto-time stepper will adjust the time step size depending on the convergence stats of the previous time step. It defines the following parameters.

| Value          | Description                                                           | Default |
|----------------|-----------------------------------------------------------------------|---------|
| dtmin          | Minimum time step size (1)                                            |         |
| dtmax          | Maximum time step size (3)                                            |         |
| opt_iter       | Optimal, or desired, number of iterations per time step (2)           | 10      |
| max_retries    | Maximum number of times a time step is restarted. (2)                 | 5       |
| aggressiveness | Algorithm for cutting the time step size after a failed time step (4) | 0       |
| cutback        | Time step scale factor, used when aggressiveness = 1                  | 0.5     |

*Comments:*

1. The *dtmin* and *dtmax* values are used to constrain the range of possible time step values. The *opt\_iter* defines the estimated optimal number of quasi-Newton iterations. If the actual number of iterations is less than or equal to this value the time step size is increased, otherwise it is decreased, as detailed further below.
2. When a time step fails (e.g. due to a negative Jacobian), FEBio will retry the time step with a smaller time step size. The *max\_retries* parameter determines the maximum number of times a timestep may be retried before FEBio error terminates. The new time step size is determined by the ratio of the time step size before restarts started and *max\_retries+1*. For example, if the time step size is 0.1 and *max\_retries* is set to 4, then the time step size is adjusted by 0.02: The first retry will have a step size of 0.08; the next will be 0.06, and so on. Specifically, the new time step size is determined as follows. (This is only for the case when the *aggressiveness* parameter is set to zero.)

$$\Delta t^{k+1} = \Delta t^k - \Delta t^0 / (m_{\max} + 1)$$

Here,  $\Delta t^k$  is the current time step size,  $\Delta t^{k+1}$  is the new time step size,  $\Delta t^0$  is the time step size before retries started, and  $m_{\max}$  is the max number of retries (i.e. *max\_retries*).

When the time step succeeds, the time step size will be adjusted by comparing the *opt\_iter* parameter to the actual number of iterations it took to converge. If the actual number of iterations is larger than *opt\_iter*, the time step size will be decreased.

$$\Delta t^{k+1} = \Delta t^k - \left( \Delta t^k - \Delta t_{\min} \right) (1 - r)$$

Here,  $\Delta t^k$  is the current time step size,  $\Delta t^{k+1}$  is the new time step size,  $\Delta t_{\min}$  is the minimum time step size and  $r = \sqrt{n_{\text{opt}}/n}$ , where  $n$  is the number of iterations and  $n_{\text{opt}}$  is the optimal number of iterations as specified by the user (i.e. *opt\_iter*).

On the other hand, if the actual number of iterations is less than *opt\_iter*, then the time step size will be increased. The exact equation follows.

$$\Delta t^{k+1} = \Delta t^k + \left( \Delta t_{\max} - \Delta t^k \right) \min(0.2, r - 1)$$

In addition, the new time step size is enforced to be less than 5 times the old time step size, to prevent the time step size from growing too fast.

After calculating the new time step size, it will be ensured that the new timestep size falls within the range defined by *dtmin* and *dtmax* and that must-points are respected.

3. The user can specify a load curve for the *dtmax* parameter. This load curve is referred to as the *must-point* curve and serves two purposes. Firstly, it defines the value of the *dtmax* parameter as a function of time. This can be useful, for instance, to enforce smaller time steps during rapid loading or larger time steps when approaching steady-state in a transient analysis. Secondly, the time points of the *dtmax* loadcurve define so-called *must-points*. A must-point is a time point where FEBio must pass through. This is useful for synchronizing the auto-time stepper with the loading scenario. For instance, when loading starts at time 0.5, adding a must-point at this time will guarantee that the timestepper evaluates the model at that time. In conjunction with the PLOT\_MUST\_POINT value of the *plot\_level* parameter, this option can also be used to only write results to the plotfile at the specified time points. Consider the following example.

```
<dtmax lc="1">0.1</dtmax>
...
<loadcontroller id="1" type="loadcurve">
  <interpolate>STEP</interpolate>
  <points>
    <point>0,0</point>
    <point>0.5, 0.1</point>
    <point>1.0, 0.2</point>
  </points>
</loadcontroller>
```

This example defines load curve 1 as the *must-point* curve. This curve defines three points where FEBio will pass through (namely 0, 0.5 and 1.0). The values of each time point is the value of the maximum time-step size (*dtmax*). Since the curve is defined as a step-function, each value is valid up to the corresponding time-point. Thus, between time 0 and time 0.5, the maximum time step value is 0.1. Between 0.5 and 1.0 the maximum time step value is 0.2. If the *plot\_level* parameter is set to PLOT\_MUST\_POINTS, then only the three defined time points will be stored to the plotfile.

Note that when specifying a loadcurve for the *dtmax* parameter, the value of this parameter will be ignored.

4. When FEBio fails to converge a time step, the time step size is reduced and then FEBio tries to solve the time step again. The algorithm for determining the reduced time step size, depends on the *aggressiveness* parameter.
  - (a) *aggressiveness* = 0 (default):  $\Delta t^{k+1} = \Delta t^k - \Delta t^0 / (m_{\max} + 1)$ , (see comment 2 for more details).
  - (b) *aggressiveness* = 1:  $\Delta t^{k+1} = \gamma \Delta t^k$ , where  $\gamma$  is the cutback factor (i.e. the *cutback* parameter).

### 3.3.3 Common Solver Parameters

Many of the solvers define the same parameters. They are listed in the table below. Additional parameters that are only defined for a specific type of analysis are listed in the following sections.

| Parameter             | Description                                                        | Default      |
|-----------------------|--------------------------------------------------------------------|--------------|
| etol                  | Convergence tolerance on energy (1)                                | 0.01         |
| rtol                  | Convergence tolerance on residual (1)                              | 0 (disabled) |
| lstol                 | Convergence tolerance on line search (2)                           | 0.9          |
| lsmin                 | minimum line search step                                           | 0.01         |
| lster                 | Maximum number of line search iterations                           | 5            |
| ls_check_jacobians    | Check for negative Jacobians during linesearch                     | 0 (disabled) |
| max_refs              | Max number of stiffness reformations (3)                           | 15           |
| qn_method             | Quasi-Newton update method (3)                                     | BFGS         |
| max_ups               | Max number of BFGS/Broyden stiffness updates (3)                   | 10           |
| diverge_reform        | Flag for reforming stiffness matrix when the solution diverges (3) | 1            |
| min_residual          | Sets minimal value for residual norm (5)                           | 1e-20        |
| max_residual          | Sets the max value for residual norm                               | 0 (disabled) |
| symmetric_stiffness   | Use symmetric stiffness matrix flag (6)                            | 1            |
| equation_scheme       | Set the equation allocation scheme                                 | 0            |
| equation_order        | Set the equation allocation ordering                               | 0            |
| optimize_bw           | Optimize bandwidth of stiffness matrix (4)                         | 0            |
| linear_solver         | Set the preferred linear solver to use. (7)                        | (default)    |
| check_zero_diagonal   | Check for zero diagonals in the stiffness matrix                   | 0 (disabled) |
| zero_diagonal_tol     | Tolerance value for checking zero diagonals                        | 0.0          |
| force_partition       | For a partition in the stiffness matrix                            | 0 (disabled) |
| reform_each_time_step | Reform the stiffness matrix at the start of each time step         | 1 (enabled)  |
| reform_augment        | Reform stiffness matrix after each augmentation                    | 0 (disabled) |

*Comments:*

1. FEBio determines convergence of a time step based on three convergence criteria: displacement, residual and energy (that is, residual multiplied by displacement). Each of these criteria requires a tolerance value that will determine convergence when the relative change will drop below this value. For example, a displacement tolerance of  $\varepsilon$  means that the ratio of the displacement increment (i.e. the solution of the linearized FE equations,  $\Delta \mathbf{U} = -\mathbf{K}_k^{-1} \mathbf{R}_k$ ) norm at the current iteration  $k+1$  to the norm of the total displacement ( $\mathbf{U}_{k+1} = \mathbf{U}_k + \Delta \mathbf{U}$ ) must be less than  $\varepsilon$ :

$$\frac{\|\Delta \mathbf{U}\|}{\|\mathbf{U}_{k+1}\|} < \varepsilon$$

For the residual and energy norms, it is the ratio of the current residual norm (resp. energy norm) to the initial one that needs to be less than the specified convergence tolerance.

To disable a specific convergence criterion, set the corresponding tolerance to zero. For example, by default, the residual tolerance is zero, so that this convergence criterion is not used.

2. The *lstol* parameter controls the scaling of the vector direction obtained from the line search. A line search method is used to improve the convergence of the nonlinear (quasi-) Newton solution algorithm. After each quasi-Newton iteration, this algorithm searches in the direction of the displacement increment for a solution that has less energy (that is, residual multiplied with the displacement increment) than the previous iteration. In many problems this will automatically be the case. However, in some problems that are very nonlinear (e.g. contact), the line search can improve convergence significantly. The line search can be disabled by setting the *lstol* parameter to zero, although this is not recommended.
3. The *qn\_method* property determines the quasi-Newton update method. The particular method is set via the *type* attribute. Additional parameters are then specified as child elements of this tag. See section 3.3.4 for more details.
4. The *optimize\_bw* parameter enables bandwidth minimization for the global stiffness matrix. This can drastically decrease the memory requirements and running times when using the skyline solver. It is highly recommended when using the skyline solver.

```
<optimize_bw>1</optimize_bw>
```

When using a different linear solver (e.g., pardiso or SuperLU), the bandwidth optimization can still be performed if so desired. However, for these solvers there will be little or no effect since these solvers are not as sensitive to the bandwidth as the skyline solver.

5. If no force is acting on the model, then convergence might be problematic due to numerical noise in the calculations. For example, this can happen in a displacement driven contact problem where one of the contacting bodies is moved before initial contact is made. When this happens, the residual norm will be very small. When it drops below the tolerance set by *min\_residual*, FEBio will assume that there is no force acting on the system and will converge the time step.
6. The *symmetric\_stiffness* flag is used to set the symmetry of the global stiffness matrix. Only specify this flag if you want to override the default symmetry, which depends somewhat on the analysis type. Forcing a symmetric matrix when the problem is non-symmetric may have negative impact on convergence. The values for this parameter are:
  - 0 = unsymmetric
  - 1 = symmetric
  - 2 = structurally symmetric
  - 3 = preferred

If the option is set to 3 (preferred), then one of the other formats will be chosen depending on the model components in the model. For instance, if a non-symmetric contact formulation is used, then a non-symmetric stiffness matrix will be used.
7. The default linear solver used by FEBio is usually configured in the FEBio configuration file. However, it can also be specified directly in the FEBio input file as part of the solver parameters using the *linear\_solver* tag. See 9.2 for more details on configuring linear solvers.

```
<linear_solver type="pardiso"/>
```

### 3.3.4 Quasi-Newton Solver Parameters

Most solvers in FEBio use the quasi-Newton method for solving the nonlinear FE equations. Several different QN algorithms are implemented as described below. The specific QN method is selected using the *qn\_method* property of the *solver*. The following QN methods are supported.

**BFGS** The Broyden-Fletcher-G-S method. (default if *qn\_method* is not specified). This method assumes the global stiffness matrix is symmetric. It can still be used with non-symmetric matrices, but the convergence might be compromised.

**Broyden** The Broyden method is similar to BFGS, but works with both symmetric and non-symmetric matrices. Therefore, it is the recommended method for problems that generate a non-symmetric stiffness matrix, such as (some) contact formulations, biphasic, multi-phasic, fluid, fluid FSI.

**JFNK** This is an implementation of the Jacobian-Free-Newton-Krylov method. When using this option, you must also use an iterative linear solver such as FGMRES.

Both the BFGS and Broyden methods calculate the global stiffness matrix at the beginning of each time step. For each iteration, a matrix update is then done. The maximum number of such updates is set with *max\_ups*. When FEBio reaches this number, or if the solution diverges and *diverge\_reform* is set to 1, it reforms the global stiffness matrix (that is, it recalculates it) and factorizes it, essentially taking a "full Newton" iteration. Then FEBio continues with BFGS/Broyden iterations. The *max\_refs* parameter is used to set the maximum of such reformations FEBio can do, before it fails the timestep. In that case, FEBio will either terminate or, if the auto-time stepper is enabled, retry with a smaller time step size.

The BFGS and Broyden take the following parameters.

| Parameter              | Description                                                                  | Default     |
|------------------------|------------------------------------------------------------------------------|-------------|
| <i>max_ups</i>         | This is the maximum number of updates between stiffness matrix reformations. | 10          |
| <i>cmax</i>            | Sets the max condition number.                                               | 1e5         |
| <i>max_buffer_size</i> | Sets the max buffer size.                                                    | 0 (ignored) |
| <i>cycle_buffer</i>    | Recycle buffers when <i>max_ups</i> is larger than <i>max_buffer_size</i>    | 1           |

Note that when *max\_ups* is set to 0, FEBio effectively will use a Full-Newton method since the stiffness matrix is reformed each iteration. In this case it is recommended to increase the number of *max\_refs* (to e.g. 50) in the solver settings, since the default value might cause FEBio to terminate prematurely when convergence is slow.

### 3.3.5 Solver Parameters for a Structural Mechanics Analysis

A structural mechanics analysis is defined by using the *solid* type in Module section. The solver parameters are given in the table below:

| Parameter        | Description                                 | Default |
|------------------|---------------------------------------------|---------|
| dtol             | convergence tolerance on displacement       | 0.001   |
| rhoi             | Spectral radius parameter $\rho_\infty$ (1) | -2      |
| alpha            | generalized alpha method parameter alpha    | 1       |
| beta             | generalized alpha method beta               | 0.25    |
| gamma            | generalized alpha method gamma              | 0.5     |
| logSolve         | Use an acceleration for Newton iterations   | 0 (off) |
| arc_length       | Specifies arc-length method (2)             | 0 (off) |
| arc_length_scale | Set the arc-length scale factor             | 0       |

*Comments:*

1. There are two time-integrators implemented in FEBio: the classical Newmark integration and the generalized alpha method. Setting rhoi to -2, will use the former, where as setting rhoi to the range [0,1] will use the latter. In the latter case, the values for gamma, beta, and gamma are determined from the spectral radius parameter, rhoi.
2. When using the arc-length method, there are some important restrictions. Only zero-prescribed boundary conditions are allowed. The loads must be specified via nodal loads and cannot be time-dependent. The time\_steps and step\_size control parameters are interpreted as arc-length steps and step\_size respectively.

When using the “CG-solid” solver, the following parameters are defined.

| Parameter      | Description                              | Default |
|----------------|------------------------------------------|---------|
| cgmethod       | Select solution algorithm (1)            | 0       |
| preconditioner | Select the preconditioner (2)            | 0       |
| lsiter         | maximum number of line search iterations | 10      |

*Comments:*

1. The solution algorithm can be:
  - (a) 0 : Hager-Zhang conjugate gradient (This is the default and should be preferred since its faster.)
  - (b) 1 : steepest descent
2. Set 0 for no preconditioner, 1 for diagonal stiffness preconditioner. The diagonal stiffness preconditioner greatly improves convergence when there are different size elements, mixed materials, or quadratic elements with midside nodes.

```

<solver type="CG-solid">
  <cgmethod>0</cgmethod>
  <preconditioner>1</preconditioner>
  <lsiter>10</lsiter>
</solver>

```

### 3.3.6 Solver Parameters for Biphasic Analysis

A biphasic analysis is defined by using the *biphasic* type in Module section. Since a biphasic analysis couples a fluid problem to a solid mechanics problem, all control parameters above can be used in a biphasic analysis. In addition, the following parameters can be defined:

| Parameter         | Description                                      | Default |
|-------------------|--------------------------------------------------|---------|
| ptol              | Specify the fluid pressure convergence tolerance | 0.01    |
| mixed_formulation | Use a mixed formulation (1)                      | 0       |

*Comment:*

1. The mixed formulation uses discontinuous shape functions for the pressure interpolation. It still uses the full shape functions for displacement.

### 3.3.7 Solver Parameters for Solute and Multiphasic Analyses

When the type attribute of the Module section is set to *solute* or *multiphasic*, an analysis is solved that includes solute transport. All parameters for a biphasic analysis can be used (including the ones for a structural mechanics analysis). In addition, the following parameters can be specified:

| Parameter                     | Description                                           | Default     |
|-------------------------------|-------------------------------------------------------|-------------|
| ctol                          | Specify the concentration convergence tolerance       | 0.01        |
| force_positive_concentrations | Force the concentration values always to be positive. | 1 (enabled) |

### 3.3.8 Solver Parameters for Fluid and Fluid-FSI Analyses

A fluid analysis is defined by using the *fluid* type in Module section (see Section 3.2). In addition to the common parameters, the following parameters can be specified:

| Parameter             | Description                                                        | Default      |
|-----------------------|--------------------------------------------------------------------|--------------|
| vtol                  | convergence tolerance on velocity                                  | 0.001        |
| ftol                  | convergence tolerance on dilatation                                | 0.001        |
| rhoi                  | Spectral radius parameter $\rho_\infty$                            | 0            |
| reform_each_time_step | Flag for reforming stiffness matrix at the start of each time step | 1            |
| predictor             | Sets the predictor method                                          | 0            |
| min_volume_ratio      | Sets the minimum value for the volume ratio.                       | 0 (disabled) |

A fluid-structure interaction analysis is defined by using the *fluid-FSI* type in Module section. It uses the parameters of *solid* and *fluid* analyses.

Transient fluid and fluid-FSI analyses may often run very efficiently using Broyden's method (*qnmethod* set to 1) with *max\_ups* set to 50; efficiency may be further increased by setting *reform\_each\_time\_step* to 0, which will postpone reforming the stiffness matrix at subsequent time steps until *max\_ups* updates have been exhausted. When using Broyden's method with fluid and fluid-FSI analyses, set *rtol* to a non-zero value to ensure an accurate solution, for example *rtol*=0.001 (see Section 3.3.1).



| parameter    | description                                 | default value |
|--------------|---------------------------------------------|---------------|
| mass_lumping | Chooses the mass lumping strategy. (1)      | 2             |
| dyn_damping  | The damping factor for dynamic damping. (2) | 1             |

Table 3.3: The parameters of the explicit-solid solver

The spectral radius parameter  $\rho_\infty$  determines the time integration scheme: Values in the range  $0 \leq \rho_\infty \leq 1$  use the generalized  $\alpha$ -method (see [FEBio Theory Manual](#)). With  $\rho_\infty = 0$ , damping of higher frequency components (such as those produced by a step increase in velocity) occurs theoretically within a single time step; in contrast, with  $\rho_\infty = 1$ , no damping occurs, potentially leading to instability in the solution process. When solving transient flows that exhibit eddies or shed vortices, a value of  $\rho_\infty = 0.5$  represents a good balance between too much and too little numerical damping.

### 3.3.9 Explicit-Solid Solver

FEBio supports an explicit-solid solver, which solves dynamic structural mechanics problems using an explicit time-integration scheme. This means that the solution at a given timestep is calculated using only the solution from the previous timestep. The advantage of this approach is that no stiffness matrix needs to be calculated and no linear system of equations needs to be solved (at least, with mass-lumping enabled). As a result, this solver can solve dynamic problems very fast. However, the downside of this approach is that it is only conditionally stable and imposes an upper limit on the time step size. This limit is often much smaller than the time step sizes that can be taken by implicit time-integration schemes (as used by the default Newton-based solvers in FEBio) and thus often will require many more time steps to solve the model. Thus, the explicit solver is most useful in dynamic problems that already require relatively small time step sizes, such as impact-contact models, or models that use non-elastic material behavior that may undergo rapid changes in their constitutive behavior (e.g. damage).

The explicit solver in FEBio uses the midpoint time integration rule and uses mass lumping. Mass lumping is a technique that replaces the consistent mass matrix with a diagonal approximation. In fact, it's this diagonalization of the mass matrix that really prevents the need to solve a linear system of equations. (Since solving a diagonal linear system of equations is trivial.)

The explicit solver can be selected by setting the **type** attribute of the **solver** element to *explicit-solid*.

```
<solver type="explicit-solid">
...
</solver>
```

The following parameters can be specified for the explicit-solid solver.

*Comments:*

1. As mentioned, mass lumping is a technique to approximate the mass matrix with a diagonal matrix. Several approaches are supported in FEBio.
  - (a) `mass_lumping = 0`: no mass-lumping (This is currently not implemented, but the value is reserved in case support will be added in the future.)

- (b) `mass_lumping = 1` : use row summation approach. Useful for linear elements, but may produce negative nodal masses for quadratic elements and thus should not be used for models containing higher-order elements.
  - (c) `mass_lumping = 2`: HRZ lumping technique. Better suited for higher-order elements, but should not be used for shells.
2. Dynamic damping is a technique to dampen the solution in order to obtain an equilibrium solution (e.g. for false-transient analyses). In FEBio, dynamic damping is applied to the velocity updates and the *`dyn_damping`* parameter specifies the damping factor. The default value is 1 so no damping is applied. Note that the lower the damping factor, the more damping, but also the longer it may take to find the equilibrium solution.

**Important!** As a final note, the explicit-solver is still considered an experimental feature in FEBio and may not work correctly with all of the features of FEBio yet. Use at your own risk and please report any problems you may encounter to the FEBio developers.

## 3.4 Globals Section

The *Globals* section is used to define some global variables, such as global constants, solute data and solid bound molecule data, as well user-defined global variables.

### 3.4.1 Constants

Global constants are predefined variables that some modules may rely on and currently include the universal gas constant  $R$  [ $\mathbf{F} \cdot \mathbf{L} / \mathbf{n} \cdot \mathbf{T}$ ], absolute temperature  $\theta$  [ $\mathbf{T}$ ], and Faraday constant  $F_c$  [ $\mathbf{Q} / \mathbf{n}$ ]. These constants must be expressed in units consistent with the rest of the analysis:

```
<Globals>
  <Constants>
    <R>8.314e-6</R>
    <T>298</T>
    <Fc>96485e-9</Fc>
  </Constants>
</Globals>
```

### 3.4.2 Solutes

In biphasic-solute, triphasic, and multiphasic analyses, a unique identifier must be associated with each solute in order to enforce consistent nodal degrees of freedom across boundaries of different materials. This unique identification is achieved by listing each solute species that appears in the entire finite element model and associating it with a unique *id*, *name*, and providing its charge number  $z^\alpha$ , molar mass  $M^\alpha$ , and density  $\rho_T^\alpha$ :

```
<Globals>
  <Solute>
    <solute id="1" name="Na">
      <charge_number>1</charge_number>
      <molar_mass>22.99</molar_mass>
      <density>0.97</density>
    </solute>
    <solute id="2" name="Cl">
      <charge_number>-1</charge_number>
      <molar_mass>35.45</molar_mass>
      <density>3.21</density>
    </solute>
    <solute id="3" name="Glc">
      <charge_number>0</charge_number>
      <molar_mass>180.16</molar_mass>
      <density>1.54</density>
    </solute>
  </Solute>
</Globals>
```

These solute identification numbers should be referenced in the *sol* attribute of solutes when defining a biphasic-solute (Section 4.15.2), triphasic or multiphasic material (Section 4.16.2).

The molar mass and density of solutes are needed only when solutes are involved in chemical reactions. When not specified, default values for these properties are set to 1.

### 3.4.3 Solid-Bound Molecules

In multiphasic analyses with chemical reactions involving solid-bound molecules, a unique identifier must be associated with each such molecule in order to enforce consistent properties across the entire model. This unique identification is achieved by listing each solid-bound species that appears in the entire finite element model and associating it with a unique *id*, *name*, charge number, molar mass and density:

```
<Globals>
  <SolidBoundMolecules>
    <solid_bound id="1" name="CS">
      <charge_number>-2</charge_number>
      <molar_mass>463.37</molar_mass>
      <density>1.5</density>
    </solid_bound >
  </SolidBoundMolecules >
</Globals>
```

The *id* number should be referenced in the *sbm* attribute of solid-bound molecules when included in the definition of a multiphasic material (Section 4.16). The charge number is used in the calculation of the fixed charge density contributed by this solid-bound molecule to the overall solid matrix fixed-charge density. The density is used in the calculation of the contribution of this molecule to the referential solid volume fraction. The density and molar mass are used in the calculation of the molar volume of this molecule in chemical reactions.

### 3.4.4 User Variables

Users can define custom, global variables, that can be referenced in other model parameters. This offers a convenient way to parameterize a model. Global user variables can also be used in optimizations. Currently, only scalar values can be defined. User variables are defined in the *Variables* child element.

To define a global variable, specify the *var* element, add the *name* attribute to name the variable, and specify the variable's value as the tag's value.

The following example defines a global variable, called A, and assigns the value 1.23 to it.

```
<Variables>
  <var name="A">1.23</var>
</Variables>
```

As mentioned, these global variables can be used to define other model variables, for example, in variables that are defined with mathematical expressions. To use a global variable, prefix the name with 'fem.'

```
<Material>
  <material name="mymat" type="neo-Hookean">
```

```
<E type="math">3*fem.A</E>  
<v>0.3</v>  
</material>  
</Globals>
```

### 3.5 Material Section

The material section is defined by the *Material* element. This section defines all the materials and material parameters that are used in the model. A material is defined by the *material* child element. This element has two attributes: *id*, which specifies a number that is used to reference the material, and *type*, which specifies the type of the material. The *material* element can also have a third optional attribute called *name*, which can be used to identify the material by a text description. A material definition might look like this:

```
<material id="1" type="neo-Hookean">
```

Or, if the optional *name* attribute is present:

```
<material id="2" type="rigid body" name="femur">
```

The material parameters that have to be entered depend on the material type. A complete list of available materials and their parameters is provided in [Chapter 4](#).

## 3.6 Mesh Section

The *Mesh* section contains all the mesh data, including nodal coordinates and element connectivity. It has the following sub-sections:

**Nodes** defines the nodal coordinates of the mesh

**Elements** defines the elements of the mesh. If the name attribute is provided, then this also defines an element set. Element sets defined this way are also referred to as *parts*.

**NodeSet** defines a node set

**Edge** defines an edge, i.e. a set of line elements

**Surface** defines a surface, i.e. a set of facets

**DiscreteSet** defines a set of discrete elements (e.g. springs)

**ElementSet** defines an element set

**SurfacePair** define a surface pair that can be used by a contact definition.

**PartList** define a named list of parts (i.e. element sets defined via the Elements tag)

At least one *Nodes* section must be defined and one *Elements* section. The other sections are optional. The *NodeSet*, *Edge*, *Surface*, *DiscreteSet*, *ElementSet*, *SurfacePair*, and *PartList* sections define sets of nodes, edges, facets, discrete elements, elements, surface pairs, and part lists, respectively, and can be referenced by other sections of the model file. For instance, boundary conditions can be defined by referencing the sets to which the boundary condition will be applied.

### 3.6.1 Nodes Section

The *Nodes* section contains nodal coordinates. It has an optional attribute called *name*. If this attribute is defined, the *Nodes* section also defines a node set.

```
<Nodes [name="<set name>"]>
```

The nodes are defined using the *node* tag which is a child of the *Nodes* section. Repeat the following XML-element for each node:

```
<node id="n">x,y,z</node>
```

The *id* attribute is the global identifier of the node and must be a unique number within the model definition. Nodal ids must be defined in increasing order, but do not have to be sequential. This *id* is used as a reference in the element connectivity section.

Multiple *Nodes* sections can be defined, but each node can only be defined once. For example:

```
<Nodes name="set01">
  <node id="1">0,0,0</node>
  ...
  <node id="101">1,1,1</node>
</Nodes>
```

```

<Nodes name ="set02">
  <node id="102">2,1,1</node>
  ...
  <node id="999">2,2,2</node>
</Nodes>

```

### 3.6.2 Elements Section

The *Elements* sections contain a list of the element connectivity data. Multiple *Elements* sections can be defined. The *Elements* section has the following attributes.

| Attribute | Description                             |
|-----------|-----------------------------------------|
| type      | element type                            |
| name      | unique name that identifies this domain |

Each *Elements* section contains multiple *elem* elements that define the element connectivities. Each *elem* tag has a *id* attribute that defines the element number. For example, the following *Elements* section defines a list of hexahedral elements:

```

<Elements type="hex8" name="part1">
  <elem id="1">1,2,3,4,5,6,7,8</elem>
  <elem id="2">9,10,11,12,13,14,15,16</elem>
  ...
</Elements>

```

FEBio classifies elements in three categories, namely *solids*, *shells*, and *beams*.

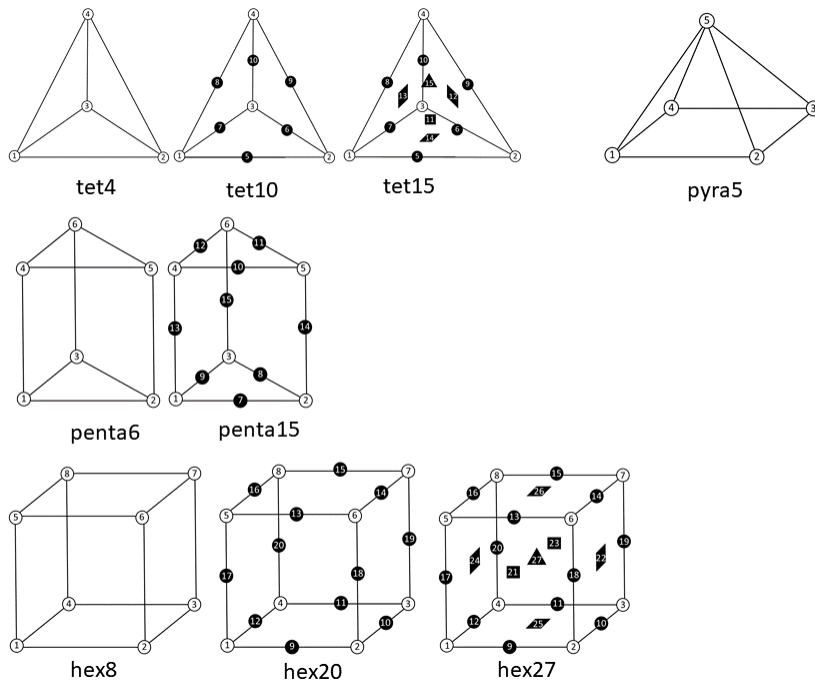
#### 3.6.2.1 Solid Elements

The following solid element types are defined:

|         |                                               |
|---------|-----------------------------------------------|
| hex8    | 8-node trilinear hexahedral element           |
| hex20   | 20-node quadratic hexahedral elements         |
| hex27   | 27-node quadratic hexahedral elements         |
| penta6  | 6-node linear pentahedral (wedge) element     |
| penta15 | 15-node quadratic pentahedral (wedge) element |
| pyra5   | 5-node linear pyramidal element               |
| pyra13  | 13-node quadratic pyramidal element           |
| tet4    | 4-node linear tetrahedral element             |
| tet10   | 10-node quadratic tetrahedral element         |
| tet15   | 15-node quadratic tetrahedral element         |

The node numbering has to be defined as in the figure below.





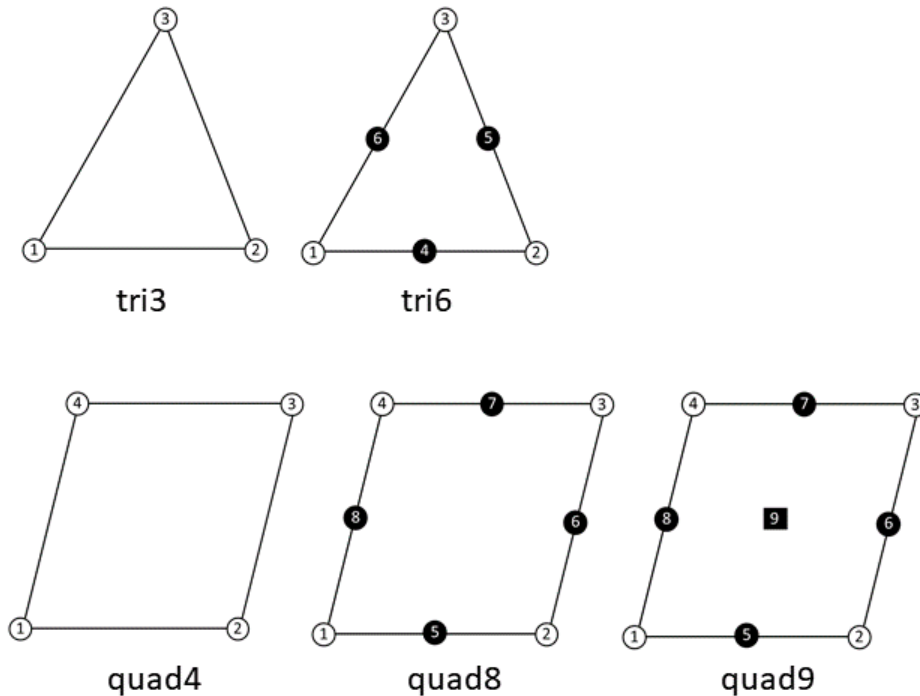
Node numbering for solid elements

### 3.6.2.2 Shell Elements

FEBio currently supports the following shell elements:

|       |                                                                     |
|-------|---------------------------------------------------------------------|
| tri3  | 3-node linear triangular shell element (compatible strain)          |
| tri6  | 6-node quadratic triangular shell element (compatible strain)       |
| quad4 | 4-node linear quadrilateral shell element (compatible strain)       |
| quad8 | 8-node quadratic quadrilateral shell element (compatible strain)    |
| quad9 | 9-node quadratic quadrilateral shell element (compatible strain)    |
| q4ans | 4-node linear quadrilateral shell element (assumed natural strain)  |
| q4eas | 4-node linear quadrilateral shell element (enhanced assumed strain) |

For shell elements that use a *compatible strain* formulation, the calculation of strain components is based on nodal displacements, similar to hexahedral or pentrahedral elements. Users should be aware that this compatible strain formulation is very susceptible to element locking when the shell thickness is much smaller than the shell size (e.g., when the aspect ratio is less than 0.01). Therefore, these shell formulations should be used with caution, keeping in mind this important constraint. Conversely, these shell elements perform very well when they are attached to solid elements (e.g., skin over muscle), or sandwiched between shell elements (e.g., cell membrane separating cytoplasm from extra-cellular matrix), as described in [48].



#### Node numbering for shell elements

The element-locking limitation of compatible strain shell formulations has motivated the development of specialized shell formulations that attempt to overcome locking. The FE literature on this subject is rather extensive and we refer the reader to the excellent review chapter by Bischoff et al. [22] on this topic. Methods for overcoming locking include the assumed natural strain (ANS) formulation for transverse shear strains [60, 18] and transverse normal strains [19, 21]. The ANS formulation may be supplemented with the enhanced assumed strain (EAS) method [82] and extended to large deformations [54, 88, 80]. FEBio includes the ANS (*q4ans*) and EAS (*q4eas*) quadrilateral shell element formulations of Vu-Quoc and Tan [88], using a seven-parameter EAS interpolation, which is otherwise substantially similar to the five-parameter interpolation presented in an earlier study by Klinkel et al. [54]. These shell elements are not suitable for attachment to a solid element, nor sandwiching between two solid elements. Since they don't experience element locking, they should be loaded more slowly than compatible strain shell elements. The most accurate of these shell formulation is *q4eas*.

By default, the nodes of a shell element define its *top* face. The location of the shell element *bottom* face is calculated from the nodal values of the shell thickness, along the opposite direction

of nodal normals. The nodal normal is evaluated by averaging the top face normal of all shell elements sharing that node. When a shell element is attached to a solid element (e.g., when the shell nodes also represent the nodes of one face of the solid element), FEBio automatically accounts for the shell thickness, reducing the height of the solid element in the direction of the shell normal. It is the user's responsibility to ensure that the shell thickness is less than that of the solid element; otherwise, a negative Jacobian error will be automatically generated.

Prior to FEBio 2.6, the nodes of a shell element defined its *middle* face (halfway between top and bottom, such that the shell element extends above and below the face defined by the shell nodes). This older formulation can be recovered by setting *shell\_formulation* to 0 in the *Control* section. This setting only works with compatible strain shell formulations.

By default, stresses and strains reported for a shell element are evaluated by averaging all the integration point values within the shell element, thus producing values that reflect the response along the middle surface. To display shell stresses or strains at the top or bottom faces, use plot variables that start with 'shell top...' or 'shell bottom...', such as 'shell top stress' or 'shell bottom nodal strain'.

When shell domains intersect such that three or more shell elements share the same edge, as in a T-connection, users should set *shell\_normal\_nodal* to 0 in the *ShellDomain* section. This setting ensures that the bottom face of a shell element is calculated using the top face normal, instead of nodal normals, since the latter would be ill-defined for nodes belonging to such an edge.

Finally, please note that shell elements cannot be stacked in FEBio.

### 3.6.2.3 Beam Elements

FEBio supports some truss and beam element formulations. Currently, only linear trusses and linear and quadratic beam elements are supported.

|       |                                     |
|-------|-------------------------------------|
| line2 | 2-node linear truss or beam element |
| line3 | 3-node quadratic beam element       |

Whether a truss or beam formulation is used, is determined in the corresponding *BeamDomain* section.

## 3.6.3 NodeSet Section

The *NodeSet* section allows users to define node sets. These node sets can then later be used in the definition of boundary conditions and loads. A node set is defined by the *NodeSet* tag. This tag takes one required attribute, *name*, which defines the name of the node set. A node set definition is followed by a list of node IDs. For example,

```
<NodeSet name="nodeset1">1, 2, 101, 102</NodeSet>
```

For larger nodesets, it is recommended to split the list across several lines.

```
<NodeSet name="nodeset1">
  1, 2, 3, 4,
  5, 6, 7, 8,
  9, 10, 11, 12
</NodeSet>
```

### 3.6.4 Edge Section

The *Edge* section allows users to define a set of edges. These edges can be used to define boundary conditions or loads.

```
<Edge name="edge1">
  <line2 lid="1">1,2</line2>
  <line2 lid="2">2,3</line2>
</Edge>
```

Here, the *lid* attribute defines the *local* id of the edge, local with respect to the edge definition. The local ids must begin at 1 and defined sequentially.

The edge elements are defined by using a tag that depends on the type of the edge element. The following edge elements are currently supported.

**line2** 2-node line element

**line3** 3-node line element

Edges cannot overlap element boundaries and must be a valid edges of elements.

### 3.6.5 Surface Section

The *Surface* section allows users to define surfaces. These surfaces can then later be used to define the boundary conditions and contact definitions. A surface definition is followed by a list of surface elements, following the format described below.

```
<Surface name="named_surface">
  <quad4 id="1">1,2,3,4</quad4>
  <...>
</Surface>
```

The *Surface* takes one required attribute, namely the *name*. This attribute sets the name of the surface. This name will be used later to refer back to this surface.

The following surface elements are available:

**quad4** 4-node quadrilateral element

**quad8** 8-node serendipity quadrilateral element

**tri3** 3-node triangular element

**tri6** 6-node quadratic triangular element

**tri7** 7-node quadratic triangular element

The value for the surface element is the nodal connectivity:

```
<quad4 id="n">n1,n2,n3,n4</quad4>
<tri3 id="n">n1,n2,n3</tri3>
```

Surface elements cannot overlap element boundaries. That is, the surface element must belong to a specific element. Surface elements do not contribute to the total number of elements in the mesh. They are also not to be confused with shell elements.

### 3.6.6 ElementSet Section

The *ElementSet* section can be used to define an element set. Element sets can be used to output data for only a subset of elements. An element set is defined through the *ElementSet* tag, which takes one attribute, namely *name* that specifies the name of the element set. The value of the tag is the list of element IDs. For example,

```
<ElementSet name="set01">
  1, 2, 3, 4
  5, 6, 7, 8
</ElementSet>
```

### 3.6.7 DiscreteSet Section

The *DiscreteSet* section defines sets of discrete elements. These sets can then be used to define e.g. springs.

```
<DiscreteSet name="springs">
  <delem>1,2</delem>
  <delem>3,4</delem>
</DiscreteSet>
```

### 3.6.8 SurfacePair Section

The *SurfacePair* section defines a pair of surfaces that can be used to define surface-to-surface interactions (e.g. contact).

```
<SurfacePair name="contact1">
  <primary>surface1</primary>
  <secondary>surface2</secondary>
</SurfacePair>
```

The surfaces (i.e. *surface1* and *surface2*) are surfaces defined in *Surface* sections. Because surfaces must already be defined before they can be referenced in the *SurfacePair* section, the *Surface* sections must be defined before the *SurfacePair* section.

### 3.6.9 PartList

A *part* is an element set that is defined via the *Elements* tag. Parts can be grouped together in part lists. A part list is defined via the *PartList* tag. The value of this tag is a list of the names of the parts that belong to the list.

```
<PartList name="allParts">part1,part2</PartList>
```

Here, *part1* and *part2* are parts defined via *Elements* sections. Because part lists depend on the *Elements* sections, the *PartList* tag must be defined after all *Elements* sections have been defined.

Part lists can be used, for instance, in body loads if the body load is to be applied to multiple parts.

### 3.6.10 Implicit Partitions

In addition to the explicit mesh partitions that can be defined in the Mesh section, as detailed above, FEBio also supports *implicit partitions*. This allows you to assign a different type of partition to a model component than the target partition. For instance, if a boundary condition requires a node set, using implicit partitioning, you can assign a surface or an element set.

Consider the following example.

```
<bc type="zero displacement" node_set="set1"/>
```

FEBio expects a node set with the name set1 defined in the Mesh section. However, if set1 is a surface, then this surface can be assigned to the boundary condition using the following syntax.

```
<bc type="zero displacement" node_set="@surface:set1"/>
```

The presence of the specifier @surface informs FEBio that set1 is a surface and the node set needs to be extracted from the surface definition.

Mesh partitions can only be assigned this way if the target partition is a subset of the source partition. For instance, it is not possible to assign a node set to a surface partition.

The following specifiers are available.

| specifier  | description                                                   | Target partition |
|------------|---------------------------------------------------------------|------------------|
| @surface   | References a surface                                          | nodeset          |
| @edge      | References an edge                                            | nodeset          |
| @elem_set  | References an element set (either via Elements or ElementSet) | nodeset          |
| @part_list | References a part list                                        | nodeset, surface |

Note that the @part\_list is the only specifier that can be used to assign a part list to a surface. This can be useful for defining contact surfaces where the surface might have a complex shape and is difficult to extract manually.

## 3.7 MeshDomains Section

The *MeshDomains* section assigns various physical attributes to the element sets of the mesh. For each of the *Elements* sections in the *Mesh* section, define a *SolidDomain*, *ShellDomain*, or *BeamDomain* section, depending on whether the elements are solid, shells, or beams (or trusses). Each domain also requires a material name that defines the material properties of the domain.

### 3.7.1 SolidDomain Section

Use the *SolidDomain* section to define a solid domain. This section takes the following parameters:

**name** The name of the element set (defined via an *Elements* section). This element set must only contain solid elements.

**mat** The name of the material that will be assigned to this domain

**type** (optional) the specific formulation used for this domain.

The optional **type** attribute can be added to specify a specific solid formulation. If omitted, FEBio will determine the proper domain type by inspecting the material and element types used. However, users can explicitly set the type attribute. The following solid domain types are defined. If the domain type is specified explicitly, the user must make sure that the assigned material is compatible with the domain type. (For instance, you cannot assign a biphasic material to an elastic-solid.)

| type                  | Description                                                                                  |
|-----------------------|----------------------------------------------------------------------------------------------|
| elastic-solid         | standard solid formulation (default for most materials) (1,2)                                |
| three-field-solid     | A three-field formulation used for modeling (nearly) incompressible materials. (default for) |
| rigid-solid           | Rigid solid domain. (default for rigid materials.)                                           |
| udg-hex               | uniform deformation gradient hex formulation                                                 |
| sri-solid             | Formulation that uses a selective reduced integration rule                                   |
| remodeling-solid      | Formulation used for elastic remodeling problems.                                            |
| elastic-mm-solid      | Formulation used for multi-scale domains. (requires the “micro-material”)                    |
| ut4-solid             | Uniform nodal strain tetrahedron. (3)                                                        |
| biphasic-solid        | The standard formulation for biphasic domains.                                               |
| biphasic-solute-solid | The standard formulation for biphasic-solute domains.                                        |
| multiphasic-solid     | The standard formulation for multiphasic domains.                                            |
| triphasic-solid       | The standard formulation for triphasic domains (deprecated, use multiphasic-solid instead)   |
| fluid-3D              | The standard formulation for fluid domains.                                                  |
| fluid-FSI-3D          | The standard formulation for fluid-FSI domains.                                              |

#### Comments:

1. For the elastic-solid domain, the *elem\_type* attribute can be specified to further specialize the quadrature rule. The values of the rule depend on the specific element type. The tables below show the available integration rules for the different element types. The values marked with an asterisk (\*) are the default.

- For the *hex8* element, the following values are defined.

| hex8    | Description                                                       |
|---------|-------------------------------------------------------------------|
| HEX8G8* | Gaussian integration using 2x2x2 integration points.              |
| HEX8G6  | Alternative integration rule for bricks using 6 integration point |

- For the *hex20* element, the following values are defined.

| hex20    | Description                                          |
|----------|------------------------------------------------------|
| HEX20G8* | Gaussian integration using 2x2x2 integration points. |

- For the *tet4* element, the following values are allowed.

| tet4    | Description                                            |
|---------|--------------------------------------------------------|
| TET4G1* | Gaussian integration rule using one integration point. |
| TET4G4  | Gaussian integration rule using 4 integration points.  |

- For the *tet10* element, the following integration rules are supported.

| tet10     | description                                                |
|-----------|------------------------------------------------------------|
| TET10G4   | Gaussian integration rule using 4 integration points       |
| TET10G8*  | Gaussian integration rule using 8 integration points       |
| TET10GL11 | Gauss-Lobatto integration rule using 11 integration points |

The Lobatto integration rule differs from a regular Gauss integration rule in that it includes the vertices of the tetrahedral element. The *Lobatto11* integration rule uses the 10 tetrahedral nodes, plus one integration rule located at the center of the element.

- For the *tet15* element, the following integration rules are defined.

| tet15     | description                                           |
|-----------|-------------------------------------------------------|
| TET15G8   | Gaussian integration rule using 8 integration points  |
| TET15G11  | Gaussian integration rule using 11 integration points |
| TET15G15* | Gaussian integration rule using 15 integration points |

- For the *penta15* element, the following values are defined.

| penta15    | Description                                          |
|------------|------------------------------------------------------|
| PENTA15G8* | Gaussian integration using 2x2x2 integration points. |

2. The “solid-elastic” defines the parameters *secant\_stress* and *secant\_tangent*. When, specified, the formulation will use a secant approximation of the corresponding variable. When setting *secant\_stress* is set to 1, the stress is calculated from a secant approximation that uses the material’s strain-energy density. Similarly, when *secant\_tangent* is specified, the material tangent is approximated using the PK2 stress.

3. The *ut4-solid* is a special formulation for tetrahedral elements that uses a nodally-averaged integration rule, as proposed by Gee et al [40]. This formulation requires additional parameters. To override the default values, use the following alternative syntax:



```

<SolidDomain name="Part1" type="ut4-solid" mat="material1">
  <alpha>0.05</alpha>
  <iso_stab>0</iso_stab>
</SolidDomain>

```

The *alpha* parameter defines the amount of “blending” between the regular tet-contribution and the nodally integrated contribution. The value must be between 0 and 1, where 0 means no contribution from the regular tet and 1 means no contribution from the nodally averaged tet. The *iso\_stab* parameter is a flag that chooses between two slightly different formulations of the nodally integrated tet. When set to 0, the stabilization is applied to the entire virtual work, whereas when set to 1 the stabilization is applied only to the isochoric part. See the [FEBio Theory Manual](#) for a detailed description of this formulation.

### 3.7.2 ShellDomain Section

Use the *ShellDomain* section to define a shell domain. This section takes two parameters:

**name** The name of the element set (defined via an *Elements* section). This element set must only contain shell elements.

**mat** The name of the material that will be assigned to this domain

**type** (optional) the specific shell formulation

Similar to solid domains, the particular shell formulation can be requested via the **type** attribute. If omitted, FEBio will determine the formulation from the material and shell element types. If specified explicitly, the user must ensure that the correct material is assigned to the domain. The following shell domain types are available.

| type                  | Description                                                                                  |
|-----------------------|----------------------------------------------------------------------------------------------|
| elastic-shell         | standard shell formulation (default for most materials) (1)                                  |
| three-field-shell     | A three-field formulation used for modeling (nearly) incompressible materials. (default for) |
| rigid-shell           | Rigid shell domain. (default for rigid materials.)                                           |
| elastic-shell-old     | The “old” shell formulation, used in FEBio version 2.0 and older.                            |
| elastic-shell-eas     | Enhanced-Assumed-Strain shell; an alternative shell formulation for incompressible shells.   |
| elastic-shell-ans     | Assumed-Natural-Strain shell; an alternative shell formulation for incompressible shells.    |
| biphasic-shell        | The standard formulation for biphasic shell domains.                                         |
| biphasic-solute-shell | The standard formulation for biphasic-solute shell domains.                                  |
| multiphasic-shell     | The standard formulation for multiphasic shell domains.                                      |

The default shell thickness can be specified as a parameter of the ShellDomain section.

```

<ShellDomain name="skin">
  <shell_thickness>0.01</shell_thickness>
</ShellDomain>

```

If the shell thickness parameter is not specified, the initial shell thickness is set to zero, and it is assumed that the actual shell thickness is specified in the MeshData section.

### 3.7.3 BeamDomain Section

Beam and trusses can be defined with the *BeamDomain* section. This tag takes the following attributes.

**name** This is the name of the part to which this domain will be assigned. The part is defined via a named *Elements* section.

**mat** The name of the material that will be assigned to this domain.

**type** The specific formulation that will be used.

The particular beam formulation that can be assigned is listed below.

| type          | Description                                                                                        |
|---------------|----------------------------------------------------------------------------------------------------|
| linear-truss  | A linear truss formulation.                                                                        |
| elastic-truss | An extension of the linear-truss formulation that allows the use of standard FEBio materials (EXP) |
| linear-beam   | A linear beam formulation.                                                                         |

## 3.8 MeshData Section

The *MeshData* section is where data is specified that can be mapped to the mesh. The data can then be applied to a parameter of the model, e.g. a material parameter, or a pressure load.

To define a mesh data section add one of the following tags.

**NodeData** Define data on a node set

**SurfaceData** Define data on a surface

**ElementData** Define data on an element set

The following attributes can be added.

**name** All mesh data sections require a name, which can be specified with the *name* attribute. This name will be used by model parameters to reference the particular mesh data.

Data maps can be defined in two ways: 1) explicitly, where the value of each node (face, element) can be specified, or 2) implicitly, via data generators that define the values of each node (face, element) procedurally.

When mesh data maps are defined explicitly, the following attributes should be specified in addition to the name attribute.

**data\_type** defines the type of data

The following table lists the supported data types.

| data type | Description                                                 |
|-----------|-------------------------------------------------------------|
| scalar    | A single floating point value                               |
| vec2      | A 2D vector defined as $x, y$                               |
| vec3      | A 3D vector defined as $x, y, z$                            |
| mat3      | A 3x3 tensor, row-major sorted                              |
| mat3s     | A symmetric 3x3 tensor, defined as $xx, yy, zz, xy, yz, xz$ |

If the data type is not defined, the *scalar* data type is assumed.

### 3.8.1 Data Generators

Data generators can be used to populate the values of a data map procedurally. To use a data generator, specify the *type* attribute. If the *type* attribute is defined, the child tags must be the parameters of the generator. For example,

```
<MeshData>
  <NodeData name="E_map" type="math">
    <math>X*X + Y*Y + Z*Z</math>
  </NodeData>
</MeshData>
```

Data generators that can be used for a specific mesh data are described in the sections below.

### 3.8.2 ElementData

This section defines data that can be mapped to the elements of an element set. The element set is specified via the *elem\_set* attribute.

**Defining maps** A map is defined by using the *name* attribute to give the map a name. This name can then be used in mapped parameters (see Appendix A).

```
<MeshData>
  <ElementData name="my_map" elem_set="part1">
    <elem lid="1">1.23</elem>
    <elem lid="2">1.25</elem>
  </ElementData>
</MeshData>
```

The following data generators are available for element data sections.

**surface-to-surface\_map** The “*surface-to-surface map*” generator defines a data field on the domain bounded by two surfaces. The data values are interpolated using a user-defined function between the surfaces.

```
<MeshData>
  <ElementData name="Emap" generator="surface-to-surface map" elem_set="Part1">
    <function type="math">x*x+1</function>
    <bottom_surface surface="surface1"/>
    <top_surface surface="surface2"/>
  </ElementData>
</MeshData>
```

**special generators** A few special data generators do not generate maps, but instead modify model data in a more direct way. The following table lists the currently supported special generators.

| Property        | Description                         | Data type/format |
|-----------------|-------------------------------------|------------------|
| fiber           | Specify a local fiber direction     | vec3/const       |
| mat_axis        | Specify local element material axes | vec3/const       |
| shell thickness | Specify the shell element thickness | float/shape      |

For example, to assign shell thicknesses to an element set, use the following syntax.

```
<MeshData>
  <ElementData type="shell thickness" elem_set="part1">
    <elem lid="1">1.0, 1.0, 1.0, 1.0</elem>
    <elem lid="2">1.0, 1.0, 1.0, 1.0</elem>
  </ElementData>
</MeshData>
```

### 3.8.3 SurfaceData

This section allows users to define data that can be mapped to the surfaces of certain boundary conditions and loads. This section only defines user-defined data maps, i.e. the *name*, *data\_type* and *data\_format* attributes must be used. The surface is defined via the *surface* attribute.

```
<SurfaceData name="values" surface="surface1">
  <face lid="1">1.23</face>
  <face lid="2">3.45</face>
</SurfaceData>
```

The surface definition is defined in the *Mesh* section. Note that the ids refer to the local ids of the surface facets. By default, the *scalar* data type and *const* data format are assumed so only one value per facet is expected. Other data types and formats can be specified with the *data\_type* and *data\_format* attributes.

```
<SurfaceData name="data" data_type="vec3" surface="surface1">
  <face lid="1">0.0, 1.0, 0.0</face>
  <face lid="2">0.0, 2.0, 0.0</face>
</SurfaceData>
```

Optionally, the surface data may be described with a data generator. The data generator is specified via the *type* attribute. The following data generators are available for surface data.

**parabolic\_map** The “*parabolic map*” defines a scalar data field with a parabolic profile on a surface. The data is determined by solving a heat-type problem with a constant source and homogeneous boundary conditions on the edge of the surface.

```
<MeshData>
  <SurfaceData name="pressure" type="parabolic map" surface="surface1">
    <value>1.0</value>
  </SurfaceData>
</MeshData>
```

### 3.8.4 EdgeData

This section allows users to define user-defined data maps that will be mapped to an edge defined in the Mesh section.

```
<EdgeData name="data" edge="edge1">
  <edge lid="1">1.0</edge>
  <edge lid="2">2.0</edge>
</EdgeData>
```

Notice that the *local* IDs are required here.

### 3.8.5 NodeData

This section allows users to define data that will be mapped to node sets.

```
<NodeData name="data" node_set="set1">  
  <node lid="1">1.0</node>  
  <node lid="2">2.0</node>  
</NodeData>
```

For node data the *data\_format* attribute, if specified, will be ignored.

## 3.9 Mesh Adaptor Section

The FEBio Adaptive Mesh Refinement framework (FEAMR, pronounced “femur”) allows users to make modifications to the mesh during the analysis of the model. There are currently two mesh adaptor applications supported, namely mesh erosion, which removes elements from the mesh, and remeshing, which can be used to adaptively refine or coarsen the mesh based on the solution. The mesh adaptors are controlled via user-defined criteria that determine the conditions under which to modify the mesh.

A mesh adaptor is applied at the end of a timestep, after the Newton-iterations and augmentations have converged. If the mesh adaptor modifies the mesh, then the timestep is solved again, using the new mesh. Several adaptors define the *criterion* property, which evaluates the variable or metric that is used by the adaptor, and optionally, can be used to filter the elements to which the adaptor will be applied. How the criterion is used, depends on the adaptor. For instance, the erosion adaptor uses it to identify the elements to deactivate. Remeshing adaptors will use it to set the new element size. It is allowed to use several mesh adaptors in a single model, but keep in mind that the order in which the mesh adaptors are listed in the input file is important. For instance, if two mesh adaptors are defined, the second adaptor will operate on the mesh created by the first adaptor. Mesh adaptors used for adaptive refinement will also map data between the old mesh and the new mesh. User data, defined in the MeshData section, is always mapped between meshes. If the model is history-dependent, it might be important to map additional data. For instance, a visco-elastic material needs to map stress-history variables. By default, mapping data (except for user-data) is not done unless the user sets a specific flag (`map_data`). New mesh adaptors and adaptor criteria can be defined via the plugin framework.

### 3.9.1 The MeshAdaptor Section

Mesh adaptors are added to the FEBio Input file (version 3.0 and above) in a new section, called **MeshAdaptor**. This section must appear after the Mesh and MeshDomains sections. The MeshAdaptor section may also be defined in a *step* section in order to apply the adaptor only during that step. Inside the MeshAdaptor section, each mesh adaptor is defined via the **mesh\_adaptor** tag. This tag allows the following attributes:

| attribute       | Description                                            |
|-----------------|--------------------------------------------------------|
| <i>type</i>     | set the type of the mesh adaptor.                      |
| <i>elem_set</i> | The element set to apply the adaptor to (1) (Optional) |

*Comments:*

1. The **elem\_set** attribute is optional, but can be used to apply the adaptor to only a part of the mesh. This is useful to limit the effect of an adaptor to a part of the mesh. The element set must reference a domain (i.e. defined via Elements) of a mesh, or a set of domains. It cannot be an element set defined through the ElementSet tag. (Note that not all mesh adaptors support this attribute.) If omitted, the adaptor is applied to the entire mesh.

Many mesh adaptors require a mesh adaptor criterion. The criterion defines the metric that will assign a value to each element. This value can then in turn be used by the adaptor to decide what to do with a particular element. Adaptor criteria are defined in section [3.9.2](#).

### 3.9.1.1 The erosion adaptor

The erosion adaptor can be used to deactivate elements in a mesh, based on a user-defined criterion. It defines the following properties:

- **max\_iters**: Set the max number of adaption iterations, for each time step.
- **remove\_islands**: If true, removes small element sets that have become disconnected from the main mesh.
- **max\_elems**: Set the max number of elements to erode, per adaptation.
- **sort**: sort the element values returned by the criterion.
- **criterion**: Set the criterion that determines the element metric to evaluate and which elements to erode.
- **erode\_surfaces**: Sets the policy of handling surface facets that attach to eroded elements. The value can be set to one of the following values:
  - **no**: don't erode surfaces (default)
  - **yes**: erode surfaces if the underlying element is eroded.
  - **grow**: add the newly exposed facets of eroded elements to the surface.
  - **reconstruct**: reconstruct surface. This only works if the surface is defined via a part.

*Example:*

In this example, elements that have an (integration point averaged) effective stress larger than 0.3, are eroded. Only three elements are eroded per adaptation cycle.

```
<mesh_adaptor type="erosion">
  <max_iters>1</max_iters>
  <max_elems>3</max_elems>
  <sort>1</sort>
  <criterion type="min-max filter">
    <min>0.3</min>
    <clamp>0</clamp>
    <data type="stress"/>
  </criterion>
</mesh_adaptor>
```

### 3.9.1.2 The mmg\_remesh adaptor

The mmg\_remesh adaptor can be used for adaptive mesh refinement of linear tetrahedral meshes.

- **max\_iters**: The maximum number of adaption iterations per time step.
- **max\_elements**: The maximum number of elements. If the number of elements of the mesh is larger, the adaptor is not applied.
- **min\_element\_size**: Set the desired minimal element size.



- **hausdorff**: The Hausdorff distance. This is used to preserve curved sections. (default = 0.01).
- **gradation**: This parameter sets the spatial gradient of element size. (default = 1.3). Value should be larger than 1.
- **mesh\_coarsen**: Allow mesh-coarsening
- **normalize\_data**: normalize the element values returned by the criterion between 0 and 1.
- **relative\_size**: Sets whether the size function is to be interpreted as a relative size function (i.e. a scale function to the current element size), or an absolute element size metric.
- **criterion**: The metric used to assign values to elements. (See section [3.9.2](#))
- **size\_function**: set the mesh size function. This function is applied to the (optionally normalized) element values returned by the criterion.

Example:

```
<mesh_adaptor type="mmg_remesh">
  <max_iters>1</max_iters>
  <criterion type="stress"/>
  <relative_size>1</relative_size>
  <normalize_data>0</normalize_data>
  <mesh_coarsen>0</mesh_coarsen>
  <size_function type="step">
    <x0>10</x0>
    <left_val>1</left_val>
    <right_val>0.5</right_val>
  </size_function>
</mesh_adaptor>
```

### 3.9.1.3 hex\_refined2d

The hex\_refined2d adaptor refines elements in the XY plane. (It assumes that the mesh consists of a single layer of hex elements, aligned in the XY coordinate plane.) Elements are split by splitting each edge into 2. This may result in a non-conforming mesh, where an element node in the refined mesh lies on an edge in the parent mesh. Such *hanging nodes* are constrained using linear constraints and are forced to remain in the same relative position as the parent edge. This ensures that the refined mesh remains “watertight”.

The following parameters are available.

| parameter       | Description                                                          |
|-----------------|----------------------------------------------------------------------|
| max_elem_refine | The max number of elements to refine per refinement step.            |
| criterion       | The adaption criterion to use                                        |
| max_value       | Elements with criterion values above this threshold will be refined. |

### 3.9.1.4 tet\_refine

Applies a uniform refinement of a tetrahedral mesh.

### 3.9.1.5 hex\_refine

The hex\_refined adaptor refines elements by splitting each edge into 2. This may result in a non-conforming mesh, where an element node in the refined mesh lies on an edge in the parent mesh. Such *hanging nodes* are constrained using linear constraints and are forced to remain in the same relative position as the parent edge. This ensures that the refined mesh remains “watertight”.

The following parameters are available.

| parameter       | Description                                                          |
|-----------------|----------------------------------------------------------------------|
| max_elem_refine | The max number of elements to refine per refinement step.            |
| criterion       | The adaption criterion to use                                        |
| max_value       | Elements with criterion values above this threshold will be refined. |

## 3.9.2 Mesh Adaptor Criteria

The purpose of the mesh adaptor criteria is to assign values to elements. Some adaptors also act as filters that limit the effect of the adaptor to certain elements. These values will then be used by the mesh adaptor to modify the mesh. How these values are used depends on the adaptor. For instance, the erosion uses it to decide which elements to erode. The mesh refinement adaptors use it to define the size function.

The following criteria are currently available. A detailed description of these adaptor criteria is given below.

| type              | Description                                                                          |
|-------------------|--------------------------------------------------------------------------------------|
| max_variable      | Evaluates average of element's nodal values of a model primary variable.             |
| element_selection | Sets the element selection to which the adaptor should be applied (only for erosion) |
| stress            | Evaluate the (integration point average) effective stress of each element.           |
| damage            | Evaluate the (integration point average) damage value.                               |
| relative error    | Evaluate a relative error measure.                                                   |
| spring force      | Evaluate the axial force of the spring.                                              |
| spring stretch    | Evaluate the stretch of a spring.                                                    |
| math              | Specify the element value based on a mathematical expression.                        |
| min-max filter    | Selects elements whose value falls within a min-max range.                           |
| contact gap       | Select elements that are in contact.                                                 |

### 3.9.2.1 max\_variable

The max\_variable criterion evaluates elements based on a primary variable's degrees of freedom.

- **dof**: The degree of freedom that defines the variable to inspect.

### 3.9.2.2 element\_selection

Sets the selected elements directly.

- **element\_list:** a comma-separated list of element indices.

### 3.9.2.3 math

Sets the scale factor for each element.

- **scale:** the scale factor to use. This can be a constant, or a mathematical expression of position (use X, Y, Z for the nodal coordinates), or a scalar map defined in the MeshData section.

Example:

```
<mesh_adaptor type="mmg_remesh">
  <max_iters>1</max_iters>
  <criterion type="math">
    <math>1*(1 - (X^2+Y^2)) + (X^2+Y^2)*(0.5*(Z-1)^2 - Z*(Z - 2))</math>
  </criterion>
</mesh_adaptor>
```

### 3.9.2.4 stress

This criterion selects elements that exceed a maximum stress value.

- **metric:** The stress measure to use: 0=effective stress, 1=max principal stress.

### 3.9.2.5 max\_damage

This criterion evaluates the maximum damage value for each element. This can only be used with materials that define the damage property.

### 3.9.2.6 relative error

This criterion evaluates a relative error metric for each element.

- **error:** If nonzero, this criterion will evaluate a relative size metric for each element. Otherwise, the actual relative error is evaluated.
- **data:** Specify the metric to evaluate the element values. Any of the other criteria defined above can be used.

### 3.9.2.7 min-max filter

This criterion evaluates a metric for each element, but then filters out the elements that have values within a min-max range.

- **min:** set the minimum value of the filter

- **max**: set the maximum value of the filter
- **clamp**: If true, the element values will be clamped to the range.
- **data**: Specify the metric to evaluate the element values. Any of the other criteria defined above can be used.

### 3.9.2.8 contact gap

This criterion evaluates the contact gap for all elements that are in contact.

- **gap**: contact gap threshold value.

### 3.9.3 Limitations

Please take into account the following limitations of the FEAMR framework.

1. Although contact can be used with the FEAMR framework, currently no contact data is mapped from the old mesh to the new mesh. It is advised to only use penalty-based contact interfaces.
2. It is not advised to erode elements that are in contact, since this may not produce the correct result. If you want to use erosion with contact, make sure to set the *erode\_surfaces* to an appropriate value.
3. Mapping history-data is supported (if the proper flag is set), but can be very time consuming since FEBio maps all material point data between meshes.
4. The FEAMR framework does not work with the auto time-stepper. It is advised to use fixed time stepping when using mesh adaptation.

## 3.10 Initial Section

The *Initial* section defines all the initial conditions that may be applied to the analysis. An initial condition allows users to set the initial values of solution variables (e.g. velocity) as well as initialize the values of some special components.

An initial condition is defined via the *ic* element and requires the *type* attribute and *node\_set* attribute. The *type* attribute defines the particular initial condition that will be applied. The *node\_set* attribute defines the set of nodes that this initial condition affects. This node set must be defined in the Mesh section of the input file. Using implicit partitions (see section 3.6.10) surfaces, element sets, and part lists can also be assigned to the *node\_set* attribute.

An optional name attribute can be defined to assign a unique name to the initial condition.

*Example:*

```
<ic name="ic01" type="velocity" node_set="set1">
  <value>1,0,0</value>
</ic>
```

### 3.10.1 The *init\_dof* Initial Condition

The *init\_dof* initial condition can be used to initialize any degree of freedom of the model. It supports the following parameters. This feature is deprecated and it is better to use one of the specific initial conditions listed in the sections below.

| parameter | description                                             | initial value |
|-----------|---------------------------------------------------------|---------------|
| dof       | The degree of freedom identifier                        | -             |
| value     | The value for the corresponding nodal degree of freedom | 0             |

*Example:*

```
<ic name="ic01" type="init_dof" node_set="set1">
  <dof>vx</dof>
  <value>1</value>
</ic>
```

### 3.10.2 The *displacement* Initial Condition

The *displacement* initial condition can be used to set the initial displacement for a solid-mechanics analysis.

| parameter | description                            | initial value |
|-----------|----------------------------------------|---------------|
| value     | The value for the initial displacement | 0,0,0         |

It should be noted that the initial displacement will be reflected in the plot file at time=0, but not the stress. However, internally, the stress at time=0 will be taken into account as the calculation is performed for subsequent times.

*Example:*

```
<ic type="displacement" node_set="set1">
  <value>1.0,0.0,0.0</value>
</ic>
```

### 3.10.3 The velocity Initial Condition

The *velocity* initial condition can be used to set the initial velocity for a dynamic solid-mechanics analysis.

| parameter | description                        | initial value |
|-----------|------------------------------------|---------------|
| value     | The value for the initial velocity | 0,0,0         |

*Example:*

```
<ic type="velocity" node_set="set1">
  <value>1.0,0.0,0.0</value>
</ic>
```

### 3.10.4 The shell velocity Initial Condition

The *shell velocity* initial condition can be used to set the initial velocity for the for the back-nodes of shells in a dynamic solid-mechanics analysis.

| parameter | description                        | initial value |
|-----------|------------------------------------|---------------|
| value     | The value for the initial velocity | 0,0,0         |

*Example:*

```
<ic type="shell velocity" node_set="set1">
  <value>1.0,0.0,0.0</value>
</ic>
```

### 3.10.5 The Initial Fluid Velocity Initial Condition

The *initial fluid velocity* initial condition can be used to set the initial velocity for a fluid mechanics analysis.

| parameter | description                              | initial value |
|-----------|------------------------------------------|---------------|
| value     | The value for the initial fluid velocity | 0,0,0         |

*Example:*

```
<ic type="initial fluid velocity" node_set="set1">
  <value>1.0,0.0,0.0</value>
</ic>
```

### 3.10.6 Initial Fluid Dilatation

The “*initial fluid dilatation*” initial condition can be used to set the initial value of the fluid dilatation in a fluid analysis.

| parameter | description                                | initial value |
|-----------|--------------------------------------------|---------------|
| value     | The value for the initial fluid dilatation | 0             |

*Example:*

```
<ic type="initial fluid dilatation" node_set="set1">
  <value>1e-8</value>
</ic>
```

### 3.10.7 The Prestrain Initial Condition

The prestrain initial condition can be used to accomplish one of two things. It can be used to convert a forward analysis in an equivalent prestrain analysis. Or it can be used to fix the prestrain gradient. In either case the current geometry is used as the new reference geometry of the following analysis step.

| parameter | description                                                         | initial value |
|-----------|---------------------------------------------------------------------|---------------|
| init      | Initialize prestrain field                                          | 1 (=true)     |
| reset     | Make current geometry the reference geometry of the following steps | 1 (=true)     |

The following example shows how a forward analysis can be converted to a prestrain analysis.

```
<ic type="prestrain">
  <init>1</init>
  <reset>1</reset>
</ic>
```

Note that this in itself may not be sufficient to define the equivalent prestrain analysis. For instance, if the forward model applied prescribed displacements, then these boundary conditions must be replaced with fixed boundary conditions.

If the *init* parameter is set to 0 (false), then the prestrain gradient will be fixated in the current geometry. This means that any remaining distortion will be applied to the prestrain correction factor before the current geometry is converted to the new reference geometry.

### 3.10.8 Fluid Pressure Initial Condition

The “*initial fluid pressure*” initial condition can be used in biphasic and multiphasic analyses, and in fluid, fluid-FSI, and fluid-solutes analyses. In all cases it sets the initial value of the effective fluid pressure  $\tilde{p}$ . If the analysis includes solutes (multiphasic or fluid-solutes), it is the user’s responsibility to evaluate  $\tilde{p}$  using eq.(4.16.7), knowing the values of the universal gas constant  $R$  and absolute temperature  $T$  specified in the Globals section (Section 3.4), and the values of initial effective solute concentrations  $\tilde{c}$  in the given multiphasic or fluid-solutes domain.

In biphasic and multiphasic analyses the value of  $\tilde{p}$  is prescribed directly as a nodal degree of freedom. In a fluid, fluid-FSI, or fluid-solutes analysis, the effective fluid pressure  $\tilde{p}$  is converted internally to the corresponding initial value of the fluid dilatation  $e^f$ , based on the constitutive models presented in eq.(4.20.2).

### 3.10.9 Initial Shell Fluid Pressure

Similar to the 3.10.8 but applied to the back nodes of shell elements.

### 3.10.10 Initial Concentration

The “*initial concentration*” initial condition sets the initial concentration values of solutes in a biphasic-solute or multiphasic analysis.

| parameter | description                           | initial value |
|-----------|---------------------------------------|---------------|
| dof       | The solute’s degree of freedom        | (none)        |
| value     | The value of the solute concentration | 0             |

### 3.10.11 Initial Shell Concentration

The “*initial shell concentration*” initial condition sets the initial concentration values of solutes in a biphasic-solute or multiphasic analysis on the back nodes of a shell element.

| parameter | description                           | initial value |
|-----------|---------------------------------------|---------------|
| dof       | The solute’s degree of freedom        | (none)        |
| value     | The value of the solute concentration | 0             |



### 3.10.12 Rigid Kinematics

The “*rigid kinematics*” initial condition sets the initial velocity of a rigid body from the following equation.

$$\mathbf{v}_0 = \mathbf{V} + \boldsymbol{\omega} \times (\mathbf{r}_0 - \mathbf{c})$$

Here,  $\mathbf{V}$  is the linear velocity,  $\boldsymbol{\omega}$  is the angular velocity,  $\mathbf{r}_0$  is the initial position of the rigid body, and  $\mathbf{c}$  is a position vector indicating the center of rotation.

| parameter          | description                                       | initial value |
|--------------------|---------------------------------------------------|---------------|
| velocity           | The velocity vector $\mathbf{V}$                  | 0,0,0         |
| angular_velocity   | The angular velocity vector $\boldsymbol{\omega}$ | 0,0,0         |
| center_of_rotation | The center or rotation vector $\mathbf{c}$        | 0,0,0         |

### 3.11 Boundary Section

The *Boundary* section defines all fixed and prescribed boundary conditions that may be applied to the model. Individual boundary conditions are defined via the *bc* tag and the particular boundary condition is defined via the *type* attribute. An optional *name* attribute can be added to uniquely name the boundary condition.

For example:

```
<bc name="bc1" type="zero displacement" node_set="nodeset1">
  <x_dof>1</x_dof>
  <y_dof>1</y_dof>
  <z_dof>1</z_dof>
</bc>
```

Most of these boundary conditions require the node set to which to apply the boundary condition to. That nodeset can be defined via the *node\_set* attribute. The value of this attribute should be set to the name of a nodeset defined in the *Mesh* section. See Section 3.6.3 for more information on how to define node sets. Nodesets can also be defined via surfaces or element sets. To define a node set via a surface, use the following syntax.

```
node_set="@surface:surface1"
```

Here, *surface1* is a surface defined in the *Mesh* section. Similarly, defining a node set via an element set is done using *@elem\_set*:. See section 3.6.10 for more information on how to implicit partitions.

#### 3.11.1 Zero Displacement

To fix the nodal displacement degrees of freedom (dof) use the *zero displacement* boundary condition. This element has two required attributes: the *type* is set to “zero displacement” and the node set to which this boundary condition applies is defined via *node\_set*. The optional *name* attribute can be used to give the boundary condition a name.

This boundary condition has three parameters, one for each degree of freedom. Set the value of each parameter to 1 to constrain the corresponding dof.

```
<bc type="zero displacement" node_set="nodeset1">
  <x_dof>1</x_dof>
  <y_dof>1</y_dof>
  <z_dof>1</z_dof>
</bc>
```

#### 3.11.2 Zero Shell Displacement

To fix the nodal back-displacement degrees of freedom (dof) of a shell use the *zero shell displacement* boundary condition. This element has two required attributes: the *type* is set to “zero shell displacement” and the node set to which this boundary condition applies is defined via *node\_set*. The optional *name* attribute can be used to give the boundary condition a name.

This boundary condition has three parameters, one for each degree of freedom. Set the value of each parameter to 1 to constrain the corresponding dof.

```
<bc type="zero shell displacement" node_set="nodeset1">
  <sx_dof>1</sx_dof>
  <sy_dof>1</sy_dof>
  <sz_dof>1</sz_dof>
</bc>
```

This boundary condition can only be applied to nodes that belong to shells.

### 3.11.3 Zero Fluid Pressure

To fix the effective fluid pressure  $\tilde{p}$  in a biphasic or multiphasic analysis, use the *zero fluid pressure* boundary condition.

```
<bc node_set="set01" type="zero fluid pressure"/>
```

### 3.11.4 Zero Concentration

To fix an effective solute concentration  $\tilde{c}$  degree of freedom (c\_dof) in a multiphasic or fluid-solutes analysis, use the *zero concentration* boundary conditions.

```
<bc node_set="set01" type="zero concentration">
  <c_dof>c1</c_dof>
</bc>
```

### 3.11.5 Zero Fluid Dilatation

To fix the fluid dilatation  $e^f$  in a fluid or fluid-FSI or fluid-solutes analysis, use the *zero fluid dilatation* boundary condition.

```
<bc node_set="set01" type="zero fluid dilatation"/>
```

Zero fluid dilatation is the same as setting the effective fluid pressure  $\tilde{p}$  to zero in fluid analyses. If you want to set the actual fluid pressure  $p$  to some value (such as zero) in a fluid analysis, use the *fluid pressure* boundary condition (Section [3.11.18](#)).

### 3.11.6 Prescribed Displacement

The prescribed displacement boundary condition can be used to prescribe the value of the nodal displacement degrees of freedom. This boundary condition takes the following parameters.

| parameter | description                                                     | default |
|-----------|-----------------------------------------------------------------|---------|
| dof       | The displacement dof to prescribe, x, y, or z                   | -       |
| value     | The prescribed value                                            | 0       |
| relative  | Flag that defines whether the value is absolute or relative (1) | 0 (off) |

*Comments:*

1. If the relative flag is set to 1, then the value is relative with respect to the current position of the node at the time the boundary condition becomes active. This only has an effect in multi-step analyses.

```
<bc type="prescribed displacement" node_set="set1">
  <dof>x</dof>
  <value lc="1">1.0</value>
  <relative>0</relative>
</bc>
```

Although the *prescribe displacement* element with a value of zero can also be used to fix a certain nodal degree of freedom, the user should use the *zero displacement* boundary condition whenever possible, since this option causes the equation corresponding to the constrained degree of freedom to be removed from the linear system of equations. This results in fewer equations that need to be solved for and thus reduces the run time of the FE analysis.

### 3.11.7 Prescribed Shell Displacement

The *prescribed shell displacement* boundary condition can be used to prescribe the value of the nodal back-displacement degrees of freedom of a shell. This boundary condition takes the following parameters.

| parameter | description                                                     | default |
|-----------|-----------------------------------------------------------------|---------|
| dof       | The displacement dof to prescribe, sx, sy, or sz                | -       |
| value     | The prescribed value                                            | 0       |
| relative  | Flag that defines whether the value is absolute or relative (1) | 0 (off) |

*Comments:*

1. If the relative flag is set to 1, then the value is relative with respect to the current position of the node at the time the boundary condition becomes active. This only has an effect in multi-step analyses.

```
<bc type="prescribed shell displacement" node_set="set1">
  <dof>sx</dof>
  <value lc="1">1.0</value>
  <relative>0</relative>
</bc>
```

Although the *prescribe shell displacement* element with a value of zero can also be used to fix a certain nodal degree of freedom, the user should use the *zero shell displacement* boundary condition whenever possible, since this option causes the equation corresponding to the constrained degree of freedom to be removed from the linear system of equations. This results in fewer equations that need to be solved for and thus reduces the run time of the FE analysis.

### 3.11.8 Prescribed Fluid Pressure

The *prescribed fluid pressure* boundary condition can be used to prescribe the effective fluid pressure  $\tilde{p}$  on the boundary of a biphasic or multiphasic material. This boundary condition takes the following parameters.

| parameter | description                                                     | default |
|-----------|-----------------------------------------------------------------|---------|
| value     | The prescribed value                                            | 0       |
| relative  | Flag that defines whether the value is absolute or relative (1) | 0 (off) |

*Comments:*

1. If the relative flag is set to 1, then the value is relative with respect to the current position of the node at the time the boundary condition becomes active. This only has an effect in multi-step analyses.

```
<bc type="prescribed fluid pressure" node_set="set1">
  <value lc="1">1.0</value>
  <relative>0</relative>
</bc>
```

Although the *prescribed fluid pressure* element with a value of zero can also be used to fix the effective fluid pressure, the user should use the *zero fluid pressure* boundary condition whenever possible, since this option causes the equation corresponding to that degree of freedom to be removed from the linear system of equations. This results in fewer equations that need to be solved for and thus reduces the run time of the FE analysis.

### 3.11.9 Prescribed Concentration

The *prescribed concentration* boundary condition can be used to prescribe the value of the effective solute concentration  $\tilde{c}$  in a multiphasic or fluid-solutes analysis. This boundary condition takes the following parameters.

| parameter | description                                                     | default |
|-----------|-----------------------------------------------------------------|---------|
| dof       | The solute concentration dof to prescribe, c1, c2, etc.         | -       |
| value     | The prescribed value                                            | 0       |
| relative  | Flag that defines whether the value is absolute or relative (1) | 0 (off) |

*Comments:*

1. If the relative flag is set to 1, then the value is relative with respect to the current effective solute concentration at the time the boundary condition becomes active. This only has an effect in multi-step analyses.

```

<bc type="prescribed concentration" node_set="set1">
  <dof>c1</dof>
  <value lc="1">1.0</value>
  <relative>0</relative>
</bc>

```

Although the *prescribed concentration* element with a value of zero can also be used to fix that degree of freedom, the user should use the *zero concentration* boundary condition whenever possible, since this option causes the equation corresponding to the constrained degree of freedom to be removed from the linear system of equations. This results in fewer equations that need to be solved for and thus reduces the run time of the FE analysis.

### 3.11.10 Prescribed Deformation

The *prescribed deformation* boundary condition can be used to prescribe the displacement of a node using the following equation.

$$\mathbf{u} = s (\mathbf{F} - \mathbf{I}) \mathbf{X}$$

Here,  $\mathbf{u}$  is the displacement vector,  $\mathbf{X}$  the reference position vector,  $\mathbf{F}$  a “deformation gradient”,  $\mathbf{I}$  the identity tensor, and  $s$  a scale factor.

The following parameters are defined.

| parameter | description              | default      |
|-----------|--------------------------|--------------|
| scale     | The scale factor $s$     | 1            |
| F         | The deformation gradient | $\mathbf{I}$ |

```

<bc type="prescribed deformation" node_set="set1">
  <scale lc="1">0.1</scale>
  <F>2,0,0, 0,2,0, 0,0.25,0</F>
</bc>

```

### 3.11.11 Prescribed Fluid Dilatation

The *prescribed fluid dilatation* boundary condition can be used to prescribe the fluid dilatation  $e^f$  on the boundary of a fluid, fluid-FSI or fluid-solutes material. This boundary condition takes the following parameters.

| parameter | description                                                     | default |
|-----------|-----------------------------------------------------------------|---------|
| value     | The prescribed value                                            | 0       |
| relative  | Flag that defines whether the value is absolute or relative (1) | 0 (off) |

*Comments:*

1. If the relative flag is set to 1, then the value is relative with respect to the current position of the node at the time the boundary condition becomes active. This only has an effect in multi-step analyses.

```
<bc type="prescribed fluid dilatation" node_set="set1">
  <value lc="1">-1e-8</value>
  <relative>0</relative>
</bc>
```

Although the *prescribed fluid dilatation* element with a value of zero can also be used to fix the effective fluid pressure, the user should use the *zero fluid dilatation* boundary condition whenever possible, since this option causes the equation corresponding to that degree of freedom to be removed from the linear system of equations. This results in fewer equations that need to be solved for and thus reduces the run time of the FE analysis.

### 3.11.12 Prescribed Fluid Velocity

The “*prescribed fluid velocity*” boundary condition can be used to to prescribe the fluid velocity on the boundary of a fluid domain. This boundary condition takes the following parameters.

| parameter | description                                                          | default |
|-----------|----------------------------------------------------------------------|---------|
| dof       | The degree of freedom to prescribe                                   | (none)  |
| value     | The prescribed values                                                | 0       |
| relative  | If set, the value is relative to the value at the last analysis step | 0 (off) |

### 3.11.13 Prescribed Shell Fluid Pressure

The “*prescribed shell fluid pressure*” boundary condition can be used to prescribe the fluid pressure on the back nodes of a shell element in a biphasic or multiphasic analysis.

| parameter | description                                                          | default |
|-----------|----------------------------------------------------------------------|---------|
| value     | The prescribed values                                                | 0       |
| relative  | If set, the value is relative to the value at the last analysis step | 0 (off) |

### 3.11.14 Normal Displacement

The *normal displacement* boundary condition prescribes the displacement of a node along the normal vector to the surface on which the node lies.

$$\mathbf{u} = s\mathbf{N}$$

Here,  $\mathbf{u}$  is the displacement vector,  $\mathbf{N}$  is the normal to the surface in the reference configuration, and  $s$  is a scale factor.

Note that this boundary condition requires a *surface* attribute, instead of the *node\_set* attribute. The following parameters are defined.

| parameter    | description                             | default |
|--------------|-----------------------------------------|---------|
| scale        | The scale factor $s$                    | 0       |
| surface_hint | Flag indicating the type of surface (1) | 0       |

*Comments:*

1. The *surface\_hint* parameter indicates the shape of the surface. Set this value to 1 if the surface is spherical. This will calculate a more accurate normal vector.

```
<bc type="normal displacement" surface="surf1">
  <scale lc="1">0.1</scale>
  <surface_hint>1</surface_hint>
</bc>
```

### 3.11.15 Rigid Deformation

The *rigid deformation* boundary condition can be used to prescribe the motion of a nodeset via a rigid transformation. The displacement is given by the following equation.

$$\mathbf{u} = \mathbf{Q}(\mathbf{X} - \mathbf{r}_0) + \mathbf{r}_t - \mathbf{X}$$

Here,  $\mathbf{u}$  is the displacement vector,  $\mathbf{Q}$  is a rotation tensor,  $\mathbf{X}$  is the initial position of the node, and  $\mathbf{r}_0$  and  $\mathbf{r}_t$  are the reference and current position respectively of the rigid coordinate system. The following parameters are defined.

| parameter | description                                        | default |
|-----------|----------------------------------------------------|---------|
| pos       | Position vector of the rigid coordinate system     | {0,0,0} |
| rot       | Rotation vector of the rigid coordinate system (1) | {0,0,0} |

*Comments:*

1. The rotation of the rigid coordinate system is given by a rotation vector. The direction of this vector defines the (instantaneous) axis of rotation, and the magnitude is the rotation angle (in radians) around this axis.



### 3.11.16 Rigid

A node set can be attached to a rigid body using the *rigid* boundary condition. Rigid nodes are not assigned degrees of freedom.

```
<bc type="rigid" node_set="set1">
  <rb>2</rb>
</bc>
```

The *rb* parameter defines the material (which must be a “rigid body” material) that in turn defines the rigid body. The node set must be defined in the *Mesh* section.

### 3.11.17 Multiphasic Fluid Pressure

In a multiphasic mixture the nodal degree of freedom for fluid pressure represents the effective fluid pressure  $\tilde{p}$ , which is related to the actual fluid pressure  $p$  according to eq.(4.16.7). One may assign the value of  $p$  directly on a boundary, by using the *actual fluid pressure* boundary condition.

```
<bc surface="FluidPressure1" type="actual fluid pressure">
  <pressure>0</pressure>
  <shell_bottom>0</shell_bottom>
</bc>
```

This surface load evaluates  $p$  from nodal values of  $\tilde{p}$  on the surface, and using the average values of  $\Phi$  and  $c^\alpha$  at the integration points of the underlying element. This surface load is particularly useful when the effective solute concentrations  $\tilde{c}^\alpha$  are not prescribed as boundary conditions on that surface, but evolve with the analysis solution. the *shell\_bottom* boolean flag (0=false, 1=true) should be set to true when prescribing this pressure on the bottom of a multiphasic shell domain.

### 3.11.18 Fluid Pressure

To set the actual fluid pressure  $p$  in a fluid, fluid-FSI, or fluid-solutes analysis, use the *fluid pressure* boundary condition.

```
<bc type="fluid pressure" surface="FluidPressure1">
  <pressure lc="1">100</pressure>
</bc>
```

The optional *lc* parameter associates a load curve (a user-defined scale factor that evolves with time) which multiplies the value of <pressure>. The surface must be defined in the *Mesh* section. For a biphasic analysis the actual fluid pressure is equal to the effective fluid pressure ( $p = \tilde{p}$ ), in which case use *prescribed fluid pressure* (Section 3.11.8).

### 3.11.19 Fluid Resistance

In a fluid, fluid-FSI or fluid-solutes analysis, this boundary condition prescribes an elastic pressure  $p = Rq + p_d$  on a surface, where  $R$  is the flow resistance and  $q$  is the volumetric flow rate across the surface (calculated internally);  $p_d$  is an optional offset pressure. The pressure  $p$  is converted to a dilatation and prescribed as an essential boundary condition at the nodes of the facets.

```
<bc type="fluid resistance" surface="surface1">
  <R lc="1">7.93e+07</R>
  <pressure_offset lc="2">10640</pressure_offset>
</bc>
```

### 3.11.20 Fluid RC

This boundary condition models a fluid surface that has an RC-equivalent circuit for outflow conditions.

### 3.11.21 Fluid RCR

This boundary condition models a three-element Windkessel outflow condition on a surface in a fluid, fluid-FSI or fluid-solutes analysis, as described in [87]. It prescribes an elastic pressure  $p$  on a surface, which satisfies the ordinary differential equation

$$p + \tau \frac{dp}{dt} = (R_d + R) q + R \tau \frac{dq}{dt} + p_d + \tau \frac{dp_d}{dt}, \quad \tau = R_d C$$

where  $R$  is the upstream flow resistance (series resistance),  $R_d$  is the downstream flow resistance and  $C$  is the downstream flow capacitance (parallel  $R_d C$  element downstream of  $R$ ), whereas  $q$  is the volumetric flow rate across the surface (calculated internally);  $p_d$  is an optional downstream offset pressure, which may be associated with a loadcurve to produce a function of time. The actual pressure  $p$  obtained by solving this differential equation is converted to a dilatation and prescribed as an essential boundary condition at the nodes of the facets. Since the pressure is the solution of an ordinary differential equation, the user may also optionally specify the initial condition for the pressure.

```
<bc name="FluidRCR1" type="fluid RCR" surface="FluidRCR1">
  <R>1</R>
  <Rd>10</Rd>
  <capacitance>0.5</capacitance>
  <pressure_offset>0</pressure_offset>
  <initial_pressure>0</initial_pressure>
</bc>
```

By setting  $C = 0$  we recover the fluid resistance surface load described in Section 3.11.19, with flow resistance equal to  $R_d + R$ . By setting  $R$  to zero we can have simulate a parallel  $R_d C$  circuit.

Internally, the ordinary differential equation presented above is solved numerically using a standard finite difference scheme,

$$p_{n+1} = \left( \frac{R_d}{1 + \frac{\tau}{\Delta t}} + R \right) q_{n+1} + \frac{\tau}{\Delta t + \tau} (p_n - p_{d,n} - R q_n) + p_{d,n+1}$$

where  $f_{n+1}$  is the value of any function  $f(t)$  at the current time step  $t_{n+1}$ ,  $f_n$  is its value at the previous time step  $t_n$ , and  $\Delta t = t_{n+1} - t_n$  is the time increment.

### 3.11.22 Linear Constraints

A linear constraint can be used to couple nodal degrees of freedom. The linear constraint has one dependent degree of freedom  $u^*$ , and several independent degrees of freedom  $u_i$ . The linear constraint is defined as follows:

$$u^* = c_0 + c_1 u_1 + \dots + c_n u_n$$

Here, the  $c_i$  are user-specified scale factors and  $c_0$  can be used to specify an offset.

The dependent degree of freedom  $u^*$  will be removed from the linear system of equations. Consequently, this nodal degree of freedom should not be used in any other boundary condition.

To define a linear constraint, set the *type* attribute to *linear constraint*. The dependent degree of freedom, i.e. the dof that will be removed, is specified via a node number and a dof identifier. Each child dof is defined via the *child\_dof* tag and similarly requires the corresponding node and dof identifier, as well as the value parameter.

```
<bc type="linear constraint">
  <node>5</node>
  <dof>x</dof>
  <child_dof>
    <node>8</node>
    <dof>x</dof>
    <value>1.0</value>
  </child_dof>
  <child_dof>
    <node>9</node>
    <dof>x</dof>
    <value>-1.0</value>
  </child_dof>
</bc>
```

### 3.11.23 Matching OSM Coefficient

The “matching\_osm\_coef” boundary condition is a surface boundary condition that imposes the same osmotic coefficient in the bath as in the multiphase material bounded by that surface.

```
<bc type="matching_osm_coef">
  <ambient_pressure>1</ambient_pressure>
  <ambient_osmolarity>1e-3</ambient_osmolarity>
  <shell_bottom>0</shell_bottom>
</bc>
```

## 3.12 Rigid Section

The *Rigid* section is used to define all rigid constraints, rigid body initial kinematics, and rigid connectors. The motion of a point  $\mathbf{x}$  connected to a rigid body is given by

$$\mathbf{x} = \mathbf{r}(t) + \mathbf{R}(t) \cdot \mathbf{Z} = \mathbf{r}(t) + \mathbf{z}(t) \quad (3.12.1)$$

where  $\mathbf{r}(t)$  is the position of the rigid body center of mass and  $\mathbf{R}(t)$  is the body's (proper orthogonal) rotation tensor, which satisfies  $\mathbf{R}(t_0) = \mathbf{I}$  at the initial time  $t_0$ ; here,  $\mathbf{z}(t) = \mathbf{R}(t) \cdot \mathbf{Z}$  is the position vector of the point relative to the body's center of mass, and  $\mathbf{X} = \mathbf{r}(t_0) + \mathbf{Z}$  is the initial position. In FEBio, the rotation is evaluated as  $\mathbf{R}(t) = \exp[\boldsymbol{\xi}(t)]$ , where  $\boldsymbol{\xi}(t)$  is the material rotation of the rigid body from its reference configuration. In practice,  $\boldsymbol{\xi}$  is stored as a quaternion to facilitate the multiplication of rotation tensors. The exponential map  $\exp[\boldsymbol{\xi}(t)]$  is given in full form as

$$\mathbf{R}(t) = \exp[\boldsymbol{\xi}(t)] = \cos \xi \mathbf{I} - \sin \xi \boldsymbol{\mathcal{E}} \cdot \mathbf{n} + (1 - \cos \xi) \mathbf{n} \otimes \mathbf{n} \quad (3.12.2)$$

where  $\boldsymbol{\mathcal{E}}$  is the third-order permutation pseudo-tensor with rectangular components  $\varepsilon_{ijk}$  (permutation symbol). Here,  $\mathbf{n}$  is the unit vector along the axis of rotation and  $\xi$  is the angle of rotation about  $\mathbf{n}$ , such that  $\boldsymbol{\xi} = \xi \mathbf{n}$ . Both  $\xi$  and  $\mathbf{n}$  may evolve over time. Either none or all the components of  $\boldsymbol{\xi}(t)$  must be prescribed (or fixed) for a rigid body.

The Rigid section can contain the following subsections.

**rigid\_bc** Define a rigid constraint.

**rigid\_ic** Define a rigid initial condition.

**rigid\_load** Define a load on a rigid body.

**rigid\_connector** Define a connection or joint between two rigid bodies.

### 3.12.1 Rigid Constraints

The rigid constraints section allows users to prescribe the kinematics applied to a single rigid body. A rigid constraint is added using the *rigid\_bc* tag in the *Rigid* parent tag. The particular constraint is defined via the *type* attribute. An optional *name* attribute can be added to uniquely name the constraint.

#### 3.12.1.1 Rigid\_fixed Constraint

The *rigid\_fixed* constraint fixes one or multiple rigid degrees of freedom. It requires the following parameters.

| parameters    | Description                          |
|---------------|--------------------------------------|
| <i>rb</i>     | The rigid material name (1)          |
| <i>Rx_dof</i> | fix x-displacement degree of freedom |
| <i>Ry_dof</i> | fix y-displacement degree of freedom |
| <i>Rz_dof</i> | fix z-displacement degree of freedom |
| <i>Ru_dof</i> | fix x-rotational degree of freedom   |
| <i>Rv_dof</i> | fix y-rotational degree of freedom   |
| <i>Rw_dof</i> | fix z-rotational degree of freedom   |

*Comments:*

1. The name of the rigid material is defined in the Material section. The referenced material must be a “rigid body” material.

The following example fixes the displacement degrees of freedom.

```
<rigid_bc type="rigid_fixed">
  <rb>RigidMaterial1</rb>
  <Rx_dof>1</Rx_dof>
  <Ry_dof>1</Ry_dof>
  <Rz_dof>1</Rz_dof>
</rigid_bc>
```

**3.12.1.2 Rigid\_displacement Constraint**

The *rigid\_displacement* constraint prescribes the value of a rigid displacement degree of freedom. It requires the following parameters.

| parameters      | Description                                    |
|-----------------|------------------------------------------------|
| <i>rb</i>       | The rigid body's material name (1)             |
| <i>dof</i>      | The rigid degree of freedom to prescribe. (2)  |
| <i>value</i>    | The prescribed value                           |
| <i>relative</i> | Defines the value as relative or absolute. (3) |

*Comments:*

1. This is the name attribute assigned to the corresponding rigid body material as defined in the Material section.
2. The values allowed for the *dof* parameter are: x, y, or z.
3. If the relative flag is set, the value is taken relative to the dof value at the start of the step.

The following example prescribes the x-displacement degree of freedom.

```
<rigid_bc type="rigid_displacement">
  <rb>rigid</rb>
  <dof>x</dof>
  <value lc="1">3.14</value>
</rigid_bc>
```

**3.12.1.3 Rigid\_rotation Constraint**

The *rigid\_rotation* constraint prescribes the value of a rigid rotational degree of freedom. It requires the following parameters.

| parameters      | Description                                    |
|-----------------|------------------------------------------------|
| <i>rb</i>       | The rigid body's material name (1)             |
| <i>dof</i>      | The rigid degree of freedom to prescribe. (2)  |
| <i>value</i>    | The prescribed value                           |
| <i>relative</i> | Defines the value as relative or absolute. (3) |

*Comments:*

1. This is the name attribute assigned to the corresponding rigid body material as defined in the Material section.
2. The values allowed for the *dof* parameter are: Ru, Rv, or Rw.
3. If the relative flag is set, the value is taken relative to the dof value at the start of the step.

The following example prescribes a rotation around the x-axis.

```
<rigid_bc type="rigid_rotation">
  <rb>rigid</rb>
  <dof>Ru</dof>
  <value lc="1">3.14</value>
</rigid_bc>
```

### 3.12.1.4 Rigid\_rotation\_vector Constraint

The *rigid\_rotation\_vector* constraint allows the user to prescribe the rotation of a rigid body using a rotation vector. In essence, this combines the effect of using three separate rigid rotation constraints, one for each component of the rotation vector, and thus might be a more convenient choice if the rotation of the rigid body is fully prescribed.

| parameters | Description                        |
|------------|------------------------------------|
| <i>rb</i>  | The rigid body's material name (1) |
| <i>vx</i>  | x-component of the rotation vector |
| <i>vy</i>  | y-component of the rotation vector |
| <i>vz</i>  | z-component of the rotation vector |

*Comments:*

1. This is the name attribute assigned to the corresponding rigid body material as defined in the Material section.

The following example prescribes a rotation around the z-axis using a load controller.

```
<rigid_bc type="rigid_rotation_vector">
  <rb>rigid</rb>
  <vx>0</vx>
  <vy>0</vy>
  <vz lc="1">3.14</vz>
</rigid_bc>
```

### 3.12.1.5 Rigid\_euler\_angles Constraint

The *rigid\_euler\_angles* constraint allows the user to prescribe the rotation of a rigid body using three Euler angles.

| parameters | Description                        |
|------------|------------------------------------|
| <i>rb</i>  | The rigid body's material name (1) |
| <i>Ex</i>  | X Euler angle (degrees)            |
| <i>Ey</i>  | Y Euler angle (degrees)            |
| <i>Ez</i>  | Z Euler angle (degrees)            |

*Comments:*

1. This is the name attribute assigned to the corresponding rigid body material as defined in the Material section.

The following example prescribes a rotation around the z-axis using a load controller.

```
<rigid_bc type="rigid_euler_vector">
  <rb>rigid</rb>
  <vx>0</vx>
  <vy>0</vy>
  <vz lc="1">180.0</vz>
</rigid_bc>
```

### 3.12.2 Rigid Initial Conditions

The *rigid\_ic* section defines the initial conditions on rigid bodies. Rigid initial conditions can be used to set the initial values for the rigid body's linear and angular velocity. The particular initial condition is selected using the *type* attribute.

#### 3.12.2.1 Initial Rigid Velocity

In dynamic analysis, the initial velocity of a rigid body can be set via the *initial\_rigid\_velocity* rigid constraint. It requires the following parameters.

| parameters   | Description                        |
|--------------|------------------------------------|
| <i>rb</i>    | The rigid body's material name (1) |
| <i>value</i> | The initial velocity vector        |

*Comments:*

1. This is the "name" attribute assigned to the corresponding rigid body material as defined in the Material section.

The following example defines an initial velocity for a rigid body.

```
<rigid_ic type="initial_rigid_velocity">
  <rb>rigid</rb>
  <value>1,0,0</value>
</rigid_ic>
```

### 3.12.2.2 Initial Rigid Angular Velocity

In dynamic analysis, the initial angular velocity of a rigid body can be set via the *initial\_rigid\_angular\_velocity* rigid constraint. It requires the following parameters.

| parameters   | Description                         |
|--------------|-------------------------------------|
| <i>rb</i>    | The rigid body's material name. (1) |
| <i>value</i> | The initial angular velocity vector |

*Comments:*

1. This is the “name” attribute assigned to the corresponding rigid body material as defined in the Material section.

The following example defines an initial angular velocity for a rigid body.

```
<rigid_ic type="initial_rigid_angular_velocity">
  <rb>rigid</rb>
  <value>1,0,0</value>
</rigid_ic>
```

### 3.12.3 Rigid Loads

The *rigid\_load* section can be used to applied loads to a rigid body.

#### 3.12.3.1 Rigid\_force Load

The *rigid\_force* load applies a load directly to the rigid degree of freedom. It requires the following parameters.

| parameters       | Description                                   |
|------------------|-----------------------------------------------|
| <i>rb</i>        | The rigid body's material name (1)            |
| <i>dof</i>       | The rigid degree of freedom to prescribe. (2) |
| <i>value</i>     | The prescribed force value                    |
| <i>load_type</i> | Defines the type of load. (3)                 |

*Comments:*

1. This is the “name” attribute assigned to the corresponding rigid body material as defined in the Material section.
2. The values allowed for the *dof* parameter are: Rx, Ry, Rz.
3. The *load\_type* allows the following values:
  - (a) 0 = a load (force/moment) is applied directly to the degree of freedom.
  - (b) 1 = a follower load applied is applied to the rigid body. The load is applied in the rigid body's coordinate system, so it rotates with the rigid body. This only works for Rx, Ry, Rz.



- (c) 2 = target load. The load's value is ramped up from its initial value at the start of the step, and ramped up linearly to the value specified in the *value* parameter.

The following example applies a force in the x-direction.

```
<rigid_load type="rigid_force">
  <rb>1</rb>
  <dof>Rx</dofs>
  <value lc="1">3.14</value>
</rigid_load>
```

### 3.12.3.2 Rigid\_moment Load

The *rigid\_moment* load applies a moment about a particular coordinate axis. It requires the following parameters.

| parameters      | Description                                   |
|-----------------|-----------------------------------------------|
| <i>rb</i>       | The rigid body's material name (1)            |
| <i>dof</i>      | The rigid degree of freedom to prescribe. (2) |
| <i>value</i>    | The prescribed moment value                   |
| <i>relative</i> | Defines the type of load. (3)                 |

*Comments:*

1. This is the "name" attribute assigned to the corresponding rigid body material as defined in the Material section.
2. The values allowed for the *dof* parameter are: Ru, Rv, Rw.
3. The relative flag indicates whether the applied value should be absolute or relative to the value at the end of the previous analysis step.

The following example applies a moment about the x-axis.

```
<rigid_load type="rigid_moment">
  <rb>rigid</rb>
  <dof>Ru</dofs>
  <value lc="1">3.14</value>
</rigid_load>
```

### 3.12.3.3 Rigid\_follower\_force Load

The *rigid\_follower\_force* load applies a force in the local rigid body coordinate system. The direction of the force in global coordinates therefore depends on orientation of the rigid body. It requires the following parameters.

| parameters       | Description                              |
|------------------|------------------------------------------|
| <i>rb</i>        | The rigid body's material name (1)       |
| <i>insertion</i> | Location where the force is applied. (2) |
| <i>force</i>     | The vector force value                   |
| <i>relative</i>  | Relative flag (3)                        |

*Comments:*

1. This is the “name” attribute assigned to the corresponding rigid body material as defined in the Material section.
2. This is the position in the reference coordinate system of the point where the force is applied.
3. The relative flag indicates whether the applied value should be absolute or relative to the value at the end of the previous analysis step.

The following example applies a moment about the x-axis.

```
<rigid_load type="rigid_follower_force">
  <rb>rigid</rb>
  <insertion>1,2,3</dofs>
  <force>100,0,0</force>
</rigid_load>
```

### 3.12.3.4 Rigid\_follower\_moment Load

The *rigid\_follower\_moment* load applies a moment in the local rigid body coordinate system. The direction of the moment in global coordinates therefore depends on orientation of the rigid body. It requires the following parameters.

| parameters    | Description                        |
|---------------|------------------------------------|
| <i>rb</i>     | The rigid body's material name (1) |
| <i>moment</i> | The vector moment value            |

*Comments:*

1. This is the “name” attribute assigned to the corresponding rigid body material as defined in the Material section.

The following example applies a moment about the x-axis.

```
<rigid_load type="rigid_follower_moment">
  <rb>rigid</rb>
  <moment>100,0,0</force>
</rigid_load>
```

### 3.12.3.5 Rigid\_cable Load

The *rigid\_cable* load emulates the effect of a force applied on one end of a cable that is connected to one or more rigid bodies. It requires the following parameters.

| parameters               | Description                                                                            | Default |
|--------------------------|----------------------------------------------------------------------------------------|---------|
| <i>force</i>             | The scalar force magnitude                                                             | 0       |
| <i>force_direction</i>   | The unit vector defining the force direction                                           | 0,0,0   |
| <i>relative</i>          | Indicates whether the through-points are defined in global or relative coordinates (1) | 1       |
| <i>rigid_cable_point</i> | A point on a rigid body that the cable runs through. (2)                               | (non)   |

*Comments:*

1. The *relative* flag indicates whether the location of the cable's via-points are defined in global coordinates (*relative*=0) or in the local coordinates of the corresponding rigid body (*relative*=1).
2. One or more rigid cable points must be defined that define the points that the cable runs through. The first point defined is the anchor, where the cable is attached to. The other points define the via points, i.e. a point where the cable runs through. Each cable point is defined by the rigid body that it connects to and a point on the rigid body that the cable runs through. The point's coordinates are either in the global or the local coordinate system of the rigid body, as determined by the *relative* flag.

The following example illustrates a rigid cable setup.

```
<rigid_load type="rigid_cable">
  <force lc="1">100</force>
  <force_direction>1,0,0</force_direction>
  <relative>1</relative>
  <rigid_cable_point>
    <rigid_body_id>rigid1</rigid_body_id>
    <position>0, -1, 1</position>
  </rigid_cable_point>
  <rigid_cable_point>
    <rigid_body_id>rigid2</rigid_body_id>
    <position>0, 1, 1</position>
  </rigid_cable_point>
</rigid_load>
```

### 3.12.4 Rigid Connectors

The rigid connector section allows users to define different type of connections between two rigid bodies. The specific connector is defined via the *type* attribute.

The rigid connectors fall into two categories. The first category define different joints that constrain the relative motion of one rigid body with respect to the second one.

Rigid joints produce nonlinear constraints between rigid bodies, which prevent relative motion except along the degrees of freedom of the joint. The term 'rigid' refers to the bodies, not to the joints. Each rigid joint needs to define two rigid bodies (*a* and *b*), a joint origin common to both bodies, and a set of axes that determine the relative orientation of the joint degrees of freedom. These axes define orthonormal basis vectors  $\{e_1^a, e_2^a, e_3^a\}$  and  $\{e_1^b, e_2^b, e_3^b\}$  on each rigid body, with both bases being coincident,  $\{e_1, e_2, e_3\}$ , at the start of the analysis, and given in world coordinates.

The rigid joint nonlinear constraints produce reaction forces and moments that are enforced with penalty parameters and Lagrange multipliers. The penalty parameters, *force\_penalty* and *moment\_penalty*, may be conceptualized as stiffnesses of linear/torsional springs that prevent relative translations/rotations of the rigid bodies along degrees of freedom that must remain constrained for that joint. The penalty values should be selected based on a rough estimation of the maximum reactions forces and moments acting at these joints, divided by the maximum amount of linear/angular separation that your analysis can tolerate for that joint. Alternatively, set the

*force\_penalty* and *moment\_penalty* parameters to 1 and turn on the *auto\_penalty*; this setting will automatically adjust the *force\_penalty* and *moment\_penalty* to an appropriate value.

The augmented Lagrangian method is used by default, where the joint reaction force and moment are treated as Lagrange multipliers, augmented at each time step by the product of the force/moment penalty parameter and the linear/angular gap. Use the parameter *maxaug* to control the maximum allowable number of augmentations at each time step (*maxaug*=0 produces the penalty method); use a non-zero value for the parameter *minaug* to ensure a minimum number of augmentations. Augmentations will proceed until the relative change in the reaction force/moment magnitude is less than *tolerance*, and/or the linear gap is less than *gaptol*, and/or the angular gap is less than *angtol*. Setting any of these parameters to zero disables that check.

The nonlinear constraints that enforce these joints produce a non-symmetric stiffness matrix. Therefore, when using rigid joints, use the analysis setting for a non-symmetric formulation, see Section 3.3.5.

The following rigid joints can be defined.

| type                           | Description                                       |
|--------------------------------|---------------------------------------------------|
| <i>rigid spherical joint</i>   | Connect rigid bodies at point.                    |
| <i>rigid revolute joint</i>    | Rotation about a single prescribed axis           |
| <i>rigid prismatic joint</i>   | Translation along a single prescribed axis.       |
| <i>rigid cylindrical joint</i> | Rotation and translation about prescribed axis.   |
| <i>rigid planar joint</i>      | 2D plane translation and rotation about normal.   |
| <i>rigid lock</i>              | Prevent any relative motion between rigid bodies. |

The second category of connectors apply a force on the rigid bodies that is proportional to some relative motion. The term 'rigid' refers to the bodies, not to the connectors.

The following rigid connectors can be defined.

| type                           | Description                                          |
|--------------------------------|------------------------------------------------------|
| <i>rigid spring</i>            | Applies a spring that connects the two rigid bodies. |
| <i>rigid damper</i>            | Applies a damper based on relative linear velocity.  |
| <i>rigid angular damper</i>    | Applies a damper based on relative angular velocity. |
| <i>rigid contractile force</i> | Constant contractile force between rigid bodies.     |
| <i>rigid planar joint</i>      | 2D plane translation and rotation about normal.      |
| <i>rigid lock</i>              | Prevent any relative motion between rigid bodies.    |

All of these joints and connectors define two parameters that reference the two rigid bodies.

| parameter     | Description                            |
|---------------|----------------------------------------|
| <i>body_a</i> | The first rigid body in the connector  |
| <i>body_b</i> | The second rigid body in the connector |

In addition, the joints defined above are all based on the Augmented Lagrangian method, and consequently, also share the following parameters.

| parameter             | Description                                     |
|-----------------------|-------------------------------------------------|
| <i>laugon</i>         | Augmentation flag                               |
| <i>tolerance</i>      | Specifies the augmentation tolerance            |
| <i>minaug</i>         | The minimum number of augmentations.            |
| <i>maxaug</i>         | The maximum number of augmentations.            |
| <i>gaptol</i>         | Gap tolerance                                   |
| <i>angtol</i>         | Angular separation tolerance                    |
| <i>force_penalty</i>  | Penalty factor applied to force augmentations.  |
| <i>moment_penalty</i> | Penalty factor applied to moment augmentations. |
| <i>auto_penalty</i>   | Flag to calculate penalty factor automatically. |

Additional parameters, unique to a specific connector, are defined in the sections below.

#### 3.12.4.1 Rigid Spherical Joint

A rigid spherical joint connects rigid bodies *a* and *b* at a point in space, allowing three degrees of freedom for rotation about that point.

In addition to the shared parameters above, it defines the following parameters.

| parameter                  | Description                                 |
|----------------------------|---------------------------------------------|
| <i>joint_origin</i>        | The position of the joint.                  |
| <i>prescribed_rotation</i> | The prescribed rotation flag.               |
| <i>rotation_x</i>          | The x-component of the prescribed rotation. |
| <i>rotation_y</i>          | The y-component of the prescribed rotation. |
| <i>rotation_z</i>          | The z-component of the prescribed rotation. |
| <i>moment_x</i>            | X-component of applied moment               |
| <i>moment_y</i>            | Y-component of applied moment               |
| <i>moment_z</i>            | Z-component of applied moment               |

The *tolerance* element defines the augmentation tolerance. That is, when the relative change in the constraint forces and moments (the Lagrange multipliers) are less than this value. The *gaptol* element defines the tolerance for spatial separation of the joint origin on the two bodies (in units of length). Setting either of these elements to zero disables the enforcement of that tolerance. The *force\_penalty* parameter (with units of force per length) represents the stiffness that prevents the joint origin on the two bodies from separating. The *body\_a* and *body\_b* elements are the material numbers of the two rigid bodies. The *joint\_origin* element defines the position of the joint (the origin of the basis  $\{e_1, e_2, e_3\}$ ) in world coordinates at the start of the analysis. Note that this point does not have to be inside or on the surface of either of the two bodies. The *rotation\_axis* element defines the orientation of the joint rotation axis in world coordinates at the start of the analysis.

Optionally, the rotation of body *b* relative to body *a* may be prescribed using the additional tags

```

<moment_penalty>1e0</moment_penalty>
<prescribed_rotation>1</prescribed_rotation>
<rotation_x lc="1">1</rotation_x>
<rotation_y lc="2">1</rotation_y>
<rotation_z>0</rotation_z>

```

The *prescribed\_rotation* element is a flag that indicates that the motion of the joint is prescribed (1 for prescribed, 0 for free). The *rotation\_x*, *rotation\_y* and *rotation\_z* elements specify the components  $(\theta_1, \theta_2, \theta_3)$  of rotation (with units of radians), with optional associated load curves. The rotation occurs about the axis directed along  $\theta_1 \mathbf{e}_1^a + \theta_2 \mathbf{e}_2^a + \theta_3 \mathbf{e}_3^a$ , with a magnitude  $\sqrt{\theta_1^2 + \theta_2^2 + \theta_3^2}$ . Either all or none of the rotation components must be prescribed, since all rotation components are needed to define a rotation tensor. The *moment\_penalty* parameter (with units of moment per radians) represents the torsional stiffness that enforces tracking of the prescribed rotations between the two bodies.

Optionally, moments may be prescribed on body *b* relative to body *a*, about the world coordinate axes, using the additional tag

```

<moment_x lc="3">1.e-3</moment_x>
<moment_y lc="4">3.e-3</moment_y>
<moment_z lc="5">2.e-3</moment_z>

```

The *moment* elements specify the components of the moment vector in world coordinates, with optional associated load curves. The *moment* elements should not be used simultaneously with a prescribed rotation.

*Example:*

```

<rigid_connector type="rigid spherical joint" name="Joint01">
  <tolerance>0.1</tolerance>
  <gaptol>0</gaptol>
  <force_penalty>1e0</force_penalty>
  <auto_penalty>1</auto_penalty>
  <body_a>1</body_a>
  <body_b>2</body_b>
  <joint_origin>0,0,0</joint_origin>
</rigid_connector>

```

### 3.12.4.2 Rigid Revolute Joint

The rigid revolute joint connector allows the user to connect two rigid bodies *a* and *b* at a point in space and allow rotation about a single prescribed axis.

In addition to the shared parameters above, it defines the following parameters.

| parameter                  | Description                      |
|----------------------------|----------------------------------|
| <i>joint_origin</i>        | The position of the joint.       |
| <i>rotation_axis</i>       | The rotation axis.               |
| <i>transverse_axis</i>     | The transverse axis.             |
| <i>prescribed_rotation</i> | Prescribed rotation flag         |
| <i>rotation</i>            | Prescribed rotation around axis. |
| <i>moment</i>              | Applied moment about axis.       |

The *tolerance* element defines the augmentation tolerance. That is, when the relative change in the constraint forces and moments (the Lagrange multipliers) are less than this value. The *gaptol* element defines the tolerance for spatial separation of the joint origin on the two bodies (in units of length). The *angtol* element defines the tolerance for angular separation of the joint axis on the two bodies (in units of radians). Setting any of these three elements to zero disables the enforcement of that tolerance. The *force\_penalty* parameter (with units of force per length) represents the stiffness that prevents the joint origin on the two bodies from separating. The *moment\_penalty* parameter (with units of moment per radians) represents the torsional stiffness that enforces parallelism of the joint axis on the two bodies. The *body\_a* and *body\_b* elements are the material numbers of the two rigid bodies. The *joint\_origin* element defines the position of the joint origin (the origin of the basis  $\{e_1, e_2, e_3\}$ ) in world coordinates at the start of the analysis. Note that this point does not have to be inside or on the surface of either of the two bodies. The *rotation\_axis* element defines the orientation of the joint rotation axis  $e_1$  in world coordinates at the start of the analysis.

Optionally, the rotation of body *b* relative to body *a* may be prescribed using the additional tags

```
<prescribed_rotation>1</prescribed_rotation>
<rotation lc="1">3.14159</rotation>
```

The *prescribed\_rotation* element is a flag that indicates that the motion of the joint is prescribed (1 for prescribed, 0 for free). The *rotation* element specifies the amount of rotation (with units of radians) with an optional associated load curve.

Optionally, a moment may be prescribed on body *b* relative to body *a*, about the joint axis, using the additional tag

```
<moment lc="1">5.e-3</moment>
```

The *moment* element specifies the magnitude of the moment, with an optional associated load curve. The *moment* element should not be used simultaneously with a prescribed rotation.

*Example:*

```
<rigid_connector type="rigid revolute joint" name="Joint 1">
  <tolerance>0</tolerance>
  <gaptol>1e-4</gaptol>
  <angtol>1e-4</angtol>
  <force_penalty>1e0</force_penalty>
  <moment_penalty>1e0</moment_penalty>
  <auto_penalty>1</auto_penalty>
  <body_a>1</body_a>
  <body_b>3</body_b>
  <joint_origin>-150,0,2</joint_origin>
  <rotation_axis>0,0,1</rotation_axis>
</rigid_connector>
```

### 3.12.4.3 Rigid Prismatic Joint

The rigid prismatic joint connector allows the user to connect two rigid bodies *a* and *b* at a point in space and allow translation along a single prescribed axis.

In addition to the shared parameters above, it defines the following parameters.

| parameter                     | Description                        |
|-------------------------------|------------------------------------|
| <i>joint_origin</i>           | The position of the joint.         |
| <i>translation_axis</i>       | The translation axis.              |
| <i>transverse_axis</i>        | The transverse axis.               |
| <i>prescribed_translation</i> | Prescribed translation flag        |
| <i>translation</i>            | Prescribed translation along axis. |
| <i>force</i>                  | Applied force along axis.          |

The *tolerance* element defines the augmentation tolerance. That is, when the relative change in the constraint forces and moments (the Lagrange multipliers) are less than this value. The *gaptol* element defines the tolerance for spatial separation of the joint origin on the two bodies (in units of length). The *angtol* element defines the tolerance for angular separation of the joint axis on the two bodies (in units of radians). Setting any of these three elements to zero disables the enforcement of that tolerance. The *force\_penalty* parameter (with units of force per length) represents the stiffness that prevents the joint origin on the two bodies from separating. The *moment\_penalty* parameter (with units of moment per radians) represents the torsional stiffness that enforces parallelism of the joint axis on the two bodies. The *body\_a* and *body\_b* elements are the material numbers of the two rigid bodies. The *joint\_origin* element defines the position of the joint (the origin of the basis  $\{e_1, e_2, e_3\}$ ) in world coordinates at the start of the analysis. Note that this point does not have to be inside or on the surface of either of the two bodies. The *translation\_axis* element defines the orientation  $e_1$  of the joint translation axis in world coordinates at the start of the analysis.

Optionally, the translation of body *b* relative to body *a* may be prescribed using the additional tags

```
<prescribed_translation>1</prescribed_translation>
<translation lc="1">5.0</translation>
```

The *prescribed\_translation* element is a flag that indicates that the motion of the joint is prescribed (1 for prescribed, 0 for free). The *translation* element specifies the amount of translation (with units of length) with an optional associated load curve.

Optionally, a force may be prescribed on body *b* relative to body *a*, along the joint axis using the additional tag

```
<force lc="1">5.e-3</force>
```

The *force* element specifies the magnitude of the force, with an optional associated load curve. The *force* element should not be used simultaneously with a prescribed translation.

*Example:*

```
<rigid_connector type="rigid prismatic joint" name="Joint02 ">
  <tolerance>0</tolerance>
  <gaptol>1e-4</gaptol>
  <angtol>1e-4</angtol>
  <force_penalty>1e0</force_penalty>
```



```

<moment_penalty>1e0</moment_penalty>
<auto_penalty>1</auto_penalty>
<body_a>4</body_a>
<body_b>1</body_b>
<joint_origin>-150,0,2</joint_origin>
<translation_axis>1,0,0</translation_axis>
<prescribed_translation>150</prescribed_translation>
<translation lc="1">1</translation>
</rigid_connector>

```

#### 3.12.4.4 Rigid Cylindrical Joint

A rigid cylindrical joint connects rigid bodies *a* and *b* at a point in space, allowing one degree of freedom for rotation about an axis through that point, and another degree of freedom for translation along that axis.

In addition to the shared parameters above, it defines the following parameters.

| parameter                     | Description                        |
|-------------------------------|------------------------------------|
| <i>joint_origin</i>           | The position of the joint.         |
| <i>joint_axis</i>             | The joint axis.                    |
| <i>transverse_axis</i>        | The transverse axis.               |
| <i>prescribed_translation</i> | Prescribed translation flag        |
| <i>translation</i>            | Prescribed translation along axis. |
| <i>force</i>                  | Applied force along axis.          |
| <i>prescribed_rotation</i>    | Prescribed rotation flag           |
| <i>rotation</i>               | Prescribed rotation around axis.   |
| <i>moment</i>                 | Applied moment about axis.         |

The *tolerance* element defines the augmentation tolerance. That is, when the relative change in the constraint forces and moments (the Lagrange multipliers) are less than this value. The *gaptol* element defines the tolerance for spatial separation of the joint origin on the two bodies (in units of length). The *angtol* element defines the tolerance for angular separation of the joint axis on the two bodies (in units of radians). Setting any of these three elements to zero disables the enforcement of that tolerance. The *force\_penalty* parameter (with units of force per length) represents the stiffness that prevents the joint origin on the two bodies from separating. The *moment\_penalty* parameter (with units of moment per radians) represents the torsional stiffness that enforces parallelism of the joint axis on the two bodies. The *body\_a* and *body\_b* elements are the material numbers of the two rigid bodies. The *joint\_origin* element defines the position of the joint (the origin of the basis  $\{e_1, e_2, e_3\}$ ) in world coordinates at the start of the analysis. Note that this point does not have to be inside or on the surface of either of the two bodies. The *joint\_axis* element defines the orientation  $e_1$  of the joint rotation and translation axis in world coordinates at the start of the analysis.

Optionally, the rotation of body *b* relative to body *a* may be prescribed using the additional tags

```
<prescribed_rotation>1</prescribed_rotation>
<rotation lc="1">1.570796327</rotation>
```

The *prescribed\_rotation* element is a flag that indicates that the motion of the joint is prescribed (1 for prescribed, 0 for free). The *rotation* element specifies the amount of rotation (with units of radians) with an optional associated load curve.

Optionally, the translation of body *b* relative to body *a* may be prescribed using the additional tags

```
<prescribed_translation>1</prescribed_translation>
<translation lc="2">10.0</translation>
```

Optionally, a moment may be prescribed about the joint axis using the additional tag

```
<moment lc="1">5.e-3</moment>
```

The *moment* element specifies the magnitude of the moment, with an optional associated load curve. The *moment* element should not be used simultaneously with a prescribed rotation.

Optionally, a force may be prescribed along the joint axis using the additional tag

```
<force lc="1">2.0</force>
```

The *force* element specifies the magnitude of the force, with an optional associated load curve. The *force* element should not be used simultaneously with a prescribed translation.

*Example:*

```
<rigid_connector type="rigid cylindrical joint" name="Joint03">
  <tolerance>0</tolerance>
  <gaptol>1e-4</gaptol>
  <angtol>1e-4</angtol>
  <force_penalty>1e0</force_penalty>
  <moment_penalty>1e0</moment_penalty>
  <auto_penalty>1</auto_penalty>
  <body_a>1</body_a>
  <body_b>2</body_b>
  <joint_origin>0,0,0</joint_origin>
  <joint_axis>0,0,1</joint_axis>
</rigid_connector>
```

### 3.12.4.5 Rigid Planar Joint

A rigid planar joint connects rigid bodies *a* and *b*, allowing one degree of freedom for rotation about the axis  $\mathbf{e}_1$  through that point, and two degrees of freedom for translations in the plane perpendicular to that axis, along  $\mathbf{e}_2^a$  and  $\mathbf{e}_3^a$ .

In addition to the shared parameters above, it defines the following parameters.

| parameter                       | Description                         |
|---------------------------------|-------------------------------------|
| <i>joint_origin</i>             | The position of the joint.          |
| <i>rotation_axis</i>            | The joint axis.                     |
| <i>translation_axis_1</i>       | The first translation axis.         |
| <i>prescribed_rotation</i>      | Prescribed rotation flag            |
| <i>rotation</i>                 | Prescribed rotation around axis.    |
| <i>prescribed_translation_1</i> | Prescribed translation 1 flag       |
| <i>translation_1</i>            | Prescribed translation along axis 1 |
| <i>prescribed_translation_2</i> | Prescribed translation 2 flag       |
| <i>translation_2</i>            | Prescribed translation along axis 2 |

The *tolerance* element defines the augmentation tolerance. That is, when the relative change in the constraint forces and moments (the Lagrange multipliers) are less than this value. The *gaptol* element defines the tolerance for spatial separation of the joint origin on the two bodies (in units of length). The *angtol* element defines the tolerance for angular separation of the joint axes on the two bodies (in units of radians). Setting any of these three elements to zero disables the enforcement of that tolerance. The *force\_penalty* parameter (with units of force per length) represents the stiffness that prevents the joint origin on the two bodies from separating along the rotation axis. The *moment\_penalty* parameter (with units of moment per radians) represents the torsional stiffness that enforces parallelism of the joint rotation axis on the two bodies. The *body\_a* and *body\_b* elements are the material numbers of the two rigid bodies. The *joint\_origin* element defines the position of the joint (the origin of the basis  $\{e_1, e_2, e_3\}$ ) in world coordinates at the start of the analysis. Note that this point does not have to be inside or on the surface of either of the two bodies. The *rotation\_axis* element defines the orientation of the joint rotation axis  $e_1$  in world coordinates at the start of the analysis. The *translation\_axis\_1* element defines the orientation of the joint translation axis  $e_2^a$  in the plane perpendicular to the joint rotation axis, in world coordinates at the start of the analysis.

Optionally, the rotation of body *b* relative to body *a* may be prescribed using the additional tags

```
<prescribed_rotation>1</prescribed_rotation>
<rotation lc="1">1.570796327</rotation>
```

The *prescribed\_rotation* element is a flag that indicates that the motion of the joint is prescribed (1 for prescribed, 0 for free). The *rotation* element specifies the amount of rotation (with units of radians) with an optional associated load curve.

Optionally, the translation of body *b* relative to body *a* may be prescribed along  $e_2^a$  using the additional tags

```
<prescribed_translation_1>1</prescribed_translation_1>
<translation_1 lc="2">10.0</translation_1>
```

Optionally, the translation of body *b* relative to body *a* may be prescribed along  $e_3^a$  using the additional tags

```
<prescribed_translation_2>1</prescribed_translation_2>
<translation_2 lc="2">10.0</translation_2>
```

*Example:*

```

<rigid_connector type="rigid planar joint" name="Joint01">
  <tolerance>0</tolerance>
  <gaptol>1e-4</gaptol>
  <angtol>1e-4</angtol>
  <force_penalty>1e0</force_penalty>
  <moment_penalty>1e0</moment_penalty>
  <auto_penalty>1</auto_penalty>
  <body_a>1</body_a>
  <body_b>2</body_b>
  <joint_origin>0,0,0</joint_origin>
  <rotation_axis>0,-0.5,0.8660254</rotation_axis>
  <translation_axis_1>1,0,0</translation_axis_1>
</rigid_connector>

```

### 3.12.4.6 Rigid Lock Joint

A rigid lock connects rigid bodies  $a$  and  $b$ , preventing relative motion between them. It requires the specification of a joint origin and two orthogonal axes  $\mathbf{e}_1$  and  $\mathbf{e}_2$ .

In addition to the shared parameters above, it defines the following parameters.

| parameter           | Description                |
|---------------------|----------------------------|
| <i>joint_origin</i> | The position of the joint. |
| <i>first_axis</i>   | The first axis.            |
| <i>second_axis</i>  | The second axis.           |

The *tolerance* element defines the augmentation tolerance. That is, when the relative change in the constraint forces and moments (the Lagrange multipliers) are less than this value. The *gaptol* element defines the tolerance for spatial separation of the joint origin on the two bodies (in units of length). The *angtol* element defines the tolerance for angular separation of the joint axes on the two bodies (in units of radians). Setting any of these three elements to zero disables the enforcement of that tolerance. The *force\_penalty* parameter (with units of force per length) represents the stiffness that prevents the joint origin on the two bodies from separating along the rotation axis. The *moment\_penalty* parameter (with units of moment per radians) represents the torsional stiffness that enforces parallelism of the joint rotation axis on the two bodies. The *body\_a* and *body\_b* elements are the material numbers of the two rigid bodies. The *joint\_origin* element defines the position of the joint (the origin of the basis  $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ ) in world coordinates at the start of the analysis. Note that this point does not have to be inside or on the surface of either of the two bodies. The *first\_axis* element defines the orientation of the axis  $\mathbf{e}_1$  in world coordinates at the start of the analysis. The *second\_axis* element defines the orientation of the second axis  $\mathbf{e}_2$  in world coordinates at the start of the analysis.

*Example:*

```

<rigid_connector type="rigid lock" name="Joint01">
  <tolerance>0</tolerance>
  <gaptol>1e-4</gaptol>

```

```

<angtol>1e-4</angtol>
<force_penalty>1e0</force_penalty>
<moment_penalty>1e0</moment_penalty>
<auto_penalty>1</auto_penalty>
<body_a>1</body_a>
<body_b>2</body_b>
<joint_origin>0,0,0</joint_origin>
<first_axis>1,0,0</first_axis>
<second_axis>0,1,0</second_axis>
</rigid_connector>

```

### 3.12.4.7 Rigid Spring

The rigid spring applies a linear spring that connects two rigid bodies  $a$  and  $b$  at arbitrary points (not necessarily nodes).

In addition to the shared parameters above, it defines the following parameters.

| parameter      | Description                                    |
|----------------|------------------------------------------------|
| $k$            | spring constant (in units of force per length) |
| $insertion\_a$ | Insertion point of the spring on body $a$      |
| $insertion\_b$ | Insertion point of the spring on body $b$      |
| $free\_length$ | The length of the unloaded spring (1)          |

*Comments:*

1. If the  $free\_length$  parameter is zero, then the initial length of the spring, as defined by the two insertion points, is taken as the free length.

*Example:*

```

<rigid_connector type="rigid spring">
  <body_a>1</body_a>
  <body_b>2</body_b>
  <insertion_a>0,0,1</insertion_a>
  <insertion_b>0,0,3</insertion_b>
  <k>1</k>
</rigid_connector>

```

### 3.12.4.8 Rigid Damper

The rigid damper applies a linear damper that connects two rigid bodies  $a$  and  $b$  at arbitrary points (not necessarily nodes).

In addition to the shared parameters above, it defines the following parameters.

| parameter          | Description                             |
|--------------------|-----------------------------------------|
| <i>c</i>           | Damping constant                        |
| <i>insertion_a</i> | Insertion point of the spring on body a |
| <i>insertion_b</i> | Insertion point of the spring on body b |

*Example:*

```
<rigid_connector type="rigid damper">
  <body_a>1</body_a>
  <body_b>2</body_b>
  <insertion_a>0,0,1</insertion_a>
  <insertion_b>0,0,3</insertion_b>
  <c>1e-7</c>
</rigid_connector>
```

#### 3.12.4.9 Rigid Angular Damper

The rigid angular damper applies an angular damper that connects two rigid bodies *a* and *b*. In addition to the shared parameters above, it defines the following parameters.

| parameter | Description              |
|-----------|--------------------------|
| <i>c</i>  | Angular damping constant |

*Example:*

```
<rigid_connector type="rigid angular damper">
  <body_a>1</body_a>
  <body_b>2</body_b>
  <c>1e-3</c>
</rigid_connector>
```

#### 3.12.4.10 Rigid Contractile Force

The rigid contractile force applies an actuator force between arbitrary points (not necessarily nodes) on two rigid bodies *a* and *b*.

In addition to the shared parameters above, it defines the following parameters.

| parameter          | Description                             |
|--------------------|-----------------------------------------|
| <i>f0</i>          | The applied force (1)                   |
| <i>insertion_a</i> | Insertion point of the spring on body a |
| <i>insertion_b</i> | Insertion point of the spring on body b |

*Comments:*

1. The `f0` parameter represents the actuator force (positive for contraction). Optionally, it may be associated with a load curve.

*Example:*

```
<rigid_connector type="rigid contractile force">
  <body_a>1</body_a>
  <body_b>2</body_b>
  <insertion_a>0,0,1</insertion_a>
  <insertion_b>0,0,3</insertion_b>
  <f0 lc="1">1</f0>
</rigid_connector>
```

### 3.13 Loads Section

The *Loads* section defines all nodal, edge, surface, and body loads that can be applied to the model.

#### 3.13.1 Nodal Loads

Nodal loads are applied by the *nodal\_load* element. The nodal load is defined by the *type* attribute. The following table lists the available nodal loads.

| type               | Description                                                                |
|--------------------|----------------------------------------------------------------------------|
| nodal_load         | A load that is added to the corresponding dofs in the global force vector. |
| nodal_force        | A force applied to each node of a node set                                 |
| nodal_target_force | A force that ramps the current nodal load to a target value.               |

##### 3.13.1.1 nodal\_load

The *nodal\_load* adds a value directly to the corresponding degrees of freedom in the global external load vector. When the loads are applied to displacement degrees of freedom, the forces always point in the same direction and do not deform with the geometry (i.e. they are non-follower forces). For other degrees of freedom they define a constant normal flux.

```
<nodal_load type="nodal_load" node_set="set1">
  <dof>x</dof>
  <scale lc="1">1.0</scale>
</nodal_load>
```

The *dof* parameter defines the degree of freedom. The following values are allowed:

**x** apply force in *x*-direction

**y** apply force in *y*-direction

**z** apply force in  $z$ -direction

**p** apply normal volumetric fluid flow rate

**c<sub>n</sub>** apply normal molar solute flow rate

**t** apply normal heat flux (heat transfer analysis)

For solutes, replace “*n*” with the solute id from the global solute table (Section 3.4.2); for example, “c2”.

An optional *loadcurve* can be specified for the *scale* parameter with the *lc* attribute.

### 3.13.1.2 nodal\_force

This nodal load defines a force that is applied to each node in the node set. The *value* parameter sets the force vector. An optional load curve can be defined to scale the vector.

```
<nodal_load type="nodal_force" node_set="set1">
  <value lc="1">1,0,0</scale>
</nodal_load>
```

### 3.13.1.3 nodal\_target\_force

This nodal load defines a force that is applied to each node in the node set. The force will ramp up from the value at the end of the last time step, to the desired value defined in the *force* variable.

```
<nodal_load type="nodal_force_target" node_set="set1">
  <force>1,0,0</force>
  <scale lc="1">1</scale>
</nodal_load>
```

Note that, in order to reach the target load, the loadcurve assigned to *scale* must start at 0 and end at 1.

### 3.13.1.4 Nodal Fluidflux

The “*nodal fluidflux*” boundary condition implements an equivalent nodal force load. This load will be applied directly to the load vector of the system.

| parameter | Description      | default |
|-----------|------------------|---------|
| value     | The applied flux | 0       |

```
<nodal_load type="nodal fluidflux" node_set="set1">
  <value lc="1">1</value>
</nodal_load>
```

## 3.13.2 Surface Loads

A surface load can be applied using the *surface\_load* element. This element takes two attributes, namely *type*, which defines the type of surface load that will be applied, and *surface* which defines the surface that this load will be applied to.



### 3.13.2.1 Pressure Load

Pressure loads are applied to the surface of the geometry and are defined by the *surface\_load* element with the type attribute set to *pressure*. These pressure forces are also known as *follower forces*; they change direction as the body is deformed and, in this case, are always oriented along the local surface normal. The sign convention is so that a positive pressure will act opposite to the normal, so it will compress the material. The surface that the load is applied to is specified with the *surface* attribute. The surface must be defined in the *Mesh* section.

| parameter           | Description                     | default |
|---------------------|---------------------------------|---------|
| pressure            | The applied pressure values (1) | 0 [P]   |
| symmetric_stiffness | symmetric stiffness flag (2)    | 1       |
| linear              | linear flag (3)                 | 0       |
| shell_bottom        | shell bottom flag (4)           | 0       |

*Comments:*

1. The *pressure* element defines the pressure value [P]. The optional parameter *lc* defines a loadcurve for the pressure evolution. If *lc* is not defined a constant pressure is applied.
2. The optional *symmetric\_stiffness* parameter (0 or 1) determines whether to use the exact form of the stiffness matrix for this surface load (0=non-symmetric), or the symmetric form (1=default). The non-symmetric form is numerically more robust, but it may also require using a globally non-symmetric stiffness matrix (see Section 3.3.5).
3. If the optional *linear* parameter is set to 1, a constant, deformation independent pressure load is applied. This is not recommend for large deformations.
4. When the *shell\_bottom* flag is set to 1, the surface is assumed to be a shell surface and the pressure is applied to the bottom part of the shell surface.

*Example:*

```
<surface_load type="pressure" surface="surface1">
  <pressure lc="1">1.0</pressure>
  <symmetric_stiffness>0</symmetric_stiffness>
</surface_load>
```

### 3.13.2.2 Traction Load

A traction load applies a traction to a surface. The direction of the traction remains unchanged as the mesh deforms.

```
<surface_load type="traction" surface="surface1">
  <scale lc="1">1.0</scale>
  <traction>0,0,1</traction>
</surface_load>
```

The traction vector is determined by two quantities. The direction and magnitude is defined by the *traction* element. In addition, the magnitude can be scaled using the *scale* element. An optional load curve can be defined for the *scale* element using the *lc* attribute. This allows the traction load to become time dependent. If the *lc* attribute is omitted a constant traction load is applied.

### 3.13.2.3 Surface Force

A surface force applies an equivalent uniform traction load to a surface. The direction of the force remains unchanged as the mesh deforms.

```
<surface_load type="force" surface="surface1">
  <scale lc="1">1.0</scale>
  <force>0,0,1</force>
</surface_load>
```

The force vector is determined by two quantities. The direction and magnitude is defined by the *force* element. In addition, the magnitude can be scaled using the *scale* element. An optional load curve can be defined for the *scale* element using the *lc* attribute. This allows the surface force to become time dependent. If the *lc* attribute is omitted a constant force is applied.

### 3.13.2.4 Bearing Load

A bearing load can be applied on a cylindrical surface that represents a bearing journal, or the portion of a shaft that is supported by a bearing journal. This load gets prescribed as a non-uniform compressive pressure distribution on that surface, representing the radial component of the bearing load. (The axial component of a bearing load may be prescribed using the surface force presented in Section 3.13.2.3.) The non-uniform pressure distribution may be sinusoidal or parabolic. It gets distributed over a  $\pm 90$  degree arc relative to the loading direction, which is specified as a force vector. The user is expected to specify the bearing force in a plane perpendicular to the axis of the cylindrical bearing surface. Otherwise, only the projection of the bearing force onto that plane will be prescribed as the radial bearing load. For example, a bearing load may be specified as

```
<surface_load type="bearing load" surface="BearingSurface1">

  <scale lc="1">1000.0</scale>
  <force>0.707107,0.707107,0</force>
  <profile>1</profile>
</surface_load>
```

In this example the bearing load is in the *xy*-plane, oriented at 45 degrees from the *x*-axis. The surface *BearingSurface1* should be a cylinder with its axis along the *z*-direction. The pressure distribution is parabolic (*profile*=1), and the alternative option is sinusoidal (*profile*=0). In general, a parabolic distribution produces a higher (often more realistic) central peak pressure on the bearing surface. An optional loadcurve can be associated with the *scale* as shown in this example, to scale the prescribed bearing force as a function of time. Alternatively, the user may assign a loadcurve to the *force*.

Additional options are available for a bearing load, including *symmetric\_stiffness*, *linear* and *shell\_bottom* as outlined in Section 3.13.2.1. When the *linear* flag is set to false, the pressure prescribed on the bearing surface is adjusted to account for the change in area and orientation of each face, caused by the finite deformation of the bearing surface. However, the bearing radial force direction remains unchanged with deformation. It is recommended to set the *symmetric\_stiffness* flag to false when performing a finite deformation analysis.

The bearing load is always prescribed as a compressive load on the selected bearing surface. Therefore, it is recommended to select a complete cylindrical surface on which the bearing load

is prescribed; the solver automatically determines which faces of the selected surface must be loaded in compression and which of those have zero pressure prescribed on them, based on the vectorial orientation of the bearing *force*. If the user selects a semi-cylindrical surface as the bearing surface, it becomes the user's responsibility to ensure that the prescribed bearing force bisects the 180 degree arc of that surface. Otherwise, the analysis may produce unexpected results, due to unevenness of the pressure distribution about the force's line of action. If the user selects a surface with a smaller arc than  $\pm 90$  degrees, the pressure will not reduce to zero at the straight edges of the surface, though the pressure will be scaled to produce the prescribed force magnitude along the bearing force direction.

### 3.13.2.5 Pipette Aspiration

Pipette aspiration is a common tool in biomechanics: A pipette is pressed against a tissue surface (contact problem) and a known negative pressure is typically applied through the pipette to aspirate the tissue. Then, some relevant material property may be extracted from the observed aspirated height of the tissue under the prescribed pressure. This pressure load is similar to the pressure load described in Section 3.13.2.1, however *it must be used on a surface which is also the primary surface* in a *sliding-elastic* contact interface (see Section 3.14.1). The contact interface must be between a *circular* pipette with inner radius  $R$  and the tissue. (In other words, the pipette has to be modeled explicitly in this contact analysis). The purpose of this *pipette aspiration* load is to prescribe the pressure only on the faces of the selected contact surface which happen to be inside the pipette footprint. Therefore, the pressure is not applied on the portion of the contact surface which has a non-zero contact pressure, or on the faces that fall outside of the pipette footprint. The parameters that need to be defined for this pressure load are:

| parameter | Description                                                                        | default   |
|-----------|------------------------------------------------------------------------------------|-----------|
| pressure  | The applied pressure values (1)                                                    | 0 [P]     |
| radius    | The inner radius $R$ of the pipette                                                | 0 [L]     |
| center    | A vector representing the position of the center of the circular pipette footprint | 0,0,0 [L] |
| normal    | A vector represent the normal direction representing the pipette axis              | 0,0,1     |

This pressure load only works with circular pipettes. However, the pipette wall thickness is only modeled via the explicit representation of the pipette, and the pipette tip need not be flat (e.g., it could be semi-circular or beveled, etc.). The radius  $R$  represents the circular edge of the contact region in the absence of deformation (in the reference condition). The model may employ planes of symmetry, since those do not affect the specification of *radius*, *center*, or *normal*.

### 3.13.2.6 Mixture Normal Traction

This section applies to biphasic, biphasic-solute, triphasic and multiphasic analyses. In a mixture of intrinsically incompressible solid and fluid constituents, the formulation adopted in FEBio implies that the total traction is a natural boundary condition ([FEBio Theory Manual](#)). If this boundary condition is not explicitly prescribed, the code automatically assumes that it is equal to zero. Therefore, boundaries of mixtures are traction-free by default.

The *mixture traction*  $\mathbf{t}$  is the traction vector corresponding to the mixture (or total) stress  $\boldsymbol{\sigma}$ ; thus  $\mathbf{t} = \boldsymbol{\sigma} \cdot \mathbf{n}$ , where  $\mathbf{n}$  is the outward unit normal to the boundary surface. Since  $\boldsymbol{\sigma} = -p\mathbf{I} + \boldsymbol{\sigma}^e$ , where  $p$  is the fluid pressure and  $\boldsymbol{\sigma}^e$  is the *effective stress* resulting from strains in the solid matrix,

it is also possible to represent the total traction as  $\mathbf{t} = -p\mathbf{n} + \mathbf{t}^e$ , where  $\mathbf{t}^e = \boldsymbol{\sigma}^e \cdot \mathbf{n}$  is the *effective traction*. Currently, FEBio allows the user to specify only the normal component of the traction, either  $t_n = \mathbf{t} \cdot \mathbf{n}$  (the normal component of the mixture traction) or  $t_n^e = \mathbf{t}^e \cdot \mathbf{n}$  (the normal component of the effective traction):

A mixture normal traction is defined by the *surface\_load* element using *normal\_traction* for the type attribute.

```
<surface_load type="normal_traction" surface="surface1">
  <traction [lc="1"]>1.0</traction>
  <effective>1</effective>
  <linear>0</linear>
</surface_load>
```

The *traction* element defines the magnitude of the traction force. The optional attribute *lc* defines a loadcurve that controls the time dependency of the traction force magnitude. If omitted a constant traction is applied.

The *effective* element defines whether the traction is applied as an effective traction or a total mixture traction.

The *linear* element defines whether the traction remains normal to the deformed surface or the reference surface. If set to true the traction remains normal to the reference surface. When false it defines a follower force that remains normal to the deformed surface.

The *surface* element defines the surface to which the traction is applied. It consists of child elements defining the individual surface facets.

Unlike the mixture and effective traction, the fluid pressure  $p$  is a nodal variable (see Section 3.11.1). There may be common situations where the user must apply a combination of related fluid pressure and traction boundary conditions. For example, if a biphasic surface is subjected to a non-zero fluid pressure  $p_0$ , the corresponding boundary conditions are  $p = p_0$  and  $t_n = -p_0$  (or  $t_n^e = 0$ ). In FEBio, both boundary conditions must be applied. For example:

```
<Boundary>
  <prescribed bc="p" node_set="set1">
    <scale lc="1">1.0</scale>
  </prescribed>
  <surface_load type="normal_traction" surface="surf1">
    <traction lc="1">-1.0</traction>
    <effective>0</effective>
    <linear>0</linear>
  </surface_load>
</Boundary>
<LoadData>
  <loadcontroller id="1" type="loadcurve">
    <points>
      <loadpoint>0,0</loadpoint>
      <loadpoint>1,2.0</loadpoint>
    </points>
  </loadcurve>
</LoadData>
```

### 3.13.2.7 Fluid Flux

In a biphasic mixture of intrinsically incompressible solid and fluid constituents, the  $\mathbf{u}-p$  formulation adopted in FEBio implies that the normal component of the relative fluid flux is a natural boundary condition. If this boundary condition is not explicitly prescribed, the code automatically assumes that it is equal to zero. Therefore, biphasic boundaries are impermeable by default. (To implement a free-draining boundary, the fluid pressure nodal degrees of freedom should be set to zero.)

The flux of fluid relative to the solid matrix is given by the vector  $\mathbf{w}$ . Since viscosity is not explicitly modeled in a biphasic material, the tangential component of  $\mathbf{w}$  on a boundary surface may not be prescribed. Only the normal component of the relative fluid flux,  $w_n = \mathbf{w} \cdot \mathbf{n}$ , represents a natural boundary condition. To prescribe a value for  $w_n$  on a surface, use:

```
<surface_load type="fluidflux" surface="surf1">
  <flux lc="1">1.0</flux>
  <linear>0</linear>
  <mixture>1</mixture>
</fluidflux>
```

The *flux* parameter defines the flux that will be applied to the surface. The optional parameter *lc* defines a loadcurve for the normal flux evolution. If omitted a constant fluid flux is applied.

When *linear* is set to zero (default) it means that the flux matches the prescribed value even if the surface on which it is applied changes in area as it deforms. Therefore, the net volumetric flow rate across the surface changes with changes in area. This type of boundary condition is useful if the fluid flux is known in the current configuration.

When *linear* is set to non-zero it means that the prescribed flux produces a volumetric flow rate based on the undeformed surface area in the reference configuration. Therefore, the flux in the current configuration does not match the prescribed value. This type of boundary condition is useful if the net volumetric flow rate across the surface is known. For example: Let  $Q$  be the known volumetric flow rate, let  $A_0$  be the surface area in the reference configuration (a constant). Using “*linear*” means that the user prescribes  $Q/A_0$  as the flux boundary condition. (However, regardless of the *type*, the fluid flux saved in the output file has a normal component equal to  $Q/A$ , where  $A$  = area in current configuration.)

Prescribing  $w_n$  on a free surface works only if the nodal displacements of the corresponding faces are also prescribed. If the nodal displacements are not known a priori, the proper boundary condition calls for prescribing the normal component of the mixture velocity,  $v_n = (\mathbf{v}^s + \mathbf{w}) \cdot \mathbf{n}$ . To prescribe the value of  $v_n$  on a surface, use

```
<surface_load type="fluidflux" surface="surf1">
  <flux lc="1">1.0</flux>
  <linear>0</linear>
  <mixture>1</mixture>
</surface_load>
```

For example, this boundary condition may be used when modeling a permeation problem through a biphasic material, when the upstream fluid velocity is prescribed,  $v_n = v_0$ . If the upstream face is free, the companion boundary condition would be to let  $t_n^e = 0$  on that face as well.

### 3.13.2.8 Multiphasic Solute Flux

In biphasic-solute and multiphasic analyses the molar flux of a solute  $\iota$ , given by the vector  $\mathbf{j}^\iota = \varphi^w c^\iota (\mathbf{v}^\iota - \mathbf{v}^s)$ , is evaluated relative to the solid matrix; it includes a diffusive component as well as a convective component contributed by the fluid flux  $\mathbf{w}$  relative to the solid, see eq.(4.15.4). Since solute viscosity is not explicitly modeled, the tangential component of  $\mathbf{j}^\iota$  on a boundary surface may not be prescribed. Only the normal component of the relative solute flux,  $j_n^\iota = \mathbf{j}^\iota \cdot \mathbf{n}$ , represents a natural boundary condition. Since a multiphasic mixture may contain electrically charged solutes, the solute flux that must be prescribed as a boundary condition is the effective solute flux  $\tilde{j}_n^\iota = j_n^\iota + \sum_\gamma z^\gamma j_n^\gamma$ , as explained in Section 4.16.1.4. To prescribe a value for  $\tilde{j}_n^\iota$  on a surface, use:

```
<surface_load type="soluteflux" surface="surface1">
  <flux lc="1">1.0</flux>
  <linear>0</linear>
  <solute_id>solute_name</solute_id>
  <shell_bottom>0</shell_bottom>
</surface_load>
```

The parameter *solute\_id* specifies to which solute this flux condition applies, referencing the corresponding list of solute names in the Globals section (Section 3.4.2).

The *flux* element defines the flux magnitude. The optional parameter *lc* defines a loadcurve for the normal flux evolution. If omitted a constant flux is applied.

When *linear* is set to 0 (default) it means that the flux matches the prescribed value even if the surface on which it is applied changes in area as it deforms. Therefore, the net molar flow rate across the surface changes with changes in area. This type of boundary condition is useful if the solute molar flux is known in the current configuration.

When *linear* is set to non-zero it means that the prescribed flux produces a molar flow rate based on the undeformed surface area in the reference configuration. Therefore, the flux in the current configuration does not match the prescribed value. This type of boundary condition is useful if the net molar flow rate across the surface is known. For example: Let  $Q$  be the known molar flow rate (in units of moles per time  $[\mathbf{n}/\mathbf{t}]$ ), let  $A_0$  be the surface area in the reference configuration (a constant). Using *linear* means that the user prescribes  $Q/A_0$  as the flux boundary condition (in units of moles per area per time  $[\mathbf{n}/\mathbf{L}^2 \cdot \mathbf{t}]$ ). However, regardless of the *type*, the solute molar flux saved in the output file has a normal component equal to  $Q/A$ , where  $A$  = area in current configuration. The *shell\_bottom* tag should be set to 1 (true) when prescribing the flux on the bottom surface of a multiphasic shell.

### 3.13.2.9 Solute Flux

In fluid-solutes analyses the molar flux of solute  $\iota$ , given by the vector  $\mathbf{j}^\iota = c^\iota (\mathbf{v}^\iota - \mathbf{v}^f)$ , is evaluated relative to the solvent (fluid); it only includes a diffusive contribution relative to the solvent. Since solute viscosity is not explicitly modeled, the tangential component of  $\mathbf{j}^\iota$  on a boundary surface may not be prescribed. Only the normal component of the relative solute flux,  $j_n^\iota = \mathbf{j}^\iota \cdot \mathbf{n}$ , represents a natural boundary condition. Since a fluid-solutes mixture may contain electrically charged solutes, the solute flux that must be prescribed as a boundary condition is the effective solute flux  $\tilde{j}_n^\iota = j_n^\iota + \sum_\gamma z^\gamma j_n^\gamma$ , such as that in eq.(8.8.1). To prescribe a value for  $\tilde{j}_n^\iota$  on a surface, use:

```
<surface_load type="solute flux" surface="surface1">
```

```

<flux lc="1">1.0</flux>
<solute_id>solute_name</solute_id>
</surface_load>

```

The parameter *solute\_id* specifies to which solute this flux condition applies, referencing the corresponding list of solute names in the Globals section (Section 3.4.2).

The *flux* element defines the flux magnitude. The optional parameter *lc* defines a loadcurve for the normal flux evolution. If omitted a constant flux is applied.

### 3.13.2.10 Multiphasic Solute Natural Flux

This surface load can be prescribed on a multiphasic boundary where a natural solute flux is desired. On this boundary, the solute is convected with the solvent according to  $j_n = cw_n$ . The implication of this surface load is that there is no diffusive hindrance to solute transport on the external side of the boundary. Here,  $c$  is the average solute concentration and  $w$  is the average solvent volumetric flux in the finite element connected to the boundary face on which this surface load is prescribed. The normal solvent flux  $w_n$  is evaluated as  $w \cdot n$  using the normal  $n$  on the boundary. There are no user-specified parameters in this surface load (other than specifying the solute for which it applies, and whether the boundary is the bottom of a shell domain). To prescribe a solute natural flux on a surface, use:

```

<surface_load type="solute natural flux" surface="surface1">
  <solute_id>solute_name</solute_id>
  <shell_bottom>0</shell_bottom>
  <update>0</update>
</surface_load>

```

The parameter *solute\_id* specifies to which solute this flux condition applies, referencing the corresponding list of solute names in the Globals section (Section 3.4.2). The parameter *update* is a boolean flag (0=*false*, 1=*true*, default is *false*) which, when set to *true*, initializes the effective solute concentration of surface nodes to the average value of concentrations within the underlying elements, at each iteration. You may set this flag to *true* when the iterative numerical solution fails to converge. This surface load is only needed for biphasic-solute and multiphasic analyses, since  $\tilde{j}_n^\nu = 0$  is the natural boundary condition for fluid-solutes analyses.

### 3.13.2.11 Heat Flux

A heat flux can be defined for heat transfer analyses using the *surface\_load* element with type set to *heatflux* element.

```

<surface_load type="heatflux" surface="surface1">
  <flux lc="1">1.0</flux>
</surface_load>

```

The heat flux element takes two parameters, namely *flux* which defines the flux that will be applied. It has an optional *lc* attribute which defines the load curve for this parameter. If omitted, a constant heat flux will be applied. The other (optional) parameter is the *value* parameter which can be used to define a different flux value for each surface facet.



### 3.13.2.12 Convective Heat Flux

A convective heat flux can be defined to a surface that is cooled (or heated) by exposure to an ambient temperature.

```
<surface_load type="convective_heatflux" surface="surf1">
  <Ta lc="1">1.0</Ta>
  <hc>1.0</hc>
</surface_load>
```

The *hc* parameter defines the heat transfer coefficient. The ambient temperature is defined by the *Ta* parameter. It takes an optional load curve defined through the *lc* attribute. The *value* parameter with the *surface\_data* attribute can be defined to define a different temperature value for each surface facet (Section 3.8.1).

### 3.13.2.13 Fluid Traction

In a fluid or fluid-FSI analysis, the Cauchy stress is given by  $\sigma = -p\mathbf{I} + \tau$ , where  $p$  is the elastic pressure and  $\tau$  is the viscous stress. The traction on a fluid surface with outward normal  $\mathbf{n}$  is given by  $\mathbf{t} = \sigma \cdot \mathbf{n} = -p\mathbf{n} + \mathbf{t}^\tau$ , where  $\mathbf{t}^\tau = \tau \cdot \mathbf{n}$  is the viscous component of the traction. The *fluid traction* surface load prescribes  $\mathbf{t}^\tau$  on a boundary surface.

```
<surface_load type="fluid traction" surface="surface1">
  <scale lc="1">1.0</scale>
  <traction>0,0,1</traction>
</surface_load>
```

### 3.13.2.14 Fluid Pressure

In a fluid or fluid-FSI analysis, the Cauchy stress is given by  $\sigma = -p\mathbf{I} + \tau$ , where  $p$  is the elastic pressure and  $\tau$  is the viscous stress. The *fluid pressure* surface load prescribes  $p$  on a boundary surface.

```
<surface_load type="fluid pressure" surface="surface1">
  <pressure>0.05</pressure>
</surface_load>
```

This boundary condition internally solves for the fluid dilatation  $e^f$  for the user-prescribed pressure  $p$ , using the user-selected equation of state from the list given in eq.(4.20.2). Then  $e^f$  is prescribed as an essential boundary condition on the selected surface.

### 3.13.2.15 Fluid Normal Traction

In a fluid or fluid-FSI analysis, the normal component  $t_n^\tau = \mathbf{t}^\tau \cdot \mathbf{n}$  of the viscous traction  $\mathbf{t}^\tau$  (Section 3.13.2.13) may be prescribed on a boundary surface in a fluid analysis. The *fluid normal traction* surface load prescribes  $t_n^\tau$  on a boundary surface.

```
<surface_load type="fluid normal traction" surface="surface1">
  <scale lc="1">1.0</scale>
  <traction>1</traction>
</surface_load>
```



### 3.13.2.16 Fluid Backflow Stabilization

In a fluid or fluid-FSI analysis, this boundary condition prescribes the normal component of the viscous traction (Section 3.13.2.13),  $t_n^\tau = \mathbf{t}^\tau \cdot \mathbf{n}$ , such that it opposes backflow, i.e., when  $v_n = \mathbf{v} \cdot \mathbf{n}$  is negative. Specifically,  $t_n^\tau = \beta \rho_r v_n^2$  when  $v_n < 0$  and 0 otherwise, where  $\beta$  is a unitless user-defined parameter (typical values may range from 0 to 1) and  $\rho_r$  is the referential fluid density.

```
<surface_load type="fluid backflow stabilization" surface="surface1">
  <beta lc="1">1</beta>
</surface_load>
```

### 3.13.2.17 Fluid Tangential Stabilization

In a fluid or fluid-FSI analysis, this boundary condition prescribes the tangential component of the viscous traction (Section 3.13.2.13),  $\mathbf{t}_t^\tau = (\mathbf{I} - \mathbf{n} \otimes \mathbf{n}) \cdot \mathbf{t}^\tau$ , such that it opposes the tangential fluid flow  $\mathbf{v}_t = (\mathbf{I} - \mathbf{n} \otimes \mathbf{n}) \cdot \mathbf{v}$  on the selected boundary surface, with  $\mathbf{n}$  representing the outward normal to this surface. Specifically,  $\mathbf{t}_t^\tau = -\beta \rho_r |\mathbf{v}_t| \mathbf{v}_t$ , where  $\beta$  is a unitless user-defined parameter (typical values may range from 0 to 1) and  $\rho_r$  is the referential fluid density.

```
<surface_load type="fluid tangential stabilization" surface="surface1">
  <beta lc="1">1</beta>
</surface_load>
```

### 3.13.2.18 Fluid Tangential Damping

The “*fluid tangential damping*” prescribes a shear traction that opposes tangential fluid velocity on a boundary surface. This can help stabilize inflow conditions.

| parameter | Description         | default |
|-----------|---------------------|---------|
| penalty   | damping coefficient | 0       |

```
<surface_load type="fluid tangential damping" surface="surface1">
  <penalty>1</penalty>
</surface_load>
```

### 3.13.2.19 Fluid Viscous Traction

The “*fluid viscous traction*” is a fluid surface that has a prescribed viscous traction vector on it.

| parameter | Description                              | default   |
|-----------|------------------------------------------|-----------|
| scale     | Scale factor for the load                | 0 []      |
| traction  | The traction vector that will be applied | 0,0,0 [P] |

### 3.13.2.20 Fluid Normal Velocity

In a fluid or fluid-FSI analysis the fluid velocity relative to the mesh,  $\mathbf{w}$ , on a boundary surface with outward normal  $\mathbf{n}$  may be decomposed into the normal component,  $w_n = \mathbf{w} \cdot \mathbf{n}$ , and tangential component,  $\mathbf{w}_\tau = (\mathbf{I} - \mathbf{n} \otimes \mathbf{n}) \cdot \mathbf{w} = \mathbf{w} - w_n \mathbf{n}$ . The surface load *fluid normal velocity* may be used to prescribe a desired value of  $w_n$  on a boundary surface.

```
<surface_load type="fluid normal velocity" surface="surface1">
  <value lc="1">1.0</value>
  <velocity>1</velocity>
</surface_load>
```

Optionally, FEBio can generate a parabolic velocity profile over the named surface. This option can be turned on using the *parabolic* element (0=default, 1=parabolic). In this case, the *velocity* element provides the average value of this parabolic velocity profile, while *value* remains a scale factor optionally tied to a loadcurve. The parabolic profile is calculated under the assumption of steady Poiseuille flow in an infinite tube whose cross-section has the shape of the named surface. The parabolic velocity distribution reduces to zero only on portions of the boundary curve of the named surface where the user has fixed all three components of the fluid velocity (no-slip condition). On remaining portions of this boundary curve, symmetry conditions are assumed for the three-dimensional parabolic normal velocity profile.

```
<surface_load type="fluid normal velocity" surface="surface1">
  <velocity lc="1">-1.0</velocity>
  <parabolic>1</parabolic>
</surface_load>
```

If an arbitrary distribution with different scalar value needs to be specified for each facet, use the following syntax,

```
<surface_load type="fluid normal velocity" surface="surface1">
  <value surface_data="inlet"/>
  <velocity lc="1">-1.0</value>
</surface_load>
```

and provide the facet values in a user-defined *SurfaceData* map with *name="inlet"* and *data\_type="scalar"* (Section 3.8.1).

By default, this condition only prescribes the natural boundary condition  $w_n$ , leaving  $\mathbf{w}_\tau$  free. Optionally, it is possible to also enforce  $\mathbf{w}_\tau = \mathbf{0}$  by prescribing essential boundary conditions on the  $w_x$ ,  $w_y$  and  $w_z$  components of  $\mathbf{w} = w_n \mathbf{n}$  at the nodes of each facet, based on the value of  $\mathbf{n}$  at each node (obtained by averaging  $\mathbf{n}$  from adjoining facets). This option may be turned on using the *prescribe\_nodal\_velocities* element (0=default, 1=prescribed), for example,

```
<surface_load type="fluid normal velocity" surface="surface1">
  <velocity>-1.0</velocity>
  <prescribe_nodal_velocities>1</prescribe_nodal_velocities>
</surface_load>
```

When the fluid is prescribed on a surface boundary, better numerical convergence is achieved if the fluid dilatation is also prescribed on the rim (the boundary curve) of that surface boundary. While this can be done explicitly, a better option is to set the fluid dilatation on the rim to match the average pressure on the surface boundary, calculated during the analysis. To select this option, use the *prescribe\_rim\_pressure* element (0=default, 1=prescribed), for example,

```
<surface_load type="fluid normal velocity" surface="surface1">
  <velocity>-1.0</velocity>
  <parabolic>1</parabolic>
  <prescribe_nodal_velocities>1</prescribe_nodal_velocities>
  <prescribe_rim_pressure>1</prescribe_rim_pressure>
</surface_load>
```

Note that this *prescribe\_rim\_pressure* option will not work if the mesh of the boundary surface has no internal nodes.

### 3.13.2.21 Fluid Velocity

In a fluid or fluid-FSI analysis, this boundary condition prescribes the fluid velocity vector relative to the mesh,  $\mathbf{w}$ , on a named surface, simultaneously satisfying the natural boundary condition for  $w_n = \mathbf{w} \cdot \mathbf{n}$  and prescribing essential boundary conditions on the  $w_x$ ,  $w_y$  and  $w_z$  components of  $\mathbf{w}$  at the nodes of each facet, based on the value of  $\mathbf{n}$  at each node (obtained by averaging  $\mathbf{n}$  from adjoining facets).

```
<surface_load type="fluid velocity" surface="surface1">
  <scale lc="1">1.0</scale>
  <velocity>0,0,1</velocity>
</surface_load>
```

If a different vector value needs to be specified for each facet, use the following syntax,

```
<surface_load type="fluid velocity" surface="surface1">
  <scale lc="1">1.0</scale>
  <velocity surface_data="inlet"/>
</surface_load>
```

and provide the facet values in a user-defined *SurfaceData* map with *name="inlet"* and *data\_type="vec3"* (Section 3.8.1).

### 3.13.2.22 Fluid Rotational Velocity

This boundary condition prescribes a rotational velocity on an axisymmetric surface. It is the user's responsibility to ensure that the surface is axisymmetric. The angular velocity  $\boldsymbol{\omega} = \omega \mathbf{n}$  is prescribed by specifying the angular speed  $\omega$  and the unit vector along the axis of rotation (axis of symmetry)  $\mathbf{n}$ . The user-defined vector  $\mathbf{n}$  is automatically normalized. If the axis does not pass through the origin, the user may specify any point  $\mathbf{p}$  on the axis. The fluid velocity relative to the mesh,  $\mathbf{w}$ , for nodes on the selected surface is evaluated from  $\mathbf{w} = \boldsymbol{\omega} \times \mathbf{r}$ , where  $\mathbf{r} = (\mathbf{I} - \mathbf{n} \otimes \mathbf{n}) \cdot (\mathbf{x} - \mathbf{p})$  is the shortest vector from the axis to the node and  $\mathbf{x}$  is the nodal position.

```

<surface_load type="fluid rotational velocity" surface="FluidRotationalVelocity01">
<angular_speed lc="1">1</angular_speed>
<axis>0,0,1</axis>
<origin>0,0,0</origin>
</surface_load>/*!

```

Though this boundary condition may be used in fluid-FSI analyses, it is most relevant for standard fluid analyses where the mesh is stationary.

### 3.13.2.23 Fluid-FSI Traction

In a fluid-FSI analysis, the fluid traction across an interface between a *fluid-FSI* material and a solid domain must be prescribed explicitly as a traction boundary condition on the solid domain. Otherwise, the solid domain cannot sense the fluid pressure and viscous traction. Users must identify each surface on which this FSI surface load must be prescribed. There are no user-defined parameters for this surface load.

```

<surface_load type="fluid-FSI traction" surface="surface1">
<surface_load>

```

This surface load must also be prescribed on free fluid surfaces (such as the surface of the fluid in open channel flow) to allow those surfaces to deform in response to the fluid stresses (e.g., producing surface waves). (Do not apply this surface load on inlet or outlet boundaries through which fluid flows.)

### 3.13.2.24 Biphasic-FSI Traction

In a fluid-FSI analysis, the fluid traction across an interface between a *biphasic-FSI* material and a solid domain must be prescribed explicitly as a traction boundary condition on the solid domain. Otherwise, the solid domain cannot sense the fluid pressure and viscous traction. Users must identify each surface on which this FSI surface load must be prescribed. There are no user-defined parameters for this surface load.

```

<surface_load type="biphasic-FSI traction" surface="surface1">
<surface_load>

```

This surface load must also be prescribed on free fluid surfaces (such as the surface of the fluid in open channel flow) to allow those surfaces to deform in response to the fluid stresses (e.g., producing surface waves). (Do not apply this surface load on inlet or outlet boundaries through which fluid flows.)

### 3.13.2.25 Multiphasic-FSI Traction

In a fluid-FSI analysis, the fluid traction across an interface between a *multiphasic-FSI* material and a solid domain must be prescribed explicitly as a traction boundary condition on the solid domain. Otherwise, the solid domain cannot sense the fluid pressure and viscous traction. Users must identify each surface on which this FSI surface load must be prescribed. There are no user-defined parameters for this surface load.

```
<surface_load type="multiphasic-FSI traction" surface="surface1">
<surface_load>
```

This surface load must also be prescribed on free fluid surfaces (such as the surface of the fluid in open channel flow) to allow those surfaces to deform in response to the fluid stresses (e.g., producing surface waves). (Do not apply this surface load on inlet or outlet boundaries through which fluid flows.)

### 3.13.2.26 Solute Backflow Stabilization

In a fluid-solutes analysis this surface load checks for the presence of solvent (fluid) backflow, i.e., when  $v_n^f = \mathbf{v}^f \cdot \mathbf{n}$  is negative. When this condition is encountered, the effective concentration of the selected solute (*solute\_id*) is set to its converged value from the previous time step. Otherwise, the natural boundary condition  $\tilde{j}_n = 0$  (zero diffusive solute flux relative to the solvent) prevails.

```
<surface_load type="solute backflow stabilization" surface="surface1">
<solute_id>neutral</solute_id>
</surface_load>
```

### 3.13.2.27 Fluid Mixture Viscous Traction

The “*fluid mixture viscous traction*” is a fluid surface that has a prescribed viscous traction vector on it.

| parameter | Description                              | default   |
|-----------|------------------------------------------|-----------|
| scale     | Scale factor for the load                | 0 []      |
| traction  | The traction vector that will be applied | 0,0,0 [P] |

### 3.13.2.28 Pressure Stabilization

The “*pressure stabilization*” surface load is a pseudo-surface load that is used to calculate the pressure stabilization time constant based on the properties of biphasic elements under that surface.

| parameter | Description                      | default |
|-----------|----------------------------------|---------|
| stabilize | Turn stabilization on or off (1) | on      |

*Comments:*

1. The *stabilize* parameter is currently ignored so adding this feature will always perform the stabilization.

## 3.13.3 Body Loads

A body load can be used to define a source term to the model. The following sections define the different type of body loads that can be applied. In the case of body forces, these body loads represent the body force per mass,  $\mathbf{b}$ , which appears in the momentum balance as

$$\rho \mathbf{a} = \text{div } \boldsymbol{\sigma} + \rho \mathbf{b},$$

where  $\rho$  is the mass density,  $\mathbf{a}$  is the acceleration (set to zero in quasi-static analyses), and  $\boldsymbol{\sigma}$  is the Cauchy stress tensor.

### 3.13.3.1 Constant Body Force

This constant body force per mass  $b$  is defined as a 3D vector. Each component can be associated with a load curve to define a time dependent body force. Only the non-zero components need to be defined. This type of body force is spatially homogeneous, though it may vary with time when associated with a load curve:

```
<body_load type="const">
  <x lc="1">1.0</x>
  <y lc="2">1.0</y>
  <z lc="3">1.0</z>
</body_load>
```

The *lc* attribute defines the load curve to use for the corresponding component. The values of the components can be used to define scale factors for the load values.

### 3.13.3.2 Non-Constant Body Force

This body force per mass  $b$  may be spatially inhomogeneous. The spatial inhomogeneity may be specified using a formula with variables  $X$ ,  $Y$ ,  $Z$ . For example:

```
<body_load type="non-const">
  <x>X+Y+Z</x>
  <y>X-Y-Z</y>
  <z>X*X+Z</z>
</body_load>
```

### 3.13.3.3 Centrifugal Body Force

A centrifugal body force per mass  $b = \omega^2 r e_r$  may be used for bodies undergoing steady-state rotation with angular speed  $\omega$  about a rotation axis directed along  $\mathbf{n}$  and passing through the rotation center  $c$ . Here,  $r$  represents the distance of each material point from the axis of rotation, and  $e_r$  is the radial unit vector from the axis of rotation, both of which are calculated internally from the knowledge of  $\mathbf{n}$  and  $c$ .

```
<body_load type="centrifugal">
  <angular_speed>1.0</angular_speed>
  <rotation_axis>0.707,0.707,0</rotation_axis>
  <rotation_center>0,0,0</rotation_center>
</body_load>
```

### 3.13.3.4 Heat Source

A heat source can be defined using the *heat\_source* type.

```
<body_load type="heat_source">
  <Q>1.0</Q>
</body_load>
```

### 3.13.3.5 Surface Attraction

A surface attraction body force can be used to attract a mesh toward a boundary surface, for the purpose of creating a mesh bias. This can be useful for fluid analyses where a finer mesh is needed in the vicinity of a no-slip boundary. The benefit of this tool over other existing tools for creating a biased mesh is its ability to work with structured and unstructured meshes, as well as linear and higher-order elements.

For each element in the mesh, its shortest distance  $r$  to the selected boundary surface is evaluated using a surface projection algorithm. Then, an attractive body force per mass,

$$\mathbf{b} = s \exp\left(-\frac{r}{\rho}\right) \mathbf{n}$$

is evaluated along the unit vector  $\mathbf{n}$  directed along the shortest projection distance. Here,  $\rho$  is a characteristic boundary layer thickness ( $b/l$ ) and  $s$  is a scale factor for the body force ( $bsf$ ), whose value may be adjusted relative to account for the stiffness of the elastic solid material assigned to the mesh.

```
<body_load type="surface attraction" elem_set="EB1" surface="PressureLoad1">
  <blt>0.5</blt>
  <bsf lc="1">20</bsf>
  <search_radius>0.1</search_radius>
  <search_tol>0.01</search_tol>
</body_load>
```

Note that this body force requires the specification of an attractive surface; optionally, it also accepts the specification of the element set subjected to this attractive body force. The *search\_radius* and *search\_tol* parameters are used for the projection algorithm. They have the same meaning as in Section 3.14.1.

### 3.13.3.6 Moving frame

The *moving frame* load can be used to model the effects of being embedded in an accelerating and rotating coordinate system. Whereas the *centrifugal body force* only emulates the effect of a centrifugal force and assumes a static analysis, the moving frame load includes Coriolis effect and the effects of an accelerating coordinate system and can be used in a dynamic analysis.

The motion of the moving frame is decomposed into a translation and a rotation. The position of a material point  $\mathbf{r}$  in the fixed inertial frame can then be related to the position in the rotating frame  $\mathbf{r}'$  via,

$$\mathbf{r} = \mathbf{Y} + \mathbf{R}\mathbf{r}'$$

Here,  $\mathbf{Y}$  is the position of the origin of the moving frame and  $\mathbf{R}$  is the rotation matrix.

Substituting this into the equations of motion (i.e. the balance of linear momentum), results in a modified equation of motion where the effects of the moving frame can be combined into a single body load.

$$\mathbf{f} = \varrho \left( \mathbf{R}^T \mathbf{A} + \dot{\mathbf{W}} \times \mathbf{r}' + \mathbf{W} \times (\mathbf{W} \times \mathbf{r}') + 2\mathbf{W} \times \mathbf{v}' \right)$$

Here,  $\mathbf{A}$  is the linear acceleration of the moving frame,  $\mathbf{W}$  is the angular velocity of the moving frame, measured in the moving frame (i.e.  $\mathbf{W} = \mathbf{R}^T \mathbf{w}$ , where  $\mathbf{w}$  is the angular velocity in the fixed frame), and  $\mathbf{v}$  is the velocity of the material point, measured in the moving frame.

The moving frame is defined via the two vectors, the angular velocity of the rotating frame  $\mathbf{w}$ , and the linear acceleration of the rotating frame's origin  $\mathbf{A}$ . The components of both vectors may be defined via load curves to make them time-dependent.

| parameter | Description                        | default |
|-----------|------------------------------------|---------|
| wx        | x-component of angular velocity    | 0       |
| wy        | y-component of angular velocity    | 0       |
| wz        | x-component of angular velocity    | 0       |
| ax        | x-component of linear acceleration | 0       |
| ay        | y-component of linear acceleration | 0       |
| az        | y-component of linear acceleration | 0       |

Note that if you wish to include the effect of gravity, do not add a separate body force load. Instead, modify the linear acceleration to include gravity:  $\mathbf{a} = \mathbf{A} - \mathbf{g}$ .

Example:

```
<Loads>
  <body_load type="moving frame">
    <wx>0</wx>
    <wy>0</wy>
    <wz lc="1">1</wz>
    <ax>0</ax>
    <ay>0</ay>
    <az>0</az>
  </body_load>
</Loads>
```

### 3.13.3.7 Mass Damping

The “*mass damping*” body load applies a body force that is proportional to the linear momentum density  $\mathbf{p} = \rho \mathbf{v}$ , where  $\rho$  is the material density and  $\mathbf{v}$  the velocity. An additional scale factor can be used to control the strength of the force,  $\mathbf{f} = C\mathbf{p}$ .

| parameter | Description         | default |
|-----------|---------------------|---------|
| C         | damping coefficient | 0       |

### 3.13.3.8 Fluid Centrifugal Force

A fluid centrifugal body force per mass  $\mathbf{b} = \omega^2 r \mathbf{e}_r$  may be used for bodies undergoing steady-state rotation with angular speed  $\omega$  about a rotation axis directed along  $\mathbf{n}$  and passing through the rotation center  $\mathbf{c}$ . Here,  $r$  represents the distance of each material point from the axis of rotation, and  $\mathbf{e}_r$  is the radial unit vector from the axis of rotation, both of which are calculated internally from the knowledge of  $\mathbf{n}$  and  $\mathbf{c}$ .



```
<body_load type="fluid centrifugal force">  
  <angular_speed>1.0</angular_speed>  
  <rotation_axis>0.707,0.707,0</rotation_axis>  
  <rotation_center>0,0,0</rotation_center>  
</body_load>
```

### 3.13.3.9 Fluid Moving Frame

This body load is identical to the moving frame load described in [3.13.3.6](#), but applied to a fluid domain.

| parameter | Description                        | default |
|-----------|------------------------------------|---------|
| wx        | x-component of angular velocity    | 0       |
| wy        | y-component of angular velocity    | 0       |
| wz        | z-component of angular velocity    | 0       |
| ax        | x-component of linear acceleration | 0       |
| ay        | y-component of linear acceleration | 0       |
| az        | z-component of linear acceleration | 0       |

### 3.14 Contact Section

The *Contact* section defines all the contact interfaces. Contact boundary conditions are defined with the *contact* sub-element. The *type* attribute specifies the type of contact interface that is defined. The *surface\_pair* attribute defines the surface pair to use for this contact definition. The surface pair is defined in the *Mesh* section. For example:

```
<contact type="sliding-elastic" surface_pair="contact1">
  <!-- parameter go here -->
</contact>
```

The *type* can be one of the following options:

| Type                                                                                                                                          | Description                                                                                                                                                                                                      |
|-----------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| sliding-node-on-facet,<br>sliding-facet-on-facet,<br>sliding-elastic,<br>sliding-biphasic,<br>sliding-biphasic-solute,<br>sliding-multiphasic | A sliding interface that may separate (with biphasic contact for <i>sliding</i> -biphasic, biphasic-solute contact for <i>sliding-biphasic</i> -solute, and multiphasic contact for <i>sliding-multiphasic</i> ) |
| rigid_wall                                                                                                                                    | A sliding interface with rigid wall as secondary surface                                                                                                                                                         |
| rigid_joint                                                                                                                                   | A joint between two rigid bodies                                                                                                                                                                                 |
| tied-elastic,<br>tied-node-on-facet,<br>tied-facet-on-facet, sticky,<br>tied-biphasic, tied-multiphasic                                       | A tied interface (solid-solid, solid-rigid, solid-biphasic, rigid-biphasic) or tied-biphasic interface (biphasic-biphasic).                                                                                      |
| contact potential                                                                                                                             | A potential based sliding contact interface                                                                                                                                                                      |

#### A note on auto-contact

The two surfaces that participate in the contact are specified via a surface pair. These surface pairs are defined in the SurfacePair subsection of the Mesh section of the input file. Typically the surfaces are chosen based on the user's knowledge of where the contact will occur. However, for complex geometries it might not be clear where the contact will occur, or identifying the surfaces may be cumbersome. For this reason, FEBio supports an auto-contact option, where users only need to specify the parts of the primary and secondary surfaces. From these part lists, FEBio will extract the surfaces automatically. To use the auto-contact, simply specify a part list instead of the surface names when defining the surface pair.

```
<PartList name="SlidingElastic1Primary">Part1</PartList>
<PartList name="SlidingElastic1Secondary">Part2</PartList>
<SurfacePair name="SlidingElastic1">
  <primary>@part_list:SlidingElastic1Primary</primary>
  <secondary>@part_list:SlidingElastic1Secondary</secondary>
</SurfacePair>
```

Note the **@part\_list**: prefix in front of the part list name.

Although using the auto-contact can greatly simplify defining contact between parts, it is important to keep in mind that there are a few drawbacks. First and foremost, the surfaces extracted from parts will likely be much larger than the contact area and this will result in a performance cost. Second, for complex geometries the extracted surfaces may have a complex topology that may confuse some of the contact detection algorithms. In those situations it might be beneficial to decrease the `search_radius` parameter (for contact algorithms that define this parameter).

### 3.14.1 Sliding Interfaces

A sliding interface can be used to setup a non-penetration constraint between two surfaces. As of version 1.2, three different sliding contact algorithms are available. Although all three are based on the same contact enforcement method, they all differ slightly in their implementation and have been shown to give different performance for different contact scenarios. Each sliding contact implementation is identified by a different *type* attribute.

**sliding-node-on-facet (N2F)** This is FEBio's original implementation of sliding contact. It is based on Laursen's contact formulation [57] which poses the contact problem as a nonlinear constrained optimization problem. In FEBio, the Lagrange multipliers that enforce the contact constraints are computed either using a penalty method or the augmented Lagrangian method.

**sliding-facet-on-facet (F2F)** This implementation is identical to the *sliding-node-on-facet* implementation but uses a more accurate integration rule: where the former method uses nodal integration, this method uses Gaussian quadrature to integrate the contact equations. This method has been demonstrated to give additional stability and often converges when the former method does not.

**sliding-elastic (SE)** This sliding contact interface also uses facet-on-facet contact but differs in the linearization of the contact forces, which results in a different contact stiffness matrix compared to the previous two methods. It can be used for frictionless or frictional contact. It may optionally be set to sustain tension to prevent contact surfaces from separating along the direction normal to the interface, while still allowing tangential sliding. This method sometimes performs better than the previous two methods for problems that are dominated by compression. However, the formulation is inherently non-symmetric and therefore will require additional memory and running time. A symmetrized version of this implementation is available (see below), but the symmetric version does not converge as well as the non-symmetric version.

**sliding-biphasic (SBP)** This method extends the *sliding-elastic* algorithm to support frictionless biphasic contact (see the next section). This method is described in detail in [12] for the frictionless case, and in [94] for the frictional algorithm.

**sliding-biphasic-solute (SBS)** This method is similar to *sliding-biphasic*. This contact implementation supports biphasic-solute contact (see the next section) [16]. When using biphasic-solute materials, the non-symmetric version must be used.

**sliding-multiphasic (SMP)** This method is similar to *sliding-biphasic-solute*. This contact implementation supports multiphasic contact (see the next section) [16]. When using multiphasic materials, the non-symmetric version must be used.

**contact potential (CP)** The “contact potential” contact interface implements the method by [51]. This method defines a contact potential that generates a (repulsing) force that prevents penetration from occurring. This contact formulation is not based on the augmented Lagrangian formulation and does not have any of the parameters in the table below. Instead, please see section 3.14.10 for more details on this method.

The following table lists the properties that are defined for sliding interfaces (except contact potential). It is important to note that the three different sliding implementations cannot be used interchangeably: not all features are available for each method. The third, fourth and fifth column indicate if a parameter is available for a particular implementation.

| Parameter           | Description                                                       | N2F | F2F | SE | SBP | SBS<br>SMP | Default   |
|---------------------|-------------------------------------------------------------------|-----|-----|----|-----|------------|-----------|
| penalty             | normal penalty scale factor (1)                                   | •   | •   | •  | •   | •          | 1.0       |
| auto_penalty        | auto-penalty calculation flag (2)                                 | •   | •   | •  | •   | •          | 0         |
| update_penalty      | updated auto-penalty calculation at each step (2)                 |     | •   | •  | •   | •          | 0         |
| two_pass            | two-pass flag (3)                                                 | •   | •   | •  | •   | •          | 0         |
| laugon              | augmented Lagrangian flag (4)                                     | •   | •   | •  | •   | •          | 0         |
| tolerance           | aug. Lagrangian convergence tolerance (4)                         | •   | •   | •  | •   | •          | 1.0       |
| gaptol              | tolerance for gap value (4)                                       | •   | •   | •  | •   | •          | 0.0 (off) |
| minaug              | minimum number of augmentations (4)                               | •   | •   | •  | •   |            | 0         |
| maxaug              | maximum number of augmentations (4)                               | •   | •   | •  | •   |            | 10        |
| fric_coeff          | frictional coefficient (5)                                        | •   |     | •  | •   |            | 0.0       |
| fric_penalty        | tangential penalty factor (5)                                     | •   |     |    |     |            | 0.0       |
| ktmult              | tangential stiffness multiplier (5)                               | •   |     | •  |     |            | 1.0       |
| seg_up              | maximum number of segment updates (6)                             | •   |     | •  | •   |            | 0 (off)   |
| smooth_aug          | smoothed Lagrangian augmentation (7)                              |     | •   | •  | •   | •          | 0 (off)   |
| smooth_fls          | smoothed fluid load support for friction calculation (7)          |     |     |    | •   |            | 0 (off)   |
| contact_frac        | solid-on-solid contact area fraction for friction calculation (5) |     |     |    | •   |            | 0         |
| symmetric_stiffness | symmetric stiffness matrix flag (8)                               |     |     | •  | •   | •          | 0 (off)   |
| search_tol          | Projection search tolerance (9)                                   | •   | •   | •  | •   | •          | 0.01      |

|                        |                                                   |  |   |   |   |   |         |
|------------------------|---------------------------------------------------|--|---|---|---|---|---------|
| search_radius          | search radius (10)                                |  | • | • | • | • | 1.0     |
| tension                | tension flag (11)                                 |  |   | • |   |   | 0       |
| flip_primary           | flip normal on primary surface                    |  |   | • | • |   | 0 (off) |
| flip_secondary         | flip normal on secondary surface                  |  |   | • | • |   | 0 (off) |
| shell_bottom_primary   | contact with bottom of shell on primary surface   |  |   | • | • |   | 0 (off) |
| shell_bottom_secondary | contact with bottom of shell on secondary surface |  |   | • | • |   | 0 (off) |
| offset                 | offset distance between contacting surfaces (12)  |  |   | • |   |   |         |

*Comments:*

1. If the augmented Lagrangian flag is turned off (see comment 4), the penalty method is used to enforce the contact constraint. In this method, the contact traction is determined by the gap (i.e. penetration distance) multiplied by the user-defined *penalty* factor. In the augmented Lagrangian method, the *penalty* parameter is also used but has a slightly different meaning. In this case, it scales the Lagrange multiplier increment. Due to the different meanings, the user might have to adjust the penalty factor when switching between penalty method and augmented Lagrangian method. In general the penalty method requires a larger penalty factor to reach the same gap than the augmented Lagrangian method. See comment 4 for more information on when to use which method.
2. Choosing a good initial penalty parameter can often be a difficult task, since this parameter depends on material properties as well as on mesh dimensions. For this reason, an algorithm has been implemented in FEBio that attempts to calculate a good initial value for the penalty factor  $\varepsilon_i$  for a particular node/integration point  $i$  on the contact interface:

$$\varepsilon_i = \frac{EA}{V}.$$

Here,  $A$  is the area of the element the integration point belongs to,  $V$  is the element volume and  $E$  is a measure of the elasticity modulus, which is calculated from the elasticity tensor of the element. Although the meaning of  $E$  depends on the precise material formulation, in general one can regard it as a measure of the small strain Young's modulus of the material, when *update\_penalty* is 0. If *update\_penalty* is set to 1,  $E$  is recalculated at each iteration to account for the current deformation.

To use this feature, add the following element to the contact section:

```
<auto_penalty>1</auto_penalty>
```

When the auto-penalty flag is on, the value of the *penalty* parameter serves as a scale factor for the automatically-calculated penalty factor.

3. Each sliding interface consists of a *primary* surface and a *secondary* surface. The primary surface is the surface over which the contact equations are integrated and on which the

contact tractions are calculated. The secondary surface is used to measure the gap function and to define the necessary kinematic quantities such as surface normals and tangents. This approach is usually referred to as the *single-pass* method. When using the single-pass algorithm, the results can be influenced by the choice of primary and secondary surfaces. It is best to use the most tessellated surface as the primary and the coarsest as the secondary surface. To resolve the bias issue, one can also use a *two-pass* algorithm for enforcement of the contact constraint. In this case, a single pass is performed first, after which the primary and secondary surfaces are swapped and another pass is performed. When using the two-pass method, the definition of primary and secondary surfaces is arbitrary. In most problems, the single pass is sufficient to enforce contact; with a judicious choice of primary-secondary pair and contact parameters, good results can be obtained. If however, the single pass does not give good answers, for example, when due to the geometry's curvature the gap cannot be small enough with a single pass, the two-pass method can be used, although at the expense of more calculations.

If one of the contacting surfaces is rigid, a slightly different approach is recommended. In this case, it is best to pick the rigid surface as the secondary surface and to use a single pass algorithm. The reason is that the nodal degrees of freedom on the rigid surface are condensed to the rigid degrees of freedom and if the rigid surface is the primary surface, the reaction forces may not propagate correctly to the secondary surface. This is especially true if the rigid degrees of freedom are fixed.

4. In the presence of a sliding interface (and other contact interfaces), FEBio needs to calculate the contact tractions that prevent the two participating surfaces from penetrating. In general these tractions can be found using the method of Lagrange multipliers. However, the direct calculation of these multipliers has several computational disadvantages and therefore FEBio approximates the multipliers using one of two alternative methods: the penalty method and the augmented Lagrangian method. In the former method, the multipliers are approximated by the gap (i.e. penetration distance) scaled by a suitably chosen penalty factor. In many cases, this method is sufficient to get good results. Since the correctness of a contact solution is directly determined by the amount of penetration at the converged state, the user has direct control over the quality of the solution. By increasing the penalty factor, the penetration is reduced. However, in some cases, especially in large compression problems, the penalty factor required to achieve an acceptable amount of penetration has to be so large that it causes numerical instabilities in the non-linear solution algorithm due to ill-conditioning of the stiffness matrix. In these cases, the augmented Lagrangian method might be a better choice. In this method, the multipliers are determined iteratively where, in each iteration, the multiplier's increments are determined with a penalty-like method. The advantage of this method is twofold: due to the iterative nature, the method will work with a smaller penalty factor, and in the limit, the exact Lagrange multipliers can be recovered.

To turn on the augmented Lagrangian method, simply add the following line to the contact section:

```
<laugon>1</laugon>
```

To turn off the augmented Lagrangian method, either set the value to 0 or remove the parameter altogether. The convergence tolerance is set as follows:

```
<tolerance>0.01</tolerance>
```

With this parameter set, the augmented Lagrangian method will iterate until the relative increment in the multipliers is less than the tolerance. For instance, setting the tolerance parameter to 0.01 implies that the augmented Lagrangian method will converge when there is less than a 1% change in the L2 norm of the augmented Lagrangian multiplier vector between successive augmentations. Alternatively, the user can also specify a tolerance for the gap value. In this case, the iterations will terminate when the gap norm, which is defined as the averaged L2 norm,  $(\sqrt{\sum_i \langle g_i \rangle^2} / N, \langle \cdot \rangle$  the Macauley Bracket) is less than the user-specified value:

```
<gaptol>0.001</gaptol>
```

However, one must be careful when specifying a gap tolerance. First note that the gap tolerance is an absolute value (unlike the *tolerance* which is a relative value), so this parameter depends on the dimensions of the model. Also, there are cases when a gap tolerance simply cannot be reached due to the geometry of the model in which case the augmentations may never converge.

Note that both convergence parameters may be defined at once. In that case, FEBio will try to satisfy both convergence criteria. On the other hand, setting a value of zero will turn off the convergence criteria. For example, the default value for *gaptol* is zero, so that FEBio will not check the gap norm by default.

Finally, the *minaug* and *maxaug* can be used to set a minimum and maximum number of augmentations. When the *maxaug* value is reached, FEBio will move on to the next timestep, regardless of whether the force and gap tolerances have been met. When specifying a value for *minaug*, FEBio will perform at least *minaug* augmentations.

5. The *sliding-node-on-facet*, *sliding-elastic*, and *sliding-biphasic* contact implementations support friction. For *node-on-facet*, three parameters control the frictional response: *fric\_coeff* is the material's friction coefficient and its value must be in the range from 0.0 to 1.0; *fric\_penalty* is the penalty factor that regulates the tangential traction forces, much like the *penalty* parameter regulates the normal traction force component; the parameter *ktmult* is a scale factor for the tangential stiffness matrix. It is default to 1.0, but it is observed that reducing this value might sometimes improve convergence. For *sliding-elastic* and *sliding-biphasic* only *fric\_coeff* is needed to use frictional contact. In addition, for *sliding-biphasic*, the friction algorithm requires the specification of the solid-on-solid contact area fraction, which is a number between 0 and 1 (see [FEBio Theory Manual](#)). Frictional contact is inherently dissipative and thus history-dependent. The solution may vary with the size of time increments, the cycle of loading, or Lagrangian augmentations (when *laugon*=1).
6. In a contact problem, FEBio calculates the projection of each node of the primary surface onto the secondary surface. As a node slides across the secondary surface, the corresponding surface facet can change. However, in some cases, switching facets is undesirable since it might cause instabilities in the solution process or a state in which the node oscillates continuously between two adjacent facets and thus prevents FEBio from meeting the displacement convergence tolerance. The parameter *seg\_up* allows the user to control the number of segment updates FEBio will perform during each time step. For example, a value of 4 tells FEBio it can do the segment updates during the first four iterations. After that, nodes will not be allowed to switch to new segments. The default value is 0, which means that FEBio will do a segment update each iteration of each timestep.

7. The augmented Lagrangian method may produce a non-smooth contact pressure distribution at the contact surface, even though stresses within the underlying elements may remain relatively smoother. Turning this flag on changes the method of updating the Lagrange multiplier to use the projection of the element stress to the corresponding contact face. This feature only works with some element types (HEX8G8, HEX8G1, TET4G4, TET4G1, PENTA6G6, TET10G1, TET10G4, TET10G8, TET10GL11, TET15G4, TET15G8, TET15G11, TET15G15, PYRA5G8, PYRA13G8).

In the *sliding-biphasic* friction algorithm the friction coefficient depends on the fluid pressure and load support. Smoothing this fluid load support may produce a smoother distribution of the friction coefficient over the contact interface.

8. The *sliding-elastic*, *sliding-biphasic*, *sliding-biphasic-solute* and *sliding-multiphasic* contact implementations for sliding contact are inherently non-symmetric formulations. Symmetrized versions of these algorithms do not perform as well as the nonsymmetric version so it is recommended to use the latter. The following line in the *contact* element controls which version of the algorithm is used.

```
<symmetric_stiffness>0</symmetric_stiffness>
```

A value of 1 uses the symmetric version, whereas a value of 0 uses the non-symmetric version. A similar line may be included in the *Control* element to use a non-symmetric global stiffness matrix. In some cases, a non-symmetric contact stiffness may converge well even when the global stiffness matrix is symmetric.

9. The *search\_tol* parameter defines the search tolerance of the algorithm that projects nodes of the primary surface onto facets of the secondary surface. A node that falls outside an element, but whose distance to the closest element's edge is less than the search tolerance is still considered inside. This can alleviate convergence problems when nodes are projected onto edges of elements and due to numerical error may be projected outside the surface.
10. The *search\_radius* is a dimensional search radius used by the algorithm that projects points onto facets. When the distance between the point and the facet exceeds the dimensional search radius, that projection is ignored. This can alleviate convergence problems when surfaces have multiple folds and the projection produces multiple solutions, only one of which (the closest distance) is valid.
11. The *tension* flag determines whether the contact interface can sustain tension and compression (*tension*=1) or only compression (*tension*=0).
12. The *offset* distance makes it possible to maintain the contact surfaces at a distance. This option is useful, for example, in fluid-structure interaction problems, where solid structures separated by a fluid domain are pushed against each other. Specifying a contact interface between the (internal) surfaces of the solid structures would prevent them from overlapping, but adding an offset distance will also prevent the fluid mesh from getting squished too thin or from inverting. These internal surfaces would also typically have FSI Traction interfaces defined on them (see Sections 3.13.2.23-3.13.2.24).



### 3.14.2 Biphasic Contact

The *sliding-biphasic* implementation for sliding interfaces can deal with biphasic contact surfaces (including biphasic-on-biphasic, biphasic-on-elastic, biphasic-on-rigid) [12, 94]. It allows for the possibility to track fluid flow across the contact interface. In other words, fluid can flow from one side of the contact interface to the other when both contact surfaces are biphasic. To use this feature, the user must define an additional contact parameter, namely:

```
<pressure_penalty>1.0</pressure_penalty>
```

In the same way that the *penalty* parameter controls the contact tractions, this parameter controls the penalty value that is used to calculate the Lagrange multipliers for the pressure constraint. If the *laugon* flag is set, the augmented Lagrangian method is used to enforce the pressure constraint. And if the *auto\_penalty* flag is defined (which is the recommended approach), an initial guess for the pressure penalty is calculated automatically using the following formula:

$$\varepsilon_p = \frac{kA}{V},$$

where  $A$  is the element's area,  $V$  is the element's volume and  $k$  is a measure of the permeability which is defined as one third of the trace of the material's initial permeability tensor.

When either contact surface is biphasic, the surface outside the contact area(s) is automatically set to free-draining conditions (equivalent to setting the fluid pressure to zero).

As detailed in [94] and the *FEBio Theory Manual*, the *sliding-biphasic* contact algorithm can include frictional contact. The effective friction coefficient  $\mu_{\text{eff}}$  in this frictional contact algorithm depends on the temporally evolving local fluid load support (the ratio of fluid pressure  $p$  to the normal component of the mixture contact traction  $t_n$ ), according to

$$\mu_{\text{eff}} = \mu_{\text{eq}} \left( 1 + (1 - \varphi) \frac{p}{t_n} \right).$$

Recall that  $t_n$  is negative whereas  $p$  is positive in compression. Here,  $\mu_{\text{eq}}$  (*fric\_coeff*) is the friction coefficient in the limit when the fluid load support has reduced to zero, and  $\varphi$  (*contact\_frac*) is the fraction of the apparent contact area over which the solid matrix of the primary surface contacts that of the secondary surface. Thus,  $1 - \varphi$  is the fraction of the apparent contact area where fluid contacts fluid or solid. For perfectly smooth contact surfaces one may assume that  $\varphi = \varphi_1^s \varphi_2^s$ , where  $\varphi_1^s$  and  $\varphi_2^s$  are the solid area (or volume) fractions of the primary and secondary contact surfaces, respectively. For example, when both contact surfaces are non-porous ( $\varphi_1^s = \varphi_2^s = 1$ ), the effective friction coefficient reduces to  $\mu_{\text{eff}} = \mu_{\text{eq}}$ .

When performing biphasic-on-rigid contact a two-pass analysis should not be used; the rigid surface should be the secondary surface.

### 3.14.3 Biphasic-Solute and Multiphasic Contact

The *sliding-biphasic-solute* implementation for sliding interfaces can deal with biphasic-solute contact surfaces (including biphasic-solute-on-biphasic-solute, biphasic-solute-on-biphasic, biphasic-solute-on-elastic, biphasic-solute-on-rigid) and the *sliding-multiphasic* contact interface can similarly deal with multiphasic contact surfaces. These contact interfaces allow for the possibility to track fluid and solute flow across the contact interface [16]. In other words, fluid and solute can flow from one side of the contact interface to the other. To use this feature, the user must define additional contact parameters, namely:

```

<pressure_penalty>1.0</pressure_penalty>
<concentration_penalty>1.0</concentration_penalty>
<ambient_pressure>0</ambient_pressure>
<ambient_concentration>0</ambient_concentration> (for sliding-biphasic-solute)

```

or

```

<ambient_concentration sol="id">0</ambient_concentration> (for sliding-multiphasic)

```

In the same way that the *penalty* parameter controls the contact tractions, these penalty parameters control the penalty values that are used to calculate the Lagrange multipliers for the pressure and concentration constraints. If the *laugon* flag is set, the augmented Lagrangian method is used to enforce the pressure and concentration constraints. And if the *auto\_penalty* flag is defined, an initial guess for the pressure and concentration penalty is calculated automatically using the following formulas:

$$\varepsilon_p = \frac{k \cdot A}{V}, \quad \varepsilon_c = \frac{d \cdot A}{V},$$

where  $A$  is the element's area,  $V$  is the element's volume,  $k$  is a measure of the fluid permeability which is defined as one third of the trace of the material's initial permeability tensor, and  $d$  is a measure of the solute diffusivity which is defined as one third of the trace of the material's initial diffusivity tensor.

When either contact surface is biphasic-solute or multiphasic, the surface outside the contact area(s) is automatically set to ambient conditions (equivalent to setting the effective fluid pressure and effective solute concentration to the `<ambient_pressure>` and `<ambient_concentration>` values, respectively). Ambient conditions may also be associated with a load curve, for example:

```

<ambient_pressure lc="2">1.0</ambient_pressure>
<ambient_concentration lc="3">1.0</ambient_concentration>

```

When performing biphasic-solute-on-rigid or multiphasic-on-rigid contact, a two-pass analysis should not be used; the rigid surface should be the secondary surface.

### 3.14.4 Rigid Wall Interfaces

A rigid wall interface is similar to a sliding interface, except that the secondary surface is a rigid wall. The following properties are defined for rigid wall interfaces:

| Parameter | Description                                                          | Default   |
|-----------|----------------------------------------------------------------------|-----------|
| laugon    | Augmented Lagrangian flag                                            | 0 (false) |
| tolerance | augmentation tolerance                                               | 0.01      |
| penalty   | penalty factor                                                       | 1.0       |
| plane     | the plane equation for the rigid wall                                | N/A       |
| offset    | the normal offset of the plane defined by the <i>plane</i> parameter | 0.0       |

The *plane* property defines the reference plane for the rigid wall. Its value is an array of four values:  $a, b, c, d_0$ . The actual plane is defined by specifying the *offset* to the reference plane. The offset parameter takes a loadcurve as an optional attribute to define the motion of the plane as a

function of time. The loadcurve defines the offset  $h$  from the initial position in the direction of the plane normal:

$$ax + by + cz + d(t) = 0, \quad d(t) = d_0 + h(t)$$

So, for example, a rigid wall that initially lies in the xy-coordinate plane and moves in the z-direction would be specified as follows:

```
<plane>0,0,1,0</plane>
<offset lc="1">1.0</offset>
```

### 3.14.5 Tied Interfaces

A tied interface can be used to connect two non-conforming meshes [58]. A tied interface requires the definition of both a primary and a secondary surface. It is assumed that the nodes of the primary surface are connected to the faces of the secondary surface. The following control parameters need to be defined:

|             |                        |
|-------------|------------------------|
| <penalty>   | penalty factor         |
| <tolerance> | augmentation tolerance |

### 3.14.6 Tied Elastic Interfaces

A tied elastic interface is similar to the tied interface. It may be used for tying solid materials. It enforces continuity of the displacement across the interface. The following control parameters need to be defined:

| Parameter           | Description                               | Default   |
|---------------------|-------------------------------------------|-----------|
| penalty             | normal penalty scale factor (1)           | 1.0       |
| pressure_penalty    | pressure penalty scale factor             | 1.0       |
| auto_penalty        | auto-penalty calculation flag (2)         | 0         |
| update_penalty      | update auto-penalty calculation flag (2)  | 0         |
| two_pass            | two-pass flag (3)                         | 0         |
| laugon              | augmented Lagrangian flag (4)             | 0         |
| tolerance           | aug. Lagrangian convergence tolerance (4) | 1.0       |
| gaptol              | tolerance for gap value (4)               | 0.0 (off) |
| symmetric_stiffness | symmetric stiffness matrix flag (7)       | 0         |
| search_tol          | Projection search tolerance (8)           | 0.01      |
| search_radius       | search radius (10)                        | 1.0       |

### 3.14.7 Tied Biphasic Interfaces

A tied biphasic interface is similar to the tied interface. It may be used for tying any combination of solid, biphasic, and rigid materials. It enforces continuity of the fluid pressure across the interface when both materials are biphasic. The following control parameters need to be defined:

| Parameter           | Description                               | Default   |
|---------------------|-------------------------------------------|-----------|
| penalty             | normal penalty scale factor (1)           | 1.0       |
| pressure_penalty    | pressure penalty scale factor             | 1.0       |
| auto_penalty        | auto-penalty calculation flag (2)         | 0         |
| update_penalty      | update auto-penalty calculation flag (2)  | 0         |
| two_pass            | two-pass flag (3)                         | 0         |
| laugon              | augmented Lagrangian flag (4)             | 0         |
| tolerance           | aug. Lagrangian convergence tolerance (4) | 1.0       |
| gaptol              | tolerance for gap value (4)               | 0.0 (off) |
| symmetric_stiffness | symmetric stiffness matrix flag (7)       | 0         |
| search_tol          | Projection search tolerance (8)           | 0.01      |
| search_radius       | search radius (10)                        | 1.0       |

### 3.14.8 Tied Multiphasic Interfaces

A tied multiphasic interface is similar to the tied biphasic interface. It may be used for tying any combination of solid, biphasic, multiphasic and rigid materials. It enforces continuity of the effective fluid pressure and effective solute concentrations across the interface when both materials are biphasic or multiphasic. The following control parameters need to be defined:

| Parameter             | Description                               | Default   |
|-----------------------|-------------------------------------------|-----------|
| penalty               | normal penalty scale factor (1)           | 1.0       |
| pressure_penalty      | pressure penalty scale factor             | 1.0       |
| concentration_penalty | concentration penalty scale factor        | 1.0       |
| auto_penalty          | auto-penalty calculation flag (2)         | 0         |
| update_penalty        | update auto-penalty calculation flag (2)  | 0         |
| two_pass              | two-pass flag (3)                         | 0         |
| laugon                | augmented Lagrangian flag (4)             | 0         |
| tolerance             | aug. Lagrangian convergence tolerance (4) | 1.0       |
| gaptol                | tolerance for gap value (4)               | 0.0 (off) |
| symmetric_stiffness   | symmetric stiffness matrix flag (7)       | 0         |
| search_tol            | Projection search tolerance (8)           | 0.01      |
| search_radius         | search radius (10)                        | 1.0       |

### 3.14.9 Sticky Interfaces

A sticky interface is similar to a tied interface except that it allows for initial separation of the tied surfaces and breaking of the tie after a user-defined normal traction is exceeded. The tie is only applied when the surfaces contact and sustained as long as the normal traction is less than the threshold.

| Parameter        | Description                                                 |
|------------------|-------------------------------------------------------------|
| laugon           | augmentation flag                                           |
| penalty          | penalty factor                                              |
| tolerance        | augmentation tolerance                                      |
| minaug           | minimum number of required augmentations                    |
| maxaug           | maximum number of augmentations                             |
| search_tolerance | tolerance for nodal projection onto secondary surface facet |
| max_traction     | threshold for normal traction (1)                           |
| snap_tol         | minimum distance for tie activation (2)                     |

The contact surfaces are defined as in the *tied* interface (see Section 3.14.5).

*Comments:*

1. The *max\_traction* parameter can be used to break the tied interface after the normal traction exceeds the specified value. Initially, this value is set to zero, in which case FEBio will ignore this value and the tie cannot be broken.
2. The *snap\_tol* parameter is used in determining the minimum distance that a primary surface node must have approached the secondary surface facet in order to snap onto the secondary surface. The initial value is zero, meaning a node must have penetrated the secondary surface before it will be tied to it.

### 3.14.10 Contact Potential

The contact potential method is based on the formulation by [51]. In this formulation, a repulsive potential is defined between two opposing surfaces. A force is generated that aims to prevent physical contact between the two surfaces. As a result, there will always be a small gap between the contacting surfaces. Note that this is the opposite of what happens for the other sliding contact interfaces, which require physical contact (and even a little bit of penetration) before a contact force can be generated.

The type attribute for this sliding interface is “*contact potential*” and the following parameters can be defined.

| Parameter | Description                                        | Default |
|-----------|----------------------------------------------------|---------|
| kc        | force scale factor (1)                             | 0.0     |
| p         | order of polynomial that defines the potential (2) | 4       |
| R_in      | Inner radius (3)                                   | 1.0     |
| R_out     | Outer radius (3)                                   | 2.0     |
| R0_min    | Minimum separation in reference frame              | 0       |
| w_tol     | threshold for angle criterion                      | 0       |

*Comments:*

1. The *kc* parameter defines the scale factor for the force. This has a similar purpose as the penalty factor in other sliding contact interfaces.
2. The parameter *p* defines the order of the potential. The paper discusses why the value 4 is in a sense an optimal value so it is recommend to use the default value.

3. The potential is divided in an “inner” and “outer” region. If  $r$  is the distance from a point on the primary contact surface to the opposing secondary surface, then the “inner” region is defined by the criterion  $0 < r < R_{in}$  and the “outer” region is defined by  $R_{in} < r < R_{out}$ . The contact force is stronger in the inner region.
4. The  $R0\_min$  parameter defines the minimum distance between two potential contacting points in the reference configuration. In other words, two points whose initial distance was smaller than  $R0\_min$  will not be considered as a valid contact pair. This parameter is important in self-contact to avoid neighboring points from generating contact forces. The default value of 0 indicates that this criterion is not used.
5. The  $w\_tol$  parameter defines a local normal criterion for a potential pair: If the absolute value of the cosine of the angle between the two local normals is **less** than  $w\_tol$ , the two points are **not** considered in contact. In other words, the two normals have to point in the “same direction” in order for the two contacting points to be in contact. This criterion can be useful in cases of self-contact in order to avoid invalid contact pairs. A value of 0 effectively ignores this criterion.

*Example:*

```
<contact type="contact potential" name="FacetOnFacetSliding1" surface_pair="FacetOnFacetSli
  <kc>1e-6</kc>
  <p>4</p>
  <R_in>0.01</R_in>
  <R_out>0.05</R_out>
</contact>
```

## 3.15 Constraints Section

The Constraints section allows the user to enforce additional constraints to the model.

### 3.15.1 Symmetry Plane

A symmetry plane enforces zero solid displacement normal to a user-selected plane. This constraint is prescribed on a deformable surface of the model and is enforced for every node on that surface. It is the user's responsibility to select only planar surfaces.

```
<constraint type="symmetry plane" surface="SymmetryPlane01">
  <laugon>1</laugon>
  <penalty>1e6</penalty>
  <tol>1e-6</tol>
  <minaug>0</minaug>
  <maxaug>50</maxaug>
</constraint>
```

The *surface* (SymmetryPlane01 in this example), is defined in the *Mesh* section. Let  $\mathbf{u}$  denote the nodal displacement vector and let  $\mathbf{n}$  denote the unit normal to the plane; the symmetry plane constraint enforces  $u_n \equiv \mathbf{u} \cdot \mathbf{n} = 0$  using a penalty method, optionally with augmented Lagrangian. The reaction force needed to enforce this constraint is evaluated as  $\varepsilon u_n$ , where  $\varepsilon$  is the user-specified *penalty* parameter (with units of force per length). To use the augmented Lagrangian method, set *laugon* to 1, and the Lagrange multiplier  $\lambda$  will be augmented as  $\lambda \leftarrow \lambda + \varepsilon u_n$ . Augmentations terminate when the relative change in  $\lambda$  is less than the user-specified tolerance *tol*, or when the number of augmentations exceeds *maxaug*.

### 3.15.2 Frictionless Fluid Wall

One of the natural boundary conditions for fluid analyses is to enforce zero fluid velocity normal to a boundary, implying that this boundary is a frictionless fluid wall. In some analyses however, this natural boundary condition may not be enforced sufficiently well. In such cases, use this frictionless fluid wall linear constraint to produce a more strict enforcement of this condition.

```
<constraint type="frictionless fluid wall" name="FrictionlessFluidWall1" surface="Frictionl
  <laugon>0</laugon>
  <tol>0.2</tol>
  <penalty>1</penalty>
  <minaug>0</minaug>
  <maxaug>10</maxaug>
</constraint>
```

In fluid-structure interaction analyses, this constraint may be used in conjunction with the *symmetry plane* described in Section 3.15.1, using a lower value of the *penalty* parameter, as may be necessary in practice.

### 3.15.3 Fixed Normal Displacement

A fixed normal displacement constraint enforces zero solid displacement normal to a user-selected surface. This constraint is prescribed on a deformable surface of the model and is enforced for every node on that surface. It uses the local surface normal in the selected surface's reference configuration. If the selected surface is planar, the functionality of this constraint is equivalent to the Symmetry Plane constraint described in Section 3.15.1.

```
<constraint type="fixed normal displacement" surface="FixedNormalDisplacement01">
  <laugon>0</laugon>
  <penalty>1000</penalty>
  <tol>0.2</tol>
  <minaug>0</minaug>
  <maxaug>10</maxaug>
  <shell_bottom>0</shell_bottom>
</constraint>
```

The *surface* (FixedNormalDisplacement01 in this example), is defined in the *Mesh* section. Let  $\mathbf{u}$  denote the nodal displacement vector and let  $\mathbf{n}$  denote the unit normal at each node of the surface; the fixed normal displacement constraint enforces  $u_n \equiv \mathbf{u} \cdot \mathbf{n} = 0$  using a penalty method, optionally with augmented Lagrangian. The reaction force needed to enforce this constraint is evaluated as  $\varepsilon u_n$ , where  $\varepsilon$  is the user-specified *penalty* parameter (with units of force per length). To use the augmented Lagrangian method, set *laugon* to 1, and the Lagrange multiplier  $\lambda$  will be augmented as  $\lambda \leftarrow \lambda + \varepsilon u_n$ . Augmentations terminate when the relative change in  $\lambda$  is less than the user-specified tolerance *tol*, or when the number of augmentations exceeds *maxaug*.

For example, this constraint may be used on a cylindrical shaft, over the portion of the shaft which is supported by a bearing journal.

### 3.15.4 Normal Fluid Velocity Constraint

A normal fluid velocity constraint forces the fluid velocity to remain normal to the selected surface. It is enforced for every node on that surface.

```
<constraint type="normal fluid velocity" surface="NormalFlowSurface01">
  <laugon>1</laugon>
  <penalty>1e6</penalty>
  <tol>1e-6</tol>
  <minaug>0</minaug>
  <maxaug>50</maxaug>
</constraint>
```

The *surface* (NormalFlowSurface01 in this example), is defined in the *Mesh* section. Let  $\mathbf{v}$  denote the nodal fluid velocity vector and let  $\mathbf{n}$  denote the unit normal at each node; the normal flow constraint enforces  $\mathbf{v} = (\mathbf{v} \cdot \mathbf{n}) \mathbf{n}$ , or equivalently,  $\mathbf{v}_\tau = (\mathbf{I} - \mathbf{n} \otimes \mathbf{n}) \cdot \mathbf{v} = \mathbf{0}$ , using a penalty method, optionally with augmented Lagrangian. The reaction force needed to enforce this constraint is evaluated as  $\varepsilon \mathbf{v}_\tau$ , where  $\varepsilon$  is the user-specified *penalty* parameter (with units of force per velocity). To use the augmented Lagrangian method, set *laugon* to 1, and the vectorial Lagrange multiplier



$\lambda$  will be augmented as  $\lambda \leftarrow \lambda + \varepsilon \mathbf{v}_\tau$ . Augmentations terminate when the relative change in  $\lambda$  is less than the user-specified tolerance *tol*, or when the number of augmentations exceeds *maxaug*.

This constraint should only be used on a surface where the fluid pressure or dilatation has been fixed or prescribed. It is not compatible with boundary conditions that prescribe the fluid velocity. To prescribe a fluid velocity that remains normal to the selected surface, use the surface load *fluid normal velocity* (Section 3.13.2.20).

### 3.15.5 Uniform Fluid Velocity Constraint

When an inlet boundary is subjected to a prescribed fluid pressure, the profile of the inlet fluid velocity remains unconstrained, and this lack of constraint often leads to poor numerical performance due to the production of velocity spikes. To avoid this problem, use this uniform fluid velocity constraint to help stabilize inlet conditions along the direction normal to the boundary. This constraint evaluates the average fluid velocity normal to the selected boundary,  $\bar{v}_n$ , then prescribes a linear constraint at every node such that  $\mathbf{v} \cdot \mathbf{n} = \bar{v}_n$  is enforced. For example,

```
<constraint name="UniformFluidFlow1" surface="UniformFluidFlow1" type="uniform fluid flow">
  <laugon>0</laugon>
  <tol>0.1</tol>
  <penalty>10000</penalty>
  <minaug>0</minaug>
  <maxaug>50</maxaug>
</constraint>
```

If a very strong enforcement of this constraint is desired, increase the *penalty* and/or turn Lagrange augmentation on (*laugon*=1). This constraint is not compatible with boundary conditions that prescribe the normal component of the fluid velocity on that same boundary.

### 3.15.6 The Prestrain Update Rules

The prestrain update rules for are implemented via non-linear constraints in FEBio. This is done because non-linear constraints automatically participate in FEBio's augmentation mechanism, which allows the user to update and rerun the time step. The following section details how to setup the update rules in the FEBio input file and discusses the currently supported rules.

#### 3.15.6.1 Using Update rules

In order to apply an update rule, a constraint definition must be added to the Constraints section of the input file. The type attribute is used to specify which update rule to use.

```
<Constraints>
  <constraint type="[name of update rule]">
    <!-- parameters go here -->
  </constraint>
</Constraints>
```

The following table lists the available update rules.

| type            | description                                 |
|-----------------|---------------------------------------------|
| prestrain       | Eliminate distortion due to incompatibility |
| in-situ stretch | Enforce the given in-situ fiber stretches   |

Below, the supported update rules are presented in more detail. All of them share parameters for controlling the convergence of the update algorithm and these shared parameters are listed in the following table.

| parameter | description                  | initial value     |
|-----------|------------------------------|-------------------|
| update    | update flag (1)              | 1                 |
| tolerance | convergence tolerance        | 0.0               |
| min_iters | minimum number of iterations | 0                 |
| max_iters | maximum number of iterations | -1 (i.e. ignored) |

*Comments:*

1. By specifying a loadcurve for the update flag, the update can be delayed. This can be useful if, for instance, the prestrain is applied incrementally and the update rule should not be applied until the full prestrain field is applied. In that case, specifying a loadcurve for the update flag that is zero while the prestrain is applied, will delay the update process.

### 3.15.6.2 prestrain update rule

The idea behind this rule is to eliminate the distortion induced by the incompatibility of the initial prestrain gradient with the reference geometry. Thus, we retain the original reference geometry at the cost of an altered effective prestrain field. The update rule is given by the following equation.

$$\mathbf{G}^{k+1} = \mathbf{G}^k \cdot \mathbf{F}_c \quad (3.15.1)$$

This update rule does not define any additional parameters aside the ones from above.

*Example:*

```
<constraint type="prestrain">
  <tolerance>0.01</tolerance>
</constraint>
```

### 3.15.6.3 The in-situ stretch update rule

The idea behind this update rule is to enforce the fiber stretch induced by the initial prestrain gradient. As with the in-situ stretch generator option, this rule has an isochoric version and an uniaxial version for the update rule.

$$\mathbf{G}_{iso} = \mathbf{Q} \begin{bmatrix} \lambda^{-1} & & \\ & \lambda^{1/2} & \\ & & \lambda^{1/2} \end{bmatrix} \mathbf{Q}^T, \quad \mathbf{G}_{uni} = \mathbf{Q} \begin{bmatrix} \lambda^{-1} & & \\ & 1 & \\ & & 1 \end{bmatrix} \mathbf{Q}^T \quad (3.15.2)$$

where  $\lambda^2 = \mathbf{a}_0 \cdot \mathbf{C}^k \cdot \mathbf{a}_0$ , is the fiber stretch induced by the distortion.

| parameter | description                             | initial value |
|-----------|-----------------------------------------|---------------|
| isochoric | Choose the isochoric update rule or not | 1(=true)      |

*Example:*

```
<constraint type="in-situ stretch">
  <tolerance>0.01</tolerance>
  <isochoric>1</isochoric>
</constraint>
```

### 3.15.7 Node Distance

A “node distance” constraint can be used to enforce a user-defined distance between two nodes. The target distance can be defined as a distance relative to the initial separation between the two nodes, or as an absolute distance. This constraint takes the following parameters.

| parameter | description                               | initial value     |
|-----------|-------------------------------------------|-------------------|
| laugon    | Turn on augmented Lagrangian flag         | 0(=false)         |
| augtol    | Augmentation tolerance.                   | 0.01              |
| penalty   | Penalty factor for constraint enforcement | 0                 |
| node      | IDs of the two nodes                      | [-1,-1] (invalid) |
| minaug    | minimum nr of augmentations               | 0                 |
| maxaug    | maximum nr of augmentations               | 10                |
| target    | Target distance between two nodes         | 0                 |
| relative  | Use relative or absolute distance         | 1 (=true)         |

The default configuration of this constraint is such that the initial distance between the two nodes is preserved.

*Example:*

This example illustrates how the *node distance* constraint can be used to reduce the distance between two nodes to zero: the target is set to zero and the relative flag is off so that the target is interpreted as an absolute distance.

```
<constraint name="NodeDistance3" type="node distance">
  <laugon>0</laugon>
  <augtol>0.01</augtol>
  <penalty lc="1">1</penalty>
  <node>17,158</node>
  <minaug>0</minaug>
  <maxaug>10</maxaug>
  <target>0</target>
  <relative>0</relative>
</constraint>
```

## 3.16 Discrete Section

This section defines the materials used by the discrete elements and assigns these materials to the discrete elements sets defined in the *Mesh* section. The materials are defined via the *discrete\_material* element and the materials are assigned to discrete element sets using the *discrete* element.

```
<Discrete>
  <discrete_material id="1" type="linear spring">
    <E>1.0</E>
  </discrete_material>
  <discrete_material id="2" type="linear spring">
    <E>2.0</E>
  </discrete_material>
  <discrete dmat="1" discrete_set="springs1"/>
  <discrete dmat="2" discrete_set="springs2">
</Discrete>
```

The *discrete\_material* and the *discrete* elements are defined in more detail below.

### 3.16.1 Discrete Materials

The *discrete\_material* section defines a material that can be assigned to a discrete element set. The *id* attribute defines the material ID and the *type* attribute defines the material type.

#### 3.16.1.1 Linear Spring

The linear spring has a linear force-displacement relation. It requires the E parameter that defines the spring constant.

For example,

```
<discrete_material id="1" type="linear spring">
  <E>3.0</E>
</discrete_material>
```

#### 3.16.1.2 Nonlinear spring

The nonlinear spring allows users to define a custom relation between a measure of stretch and the spring force. The following parameters can be defined.

| parameter | Description                                             |
|-----------|---------------------------------------------------------|
| scale     | Force scale factor.                                     |
| measure   | Measure of stretch. (1)                                 |
| force     | 1D function that defines the stretch-force relationship |

*Comments:*

1. The *measure* parameter can be set to one of the following values:

- (a) **elongation**: Use the spring elongation, the difference between current and original length, as the measure of stretch.
- (b) **strain**: Use the strain, i.e. the ratio of elongation over original length, as the measure of stretch
- (c) **stretch**: Use the ratio of current length over original length as measure of stretch.

Example:

```
<discrete_material id="1" name="set1" type="nonlinear spring">
<scale>1</scale>
<measure>elongation</measure>
<force type="math">
<math>x</math>
</force>
</discrete_material>
```

### 3.16.1.3 Hill

The Hill discrete material defines the following parameters.

| parameter | Description                                     |
|-----------|-------------------------------------------------|
| Vmax      | maximum shortening velocity                     |
| ac        | activation level                                |
| Fmax      | maximum force                                   |
| Lmax      | strain (i.e. stretch - 1) at which Fmax occurs. |
| L0        | initial reference length                        |
| Sv        | max velocity scale                              |
| Ftl       | normalized tension-length curve                 |
| Ftv       | normalized tension-velocity curve               |

The force in the Hill discrete element is the sum of the passive element and the active element.

$$F = F_p + F_a$$

The passive force is given by,

$$F_p = \begin{cases} F_{max} (\exp(Ksh(l-1)/Lmax - 1) / \exp(Ksh - 1)) & , l > 1 \\ 0 & , l \leq 1 \end{cases}$$

where  $l$  is the relative stretch defined by,  $l = l_m/l_0$  and  $l_m$  is the discrete element length and  $l_0$  is either the L0 parameter, or the initial discrete element length (if L0 is set to zero).

The active force is given by,

$$F_a = ac * F_{max} * F_{tl}(l) * F_{tv}(v)$$

Here,  $v$ , a measure of the relative discrete element's growth speed, is defined by,

$$v = v_m / (V_{max} * S_v(ac))$$

where  $v_m$  is the actual discrete element's growth speed.

The properties Sv, Ftl, and Ftv, are optional and will evaluate to 1 if omitted. They can be defined as load curves. See the example below.

*Example:*

```
<discrete_material id="1" name="test" type="Hill">
  <Vmax>1</Vmax>
  <ac>0.1</ac>
  <Fmax>50</Fmax>
  <Ksh>5</Ksh>
  <Lmax>1.5</Lmax>
  <L0>10</L0>
  <Ftl type="point">
    <interpolate>smooth</interpolate>
    <points>
      <pt>0.0, 0</pt>
      <pt>0.1, 0.000258139</pt>
      <pt>0.2, 0.00161616</pt>
      <pt>0.3, 0.00740118</pt>
      <pt>0.4, 0.0272783</pt>
      <pt>0.5, 0.0820396</pt>
      <pt>0.6, 0.201851</pt>
      <pt>0.7, 0.406524</pt>
      <pt>0.8, 0.670275</pt>
      <pt>0.9, 0.904792</pt>
      <pt>1.0, 0.999955</pt>
      <pt>1.1, 0.904792</pt>
      <pt>1.2, 0.670275</pt>
      <pt>1.3, 0.406524</pt>
      <pt>1.4, 0.201851</pt>
      <pt>1.5, 0.0820396</pt>
      <pt>1.6, 0.0272783</pt>
      <pt>1.7, 0.00740118</pt>
      <pt>1.8, 0.00161616</pt>
      <pt>1.9, 0.000258139</pt>
      <pt>2.0, 0</pt>
    </points>
  </Ftl>
</discrete_material>
```

### 3.16.2 Discrete Section

After the discrete materials are defined, the materials are assigned to the discrete element sets that are defined in the *Mesh* section using the *discrete* element. This element requires two attributes:

| attribute    | Description                                              |
|--------------|----------------------------------------------------------|
| dmat         | discrete material ID                                     |
| discrete_set | discrete element set defined in the <i>Mesh</i> section. |

*Example:*

```
<discrete dmat="1" discrete_set="set1"/>
```

### 3.16.3 Rigid Cable

A rigid cable can be used to apply a load to a series of rigid bodies that are connected at fixed points (fixed with respect to the rigid body). The cable runs through these points and the user can prescribe the force at the end of the cable. The rigid cable requires the following parameters.

**force** The magnitude of the force applied at the end of the cable.

**force\_direction** The direction of the force (FEBio will normalize this vector if needed.)

**relative** If set to 1 the coordinates of the fixed points are relative to the rigid body frame of reference, otherwise they are in global coordinates.

**point** For each fixed point, enter the coordinates of the point. This tag requires the rb attribute to denote the rigid body it is attached to.

The following shows an example.

```
<rigid_cable>
  <force lc="1">1000</force>
  <force_direction>0,0,-1</force_direction>
  <relative>1</relative>
  <point rb="1">0,0,0</point>
  <point rb="2">0,0,0</point>
</rigid_cable>
```

This example defines a cable that runs through the centers of mass of two rigid bodies. The force is applied in the -z direction at the end of the cable (i.e. point 2).

### 3.17 Step Section

The analysis can be divided into separate steps, where in each step, different boundary and loading conditions can be applied. Each analysis step is defined via a *step* section. In each step, the following sections can be defined:

- Control
- Initial
- Boundary
- Loads
- Constraints
- Contact
- Rigid

When a boundary condition, (load, constraint, contact, etc.) is applied to a step, then this boundary condition (load, constraint, contact, etc.) is only active during that step. Boundary conditions (loads, constraints, contacts, etc.) that are defined in the main body of the file will persist through all the step.

```
<!-- main body of file -->
...
<Step>
  <step id="1">
    <Control>
      ...
    </Control>
  </step>
  <step id="2">
    <Control>
      ...
    </Control>
  </step>
  ...
</Step>
```



## 3.18 LoadData Section

The *LoadData* section is used to define load controllers. A load controller allows users to manipulate the value of most model parameters as an explicit or implicit function of time. A specific load controller is defined using the *type* attribute.

In order to make a model parameter time-dependent, first define a load controller in the LoadData section. Then, use the *id* attribute assigned to the load controller as the value of the *lc* attribute. For example, consider the following load controller was defined in the LoadData section.

```
<load_controller id="1" type="math">
  <math>2.0*sin(pi*t)</math>
</load_controller>
```

The *id* assigned to this load controller is 1. Then, this load controller can be used, for instance, in a boundary condition.

```
<bc type="prescribed displacement" node_set="set1">
  <dof>x</dof>
  <value lc="1">3.14</value>
  <relative>0</relative>
</bc>
```

The *value* parameter of this boundary condition is now made time dependent. Note that the actual value of this parameter will be the value of the load controller at a particular point in time, multiplied with the value specified by the tag's value (3.14 in the example).

### 3.18.1 The loadcurve controller

A loadcurve controller is defined by the *loadcurve* type. Each loadcurve is defined by repeating the *point* element for all data points:

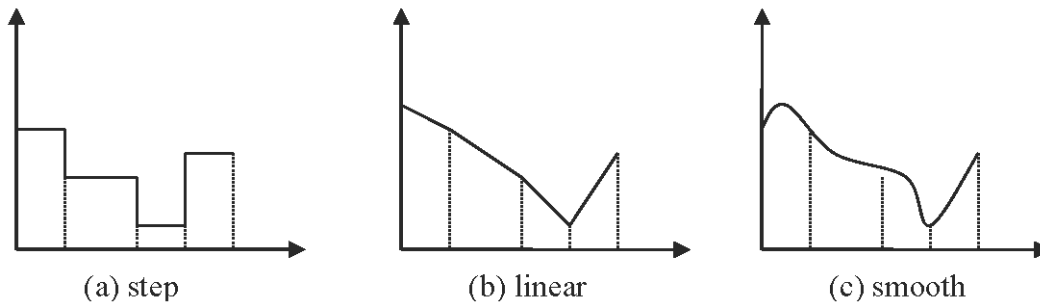
```
<load_controller id="1" type="loadcurve">
  <interpolate>LINEAR</interpolate>
  <extend>CONSTANT</extend>
  <points>
    <point> 0, 0 </point>
    ...
    <point> 1, 1 </point>
  </points>
</loadcurve>
```

For a loadcurve, the *type* is optional, since it is the default controller if the *type* attribute is omitted.

The *id* attribute is the loadcurve number and is used in other sections of the input file as a means to reference this curve.

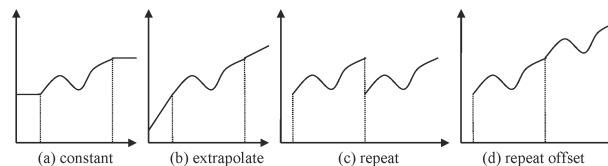
The optional parameters *interpolate* and *extend* define how the value of the loadcurve is interpolated from the data points. The *interpolate* defines the interpolation function and *extend* defines how the values of the loadcurve are determined outside of the interval defined by the first and last data point. The following tables list the possible values. The default values are marked with an asterisk (\*).

| interpolate | Description                                           |
|-------------|-------------------------------------------------------|
| STEP        | Use a step interpolation function                     |
| LINEAR*     | Use a linear interpolation function                   |
| SMOOTH      | The values are interpolated using a cubic polynomial. |



The different values for the *interpolate* parameter of load curves

| Extend        | Description                                                 |
|---------------|-------------------------------------------------------------|
| CONSTANT*     | The value of the curve is the value of the closest endpoint |
| EXTRAPOLATE   | The value is extrapolated linearly from the endpoints       |
| REPEAT        | The curve is repeated                                       |
| REPEAT OFFSET | The curve is repeated but offset from the endpoints         |



The different values for the *extend* parameter of the load curve.

### 3.18.2 The math controller

Use the “math” value for the *type* attribute to define a math controller. This controller allows users to use a mathematical expression that defines a function of time. Use the “t” parameter to reference time.

```
<load_controller id="1" type="math">
  <math>2.0*sin(pi*t)</math>
</load_controller>
```

### 3.18.3 The math-interval controller

The math-interval controller is similar to the math controller, but the math expression is defined only over a time interval. The behavior outside the interval can be specified separately. The following parameters can be defined.

| parameter    | Description                                               |
|--------------|-----------------------------------------------------------|
| interval     | min and max value of the time interval.                   |
| left_extend  | How the function is extrapolated before the min value.(1) |
| right_extend | How the function is extrapolated after the max value. (1) |
| math         | The mathematical expression. Use "t" to reference time.   |

*Comment:*

1. The values for *left\_extend* and *right\_extend* can be set to one of the following values.

| Extend          | Description                                      |
|-----------------|--------------------------------------------------|
| <i>zero</i>     | The value is set to zero.                        |
| <i>constant</i> | The value is the constant value at the end point |
| <i>repeat</i>   | The curve is repeated                            |

*Example:*

```
<load_controller id="1" name="LC1" type="math-interval">
<interval>0,1</interval>
<left_extend>constant</left_extend>
<right_extend>constant</right_extend>
<math>sin(t)</math>
</load_controller>
```

### 3.18.4 The PID controller

The PID controller allows users to create simple control systems where the value of one model parameter is used to control the output of another model parameter. The PID controller calculates the output value as a sum of three terms: a term proportional to the error (i.e. the difference between a user-defined target value and the measurement), a derivative term, and an integral term.

This controller requires the following parameters:

| parameter     | Description                                                                         |
|---------------|-------------------------------------------------------------------------------------|
| <i>var</i>    | specify the model parameter that is used as process variable, i.e. the measurement. |
| <i>target</i> | the target value for the process variable.                                          |
| <i>Kp</i>     | weight factor for the derivative term                                               |
| <i>Kd</i>     | weight factor for the derivative term                                               |
| <i>Ki</i>     | weight factor for the integral term.                                                |

For example:

```
<load_controller id="1" type="PID">
  <var>fem.rigid_body[1].euler.z</var>
  <target>1.5708</target>
  <Kp>5</Kp>
  <Kd>1</Kd>
  <Ki>4</Ki>
</load_controller>
```

## 3.19 Output Section

FEBio usually splits the output in two files: the *logfile*, which contains the same information that was written to the screen during the analysis, and the *plotfile*, which contains the results. The contents of these output files can be customized in the *Output* section.

### 3.19.1 Logfile

The logfile records the same output that is printed to the screen. In addition, the user can request FEBio to output additional data to the logfile. This feature is called *data logging*. To use this feature, simply define the following element in the *Output* section of the input file:

```
<Output>
  <logfile [file="<log file>"]>
    <node_data [attributes]>item list</node_data>
    <element_data [attributes]>item list</element_data>
    <rigid_body_data [attributes]>item list</rigid_body_data>
  </logfile>
</Output>
```

The optional attribute *file* defines the name of the logfile. If omitted, a default name is used that is derived from the FEBio input file. See Section 2.7 for details on default naming conventions for output files.

Additional data is stored to the logfile by adding one or more of the following elements:

**node\_data** request nodal data

**face\_data** request surface data

**element\_data** request element data

**domain\_data** request domain data (i.e. element data integrated over a domain)

**surface\_data** request surface data (i.e. face data evaluated over a surface)

**rigid\_body\_data** request rigid body data

**rigid\_connector\_data** request data on a rigid connector

**model\_data** request data on the model

Each of these data classes takes the following attributes:

**data** an expression defining the data that is to be stored

**name** a descriptive name for the data (optional; default = data expression)

**file** the name of the output file where the data is stored. (optional; default = logfile)

**delim** the delimiter used to separate data in multi-column format (optional; default = space)

**format** an optional format string (optional; default = not used)

The *data* attribute is the most important one and is mandatory. It contains a list of variable names, separated by a semi-colon. The available variable names depend on the data class and are defined below. For example, the data expression:

```
data="x;y;z"
```

will store the variables *x*, *y* and *z* in separate columns. See below for more examples.

The optional *name* attribute is a descriptive name for the data. It is used in the logfile to refer to this data and can be used to quickly find the data record in the logfile. If omitted, the data expression is used as the name.

The *file* attribute defines the name of the output file where the data is to be stored. This attribute is optional and when not specified the data will be stored in the logfile.

The optional *delim* attribute defines the delimiter that is used in multi-column format. As described above, data can be stored in multiple columns and the delimiter is used to separate the columns. The default is a single space.

The optional *format* attribute defines a format string that will be used to format the output. If this attribute is present, the *delim* attribute will be ignored. The format string is composed of literal characters and special formatting characters. The special formatting characters are preceded by the percentage character (%). The following formatting characters are currently defined.

**%i** replace with the index of the corresponding item (i.e. node numbers for node data)

**%g** replace with a data value. Use a %g for each data item.

**%t** insert tab character in output.

**%n** insert newline character in output.

The following example,

```
<node_data data="x;y;z" format='<node id="%i">%g,%g,%g</node>'></node_data>
```

will print the following output (e.g. for node 1).

```
<node id="1">0.1,0.2,0.3</node>
```

Notice the use of the apostrophe (') in the format string. This is necessary in order to include the quotation marks as part of the format string. Also note that each data string will automatically be printed on a new line, so there is no need to end the format string with a newline character.

The value of the data elements is a list of items for which the data is to be stored. For example, for the *node\_data* element the value is a list of nodes, for the *element\_data* element it is a list of FE elements and for the *rigid\_body* element it is a list of rigid bodies. The value may be omitted in which case the data for all items will be stored. For instance, omitting the value for the *node\_data* element will store the data for all nodes.

As stated above, the data is either stored in the logfile or in a separate file. In any case, a record is made in the logfile. When storing the data in the logfile, the following entry will be found in the logfile at the end of each converged timestep for each data element:

```

Data Record #<n>
Step = <time step>
Time = <time value>
Data = <data name>
<actual data goes here>

```

The record number  $n$  corresponds to the  $n$ th data element in the input file. The *Step* value is the current time step. The *Time* value is the current solution time. The *Data* value is the name of the data element as provided by the *name* attribute (or the *data* attribute if *name* is omitted). The actual data immediately follows this record. If multiple column output is used, the columns are separated by the *delim* attribute of the data element.

When storing the data in a separate file, the format is slightly different:

```

Data Record #<n>
Step = <time step>
Time = <time value>
Data = <data name>
File = <file name>

```

The *File* value is the name of the physical file. Note that this is the name to which the time step number is appended. In addition, the physical file that stores the data contains the following header:

```

*Step = <time step>
*Time = <time value>
*Data = <data name>
<actual data goes here>

```

In either case, the actual data is a multi-column list, separated by the delimiter specified with the *delim* attribute (or a space when omitted). The first column always contains the item number. For example, the following data element:

```

<node_data
data="x;y;z" name="nodal coordinates" delim=",">1:4:1</node_data>

```

will result in the following record in the logfile:

```

Data Record #1
Step = 1
Time = 0.1
Data = "nodal coordinates"
1,0.000,0.000,0.000
2,1.000,0.000,0.000
3,1.000,1.000,0.000
4,0.000,1.000,0.000

```

This data record is repeated for each converged time step. Please see [E.2](#) for a list of available log variables.

### 3.19.2 Plotfile

By default, all the results are stored in a binary database, referred to as the *plotfile*. The preferred storage format is the FEBio binary database format (referred to as the *xplt* format)<sup>4</sup>. This section describes how to customize the data that is stored to the plotfile.

To define the contents of the plotfile the *plotfile* element needs to be added in the *Output* section of the FEBio input file.

```
<plotfile [type="febio"] [file="name.xplt"]/>
```

The *type* attribute defines the output format. As of version 4.4, the user can choose between “febio” and “vtk”. When choosing “febio”, which is the default, the results will be stored in the standard binary *xplt* format. When choosing “vtk”, the output is stored as individual VTK (legacy format) files, one for each completed time step. Note that due to limitations of the vtk format, not all plot variables can be exported to VTK file.

The *file* attribute is also optional and allows the user to define the file name of the plotfile. If this attribute is omitted, FEBio will use a default file name for the plotfile.

By default, FEBio will store the most common data variables to the plot file. However, it is advised to always specify the specific contents of the plotfile. This can be done by adding *var* elements in the *plotfile* section as described below.

Plotfile variables are defined using the *var* keyword. This tag takes one attribute, namely the *type* of the variable. The *type* defines the specific output variable that will be stored to the plot file.

```
<var type="name"/>
```

where *name* is the name of the output variable. (For a complete list of supported output variables, please see [E.1](#))

As of FEBio 2.4, additional information can be added in the *type* attribute of the variable definition. The general format is as follows.

```
<var type="name [filter]=alias"/>
```

The filter is defined by appending the name of the plot variable with the filter inside square brackets. The filter can be either an integer, or a string. If the filter is a string, it must be surrounded by single quotes.

```
<var type="name ['filter string']=alias"/>
```

The optional alias can be used to rename the variable. The alias will be used as the name of the plot variable in the plot file, and this is the name that will show up in post-processing software such as FEBio Studio.

Some plot variables use filters to resolve possible ambiguities. For example,

```
<var type="solute concentration['solute1']"/>
```

This example will store the solute concentration of a solute named 'solute1' to the plot file.

In addition, an alias can be used to define a more descriptive name of the plot variable.

---

<sup>4</sup>As of FEBio version 2.0, the LSDYNA database is no longer supported. The FEBio database format is the only format that will be supported from now on.

```
<var type="solute concentration['Na']=Na concentration"/>
```

This variable will store the solute concentration of a solute named 'Na' to the plot file using the name 'Na concentration'.

The following example shows how to export a particular fiber vector from a solid mixture. Since solid mixtures can have several fiber distributions associated, it is necessary to specify which component of the mixture the plot variable refers to.

```
<var type="fiber vector['solid[0]']=v1"/>
<var type="fiber vector['solid[1]']=v2"/>
```

Some plot variables require the specification of a named surface (\* in table below). For example,

```
<var type="fluid surface force" surface="airfoil"/>
```

The named surface should be described in a *Surface* element (Section 3.6.5) within *Mesh* (Section 3.6).

The following example shows a complete definition of the plotfile section and stores nodal displacements and element stresses.

*Example:*

```
<plotfile type="febio">
  <var type="displacement"/>
  <var type="stress"/>
</plotfile>
```



# Chapter 4

## Materials

The following sections describe the material parameters for each of the available constitutive models, along with a short description of each material. A more detailed theoretical description of the constitutive models can be found in the [FEBio Theory Manual](#).

### 4.1 Elastic Solids

This section describes the elastic materials, which are materials defined by a hyperelastic strain-energy function. A distinction will be made between so-called *unconstrained* and *uncoupled* materials. The former describe materials that can undergo volumetric compression. The latter are used for modeling (nearly-) incompressible materials, thus their constitutive models are constrained to produce an isochoric deformation, as long as the user has selected a sufficiently large bulk modulus to enforce that constraint.

#### 4.1.1 Specifying Fiber Orientation or Material Axes

Some of the materials are transversely isotropic, requiring the specification of an initial material direction, which is called a *fiber* direction in FEBio. Other materials are orthotropic, requiring the specification of initial material axes that define the three planes of symmetry for those materials. Only one of these specifications should be provided. By default, material axes are aligned with the global Cartesian basis  $\{e_1, e_2, e_3\}$  and the fiber direction is along  $e_1$ . Local fiber or material axes orientation may be specified in several ways. FEBio gives the option to automatically generate the orientation, based on some user-specified parameters. However, the user can override this feature and specify the fiber or axes directions for each element manually in the *ElementData* section. See Section 3.8 for more details on how to do this.

##### 4.1.1.1 Transversely Isotropic Materials

For transversely isotropic materials fiber orientation is specified with the *fiber* element. This element takes one attribute, namely *type*, which specifies the type of the fiber generator. The possible values are specified in the following table.

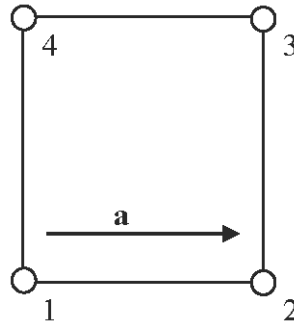
| <b>type Value</b> | <b>Description</b>                                       |
|-------------------|----------------------------------------------------------|
| local             | Use local element numbering (1)                          |
| vector            | Define the fiber orientation as a constant vector (2)    |
| spherical         | specifies a spherical fiber distribution (3)             |
| cylindrical       | specifies a cylindrical fiber distribution (4)           |
| angles            | Specifies the fiber direction using spherical angles (5) |

If the *type* attribute is omitted, the fiber distribution will follow the local element nodes 1 and 2. This would be the same as setting the fiber attribute to *local* and setting the value to "1,2".

*Comments:*

1. In this case, the fiber direction is determined by local element node numbering. The value is interpreted as the local node numbers of the nodes that define the direction of the fiber. The following example defines a local fiber axis by local element nodes 1 and 2. This option is very useful when the local fiber direction can be interpreted as following one of the mesh edges.

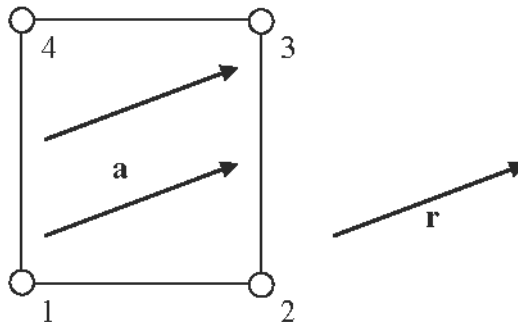
```
<fiber type="local">1,2</fiber>
```



**local fiber direction option**

2. The fiber orientation is specified by a vector. The value is the direction of the fiber. The following defines all element fiber directions in the direction of the vector  $[1,0,0]$ :

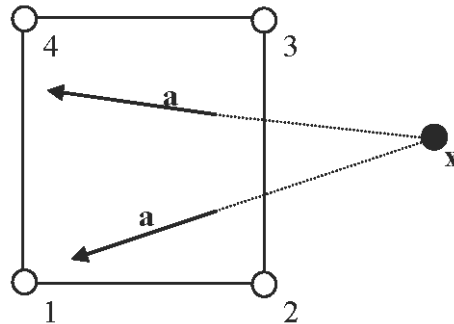
```
<fiber type="vector">1,0,0</fiber>
```



**vector fiber direction option**

3. The fiber orientation is determined by a point in space and the global location of each element integration point. The value is the location of the point. The following example defines a spherical fiber distribution centered at  $[0,0,1]$ :

```
<fiber type="spherical">0,0,1</fiber>
```



**spherical fiber direction option**

4. *cylindrical*: This type generates a fiber distribution that is cylindrical. The following subparameters must be defined.

**center** defines the center of the cylinder

**axis** defines the axis of the cylinder

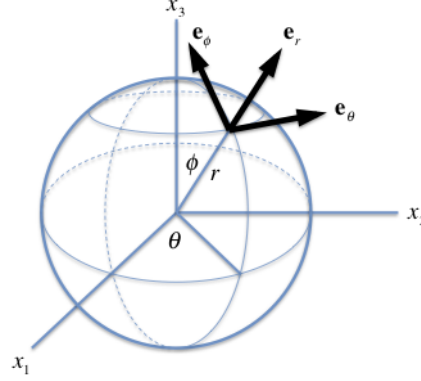
**vector** defines a vector that will be transported around the cylinder

```
<fiber type="cylindrical">
  <center>0,0,0</center>
  <axis>0,0,1</axis>
  <vector>0,1,0</vector>
</fiber>
```

5. *angles*: This type generates a fiber orientation via the specification of spherical angles (azimuth and declination) relative to the local material axes (or global coordinate system, if no local material axes are defined). The following subparameters must be defined.

**theta** azimuth angle (in degrees)

**phi** declination angle (in degrees)



### Spherical angles

The fiber is oriented along

$$\mathbf{e}_r = \cos \theta \sin \phi \mathbf{e}_1 + \sin \theta \sin \phi \mathbf{e}_2 + \cos \phi \mathbf{e}_3, 0 \leq \theta < 2\pi, 0 \leq \phi \leq \pi,$$

where  $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$  are orthonormal vectors representing the local element coordinate system (when specified, Section 4.1.1.2), or global coordinate system.

```
<fiber type="angles">
  <theta>20</theta>
  <phi>90</phi>
</fiber>
```

When specifying a fiber direction  $\mathbf{a}$ , FEBio generates a set of orthogonal material axes as described in Section 4.1.1.2. generated with  $\mathbf{d} = \mathbf{j}$ , or else  $\mathbf{d} = \mathbf{k}$  if  $\mathbf{a}$  is collinear with  $\mathbf{j}$  (where the triple  $i, j, k$  refers to unit vectors defining the global coordinate system, i.e.  $i = (1, 0, 0)$ , etc.). Because of the non-uniqueness of these material axes (only  $\mathbf{e}_1$  is along a uniquely defined direction in a *fiber* element), caution should be used when material axes are compounded, as may occur in nested materials such as solid mixtures described in Sections 4.1.2.16 & 4.1.4.27. To enforce uniqueness, use the *mat\_axis* element instead of the *fiber* element.

#### 4.1.1.2 Orthotropic Materials

For orthotropic materials, the user needs to specify two fiber directions  $\mathbf{a}$  and  $\mathbf{d}$ . From these FEBio will generate an orthonormal set of material axes vectors as follows:

$$\mathbf{e}_1 = \frac{\mathbf{a}}{\|\mathbf{a}\|}, \quad \mathbf{e}_2 = \mathbf{e}_3 \times \mathbf{e}_1, \quad \mathbf{e}_3 = \frac{\mathbf{a} \times \mathbf{d}}{\|\mathbf{a} \times \mathbf{d}\|}.$$

The vectors  $\mathbf{a}$  and  $\mathbf{d}$  are defined using the *mat\_axis* element. This element takes a *type* attribute, which can take on the following values:

| Value  | Description                                                     |
|--------|-----------------------------------------------------------------|
| local  | Use local element numbering (1)                                 |
| vector | Specify the vectors $\mathbf{a}$ and $\mathbf{d}$ directly. (2) |
| angles | Specify the angles $\theta$ and $\varphi$ [deg]. (3)            |

*Comments:*

1. When specifying *local* as the material axis type, the value is interpreted as a list of three local element node numbers. When specifying zero for all three, the default (1,2,4) is used.

```
<mat_axis type="local">0,0,0</mat_axis>
```

2. When using the *vector* type, you need to define the two generator vectors  $a$  and  $d$ . These are specified as child elements of the *mat\_axis* element:

```
<mat_axis type="vector">
  <a>1,0,0</a>
  <d>0,1,0</d>
</mat_axis>
```

3. When using the *angle* type, you need to define the two angles  $\theta$  and  $\varphi$  in degrees. These are specified as child elements of the *mat\_axis* element:

```
<mat_axis type="angles">
  <theta>0</theta>
  <phi>90</phi>
</mat_axis>
```

The material axes  $\{\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{e}_\varphi\}$  are related to the global Cartesian basis (or local element axes)  $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$  via

$$\begin{aligned}\mathbf{e}_r &= \cos \theta \sin \phi \mathbf{e}_1 + \sin \theta \sin \phi \mathbf{e}_2 + \cos \phi \mathbf{e}_3 \\ \mathbf{e}_\theta &= -\sin \theta \mathbf{e}_1 + \cos \theta \mathbf{e}_2 \\ \mathbf{e}_\varphi &= -\cos \theta \cos \phi \mathbf{e}_1 - \sin \theta \cos \phi \mathbf{e}_2 + \sin \phi \mathbf{e}_3\end{aligned}.$$

### 4.1.2 Uncoupled Materials

Uncoupled, nearly-incompressible hyperelastic materials are described by a strain energy function that features an additive decomposition of the hyperelastic strain energy into deviatoric and dilational parts [81]:

$$\Psi(\mathbf{C}) = \tilde{\Psi}(\tilde{\mathbf{C}}) + U(J), \quad (4.1.1)$$

where  $\tilde{\mathbf{C}} = \tilde{\mathbf{F}}^T \cdot \tilde{\mathbf{F}}$  and  $\tilde{\mathbf{F}} = J^{-1/3} \mathbf{F}$  is the deviatoric part of the deformation gradient. The resulting 2<sup>nd</sup> Piola-Kirchhoff stress is given by

$$\mathbf{S} = J^{-2/3} \text{Dev}[\tilde{\mathbf{S}}] + pJ\mathbf{C}^{-1}, \quad (4.1.2)$$

where

$$\tilde{\mathbf{S}} = 2 \frac{\partial \tilde{\Psi}}{\partial \tilde{\mathbf{C}}}, \quad (4.1.3)$$

and

$$p := \frac{dU}{dJ}, \quad (4.1.4)$$

and  $\text{Dev}[\cdot]$  is the deviatoric operator in the material frame.

The corresponding Cauchy stress is given by

$$\boldsymbol{\sigma} = \text{dev}[\tilde{\boldsymbol{\sigma}}] + p\mathbf{I}, \quad (4.1.5)$$

where  $\tilde{\boldsymbol{\sigma}} = J^{-1} \tilde{\mathbf{F}} \cdot \tilde{\mathbf{S}} \cdot \tilde{\mathbf{F}}^T$  and  $\text{dev}[\cdot]$  is the deviatoric operator in the spatial frame.

For these materials, the entire bulk (volumetric) behavior is determined by the function  $U(J)$ , and  $p$  represents the entire hydrostatic stress. The function  $U(J)$  is constructed to have a value of 0 for  $J=1$  and to have a positive value for all other values of  $J > 0$ . The computational literature has employed different functional forms for  $U(J)$ . To compare one's results to literature references that employ a  $U(J)$  that differs from the default FEBio formulation, use one of the “pressure\_model” options listed below.

By default, all of these materials make use of three-field elements (when available for a particular element type), as described by Simo and Taylor [81]. This element uses a trilinear interpolation of the displacement field and piecewise-constant interpolations for the pressure and volume ratio.

The properties that should be specified for all uncoupled materials are:

| Parameter        | Description                               | Default |
|------------------|-------------------------------------------|---------|
| <k>              | Bulk modulus $k$ appearing in $U(J)$ (1)  | 0 [P]   |
| <pressure_model> | Choose from 0 to 3 for various models (1) | 0       |

*Comments:*

1. The pressure models are as follows:

- (a) 0:  $U(J) = \frac{k}{2} (\ln J)^2$  (FEBio default)
- (b) 1:  $U(J) = \frac{k}{4} (J^2 - 2 \ln J - 1)$  (Nike3D's Ogden material)
- (c) 2:  $U(J) = \frac{k}{2} (J - 1)^2$  (ABAQUS)
- (d) 3:  $U(J) = \frac{k}{2} \left( \frac{1}{2} (J^2 - 1) - \ln J \right)$  (Gasser-Ogden-Holzapfel material implementation in ABAQUS)

The uncoupled materials and the associated three-field element are very effective for representing nearly incompressible material behavior. Fully incompressible behavior can be obtained (for all uncoupled materials) using an augmented Lagrangian method. To use the three-field element formulation and augmented Lagrangian method, please see Section 3.7.1, “three-field-solid”. For enforcement of the incompressibility constraint to a user-defined tolerance, the user must define two material parameters for the three-field-solid:

| Parameter | Description                                               | Default |
|-----------|-----------------------------------------------------------|---------|
| <laugon>  | Turn augmented Lagrangian on for this material or off (1) | 0 (off) |
| <atol>    | Augmentation tolerance (2)                                | 0.01    |

1. A value of 1 (one) turns augmentation on, where a value of 0 (zero) turns it off.
2. The augmentation tolerance determines the convergence condition that is used for the augmented Lagrangian method: convergence is reached when the relative ratio of the Lagrange multiplier norm of the previous augmentation  $\|\lambda_k\|$  to the current one  $\|\lambda_{k+1}\|$  is less than the specified value:

$$\left| \frac{\|\lambda_{k+1}\| - \|\lambda_k\|}{\|\lambda_{k+1}\|} \right| < \varepsilon$$

Thus, a value of 0.01 implies that the change in norm between the previous augmentation loop and the current loop is less than 1%.

*Example:*

```
<material id="1" type="Mooney-Rivlin">
  <c1>5</c1>
  <c2>0.4</c2>
  <k>10000</k>
  <laugon>1</laugon> ← turns on augmented Lagrangian iterations
  <atol>0.05</atol> ← sets the augmentation tolerance
</material>
```

#### 4.1.2.1 Arruda-Boyce

This material describes an incompressible Arruda-Boyce model [4]. The following material parameters are required:

|      |                          |     |
|------|--------------------------|-----|
| <mu> | initial modulus          | [P] |
| <N>  | number of links in chain | [ ] |

The uncoupled strain energy function for the Arruda-Boyce material is given by:

$$\Psi = \mu \sum_{i=1}^5 \frac{C_i}{N^{i-1}} \left( \tilde{I}_1^i - 3^i \right) + U(J),$$

where,  $C_1 = \frac{1}{2}$ ,  $C_2 = \frac{1}{20}$ ,  $C_3 = \frac{11}{1050}$ ,  $C_4 = \frac{19}{7000}$ ,  $C_5 = \frac{519}{673750}$  and  $I_1$  the first invariant of the right Cauchy-Green tensor. The volumetric strain function  $U$  is defined as follows,

$$U(J) = \frac{1}{2}k (\ln J)^2.$$

This material model was proposed by Arruda and Boyce [4] and is based on an eight-chain representation of the macromolecular network structure of polymer chains. The strain energy form represents a truncated Taylor series of the inverse Langevin function, which arises in the statistical treatment of non-Gaussian chains. The parameter  $N$  is related to the locking stretch  $\lambda_L$ , the stretch at which the chains reach their full extended state, by  $\lambda_L = \sqrt{N}$ .

*Example:*

```
<material id="1" type="Arruda-Boyce">
  <mu>0.09</mu>
  <N>26.5</N>
  <k>100</k>
</material>
```



#### 4.1.2.2 Ellipsoidal Fiber Distribution Uncoupled

The material type for an ellipsoidal continuous fiber distribution in an uncoupled formulation is “*EFD uncoupled*”. Since fibers can only sustain tension, this material is not stable on its own. It must be combined with a stable uncoupled material that acts as a ground matrix, using a “*uncoupled solid mixture*” container as described in Section 4.1.2.16. The following material parameters need to be defined:

|        |                                            |     |
|--------|--------------------------------------------|-----|
| <beta> | parameters ( $\beta_1, \beta_2, \beta_3$ ) | [ ] |
| <ksi>  | parameters ( $\xi_1, \xi_2, \xi_3$ )       | [P] |

The stress  $\tilde{\sigma}$  for this material is given by [56, 5, 7]:

$$\tilde{\sigma} = \int_0^{2\pi} \int_0^\pi H(\tilde{I}_n - 1) \tilde{\sigma}_n(\mathbf{n}) \sin \varphi d\varphi d\theta.$$

$\tilde{I}_n = \tilde{\lambda}_n^2 = \mathbf{N} \cdot \tilde{\mathbf{C}} \cdot \mathbf{N}$  is the square of the fiber stretch  $\tilde{\lambda}_n$ ,  $\mathbf{N}$  is the unit vector along the fiber direction (in the reference configuration), which in spherical angles is directed along  $(\theta, \varphi)$ ,  $\mathbf{n} = \tilde{\mathbf{F}} \cdot \mathbf{N} / \tilde{\lambda}_n$ , and  $H(\cdot)$  is the unit step function that enforces the tension-only contribution. The fiber stress is determined from a fiber strain energy function in the usual manner,

$$\tilde{\sigma}_n = \frac{2\tilde{I}_n}{J} \frac{\partial \tilde{\Psi}}{\partial \tilde{I}_n} \mathbf{n} \otimes \mathbf{n},$$

where in this material,

$$\tilde{\Psi}(\mathbf{n}, \tilde{I}_n) = \xi(\mathbf{n}) (\tilde{I}_n - 1)^{\beta(\mathbf{n})}.$$

The materials parameters  $\beta$  and  $\xi$  are determined from:

$$\xi(\mathbf{n}) = \left( \frac{\cos^2 \theta \sin^2 \varphi}{\xi_1^2} + \frac{\sin^2 \theta \sin^2 \varphi}{\xi_2^2} + \frac{\cos^2 \varphi}{\xi_3^2} \right)^{-1/2}$$

$$\beta(\mathbf{n}) = \left( \frac{\cos^2 \theta \sin^2 \varphi}{\beta_1^2} + \frac{\sin^2 \theta \sin^2 \varphi}{\beta_2^2} + \frac{\cos^2 \varphi}{\beta_3^2} \right)^{-1/2}.$$

The orientation of the material axis can be defined as explained in detail in Section 4.1.1.

*Example:*

```
<material id="1" type="uncoupled solid mixture">
  <mat_axis type="local">0,0,0</mat_axis>
  <k>1000</k>
  <solid type="Mooney-Rivlin">
    <c1>1</c1>
    <c2>0</c2>
  </solid>
  <solid type="EFD uncoupled">
    <ksi>10, 12, 15</ksi>
    <beta>2.5, 3, 3</beta>
  </solid>
</material>
```

### 4.1.2.3 Ellipsoidal Fiber Distribution Mooney-Rivlin

The material type for a Mooney-Rivlin material with an ellipsoidal continuous fiber distribution is “*EFD Mooney-Rivlin*”. The following material parameters need to be defined:

|        |                                          |     |
|--------|------------------------------------------|-----|
| <c1>   | Mooney-Rivlin parameter c1               | [P] |
| <c2>   | Mooney-Rivlin parameter c2               | [P] |
| <k>    | bulk modulus                             | [P] |
| <beta> | parameters $(\beta_1, \beta_2, \beta_3)$ | [ ] |
| <ksi>  | parameters $(\xi_1, \xi_2, \xi_3)$       | [P] |

The stress  $\tilde{\sigma}$  for this material is given by,

$$\tilde{\sigma} = \tilde{\sigma}_{MR} + \tilde{\sigma}_f .$$

Here,  $\tilde{\sigma}_{MR}$  is the stress from the Mooney-Rivlin basis (Section 4.1.2.9), and  $\tilde{\sigma}_f$  is the stress contribution from the ellipsoidal fiber distribution (Section 4.1.2.2). The orientation of the material axes can be defined as explained in detail in Section 4.1.1.

*Example:*

```
<material id="1" type="EFD Mooney-Rivlin">
  <c1>1</c1>
  <c2>0</c2>
  <beta>4.5,4.5,4.5</beta>
  <ksi>1,1,1</ksi>
  <k>20000</k>
  <mat_axis type="local">0,0,0</mat_axis>
</material>
```

#### 4.1.2.4 Ellipsoidal Fiber Distribution Veronda-Westmann

The material type for a Veronda-Westmann material with an ellipsoidal continuous fiber distribution is “*EFD Veronda-Westmann*”. The following material parameters need to be defined:

|        |                                          |     |
|--------|------------------------------------------|-----|
| <c1>   | First VW coefficient                     | [P] |
| <c2>   | Second VW coefficient                    | [ ] |
| <k>    | Bulk modulus                             | [P] |
| <beta> | parameters $(\beta_1, \beta_2, \beta_3)$ | [ ] |
| <ksi>  | parameters $(\xi_1, \xi_2, \xi_3)$       | [P] |

The stress  $\tilde{\sigma}$  for this material is given by,

$$\tilde{\sigma} = \tilde{\sigma}_{VW} + \tilde{\sigma}_f.$$

Here,  $\tilde{\sigma}_{VW}$  is the stress from the Veronda-Westmann basis (Section 4.1.2.17), and  $\tilde{\sigma}_f$  is the stress contribution from the ellipsoidal fiber distribution (Section 4.1.2.2).

*Example:*

```
<material id="1" type="EFD Veronda-Westmann">
  <c1>1</c1>
  <c2>0.5</c2>
  <k>1000</k>
  <beta>4.5,4.5,4.5</beta>
  <ksi>1,1,1</ksi>
  <mat_axis type="local">0,0,0</mat_axis>
</material>
```

#### 4.1.2.5 Fung Orthotropic

The material type for orthotropic Fung elasticity [37, 36] is “*Fung orthotropic*”. The following material parameters must be defined:

|       |                            |     |
|-------|----------------------------|-----|
| <E1>  | $E_1$ Young's modulus      | [P] |
| <E2>  | $E_2$ Young's modulus      | [P] |
| <E3>  | $E_3$ Young's modulus      | [P] |
| <G12> | $G_{12}$ shear modulus     | [P] |
| <G23> | $G_{23}$ shear modulus     | [P] |
| <G31> | $G_{31}$ shear modulus     | [P] |
| <v12> | $\nu_{12}$ Poisson's ratio | [ ] |
| <v23> | $\nu_{23}$ Poisson's ratio | [ ] |
| <v31> | $\nu_{31}$ Poisson's ratio | [ ] |
| <c>   | $c$ coefficient            | [P] |

The hyperelastic strain energy function is given by [6],

$$\Psi = \frac{1}{2}c \left( e^{\tilde{Q}} - 1 \right) + U(J), \quad (4.1.6)$$

where,

$$\tilde{Q} = c^{-1} \sum_{a=1}^3 \left[ 2\mu_a \mathbf{M}_a : \tilde{\mathbf{E}}^2 + \sum_{b=1}^3 \lambda_{ab} \left( \mathbf{M}_a : \tilde{\mathbf{E}} \right) \left( \mathbf{M}_b : \tilde{\mathbf{E}} \right) \right].$$

Here,  $\tilde{\mathbf{E}} = (\tilde{\mathbf{C}} - \mathbf{I})/2$  and  $\mathbf{M}_a = \mathbf{V}_a \otimes \mathbf{V}_a$  where  $\mathbf{V}_a$  defines the initial direction of material axis  $a$ . See Section 4.1.1.2 on how to define the material axes for orthotropic materials. The Lamé constants  $\mu_a$  ( $a = 1, 2, 3$ ) and  $\lambda_{ab}$  ( $a, b = 1, 2, 3$ ,  $\lambda_{ba} = \lambda_{ab}$ ) are related to Young's moduli  $E_a$ , shear moduli  $G_{ab}$  and Poisson's ratios  $\nu_{ab}$  via

$$= \begin{bmatrix} \lambda_{11} + 2\mu_1 & \lambda_{12} & \lambda_{13} & 0 & 0 & 0 \\ \lambda_{12} & \lambda_{22} + 2\mu_2 & \lambda_{23} & 0 & 0 & 0 \\ \lambda_{13} & \lambda_{23} & \lambda_{33} + 2\mu_3 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{2}(\mu_1 + \mu_2) & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{2}(\mu_2 + \mu_3) & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{2}(\mu_1 + \mu_3) \end{bmatrix}^{-1}.$$

$$= \begin{bmatrix} \frac{1}{E_1} & -\frac{\nu_{12}}{E_1} & -\frac{\nu_{13}}{E_1} & 0 & 0 & 0 \\ -\frac{\nu_{21}}{E_2} & \frac{1}{E_2} & -\frac{\nu_{23}}{E_2} & 0 & 0 & 0 \\ -\frac{\nu_{31}}{E_3} & -\frac{\nu_{32}}{E_3} & \frac{1}{E_3} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{G_{12}} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{G_{23}} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{G_{31}} \end{bmatrix}.$$

The orthotropic Lamé parameters should produce a positive definite stiffness matrix.

*Example:*

```
<material id="3" type="Fung orthotropic">
  <E1>124</E1>
```

```
<E2>124</E2>  
<E3>36</E3>  
<G12>67</G12>  
<G23>40</G23>  
<G31>40</G31>  
<v12>-0.075</v12>  
<v23>0.87</v23>  
<v31>0.26</v31>  
<c>1</c>  
<k>120000</k>  
</material>
```

#### 4.1.2.6 Gent Material

The uncoupled Gent material is defined using the “Gent” type string. It defines the following parameters.

|      |                                                                  |     |
|------|------------------------------------------------------------------|-----|
| <G>  | Shear modulus                                                    | [P] |
| <Jm> | $J_m = I_m - 1$ , with $I_m$ max value for first invariant $I_1$ | [P] |

The strain energy of this material is defined via an uncoupled formulation,

$$\Psi_r = \tilde{\Psi}_r(\tilde{\mathbf{C}}) + U(J)$$

Here, the deviatoric strain energy is given by,

$$\tilde{\Psi} = -\frac{\mu J_m}{2} \ln \left( 1 - \frac{\tilde{I}_1 - 3}{J_m} \right)$$

*Example:*

```
<material id="2" type="Gent">
  <G>3.14</G>
  <Jm>1.5</Jm>
  <k>1e5</k>
</material>
```

#### 4.1.2.7 Uncoupled Holmes-Mow

The material type for the uncoupled Holmes-Mow material is *uncoupled Holmes-Mow*. The following material parameters must be defined:

|        |                                            |     |
|--------|--------------------------------------------|-----|
| <mu>   | Shear modulus $\mu$                        | [P] |
| <beta> | Exponential stiffening coefficient $\beta$ | []  |

This material model uncouples deviatoric and volumetric behaviors,

$$\Psi_r = \tilde{\Psi}_r(\tilde{\mathbf{C}}) + U(J)$$

The deviatoric strain-energy function is given by

$$\tilde{\Psi}_r = \frac{1}{2} \frac{\mu}{\beta} \left( e^{\tilde{Q}} - 1 \right)$$

where  $\mu$  is the shear modulus and

$$\tilde{Q} = \beta \left( \tilde{I}_1 - 3 \right)$$

where  $\beta$  is the exponential stiffening coefficient.

*Example:*

```
<material id="1" name="Material1" type="uncoupled Holmes-Mow">
  <density>1</density>
  <mu>0.5</mu>
  <beta>2</beta>
  <k>5000</k>
</material>
```

#### 4.1.2.8 Holzapfel-Gasser-Ogden

The material type for the uncoupled Holzapfel-Gasser-Ogden material [39] is *Holzapfel-Gasser-Ogden*. The following material parameters must be defined:

|         |                                       |       |
|---------|---------------------------------------|-------|
| <c>     | Shear modulus of ground matrix        | [P]   |
| <k1>    | Fiber modulus                         | [P]   |
| <k2>    | Fiber exponential coefficient         | [P]   |
| <gamma> | Fiber mean orientation angle $\gamma$ | [deg] |
| <kappa> | Fiber dispersion                      | []    |

This material model uncouples deviatoric and volumetric behaviors. The deviatoric strain-energy function is given by:

$$\Psi_r = \tilde{\Psi}_r(\tilde{\mathbf{C}}) + U(J)$$

where

$$\tilde{\Psi}_r = \frac{c}{2} (\tilde{I}_1 - 3) + \frac{k_1}{2k_2} \sum_{\alpha=1}^2 \left( \exp \left( k_2 \langle \tilde{E}_\alpha \rangle^2 \right) - 1 \right)$$

and the default volumetric strain energy function is

$$U(J) = \frac{k}{2} \left( \frac{J^2 - 1}{2} - \ln J \right)$$

The fiber strain is

$$\tilde{E}_\alpha = \kappa (\tilde{I}_1 - 3) + (1 - 3\kappa) (\tilde{I}_{4\alpha} - 1)$$

where  $\tilde{I}_1 = \text{tr } \tilde{\mathbf{C}}$  and  $\tilde{I}_{4\alpha} = \mathbf{a}_{\alpha r} \cdot \tilde{\mathbf{C}} \cdot \mathbf{a}_{\alpha r}$ . The Macaulay brackets around  $\langle \tilde{E}_\alpha \rangle$  indicate that this term is zero when  $\tilde{E}_\alpha < 0$  and equal to  $\tilde{E}_\alpha$  when this strain is positive.

There are two fiber families along the vectors  $\mathbf{a}_{\alpha r}$  ( $\alpha = 1, 2$ ), lying in the  $\{\mathbf{e}_1, \mathbf{e}_2\}$  plane of the local material axes  $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ , making an angle  $\pm\gamma$  with respect to  $\mathbf{e}_1$ . Each fiber family has a dispersion  $\kappa$ , where  $0 \leq \kappa \leq \frac{1}{3}$ . When  $\kappa = 0$  there is no fiber dispersion, implying that all the fibers in that family act along the angle  $\pm\gamma$ ; the value  $\kappa = \frac{1}{3}$  represents an isotropic fiber dispersion. All other intermediate values of  $\kappa$  produce a  $\pi$ -periodic von Mises fiber distribution, as described in [39].  $c$  is the shear modulus of the ground matrix;  $k_1$  is the fiber modulus and  $k_2$  is the exponential coefficient.

*Example:*

```
<material id="2" type="Holzapfel-Gasser-Ogden">
  <c>7.64</c>
  <k1>996.6</k1>
  <k2>524.6</k2>
  <gamma>49.98</gamma>
  <kappa>0.226</kappa>
  <k>1e5</k>
</material>
```

#### 4.1.2.9 Mooney-Rivlin

The material type for uncoupled Mooney-Rivlin materials is *Mooney-Rivlin*. The following material parameters must be defined:

|      |                                      |     |
|------|--------------------------------------|-----|
| <c1> | Coefficient of first invariant term  | [P] |
| <c2> | Coefficient of second invariant term | [P] |

This material model is a hyperelastic Mooney-Rivlin type with uncoupled deviatoric and volumetric behavior. The strain-energy function is given by:

$$\Psi = C_1 \left( \tilde{I}_1 - 3 \right) + C_2 \left( \tilde{I}_2 - 3 \right) + \frac{1}{2} K (\ln J)^2 .$$

$C_1$  and  $C_2$  are the Mooney-Rivlin material coefficients. The variables  $\tilde{I}_1$  and  $\tilde{I}_2$  are the first and second invariants of the deviatoric right Cauchy-Green deformation tensor  $\tilde{\mathbf{C}}$ . The coefficient  $K$  is a bulk modulus-like penalty parameter and  $J$  is the determinant of the deformation gradient tensor. When  $C_2 = 0$ , this model reduces to an uncoupled version of the neo-Hookean constitutive model.

This material model uses a three-field element formulation, interpolating displacements as linear field variables and pressure and volume ratio as piecewise constant on each element [81].

This material model is useful for modeling certain types of isotropic materials that exhibit some limited compressibility, i.e.  $100 < (K/C_1) < 10000$ .

*Example:*

```
<material id="2" type="Mooney-Rivlin">
  <c1>10.0</c1>
  <c2>20.0</c2>
  <k>1000</k>
</material>
```



#### 4.1.2.10 Muscle Material

This material model implements the constitutive model developed by Silvia S. Blemker [23]. The material type for the muscle material is *muscle material*. The model is designed to simulate the passive and active material behavior of skeletal muscle. It defines the following parameters:

|              |                                       |     |
|--------------|---------------------------------------|-----|
| <g1>         | along fiber shear modulus             | [P] |
| <g2>         | cross fiber shear modulus             | [P] |
| <p1>         | exponential stress coefficients       | [ ] |
| <p2>         | fiber uncrimping factor               | [ ] |
| <Lofl>       | optimal fiber length                  | [ ] |
| <smax>       | maximum isometric stress              | [P] |
| <lambda>     | fiber stretch for straightened fibers | [ ] |
| <activation> | activation level                      |     |

The main difference between this material formulation compared to other transversely hyperelastic materials is that it is formulated using a set of new invariants, originally due to Criscione [28], instead of the usual five invariants proposed by A.J.M. Spencer [84]. For this particular material, only two of the five Criscione invariants are used. The strain energy function is defined as follows:

$$\Psi(B_1, B_2, \lambda) = G_1 \tilde{B}_1^2 + G_2 \tilde{B}_2^2 + F_m(\tilde{\lambda}) + U(J).$$

The function  $F_m$  is the strain energy contribution of the muscle fibers. It is defined as follows:

$$\lambda \frac{\partial F_m}{\partial \lambda} = \sigma_{\max} \left( f_m^{\text{passive}}(\lambda) + \alpha f_m^{\text{active}}(\lambda) \right) \frac{\lambda}{\lambda_{ofl}},$$

where,

$$f_m^{\text{passive}}(\lambda) = \begin{cases} 0 & \lambda \leq \lambda_{ofl} \\ P_1 \left( e^{P_2(\lambda/\lambda_{ofl}-1)} - 1 \right) & \lambda_{ofl} < \lambda < \lambda^* \\ P_3 \lambda / \lambda_{ofl} + P_4 & \lambda \geq \lambda^* \end{cases},$$

and

$$f_m^{\text{active}}(\lambda) = \begin{cases} 9(\lambda/\lambda_{ofl} - 0.4)^2 & \lambda \leq 0.6\lambda_{ofl} \\ 9(\lambda/\lambda_{ofl} - 1.6)^2 & \lambda \geq 1.4\lambda_{ofl} \\ 1 - 4(1 - \lambda/\lambda_{ofl})^2 & 0.6\lambda_{ofl} < \lambda < 1.4\lambda_{ofl} \end{cases}.$$

The values  $P_3$  and  $P_4$  are determined by requiring  $C^0$  and  $C^1$  continuity at  $\lambda = \lambda^*$ .

The parameter  $\alpha$  is the activation level and can be specified using the *activation* element. You can specify a loadcurve using the *lc* attribute. The value is interpreted as a scale factor when a loadcurve is defined or as the constant activation level when no loadcurve is defined.

The muscle fiber direction is specified similarly to the transversely isotropic Mooney-Rivlin model.

*Example:*

```
<material id="1" type="muscle material">
  <g1>500</g1>
  <g2>500</g2>
  <p1>0.05</p1>
```

```
<p2>6.6</p2>  
<smax>3e5</smax>  
<Lofl>1.07</Lofl>  
<lambda>1.4</lambda>  
<k>1e6</k>  
<fiber type="vector">1,0,0</fiber>  
</material>
```

#### 4.1.2.11 Ogden

This material describes an incompressible hyperelastic Ogden material [81]. The following material parameters must be defined:

|        |                                                                    |     |
|--------|--------------------------------------------------------------------|-----|
| <c[n]> | Coefficient of $n^{\text{th}}$ term, where n can range from 1 to 6 | [P] |
| <m[n]> | Exponent of $n^{\text{th}}$ term, where n can range from 1 to 6    | [ ] |

The uncoupled hyperelastic strain energy function for this material is given in terms of the eigenvalues of the deformation tensor:

$$\Psi = \sum_{i=1}^N \frac{c_i}{m_i^2} \left( \tilde{\lambda}_1^{m_i} + \tilde{\lambda}_2^{m_i} + \tilde{\lambda}_3^{m_i} - 3 \right) + U(J).$$

Here,  $\tilde{\lambda}_i^2$  are the eigenvalues of  $\tilde{\mathbf{C}}$ ,  $c_i$  and  $m_i$  are material coefficients and  $N$  ranges from 1 to 6. Note that you only have to include the material parameters for the terms you intend to use.

*Example:*

```
<material id="1" type="Ogden">
  <m1>2.4</m1>
  <c1>1</c1>
  <k>100</k>
</material>
```

#### 4.1.2.12 Tendon Material

The material type for the tendon material is *tendon material*. The tendon material is similar to the muscle material [23]. The only difference is the fiber function. For tendon material this is defined as:

$$\lambda \frac{\partial F_t}{\partial \lambda} = \sigma(\lambda),$$

where

$$\sigma(\lambda) = \begin{cases} 0 & \lambda \leq 1 \\ L_1 (e^{L_2(\lambda-1)} - 1) & 1 < \lambda < \lambda^* \\ L_3\lambda + L_4 & \lambda \geq \lambda^* \end{cases}.$$

The parameters  $L_3$  and  $L_4$  are determined by requiring  $C^0$  and  $C^1$  continuity at  $\lambda^*$ .

The material parameters for this material are listed below.

|          |                                       |     |
|----------|---------------------------------------|-----|
| <g1>     | along fiber shear modulus             | [P] |
| <g2>     | cross fiber shear modulus             | [P] |
| <l1>     | exponential stress coefficients       | [P] |
| <l2>     | fiber uncrimping factor               | [ ] |
| <lambda> | fiber stretch for straightened fibers | [ ] |

The tendon fiber direction is specified similarly to the transversely isotropic Mooney-Rivlin model.

*Example:*

```
<material id="1" type="tendon material">
  <g1>5e4</g1>
  <g2>5e4</g2>
  <l1>2.7e6/l1>
  <l2>46.4</l2>
  <lambda>1.03</lambda>
  <k>1e7</k>
  <fiber type="vector">1,0,0</fiber>
</material>
```

#### 4.1.2.13 Tension-Compression Nonlinear Orthotropic

The material type for the tension-compression nonlinear orthotropic material is “*TC nonlinear orthotropic*”. The following material parameters are defined:

|            |                                         |     |
|------------|-----------------------------------------|-----|
| <c1>       | First Mooney-Rivlin material parameter  | [P] |
| <c2>       | Second Mooney-Rivlin material parameter | [P] |
| <beta>     | the $\beta$ parameter (see below)       | [ ] |
| <ksi>      | the $\xi$ parameter (see below)         | [P] |
| <mat_axis> | defines the material axes               |     |

This material is based on the following uncoupled hyperelastic strain energy function [14]:

$$\Psi(\mathbf{C}, \lambda_1, \lambda_2, \lambda_3) = \tilde{\Psi}_{iso}(\tilde{\mathbf{C}}) + \sum_{i=1}^3 \tilde{\Psi}_i^{TC}(\tilde{\lambda}_i) + U(J) .$$

The isotropic strain energy  $\tilde{\Psi}_{iso}$  and the dilatational energy  $U$  are the same as for the Mooney-Rivlin material. The tension-compression term is defined as follows:

$$\tilde{\Psi}_i^{TC}(\tilde{\lambda}_i) = \begin{cases} \xi_i (\tilde{\lambda}_i - 1)^{\beta_i} & \tilde{\lambda}_i > 1 \\ 0 & \tilde{\lambda}_i \leq 1 \end{cases} \quad \xi_i \geq 0 \quad (\text{no sum over } i) .$$

The  $\tilde{\lambda}_i$  parameters are the deviatoric fiber stretches of the local material fibers:

$$\tilde{\lambda}_i = \left( \mathbf{a}_i^0 \cdot \tilde{\mathbf{C}} \cdot \mathbf{a}_i^0 \right)^{1/2} .$$

The local material fibers are defined (in the reference frame) as an orthonormal set of vectors  $\mathbf{a}_i^0$ . See Section 4.1.1 for more information. As with all uncoupled materials, this material uses the three-field element formulation.

A complete example for this material follows.

```
<material id="7" name="cartilage" type="TC nonlinear orthotropic">
  <c1>1.0</c1>
  <c2>0.0</c2>
  <k>100</k>
  <beta>4.3,4.3,4.3</beta>
  <ksi>4525, 4525, 4525</ksi>
  <mat_axis type="local">0,0,0</mat_axis>
</material>
```

#### 4.1.2.14 Transversely Isotropic Mooney-Rivlin

The material type for transversely isotropic Mooney-Rivlin materials is “*trans iso Mooney-Rivlin*”. The following material parameters must be defined:

|                      |                                       |     |
|----------------------|---------------------------------------|-----|
| <c1>                 | Mooney-Rivlin coefficient 1           | [P] |
| <c2>                 | Mooney-Rivlin coefficient 2           | [P] |
| <c3>                 | Exponential stress coefficient        | [P] |
| <c4>                 | Fiber uncrimping coefficient          | [ ] |
| <c5>                 | Modulus of straightened fibers        | [P] |
| <lam_max>            | Fiber stretch for straightened fibers | [ ] |
| <fiber>              | Fiber distribution option             |     |
| <active_contraction> | Optional active contraction           |     |

This constitutive model can be used to represent a material that has a single preferred fiber direction and was developed for application to biological soft tissues [70, 71, 91]. It can be used to model tissues such as tendons, ligaments and muscle. The elastic response of the tissue is assumed to arise from the resistance of the fiber family and an isotropic matrix. It is assumed that the uncoupled strain energy function can be written as follows:

$$\Psi = F_1(\tilde{I}_1, \tilde{I}_2) + F_2(\tilde{\lambda}) + \frac{K}{2} [\ln(J)]^2.$$

Here  $\tilde{I}_1$  and  $\tilde{I}_2$  are the first and second invariants of the deviatoric version of the right Cauchy Green deformation tensor  $\tilde{\mathbf{C}}$  and  $\tilde{\lambda}$  is the deviatoric part of the stretch along the fiber direction ( $\tilde{\lambda}^2 = \mathbf{a}_0 \cdot \tilde{\mathbf{C}} \cdot \mathbf{a}_0$ , where  $\mathbf{a}_0$  is the initial fiber direction), and  $J = \det(\mathbf{F})$  is the Jacobian of the deformation (volume ratio). The function  $F_1$  represents the material response of the isotropic ground substance matrix and is the same as the Mooney-Rivlin form specified above, while  $F_2$  represents the contribution from the fiber family. The strain energy of the fiber family is as follows:

$$F_2(\tilde{\lambda}) = \begin{cases} 0 & \tilde{\lambda} \leq 1 \\ C_3 \left( e^{-C_4} \left( \text{Ei}(C_4 \tilde{\lambda}) - \text{Ei}(C_4) \right) - \ln \tilde{\lambda} \right) & 1 < \tilde{\lambda} < \lambda_m \\ C_5 (\tilde{\lambda} - 1) + C_6 \ln \tilde{\lambda} & \tilde{\lambda} \geq \lambda_m \end{cases}$$

where  $\text{Ei}(\cdot)$  is the exponential integral function. The resulting fiber stress is evaluated from

$$\tilde{\lambda} \frac{\partial F_2}{\partial \tilde{\lambda}} = \begin{cases} 0 & \tilde{\lambda} \leq 1 \\ C_3 \left( e^{C_4(\tilde{\lambda}-1)} - 1 \right) & 1 < \tilde{\lambda} < \lambda_m \\ C_5 \tilde{\lambda} + C_6 & \tilde{\lambda} \geq \lambda_m \end{cases}.$$

Here,  $C_1$  and  $C_2$  are the Mooney-Rivlin material coefficients,  $\lambda_m$  is the stretch at which the fibers are straightened,  $C_3$  scales the exponential stresses,  $C_4$  is the rate of uncrimping of the fibers, and  $C_5$  is the modulus of the straightened fibers.  $C_6$  is determined from the requirement that the stress is continuous at  $\lambda_m$ .

This material model uses a three-field element formulation, interpolating displacements as linear field variables and pressure and volume ratio as piecewise constant on each element [81].

The fiber orientation can be specified as explained in Section 4.1.1. An optional active contraction model can also be defined for this material. See Section 4.1.3 for more details.

*Example:*

This example defines a transversely isotropic material with a Mooney-Rivlin basis. It defines a homogeneous fiber direction and uses the active fiber contraction feature.

```
<material id="3" type="trans iso Mooney-Rivlin">  
  <c1>13.85</c1>  
  <c2>0.0</c2>  
  <c3>2.07</c3>  
  <c4>61.44</c4>  
  <c5>640.7</c5>  
  <k>100.0</k>  
  <lam_max>1.03</lam_max>  
  <fiber type="vector">1,0,0</fiber>  
</material>
```

#### 4.1.2.15 Transversely Isotropic Veronda-Westmann

The material type for transversely isotropic Veronda-Westmann materials is “*trans iso Veronda-Westmann*” [90]. The following material parameters must be defined:

|                      |                                       |     |
|----------------------|---------------------------------------|-----|
| <c1>                 | Veronda-Westmann coefficient 1        | [P] |
| <c2>                 | Veronda-Westmann coefficient 2        | [ ] |
| <c3>                 | Exponential stress coefficient        | [P] |
| <c4>                 | Fiber uncrimping coefficient          | [ ] |
| <c5>                 | Modulus of straightened fibers        | [P] |
| <lam_max>            | Fiber stretch for straightened fibers | [ ] |
| <fiber>              | Fiber distribution option.            |     |
| <active_contraction> | Optional active contraction           |     |

This uncoupled hyperelastic material differs from the Transversely Isotropic Mooney-Rivlin model in that it uses the Veronda-Westmann model for the isotropic matrix. The interpretation of the material parameters, except  $C_1$  and  $C_2$  is identical to this material model.

The fiber distribution option is explained in Section 4.1.1. An optional active contraction model can also be defined for this material. See Section 4.1.3 for more details.

*Example:*

This example defines a transversely isotropic material model with a Veronda-Westmann basis. The fiber direction is implicitly implied as *local*.

```
<material id="3" type="trans iso Veronda-Westmann">
  <c1>13.85</c1>
  <c2>0.0</c2>
  <c3>2.07</c3>
  <c4>61.44</c4>
  <c5>640.7</c5>
  <lam_max>1.03</lam_max>
</material>
```



#### 4.1.2.16 Uncoupled Solid Mixture

The material type for a constrained mixture of uncoupled solids is “*uncoupled solid mixture*”. This material describes a mixture of quasi-incompressible elastic solids. It is a container for any combination of the materials described in Section 4.1.2.

|         |                                  |  |
|---------|----------------------------------|--|
| <solid> | Container tag for solid material |  |
|---------|----------------------------------|--|

The mixture may consist of any number of solids. The stress tensor for the solid mixture is the sum of the stresses for all the solids. The bulk modulus of the uncoupled solid mixture is the sum of the bulk moduli of the individual <solid> materials. A bulk modulus specified outside of the <solid> materials will be ignored.

Material axes may be optionally specified within the <material> level, as well as within each <solid>. Within the <material> level, these represent the local element axes relative to the global coordinate system. Within the <solid>, they represent local material axes relative to the element. If material axes are specified at both levels, they are properly compounded to produce local material axes relative to the global coordinate system. Material axes specified in the <ElementData> section are equivalent to a specification at the <material> level: they correspond to local element axes relative to the global system.

*Example:*

```
<material id="1" type="uncoupled solid mixture">
  <k>30e3</k>
  <mat_axis type="vector">
    <a>1,0,0</a>
    <d>0,1,0</d>
  </mat_axis>
  <solid type="Mooney-Rivlin">
    <c1>2.0</c1>
    <c2>0.0</c2>
  </solid>
  <solid type="EFD uncoupled">
    <mat_axis type="vector">
      <a>0.8660254,0.5,0</a>
      <d>0,0,1</d>
    </mat_axis>
    <ksi>5, 1, 1</ksi>
    <beta>2.5, 3, 3</beta>
  </solid>
  <solid type="EFD uncoupled">
    <mat_axis type="vector">
      <a>0.8660254,-0.5,0</a>
      <d>0,0,1</d>
    </mat_axis>
    <ksi>5, 1, 1</ksi>
    <beta>2.5, 3, 3</beta>
  </solid>
</material>
```

#### 4.1.2.17 Veronda-Westmann

The material type for incompressible Veronda-Westmann materials is *Veronda-Westmann* [86]. The following material parameters must be defined:

|      |                       |     |
|------|-----------------------|-----|
| <c1> | First VW coefficient  | [P] |
| <c2> | Second VW coefficient | [ ] |

This model is similar to the Mooney-Rivlin model in that it also uses an uncoupled deviatoric dilatational strain energy:

$$\Psi = C_1 \left[ e^{(C_2(\tilde{I}_1-3))} - 1 \right] - \frac{C_1 C_2}{2} (\tilde{I}_2 - 3) + U(J) .$$

The dilatational term is identical to the one used in the Mooney-Rivlin model. This model can be used to describe certain types of biological materials that display exponential stiffening with increasing strain. It has been used to describe the response of skin tissue [86].

*Example:*

```
<material id="2" type="Veronda-Westmann">
  <c1>1000.0</c1>
  <c2>2000.0</c2>
  <k>1000</k>
</material>
```

#### 4.1.2.18 Mooney-Rivlin Von Mises Distributed Fibers

##### (Sclera and other thin soft tissues)

Authors: Cécile L.M. Gouget and Michaël J.A. Girard

The material type for a thin material where fiber orientation follows a von Mises distribution is “Mooney-Rivlin von Mises Fibers”. The following parameters must be defined:

|            |                                                                                                                                             |       |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <c1>       | Mooney-Rivlin coefficient 1                                                                                                                 | [P]   |
| <c2>       | Mooney-Rivlin coefficient 2                                                                                                                 | [P]   |
| <c3>       | Exponential stress coefficient                                                                                                              | [P]   |
| <c4>       | Fiber uncrimping coefficient                                                                                                                | [ ]   |
| <c5>       | Modulus of straightened fibers                                                                                                              | [P]   |
| <lam_max>  | Fiber stretch for straightened fibers                                                                                                       | [ ]   |
| <tp>       | Preferred fiber orientation in radian (angle within the plane of the fibers, defined with respect to the first direction given in mat_axis) | [rad] |
| <kf>       | Fiber concentration factor                                                                                                                  | [ ]   |
| <vmc>      | Choice of von Mises distribution                                                                                                            |       |
|            | =1 semi-circular von Mises distribution                                                                                                     |       |
|            | =2 constrained von Mises mixture distribution                                                                                               |       |
| <var_n>    | Exponent (only for vmc=2)                                                                                                                   |       |
| <gip>      | Number of integration points (value is a multiple of 10)                                                                                    |       |
| <mat_axis> | Reference frame of the plane of the fibers                                                                                                  |       |

This constitutive model is designed for thin soft tissues. Fibers are multi-directional: they are distributed within the plane tangent to the tissue surface and they follow a unimodal distribution. It is a constitutive model that is suitable for ocular tissues (e.g. sclera and cornea) but can be used for other thin soft tissues. The proposed strain energy function is as follows:

$$\Psi = F_1(\tilde{I}_1, \tilde{I}_2) + \int_{\theta_P - \pi/2}^{\theta_P + \pi/2} P(\theta) F_2(\tilde{\lambda}[\theta]) d\theta + \frac{K}{2} [\ln(J)]^2,$$

where  $P$  is the 2D unimodal distribution function of the fibers which satisfies the normalization condition:

$$\int_{\theta_P - \pi/2}^{\theta_P + \pi/2} P(\theta) d\theta = 1.$$

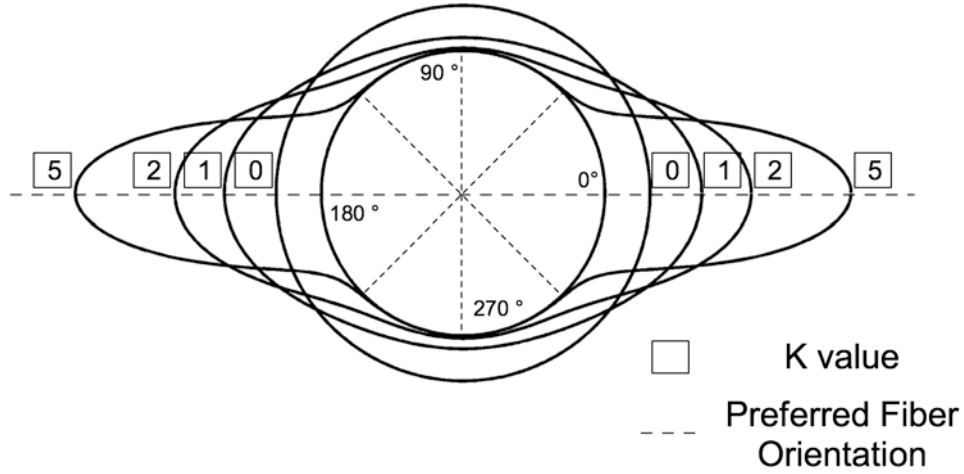
$\theta_P$  is the preferred fiber orientation relative to a local coordinate system (parameter tp). The functions  $F_1$  and  $F_2$  are described in the “Transversely Isotropic Mooney-Rivlin” model. The user can choose between 2 distribution functions using the parameter vmc.

##### 1. Semi-circular von Mises distribution (vmc = 1) [41]

The semi-circular von Mises distribution is one of the simplest unimodal distribution in circular statistics. It can be expressed as

$$P(\theta) = \frac{1}{\pi I_0(k_f)} \exp[k_f \cos(2(\theta - \theta_p))],$$

where  $I_0$  is the modified Bessel function of the first kind (order 0), and  $k_f$  is the fiber concentration factor.  $k_f$  controls the amount of fibers that are concentrated along the orientation  $\theta_P$  as illustrated in the figure below.



Polar representation of the semi-circular von-Mises distribution describing in-plane collagen fiber alignment. In this case, the preferred fiber orientation  $\theta_P$  is equal to zero degrees. When the fiber concentration factor  $k$  is equal to zero, the collagen fibers have an isotropic distribution in a plane tangent to the scleral wall. As  $k$  increases, the collagen fibers align along the preferred fiber orientation  $\theta_P$ . Note that the distributions were plotted on a circle of unit one to ease visualization.

## 2. Constrained von Mises Mixture Distribution (vmc = 2) [42]

The semi-circular von Mises distribution is ideal for its simplicity but in some instance it fails to accurately describe the isotropic subpopulation of fibers present in thin soft tissues. An improved mathematical description is proposed here as a weighted mixture of the semi-circular uniform distribution and the semi-circular von Mises distribution. It can be expressed as:

$$P(\theta) = \frac{1 - \beta}{\pi} + \frac{\beta}{\pi I_0(k_f)} \exp[k_f \cos(2(\theta - \theta_p))],$$

where  $\beta$  needs to be constrained for uniqueness and stability.  $\beta$  is expressed as:

$$\beta = \left( \frac{I_1(k_f)}{I_0(k_f)} \right)^n,$$

where  $I_1$  is the modified Bessel function of the first kind (order 1), and  $n$  is an exponent to be determined experimentally (parameter  $\text{var}_n$ ).  $n=2$  has been found to be suitable for the sclera based on fiber distribution measurements using small angle light scattering.

### Note about Numerical Integration

All numerical integrations are performed using a 10-point Gaussian quadrature rule. The number of Gaussian integration points can be increased for numerical stability. This is controlled through the parameter *gip*. We recommend to use at least *gip* = 20 ( $2 \times 10$  integration points). Note that the more integration points are used, the slower the model will run. By increasing the number of integration points, one should observe convergence in the numerical accuracy. The parameter *gip* is required to be a multiple of 10.

### Definition of the Fiber Plane

The user must specify two directions to define a local coordinate system of the plane in which the fibers lay. As explained in Section 4.1.1.2, this can be done by defining two directions with

the parameter *mat\_axis*. The first direction (vector *a* of *mat\_axis*) must be the reference direction used to define the angle *tp* ( $\theta_P$ ). The second direction (vector *d* of *mat\_axis*) can be any other direction in the plane of the fibers. Thus, a fiber at angle  $\theta_P$  will be along the vector  $\mathbf{v} = \cos \theta_P \mathbf{e}_1 + \sin \theta_P \mathbf{e}_2$  where  $\mathbf{e}_1 = \mathbf{a} / \|\mathbf{a}\|$ ,  $\mathbf{e}_2 = (\mathbf{e}_3 \times \mathbf{a}) / \|\mathbf{e}_3 \times \mathbf{a}\|$ ,  $\mathbf{e}_3 = (\mathbf{a} \times \mathbf{d}) / \|\mathbf{a} \times \mathbf{d}\|$ , as was defined in Section 4.1.1.2.

#### Note on parameters *kf* and *tp*

The parameters *kf* and *tp* can be specified either in the Material section or in the *ElementData* section with the tag *MRVonMisesParameters*. With this second option the user can define different fiber characteristics for each element separately, which can be useful if fiber orientations or concentrations vary spatially. The parameter *mat\_axis* can also be defined either in the Material section or in the *ElementData* section.

#### Example

This example defines a thin soft tissue material where fibers are distributed according to a constrained von Mises mixture distribution.

```
<material id="1" name="Material 1" type="Mooney-Rivlin von Mises Fibers">
  <c1>10.0</c1>
  <c2>0.0</c2>
  <c3>50.0</c3>
  <c4>5.0</c4>
  <c5>1</c5>
  <lam_max>10</lam_max>
  <k>1000000.0</k>
  <kf>1</kf>
  <vmc>2</vmc>
  <var_n>2.0</var_n>
  <tp>0.0</tp>
  <gipt>40</gipt>
  <mat_axis type="local">4,1,3</mat_axis>
</material>
```

#### 4.1.2.19 Isotropic Lee-Sacks uncoupled

This material implements a Fung-type material, as presented in Kamensy, CMAE 2018. The material formulation is selected by setting the type attribute to "uncoupled isotropic Lee-Sacks". It defines the following parameters:

|                 |                                                        |     |
|-----------------|--------------------------------------------------------|-----|
| <c0>            | shear-modulus of neo-Hookean matrix                    | [P] |
| <c1>            | c1 parameter                                           | [P] |
| <c2>            | c2 parameter                                           | [ ] |
| <tangent_scale> | scale factor for c0 when used in tangent (default = 1) | [ ] |

The strain-energy density function for this material combines a neo-Hookean matrix with a Fung-type exponential term, and is defined as follows.

$$\Psi = \frac{c_0}{2} (I_1 - 3) + \frac{c_1}{2} \left( e^{c_2(I_1-3)^2} - 1 \right)$$

As reported in Kamensky, the exponential term may cause convergence difficulties under certain loading conditions. It was reported that increasing the value of  $c_0$  during the stiffness evaluation may improve convergence. The default value for *tangent\_scale* is 1, and thus tangent scaling is not applied. A tangent scale factor of 20 was used in Kamensky.

*Example:*

```
<material id="1" name="material1" type="uncoupled isotropic Lee-Sacks">
  <k>1000</k>
  <c0>10</c0>
  <c1>0.209</c1>
  <c2>9.046</c2>
</material>
```

#### 4.1.2.20 Yeoh Material

This material implements a Yeoh material, as presented online ([https://en.wikipedia.org/wiki/Yeoh\\_hyperelastic\\_m](https://en.wikipedia.org/wiki/Yeoh_hyperelastic_m)). The material formulation is selected by setting the type attribute to "Yeoh". It defines the following parameters:

|             |                                           |     |
|-------------|-------------------------------------------|-----|
| <c1>...<c6> | Elastic moduli of Yeoh material (up to 6) | [P] |
|-------------|-------------------------------------------|-----|

The strain-energy density function for this material combines a neo-Hookean matrix with a Fung-type exponential term, and is defined as follows.

$$\Psi = \sum_{i=1}^6 c_i \left( \tilde{I}_1 - 3 \right)^i$$

*Example:*

```
<material id="1" name="material1" type="Yeoh">
  <k>100</k>
  <c1>0.75</c0>
  <c1>-0.04</c1>
  <c2>0.08</c2>
</material>
```

### 4.1.3 Fiber Active Contraction

Some of the material models described in Section 4.1.2 which include a fiber model may provide the option to specify an active contraction along that fiber. This section lists the types of fiber active contraction models.

#### 4.1.3.1 Active Contraction

The default active contraction model is “*active\_contraction*”. It is based on the formulation of Guccione et al. [43] and reviewed in the *FEBio Theory Manual*. The active stress is evaluated as  $T^a \mathbf{a} \otimes \mathbf{a}$ , where  $\mathbf{a}$  is the unit vector along the fiber in the current configuration, and

$$T^a = T_{\max} \frac{Ca_0^2}{Ca_0^2 + ECa_{50}^2} C(t)$$

where

$$ECa_{50} = \frac{(Ca_0)_{\max}}{\sqrt{\exp[\beta(\lambda(t)l_r - l_0)] - 1}}$$

In this expression,  $\lambda(t)$  is the fiber stretch ratio at the current time. The activation curve  $C(t)$  is represented by the *ascl* property that takes an optional attribute, *lc*, which defines the loadcurve. The list of parameters required for this model is as follows.

|         |                                                                                |
|---------|--------------------------------------------------------------------------------|
| <ascl>  | Activation scale factor $C(t)$                                                 |
| <ca0>   | Intracellular calcium concentration $Ca_0$ (default=1)                         |
| <camax> | Maximum peak intracellular calcium concentration $(Ca_0)_{\max}$ (default=ca0) |
| <beta>  | tension-sarcomere length relation constant $\beta$                             |
| <l0>    | No tension sarcomere length $l_0$                                              |
| <refl>  | Unloaded sarcomere length $l_r$                                                |
| <Tmax>  | Isometric tension under maximal activation at camax $T_{\max}$ (default=1)     |

*Example:*

This example defines a transversely isotropic material with a Mooney-Rivlin basis. It defines a homogeneous fiber direction and uses the active fiber contraction feature.

```
<material id="3" type="trans iso Mooney-Rivlin">
  <c1>13.85</c1>
  <c2>0.0</c2>
  <c3>2.07</c3>
  <c4>61.44</c4>
  <c5>640.7</c5>
  <k>100.0</k>
  <lam_max>1.03</lam_max>
  <fiber type="vector">1,0,0</fiber>
  <active_contraction type="active_contraction">
    <ascl lc="1">1</ascl>
    <ca0>4.35</ca0>
    <beta>4.75</beta>
    <l0>1.58</l0>
    <refl>2.04</refl>
  </active_contraction>
</material>
```

### 4.1.3.2 Force-Velocity Active Contraction

This material model was formulated by Estrada et al. [34] and is called “*force-velocity-Estrada*”. It is a modification of the “*active\_contraction*” material based on the formulation of Guccione et al. [43] described in Section 4.1.3.1, based on the fading-memory formulation proposed by Hunter et al. [49]. The active stress is evaluated as  $T^a \mathbf{a} \otimes \mathbf{a}$ , where  $\mathbf{a}$  is the unit vector along the fiber in the current configuration, and

$$T^a = e(t) \underbrace{T_{\max} \frac{Ca_0^2}{Ca_0^2 + ECa_{50}^2}}_{\text{instantaneous length dependence}} \underbrace{\frac{1 + aQ(t, \lambda(t))}{1 - Q(t, \lambda(t))}}_{\text{force-velocity}}$$

where

$$ECa_{50} = \frac{(Ca_0)_{\max}}{\sqrt{\exp[\beta(\lambda(t)l_r - l_0)] - 1}}$$

and

$$Q(t, \lambda(t)) = \sum_{k=1}^3 A_k \int_{-\infty}^t e^{-\alpha_k(t-\tau)} \dot{\lambda}(\tau) d\tau$$

In these expressions,  $\lambda(t)$  is the fiber stretch ratio at the current time. Here,  $T^a$  “is the product of three distinct components: a time-varying normalized activation,  $e(t)$ , that defines the time course of force generation throughout the cardiac cycle; an instantaneous length-dependent term that scales the peak possible isometric tension based on the current fiber stretch; and a force-velocity term that dampens the instantaneous force generation based on the rate of shortening of the fibers” [34]. The activation curve  $e(t)$  is represented by the *ascl* property that takes an optional attribute, *lc*, which defines the time-dependent loadcurve. The list of parameters required for this model is as follows.

|                              |                                                                                  |
|------------------------------|----------------------------------------------------------------------------------|
| <ascl>                       | Activation scale factor $e(t)$                                                   |
| <ca0>                        | Intracellular calcium concentration $Ca_0$ (default=1)                           |
| <camax>                      | Maximum peak intracellular calcium concentration $(Ca_0)_{\max}$ (default=ca0)   |
| <beta>                       | tension-sarcomere length relation constant $\beta$                               |
| <l0>                         | No tension sarcomere length $l_0$                                                |
| <refl>                       | Unloaded sarcomere length $l_r$                                                  |
| <Tmax>                       | Isometric tension under maximal activation at camax $T_{\max}$ (default=1)       |
| <A1>, <A2>, <A3>             | Parameters $A_k$ for $k = 1, 2, 3$                                               |
| <alpha1>, <alpha2>, <alpha3> | Parameters $\alpha_k$ for $k = 1, 2, 3$                                          |
| <a_t>                        | Parameter $a$                                                                    |
| <force_velocity>             | Set to true (1) by default. Set to false (0) to turn off the force-velocity term |

*Example:*

This example defines a transversely isotropic material with a Mooney-Rivlin basis. It defines a homogeneous fiber direction and uses the active fiber contraction feature.

```
<material id="3" type="trans iso Mooney-Rivlin">
  <c1>13.85</c1>
  <c2>0.0</c2>
```



```
<c3>2.07</c3>
<c4>61.44</c4>
<c5>640.7</c5>
<k>100.0</k>
<lam_max>1.03</lam_max>
<fiber type="vector">1,0,0</fiber>
<active_contraction type="force-velocity-Estrada">
  <ascl lc="1">1</ascl>
  <ca0>4.35</ca0>
  <beta>4.75</beta>
  <l0>1.58</l0>
  <refl>1.58</refl>
<Tmax>135.7</Tmax>
<alpha1>30.30303</alpha1>
<A1>50</A1>
<a_t>0.5</a_t>
<force_velocity>1</force_velocity>
</active_contraction>
</material>
```

#### 4.1.4 Unconstrained Materials

Unlike the materials in Section 4.1.2, these materials do not necessarily assume an additive decomposition of the bulk and deviatoric parts of the strain energy or stress. Further, these materials can only be used with the standard displacement-based finite element formulation, rather than the three-field element formulation. They should not be used for nearly-incompressible material behavior due to the potential for element locking.

##### 4.1.4.1 Arruda-Boyce Unconstrained

The material type for an unconstrained Arruda-Boyce material is *Arruda-Boyce unconstrained*. This isotropic Arruda-Boyce [4] 8-chain model has been used in the field of polymer modeling and in cervical biomechanics [77]. It is based on the simple freely-jointed-chain model and can be derived using the Langevin equation of statistical mechanics. The strain energy density for this material takes the form

$$\Psi(I_1, J) = \xi \left( \sqrt{\frac{I_1}{3N}} \beta + \ln \frac{\beta}{\sinh \beta} \right) - \frac{\xi}{3} \sqrt{N} \beta_0 \ln J + \frac{1}{2} \kappa (\ln J^2 - 1 - 2 \ln J) \quad (4.1.7)$$

where  $I_1 = \text{tr } \mathbf{C}$  is the first invariant of  $\mathbf{C}$ ,  $\beta = \mathcal{L}^{-1} \left[ \sqrt{\frac{I_1}{3N}} \right]$  is the inverse Langevin equation, and

$\beta_0 = \mathcal{L}^{-1} \left[ \sqrt{\frac{1}{N}} \right]$ . Here,  $\xi = nk\Theta$  is the initial fiber modulus with  $n, k, \Theta$  respectively representing the number of chains per unit volume, Boltzmann's constant, and the absolute temperature. The parameter  $N = \zeta^2$  is the number of chain segments, and  $\zeta$  is the locking stretch that represents the extensibility of the material.

A Taylor series expansion is used to evaluate the inverse Langevin equation. The parameter  $n_{\text{term}}$  is used to control the number of terms used to evaluate the series;  $n_{\text{term}}$  must be an integer between 3 and 30.

Langevin mechanics describes a stochastic process with non-Gaussian distribution. It will reduce to a Gaussian distribution when the system approaches equilibrium. Thus, when the material is far from its maximum extensibility, i.e., when  $\zeta = \sqrt{N}$  is much larger than the stretch, this model will reduce to a neo-Hookean material.

This material The following parameters must be defined:

|          |                                                                                                       |     |
|----------|-------------------------------------------------------------------------------------------------------|-----|
| <ksi>    | Initial fiber modulus $\xi$                                                                           | [P] |
| <N>      | Number of chain segments $N$                                                                          | [ ] |
| <n_term> | Number of terms in the Taylor series expansion $n_{\text{term}}$ ( $3 \leq n_{\text{term}} \leq 30$ ) | [ ] |
| <kappa>  | Bulk modulus $\kappa$ of ground matrix                                                                | [P] |

*Example:*

```
<material id="1" name="Soft Tissue" type="Arruda-Boyce unconstrained">
<N>5</N>
<ksi>1</ksi>
<n_term>30</n_term>
<kappa>1</kappa>
</material>
```

#### 4.1.4.2 Carter-Hayes

The material type for a Carter-Hayes material is *Carter-Hayes*. The following parameters must be defined:

|         |                                                                                      |                     |
|---------|--------------------------------------------------------------------------------------|---------------------|
| <E0>    | Young's modulus at reference density $E_0$                                           | [P]                 |
| <rho0>  | reference density $\rho_0$                                                           | [M/L <sup>3</sup> ] |
| <gamma> | exponent of solid-bound molecule density for calculation of Young's modulus $\gamma$ | [ ]                 |
| <v>     | Poisson's ratio $\nu$                                                                | [ ]                 |
| <sbm>   | index of solid bound molecule                                                        | [ ]                 |

This model describes an unconstrained neo-Hookean material [24] whose Young's modulus is a power-law function of the referential apparent density  $\rho_r^\sigma$  of a solid-bound molecule. It is derived from the following hyperelastic strain-energy function:

$$\Psi_r = \frac{E_Y}{2(1+\nu)} \left[ \frac{1}{2} (\text{tr } \mathbf{C} - 3) - \ln J + \frac{\nu}{1-2\nu} (\ln J)^2 \right].$$

Here,  $\mathbf{C}$  is the right Cauchy-Green deformation tensor and  $J$  is the determinant of the deformation gradient tensor.

Young's modulus depends on  $\rho_r^\sigma$  according to a power law [25, 26],

$$E_Y = E_0 \left( \frac{\rho_r^\sigma}{\rho_0} \right)^\gamma.$$

This type of material references a solid-bound molecule that belongs to a multiphasic mixture. Therefore this material may only be used as the solid (or a component of the solid) in a multiphasic mixture (Section 4.16). The solid-bound molecule must be defined in the <Globals> section (Section 3.4.3) and must be included in the multiphasic mixture using a <solid\_bound> tag. The parameter *sbm* must refer to the global index of that solid-bound molecule. The value of  $\rho_r^\sigma$  is specified within the <solid\_bound> tag. If a chemical reaction is defined within that multiphasic mixture that alters the value of  $\rho_r^\sigma$ , lower and upper bounds may be specified for this referential density within the <solid\_bound> tag to prevent  $E_Y$  from reducing to zero or achieving excessively elevated values.

*Example:*

```
<material id="1" name="solid matrix" type="multiphasic">
  <phi0>0</phi0>
  <solid_bound sbm="1">
    <rho0>1</rho0>
    <rhomin>0.1</rhomin>
    <rhomax>5</rhomax>
  </solid_bound>
  <fixed_charge_density>0</fixed_charge_density>
  <solid type="Carter-Hayes">
    <sbm>1</sbm>
    <E0>10000</E0>
    <rho0>1</rho0>
```

```

    <gamma>2.0</gamma>
    <v>0</v>
</solid>
<permeability type="perm-const-iso">
    <perm>1</perm>
</permeability>
<osmotic_coefficient type="osm-coef-const">
    <osmcoef>1</osmcoef>
</osmotic_coefficient>
<reaction name="solid remodeling" type="mass-action-forward">
    <Vbar>0</Vbar>
    <vP sbm="1">1</vP>
    <forward_rate type="Huiskes reaction rate">
        <B>1</B>
        <psi0>0.01</psi0>
    </forward_rate>
</reaction>
</material>

```

#### 4.1.4.3 Cell Growth

The material type for cell growth is “*cell growth*” [11]. The following material parameters need to be defined:

|        |                                                                                                                                |                     |
|--------|--------------------------------------------------------------------------------------------------------------------------------|---------------------|
| <phir> | intracellular solid volume fraction in reference (strain-free) configuration, $\varphi_r^s$                                    | [ ]                 |
| <cr>   | intracellular molar content of membrane-impermeant solute (moles per volume of the cell in the reference configuration), $c_r$ | [n/L <sup>3</sup> ] |
| <ce>   | extracellular osmolarity, $c_e$                                                                                                | [n/L <sup>3</sup> ] |

The Cauchy stress for this material is

$$\boldsymbol{\sigma} = -\pi \mathbf{I},$$

where  $\pi$  is the osmotic pressure, given by

$$\pi = RT \left( \frac{c_r}{J - \varphi_r^s} - c_e \right),$$

where  $J = \det \mathbf{F}$  is the relative volume. The values of the universal gas constant  $R$  and absolute temperature  $T$  must be specified as global constants.

Cell growth may be modeled by simply increasing the mass of the intracellular solid matrix and membrane-impermeant solute. This is achieved by allowing the parameters  $\varphi_r^s$  and  $c_r$  to increase over time as a result of growth, by associating them with user-defined load curves. Since cell growth is often accompanied by cell division, and since daughter cells typically achieve the same solid and solute content as their parent, it may be convenient to assume that  $\varphi_r^s$  and  $c_r$  increase proportionally, though this is not an obligatory relationship. To ensure that the initial configuration is a stress-free reference configuration, let  $c_r = (1 - \varphi_r^s) c_e$  in the initial state prior to growth.

*Example (using units of mm, N, s, nmol, K):*

```

<Globals>
  <Constants>
    <T>298</T>
    <R>8.314e-06</R>
    <Fc>0</Fc>
  </Constants>
</Globals>
<Material>
  <material id="1" name="Cell" type="cell growth">
    <phir lc="1">1</phir>
    <cr lc="2">1</cr>
    <ce>300</ce>
  </material>
</Material>
<LoadData>
  <loadcurve id="1" type="smooth">

```

```
<loadpoint>0,0.3</loadpoint>
<loadpoint>1,0.6</loadpoint>
</loadcurve>
<loadcurve id="2" type="smooth">
  <loadpoint>0,210</loadpoint>
  <loadpoint>1,420</loadpoint>
</loadcurve>
</LoadData>
```

#### 4.1.4.4 Kinematic Growth

The material type for kinematic growth is *kinematic growth*. The following parameters must be defined:

|           |                                                                                                       |  |
|-----------|-------------------------------------------------------------------------------------------------------|--|
| <elastic> | Specification of the elastic material undergoing kinematic growth                                     |  |
| <growth>  | Specification of the growth model ( <i>volume growth</i> , <i>area growth</i> , <i>fiber growth</i> ) |  |

Kinematic growth, based on the framework first proposed in [75], uses the multiplicative decomposition of the deformation gradient into

$$\mathbf{F} = \mathbf{F}^e \cdot \mathbf{F}^g,$$

where  $\mathbf{F}^e$  represents the elastic deformation and  $\mathbf{F}^g$  represents the growth. The related Jacobians are

$$J^e = \det \mathbf{F}^e \quad J^g = \det \mathbf{F}^g.$$

A constitutive model for the growth tensor was provided by [63],

$$\mathbf{F}^g = \vartheta^{\text{iso}} \mathbf{I} + (\vartheta^{\text{ani}} - 1) \mathbf{n}_r \otimes \mathbf{n}_r$$

where  $\mathbf{n}_r$  is the referential unit vector for a fiber direction (or normal to an area),  $\vartheta^{\text{iso}}$  represents isotropic growth, and  $\vartheta^{\text{ani}}$  represents anisotropic growth.

For *volume growth*, let  $\vartheta^{\text{iso}} = \vartheta^g$  and  $\vartheta^{\text{ani}} = 1$ , where  $\vartheta^g$  is called the growth *multiplier* in FEBio, which is typically prescribed using a loadcurve. For *area growth* in the plane transverse to the normal direction  $\mathbf{n}$ , let  $\vartheta^{\text{iso}} = \sqrt{\vartheta^g}$  and  $\vartheta^{\text{ani}} = 2 - \sqrt{\vartheta^g}$ . Finally, for *fiber growth*, let  $\vartheta^{\text{iso}} = 1$  and  $\vartheta^{\text{ani}} = \vartheta^g$ . For *general growth* the user specifies the desired values of  $\vartheta^{\text{iso}}$  and  $\vartheta^{\text{ani}}$ .

Internally, the calculations within FEBio require the evaluation of the inverse of  $\mathbf{F}^g$ , which is generally given by

$$(\mathbf{F}^g)^{-1} = \frac{1}{\vartheta^{\text{iso}} + \vartheta^{\text{ani}} - 1} \left( \left( 1 + \frac{\vartheta^{\text{ani}} - 1}{\vartheta^{\text{iso}}} \right) \mathbf{I} - \frac{\vartheta^{\text{ani}} - 1}{\vartheta^{\text{iso}}} \mathbf{n}_r \otimes \mathbf{n}_r \right)$$

The mass density of a growing material is evaluated from  $\rho = \rho_r / J$  where  $\rho_r$  is the (invariant) referential mass density of the growing solid (when there is neither growth nor deformation) and  $J = \det \mathbf{F} = \det \mathbf{F}^e \det \mathbf{F}^g \equiv J^e J^g$ . It follows that the mass density change produced only by the growth is  $\rho J^e \equiv \rho_{rg} = \rho_r / J^g$ , thus  $J^g = \det \mathbf{F}^g$  may be used to evaluate the evolving solid mass density  $\rho_{rg}$  due only to growth. From the general expression for  $\mathbf{F}^g$  above it follows that

$$J^g = \det \mathbf{F}^g = (\vartheta^{\text{iso}})^2 (\vartheta^{\text{iso}} + \vartheta^{\text{ani}} - 1)$$

#### 4.1.4.5 Cubic CLE

The material type for a conewise linear elastic (CLE) material with cubic symmetry is *cubic CLE*. The following parameters must be defined:

|       |                                                            |     |
|-------|------------------------------------------------------------|-----|
| <lp1> | Tensile diagonal first Lamé coefficient $\lambda_{+1}$     | [P] |
| <lm1> | Compressive diagonal first Lamé coefficient $\lambda_{-1}$ | [P] |
| <l2>  | Off-diagonal first Lamé coefficient $\lambda_2$            | [P] |
| <mu>  | Second Lamé coefficient $\mu$                              | [P] |

This bimodular elastic material is specialized from the orthotropic conewise linear elastic material described by Curnier et al. [30], to the case of cubic symmetry. It is derived from the following hyperelastic strain-energy function:

$$\Psi_r = \mu \mathbf{I} : \mathbf{E}^2 + \sum_{a=1}^3 \frac{1}{2} \lambda_1 [\mathbf{A}_a^r : \mathbf{E}] (\mathbf{A}_a^r : \mathbf{E})^2 + \frac{1}{2} \lambda_2 \sum_{\substack{b=1 \\ b \neq a}}^3 (\mathbf{A}_a^r : \mathbf{E}) (\mathbf{A}_b^r : \mathbf{E}) ,$$

where

$$\lambda_1 [\mathbf{A}_a^r : \mathbf{E}] = \begin{cases} \lambda_{+1} & \mathbf{A}_a^r : \mathbf{E} \geq 0 \\ \lambda_{-1} & \mathbf{A}_a^r : \mathbf{E} < 0 \end{cases}$$

Here,  $\mathbf{E}$  is the Lagrangian strain tensor and  $\mathbf{A}_a^r = \mathbf{a}_a^r \otimes \mathbf{a}_a^r$ , where  $\mathbf{a}_a^r$  ( $a = 1, 2, 3$ ) are orthonormal vectors aligned with the material axes. This material response was originally formulated for infinitesimal strain analyses; its behavior under finite strains may not be physically realistic.

*Example:*

```
<material id="1" type="cubic CLE">
  <density>1</density>
  <lp1>13.01</lp1>
  <lm1>0.49</lm1>
  <l2>0.66</l2>
  <mu>0.16</mu>
</material>
```



#### 4.1.4.6 Donnan Equilibrium Swelling

The material type for a Donnan equilibrium swelling pressure is “*Donnan equilibrium*”. The swelling pressure is described by the equations for ideal Donnan equilibrium, assuming that the material is porous, with a charged solid matrix, and the external bathing environment consists of a salt solution of monovalent counter-ions [67, 55]. Since osmotic swelling must be resisted by a solid matrix, this material is not stable on its own. It must be combined with an elastic material that resists the swelling, using a “*solid mixture*” container as described in Section 4.1.4.27. The following material parameters need to be defined:

|         |                                                                                              |                     |
|---------|----------------------------------------------------------------------------------------------|---------------------|
| <phiw0> | gel porosity (fluid volume fraction) in reference (strain-free) configuration, $\varphi_0^w$ | [ ]                 |
| <cF0>   | fixed-charge density in reference (strain-free) configuration, $c_0^F$                       | [n/L <sup>3</sup> ] |
| <bosm>  | external bath osmolarity, $\bar{c}^*$                                                        | [n/L <sup>3</sup> ] |
| <Phi>   | osmotic coefficient, $\Phi$                                                                  | [ ]                 |

The Cauchy stress for this material is the stress from the Donnan equilibrium response [7]:

$$\boldsymbol{\sigma} = -\pi \mathbf{I},$$

where  $\pi$  is the osmotic pressure, given by

$$\pi = RT\Phi \left( \sqrt{(c^F)^2 + (\bar{c}^*)^2} - \bar{c}^* \right),$$

and  $c^F$  is the fixed-charge density in the current configuration, related to the reference configuration via

$$c^F = \frac{\varphi_0^w}{J - 1 + \varphi_0^w} c_0^F,$$

where  $J = \det \mathbf{F}$  is the relative volume. The values of the universal gas constant  $R$  and absolute temperature  $T$  must be specified as global constants. For ideal Donnan law, use  $\Phi = 1$ .

Note that  $c_0^F$  may be negative or positive; the gel porosity is unitless and must be in the range  $0 < \varphi_0^w < 1$ . A self-consistent set of units must be used for this model. For example:

|             | (m, N, s, mol, K)         | (mm, N, s, nmol, K)              |
|-------------|---------------------------|----------------------------------|
| $R$         | 8.314 J/mol·K             | 8.314×10 <sup>-6</sup> mJ/nmol·K |
| $T$         | K                         | K                                |
| $c_0^F$     | Eq/m <sup>3</sup> = mEq/L | nEq/mm <sup>3</sup> = mEq/L      |
| $\bar{c}^*$ | mol/m <sup>3</sup> = mM   | nmol/mm <sup>3</sup> = mM        |
| $\xi_i$     | Pa                        | MPa                              |
| $\pi$       | Pa                        | MPa                              |

Though this material is porous, this is not a full-fledged biphasic material as described in Section 4.14 for example. The behavior described by this material is strictly valid only after the transient response of interstitial fluid and ion fluxes has subsided (thus *Donnan equilibrium*).

Donnan osmotic swelling reduces to zero when either  $c_0^F = 0$  or  $\bar{c}^* \rightarrow \infty$ . Therefore, entering any other values for  $c_0^F$  and  $\bar{c}^*$  at the initial time point of an analysis produces an instantaneous, non-zero swelling pressure. Depending on the magnitude of this pressure relative to the solid

matrix stiffness, the nonlinear analysis may not converge due to this sudden swelling. Therefore, it is recommended to prescribe a load curve for either `<cF0>` or `<bosm>`, to ease into the initial swelling prior to the application of other loading conditions.

*Example (using units of mm, N, s, nmol, K):*

```
<material id="1" type="solid mixture">
  <mat_axis type="local">0,0,0</mat_axis>
  <solid type="Donnan equilibrium">
    <phiw0>0.8</phiw0>
    <cF0 lc="1">1</cF0>
    <bosm>300</bosm>
  </solid>
  <solid type="ellipsoidal fiber distribution">
    <ksi>0.01,0.01,0.01</ksi>
    <beta>3,3,3</beta>
  </solid>
</material>
<LoadData>
  <loadcurve id="1">
    <loadpoint>0,0</loadpoint>
    <loadpoint>1,150</loadpoint>
  </loadcurve>
</LoadData>
<Globals>
  <Constants>
    <R>8.314e-6</R>
    <T>310</T>
  </Constants>
</Globals>
```

#### 4.1.4.7 Ellipsoidal Fiber Distribution

The material type for an ellipsoidal continuous fiber distribution is “*ellipsoidal fiber distribution*”. Since fibers can only sustain tension, this material is not stable on its own. It must be combined with a stable compressible material that acts as a ground matrix, using a “*solid mixture*” container as described in Section 4.1.4.27. The following material parameters need to be defined:

|        |                                            |     |
|--------|--------------------------------------------|-----|
| <beta> | parameters ( $\beta_1, \beta_2, \beta_3$ ) | [ ] |
| <ksi>  | parameters ( $\xi_1, \xi_2, \xi_3$ )       | [P] |

The Cauchy stress for this fibrous material is given by [56, 5, 7]:

$$\boldsymbol{\sigma} = \int_0^{2\pi} \int_0^\pi H(I_n - 1) \sigma_n(\mathbf{n}) \sin \varphi \, d\varphi \, d\theta.$$

Here,  $I_n = \lambda_n^2 = \mathbf{N} \cdot \mathbf{C} \cdot \mathbf{N}$  is the square of the fiber stretch  $\lambda_n$ ,  $\mathbf{N}$  is the unit vector along the fiber direction, in the reference configuration, which in spherical angles is directed along  $(\theta, \varphi)$ ,  $\mathbf{n} = \mathbf{F} \cdot \mathbf{N} / \lambda_n$ , and  $H(\cdot)$  is the unit step function that enforces the tension-only contribution.

The fiber stress is determined from a fiber strain energy function,

$$\sigma_n = \frac{2I_n}{J} \frac{\partial \Psi}{\partial I_n} \mathbf{n} \otimes \mathbf{n},$$

where in this material,

$$\Psi(\mathbf{n}, I_n) = \xi(\mathbf{n}) (I_n - 1)^{\beta(\mathbf{n})}.$$

The materials parameters  $\beta$  and  $\xi$  are assumed to vary ellipsoidally with  $\mathbf{n}$ , according to

$$\xi(\mathbf{n}) = \left( \frac{\cos^2 \theta \sin^2 \varphi}{\xi_1^2} + \frac{\sin^2 \theta \sin^2 \varphi}{\xi_2^2} + \frac{\cos^2 \varphi}{\xi_3^2} \right)^{-1/2}$$

$$\beta(\mathbf{n}) = \left( \frac{\cos^2 \theta \sin^2 \varphi}{\beta_1^2} + \frac{\sin^2 \theta \sin^2 \varphi}{\beta_2^2} + \frac{\cos^2 \varphi}{\beta_3^2} \right)^{-1/2}.$$

The orientation of the material axis can be defined as explained in detail in Section 4.1.1.

*Example:*

```
<material id="1" type="solid mixture">
  <mat_axis type="local">0,0,0</mat_axis>
  <solid type="neo-Hookean">
    <E>1000.0</E>
    <v>0.45</v>
  </solid>
  <solid type="ellipsoidal fiber distribution">
    <ksi>10, 12, 15</ksi>
    <beta>2.5, 3, 3</beta>
  </solid>
</material>
```

#### 4.1.4.8 Ellipsoidal Fiber Distribution Neo-Hookean

The material type for a Neo-Hookean material with an ellipsoidal continuous fiber distribution is “*EFD neo-Hookean*”. The following material parameters need to be defined:

|        |                                          |     |
|--------|------------------------------------------|-----|
| <E>    | Young’s modulus                          | [P] |
| <v>    | Poisson’s ratio                          | [ ] |
| <beta> | parameters $(\beta_1, \beta_2, \beta_3)$ | [ ] |
| <ksi>  | parameters $(\xi_1, \xi_2, \xi_3)$       | [P] |

The Cauchy stress for this material is given by,

$$\boldsymbol{\sigma} = \boldsymbol{\sigma}_{NH} + \boldsymbol{\sigma}_f.$$

Here,  $\boldsymbol{\sigma}_{NH}$  is the stress from the Neo-Hookean basis (Section 4.1.4.20), and  $\boldsymbol{\sigma}_f$  is the stress contribution from the fibers (Section 4.1.4.7).

*Example:*

```
<material id="1" type="EFD neo-Hookean">
  <E>1</E>
  <v>0.3</v>
  <beta>4.5,4.5,4.5</beta>
  <ksi>1,1,1</ksi>
  <mat_axis type="local">0,0,0</mat_axis>
</material>
```

#### 4.1.4.9 Ellipsoidal Fiber Distribution with Donnan Equilibrium Swelling

The material type for a swelling pressure combined with an ellipsoidal continuous fiber distribution is “*EFD Donnan equilibrium*”. The swelling pressure is described by the equations for ideal Donnan equilibrium, assuming that the material is porous, with a charged solid matrix, and the external bathing environment consists of a salt solution of monovalent counter-ions. The following material parameters need to be defined:

|         |                                                                                              |                     |
|---------|----------------------------------------------------------------------------------------------|---------------------|
| <phiw0> | gel porosity (fluid volume fraction) in reference (strain-free) configuration, $\varphi_0^w$ | [ ]                 |
| <cF0>   | fixed-charge density in reference (strain-free) configuration, $c_0^F$                       | [n/L <sup>3</sup> ] |
| <bosm>  | external bath osmolarity, $\bar{c}^*$                                                        | [n/L <sup>3</sup> ] |
| <beta>  | parameters ( $\beta_1, \beta_2, \beta_3$ )                                                   | [ ]                 |
| <ksi>   | parameters ( $\xi_1, \xi_2, \xi_3$ )                                                         | [P]                 |

The Cauchy stress for this material is given by,

$$\boldsymbol{\sigma} = \boldsymbol{\sigma}_{DE} + \boldsymbol{\sigma}_f.$$

$\boldsymbol{\sigma}_f$  is the stress contribution from the fibers, as described in Section 4.1.1.  $\boldsymbol{\sigma}_{DE}$  is the stress from the Donnan equilibrium response, as described in Section 4.1.4.6

*Example (using units of mm, N, s, nmol, K):*

```

<material id="1" type="EFD Donnan equilibrium">
  <phiw0>0.8</phiw0>
  <cF0 lc="1">1</cF0>
  <bosm>300</bosm>
  <beta>3,3,3</beta>
  <ksi>0.01,0.01,0.01</ksi>
  <mat_axis type="local">0,0,0</mat_axis>
</material>
<LoadData>
  <loadcurve id="1">
    <loadpoint>0,0</loadpoint>
    <loadpoint>1,150</loadpoint>
  </loadcurve>
</LoadData>
<Globals>
  <Constants>
    <R>8.314e-6</R>
    <T>310</T>
  </Constants>
</Globals>

```

#### 4.1.4.10 Fung Orthotropic Compressible

The material type for unconstrained orthotropic Fung elasticity [37, 36] is “*Fung-ortho-compressible*”. The following material parameters must be defined:

|       |                            |     |
|-------|----------------------------|-----|
| <E1>  | $E_1$ Young's modulus      | [P] |
| <E2>  | $E_2$ Young's modulus      | [P] |
| <E3>  | $E_3$ Young's modulus      | [P] |
| <G12> | $G_{12}$ shear modulus     | [P] |
| <G23> | $G_{23}$ shear modulus     | [P] |
| <G31> | $G_{31}$ shear modulus     | [P] |
| <v12> | $\nu_{12}$ Poisson's ratio | [ ] |
| <v23> | $\nu_{23}$ Poisson's ratio | [ ] |
| <v31> | $\nu_{31}$ Poisson's ratio | [ ] |
| <c>   | $c$ coefficient            | [P] |
| <k>   | $\kappa$ bulk modulus      | [P] |

The hyperelastic strain energy function is given by [6],

$$\Psi = \frac{1}{2}c(e^Q - 1) + U(J),$$

where,

$$Q = c^{-1} \sum_{a=1}^3 \left[ 2\mu_a \mathbf{M}_a : \mathbf{E}^2 + \sum_{b=1}^3 \lambda_{ab} (\mathbf{M}_a : \mathbf{E}) (\mathbf{M}_b : \mathbf{E}) \right],$$

and

$$U(J) = \frac{\kappa}{2} (\ln J)^2.$$

Here,  $\mathbf{E} = (\mathbf{C} - \mathbf{I})/2$  and  $\mathbf{M}_a = \mathbf{V}_a \otimes \mathbf{V}_a$  where  $\mathbf{V}_a$  defines the initial direction of material axis  $a$ . See Section 4.1.1.2 on how to define the material axes for orthotropic materials. The Lamé constants  $\mu_a$  ( $a = 1, 2, 3$ ) and  $\lambda_{ab}$  ( $a, b = 1, 2, 3$ ,  $\lambda_{ba} = \lambda_{ab}$ ) are related to Young's moduli  $E_a$ , shear moduli  $G_{ab}$  and Poisson's ratios  $\nu_{ab}$  via

$$= \begin{bmatrix} \lambda_{11} + 2\mu_1 & \lambda_{12} & \lambda_{13} & 0 & 0 & 0 \\ \lambda_{12} & \lambda_{22} + 2\mu_2 & \lambda_{23} & 0 & 0 & 0 \\ \lambda_{13} & \lambda_{23} & \lambda_{33} + 2\mu_3 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{2}(\mu_1 + \mu_2) & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{2}(\mu_2 + \mu_3) & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{2}(\mu_1 + \mu_3) \end{bmatrix}^{-1}$$

$$= \begin{bmatrix} \frac{1}{E_1} & -\frac{\nu_{12}}{E_1} & -\frac{\nu_{13}}{E_1} & 0 & 0 & 0 \\ -\frac{\nu_{21}}{E_2} & \frac{1}{E_2} & -\frac{\nu_{23}}{E_2} & 0 & 0 & 0 \\ -\frac{\nu_{31}}{E_3} & -\frac{\nu_{32}}{E_3} & \frac{1}{E_3} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{G_{12}} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{G_{23}} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{G_{31}} \end{bmatrix}$$

The orthotropic Lamé parameters should produce a positive definite stiffness matrix.

*Example:*

```

<material id="3" type="Fung-ortho-compressible">
  <E1>124</E1>
  <E2>124</E2>
  <E3>36</E3>
  <G12>67</G12>
  <G23>40</G23>
  <G31>40</G31>
  <v12>-0.075</v12>
  <v23>0.87</v23>
  <v31>0.26</v31>
  <c>1</c>
  <k>120</k>
</material>

```

#### 4.1.4.11 Gent Compressible

The compressible Gent material is defined using the “*compressible Gent*” type string. It defines the following parameters.

|      |                                                                  |     |
|------|------------------------------------------------------------------|-----|
| <G>  | Shear modulus                                                    | [P] |
| <Jm> | $J_m = I_m - 1$ , with $I_m$ max value for first invariant $I_1$ | [P] |
| <k>  | Bulk modulus                                                     | [P] |

The strain energy of this material is defined as follows,

$$\Psi_r = -\frac{\mu J_m}{2} \ln \left( 1 - \frac{I_1 - 3}{J_m} \right) + \frac{\kappa}{2} \left( \frac{J^2 - 1}{2} - \ln J \right)^4$$

Here,  $\mu$  is the shear modulus.

*Example:*

```

<material id="2" type="compressible Gent">
  <G>3.14</G>
  <Jm>1.5</Jm>
  <k>1e5</k>
</material>

```

#### 4.1.4.12 Isotropic Hencky

The material type for isotropic Hencky hyperelasticity [92] is *isotropic Hencky*. The following material parameters must be defined:

|     |                 |     |
|-----|-----------------|-----|
| <E> | Young's modulus | [P] |
| <v> | Poisson's ratio | [ ] |

This material is an implementation of a hyperelastic constitutive material that reduces to the classical linear elastic material for small strains, but is objective for large deformations and rotations. The hyperelastic strain-energy function is given by:

$$W = \frac{1}{2} \lambda (\text{tr } \mathbf{H})^2 + \mu \mathbf{H} : \mathbf{H}$$

Here,  $\mathbf{H}$  is the right Hencky strain tensor and  $\lambda$  and  $\mu$  are the Lamé parameters, which are related to the more familiar Young's modulus  $E$  and Poisson's ratio  $\nu$  as follows:

$$\lambda = \frac{\nu E}{(1 + \nu)(1 - 2\nu)}, \mu = \frac{E}{2(1 + \nu)}.$$

The Cauchy stress for this material takes the form

$$\boldsymbol{\sigma} = \frac{\lambda}{J} (\mathbf{I} : \mathbf{h}) \mathbf{I} + \frac{2\mu}{J} \mathbf{h}$$

where  $\mathbf{h}$  is the left Hencky strain tensor. In the limit of infinitesimal strains and rotations,  $J \rightarrow 1$  and  $\mathbf{h} \rightarrow \boldsymbol{\varepsilon}$  where  $\boldsymbol{\varepsilon}$  is the infinitesimal strain tensor.

It is often convenient to express the material properties using the bulk modulus  $K$  and shear modulus  $G$ . To convert to Young's modulus and Poisson's ratio, use the following formulas:

$$E = \frac{9KG}{3K + G}, \quad \nu = \frac{3K - 2G}{6K + 2G}.$$

*Example:*

```
<material id="1" type="isotropic Hencky">
  <E>1000.0</E>
  <v>0.45</v>
</material>
```

#### 4.1.4.13 Holmes-Mow

The material type for the Holmes-Mow material [46] is *Holmes-Mow*. This isotropic hyperelastic material has been used to represent the solid matrix of articular cartilage [46, 9] and intervertebral disc [50]. The following material parameters must be defined:

|        |                                    |     |
|--------|------------------------------------|-----|
| <E>    | Young's modulus                    | [P] |
| <v>    | Poisson's ratio                    | [ ] |
| <beta> | Exponential stiffening coefficient | [ ] |

The coupled hyperelastic strain-energy function for this material is given by [46]:

$$W(I_1, I_2, J) = \frac{1}{2}c(e^Q - 1),$$

where  $I_1$  and  $I_2$  are the first and second invariants of the right Cauchy-Green tensor and  $J$  is the jacobian of the deformation gradient. Furthermore,

$$Q = \frac{\beta}{\lambda + 2\mu} [(2\mu - \lambda)(I_1 - 3) + \lambda(I_2 - 3) - (\lambda + 2\mu) \ln J^2]$$

$$c = \frac{\lambda + 2\mu}{2\beta},$$

and  $\lambda$  and  $\mu$  are the Lamé parameters which are related to the more familiar Young's modulus and Poisson's ratio in the usual manner:

$$\lambda = \frac{E\nu}{(1 + \nu)(1 - 2\nu)}$$

$$\mu = \frac{E}{2(1 + \nu)}.$$

*Example:*



```
<material id="3" type="Holmes-Mow">  
  <E>1</E>  
  <v>0.35</v>  
  <beta>0.25</beta>  
</material>
```

#### 4.1.4.14 Holzapfel-Gasser-Ogden Unconstrained

The material type for the unconstrained Holzapfel-Gasser-Ogden material [39] is *HGO unconstrained*. The following material parameters must be defined:

|         |                                       |       |
|---------|---------------------------------------|-------|
| <c>     | Shear modulus of ground matrix        | [P]   |
| <k1>    | Fiber modulus                         | [P]   |
| <k2>    | Fiber exponential coefficient         | [P]   |
| <gamma> | Fiber mean orientation angle $\gamma$ | [deg] |
| <kappa> | Fiber dispersion                      | []    |
| <k>     | Bulk modulus                          | [P]   |

The strain-energy function is given by:

$$\Psi_r = \frac{c}{2} (I_1 - 3) - c \ln J + \frac{k_1}{2k_2} \sum_{\alpha} \left( \exp \left( k_2 \langle E_{\alpha} \rangle^2 \right) - 1 \right) + \frac{K_0}{2} \left( \frac{J^2 - 1}{2} - \ln J \right)$$

The fiber strain is

$$E_{\alpha} = \kappa (I_1 - 3) + (1 - 3\kappa) (I_{4\alpha} - 1)$$

where  $I_1 = \text{tr } \mathbf{C}$  and  $I_{4\alpha} = \mathbf{a}_{\alpha r} \cdot \mathbf{C} \cdot \mathbf{a}_{\alpha r}$ . The Macaulay brackets around  $\langle \tilde{E}_{\alpha} \rangle$  indicate that this term is zero when  $E_{\alpha} < 0$  and equal to  $E_{\alpha}$  when this strain is positive.

There are two fiber families along the vectors  $\mathbf{a}_{\alpha r}$  ( $\alpha = 1, 2$ ), lying in the  $\{\mathbf{e}_1, \mathbf{e}_2\}$  plane of the local material axes  $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ , making an angle  $\pm\gamma$  with respect to  $\mathbf{e}_1$ . Each fiber family has a dispersion  $\kappa$ , where  $0 \leq \kappa \leq \frac{1}{3}$ . When  $\kappa = 0$  there is no fiber dispersion, implying that all the fibers in that family act along the angle  $\pm\gamma$ ; the value  $\kappa = \frac{1}{3}$  represents an isotropic fiber dispersion. All other intermediate values of  $\kappa$  produce a  $\pi$ -periodic von Mises fiber distribution, as described in [39].  $c$  is the shear modulus of the ground matrix;  $k_1$  is the fiber modulus and  $k_2$  is the exponential coefficient.

Unlike the uncoupled Holzapfel-Gasser-Ogden material presented in Section 4.1.2.8, this unconstrained version does not enforce isochoric deformation. This unconstrained model may be used to describe the porous solid matrix of a biphasic or multiphasic tissue model, where pore volume may change in response to influx or efflux of interstitial fluid.

*Example:*

```
<material id="2" type="HGO unconstrained">
  <c>7.64</c>
  <k1>996.6</k1>
  <k2>524.6</k2>
  <gamma>49.98</gamma>
  <kappa>0.226</kappa>
  <k>7.64e3</k>
</material>
```

#### 4.1.4.15 Isotropic Elastic

The material type for isotropic elasticity is *isotropic elastic*<sup>1</sup>. The following material parameters must be defined:

|     |                 |     |
|-----|-----------------|-----|
| <E> | Young's modulus | [P] |
| <v> | Poisson's ratio | [ ] |

This material is an implementation of a hyperelastic constitutive material that reduces to the classical linear elastic material for small strains, but is objective for large deformations and rotations. The hyperelastic strain-energy function is given by:

$$W = \frac{1}{2} \lambda (\text{tr } \mathbf{E})^2 + \mu \mathbf{E} : \mathbf{E}.$$

Here,  $\mathbf{E}$  is the Euler-Lagrange strain tensor and  $\lambda$  and  $\mu$  are the Lamé parameters, which are related to the more familiar Young's modulus  $E$  and Poisson's ratio  $\nu$  as follows:

$$\lambda = \frac{\nu E}{(1 + \nu)(1 - 2\nu)}, \mu = \frac{E}{2(1 + \nu)}.$$

It is often convenient to express the material properties using the bulk modulus  $K$  and shear modulus  $G$ . To convert to Young's modulus and Poisson's ratio, use the following formulas:

$$E = \frac{9KG}{3K + G}, \quad \nu = \frac{3K - 2G}{6K + 2G}.$$

*Remark:* Note that although this material is objective, it is not advised to use this model for large strains since the behavior may be unphysical. For example, it can be shown that for a uniaxial tension the stress grows unbounded and the volume tends to zero for finite strains. Also for large strains, the Young's modulus and Poisson's ratio input values have little relationship to the actual material parameters. Therefore it is advisable to use this material only for small strains and/or large rotations. To represent isotropic elastic materials under large strain and rotation, it is best to use some of the other available nonlinear material models described in this chapter, such as the Holmes-Mow material in Section 4.1.4.13, the neo-Hookean material in Section 4.1.4.20, or the unconstrained Ogden material in Section 4.1.4.23.

*Example:*

```
<material id="1" type="isotropic elastic">
  <E>1000.0</E>
  <v>0.45</v>
</material>
```

<sup>1</sup>This material replaces the now-obsolete *linear elastic* and *St.Venant-Kirchhoff* materials. These materials are still available for backward compatibility although it is recommended to use the *isotropic elastic* material instead.

#### 4.1.4.16 Orthotropic Elastic

The material type for orthotropic elasticity is “*orthotropic elastic*”. The following material parameters must be defined:

|       |                                            |     |
|-------|--------------------------------------------|-----|
| <E1>  | Young's modulus in the x-direction         | [P] |
| <E2>  | Young's modulus in the y-direction         | [P] |
| <E3>  | Young's modulus in the z-direction         | [P] |
| <G12> | Shear modulus in the xy-plane              | [P] |
| <G23> | Shear modulus in the yz-plane              | [P] |
| <G31> | Shear modulus in the xz-plane              | [P] |
| <v12> | Poisson's ratio between x- and y-direction | [ ] |
| <v23> | Poisson's ratio between y- and z-direction | [ ] |
| <v31> | Poisson's ratio between z- and x-direction | [ ] |

The stress-strain relation for this material is given by

$$\begin{bmatrix} E_{11} \\ E_{22} \\ E_{33} \\ 2E_{23} \\ 2E_{31} \\ 2E_{12} \end{bmatrix} = \begin{bmatrix} 1/E_1 & -\nu_{21}/E_2 & -\nu_{31}/E_3 & 0 & 0 & 0 \\ -\nu_{12}/E_1 & 1/E_2 & -\nu_{32}/E_3 & 0 & 0 & 0 \\ -\nu_{13}/E_1 & -\nu_{23}/E_2 & 1/E_3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1/G_{23} & 0 & 0 \\ 0 & 0 & 0 & 0 & 1/G_{31} & 0 \\ 0 & 0 & 0 & 0 & 0 & 1/G_{12} \end{bmatrix} \begin{bmatrix} T_{11} \\ T_{22} \\ T_{33} \\ T_{23} \\ T_{31} \\ T_{12} \end{bmatrix}$$

Material axes may be specified as described in Section 4.1.1.2.

*Example:*

```
<material id="3" type="orthotropic elastic">
  <mat_axis type="vector">
    <a>0.866,0.5,0</a>
    <d>-0.5,0.866,0</d>
  </mat_axis>
  <E1>1</E1>
  <E2>2</E2>
  <E3>3</E3>
  <v12>0</v12>
  <v23>0</v23>
  <v31>0</v31>
  <G12>1</G12>
  <G23>1</G23>
  <G31>1</G31>
</material>
```

#### 4.1.4.17 Orthotropic CLE

The material type for a conewise linear elastic (CLE) material with orthotropic symmetry is *orthotropic CLE*. The following parameters must be defined:

|        |                                                                               |     |
|--------|-------------------------------------------------------------------------------|-----|
| <lp11> | Tensile diagonal first Lamé coefficient along direction 1 $\lambda_{+11}$     | [P] |
| <lp22> | Tensile diagonal first Lamé coefficient along direction 2 $\lambda_{+22}$     | [P] |
| <lp33> | Tensile diagonal first Lamé coefficient along direction 3 $\lambda_{+33}$     | [P] |
| <lm11> | Compressive diagonal first Lamé coefficient along direction 1 $\lambda_{-11}$ | [P] |
| <lm22> | Compressive diagonal first Lamé coefficient along direction 2 $\lambda_{-22}$ | [P] |
| <lm33> | Compressive diagonal first Lamé coefficient along direction 3 $\lambda_{-33}$ | [P] |
| <l12>  | Off-diagonal first Lamé coefficient in 1-2 plane $\lambda_{12}$               | [P] |
| <l23>  | Off-diagonal first Lamé coefficient in 2-3 plane $\lambda_{23}$               | [P] |
| <l31>  | Off-diagonal first Lamé coefficient in 3-1 plane $\lambda_{31}$               | [P] |
| <mu1>  | Second Lamé coefficient along direction 1 $\mu_1$                             | [P] |
| <mu2>  | Second Lamé coefficient along direction 2 $\mu_2$                             | [P] |
| <mu3>  | Second Lamé coefficient along direction 3 $\mu_3$                             | [P] |

This bimodular elastic material is the orthotropic conewise linear elastic material described by Curnier et al. [30]. It is derived from the following hyperelastic strain-energy function:

$$\Psi_r = \sum_{a=1}^3 \mu_a \mathbf{A}_a^r : \mathbf{E}^2 + \frac{1}{2} \lambda_{aa} [\mathbf{A}_a^r : \mathbf{E}] (\mathbf{A}_a^r : \mathbf{E})^2 + \sum_{\substack{b=1 \\ b \neq a}}^3 \frac{1}{2} \lambda_{ab} (\mathbf{A}_a^r : \mathbf{E}) (\mathbf{A}_b^r : \mathbf{E})$$

where  $\lambda_{ba} = \lambda_{ab}$  and

$$\lambda_{aa} [\mathbf{A}_a^r : \mathbf{E}] = \begin{cases} \lambda_{+aa} & \mathbf{A}_a^r : \mathbf{E} \geq 0 \\ \lambda_{-aa} & \mathbf{A}_a^r : \mathbf{E} < 0 \end{cases}, \quad a = 1, 2, 3$$

Here,  $\mathbf{E}$  is the Lagrangian strain tensor and  $\mathbf{A}_a^r = \mathbf{a}_a^r \otimes \mathbf{a}_a^r$ , where  $\mathbf{a}_a^r$  ( $a = 1, 2, 3$ ) are orthonormal vectors aligned with the material axes. This material response was originally formulated for infinitesimal strain analyses; its behavior under finite strains may not be physically realistic.

*Example:*

```
<material id="1" type=" orthotropic CLE">
  <density>1</density>
  <lp11>13.01</lp11>
  <lp22>13.01</lp22>
  <lp33>13.01</lp33>
  <lm11>0.49</lm11>
  <lm22>0.49</lm22>
  <lm33>0.49</lm33>
  <l12>0.66</l12>
  <l23>0.66</l23>
  <l31>0.66</l31>
  <mu1>0.16</mu1>
  <mu2>0.16</mu2>
  <mu3>0.16</mu3>
</material>
```

#### 4.1.4.18 Osmotic Pressure from Virial Expansion

The material type for osmotic pressure from virial expansion is “*osmotic virial expansion*”. The following material parameters need to be defined:

|         |                                                                                                                                           |                                     |
|---------|-------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| <phiw0> | Fluid volume fraction in reference (strain-free) configuration, $\varphi_r^w$                                                             | [ ]                                 |
| <cr>    | Concentration of interstitial solute causing the osmotic pressure (moles per volume of the mixture in the reference configuration), $c_r$ | [n/L <sup>3</sup> ]                 |
| <c1>    | First virial coefficient $c_1$                                                                                                            | [F·L/n]                             |
| <c2>    | Second virial coefficient $c_2$                                                                                                           | [F·L <sup>4</sup> /n <sup>2</sup> ] |
| <c3>    | Third virial coefficient $c_3$                                                                                                            | [F·L <sup>7</sup> /n <sup>3</sup> ] |

The Cauchy stress for this material is

$$\boldsymbol{\sigma} = -\pi \mathbf{I},$$

where  $\pi$  is the osmotic pressure, given by

$$\pi = c_1 c + c_2 c^2 + c_3 c^3, \quad c = \frac{\varphi_r^w c_r}{J - 1 + \varphi_r^w},$$

$c$  is the solute concentration in the current configuration, and  $J = \det \mathbf{F}$  is the relative volume.

This osmotic swelling pressure in the interstitial fluid of a porous material represents an entropic mechanism whose magnitude is independent of the external bath osmolarity. Typically, this material should be used in a solid mixture where the swelling pressure is resisted by a solid matrix in tension.

*Example:*

```

<Material>
  <material id="1" type="solid mixture">
    <solid type="osmotic virial expansion">
      <phiw0>0.8</phiw0>
      <cr lc="1">100</cr>
      <c1>2.436e-6</c1>
      <c2>0</c2>
      <c3>0</c3>
    </solid>
    <solid type="spherical fiber distribution"/>
  </material>
</Material>
<LoadData>
  <loadcurve id="1" type="smooth">
    <loadpoint>0,0</loadpoint>
    <loadpoint>1,1</loadpoint>
  </loadcurve>
</LoadData>

```

#### 4.1.4.19 Natural Neo-Hookean

The material type for a natural Neo-Hookean material is *natural neo-Hookean*. The following parameters must be defined:

|     |                       |     |
|-----|-----------------------|-----|
| <E> | Young's modulus $E$   | [P] |
| <v> | Poisson's ratio $\nu$ | [ ] |

This model describes an unconstrained Neo-Hookean material using the natural strain tensor [29]. It has a non-linear stress-strain behavior, but reduces to the classical linear elasticity model for small strains *and* small rotations. It is derived from the following hyperelastic strain-energy function:

$$W = \frac{\kappa}{2} K_1^2 + \mu K_2^2,$$

where  $\kappa$  is the bulk modulus and  $\mu$  is the shear modulus,

$$\kappa = \frac{E}{3(1-2\nu)}, \quad \mu = \frac{E}{2(1+\nu)}$$

Here,  $K_1$  and  $K_2$  are the first and second invariants of the left natural (Hencky) strain tensor  $\boldsymbol{\eta} = \ln \mathbf{V}$  where  $\mathbf{V}$  is the left stretch tensor in the polar decomposition  $\mathbf{F} = \mathbf{V} \cdot \mathbf{R}$ .

This material model uses a standard displacement-based element formulation, so care must be taken when modeling materials with nearly-incompressible material behavior to avoid element locking. For this case, use the *Mooney-Rivlin* material described in Section 4.1.2.9.

*Example:*

```
<material id="1" type="natural neo-Hookean">
  <E>200e3</E>
  <v>0.3</v>
</material>
```

#### 4.1.4.20 Neo-Hookean

The material type for a Neo-Hookean material is *neo-Hookean*. The following parameters must be defined:

|     |                 |     |
|-----|-----------------|-----|
| <E> | Young's modulus | [P] |
| <v> | Poisson's ratio | [ ] |

This model describes an unconstrained Neo-Hookean material [24]. It has a non-linear stress-strain behavior, but reduces to the classical linear elasticity model for small strains *and* small rotations. It is derived from the following hyperelastic strain-energy function:

$$W = \frac{\mu}{2} (I_1 - 3) - \mu \ln J + \frac{\lambda}{2} (\ln J)^2.$$

Here,  $I_1$  and  $I_2$  are the first and second invariants of the right Cauchy-Green deformation tensor  $\mathbf{C}$  and  $J$  is the determinant of the deformation gradient tensor. The relationship between the material parameters,  $E$ , and  $\nu$ , and the parameters used in the strain-energy function, is as follows.

$$\mu = \frac{E}{2(1 + \nu)}$$

$$\lambda = \frac{\nu E}{(1 + \nu)(1 - 2\nu)}$$

This material model uses a standard displacement-based element formulation, so care must be taken when modeling materials with nearly-incompressible material behavior to avoid element locking. For this case, use the *Mooney-Rivlin* material described in Section 4.1.2.9.

*Example:*

```
<material id="1" type="neo-Hookean">
  <E>1000.0</E>
  <v>0.45</v>
</material>
```



#### 4.1.4.21 Coupled Mooney-Rivlin

The coupled Mooney-Rivlin material describes an unconstrained formulation of the Mooney-Rivlin material. The material type for this material is *coupled Mooney-Rivlin*. The following material parameters can be defined.

|      |                            |     |
|------|----------------------------|-----|
| <c1> | Mooney-Rivlin c1 parameter | [P] |
| <c2> | Mooney-Rivlin c2 parameter | [P] |
| k    | "Bulk-modulus"             | [P] |

The strain-energy function is given by the following expression.

$$W = c_1 (I_1 - 3) + c_2 (I_2 - 3) - 2(c_1 + 2c_2) \ln J + \frac{k}{2} (\ln J)^2$$

Here,  $I_1$  and  $I_2$  are the first and second invariants of the right Cauchy-Green deformation tensor  $\mathbf{C}$  and  $J$  is the determinant of the deformation gradient tensor.

This material model uses a standard displacement-based element formulation, so care must be taken when modeling materials with nearly-incompressible material behavior to avoid element locking. For (nearly-) incompressible materials, use the *Mooney-Rivlin* material described in Section [4.1.2.9](#).

*Example:*

```
<material id="1" type="coupled Mooney-Rivlin">
  <c1>10.0</c1>
  <c2>1.0</c2>
  <k>100.0</k>
</material>
```

#### 4.1.4.22 Coupled Veronda-Westmann

The material type for the coupled Veronda-Westmann material is *coupled Veronda-Westmann*. The following material parameters can be defined.

|      |                               |     |
|------|-------------------------------|-----|
| <c1> | Veronda-Westmann c1 parameter | [P] |
| <c2> | Veronda-Westmann c2 parameter | [P] |
| k    | "Bulk-modulus"                | [P] |

The coupled Veronda-Westmann material is an unconstrained formulation of the Veronda-Westmann material and is defined by the following strain-energy function.

$$W = c_1 \left( e^{c_2(I_1-3)} - 1 \right) - \frac{c_1 c_2}{2} (I_2 - 3) + \frac{\lambda}{2} (\ln J)^2$$

Here,  $I_1$  and  $I_2$  are the first and second invariants of the right Cauchy-Green deformation tensor  $\mathbf{C}$  and  $J$  is the determinant of the deformation gradient tensor.

This material model uses a standard displacement-based element formulation, so care must be taken when modeling materials with nearly-incompressible material behavior to avoid element locking. For (nearly-) incompressible materials, use the *Veronda-Westmann* material described in Section 4.1.2.17.

*Example:*

```
<material id="1" type="coupled Veronda-Westmann">
  <c1>10.0</c1>
  <c2>1.0</c2>
  <k>100.0</k>
</material>
```

#### 4.1.4.23 Ogden Unconstrained

This material describes an unconstrained hyperelastic Ogden material [81]. The following material parameters must be defined:

|        |                                                                    |     |
|--------|--------------------------------------------------------------------|-----|
| <c[n]> | Coefficient of $n^{\text{th}}$ term, where n can range from 1 to 6 | [P] |
| <m[n]> | Exponent of $n^{\text{th}}$ term, where n can range from 1 to 6    | [ ] |
| <cp>   | Bulk-like modulus                                                  | [P] |

The hyperelastic strain energy function for this material is given in terms of the eigenvalues of the right or left stretch tensor:

$$W(\lambda_1, \lambda_2, \lambda_3) = \frac{1}{2}c_p(J-1)^2 + \sum_{i=1}^N \frac{c_i}{m_i^2} (\lambda_1^{m_i} + \lambda_2^{m_i} + \lambda_3^{m_i} - 3 - m_i \ln J).$$

Here,  $\lambda_i^2$  are the eigenvalues of the right or left Cauchy deformation tensor,  $c_p$ ,  $c_i$  and  $m_i$  are material coefficients and  $N$  ranges from 1 to 6. Any material parameters that are not specified by the user are assumed to be zero.

*Example:*

```
<material id="1" type="Ogden unconstrained">
  <m1>2.4</m1>
  <c1>1</c1>
  <cp>2</cp>
</material>
```

#### 4.1.4.24 Perfect Osmometer Equilibrium Osmotic Pressure

The material type for a perfect osmometer equilibrium swelling pressure is “*perfect osmometer*”. The swelling pressure is described by the equations for a perfect osmometer, assuming that the material is porous, containing an interstitial solution whose solutes cannot be exchanged with the external bathing environment; similarly, solutes in the external bathing solution cannot be exchanged with the interstitial fluid of the porous material. Therefore, osmotic pressurization occurs when there is an imbalance between the interstitial and bathing solution osmolarities. Since osmotic swelling must be resisted by a solid matrix, this material is not stable on its own. It must be combined with an elastic material that resists the swelling, using a “*solid mixture*” container as described in Section 4.1.4.27. The following material parameters need to be defined:

|         |                                                                                              |                     |
|---------|----------------------------------------------------------------------------------------------|---------------------|
| <phiw0> | gel porosity (fluid volume fraction) in reference (strain-free) configuration, $\varphi_0^w$ | [ ]                 |
| <iosm>  | interstitial fluid osmolarity in reference configuration, $\bar{c}_0$                        | [n/L <sup>3</sup> ] |
| <bosm>  | external bath osmolarity, $\bar{c}^*$                                                        | [n/L <sup>3</sup> ] |

The Cauchy stress for this material is the stress from the perfect osmometer equilibrium response:

$$\boldsymbol{\sigma} = -\pi \mathbf{I},$$

where  $\pi$  is the osmotic pressure, given by

$$\pi = RT (\bar{c} - \bar{c}^*) .$$

$\bar{c}$  is the interstitial fluid in the current configuration, related to the reference configuration via

$$\bar{c} = \frac{\varphi_0^w}{J - 1 + \varphi_0^w} \bar{c}_0 ,$$

where  $J = \det \mathbf{F}$  is the relative volume. The values of the universal gas constant  $R$  and absolute temperature  $T$  must be specified as global constants.

Though this material is porous, this is not a full-fledged biphasic material as described in Section 4.14 for example. The behavior described by this material is strictly valid only after the transient response of interstitial fluid and solute fluxes has subsided.

*Example (using units of mm, N, s, nmol, K):*

Hyperosmotic loading of a cell with a membrane-impermeant solute, starting from isotonic conditions.

```
<material id="1" type="solid mixture">
  <solid type="perfect osmometer">
    <phiw0>0.8</ phiw0>
    <iosm>300</cF0>
    <bosm lc="1">1</bosm>
  </solid>
  <solid type="neo-Hookean">
    <E>1.0</E>
    <v>0</v>
  </solid>
```

```
</material>
<LoadData>
  <loadcurve id="1">
    <loadpoint>0,300</loadpoint>
    <loadpoint>1,1500</loadpoint>
  </loadcurve>
</LoadData>
<Globals>
  <Constants>
    <R>8.314e-6</R>
    <T>310</T>
  </Constants>
</Globals>
```

#### 4.1.4.25 Porous Neo-Hookean

The material type for a porous neo-Hookean solid is “*porous neo-Hookean*”. This material describes a porous neo-Hookean material with referential solid volume fraction  $\varphi_r^s$  (i.e., porosity  $\varphi_r^w = 1 - \varphi_r^s$ ). The pores are compressible but the skeleton is intrinsically incompressible. Thus, upon pore closure, the material behavior needs to switch from compressible to incompressible. This material may be used to model the porous solid matrix of a biphasic or multiphasic material, in which case it inherits the value of  $\varphi_r^s$  from that parent material.

The material type for a porous Neo-Hookean material is *porous neo-Hookean*. The following parameters must be defined:

|                          |                            |     |
|--------------------------|----------------------------|-----|
| $\langle E \rangle$      | Young's modulus            | [P] |
| $\langle \phi_0 \rangle$ | $(0 \leq \varphi_r^s < 1)$ | [ ] |

This model is derived from the following hyperelastic strain-energy function:

$$W = \frac{\mu}{2} (\bar{I}_1 - 3) - \mu \ln \bar{J},$$

where  $\bar{J}$  represents the pore volume ratio, evaluated from

$$\bar{J} = \frac{J - \varphi_r^s}{1 - \varphi_r^s},$$

and  $\bar{I}_1 = \text{tr} \bar{\mathbf{C}}$ , with

$$\bar{\mathbf{F}} = \left( \frac{\bar{J}}{J} \right)^{1/3} \mathbf{F}, \quad \bar{\mathbf{C}} = \bar{\mathbf{F}}^T \cdot \bar{\mathbf{F}} = \left( \frac{\bar{J}}{J} \right)^{2/3} \mathbf{C},$$

and  $\bar{J} = \det \bar{\mathbf{F}}$ . The porosity  $\varphi^w$  in the current configuration evolves with the volume ratio  $J$  according to

$$\varphi^w = \frac{J - 1 + \varphi_r^w}{J} = \frac{J - \varphi_r^s}{J} = \bar{J} (1 - \varphi_r^s).$$

Thus, pore closure occurs when  $J = \varphi_r^s$  and  $\bar{J} = 0$ .

By comparison to a standard neo-Hookean material, this porous neo-Hookean material has an effective Young's modulus equal to

$$E = \frac{3\mu}{1 + \frac{1}{2} (\varphi_r^w)^2},$$

and an effective Poisson's ratio equal to

$$\nu = \frac{1 - (\varphi_r^w)^2}{2 + (\varphi_r^w)^2}.$$

Therefore, the two material properties that need to be provided are  $E$  and the referential porosity  $\varphi_r^s = 1 - \varphi_r^w$ . Poisson's ratio in the limit of infinitesimal strains is dictated by the porosity according to the above formula. In particular, a highly porous material ( $\varphi_r^w \rightarrow 1$ ) has an effective Poisson ratio that approaches zero ( $\nu \rightarrow 0$ ) and  $E \rightarrow 2\mu$  in the range of infinitesimal strains. A low porosity material ( $\varphi_r^w \rightarrow 0$ ) has  $\nu \rightarrow \frac{1}{2}$  and  $E \rightarrow 3\mu$ , which is the expected behavior of an incompressible neo-Hookean solid. Note that setting  $\varphi_r^w = 0$  ( $\varphi_r^s = 1$ ) would not produce good numerical behavior, since the Cauchy stress in an incompressible material would need to be supplemented by a hydrostatic pressure term (a Lagrange multiplier that enforces the incompressibility constraint). Nevertheless, this compressible porous neo-Hookean material behaves well even for values of  $\varphi^w$  as low as  $\sim 0.015$ .

*Example:*

```
<material id="1" type="porous neo-Hookean">  
  <density>1.0</density>  
  <E>1.0</E>  
  <phi0>0.2</phi0>  
</material>
```

#### 4.1.4.26 Lung Material

This is a material used for modeling human lung tissue as described by Birzle[20]. The following parameters must be defined:

|      |                                             |     |
|------|---------------------------------------------|-----|
| <E>  | Young's modulus, $E$                        | [P] |
| <v>  | Poisson's ratio, $\nu$                      | []  |
| <c1> | Birzle-Wall c1 parameter, $c_1$             | [P] |
| <c3> | Birzle-Wall c3 parameter, $c_3$             | [P] |
| <d1> | Birzle-Wall exponential d1 parameter, $d_1$ | []  |
| <d3> | Birzle-Wall exponential d3 parameter, $d_3$ | []  |

This model is derived from the following hyperelastic strain-energy function:

$$W = c(I_1 - 3) + \frac{c}{\beta} \left( I_3^{-\beta} - 1 \right) + c_1 \left( I_3^{-\frac{1}{3}} I_1 - 3 \right)^{d_1} + c_3 \left( I_3^{\frac{1}{3}} - 1 \right)^{d_3},$$

where  $I_1$  and  $I_2$  are the first and second invariants of the right Cauchy-Green deformation tensor  $\mathbf{C}$ ,

$$c = \frac{E}{4(1 + \nu)},$$

and

$$\beta = \frac{\nu}{1 - 2\nu}.$$

*Example:*

```
<material id="1" type="lung">
  <density>1</density>
  <E>1913.7</E>
  <v>0.3413</v>
  <c1>278.2</c1>
  <c3>5.766</c3>
  <d1>3</d1>
  <d3>6</d3>
</material>
```



#### 4.1.4.27 Solid Mixture

The material type for a constrained mixture of solids is “*solid mixture*”. This material describes a mixture of elastic solids that share the same reference configuration and same deformation gradient. It is a container for any combination of the unconstrained elastic materials described in Section 4.1.4.

|         |                                                |
|---------|------------------------------------------------|
| <solid> | Container tag for unconstrained solid material |
|---------|------------------------------------------------|

The mixture may consist of any number of solids. The stress tensor for the solid mixture is the sum of the stresses for all the solids.

Material axes may be optionally specified within the <material> level, as well as within each <solid>. Within the <material> level, these represent the local element axes relative to the global coordinate system. Within the <solid>, they represent local material axes relative to the element. If material axes are specified at both levels, they are properly compounded to produce local material axes relative to the global coordinate system. Material axes specified in the <ElementData> section are equivalent to a specification at the <material> level: they correspond to local element axes relative to the global system.

*Example:*

```
<material id="1" type="solid mixture">
  <solid type="neo-Hookean">
    <E>1000.0</E>
    <v>0.45</v>
  </solid>
  <solid type="ellipsoidal fiber distribution">
    <mat_axis type="vector">
      <a>0.8660254, 0.5, 0</a>
      <d>0,0,1</d>
    </mat_axis>
    <ksi>5, 1, 1</ksi>
    <beta>2.5, 3, 3</beta>
  </solid>
  <solid type="ellipsoidal fiber distribution">
    <mat_axis type="vector">
      <a>0.8660254,-0.5, 0</a>
      <d>0,0,1</d>
    </mat_axis>
    <ksi>5, 1, 1</ksi>
    <beta>2.5, 3, 3</beta>
  </solid>
</material>
```

#### 4.1.4.28 Spherical Fiber Distribution

The material type for a spherical (isotropic) continuous fiber distribution is “*spherical fiber distribution*”. Since fibers can only sustain tension, this material is not stable on its own. It must be combined with a stable compressible material that acts as a ground matrix, using a “*solid mixture*” container as described in Section 4.1.4.27. The following material parameters need to be defined:

|         |                    |     |
|---------|--------------------|-----|
| <alpha> | parameter $\alpha$ | [ ] |
| <beta>  | parameter $\beta$  | [ ] |
| <ksi>   | parameters $\xi$   | [P] |

The Cauchy stress for this fibrous material is given by [56, 5, 7]:

$$\boldsymbol{\sigma} = \int_0^{2\pi} \int_0^\pi H(I_n - 1) \sigma_n(\mathbf{n}) \sin \varphi \, d\varphi \, d\theta.$$

Here,  $I_n = \lambda_n^2 = \mathbf{N} \cdot \mathbf{C} \cdot \mathbf{N}$  is the square of the fiber stretch  $\lambda_n$ ,  $\mathbf{N}$  is the unit vector along the fiber direction, in the reference configuration, which in spherical angles is directed along  $(\theta, \varphi)$ ,  $\mathbf{n} = \mathbf{F} \cdot \mathbf{N} / \lambda_n$ , and  $H(\cdot)$  is the unit step function that enforces the tension-only contribution.

The fiber stress is determined from a fiber strain energy function,

$$\sigma_n = \frac{2I_n}{J} \frac{\partial \Psi}{\partial I_n} \mathbf{n} \otimes \mathbf{n},$$

where in this material, the fiber strain energy density is given by

$$\Psi = \frac{\xi}{\alpha} \left( \exp \left[ \alpha (I_n - 1)^\beta \right] - 1 \right),$$

where  $\xi > 0$ ,  $\alpha \geq 0$ , and  $\beta \geq 2$ .

Note: In the limit when  $\alpha \rightarrow 0$ , this expressions produces a power law,

$$\lim_{\alpha \rightarrow 0} \Psi = \xi (I_n - 1)^\beta$$

Note: When  $\beta > 2$ , the fiber modulus is zero at the strain origin ( $I_n = 1$ ). Therefore, use  $\beta > 2$  when a smooth transition in the stress is desired from compression to tension.

*Example:*

```
<material id="1" type="solid mixture">
  <solid type="neo-Hookean">
    <E>1000.0</E>
    <v>0.45</v>
  </solid>
  <solid type="spherical fiber distribution">
    <ksi>10</ksi>
    <alpha>0</alpha>
    <beta>2.5</beta>
  </solid>
</material>
```

#### 4.1.4.29 Spherical Fiber Distribution from Solid-Bound Molecule

The material type for a spherical (isotropic) continuous fiber distribution with fiber modulus dependent on solid-bound molecule content is “*spherical fiber distribution sbm*”. Since fibers can only sustain tension, this material is not stable on its own. It must be combined with a stable compressible material that acts as a ground matrix, using a “*solid mixture*” container as described in Section 4.1.4.27. The following material parameters need to be defined:

|         |                                        |                     |
|---------|----------------------------------------|---------------------|
| <alpha> | parameter $\alpha$                     | [ ]                 |
| <beta>  | parameter $\beta$                      | [ ]                 |
| <ksi0>  | fiber modulus $\xi_0$                  | [P]                 |
| <gamma> | fiber modulus exponent $\gamma$        | [ ]                 |
| <rho0>  | fiber mass density $\rho_0$            | [M/L <sup>3</sup> ] |
| sbm     | index of solid-bound molecule $\sigma$ | [ ]                 |

The Cauchy stress for this fibrous material is given by [56, 5, 7]:

$$\boldsymbol{\sigma} = \int_0^{2\pi} \int_0^\pi H(I_n - 1) \sigma_n(\mathbf{n}) \sin \varphi \, d\varphi \, d\theta.$$

Here,  $I_n = \lambda_n^2 = \mathbf{N} \cdot \mathbf{C} \cdot \mathbf{N}$  is the square of the fiber stretch  $\lambda_n$ ,  $\mathbf{N}$  is the unit vector along the fiber direction, in the reference configuration, which in spherical angles is directed along  $(\theta, \varphi)$ ,  $\mathbf{n} = \mathbf{F} \cdot \mathbf{N} / \lambda_n$ , and  $H(\cdot)$  is the unit step function that enforces the tension-only contribution.

The fiber stress is determined from a fiber strain energy function,

$$\sigma_n = \frac{2I_n}{J} \frac{\partial \Psi}{\partial I_n} \mathbf{n} \otimes \mathbf{n},$$

where in this material, the fiber strain energy density is given by

$$\Psi = \frac{\xi}{\alpha} \left( \exp \left[ \alpha (I_n - 1)^\beta \right] - 1 \right),$$

where  $\xi > 0$ ,  $\alpha \geq 0$ , and  $\beta \geq 2$ . The fiber modulus is dependent on the solid-bound molecule referential density  $\rho_r^\sigma$  according to the power law relation

$$\xi = \xi_0 \left( \frac{\rho_r^\sigma}{\rho_0} \right)^\gamma,$$

where  $\rho_0$  is the density at which  $\xi = \xi_0$ .

This type of material references a solid-bound molecule that belongs to a multiphasic mixture. Therefore this material may only be used as the solid (or a component of the solid) in a multiphasic mixture (Section 4.16). The solid-bound molecule must be defined in the <Globals> section (Section 3.4.3) and must be included in the multiphasic mixture using a <solid\_bound> tag. The parameter *sbm* must refer to the global index of that solid-bound molecule. The value of  $\rho_r^\sigma$  is specified within the <solid\_bound> tag. If a chemical reaction is defined within that multiphasic mixture that alters the value of  $\rho_r^\sigma$ , lower and upper bounds may be specified for this referential density within the <solid\_bound> tag to prevent  $\xi$  from reducing to zero or achieving excessively elevated values.

Note: In the limit when  $\alpha \rightarrow 0$ , the expression for  $\Psi$  produces a power law,

$$\lim_{\alpha \rightarrow 0} \Psi = \xi (I_n - 1)^\beta$$

Note: When  $\beta > 2$ , the fiber modulus is zero at the strain origin ( $I_n = 1$ ). Therefore, use  $\beta > 2$  when a smooth transition in the stress is desired from compression to tension.

*Example:*

```
<solid type="solid mixture">
  <solid type="neo-Hookean">
    <E>1000.0</E>
    <v>0.45</v>
  </solid>
  <solid type="spherical fiber distribution sbm">
    <alpha>0</alpha>
    <beta>2.5</beta>
    <ksi0>10</ksi0>
    <gamma>2</gamma>
    <rho0>1</rho0>
    <sbm>1</sbm>
  </solid>
</solid>
```

#### 4.1.4.30 Coupled Transversely Isotropic Mooney-Rivlin

This material describes a transversely isotropic Mooney-Rivlin material using a fully-coupled formulation. It is defined through the *coupled trans-iso Mooney-Rivlin* material type. The following material parameters must be defined.

|         |                                     |     |
|---------|-------------------------------------|-----|
| c1      | Mooney-Rivlin $c_1$ parameter.      | [P] |
| c2      | Mooney-Rivlin $c_2$ parameter.      | [P] |
| c3      | exponential multiplier              | [P] |
| c4      | fiber scale factor                  | [ ] |
| c5      | fiber modulus in linear region      | [P] |
| lam_max | maximum fiber straightening stretch | [ ] |
| k       | bulk-like modulus                   | [P] |

The strain-energy function for this constitutive model is defined by

$$W = c_1 (I_1 - 3) + c_2 (I_2 - 3) - 2(c_1 + 2c_2) \ln J + F(\lambda) + U(J)$$

The first three terms define the coupled Mooney-Rivlin matrix response. The third term is the fiber response which is a function of the fiber stretch  $\lambda$  and is defined as in [91]. For  $U$ , the following form is chosen in FEBio.

$$U(J) = \frac{1}{2} k (\ln J)^2$$

where  $J = \det \mathbf{F}$  is the Jacobian of the deformation.

*Example:*

```
<material id="1" type="coupled trans-iso Mooney-Rivlin">
  <c1>1</c1>
  <c2>0.1</c2>
  <c3>1</c3>
  <c4>1</c4>
  <c5>1.34</c5>
  <lam_max>1.3</lam_max>
  <k>100</k>
</material>
```

#### 4.1.4.31 Coupled Transversely Isotropic Veronda-Westmann

This material describes a transversely isotropic Veronda-Westmann material using a fully-coupled formulation. It is define through the *coupled trans-iso Veronda-Westmann* material type. The following material parameters must be defined.

|         |                                     |     |
|---------|-------------------------------------|-----|
| c1      | Veronda-Westmann $c_1$ parameter.   | [P] |
| c2      | Veronda-Westmann $c_2$ parameter.   | [ ] |
| c3      | exponential multiplier              | [P] |
| c4      | fiber scale factor                  | [ ] |
| c5      | fiber modulus in linear region      | [P] |
| lam_max | maximum fiber straightening stretch | [ ] |
| k       | bulk-like modulus                   | [P] |

The strain-energy function for this constitutive model is defined by

$$W = c_1 \left( e^{c_2(I_1-3)} - 1 \right) - \frac{1}{2} c_1 c_2 (I_2 - 3) + F(\lambda) + U(J) .$$

The first two terms define the coupled Veronda-Westmann matrix response. The third term is the fiber response which is a function of the fiber stretch  $\lambda$  and is defined as in [91]. For  $U$ , the following form is chosen in FEBio.

$$U(J) = \frac{1}{2} k (\ln J)^2$$

where  $J = \det \mathbf{F}$  is the Jacobian of the deformation.

*Example:*

```
<material id="1" type="coupled trans-iso Veronda-Westmann">
  <c1>1</c1>
  <c2>0.1</c2>
  <c3>1</c3>
  <c4>1</c4>
  <c5>1.34</c5>
  <lam_max>1.3</lam_max>
  <k>100</k>
</material>
```

#### 4.1.4.32 Large Poisson's Ratio Ligament

This material describes a coupled, transversely isotropic material that will conform to a particular Poisson's ratio when stretched along the fiber direction [85]. The following material parameters must be defined:

|    |                                         |
|----|-----------------------------------------|
| c1 | Fiber coefficient $c_1$                 |
| c2 | Fiber coefficient $c_2$                 |
| mu | Matrix coefficient $\mu$                |
| v0 | Poisson's ratio parameter $v_0$         |
| m  | Poisson's ratio parameter $m$           |
| k  | Volumetric penalty coefficient $\kappa$ |

The strain energy function for this constitutive model is a three part expression:

$$W = W_{\text{fiber}} + W_{\text{matrix}} + W_{\text{vol}},$$

where:

$$\begin{aligned} W_{\text{fiber}} &= \frac{1}{2} \frac{c_1}{c_2} \left( e^{c_2(\lambda-1)^2} - 1 \right), \\ W_{\text{matrix}} &= \frac{\mu}{2} (I_1 - 3) - \mu \ln \left( \sqrt{I_3} \right), \\ W_{\text{vol}} &= \frac{\kappa}{2} \left( \ln \left( \frac{I_5 - I_1 I_4 + I_2}{I_4^{2(m-v_0)} e^{-4m(\lambda-1)}} \right) \right)^2. \end{aligned}$$

In the equations above,  $\lambda$  is the stretch ratio of the material along the fiber direction. The desired Poisson's ratio must first be selected based on available data for uniaxial tension along the fiber direction. The function with which to fit the Poisson's ratio data is:

$$v = - \frac{\lambda^{m-v_0} e^{-m(\lambda-1)} - 1}{\lambda - 1}.$$

The volumetric penalty coefficient  $\kappa$  must be selected to be large enough to enforce the Poisson's function above. If this material is to be used in a biphasic representation,  $\kappa$  must be selected based on experimental stress-relaxation data, since  $\kappa$  has an effect on the biphasic behavior of the material. Once  $\kappa$ ,  $v_0$ , and  $m$  are chosen,  $c_1$ ,  $c_2$  and  $\mu$  should be selected by fitting the stress-strain behavior of the material to experimental data. The Cauchy stress of the material is given by:

$$\boldsymbol{\sigma} = \frac{2}{J} \left( (W_1 + I_1 W_2) \mathbf{B} - W_2 \mathbf{B}^2 + W_3 (I_3 \mathbf{1}) + W_4 I_4 (\mathbf{a} \otimes \mathbf{a}) + W_5 I_4 (\mathbf{a} \otimes \mathbf{a} \cdot \mathbf{B} + \mathbf{B} \cdot \mathbf{a} \otimes \mathbf{a}) \right)$$

where  $J$  is the jacobian of the deformation gradient  $\mathbf{F}$ ,  $\mathbf{B} = \mathbf{F} \cdot \mathbf{F}^T$  is the left Cauchy-Green deformation tensor,  $\mathbf{1}$  is the  $3 \times 3$  identity tensor,  $\mathbf{a}$  is the fiber orientation vector in the deformed configuration.

*Example:*

```
<material id="1" name="Material1" type="PRLig">
  <c1>90</c1>
  <c2>160</c2>
  <mu>0.025</mu>
```

```
<v0>5.85</v0>  
<m>-100</m>  
<k>1.55</k>  
</material>
```



#### 4.1.4.33 Shenoy-Wang material

This material implements the constitutive model by Wang et al. (Biophysical Journal, 107, 2014, pp:2592 - 2603), which proposes a mechanism for long-range force transmission in fibrous matrices enabled by tension-driven alignment of fibers.

It defines the following parameters.

|       |                                      |
|-------|--------------------------------------|
| mu    | matrix shear modulus $\mu$           |
| k     | matrix bulk modulus $k$              |
| Ef    | Fiber stiffness modulus $E_f$        |
| lam_c | transition point $\lambda_c$         |
| lam_t | transition interval size $\lambda_t$ |
| n     | strain hardening exponent $n$        |
| m     | strain hardening exponent $m$        |

The strain-energy density function is given by,

$$W = W_b + W_f,$$

where:

$$W_b = \frac{\mu}{2} (\bar{I}_1 - 3) + \frac{\kappa}{2} (J - 1)^2,$$

$$W_f = \sum_{a=1}^3 f(\lambda_a),$$

and  $f$  is defined via its derivative,

$$\frac{\partial f}{\partial \lambda_a}(\lambda_a) = \begin{cases} 0, & \lambda_a < \lambda_1 \\ \frac{E_f \left( \frac{\lambda_a - \lambda_1}{\lambda_2 - \lambda_1} \right)^n (\lambda_a - \lambda_1)}{n+1}, & \lambda_1 \leq \lambda_a < \lambda_2 \\ E_f \left[ \frac{\lambda_2 - \lambda_1}{n+1} + \frac{(1 + \lambda_a - \lambda_2)^{m+1} - 1}{m+1} \right], & \lambda_a \geq \lambda_2 \end{cases}$$

Finally, the parameters  $\lambda_1$  and  $\lambda_2$  are given as follows.

$$\lambda_1 = \lambda_c - \lambda_t/2$$

$$\lambda_2 = \lambda_c + \lambda_t/2$$

Example:

```
<material id="1" name="Material1" type="Shenoy">
  <density>1</density>
  <mu>0.7692</mu>
  <k>1.667</k>
  <Ef>134.6</Ef>
  <lam_c>1.02</lam_c>
  <lam_t>0.255</lam_t>
  <n>5</n>
  <m>30</m>
</material>
```

## 4.2 Fibers

Fiber materials are used to model one-dimensional structures oriented along a unit vector. The associated strain energy density is a function of the normal strain along that vector. These fiber models are constructed such that the strain energy is non-zero only when the fiber is in tension, under the idealized assumption that fibers do not sustain any load in compression. This assumption produces an inherent instability in the material response of fibers, therefore such models must be combined with elastic materials that can sustain compression and tension, thus serving as models of a ground matrix.

FEBio accommodates fiber models that can be combined with unconstrained or uncoupled models of the ground matrix. Historically, unconstrained models of a ground matrix have been combined with unconstrained fiber models, whereas uncoupled models of a ground matrix have been combined with uncoupled fiber models. The manual sections presented below follow this conventional approach.

However, it should be noted that some authors have expressed concerns about using uncoupled fiber formulations in fiber-matrix material models [76, 45, 44], due to two fundamental concerns: When highly nonlinear fibers become much stiffer than the ground matrix upon loading, it may become difficult to enforce an isochoric deformation, as would be expected in an uncoupled formulation. Furthermore, in an uncoupled formulation, the strain is split into its deviatoric (isochoric) and volumetric parts and the fiber strain is evaluated only from the deviatoric part. This deviatoric fiber strain is not the true fiber strain, yet it is used to determine whether the fiber is in tension. These factors may result in non-physical deformations, as described by Helfenstein et al. [45].

The solution advocated by these investigators has been to use an uncoupled formulation for the ground matrix only, while the fiber models should remain unconstrained [76, 45, 44]. This can be done in FEBio by using a “*uncoupled solid mixture*” container as described in Section 4.1.2.16, where the material representing the ground matrix is taken from the list of uncoupled materials in Section 4.1.2, whereas fiber models are taken from the list of unconstrained models in Section 4.2.1 or Section 4.3.1.

### 4.2.1 Unconstrained Fiber Models

Since fibers can only sustain tension, the materials listed here are not stable on their own. They must be combined with a stable material that acts as a ground matrix, using a “*solid mixture*” container as described in Section [4.1.4.27](#), within a <solid> tag. These fiber models may also be used in continuous fiber distribution models as described in Section [4.3.1](#), within a <fibers> tag.

#### 4.2.1.1 Fiber with Exponential-Power Law

The material type for a single fiber with an exponential-power law is “*fiber-exp-pow*”. Since fibers can only sustain tension, this material is not stable on its own. It must be combined with a stable compressible material that acts as a ground matrix, using a “*solid mixture*” container as described in Section 4.1.4.27. The following material parameters need to be defined:

|         |                                                      |     |
|---------|------------------------------------------------------|-----|
| <ksi>   | $\xi$ , representing a measure of the fiber modulus  | [P] |
| <alpha> | $\alpha$ , coefficient of exponential argument       | [ ] |
| <beta>  | $\beta$ , power of exponential argument              | [ ] |
| <lam0>  | $\lambda_0$ , stretch threshold for tensile response | [ ] |

The fiber is oriented along the unit vector  $\mathbf{e}_1$ , where  $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$  are orthonormal basis vectors representing the local element coordinate system when specified (Section 4.1.1), or else the global Cartesian coordinate system. The Cauchy stress for this fibrous material is given by

$$\boldsymbol{\sigma} = H(I_n - I_0) \frac{2I_n}{J} \frac{\partial \Psi}{\partial I_n} \mathbf{n} \otimes \mathbf{n},$$

where  $I_n = \lambda_n^2 = \mathbf{n}_r \cdot \mathbf{C} \cdot \mathbf{n}_r$  is the square of the fiber stretch,  $\mathbf{n} = \mathbf{F} \cdot \mathbf{n}_r / \lambda_n$ ,  $I_0 = \lambda_0^2$  is the square of the stretch threshold for the tensile response ( $\lambda_0 = 1$  by default) and  $H(\cdot)$  is the unit step function that enforces the tension-only contribution. The fiber strain energy density is given by

$$\Psi = \frac{\xi}{\alpha\beta} \left( \exp \left[ \alpha (I_n - I_0)^\beta \right] - 1 \right),$$

where  $\xi > 0$ ,  $\alpha \geq 0$ , and  $\beta \geq 2$ .

Note: In the limit when  $\alpha \rightarrow 0$ , this expressions produces a power law,

$$\lim_{\alpha \rightarrow 0} \Psi = \frac{\xi}{\beta} (I_n - I_0)^\beta$$

Note: When  $\beta > 2$ , the fiber modulus is zero at the strain origin ( $I_n = I_0$ ). Therefore, use  $\beta > 2$  when a smooth transition in the stress is desired from compression to tension.

*Example:*

Single fiber oriented along  $\mathbf{e}_1$ , embedded in a neo-Hookean ground matrix.

```
<material id="1" type="solid mixture">
  <mat_axis type="local">0,0,0</mat_axis>
  <solid type="neo-Hookean">
    <E>1000.0</E>
    <v>0.45</v>
  </solid>
  <solid type="fiber-exp-pow">
    <ksi>5</ksi>
    <alpha>20</alpha>
    <beta>3</beta>
    <mat_axis type="angles">
      <theta>0</theta>
      <phi>90</phi>
    </mat_axis>
  </solid>
</material>
```

*Example:*

Two fibers in the plane orthogonal to  $e_1$ , oriented at  $\pm 25$  degrees relative to  $e_3$ , embedded in a neo-Hookean ground matrix.

```
<material id="1" type="solid mixture">
  <mat_axis type="local">0,0,0</mat_axis>
  <solid type="neo-Hookean">
    <E>1000.0</E>
    <v>0.45</v>
  </solid>
  <solid type="fiber-exp-pow">
    <ksi>5</ksi>
    <alpha>20</alpha>
    <beta>3</beta>
    <fiber type="angles">
      <theta>90</theta>
      <phi>25</phi>
    </fiber>
  </solid>
  <solid type="fiber-exp-pow">
    <ksi>5</ksi>
    <alpha>20</alpha>
    <beta>3</beta>
    <fiber type="angles">
      <theta>-90</theta>
      <phi>25</phi>
    </fiber>
  </solid>
</material>
```

#### 4.2.1.2 Fiber with Neo-Hookean Law

This material type is “*fiber-NH*”. The following material parameters need to be defined:

|                         |                                                     |            |
|-------------------------|-----------------------------------------------------|------------|
| <code>&lt;mu&gt;</code> | $\mu$ , representing a measure of the fiber modulus | <b>[P]</b> |
|-------------------------|-----------------------------------------------------|------------|

The fiber strain energy density is given by

$$\Psi_n(I_n) = \frac{\mu}{4} (I_n - 1)^2 ,$$

where  $\mu > 0$ .

*Example:*

Fiber model as specified in a continuous fiber distribution (Section [4.3.1](#))

```
<fibers type="fiber-NH">
  <mu>1</mu>
</fibers>
```

### 4.2.1.3 Fiber with Natural Neo-Hookean Law

This material type is “*fiber-natural-NH*”. The following material parameters need to be defined:

|        |                                                      |     |
|--------|------------------------------------------------------|-----|
| <ksi>  | $\xi$ , fiber modulus                                | [P] |
| <lam0> | $\lambda_0$ , stretch threshold for tensile response | [ ] |

The fiber strain energy density is given by

$$\Psi_n(\lambda_n) = \frac{\xi}{2} H \left( \ln \frac{\lambda_n}{\lambda_0} \right) \left( \ln \frac{\lambda_n}{\lambda_0} \right)^2,$$

where  $\xi > 0$  and  $\lambda_0 \geq 1$ . The tensile response engages at the tensile stretch ratio  $\lambda_0$  ( $\lambda_0 = 1$  by default). The natural (logarithmic or Hencky) strain along the fiber is  $\ln \lambda_n$ , whereas  $\ln \frac{\lambda_n}{\lambda_0} = \ln \lambda_n - \ln \lambda_0$  is the natural strain relative to the tensile stretch threshold. This model produces a stress response which varies linearly with the natural strain.

*Example:*

Fiber model as specified in a solid mixture (Section [4.1.4.27](#))

```
<material id="1" name="Material1" type="solid mixture">
  <density>1</density>
  <solid type="neo-Hookean">
    <E>0.13</E>
    <v>0.3</v>
  </solid>
  <solid type="fiber-natural-NH">
    <ksi>1</ksi>
    <lam0>1.25</lam0>
    <fiber type="vector">0,0,1</fiber>
  </solid>
</material>
```

#### 4.2.1.4 Fiber with Toe-Linear Response

This material type is “*fiber-pow-linear*”. The following material parameters need to be defined:

|        |                                                                                                             |     |
|--------|-------------------------------------------------------------------------------------------------------------|-----|
| <E>    | $E$ , the fiber modulus in the linear range<br>( $E \geq 0$ )                                               | [P] |
| <beta> | $\beta$ , the power-law exponent in the toe region<br>( $\beta \geq 2$ )                                    | [ ] |
| <lam0> | $\lambda_0$ , the stretch ratio when the toe region<br>transitions to the linear region ( $\lambda_0 > 1$ ) | [ ] |

The fiber strain energy density is given by

$$\Psi_n(I_n) = \begin{cases} 0 & I_n < 1 \\ \frac{\xi}{\beta} (I_n - 1)^\beta & 1 \leq I_n \leq I_0 \\ B (I_n - I_0) - E \left( I_n^{1/2} - I_0^{1/2} \right) + \frac{\xi}{\beta} (I_0 - 1)^\beta & I_0 < I_n \end{cases},$$

where  $I_0 = \lambda_0^2$ ,

$$\xi = \frac{E}{4(\beta - 1)} I_0^{-3/2} (I_0 - 1)^{2-\beta}, \quad B = \xi (I_0 - 1)^{\beta-1} + \frac{E}{2} I_0^{-1/2}$$

For this material type, the fiber elasticity at the strain origin reduces to zero unless  $\beta = 2$ .

*Example:*

Fiber model as specified in a solid mixture (Section 4.1.4.27)

```
<solid type="fiber-pow-linear">
  <fiber type="angles">
    <theta>20</center>
    <phi>90</phi>
  </fiber>
  <E>1</E>
  <beta>2.5</beta>
  <lam0>1.06</lam0>
</solid>
```



#### 4.2.1.5 Fiber with Exp-Pow-Linear Response

This material type is “*fiber-exp-pow-linear*”. The following material parameters need to be defined:

|         |                                                                                                          |     |
|---------|----------------------------------------------------------------------------------------------------------|-----|
| <E>     | $E$ , the fiber modulus in the linear range ( $E \geq 0$ )                                               | [P] |
| <alpha> | $\alpha$ , coefficient of exponential argument                                                           | [ ] |
| <beta>  | $\beta$ , the power-law exponent in the toe region ( $\beta \geq 2$ )                                    | [ ] |
| <lam0>  | $\lambda_0$ , the stretch ratio when the toe region transitions to the linear region ( $\lambda_0 > 1$ ) | [ ] |

The fiber strain energy density is given by

$$\Psi_n = \begin{cases} 0 & I_n < 1 \\ \frac{\xi}{\alpha\beta} \left( \exp \left[ \alpha (I_n - 1)^\beta \right] - 1 \right) & 1 \leq I_n \leq I_0 \\ B (I_n - I_0) - E \left( I_n^{1/2} - I_0^{1/2} \right) + \frac{\xi}{\alpha\beta} \left( \exp \left[ \alpha (I_0 - 1)^\beta \right] - 1 \right) & I_n > I_0 \end{cases}$$

where  $I_0 = \lambda_0^2$ ,

$$\xi = \frac{E (I_0 - 1)^{2-\beta} \exp \left[ -\alpha (I_0 - 1)^\beta \right]}{4I_0^{3/2} \left( \beta - 1 + \alpha\beta (I_0 - 1)^\beta \right)}$$

and

$$B = E \frac{2I_0 \left( \beta - \frac{1}{2} + \alpha\beta (I_0 - 1)^\beta \right) - 1}{4I_0^{3/2} \left( \beta - 1 + \alpha\beta (I_0 - 1)^\beta \right)}$$

For this material type, the fiber elasticity at the strain origin reduces to zero unless  $\beta = 2$ . This model reduces to “*fiber-pow-linear*” when  $\alpha = 0$ . Alternatively, in the limit when  $I_0 = 1$ , the above parameters reduce to  $\xi = 0$  and  $B = E/2$  and the strain energy density takes the quadratic form

$$\Psi_n = \begin{cases} 0 & I_n < 1 \\ \frac{E}{2} \left( I_n^{1/2} - 1 \right)^2 & I_n > 1 \end{cases}$$

where  $I_n^{1/2} = \lambda_n$  is the stretch ratio along the fiber.

*Example:*

Fiber model as specified in a solid mixture (Section 4.1.4.27)

```
<solid type="fiber-exp-pow-linear">
  <E>1080</E>
  <alpha>1400</alpha>
  <beta>2.73</beta>
  <lam0>1.012</lam0>
  <fiber type="vector">0,0,1</fiber>
</solid>
```

#### 4.2.1.6 Fiber Exp-Linear

This material type is “*fiber-exp-linear*”. This constitutive fiber model has an initial exponential rise and then grows linearly after a user-specified stretch transition point. This fiber material is based on the trans-iso Mooney-Rivlin model introduced in [91]. This material by itself is not stable, so it is recommend to use it as part of a solid mixture.

The strain energy is as follows:

$$F_2(\lambda) = \begin{cases} 0 & \lambda < 1 \\ C_3 (e^{-C_4} (\text{Ei}(C_4\lambda) - \text{Ei}(C_4)) - \ln \lambda) & 1 \leq \lambda < \lambda_m \\ C_5 (\lambda - \lambda_m) + C_6 \ln \frac{\lambda}{\lambda_m} + C_3 (e^{-C_4} \text{Ei}(C_4\lambda_m) - \text{Ei}(C_4) - \ln \lambda_m) & \lambda_m \leq \lambda \end{cases}$$

where  $\text{Ei}(\cdot)$  is the exponential integral function. The resulting fiber stress is evaluated from

$$\lambda \frac{\partial F_2}{\partial \lambda} = \begin{cases} 0 & \tilde{\lambda} \leq 1 \\ C_3 (e^{C_4(\lambda-1)} - 1) & 1 < \tilde{\lambda} < \lambda_m \\ C_5 \lambda + C_6 & \tilde{\lambda} \geq \lambda_m \end{cases}.$$

Here,  $\lambda_m$  is the stretch at which the fibers are straightened,  $C_3$  scales the exponential stresses,  $C_4$  is the rate of uncrimping of the fibers, and  $C_5$  is the modulus of the straightened fibers.  $C_6$  is determined from the requirement that the stress is continuous at  $\lambda_m$  (see below).

|          |                                       |     |
|----------|---------------------------------------|-----|
| <c3>     | Exponential stress coefficient        | [P] |
| <c4>     | Fiber uncrimping coefficient          | [ ] |
| <c5>     | Modulus of straightened fibers        | [P] |
| <lambda> | Fiber stretch for straightened fibers | [ ] |

While this material enforces continuity of the strain energy density and stress at  $\lambda_m$ , it does not guarantee continuity of the elasticity. The continuity of the elasticity is only satisfied if the parameter  $C_3$  satisfies

$$C_3 = \frac{C_5}{C_4} e^{-C_4(\lambda_m-1)}$$

The parameter  $C_6$  is chosen so that the stress defined above is continuous  $\lambda_m$  and is determined by,

$$C_6 = C_3 (e^{C_4(\lambda_m-1)} - 1) - C_5 \lambda_m$$

To use the form of  $F_2$  that satisfies continuity of the elasticity at  $\lambda_m$  the user may set  $C_3 = 0$  in the input file, and specify  $C_4$  and  $C_5$  as explained above. When  $C_3 = 0$  the code will automatically recalculate  $C_3$  internally using the formula above.

*Example:*

```
<material id="1" name="Material" type="solid mixture">
  <solid type="neo-Hookean">
    <E>1e-6</E>
    <v>0</v>
  </solid>
  <solid type="fiber-exp-linear">
    <c3>0.0023</c3>
```

```
<c4>43</c4>  
<c5>3</c5>  
<lambda>1.05</lambda>  
<fiber type="vector">1,0,0</fiber>  
</solid>  
</material>
```

#### 4.2.1.7 Fiber as Entropy Chain

This material type is “*fiber-entropy-chain*”. It was proposed by [77]. This fiber model is based on statistical mechanics to reflect the entropy-driven nature of a biological fiber. The model is derived from the freely-jointed-chain mechanism. Its strain energy is directly related to the entropic change of the chains in the material, given by

$$\Psi_n(I_n) = \xi N \left( \sqrt{\frac{I_n}{N}} \beta + \ln \frac{\beta}{\sinh \beta} \right) - \frac{\xi \sqrt{N}}{2} \beta_0 I_n - \alpha_{00}$$

where  $I_n = \mathbf{n}_r \cdot \mathbf{C} \cdot \mathbf{n}_r$  is the square of the stretch ratio along the referential fiber direction  $\mathbf{n}_r$ .

The inverse Langevin equation relates the parameter  $\beta$  to  $I_n$  and  $N$  according to  $\beta = \mathcal{L}^{-1} \left[ \sqrt{\frac{I_n}{N}} \right]$ .

Here,  $\beta_0$  is the value of  $\beta$  when  $I_n = 1$ , and  $\alpha_{00}$  is needed to produce a state of zero energy at  $I_n = 1$ . The parameter  $\xi = nk\Theta$  is the initial fiber stiffness, with  $n$ ,  $k$ , and  $\Theta$  respectively representing the number of chains per unit volume, Boltzmann's constant, and the absolute temperature. The parameter  $N = \zeta^2$  is the number of chain segments, and  $\zeta$  is the locking stretch, representing the extensibility of the fiber. The Langevin function is given by  $\mathcal{L}(x) = \coth x - \frac{1}{x}$ .

When evaluating the inverse Langevin equation, a Taylor series expansion is applied for computational stability. The parameter  $n_{\text{term}}$  is used to control the number of terms used to evaluate the equation; it must be an integer between 3 and 30.

The following material parameters must be defined:

|          |                                            |     |
|----------|--------------------------------------------|-----|
| <ksi>    | Initial modulus                            | [P] |
| <N>      | Square of locking stretch                  | [ ] |
| <n_term> | Number of Taylor series terms              |     |
| <mu>     | Shear modulus for fiber-matrix interaction | [P] |

*Example:*

```
<material id="1" name="Soft Tissue" type="solid mixture">
  <solid type="Arruda-Boyce unconstrained">
    <N>2</N>
    <ksi>1</ksi>
    <n_term>30</n_term>
    <kappa>1</kappa>
  </solid>
  <solid type="continuous fiber distribution">
    <mat_axis type="angles">
      <theta>0</theta>
      <phi>0</phi>
    </mat_axis>
    <fibers type="fiber-entropy-chain">
      <ksi>1</ksi>
      <N>2</N>
      <n_term>30</n_term>
    </fibers>
    <distribution type="von-Mises-3d">
```

```
        <b>0.3</b>
    </distribution>
    <scheme type="fibers-3d-fei">
        <resolution>196</resolution>
    </scheme>
</solid>
</material>
```

### 4.2.2 Uncoupled Fiber Models

Since fibers can only sustain tension, the materials listed here are not stable on their own. They must be combined with a stable material that acts as a ground matrix, using a “*uncoupled solid mixture*” container as described in Section [4.1.2.16](#), within a <solid> tag. These fiber models may also be used in continuous fiber distribution models as described in Section [4.3.2](#), using a <fibers> tag.

#### 4.2.2.1 Fiber with Exponential-Power Law, Uncoupled Formulation

The material type for a single fiber with an exponential-power law, in an uncoupled strain energy formulation, is “*fiber-exp-pow-uncoupled*”. Since fibers can only sustain tension, this material is not stable on its own. It must be combined with a stable uncoupled material that acts as a ground matrix, using a “*uncoupled solid mixture*” container as described in Section 4.1.2.16. The following material parameters need to be defined:

|         |                                                     |     |
|---------|-----------------------------------------------------|-----|
| <ksi>   | $\xi$ , representing a measure of the fiber modulus | [P] |
| <alpha> | $\alpha$ , coefficient of exponential argument      | [ ] |
| <beta>  | $\beta$ , power of exponential argument             | [ ] |

The fiber is oriented along the unit vector  $\mathbf{e}_1$ , where  $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$  are orthonormal basis vectors representing the local element coordinate system when specified (Section 4.1.1), or else the global Cartesian coordinate system. The stress  $\tilde{\sigma}$  for this fibrous material is given by

$$\tilde{\sigma} = H\left(\tilde{I}_n - 1\right) \frac{2\tilde{I}_n}{J} \frac{\partial \tilde{\Psi}}{\partial \tilde{I}_n} \mathbf{n} \otimes \mathbf{n},$$

where  $\tilde{I}_n = \tilde{\lambda}_n^2 = \mathbf{n}_r \cdot \tilde{\mathbf{C}} \cdot \mathbf{n}_r$  is the square of the fiber stretch,  $\mathbf{n} = \tilde{\mathbf{F}} \cdot \mathbf{n}_r / \tilde{\lambda}_n$ , and  $H(\cdot)$  is the unit step function that enforces the tension-only contribution.. The fiber strain energy density is given by

$$\tilde{\Psi} = \frac{\xi}{\alpha\beta} \left( \exp \left[ \alpha \left( \tilde{I}_n - 1 \right)^\beta \right] - 1 \right),$$

where  $\xi > 0$ ,  $\alpha \geq 0$ , and  $\beta \geq 2$ .

Note: In the limit when  $\alpha \rightarrow 0$ , this expressions produces a power law,

$$\lim_{\alpha \rightarrow 0} \tilde{\Psi} = \frac{\xi}{\beta} \left( \tilde{I}_n - 1 \right)^\beta.$$

Note: When  $\beta > 2$ , the fiber modulus is zero at the strain origin ( $\tilde{I}_n = 1$ ). Therefore, use  $\beta > 2$  when a smooth transition in the stress is desired from compression to tension.

*Example:*

Single fiber oriented along  $\mathbf{e}_1$ , embedded in a Mooney-Rivlin ground matrix.

```
<material id="1" type="uncoupled solid mixture">
  <mat_axis type="local">0,0,0</mat_axis>
  <k>10e3</k>
  <solid type="Mooney-Rivlin">
    <c1>10.0</c1>
    <c2>0</c2>
  </solid>
  <solid type="fiber-exp-pow-uncoupled">
    <ksi>5</ksi>
    <alpha>20</alpha>
    <beta>3</beta>
    <mat_axis type="angles">
      <theta>0</theta>
      <phi>90</phi>
```

```

    </mat_axis>
  </solid>
</material>

```

*Example:*

Two fibers in the plane orthogonal to  $e_1$ , oriented at  $\pm 25$  degrees relative to  $e_3$ , embedded in a Mooney-Rivlin ground matrix.

```

<material id="1" type="uncoupled solid mixture">
  <mat_axis type="local">0,0,0</mat_axis>
  <k>10e3</k>
  <solid type="Mooney-Rivlin">
    <c1>10.0</c1>
    <c2>0</c2>
  </solid>
  <solid type="fiber-exp-pow-uncoupled">
    <ksi>5</ksi>
    <alpha>20</alpha>
    <beta>3</beta>
    <mat_axis type="angles">
      <theta>90</theta>
      <phi>25</phi>
    </mat_axis>
  </solid>
  <solid type="fiber-exp-pow-uncoupled">
    <ksi>5</ksi>
    <alpha>20</alpha>
    <beta>3</beta>
    <mat_axis type="angles">
      <theta>-90</theta>
      <phi>25</phi>
    </mat_axis>
  </solid>
</material>

```



#### 4.2.2.2 Fiber Kiouisis Uncoupled

This material type is “*fiber-Kiouisis-uncoupled*”. It is based on the material model proposed by Kiouisis et al. [52, 53] for modeling the balloon in a balloon-stent +system. The following material parameters need to be defined:

|      |                                                            |     |
|------|------------------------------------------------------------|-----|
| <d1> | $d_1$ , represents a measure of the fiber modulus          | [P] |
| <d2> | $d_2$ , square of the stretch ratio when the fiber engages |     |
| <n>  | $n$ , power exponent                                       |     |

The fiber strain energy density is given by

$$\tilde{\Psi}_n(I_n) = \begin{cases} \frac{d_1}{n} (\tilde{I}_n - d_2)^n & \tilde{I}_n \geq d_2 \\ 0 & \tilde{I}_n < d_2 \end{cases},$$

where  $\tilde{I}_n = \tilde{\lambda}_n^2 = \mathbf{n}_r \cdot \tilde{\mathbf{C}} \cdot \mathbf{n}_r$  is the square of the fiber stretch and  $\mathbf{n}_r$  is the unit vector along the fiber direction in its reference configuration.

*Example:*

Fiber model as specified in a continuous fiber distribution (Section 4.3.2)

```
<fibers type="fiber-Kiouisis-uncoupled">
  <d1>500</d1>
  <d2>2.25</d2>
  <n>3</n>
</fibers>
```

#### 4.2.2.3 Fiber with Toe-Linear Response, Uncoupled Formulation

This material type is “*fiber-pow-linear-uncoupled*”. The following material parameters need to be defined:

|        |                                                                                                             |     |
|--------|-------------------------------------------------------------------------------------------------------------|-----|
| <E>    | $E$ , the fiber modulus in the linear range<br>( $E \geq 0$ )                                               | [P] |
| <beta> | $\beta$ , the power-law exponent in the toe region<br>( $\beta \geq 2$ )                                    | [ ] |
| <lam0> | $\lambda_0$ , the stretch ratio when the toe region<br>transitions to the linear region ( $\lambda_0 > 1$ ) | [ ] |

The fiber strain energy density is given by

$$\tilde{\Psi}_n(\tilde{I}_n) = \begin{cases} 0 & \tilde{I}_n < 1 \\ \frac{\xi}{\beta} (\tilde{I}_n - 1)^\beta & 1 \leq \tilde{I}_n \leq I_0 \\ B (\tilde{I}_n - I_0) - E (\tilde{I}_n^{1/2} - I_0^{1/2}) + \frac{\xi}{\beta} (I_0 - 1)^\beta & I_0 < \tilde{I}_n \end{cases},$$

where  $I_0 = \lambda_0^2$ ,

$$\xi = \frac{E}{4(\beta - 1)} I_0^{-3/2} (I_0 - 1)^{2-\beta}, \quad B = \xi (I_0 - 1)^{\beta-1} + \frac{E}{2} I_0^{-1/2}.$$

*Example:*

Fiber model as specified in a solid mixture (Section 4.1.2.16)

```
<solid type="fiber-pow-linear-uncoupled">
  <fiber type="angles">
    <theta>20</center>
    <phi>90</phi>
  </fiber>
  <E>1</E>
  <beta>2.5</beta>
  <lam0>1.06</lam0>
</solid>
```

#### 4.2.2.4 Uncoupled Fiber Exp-Linear

This constitutive fiber model has an initial exponential rise and then grows linearly after a user-specified stretch transition point. This fiber material is based on the trans-iso Mooney-Rivlin model introduced in [91]. This material by itself is not stable, so it is recommend to use it as part of a solid mixture.

The strain energy is as follows:

$$F_2(\tilde{\lambda}) = \begin{cases} 0 & \tilde{\lambda} < 1 \\ C_3 \left( e^{-C_4} \left( \text{Ei}(C_4 \tilde{\lambda}) - \text{Ei}(C_4) \right) - \ln \tilde{\lambda} \right) & 1 \leq \tilde{\lambda} < \lambda_m \\ C_5 \left( \tilde{\lambda} - \lambda_m \right) + C_6 \ln \frac{\tilde{\lambda}}{\lambda_m} + C_3 \left( e^{-C_4} \text{Ei}(C_4 \lambda_m) - \text{Ei}(C_4) - \ln \lambda_m \right) & \lambda_m \leq \tilde{\lambda} \end{cases}$$

where  $\text{Ei}(\cdot)$  is the exponential integral function. The resulting fiber stress is evaluated from

$$\tilde{\lambda} \frac{\partial F_2}{\partial \tilde{\lambda}} = \begin{cases} 0 & \tilde{\lambda} \leq 1 \\ C_3 \left( e^{C_4(\tilde{\lambda}-1)} - 1 \right) & 1 < \tilde{\lambda} < \lambda_m \\ C_5 \tilde{\lambda} + C_6 & \tilde{\lambda} \geq \lambda_m \end{cases}.$$

Here,  $\lambda_m$  is the stretch at which the fibers are straightened,  $C_3$  scales the exponential stresses,  $C_4$  is the rate of uncrimping of the fibers, and  $C_5$  is the modulus of the straightened fibers.  $C_6$  is determined from the requirement that the stress is continuous at  $\lambda_m$  (see below).

|          |                                       |     |
|----------|---------------------------------------|-----|
| <c3>     | Exponential stress coefficient        | [P] |
| <c4>     | Fiber uncrimping coefficient          | [ ] |
| <c5>     | Modulus of straightened fibers        | [P] |
| <k>      | Bulk modulus                          | [P] |
| <lambda> | Fiber stretch for straightened fibers | [ ] |
| <fiber>  | Fiber distribution option             |     |

While this material enforces continuity of the strain energy density and stress at  $\lambda_m$ , it does not enforce continuity of the elasticity. To enforce continuity of all three parameters, we need to let

$$C_3 = \frac{C_5}{C_4} e^{-C_4(\lambda_m-1)}$$

The parameter  $C_6$  is chosen so that the stress defined above is continuous at  $\lambda_m$  and is determined by,

$$C_6 = C_3 \left( e^{C_4(\lambda_m-1)} - 1 \right) - C_5 \lambda_m$$

To enforce continuity of the elasticity at  $\lambda_m$  the user may also set  $C_3 = 0$  in the input file, and specify  $C_4$  and  $C_5$  as explained above. When  $C_3 = 0$  the code will automatically recalculate  $C_3$  internally. Using the above formula to calculate  $C_3$  manually for use in the first form can also enforce continuity of the elastic modulus at  $\lambda_m$ .

*Example:*

```
<material id="1" name="Material" type="uncoupled solid mixture">
  <k>10</k>
  <solid type="Mooney-Rivlin">
    <c1>2.5e-07</c1>
    <c2>0</c2>
  </solid>
  <solid type="uncoupled fiber-exp-linear">
    <c3>0.00234</c3>
    <c4>43</c4>
    <c5>3</c5>
    <lambda>1.05</lambda>
    <fiber type="vector">1,0,0</fiber>
  </solid>
</material>
```

#### 4.2.2.5 Uncoupled Fiber as Entropy Chain

This material type is “*uncoupled fiber-entropy-chain*”. It was proposed by [77]. This fiber model is based on statistical mechanics to reflect the entropy-driven nature of a biological fiber. The model is derived from the freely-jointed-chain mechanism. Its strain energy is directly related to the entropic change of the chains in the material, given by

$$\Psi_n(\tilde{I}_n) = \xi N \left( \sqrt{\frac{\tilde{I}_n}{N}} \beta + \ln \frac{\beta}{\sinh \beta} \right) - \frac{\xi \sqrt{N}}{2} \beta_0 \tilde{I}_n - \alpha_{00}$$

where  $\tilde{I}_n = \mathbf{n}_r \cdot \tilde{\mathbf{C}} \cdot \mathbf{n}_r$  is the deviatoric measure of the stretch ratio along the referential fiber direction  $\mathbf{n}_r$ . The inverse Langevin equation relates the parameter  $\beta$  to  $I_n$  and  $N$  according to  $\beta = \mathcal{L}^{-1} \left( \sqrt{\frac{I_n}{N}} \right)$ . Here,  $\beta_0$  is the value of  $\beta$  when  $I_n = 1$ , and  $\alpha_{00}$  is needed to produce a state of zero energy at  $I_n = 1$ . The parameter  $\xi = nk\Theta$  is the initial fiber stiffness, with  $n$ ,  $k$ , and  $\Theta$  respectively representing the number of chains per unit volume, Boltzmann's constant, and the absolute temperature. The parameter  $N = \zeta^2$  is the number of chain segments, and  $\zeta$  is the locking stretch, representing the extensibility of the fiber. The Langevin function is given by  $\mathcal{L}(x) = \coth x - \frac{1}{x}$ .

When evaluating the inverse Langevin equation, a Taylor series expansion is applied for computational stability. The parameter  $n_{\text{term}}$  is used to control the number of terms used to evaluate the equation; it must be an integer between 3 and 30.

The following material parameters must be defined:

|          |                               |     |
|----------|-------------------------------|-----|
| <ksi>    | Initial modulus               | [P] |
| <N>      | Square of locking stretch     | [ ] |
| <n_term> | Number of Taylor series terms |     |

*Example:*

```
<material id="1" name="Soft Tissue" type="uncoupled solid mixture">
  <k>1e4</k>
  <solid type="Mooney-Rivlin">
    <c1>10.0</c1>
    <c2>0</c2>
  </solid>
  <solid type="uncoupled fiber-entropy-chain">
    <N>2.3</N>
    <ksi>1</ksi>
    <n_term>25</n_term>
    <fiber type="vector">0,0,1</fiber>
  </solid>
</material>
```

### 4.3 Continuous Fiber Distribution

A continuous fiber distribution has a strain energy density that integrates the contributions from fiber bundles oriented along all directions emanating from a point in the continuum,

$$\Psi_r(\mathbf{C}) = \int_A H(I_n - 1) R(\mathbf{n}_r) \Psi_n(I_n) dA,$$

where  $\mathbf{n}_r$  is the unit vector along the fiber orientation in the reference configuration,  $I_n = \mathbf{n}_r \cdot \mathbf{C} \cdot \mathbf{n}_r$  is the normal component of  $\mathbf{C}$  along  $\mathbf{n}_r$  (also the square of the stretch ratio along that direction), and  $A$  represents the unit sphere (for 3D fiber distributions) or unit circle (for 2D fiber distributions) over which the integration is performed [47]. Thus,  $\mathbf{n}_r$  spans all directions from the origin to points on the unit sphere or unit circle. In the integrand,  $\Psi_n$  represents the strain energy density of the fiber bundle oriented along  $\mathbf{n}_r$ ;  $H(\cdot)$  is the Heaviside unit step function that includes only fibers that are in tension; and  $R(\mathbf{n}_r)$  is the fiber density distribution function that specifies the spatial fractional distribution of fibers. This function satisfies the constraint

$$\int_A R(\mathbf{n}_r) dA = 1.$$

Fiber constitutive models may be taken from the list given in Section 4.2.1.

For a material with an uncoupled strain energy density the corresponding expression is

$$\tilde{\Psi}_r(\tilde{\mathbf{C}}) = \int_A H(\tilde{I}_n - 1) R(\mathbf{n}_r) \tilde{\Psi}_n(\tilde{I}_n) dA,$$

where  $\tilde{I}_n = \mathbf{n}_r \cdot \tilde{\mathbf{C}} \cdot \mathbf{n}_r$ . In this case, fiber constitutive models may be taken from the list given in Section 4.2.2.

### 4.3.1 Unconstrained Continuous Fiber Distribution

The material type for an unconstrained continuous fiber distribution material is “*continuous fiber distribution*”. The following parameters must be defined:

|                |                                                                 |  |
|----------------|-----------------------------------------------------------------|--|
| <fibers>       | Specification of the fiber material response $\Psi_n(I_n)$      |  |
| <distribution> | Specification of the fiber density distribution $R(\mathbf{n})$ |  |
| <scheme>       | Numerical integration scheme                                    |  |

The <fibers> tag encloses a description of the fiber constitutive relation and associated material properties, as may be selected from the list provided in Section 4.2.1. The <distribution> tag encloses a description of the fiber density distribution function, as may be selected from the list presented in Section 4.3.4. The <scheme> tag specifies the numerical integration scheme, as may be selected from the list presented in Section 4.3.5.

*Example:*

```
<material id="1" name="Material" type="solid mixture">
  <solid type="neo-Hookean">
    <density>1</density>
    <E>1</E>
    <v>0.3</v>
  </solid>
  <solid type="continuous fiber distribution">
    <fibers type="fiber-exponential-power-law">
      <alpha>0</alpha>
      <beta>2</beta>
      <ksi>1</ksi>
    </fibers>
    <distribution type="spherical">
    </distribution>
    <scheme type="fibers-3d-gkt">
      <nph>7</nph>
      <nth>11</nth>
    </scheme>
  </solid>
</material>
```

### 4.3.2 Uncoupled Continuous Fiber Distribution

The material type for an uncoupled continuous fiber distribution material is “*continuous fiber distribution uncoupled*”. The following parameters must be defined:

|                |                                                                            |
|----------------|----------------------------------------------------------------------------|
| <fibers>       | Specification of the fiber material response $\tilde{\Psi}_n(\tilde{I}_n)$ |
| <distribution> | Specification of the fiber density distribution $R(\mathbf{n})$            |
| <scheme>       | Numerical integration scheme                                               |

The <fibers> tag encloses a description of the fiber constitutive relation and associated material properties, as may be selected from the list provided in Section 4.2.2. The <distribution> tag encloses a description of the fiber density distribution function, as may be selected from the list presented in Section 4.3.4. The <scheme> tag specifies the numerical integration scheme, as may be selected from the list presented in Section 4.3.5.

*Example:*

```
<material id="1" name="Material" type="uncoupled solid mixture">
  <solid type="Mooney-Rivlin">
    <density>1</density>
    <c1>1</c1>
    <c2>0.3</c2>
  </solid>
  <solid type="continuous fiber distribution uncoupled">
    <fibers type="fiber-exponential-power-law-uncoupled">
      <alpha>0</alpha>
      <beta>2</beta>
      <ksi>1</ksi>
    </fibers>
    <distribution type="spherical">
    </distribution>
    <scheme type="fibers-3d-gkt">
      <nph>7</nph>
      <nth>11</nth>
    </scheme>
  </solid>
</material>
```

### 4.3.3 Fiber ODF Constitutive Model

The *Fiber ODF* constitutive model is a special version of the *Continuous Fiber Distribution* constitutive model, with its own distribution type, and a unique integration scheme. It utilizes a set of non-parametric orientation distribution functions which are each defined by a position in space, and a series of spherical harmonic coefficients. This allows for the definition of extremely detailed ODFs of any shape.

This constitutive model is intended to be used in conjunction with the *Fiber ODF Analysis* tool in FEBio Studio, which directly analyzes 3D image data and generates ODFs based on fibers in the image. For more information on this tool and the algorithms involved please see the Fiber ODF Analysis section in the FEBio Studio User Manual, and the manuscripts associated with this tool [73, 74].

When an FE simulation is run using this material, some preprocessing is done on the ODFs in order to reduce the computational load during the simulation. The following is a description of these preprocessing steps. For a more detailed description, please see the associated manuscript [73].

In order to prevent sharp changes in the fiber direction between elements, a unique ODF is interpolated for each element in the mesh based on the element's physical distance from the nearby ODFs in the discrete ODF field. This interpolation is done using a weighted-mean approach utilizing the objective distance metric afforded by the Fisher-Rao inner product space [73]. This results in a unique ODF for each FE element in the domain of the constitutive model which smoothly transition between ODFs defined in each of the original image's subdivisions. If only a single ODF is defined for the domain (in other words, if the image was not subdivided before the analysis) then this step is skipped and the original ODF is used for all elements in the domain.

Constitutive models using continuous fiber distributions require an efficient integration scheme over the unit sphere to compute stress contributions from the fiber family. Because there is no analytical representation of these non-parametric ODFs, this integration is performed by sampling the values of the ODFs at its defined points. To ensure faithful representation of the ODF, all ODF calculations up to this point in the process use a high resolution, pre-defined set evenly-spaced points on the unit sphere. Integrating the stress contributions across the ODFs at their full resolution proved to be prohibitively computationally expensive, as this integration must occur at every integration point in the FE mesh for each stress calculation in the analysis.

To address this issue, we developed a mechanism to reduce the number of sampling points on the ODF using an approach based on finite element remeshing technology. The gradient of each ODF is calculated, and a triangulated mesh of the ODF probability surface and its gradient is passed to the mmg remeshing library. The mmg algorithm remeshes a surface of triangular elements, adjusting the relative nodal density of the ODF mesh based on the magnitude of the gradient, resulting in a higher sampling density in areas of high curvature while reducing the overall number of sample points, thereby preserving sharp changes in orientation (Figure 4.3.1). The number of integration points in the remeshed ODFs varies depending on the ODF topology, but the number points is generally reduced by about two orders of magnitude. Since the resampling of the ODFs only takes place once, at the beginning of the FE analysis, the time required for the remeshing step is small relative to the overall time involved in a FE analysis. The reduction in points dramatically increases the speed of stress computations without significantly altering the results. After the interpolation and remeshing steps, the material initialization is complete, and the FE analysis continues normally.

The material type for a fiber ODF distribution material is *"fiberODF"*. The following parameters must be defined:



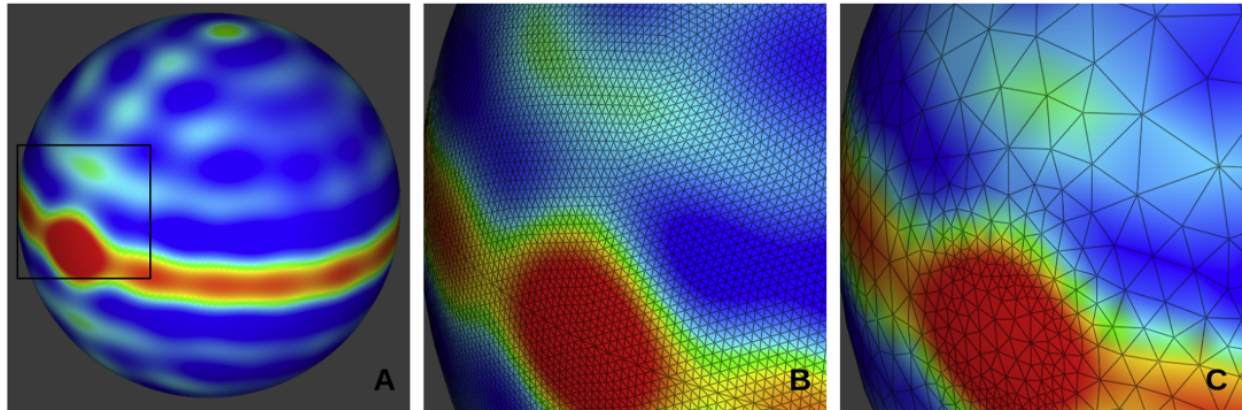


Figure 4.3.1: Spherical representations of an ODF showing the results of the remeshing algorithm. (A) A full-resolution, spherical representation of an ODF. (B) An enlarged portion of the same ODF showing the area inside the black square in panel A. In this panel, the ODF's mesh has been made visible to show the density of sampling points at the full-resolution of 40,962 points. (C) The remeshed surface of the same ODF showing the area inside the black square in panel A. The density of the ODF's sampling points has been greatly reduced in the regions of the ODF where there is little variation in the ODF's value, while maintaining high density in regions where there are sharp changes in value. This preserves the general shape of the ODF while significantly reducing the number of sampling points.

|             |                                                                                                                                                             |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <fibers>    | Specification of the fiber material response $\tilde{\Psi}_n(\tilde{I}_n)$                                                                                  |
| <fiber-odf> | A fiber-odf object specifying the position and spherical harmonic coefficients defining an ODF. This material may have any number of these objects defined. |

The <fibers> tag encloses a description of the fiber constitutive relation and associated material properties, as may be selected from the list provided in Section 4.2.2. The Fiber ODF material takes any number of <fiber-odf> tags as parameters, each with a position in space, and a list of spherical harmonic coefficients.

*Example:*

```
<material id="1" name="Material" type="uncoupled solid mixture">
  <solid type="Mooney-Rivlin">
    <density>1</density>
    <c1>1</c1>
    <c2>0.3</c2>
  </solid>
  <solid type="fiberODF">
    <fibers type="fiber-exponential-power-law-uncoupled">
      <alpha>0</alpha>
      <beta>2</beta>
      <ksi>1</ksi>
    </fibers>
    <fiber-odf>
      <shp_harmonics>8.61001e-05,6.31376e-05, etc...</shp_harmonics>
      <position>340,300,224</position>
    </fiber-odf>
  </solid>
</material>
```

```
</fiber-odf>
<fiber-odf>
  <shp_harmonics>-3.28478e-05,3.90659e-06, etc...</shp_harmonics>
  <position>400,300,224</position>
</fiber-odf>
</solid>
</material>
```

#### 4.3.4 Distribution

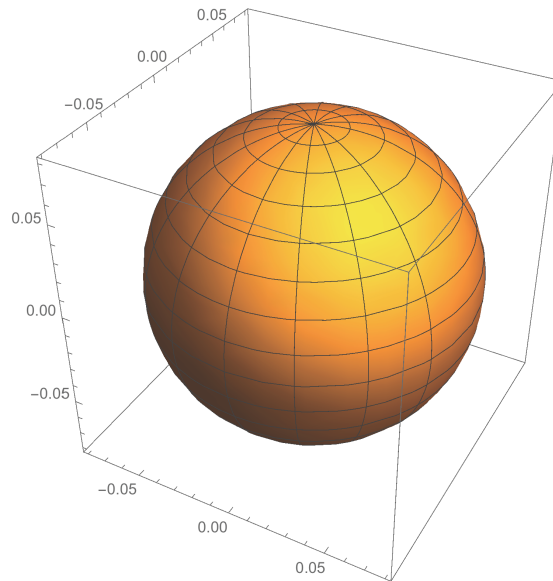
A fiber density distribution function is needed in the specification of a continuous fiber distribution.

#### 4.3.4.1 Spherical

The fiber density distribution type “*spherical*” models an isotropic 3D distribution. This distribution corresponds to

$$R(\mathbf{n}) = \frac{1}{4\pi}.$$

It requires no additional parameters.



*Example:*

```
<distribution type="spherical">
</distribution>
```

#### 4.3.4.2 Ellipsoidal

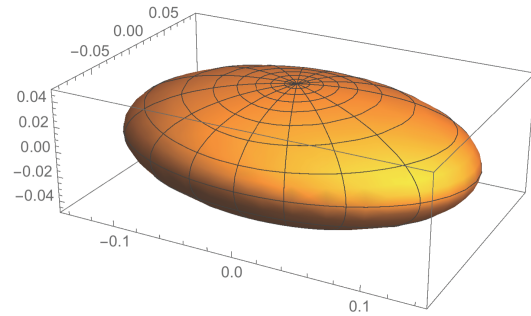
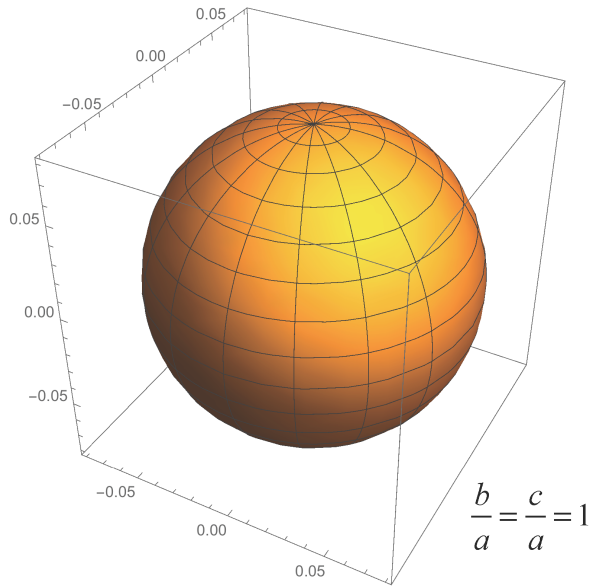
The fiber density distribution type “*ellipsoidal*” models a generally orthotropic 3D distribution. It corresponds to

$$R(\mathbf{n}) = C^{-1} \left[ \left( \frac{n_1}{a} \right)^2 + \left( \frac{n_2}{b} \right)^2 + \left( \frac{n_3}{c} \right)^2 \right]^{-1/2},$$

where  $(n_1, n_2, n_3)$  are the components of  $\mathbf{n}$  in the local Cartesian basis  $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$  of each element, when specified (see Section 4.1.1) or the global basis by default; and  $C$  is calculated to satisfy the integration constraint on  $R(\mathbf{n})$ . The parameters  $(a, b, c)$  represents the semi-principal axes of the ellipsoid and must be positive. The following material parameters need to be defined:

|       |                                                      |     |
|-------|------------------------------------------------------|-----|
| <spa> | The semi-principal axes $(a, b, c)$ of the ellipsoid | [ ] |
|-------|------------------------------------------------------|-----|

The value of  $C$  is automatically adjusted to account for the values of the semi-principal axes  $(a, b, c)$ . Therefore, only the relative ratios of these parameters matter.



$$\frac{b}{a} = \frac{2}{3}, \frac{c}{a} = \frac{1}{3}$$

*Example:*

```
<distribution type="ellipsoidal">
  <spa>3,2,1</spa>
</distribution>
```

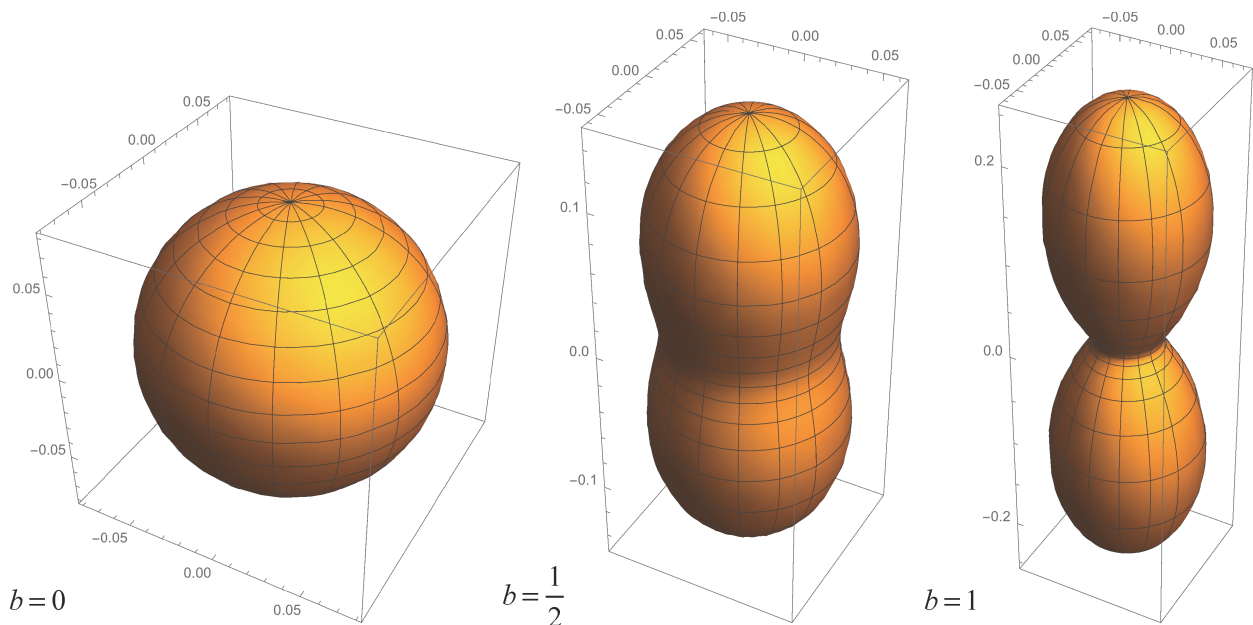
#### 4.3.4.3 $\pi$ -Periodic von Mises Distribution

The fiber density distribution type “*von-Mises-3d*” models a transversely isotropic 3D distribution [39]. It corresponds to

$$R(\mathbf{n}) = \frac{1}{\pi} \sqrt{\frac{b}{2\pi}} \frac{\exp(2bn_1^2)}{\operatorname{erfi}(\sqrt{2b})},$$

where  $(n_1, n_2, n_3)$  are the components of  $\mathbf{n}$  in the local Cartesian basis  $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$  of each element, when specified (see Section 4.1.1) or the global basis by default; and  $b$  is the concentration parameter ( $b > 0$ ). The following material parameters need to be defined:

|     |                                 |     |
|-----|---------------------------------|-----|
| <b> | The concentration parameter $b$ | [ ] |
|-----|---------------------------------|-----|



*Example:*

```
<distribution type="von-Mises-3d">
  <b>0.5</b>
</distribution>
```

#### 4.3.4.4 Circular

The fiber density distribution type “*circular*” models a transversely isotropic 2D distribution. This distribution corresponds to

$$R(\mathbf{n}) = \frac{1}{2\pi}.$$

It requires no additional parameters.

*Example:*

```
<distribution type="circular">  
</distribution>
```

#### 4.3.4.5 Elliptical

The fiber density distribution type “*elliptical*” models an orthotropic 2D distribution. This distribution corresponds to

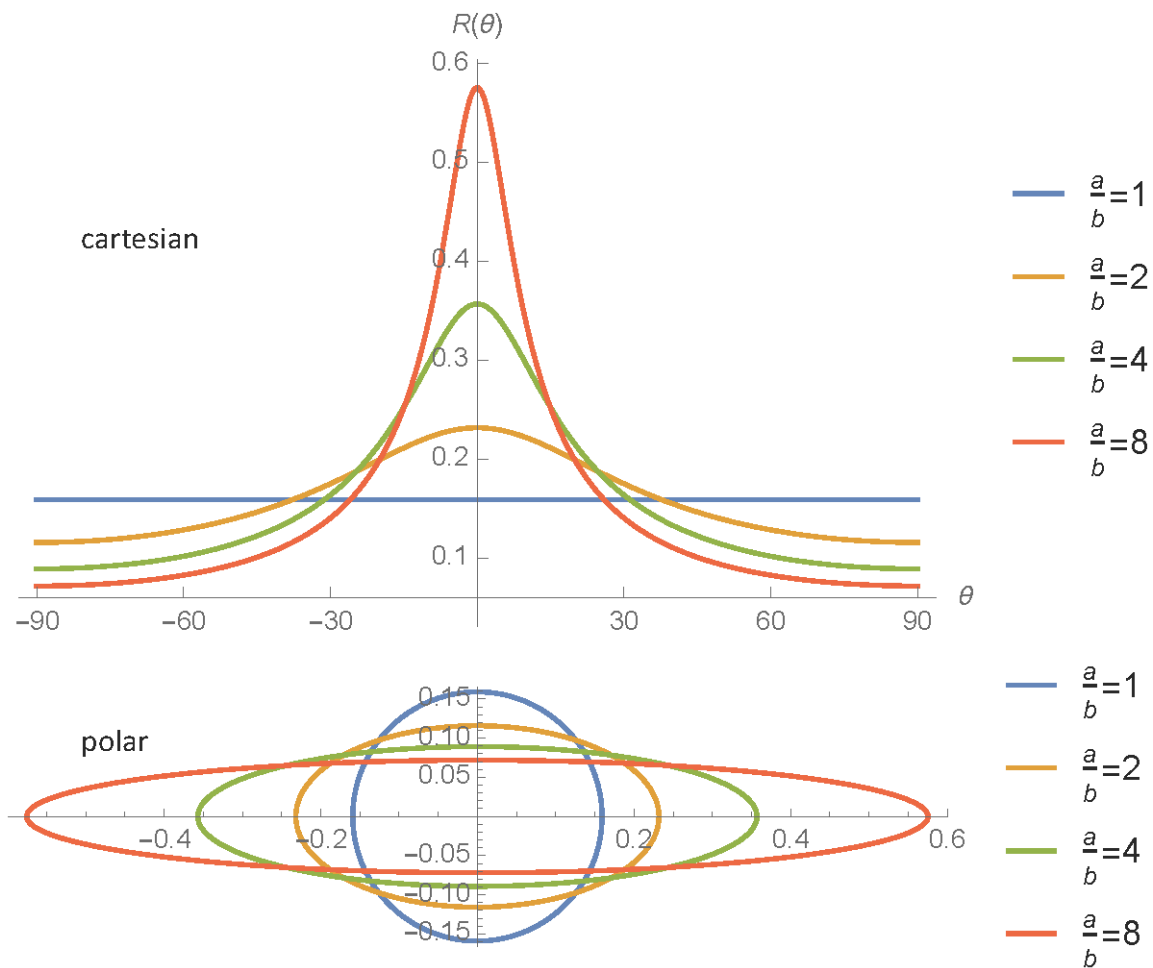
$$R(\mathbf{n}) = C^{-1} \left[ \left( \frac{n_1}{a} \right)^2 + \left( \frac{n_2}{b} \right)^2 \right]^{-1/2}$$

where  $(n_1, n_2, n_3)$  are the components of  $\mathbf{n}$  in the local Cartesian basis  $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$  of each element, when specified (see Section 4.1.1) or the global basis by default, implying that the elliptical distribution lies in the  $\{\mathbf{e}_1, \mathbf{e}_2\}$  plane; and  $(a, b)$  are the semi-principal axes of the ellipse. Here,  $C = 4bK(e)$  where  $K$  is the complete elliptic integral of the first kind and

$$e = \sqrt{1 - \frac{b^2}{a^2}}$$

is the ellipse eccentricity. The following material parameters need to be defined:

|        |                                            |     |
|--------|--------------------------------------------|-----|
| <spa1> | The semi-principal axis $a$ of the ellipse | [ ] |
| <spa2> | The semi-principal axis $b$ of the ellipse | [ ] |



*Example:*

```
<distribution type="elliptical">  
  <spa1>8</spa1>  
  <spa2>1</spa2>  
</distribution>
```



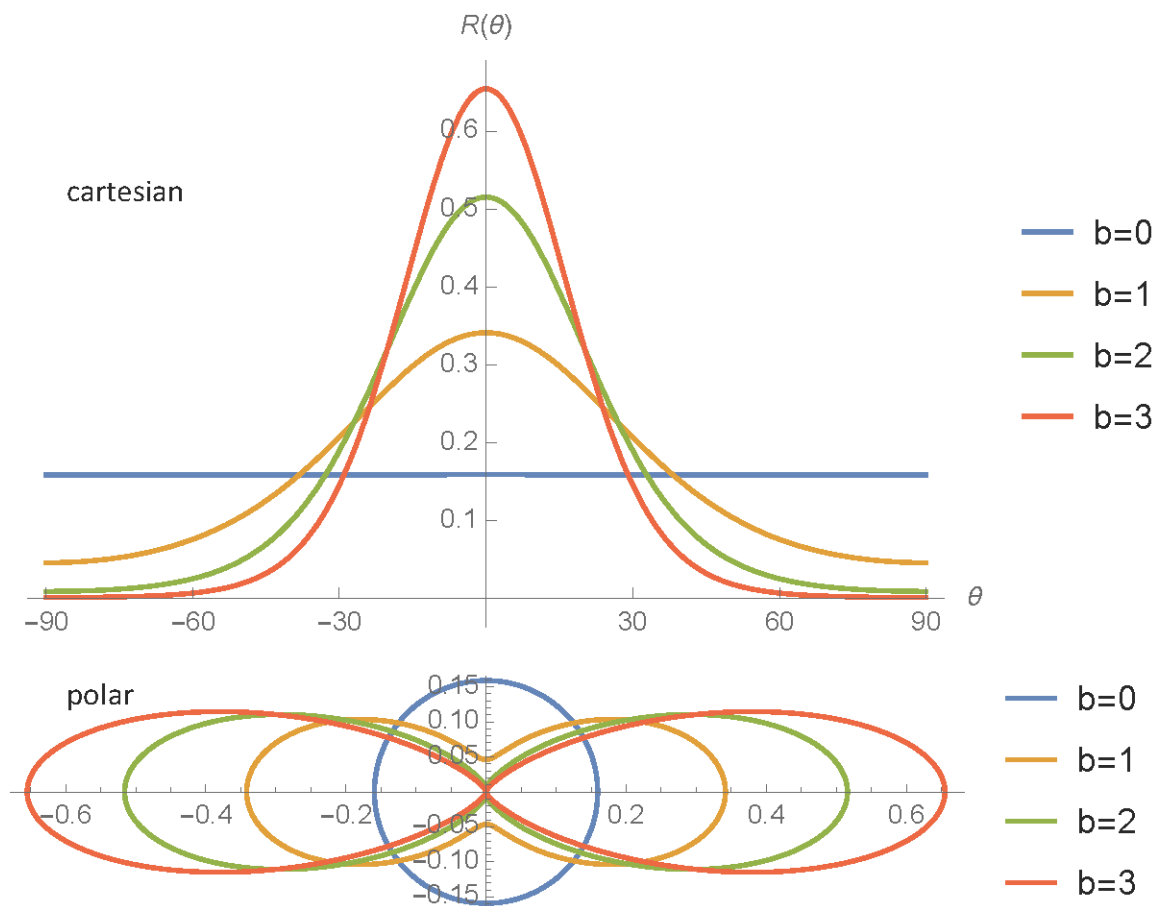
#### 4.3.4.6 von Mises Distribution

The fiber density distribution type “*von-Mises-2d*” models an orthotropic 2D distribution. This distribution corresponds to

$$R(\mathbf{n}) = \frac{\exp[b(2n_1^2 - 1)]}{2\pi I_0(b)}$$

where  $(n_1, n_2, n_3)$  are the components of  $\mathbf{n}$  in the local Cartesian basis  $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$  of each element, when specified (see Section 4.1.1) or the global basis by default, with the distribution lying in the  $\{\mathbf{e}_1, \mathbf{e}_2\}$  plane; and  $b$  is the concentration parameter ( $b > 0$ ).  $I_0$  is the modified Bessel function of the first kind of order 0. The following material parameters need to be defined:

|     |                                 |     |
|-----|---------------------------------|-----|
| <b> | The concentration parameter $b$ | [ ] |
|-----|---------------------------------|-----|



*Example:*

```
<distribution type="von-Mises-2d">
  <b>3</b>
</distribution>
```

#### **4.3.5 Scheme**

A numerical integration scheme is needed in the specification of a continuous fiber distribution to perform the integration over the unit sphere (3D) or the unit circle (2D). Use the uncoupled version of the scheme when modeling an uncoupled continuous fiber distribution.

#### 4.3.5.1 Gauss-Kronrod Trapezoidal Rule

The scheme type for the Gauss-Kronrod trapezoidal rule is “*fibers-3d-gkt*” for unconstrained and uncoupled continuous fiber distributions. This integration rule should only be used with 3D fiber density distributions. This scheme automatically limits the range of integration to fibers that are in tension [47]. A Gauss-Kronrod quadrature rule is employed for integration across latitudes of the unit sphere. A trapezoidal rule is used for integration across longitudes. The following material parameters need to be defined:

|       |                                                |     |
|-------|------------------------------------------------|-----|
| <nph> | Number of integration points across latitudes  | [ ] |
| <nth> | Number of integration points across longitudes | [ ] |

The parameter <nph> must be one of the values 7, 11, 15, 19, 23 and 27. The parameter <nth> may be any number greater than 0. Odd values for <nth> produce more accurate results than even values. A recommended combination is nph=7 and nth=31. The total number of integration points is  $N = \text{nph} \times \text{nth}$ . Increasing values of  $N$  require increasing computational time.

*Example:*

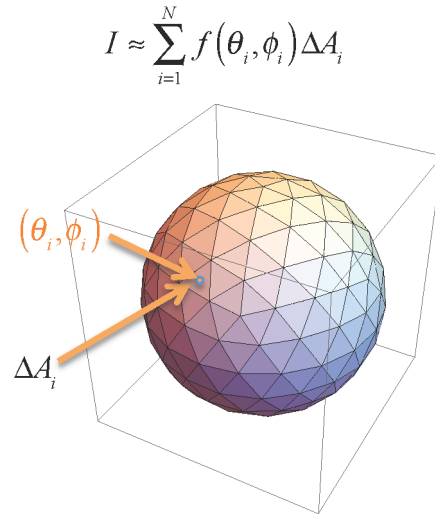
```
<scheme type="fibers-3d-gkt">
  <nph>7</nph>
  <nth>31</nth>
</scheme>
```

#### 4.3.5.2 Finite Element Integration Rule

The scheme type for the finite element integration rule is “*fibers-3d-fei*” for unconstrained and uncoupled continuous fiber distributions. This integration rule should only be used with 3D fiber density distributions. This scheme discretizes the unit sphere into a set of  $N$  spherical triangles of nearly identical surface areas. The unit normal  $\mathbf{n}$  passes through the centroid of each surface element. The integration is performed as a summation over  $N$ . For each direction  $\mathbf{n}$  the stress is evaluated only if the fiber bundle is in tension along that direction. The following material parameters need to be defined:

|              |                                      |     |
|--------------|--------------------------------------|-----|
| <resolution> | the number of integration points $N$ | [ ] |
|--------------|--------------------------------------|-----|

The parameter <resolution> must be one of the values 20, 34, 60, 74, 196, 210, 396, 410, 596, 610, 796, 810, 996, 1010, 1196, 1210, 1396, 1410, 1596, 1610, and 1796. A conservative choice for producing accurate results under general loading conditions is  $N = 1610$ . Increasing values of  $N$  require increasing computational time.



*Example :*

```
<scheme type="fibers-3d-fei">
  <resolution>1610</resolution>
</scheme>
```

#### 4.3.5.3 Trapezoidal Rule

The scheme type for the trapezoidal integration rule is “*fibers-2d-trapezoidal*” for unconstrained and uncoupled continuous fiber distributions. This integration rule should only be used with 2D fiber density distributions. This scheme discretizes the unit circle into a set of  $N$  circular arcs of identical lengths. The unit normal  $\mathbf{n}$  passes through the centroid of each arc element. The integration is performed as a summation over  $N$ . For each direction  $\mathbf{n}$  the stress is evaluated only if the fiber bundle is in tension along that direction. The following material parameters need to be defined:

|       |                                  |     |
|-------|----------------------------------|-----|
| <nth> | Number of integration points $N$ | [ ] |
|-------|----------------------------------|-----|

The parameter <nth> may be any number greater than 0. Odd values for <nth> produce more accurate results than even values.

*Example:*

```
<scheme type="fibers-2d-trapezoidal">
  <nth>31</nth>
</scheme>
```

## 4.4 Viscoelastic Solids

### 4.4.1 Uncoupled Viscoelastic Materials

These materials produce a viscoelastic response only for the deviatoric stress response. They must be used whenever the elastic response is uncoupled, as in the materials described in Section 4.1.2.

The material type for these materials is “uncoupled viscoelastic”. The following parameters need to be defined:

| parameter | description                                            | default value | units |
|-----------|--------------------------------------------------------|---------------|-------|
| <g0>      | viscoelastic coefficient $\gamma_0$                    | 1             | [ ]   |
| <t1>-<t6> | relaxation times                                       | 1             | [t]   |
| <g1>-<g6> | viscoelastic coefficients                              | 0             | [ ]   |
| <elastic> | elastic component (must be an uncoupled elastic solid) |               |       |

$$\mathbf{S}(t) = \int_{-\infty}^t G(t-s) \frac{d\mathbf{S}^e}{ds} ds$$

For a uncoupled viscoelastic material, the second Piola Kirchhoff stress can be written as follows [70]:

$$\mathbf{S}(t) = pJ \mathbf{C}^{-1} + J^{-2/3} \int_{-\infty}^t G(t-s) \frac{d(\text{Dev} [\tilde{\mathbf{S}}^e])}{ds} ds,$$

where  $\tilde{\mathbf{S}}^e$  is the elastic stress derived from  $\tilde{W}(\tilde{\mathbf{C}})$  (see Section 4.1.2) and  $G$  is the relaxation function. It is assumed that the relaxation function is given by the following discrete relaxation spectrum:

$$G(t) = \gamma_0 + \sum_{i=1}^N \gamma_i \exp(-t/\tau_i),$$

Note that the user does not have to enter all the  $\tau_i$  and  $\gamma_i$  coefficients. Instead, only the values that are used need to be entered. So, if  $N$  is 2, only  $\tau_1, \tau_2, \gamma_1$  and  $\gamma_2$  have to be entered.

The *elastic* parameter describes the elastic material as given in Section 4.1.2.

*Example:*

```
<material id="1" name="Material 1" type="uncoupled viscoelastic">
  <g1>0.95</g1>
  <t1>0.01</t1>
  <k>100</k>
  <elastic type="Mooney-Rivlin">
    <c1>1</c1>
    <c2>0.0</c2>
  </elastic>
</material>
```

### 4.4.2 Unconstrained Viscoelastic Materials

The material type for viscoelastic materials is “*viscoelastic*”. The following parameters need to be defined:

| parameter | description                                                | default value | units |
|-----------|------------------------------------------------------------|---------------|-------|
| <g0>      | viscoelastic coefficient $\gamma_0$                        | 1             | [ ]   |
| <t1>-<t6> | relaxation times                                           | 1             | [t]   |
| <g1>-<g6> | viscoelastic coefficients                                  | 0             | [ ]   |
| <elastic> | elastic component (must be an unconstrained elastic solid) |               |       |

For a viscoelastic material, the second Piola Kirchhoff stress can be written as follows [70]:

$$\mathbf{S}(t) = \int_{-\infty}^t G(t-s) \frac{d\mathbf{S}^e}{ds} ds,$$

where  $\mathbf{S}^e$  is the elastic stress and  $G$  is the relaxation function. It is assumed that the relaxation function is given by the following discrete relaxation spectrum:

$$G(t) = \gamma_0 + \sum_{i=1}^N \gamma_i \exp(-t/\tau_i),$$

Note that the user does not have to enter all the  $\tau_i$  and  $\gamma_i$  coefficients. Instead, only the values that are used need to be entered. So, if  $N$  is 2, only  $\tau_1$ ,  $\tau_2$ ,  $\gamma_1$  and  $\gamma_2$  have to be entered.

The *elastic* parameter describes the elastic material as given in Section 4.1.4.

*Example:*

```
<material id="1" name="Material 1" type="viscoelastic">
  <g1>0.95</g1>
  <t1>0.01</t1>
  <elastic type="neo-Hookean">
    <E>1</E>
    <v>0.0</v>
  </elastic>
</material>
```

## 4.5 Reactive Viscoelastic Solid

Reactive viscoelasticity models a material as a mixture of strong bonds, which are permanent, and weak bonds, which break and reform in response to loading [13]. Strong bonds produce the equilibrium elastic response, whereas weak bonds produce the transient viscous response. For a compressive reactive viscoelastic solid, the strain energy density is given by

$$\Psi_r(\mathbf{F}) = \Psi_r^e(\mathbf{F}) + \sum_u w^u \Psi_0^b(\mathbf{F}^u) \quad (4.5.1)$$

where  $\Psi_r^e$  is the strain energy density of strong bonds and  $\Psi_0^b$  is the strain energy density of weak bonds, when they all belong to the same generation.  $\mathbf{F}$  is the deformation gradient of the strong bonds and the initial weak bond generation, whereas  $\mathbf{F}^u$  is the relative deformation gradient for the  $u$ -generation weak bonds, such that  $\mathbf{F}^u(u) = \mathbf{R}(u)$  (the rotation tensor in the polar decomposition of  $\mathbf{F}$ ) at time  $u$ , thus  $\mathbf{F}^u(t) = \mathbf{F}(t) \cdot \mathbf{U}^{-1}(u)$ , where  $\mathbf{U}$  is the right-stretch tensor of  $\mathbf{F} = \mathbf{R} \cdot \mathbf{U}$ . In this expression,  $w^u(\mathbf{X}, t)$  is the mass fraction of  $u$ -generation weak bonds, which evolves over time as described next. The summation is taken over all generations  $u$  that were created prior to the current time  $t$ .

Any number of valid solutions may exist for  $w^u$ , based on constitutive assumptions for the weak bond mass fraction supply  $\hat{w}^u$ . In particular, for  $u$ -generation bonds reforming in an unloaded state during the time interval  $u \leq t < v$ , and subsequently breaking in response to loading at  $t = v$ , Type I bond kinetics provides a solution of the form

$$w^u(\mathbf{X}, t) = \begin{cases} 0 & t < u \\ f^u(\mathbf{X}, t) & u \leq t < v \\ f^u(\mathbf{X}, v) g(\mathbf{F}(v); \mathbf{X}, t - v) & v \leq t \end{cases} \quad (4.5.2)$$

where

$$f^u(\mathbf{X}, t) = 1 - \sum_{\gamma < u} w^\gamma(\mathbf{X}, t) \quad (4.5.3)$$

and  $g(\mathbf{F}(v); \mathbf{X}, t - v)$  is a reduced relaxation function which may assume any number of valid forms. (A reduced relaxation function  $g(t)$  satisfies  $g(0) = 1$  and  $g(t \rightarrow \infty) = 0$ , and decreases monotonically with  $t$ .) In particular,  $g$  may depend on the strain at time  $v$  relative to the reference configuration of the  $u$ -generation. In the recursive expression above, the earliest generation  $u = -\infty$ , which is initially at rest, produces  $w^u(t) = 1$  for  $t < v$  and  $w^u(t) = g(\mathbf{F}^u(v); \mathbf{X}, t - v)$  for  $t \geq v$ ; this latter expression seeds the recursion for subsequent generations. Therefore, providing a functional form for  $g$  suffices to produce the solution for all bond generations  $u$ .

For Type II bond kinetics, the solution for the mass fractions is given by

$$w^u(t) = \begin{cases} 0 & t < u \\ 1 - g(t - u) & u \leq t < v \\ g(t - v) - g(t - u) & v \leq t \end{cases} \quad (4.5.4)$$

For this type of bond kinetics, the reduced relaxation function  $g$  cannot depend on the magnitude of the strain, because strain-dependence might violate the constraint  $0 \leq w^u \leq 1$ . Thus, type II bond kinetics is only valid for quasilinear viscoelasticity, whereas type I bond kinetics also encompasses nonlinear viscoelasticity.



For all bond kinetics, it is also possible to constrain the occurrence of the breaking-and-reforming reaction to specific forms of the strain. For example, the reaction may be allowed to proceed only in the case of dilatational strain, or only in the case of distortional strain.

For a material with an uncoupled formulation, the strain energy density has the form

$$\Psi_r(\mathbf{F}) = U(J) + \tilde{\Psi}_r^e(\tilde{\mathbf{F}}) + \sum_u w^u \tilde{\Psi}_0^b(\tilde{\mathbf{F}}^u) \quad (4.5.5)$$

where  $\tilde{\mathbf{F}} = J^{-1/3}\mathbf{F}$ .

The material type for a compressive reactive viscoelastic solid is “*reactive viscoelastic*”. For the uncoupled formulation the material type is “uncoupled reactive viscoelastic”. The following parameters need to be defined:

|               |                                                                                                                |     |
|---------------|----------------------------------------------------------------------------------------------------------------|-----|
| <kinetics>    | Bond kinetics type                                                                                             | [ ] |
| <trigger>     | Strain invariants that trigger weak bond breakage and reformation                                              | [ ] |
| <wmin>        | Bond mass fraction threshold $w_{\min}$ for triggering new generation or merging and culling oldest generation | [ ] |
| <emin>        | Strain threshold $e_{\min}$ for triggering new generation ( $e_{\min} \geq 0$ )                                | [ ] |
| <elastic>     | Elastic (strong bond) material                                                                                 |     |
| <bond>        | Weak bond material                                                                                             |     |
| <relaxation>  | Reduced relaxation function                                                                                    |     |
| <recruitment> | Weak bond recruitment function                                                                                 |     |

The <kinetics> parameter should be set to 1 for Type I bond kinetics or 2 for Type II bond kinetics. The <trigger> parameter should be set 0 when weak bonds break and reform in response to any form of the strain; it should be set to 1 when the trigger is distortional strain; and it should be set to 2 when the trigger is dilatational strain,

$$\text{trigger} = \begin{cases} 0 & \text{for any strain} & \|\Delta\mathbf{E}\| \\ 1 & \text{for distortional strain} & \|\text{dev } \Delta\boldsymbol{\eta}\| \\ 2 & \text{for dilatational strain} & |\ln(\det \Delta\mathbf{F})| \end{cases} \quad (4.5.6)$$

where  $\Delta\mathbf{E} = \frac{1}{2}(\Delta\mathbf{F}^T \cdot \Delta\mathbf{F} - \mathbf{I})$  is the incremental Lagrange strain,  $\Delta\boldsymbol{\eta} = \ln \Delta\mathbf{V}$  is the incremental natural strain, where  $\Delta\mathbf{V}$  is the left stretch tensor of  $\Delta\mathbf{F}$ , and  $\Delta\mathbf{F} = \mathbf{F}(t_{n+1}) \cdot \mathbf{F}^{-1}(t^m)$ , where  $t_{n+1}$  is the current time and  $t^m$  is the birth time of the most recent weak bond generation.

The <elastic> and <bond> materials may be selected from the list of unconstrained elastic materials given in Section 4.1.4 (for “reactive viscoelastic”) or from the list of uncoupled elastic materials in Section 4.1.2 (for “uncoupled reactive viscoelastic”). The <relaxation> material may be selected from the list of reduced relaxation functions provided in Section 4.5.1.

The <recruitment> material may be selected from the list of weak bond recruitment functions  $F(\Xi)$  provided in Section 4.6.2. This optional material specification can be used to model nonlinear viscoelasticity where the weak bond response becomes more significant with increasing strain, substituting eq.(4.5.1) with

$$\Psi_r(\mathbf{F}) = \Psi_r^e(\mathbf{F}) + \sum_u w^u F(\Xi(u)) \Psi_0^b(\mathbf{F}^u) . \quad (4.5.7)$$

Here,  $\Xi$  is a measure of the current state of strain, similar to the measures shown in eq.(4.5.6). Thus,  $\Xi = \|\mathbf{E}\|$ , or  $\Xi = \|\text{dev } \boldsymbol{\eta}\|$ , or  $\Xi = |\ln(\det \mathbf{F})|$ , based on the choice of the <trigger> parameter. In effect,  $F(\Xi(u))$  enhances the weak bond response of generation  $u$ .

The parameters <wmin> and <emin> can be adjusted to improve computational efficiency by reducing the number of generations  $u$  in an analysis. In practice, the value of <emin> should be no greater than one-hundredth of the equilibrium strain in a loading configuration. The parameter <wmin> must be in the range  $0 \leq w_{\min} \leq 1$ . In practice, a good balance between accuracy and efficiency is achieved by using  $w_{\min} \leq 0.05$ .

*Example:*

```
<material id="1" name="RV solid" type="reactive viscoelastic">
  <kinetics>1</kinetics>
  <trigger>0</trigger>
  <wmin>0.05</wmin>
  <emin>0.001</emin>
  <elastic type="Holmes-Mow">
    <density>1</density>
    <E>1</E>
    <v>0.3</v>
    <beta>0.5</beta>
  </elastic>
  <bond type="Holmes-Mow">
    <density>1</density>
    <E>1</E>
    <v>0.3</v>
    <beta>0.5</beta>
  </bond>
  <relaxation type="relaxation-exponential">
    <tau>4</tau>
  </relaxation>
  <recruitment type="recruitment power">
    <mu0>1</mu0>
    <mu1>30</mu1>
    <alpha>2</alpha>
    <scale>1</scale>
  </recruitment>
</material>
```

## 4.5.1 Relaxation Functions

### 4.5.1.1 Exponential

The material type for this relaxation function is “relaxation-exponential”. The following material parameters need to be defined:

|       |                                       |     |
|-------|---------------------------------------|-----|
| <tau> | Characteristic relaxation time $\tau$ | [t] |
|-------|---------------------------------------|-----|

The reduced relaxation function for this material type is given by

$$g(t) = e^{-t/\tau}$$

#### 4.5.1.2 Exponential Distortional

The material type for this relaxation function is “relaxation-exp-distortion”. The following material parameters need to be defined:

|         |                                         |     |
|---------|-----------------------------------------|-----|
| <tau0>  | Characteristic relaxation time $\tau_0$ | [t] |
| <tau1>  | Characteristic time $\tau_1$            |     |
| <alpha> | Power exponent $\alpha$                 |     |

The reduced relaxation function for this material type is given by

$$g(\mathbf{F}(v); t - v) = e^{-(t-v)/\tau[K_2(v)]}$$

where

$$\tau(K_2(v)) = \tau_0 + \tau_1 \cdot (K_2(v))^\alpha$$

In general,  $K_2 = |\text{dev } \boldsymbol{\eta}|$  where  $\boldsymbol{\eta} = \ln \mathbf{V}$  is the spatial natural (left Hencky) strain tensor and  $\mathbf{V}$  is the left stretch tensor.  $K_2$  is evaluated at the time  $v$  when weak bonds from the  $u$ -generation start breaking.

#### 4.5.1.3 Exponential Distortional User-Specified

The material type for this relaxation function is “relaxation-exp-dist-user”. The following material parameters need to be defined:

|                           |                                         |     |
|---------------------------|-----------------------------------------|-----|
| <tau type="math   point"> | Characteristic relaxation time $\tau_0$ | [t] |
|---------------------------|-----------------------------------------|-----|

The reduced relaxation function for this material type is given by

$$g(\mathbf{F}(v); t - v) = e^{-(t-v)/\tau[K_2(v)]}$$

where  $\tau(K_2(v))$  is given as a mathematical expression (type="math"), or as a loadcurve (type="point").

*Example:*

```
<material id="1" name="RVE" type="reactive viscoelastic">
  <kinetics>1</kinetics>
  <trigger>1</trigger>
  <elastic type="neo-Hookean">
    <density>1</density>
    <E>0.13</E>
    <v>0.3</v>
  </elastic>
  <bond type="neo-Hookean">
    <density>1</density>
    <E>0.52</E>
    <v>0.3</v>
  </bond>
  <relaxation type="relaxation-exp-dist-user">
    <tau type="math">1.0+2.0*(K2^0.5)</tau>
  </relaxation>
</material>
```

Example:

```
<material id="1" name="RVE UC" type="uncoupled reactive viscoelastic">
  <kinetics>1</kinetics>
  <trigger>1</trigger>
  <k>25</k>
  <elastic type="Mooney-Rivlin">
    <density>1</density>
    <c1>0.025</c1>
    <c2>0</c2>
  </elastic>
  <bond type="Mooney-Rivlin">
    <density>1</density>
    <c1>0.025</c1>
    <c2>0</c2>
  </bond>
  <relaxation type="relaxation-exp-dist-user">
    <tau type="point">
      <interpolate>SMOOTH</interpolate>
      <points>
        <point>0.00, 1.00</point>
        <point>0.25, 2.00</point>
        <point>0.50, 2.41</point>
        <point>0.75, 2.73</point>
        <point>1.00, 3.00</point>
      </points>
    </tau>
  </relaxation>
</material>
```

#### 4.5.1.4 Exponential Continuous Spectrum

This relaxation function is derived from a continuous exponential relaxation spectrum, see [FEBio Theory Manual](#). The material type for this relaxation function is “relaxation-CSexp”. The following material parameter needs to be defined:

|       |                                       |     |
|-------|---------------------------------------|-----|
| <tau> | Characteristic relaxation time $\tau$ | [t] |
|-------|---------------------------------------|-----|

This parameter must satisfy  $\tau > 0$ . The reduced relaxation function for this material type is given by

$$g(t) = 2\sqrt{\frac{t}{\tau}} K_1 \left( 2\sqrt{\frac{t}{\tau}} \right)$$

where  $K_1(z)$  is the modified Bessel function of the second kind, of order 1.

#### 4.5.1.5 Exponential Continuous Spectrum Distortional User-Specified

See Section 4.5.1.4 for the description of this relaxation function. When the material parameters vary with the distortional strain  $K_2$ , the material type is “relaxation-CSexp-dist-user”. The following

material parameter needs to be defined:

|                           |                                       |     |
|---------------------------|---------------------------------------|-----|
| <tau type="math   point"> | Characteristic relaxation time $\tau$ | [t] |
|---------------------------|---------------------------------------|-----|

The definition of  $K_2$  is given in Section 4.5.1.3. See Section 4.5.1.3 for examples of how to specify these functions.

#### 4.5.1.6 Fung

This relaxation function, which is derived from a continuous relaxation spectrum, was introduced by Fung [35]. The material type for this relaxation function is “relaxation-Fung”. The following material parameters need to be defined:

|        |                                         |     |
|--------|-----------------------------------------|-----|
| <tau1> | Characteristic relaxation time $\tau_1$ | [t] |
| <tau2> | Characteristic relaxation time $\tau_2$ | [t] |

These parameters must satisfy  $\tau_2 > \tau_1$ . The reduced relaxation function for this material type is given by

$$g(t) = \frac{\tau_2 e^{-t/\tau_2} - \tau_1 e^{-t/\tau_1} + t \left[ \text{Ei} \left( -\frac{t}{\tau_2} \right) - \text{Ei} \left( -\frac{t}{\tau_1} \right) \right]}{\tau_2 - \tau_1}$$

where  $\text{Ei}(\cdot)$  is the exponential integral function.

#### 4.5.1.7 Malkin

This relaxation function is derived from the continuous relaxation spectrum given in [62]. The material type for this relaxation function is “relaxation-Malkin”. The following material parameters need to be defined:

|        |                                         |     |
|--------|-----------------------------------------|-----|
| <tau1> | Characteristic relaxation time $\tau_1$ | [t] |
| <tau2> | Characteristic relaxation time $\tau_2$ | [t] |
| <beta> | Power exponent $\beta (\geq 1)$         | [ ] |

These parameters must satisfy  $\tau_2 > \tau_1 > 0$ . When  $\beta > 1$ , the reduced relaxation function for this material type is given by

$$g(t) = \frac{(\beta - 1) t^{1-\beta}}{\tau_1^{1-\beta} - \tau_2^{1-\beta}} \left[ \Gamma \left( \beta - 1, \frac{t}{\tau_2} \right) - \Gamma \left( \beta - 1, \frac{t}{\tau_1} \right) \right]$$

where  $\Gamma(a, z)$  is the incomplete gamma function. In the limit as  $\beta \rightarrow 1$  this function reduces to the form given by [38],

$$g(t) = \frac{\Gamma \left( 0, \frac{t}{\tau_2} \right) - \Gamma \left( 0, \frac{t}{\tau_1} \right)}{\ln \frac{\tau_2}{\tau_1}} = \frac{\text{Ei} \left( -\frac{t}{\tau_2} \right) - \text{Ei} \left( -\frac{t}{\tau_1} \right)}{\ln \frac{\tau_1}{\tau_2}}$$

where  $\text{Ei}(z)$  is the exponential integral function.

#### 4.5.1.8 Malkin Distortional

See Section 4.5.1.7 for the description of this relaxation function. When the material parameters vary with the distortional strain  $K_2$  according to

$$\begin{aligned}\tau_1(K_2) &= \tau_{10} + \tau_{11} \exp\left(-\frac{K_2}{s_1}\right) \\ \tau_2(K_2) &= \tau_{20} + \tau_{21} \exp\left(-\frac{K_2}{s_2}\right)\end{aligned}$$

the material type is “relaxation-Malkin-distortion”. The following material parameters need to be defined:

|        |                         |     |
|--------|-------------------------|-----|
| <t1c0> | Parameter $\tau_{10}$   | [t] |
| <t1c1> | Parameter $\tau_{11}$   | [t] |
| <t1s0> | Parameter $s_1$         | []  |
| <t2c0> | Parameter $\tau_{20}$   | [t] |
| <t2c1> | Parameter $\tau_{21}$   | [t] |
| <t2s0> | Parameter $s_2$         | []  |
| <beta> | Poiwer exponent $\beta$ | []  |

The definition of  $K_2$  is given in Section 4.5.1.3. See Section 4.5.1.3 for examples of how to specify these functions.

#### 4.5.1.9 Malkin Distortional User-Specified

See Section 4.5.1.7 for the description of this relaxation function. When the material parameters vary with the distortional strain  $K_2$ , the material type is “relaxation-Malkin-dist-user”. The following material parameters need to be defined:

|                            |                                         |     |
|----------------------------|-----------------------------------------|-----|
| <tau1 type="math   point"> | Characteristic relaxation time $\tau_1$ | [t] |
| <tau2 type="math   point"> | Characteristic relaxation time $\tau_2$ | [t] |
| <beta type="math   point"> | Power exponent $\beta$                  | []  |

The definition of  $K_2$  is given in Section 4.5.1.3. See Section 4.5.1.3 for examples of how to specify these functions.

#### 4.5.1.10 Park

The material type for this relaxation function is “relaxation-Park”. The following material parameters need to be defined:

|        |                                       |     |
|--------|---------------------------------------|-----|
| <tau>  | Characteristic relaxation time $\tau$ | [t] |
| <beta> | Power exponent $\beta$                | []  |

The reduced relaxation function for this material type is given by

$$g(t) = \frac{1}{1 + \left(\frac{t}{\tau}\right)^\beta}$$

#### 4.5.1.11 Park Distortional

The material type for this relaxation function is “relaxation-Park-distortion”. The following material parameters need to be defined:

|         |                                         |     |
|---------|-----------------------------------------|-----|
| <tau0>  | Characteristic relaxation time $\tau_0$ | [t] |
| <beta0> | Power exponent at zero strain $\beta_0$ | [ ] |
| <tau1>  | Characteristic relaxation time $\tau_1$ |     |
| <beta1> | Power exponent at zero strain $\beta_1$ |     |
| <alpha> | Power exponent $\alpha$                 |     |

The reduced relaxation function for this material type is given by

$$g(\mathbf{F}(v); t - v) = \frac{1}{1 + \left(\frac{t-v}{\tau}\right)^\beta}$$

where

$$\tau = \tau_0 + \tau_1 \cdot (K_2(v))^\alpha$$

and

$$\beta = \beta_0 + \beta_1 \cdot (K_2(v))^\alpha$$

The definition of  $K_2(v)$  is given in Section 4.5.1.3.

#### 4.5.1.12 Park Distortional User-Specified

The material type for this relaxation function is “relaxation-Park-dist-user”. The following material parameters need to be defined:

|                            |                                       |     |
|----------------------------|---------------------------------------|-----|
| <tau type="math   point">  | Characteristic relaxation time $\tau$ | [t] |
| <beta type="math   point"> | Power exponent $\beta$                | [ ] |

The reduced relaxation function for this material type is given by

$$g(\mathbf{F}(v); t - v) = \frac{1}{1 + \left(\frac{t-v}{\tau}\right)^\beta}$$

where  $\tau(K_2(v))$  and  $\beta(K_2(v))$  are user-specified functions, given either as mathematical expressions or loadcurves. The definition of  $K_2(v)$  is given in Section 4.5.1.3. See Section 4.5.1.3 for examples of how to specify these functions.

#### 4.5.1.13 Power

The material type for this relaxation function is “relaxation-power”. The following material parameters need to be defined:

|        |                                       |     |
|--------|---------------------------------------|-----|
| <tau>  | Characteristic relaxation time $\tau$ | [t] |
| <beta> | Power exponent $\beta$                | [ ] |

The reduced relaxation function for this material type is given by

$$g(t) = \frac{1}{\left(1 + \frac{t}{\tau}\right)^\beta}$$

#### 4.5.1.14 Power Distortional

The material type for this relaxation function is “relaxation-power-distortion”. The following material parameters need to be defined:

|         |                                         |     |
|---------|-----------------------------------------|-----|
| <tau0>  | Characteristic relaxation time $\tau_0$ | [t] |
| <beta0> | Power exponent at zero strain $\beta_0$ | [ ] |
| <tau1>  | Characteristic relaxation time $\tau_1$ |     |
| <beta1> | Power exponent at zero strain $\beta_1$ |     |
| <alpha> | Power exponent $\alpha$                 |     |

The reduced relaxation function for this material type is given by

$$g(\mathbf{F}(v); t - v) = \frac{1}{\left(1 + \frac{t-v}{\tau}\right)^\beta}$$

where

$$\tau = \tau_0 + \tau_1 \cdot (K_2(v))^\alpha$$

and

$$\beta = \beta_0 + \beta_1 \cdot (K_2(v))^\alpha$$

The definition of  $K_2(v)$  is given in Section 4.5.1.3.

#### 4.5.1.15 Power

The material type for this relaxation function is “relaxation-power-dist-user”. The following material parameters need to be defined:

|                            |                                       |     |
|----------------------------|---------------------------------------|-----|
| <tau type="math   point">  | Characteristic relaxation time $\tau$ | [t] |
| <beta type="math   point"> | Power exponent $\beta$                | [ ] |

The reduced relaxation function for this material type is given by

$$g(\mathbf{F}(v); t - v) = \frac{1}{\left(1 + \frac{t-v}{\tau}\right)^\beta}$$

where  $\tau(K_2(v))$  and  $\beta(K_2(v))$  are user-specified functions, given either as mathematical expressions or loadcurves. The definition of  $K_2(v)$  is given in Section 4.5.1.3. See Section 4.5.1.3 for examples of how to specify these functions.

#### 4.5.1.16 Prony

The material type for this relaxation function is “relaxation-Prony”. The following material parameters need to be defined:

|      |                                         |     |
|------|-----------------------------------------|-----|
| <g1> | Weight factor $\gamma_1$                | [ ] |
| ⋮    | ⋮                                       | ⋮   |
| <g6> | Weight factor $\gamma_6$                | [ ] |
| <t1> | Characteristic relaxation time $\tau_1$ | [t] |
| ⋮    | ⋮                                       | ⋮   |
| <t6> | Characteristic relaxation time $\tau_6$ | [t] |



The reduced relaxation function for this material type is given by

$$g(t) = \frac{\sum_{i=1}^6 \gamma_i e^{-t/\tau_i}}{\sum_{i=1}^6 \gamma_i}$$

The coefficients  $\gamma_i$  are normalized by  $\sum_i \gamma_i$  to enforce  $g(0) = 1$ .

### 4.5.2 Weak Bond Recruitment Functions

Weak bond recruitment functions make it possible to model highly nonlinear viscoelastic responses, where the viscous part of the response increases with increasing strain.

#### 4.5.2.1 Power

The material type for a power weak bond recruitment function is “*recruitment power*”. The following material parameters must be defined:

|         |                                     |     |
|---------|-------------------------------------|-----|
| <mu0>   | Parameter $\mu_0$ (default=1)       | [ ] |
| <mu1>   | Parameter $\mu_1$ (default=0)       | [ ] |
| <alpha> | Power exponent $\alpha$ (default=2) | [ ] |
| <scale> | Scale factor $s$ (default=1)        | [ ] |

For this material the recruitment function is given by

$$F(\Xi) = \mu_0 + \mu_1 \left( \frac{\Xi}{s} \right)^\alpha,$$

where  $\Xi$  is the measure of strain that triggers a new reactive weak bond generation. This function is unbounded with increasing  $\Xi$ . Users should typically employ  $\mu_0 = 1$ .

*Example:*

```
<recruitment type="recruitment power">
  <mu0>1</mu0>
  <mu1>10</mu1>
  <alpha>2</alpha>
  <scale>0.06</scale>
</recruitment>
```

#### 4.5.2.2 Exponential

The material type for an exponential weak bond recruitment function is “*recruitment exponential*”. The following material parameters must be defined:

|         |                                     |     |
|---------|-------------------------------------|-----|
| <mu0>   | Parameter $\mu_0$ (default=1)       | [ ] |
| <mu1>   | Parameter $\mu_1$ (default=0)       | [ ] |
| <alpha> | Power exponent $\alpha$ (default=1) | [ ] |
| <scale> | Scale factor $s$ (default=1)        | [ ] |

For this material the recruitment function is given by

$$F(\Xi) = \mu_0 \exp \left( \mu_1 \left( \frac{\Xi}{s} \right)^\alpha \right)$$

where  $\Xi$  is the measure of strain that triggers a new reactive weak bond generation. This function is unbounded with increasing  $\Xi$ . Users should typically employ  $\mu_0 = 1$ .

*Example:*

```

<recruitment type="recruitment exponential">
  <mu0>1</mu0>
  <mu1>2</mu1>
  <alpha>0.5</alpha>
  <scale>0.06</scale>
</recruitment>

```

#### 4.5.2.3 Polynomial

The material type for a polynomial weak bond recruitment function is “*recruitment polynomial*”. The following material parameters must be defined:

|       |                               |     |
|-------|-------------------------------|-----|
| <mu0> | Parameter $\mu_0$ (default=1) | [ ] |
| <mu1> | Parameter $\mu_1$ (default=0) | [ ] |
| <mu2> | Parameter $\mu_2$ (default=0) | [ ] |

For this material the recruitment function is given by

$$F(\Xi) = \mu_0 + \mu_1 \Xi + \mu_2 \Xi^2$$

where  $\Xi$  is the measure of strain that triggers a new reactive weak bond generation. This function is unbounded with increasing  $\Xi$ . Users should typically employ  $\mu_0 = 1$ .

*Example:*

```

<recruitment type="recruitment poly">
  <mu0>1</mu0>
  <mu1>0</mu1>
  <mu3>50</mu3>
</recruitment>

```

#### 4.5.2.4 User

The material type for a user-defined bond recruitment function is “*recruitment user*”. The following material parameters must be defined:

|            |                              |     |
|------------|------------------------------|-----|
| <function> | Specify a user function type | [ ] |
|------------|------------------------------|-----|

For this material the recruitment function  $F(\Xi)$  is specified by the user. It can be a constant, linear ramp, mathematical expression, a series of points, or a step function. It is the user’s responsibility to ensure that the function starts at 1 and rises monotonically.

*Example:*

```

<recruitment type="recruitment user">
  <function type="point">
    <interpolate>linear</interpolate>
    <extend>constant</extend>
    <points>
      <pt>0,1</pt>

```

```

    <pt>0.063,1</pt>
    <pt>0.13,1.8</pt>
  </points>
</function>
</recruitment>

```

#### 4.5.2.5 Log-Normal

The material type for a log-normal recruitment function is “*recruitment log-normal*”. The following material parameters must be defined:

|         |                                                                |           |
|---------|----------------------------------------------------------------|-----------|
| <mu>    | Parameter $\mu$ (same units as $\Xi$ , $\mu > 0$ )             | [ $\Xi$ ] |
| <sigma> | Parameter $\sigma$ ( $\sigma > 0$ )                            | [ ]       |
| <max>   | Maximum increase $r$ in bond recruitment from 1 (default is 0) | [ ]       |

For this material the recruitment function is given by

$$F(\Xi) = 1 + \frac{r}{2} \operatorname{erfc} \left[ -\frac{\ln(\Xi/\mu)}{\sigma\sqrt{2}} \right].$$

Its minimum value is 1 and its maximum value is  $1 + r$  as  $\Xi/\mu \rightarrow \infty$ .

*Example:*

```

<recruitment type="recruitment log-normal">
  <mu>0.3</mu>
  <sigma>0.25</sigma>
  <max>4</max>
</recruitment>

```

#### 4.5.2.6 Weibull

The material type for a Weibull recruitment function is “*recruitment Weibull*”. The following material parameters must be defined:

|         |                                                                |           |
|---------|----------------------------------------------------------------|-----------|
| <mu>    | Parameter $\mu$ (same units as $\Xi$ , $\mu > 0$ )             | [ $\Xi$ ] |
| <alpha> | Parameter $\alpha$ ( $\alpha > 0$ )                            | [ ]       |
| <max>   | Maximum increase $r$ in bond recruitment from 1 (default is 0) | [ ]       |

For this material the recruitment functions is given by

$$F(\Xi) = 1 + r(1 - \exp[-(\Xi/\mu)^\alpha]).$$

Its minimum value is 1 and its maximum value is  $1 + r$  as  $\Xi/\mu \rightarrow \infty$ .

*Example:*

```

<recruitment type="recruitment Weibull">
  <mu>0.3</mu>
  <alpha>2.0</alpha>
  <max>4</max>
</recruitment>

```

#### 4.5.2.7 Quintic Polynomial

The material type for a quintic polynomial bond recruitment function is “*recruitment quintic*”. The following material parameters must be defined:

|         |                                                                           |           |
|---------|---------------------------------------------------------------------------|-----------|
| <mumin> | Parameter $\mu_{\min}$ (same units as $\Xi$ , $\mu_{\min} > 0$ )          | [ $\Xi$ ] |
| <mumax> | Parameter $\mu_{\max}$ (same units as $\Xi$ , $\mu_{\max} > \mu_{\min}$ ) | [ ]       |
| <max>   | Maximum increase $r$ in bond recruitment from 1 (default is 0)            | [ ]       |

For this material the bond recruitment function is given by

$$F(\Xi) = \begin{cases} 1 & \Xi \leq \mu_{\min} \\ 1 + rx^3(10 - 15x + 6x^2) & \mu_{\min} < \Xi \leq \mu_{\max} \\ 1 + r & \mu_{\max} < \Xi \end{cases}, \quad x = \frac{\Xi - \mu_{\min}}{\mu_{\max} - \mu_{\min}}.$$

Its minimum value is 1 and its maximum value is  $1 + r$  when  $\Xi \geq \mu_{\max}$ .

*Example:*

```
<recruitment type="recruitment quintic">
  <mumin>0.2</mumin>
  <mumax>0.4</mumax>
  <Dmax>4</Dmax>
</recruitment>
```

#### 4.5.2.8 Gamma

The material type for a gamma recruitment function is “*recruitment gamma*”. The following material parameters must be defined:

|         |                                                                |           |
|---------|----------------------------------------------------------------|-----------|
| <alpha> | Parameter $\alpha$ ( $\alpha > 0$ )                            | [ ]       |
| <mu>    | Parameter $\mu$ (same units as $\Xi$ , $\mu > 0$ )             | [ $\Xi$ ] |
| <max>   | Maximum increase $r$ in bond recruitment from 1 (default is 0) | [ ]       |

For this material the bond recruitment functions is given by

$$F(\Xi) = 1 + r \frac{\gamma(\alpha, \Xi/\mu)}{\Gamma(\alpha, 0)},$$

where  $\gamma(\alpha, z)$  and  $\Gamma(\alpha, z)$  are the lower and upper incomplete gamma functions, respectively. The normalization by  $\Gamma(\alpha, 0)$  ensures that this function is bounded by  $1 \leq F(\Xi) \leq 1 + r$ .

*Example:*

```
<recruitment type="recruitment gamma">
  <alpha>3</alpha>
  <mu>0.4</mu>
  <max>4</max>
</recruitment>
```

## 4.6 Reactive Damage Mechanics

A material may accumulate damage over a single cycle or multiple cycles of loading which alters its properties. In the classical framework of damage mechanics this attenuation of the material properties is described by a single scalar damage variable  $D$  when the material is isotropic ( $0 \leq D \leq 1$ ). For anisotropic materials however, classical frameworks require that we introduce a function of the fourth-order damage tensor  $\mathcal{D}$  to account for anisotropic damage. In FEBio, we use a reactive damage mechanics framework where the elastic response is proportional to the total number of intact bonds in the material and where, at any given time in the loading history,  $D$  represents the mass fraction of bonds that have broken. In this reactive framework, it is possible to also model damage in anisotropic materials by assuming that multiple bond types exist in the material, each of which may get damaged under different circumstances. Each bond type  $b$  may be described by a distinct solid constituent within a solid mixture (see Sections 4.1.2.16 and 4.1.4.27), each having its own scalar damage variable  $D^b$ .

For a given bond type, the strain energy density  $\Psi_r(D, \mathbf{F})$  of a damaged material is given by

$$\Psi_r = (1 - D) \Psi_0,$$

where  $\Psi_0(\mathbf{F})$  is the strain energy density when all bonds of that type are intact. Here,  $1 - D$  represents the mass fraction of bonds that remains intact. Similarly, the Cauchy stress  $\sigma(D, \mathbf{F})$  of the damaged material is given by

$$\sigma = (1 - D) \sigma_0,$$

where  $\sigma_0(\mathbf{F})$  is the stress in the intact material, at a given strain, as derived from  $\Psi_0(\mathbf{F})$ . The intact material may be based on any of the elastic materials described in Sections 4.1.2 and 4.1.4.

The evolution of the damage variable  $D$  is determined by a user-selected scalar damage criterion measure  $\Xi(\mathbf{F})$  ( $\Xi$  is the capital form of  $\xi$ ). For example,  $\Xi(\mathbf{F})$  may represent the strain energy density, or von Mises stress, or maximum principal normal strain, etc. If  $\Xi(\mathbf{F})$  exceeds a given threshold at some state of deformation  $\mathbf{F}$ , then damage may initiate or progress further. If all bonds fail at a single threshold value  $\Xi_m$ , the material undergoes fracture. More commonly, bonds may fail with increasing probability as  $\Xi$  increases over a given range. Consequently, the evolution of damage may be based on a user-selected cumulative distribution function (c.d.f.)  $F(\Xi)$ , such that  $D(t) = F(\Xi_m)$  where  $\Xi_m$  is the maximum value of  $\Xi$  achieved over the loading history up until the current time  $t$ .

### 4.6.1 General Specification of Damage Materials

The material types for damage materials are “*elastic damage*” and “*uncoupled elastic damage*”. The following parameters must be defined:

|             |                                                                |
|-------------|----------------------------------------------------------------|
| <elastic>   | Specification of the elastic material                          |
| <damage>    | Specification of the cumulative distribution function $F(\Xi)$ |
| <criterion> | Specification of the damage criterion $\Xi$                    |

The <elastic> tag encloses a description of the constitutive relation of the intact elastic material and associated material properties, as may be selected from the list provided in Section 4.1.4 for unconstrained materials (used with “*elastic damage*”) and Section 4.1.2 for uncoupled materials (used with “*uncoupled elastic damage*”). The <damage> tag encloses a description of the cumulative distribution function and associated material properties, as may be selected from the list presented in Section 4.6.2. The <criterion> tag encloses a description of the damage criterion, as may be selected from the list presented in Section 4.6.3.

*Example:*

```
<material id="1" type="elastic damage">
  <elastic type="neo-Hookean">
    <density>1</density>
    <E>0.13</E>
    <v>0.3</v>
  </elastic>
  <damage type="CDF Weibull">
    <alpha>8</alpha>
    <mu>0.3</mu>
    <Dmax>1</Dmax>
  </damage>
  <criterion type="DC max normal Lagrange strain">
  </criterion>
</material>
```

### 4.6.2 Cumulative Distribution Functions

Cumulative distribution functions provide the function  $F(\Xi)$  that determines the evolution of the damage variable  $D$  based on the maximum value of the failure criterion  $\Xi$  up until the current time.



#### 4.6.2.1 Simo

The material type for Simo's c.d.f. [79] is "*CDF Simo*". The following material parameters must be defined:

|     |                                                          |           |
|-----|----------------------------------------------------------|-----------|
| <a> | Parameter $\alpha$ (same units as $\Xi$ , $\alpha > 0$ ) | [ $\Xi$ ] |
| <b> | Parameter $\beta$ ( $0 \leq \beta \leq 1$ )              | [ ]       |

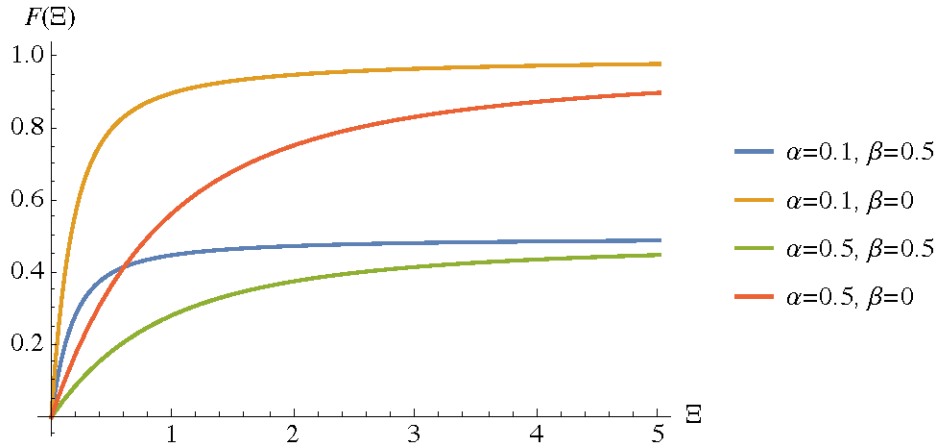
For this material the c.d.f. is given by

$$F(\Xi) = 1 - \beta - (1 - \beta) \frac{\alpha}{\Xi} \left( 1 - e^{-\Xi/\alpha} \right).$$

Note that

$$\lim_{\Xi \rightarrow \infty} F(\Xi) = 1 - \beta$$

represents the maximum allowable damage. Therefore,  $\beta$  regulates the maximum allowable damage, whereas  $\alpha$  controls the rate at which  $F(\Xi)$  increases with increasing  $\Xi$ .



*Example:*

```
<damage type="CDF Simo">
  <a>0.1</a>
  <b>0</b>
  <Dmax>1</Dmax>
</damage>
```

### 4.6.2.2 Log-Normal

The material type for a log-normal c.d.f. is “*CDF log-normal*”. The following material parameters must be defined:

|         |                                                    |           |
|---------|----------------------------------------------------|-----------|
| <mu>    | Parameter $\mu$ (same units as $\Xi$ , $\mu > 0$ ) | [ $\Xi$ ] |
| <sigma> | Parameter $\sigma$ ( $\sigma > 0$ )                | [ ]       |
| <Dmax>  | Maximum allowable damage (optional, default is 1)  | [ ]       |

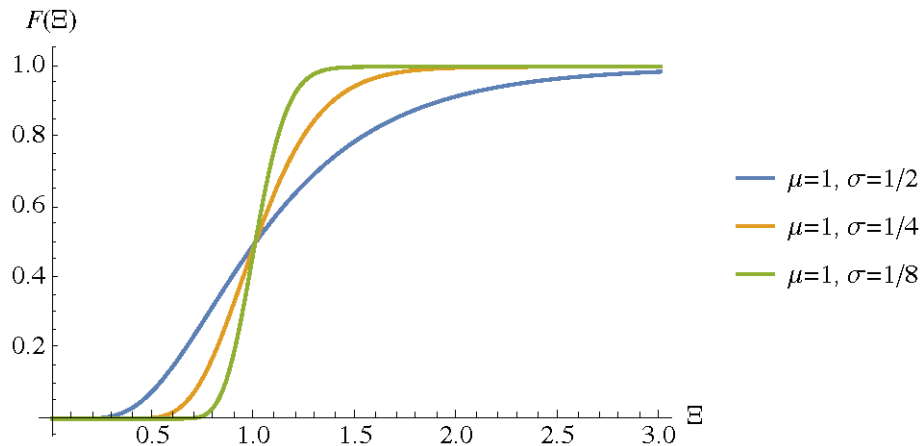
For this material the c.d.f. is given by

$$F(\Xi) = \frac{1}{2} \operatorname{erfc} \left[ -\frac{\ln(\Xi/\mu)}{\sigma\sqrt{2}} \right].$$

Note that

$$F(\mu) = 1/2,$$

which shows that  $\mu$  is the value of  $\Xi$  at which half of the bonds break. Here,  $\sigma$  regulates the rate at which damage increases with increasing  $\Xi$ , with smaller values of  $\sigma$  producing a more rapid increase.



*Example:*

```
<damage type="CDF log-normal">
  <mu>1</mu>
  <sigma>0.5</sigma>
  <Dmax>1</Dmax>
</damage>
```

### 4.6.2.3 Weibull

The material type for a Weibull c.d.f. is “*CDF Weibull*”. The following material parameters must be defined:

|         |                                                    |           |
|---------|----------------------------------------------------|-----------|
| <mu>    | Parameter $\mu$ (same units as $\Xi$ , $\mu > 0$ ) | [ $\Xi$ ] |
| <alpha> | Parameter $\alpha$ ( $\alpha > 0$ )                | [ ]       |
| <Dmax>  | Maximum allowable damage (optional, default is 1)  | [ ]       |

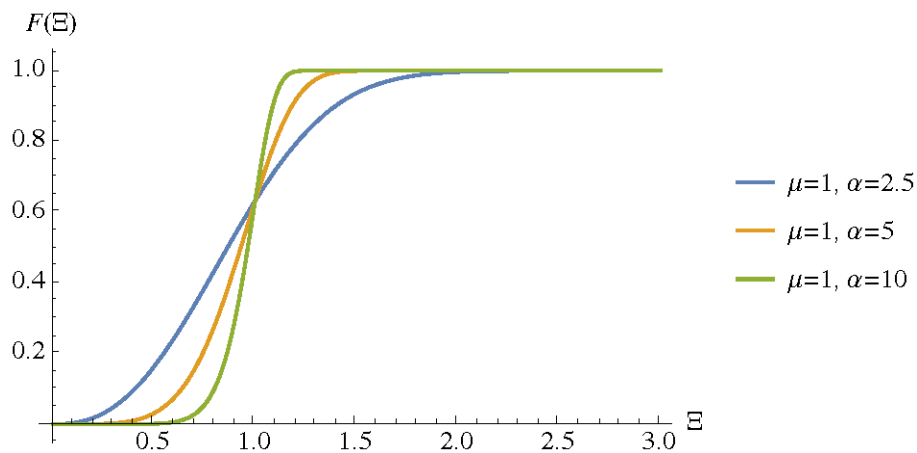
For this material the c.d.f. is given by

$$F(\Xi) = 1 - \exp[-(\Xi/\mu)^\alpha].$$

Note that

$$F(\mu) = 1 - e^{-1} \approx 0.63,$$

which shows that  $\mu$  is the value of  $\Xi$  at which the fraction  $1 - e^{-1}$  of bonds break. Here,  $\alpha$  regulates the rate at which damage increases with increasing  $\Xi$ , with higher values of  $\alpha$  producing a more rapid increase.



*Example:*

```
<damage type="CDF Weibull">
  <mu>1</mu>
  <alpha>5.0</alpha>
  <Dmax>1</Dmax>
</damage>
```

#### 4.6.2.4 Quintic Polynomial

The material type for a quintic polynomial c.d.f. is “*CDF quintic*”. The following material parameters must be defined:

|         |                                                                           |           |
|---------|---------------------------------------------------------------------------|-----------|
| <mumin> | Parameter $\mu_{\min}$ (same units as $\Xi$ , $\mu_{\min} > 0$ )          | [ $\Xi$ ] |
| <mumax> | Parameter $\mu_{\max}$ (same units as $\Xi$ , $\mu_{\max} > \mu_{\min}$ ) | [ ]       |
| <Dmax>  | Maximum allowable damage (optional, default is 1)                         | [ ]       |

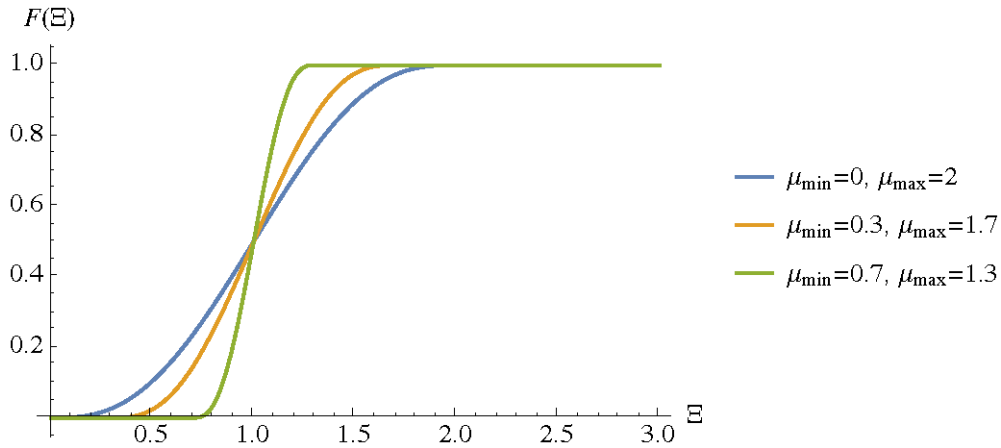
For this material the c.d.f. is given by

$$F(\Xi) = \begin{cases} 0 & \Xi \leq \mu_{\min} \\ x^3 (10 - 15x + 6x^2) & \mu_{\min} < \Xi \leq \mu_{\max} \\ 1 & \mu_{\max} < \Xi \end{cases}, \quad x = \frac{\Xi - \mu_{\min}}{\mu_{\max} - \mu_{\min}}.$$

Note that

$$F\left(\frac{1}{2}(\mu_{\min} + \mu_{\max})\right) = \frac{1}{2},$$

which shows that  $(\mu_{\min} + \mu_{\max})/2$  is the value of  $\Xi$  at which half of the bonds break. The range  $\mu_{\max} - \mu_{\min}$  regulates the rate at which damage increases with increasing  $\Xi$ , with a narrower range producing a more rapid increase.



*Example:*

```
<damage type="CDF quintic">
  <mumin>0.3</mumin>
  <mumax>1.7</mumax>
  <Dmax>1</Dmax>
</damage>
```

#### 4.6.2.5 Gamma

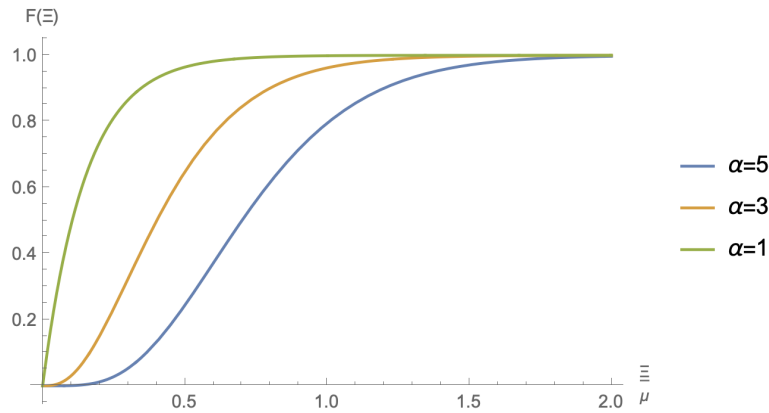
The material type for a gamma c.d.f. is “*CDF gamma*”. The following material parameters must be defined:

|         |                                                    |           |
|---------|----------------------------------------------------|-----------|
| <alpha> | Parameter $\alpha$ ( $\alpha > 0$ )                | [ ]       |
| <mu>    | Parameter $\mu$ (same units as $\Xi$ , $\mu > 0$ ) | [ $\Xi$ ] |

For this material the c.d.f. is given by

$$F(\Xi) = \frac{\gamma(\alpha, \Xi/\mu)}{\Gamma(\alpha, 0)},$$

where  $\gamma(\alpha, z)$  and  $\Gamma(\alpha, z)$  are the lower and upper incomplete gamma functions, respectively. The normalization by  $\Gamma(\alpha, 0)$  ensures that this function is bounded by  $0 \leq F(\Xi) \leq 1$ . The figure below uses  $\mu = 0.15$ .



*Example:*

```
<damage type="CDF gamma">
  <alpha>3</alpha>
  <mu>0.15</mu>
</damage>
```

#### 4.6.2.6 Step

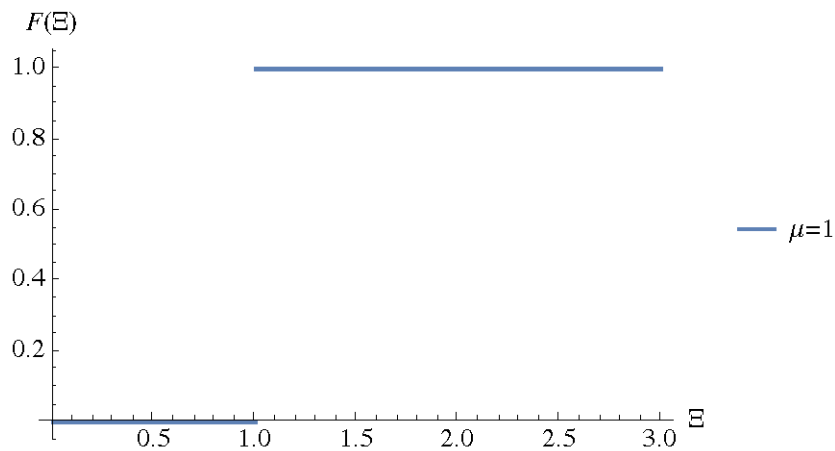
The material type for a step c.d.f. is “*CDF step*”. The following material parameters must be defined:

|        |                                                   |           |
|--------|---------------------------------------------------|-----------|
| <mu>   | Parameter $\mu$ (same units as $\Xi$ )            | [ $\Xi$ ] |
| <Dmax> | Maximum allowable damage (optional, default is 1) | [ ]       |

For this material the c.d.f. is given by

$$F(\Xi) = H(\Xi) \quad ,$$

where  $H(\cdot)$  is the Heaviside unit step function. The step c.d.f. may be used to model fracture.



*Example:*

```
<damage type="CDF step">
  <mu>1.0</mu>
  <Dmax>1</Dmax>
</damage>
```

#### 4.6.2.7 User

The material type for a user-defined CDF is “*CDF user*”. The following material parameters must be defined:

|       |                              |     |
|-------|------------------------------|-----|
| <cdf> | Specify a user function type | [ ] |
|-------|------------------------------|-----|

For this material the CDF  $F(\Xi)$  is specified by the user. It can be a constant, linear ramp, mathematical expression, a series of points, or a step function. It is the user’s responsibility to ensure that the function rises monotonically from 0 to 1 and does not exceed unity.

*Example:*

```
<damage type="CDF user">
  <cdf type="point">
    <interpolate>control points</interpolate>
    <extend>constant</extend>
    <points>
      <pt>0,0</pt>
      <pt>0.49,0</pt>
    <pt>0.5,0</pt>
      <pt>0.99,0</pt>
      <pt>1.01,1</pt>
      <pt>1.49,1</pt>
      <pt>1.5,1</pt>
      <pt>2.5,1</pt>
    </points>
  </cdf>
</recruitment>
```

### 4.6.3 Damage or Yield Criterion

The damage criterion provides the functional form of  $\Xi(\mathbf{F})$  that determines the evolution of damage. It can also be used to define the yield criterion in the reactive plasticity framework. All the functions currently modeled in FEBio are defined over the range  $\Xi \in [0, \infty[$ .

#### 4.6.3.1 Simo

The material type for Simo's damage criterion [79] is “*DC Simo*”. For this criterion,

$$\Xi(\mathbf{F}) = \sqrt{2\Psi_0(\mathbf{F})}$$

*Example:*

```
<criterion type="DC Simo"/>
```

#### 4.6.3.2 Strain Energy Density

The material type for strain energy density damage criterion is “*DC strain energy density*”. For this criterion,

$$\Xi(\mathbf{F}) = \Psi_0(\mathbf{F})$$

*Example:*

```
<criterion type="DC strain energy density"/>
```

#### 4.6.3.3 Specific Strain Energy

The material type for specific strain energy damage criterion is “*DC specific strain energy*”. For this criterion,

$$\Xi(\mathbf{F}) = \Psi_0(\mathbf{F}) / \rho$$

where  $\rho$  is the elastic material's density.

*Example:*

```
<criterion type="DC specific strain energy"/>
```

#### 4.6.3.4 Von Mises Stress

The material type for von Mises stress damage or yield criterion is “*DC von Mises stress*”. For this criterion,

$$\Xi(\mathbf{F}) = \sqrt{\frac{3}{2} \text{dev } \boldsymbol{\sigma} : \text{dev } \boldsymbol{\sigma}} = \sqrt{\frac{1}{2} \left[ (\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2 \right]}$$

where  $\sigma_1, \sigma_2, \sigma_3$  are the principal values of  $\boldsymbol{\sigma}_0(\mathbf{F})$ .

*Example:*

```
<criterion type="DC von Mises stress"/>
```



#### 4.6.3.5 Maximum Shear Stress

The material type for maximum shear stress damage or yield criterion is “*DC max shear stress*”. For this criterion,

$$\Xi(\mathbf{F}) = \max \left( \frac{|\sigma_1 - \sigma_2|}{2}, \frac{|\sigma_2 - \sigma_3|}{2}, \frac{|\sigma_3 - \sigma_1|}{2} \right)$$

where  $\sigma_1, \sigma_2, \sigma_3$  are the principal values of  $\sigma_0(\mathbf{F})$ .

*Example:*

```
< criterion type="DC max shear stress"/>
```

#### 4.6.3.6 Maximum Normal Stress

The material type for maximum normal stress damage or yield criterion is “*DC max normal stress*”. For this criterion,

$$\Xi(\mathbf{F}) = \max(\sigma_1, \sigma_2, \sigma_3)$$

where  $\sigma_1, \sigma_2, \sigma_3$  are the principal values of  $\sigma_0(\mathbf{F})$ .

*Example:*

```
< criterion type="DC max normal stress"/>
```

#### 4.6.3.7 Drucker Shear Stress

The material type for the Drucker shear stress criterion is “*DC Drucker shear stress*”. It is based on the yield criterion for plasticity introduced in [32]. For this criterion,

$$\Xi(\mathbf{F}) = k = (J_2^3 - cJ_3^2)^{1/6}$$

where  $J_2 = \frac{1}{2} \text{dev } \sigma_0 : \text{dev } \sigma_0$ ,  $J_3 = \det(\text{dev } \sigma_0)$ ,  $k$  is the yield limit in simple shear and  $c$  is a user-specified non-dimensional material constant which must lie in the range  $-\frac{27}{8} \leq c \leq \frac{9}{4}$ . To better understand the meaning of  $k$ , consider uniaxial loading of a bar which yields at the normal stress value of  $\sigma_y$ . In this case,

$$k = \frac{\sigma_y}{\sqrt{3}} \left( 1 - \frac{4}{27}c \right)^{1/6} \quad \frac{\sigma_y}{\sqrt{3}} \left( \frac{2}{3} \right)^{1/6} \leq k \leq \frac{\sigma_y}{\sqrt{3}} \left( \frac{3}{2} \right)^{1/6}$$

In the special case when  $c = 0$  the Drucker criterion reduces to the von Mises criterion, with  $k = \sigma_y/\sqrt{3}$ .

*Example:*

```
< criterion type="DC Drucker shear stress">
  <c>2.25</c>
</ criterion>
```

#### 4.6.3.8 Drucker-Prager Criterion

The material type for the Drucker-Prager criterion is “*DC Drucker-Prager*”. It is based on the yield criterion for plasticity introduced in [33]. For this criterion,

$$\Xi(\mathbf{F}) = \sigma_e - b\sigma_m$$

where  $\sigma_e$  is the von Mises stress,  $\sigma_m = \frac{1}{3} \text{tr } \sigma_0$  is the hydrostatic stress,  $b$  is a parameter that depends on the yield (or failure) stress  $\sigma_t$  in tension, and  $\sigma_c$  in compression ( $\sigma_c \geq 0$ ),

$$b = 3 \frac{\sigma_t - \sigma_c}{\sigma_c + \sigma_t}$$

This parameter  $b$  is negative when  $\sigma_c > \sigma_t$ . In the special case when  $b = 0$  the Drucker-Prager criterion reduces to the von Mises criterion. When used as a plastic yield criterion (see Section 4.8), the yield stress  $\sigma_y$  for this type of material is given by

$$\sigma_y = 2 \frac{\sigma_c \sigma_t}{\sigma_c + \sigma_t}$$

which reduces to  $\sigma_y = \sigma_t$  for uniaxial tension,  $\sigma_y = -\sigma_c$  in uniaxial compression, and more generally,  $\sigma_y = \sigma_c = \sigma_t$  when  $\sigma_t = \sigma_c$ . The parameter  $a$  and  $b$  are related to the parameters  $A$  and  $B$  appearing on Wikipedia’s description of the [Drucker-Prager yield criterion](#) via

$$a = \sqrt{3}A \quad b = 3\sqrt{3}B$$

Given these relationships, users may consult that web page for finding the relation between these parameters and cohesion  $c$  and angle of internal friction  $\phi$  (as well as the [Mohr-Coulomb yield surface](#)).

*Example:*

```
<criterion type="DC Drucker-Prager">
  <b>-1</b>
</criterion>
```

#### 4.6.3.9 Deshpande-Fleck Criterion

The material type for the Deshpande-Fleck criterion is “*DC Deshpande-Fleck*”. It is based on the yield criterion for plasticity introduced in [31]. For this criterion, we let

$$\Xi(\mathbf{F}) = \sqrt{\frac{\sigma_e^2 + (3\beta\sigma_m)^2}{1 + \beta^2}}$$

where  $\sigma_e$  is the von Mises stress,  $\sigma_m = \frac{1}{3} \text{tr } \sigma_0$  is the hydrostatic stress,  $\beta$  is a material parameter that controls the contribution of the hydrostatic stress to this criterion ( $\beta = 0$  recovers the von Mises yield criterion). This criterion was introduced to model yielding of polymer foams, for which Deshpande and Fleck recommended using  $\beta = 1/3$ . On [Wikipedia](#), this criterion is described as a modification of the Drucker-Prager criterion, but note that Deshpande and Fleck describe this material model as “phenomenological”, i.e., intended to fit experimental data well, though they do suggest that  $3\beta$  represents the ratio of hydrostatic strength to shear strength.

*Example:*

```
<criterion type="DC Deshpande-Fleck">
  <beta>0.3333</beta>
</criterion>
```

**4.6.3.10 Maximum Normal Lagrange Strain**

The material type for maximum normal Lagrange strain damage criterion is “*DC max normal Lagrange strain*”. For this criterion,

$$\Xi(\mathbf{F}) = \max(E_1, E_2, E_3)$$

where  $E_1, E_2, E_3$  are the principal values of  $\mathbf{E} = (\mathbf{F}^T \cdot \mathbf{F} - \mathbf{I}) / 2$ .

*Example:*

```
<criterion type="DC max normal Lagrange strain"/>
```

**4.6.3.11 Octahedral Shear Strain**

The material type for octahedral strain damage criterion is “*DC octahedral shear strain*”. For this criterion,

$$\Xi(\mathbf{F}) = \sqrt{\frac{2}{3} \text{dev } \mathbf{E} : \text{dev } \mathbf{E}}$$

where  $\mathbf{E}$  is the Lagrange strain tensor.

## 4.7 Reactive Fatigue

Reactive fatigue is a framework that leverages the reactive damage framework described in Section 4.6. In reactive damage, the intact bonds of an elastic material may break upon loading, causing damage. This single reactive process governs reactive damage. In reactive fatigue we assume that three reactions may take place simultaneously: (1) Intact bonds of an elastic material may break in response to loading (*elastic damage*), (2) Intact bonds of an elastic material may fatigue in response to repetitive loading, thereby reducing their damage threshold but maintaining their elasticity, and (3) Fatigued bonds of elastic material may break in response to loading (*fatigue damage*). The two damage reactions (1 and 3 in this list) behave identically with reactive damage materials described in Section 4.6. Thus, each of these reactions requires a cumulative distribution function (CDF, Section 4.6.2) which provides the probability that bonds will break at a given threshold of the damage criterion  $\Xi$  (Section 4.6.3). It is the user's responsibility to ensure that the CDF of fatigued bonds produces failure at a lesser threshold than the CDF of elastic bonds.

The fatigue reaction (2 in the list given above) progresses with a reaction rate

$$k^{if} = k_0^{if} \left( \left| \dot{\Xi}^i \right| w^b \right)^\beta$$

where  $k_0^{if}$  and  $\beta$  are user-specified parameters.  $\dot{\Xi}^i$  is the time rate of change of the damage criterion  $\Xi^i$  of intact bonds of the elastic solid, whereas  $w^b$  is the mass fraction of broken bonds in the elastic solid (also known as the *damage D*).

*Remarks:* The parameter  $k_0^{if}$  determines the rate at which the fatigue reaction proceeds, for a given  $\beta$ . Decreasing  $k_0^{if}$  shifts the  $S - N$  curve (failure stress versus number of cycles) to the right. The parameter  $\beta$  affects the shape of the  $S - N$  curve, for a given  $k_0^{if}$ . Increasing  $\beta$  causes the  $S - N$  curve to become more sigmoidal. It is acceptable to use the same criterion for elastic and fatigue damage. To model fatigue failure, the CDF for fatigue damage should be to the left of the CDF for elastic damage on a CDF vs  $\Xi$  graph (i.e., amount of damage at a given  $\Xi$  should be lower for elastic damage than for fatigue damage,  $F^i(\Xi) < F^f(\Xi)$ ). The fatigue reaction may not be initiated unless some elastic damage has taken place, since  $k^{if} = 0$  when  $w^b = 0$ . However, the amount of elastic damage needed to initiate the fatigue reaction may be very small,  $w^b \ll 1$ .

### 4.7.1 General Specification of Reactive Fatigue Material

The material types for reactive fatigue materials are “*reactive fatigue*” and “*uncoupled reactive fatigue*”. The following parameters must be defined:

|                     |                                                                                                |
|---------------------|------------------------------------------------------------------------------------------------|
| <k0>                | Specification of the reaction rate constant $k_0^{if}$ for the fatigue reaction (units vary)   |
| <beta>              | Specification of the power exponent $\beta$ for the fatigue reaction ( $\beta > 0$ )           |
| <elastic>           | Specification of the elastic material                                                          |
| <elastic_damage>    | Specification of the cumulative distribution function $F^i(\Xi^i)$ for damage of intact bonds  |
| <fatigue_damage>    | Specification of the cumulative distribution function $F^f(\Xi^f)$ for damage of fatigue bonds |
| <elastic_criterion> | Specification of the elastic damage criterion $\Xi^i$                                          |
| <fatigue_criterion> | Specification of the fatigue damage criterion $\Xi^f$                                          |

The <elastic> tag encloses a description of the constitutive relation of the intact elastic material and associated material properties, as may be selected from the list provided in Section 4.1.4 for

unconstrained materials (used with “*elastic damage*”) and Section 4.1.2 for uncoupled materials (used with “*uncoupled elastic damage*”). The <elastic\_damage> or <fatigue\_damage> tags enclose a description of the cumulative distribution function and associated material properties for elastic or fatigue damage, as may be selected from the list presented in Section 4.6.2. The <elastic\_criterion> and <fatigue\_criterion> tags enclose a description of the damage criterion for intact and fatigued bonds of the elastic solid, as may be selected from the list presented in Section 4.6.3.

*Example:*

```
<material id="1" type="elastic damage">
  <k0>0.001</k0>
  <beta>2</beta>
  <elastic type="neo-Hookean">
    <density>1</density>
    <E>200000</E>
    <v>0.3</v>
  </elastic>
  <elastic_damage type="CDF Weibull">
    <alpha>20</alpha>
    <mu>200</mu>
    <Dmax>1</Dmax>
  </elastic_damage>
  <fatigue_damage type="CDF Weibull">
    <alpha>10</alpha>
    <mu>100</mu>
    <Dmax>1</Dmax>
  </fatigue_damage>
  <elastic_criterion type="DC von Mises stress"/>
  <fatigue_criterion type="DC von Mises stress"/>
</material>
```

## 4.8 Reactive Plasticity

### 4.8.1 Kinematic “Hardening” Response

Reactive plasticity models a material as a mixture of bonds that break in response to loading and reform in a stressed state [93]. This framework is based on constrained reactive mixtures of solids [8]. A reactively plastic solid consists of a mixture of  $n_f$  bond families, all of which share the same elastic response characterized by the user-defined strain energy density  $\Psi_0$  (Section 4.1.4), each of which yields at a different threshold  $\Phi_{m\beta}$  of a user-defined yield measure  $\Phi$  (such as the von Mises stress, see Section 4.6.3). When all  $n_f$  bond families have yielded, the response becomes perfectly plastic, implying that the yield measure no longer increases with increasing strain. When  $n_f > 1$  the stress-strain response exhibits the characteristic kinematic “hardening” response consistent with the Bauschinger effect of classical plasticity theory [83]. Optionally, an additional bond family may be included that never yields, remaining elastic over the entire loading history and imparting a linear “hardening” regime.

The deformation gradient of a reactive plasticity material is  $\mathbf{F}^s$  and its associated mixture stress  $\boldsymbol{\sigma}(\mathbf{F}^s)$  is evaluated as

$$\boldsymbol{\sigma}(\mathbf{F}^s) = \sum_{\beta} w_{\beta} \left( w_{\beta}^s \boldsymbol{\sigma}_0(\mathbf{F}^s) + w_{\beta}^y \boldsymbol{\sigma}_0(\mathbf{F}_{\beta}^y) \right), \quad (4.8.1)$$

where  $w_{\beta}$  is the user-defined mass fraction of each bond family  $\beta$  in the mixture,  $w_{\beta}^s$  is the evolving mass fraction of intact bonds in family  $\beta$  which still exhibit an elastic response, and  $w_{\beta}^y$  is the evolving mass fraction of bonds in family  $\beta$  which have yielded (broken and reformed in a new reference configuration). These mass fractions satisfy  $\sum_{\beta} w_{\beta} = 1$  and  $w_{\beta}^s + w_{\beta}^y = 1$ . The bond breaking-and-reforming reaction is denoted by  $\mathcal{E}_{\beta}^s \rightarrow \mathcal{E}_{\beta}^y$ , where  $\mathcal{E}_{\beta}^s$  is the species associated with intact bonds while  $\mathcal{E}_{\beta}^y$  is associated with yielded bonds, in bond family  $\beta$ . The mass fractions of bond family  $\beta$  satisfy  $w_{\beta}^s = 1$  and  $w_{\beta}^y = 0$  prior to yielding, and  $w_{\beta}^s = 0$  and  $w_{\beta}^y = 1$  after yielding. Here,  $\mathbf{F}_{\beta}^y$  is the deformation gradient of yielded bonds  $\beta$  relative to the deformation when the yield threshold  $\Phi_{m\beta}$  was first reached; it is calculated from the relation  $\mathbf{F}^s = \mathbf{F}_{\beta}^y \cdot \mathbf{F}_{\beta}^{ys}$ , where  $\mathbf{F}_{\beta}^{ys}$  is a function of state that maps the yielded configuration  $\mathbf{X}^y$  relative to the global reference configuration  $\mathbf{X}^s$ . The specific form of the constitutive model for  $\mathbf{F}_{\beta}^{ys}$  used in FEBio can be found in [93] and the *FEBio Theory Manual*. Currently that constitutive model is not user-definable, though the user has the option to enforce isochoric plastic flow, implying that  $\det \mathbf{F}_{\beta}^{ys} = 1$ . The stress  $\boldsymbol{\sigma}_0$  in the above relation for  $\boldsymbol{\sigma}(\mathbf{F}^s)$  may be evaluated for any argument  $\mathbf{F}$  as

$$\boldsymbol{\sigma}_0(\mathbf{F}) = \frac{1}{J} \frac{\partial \Psi_0}{\partial \mathbf{F}} \cdot \mathbf{F}^T, \quad (4.8.2)$$

where  $J = \det \mathbf{F}$ .

### 4.8.2 Plastic Yield Surface

The plastic yield surface is defined as

$$\varphi(\mathbf{F}) = \Phi(\mathbf{F}) - \Phi_m \leq 0$$

where  $\Phi(\mathbf{F})$  represents the yield measure (e.g., the von Mises stress), as given in Section 4.6.3. The value of  $\Phi_m$  may be a constant (for elastic-perfectly-plastic behavior), or it may evolve along

a plastic flow curve, as described in the next section (Section 4.8.3). (In a strict theoretical sense,  $\Phi_m$  is a constant in the framework of constrained reactive mixtures, however FEBio uses an implementation where multiple plastically-yielding bond families  $\beta$  may coexist, each having its own value  $\Phi_{m\beta}$ .)

### 4.8.3 Plastic Flow Curve

Since each bond family behaves elastically until it yields, a family's yield threshold  $\Phi_{m\beta}$  is generally not the value recorded on a stress-strain curve when the slope changes. That value may be called the *apparent yield threshold*  $\Upsilon_\beta$ , which can be related to the true yield threshold  $\Phi_{m\beta} = \Phi(\varepsilon_\beta)$ , where  $\varepsilon_\beta$  is the true (logarithmic or Hencky) strain at which bond family  $\beta$  yields, by assuming a linear elastic stress-strain relationship prior to yielding,  $\Phi(\varepsilon) = E\varepsilon$ , where  $E$  is the elastic material's Young's modulus. This assumption makes it possible to relate the bond family mass fractions  $w_\beta$  to the values of  $\Phi_{m\beta}$  and  $\Upsilon_\beta$  via

$$\Phi_{m\beta} = \Phi_{m,\beta-1} + \frac{\Upsilon_\beta - \Upsilon_{\beta-1}}{1 - \sum_{b=0}^{\beta-1} w_b}, \quad \Phi_{m0} = \Upsilon_0.$$

More details can be found in [93]. FEBio offers several different methods for specifying the plastic flow curve parameters  $\Upsilon_\beta$ , which are used internally to evaluate  $\Phi_{m\beta}$  and  $w_\beta$ :

#### 4.8.3.1 User-Specified Flow Curve

A user-specified flow curve may be provided using the plastic flow curve material type “*PFC user*”. The user enters a list of points representing  $(\varepsilon_\beta, \Upsilon_\beta)$ , where  $\varepsilon_\beta$  is the true strain corresponding to the apparent yield threshold  $\Upsilon_\beta$ . Here,  $\varepsilon_\beta$  represents the total strain, not the plastic strain. The first point provided in this list,  $(\varepsilon_0, \Upsilon_0)$ , corresponds to the yield strain  $\varepsilon_0$  and yield stress  $\Upsilon_0$  for this material, such that the ratio  $\Upsilon_0/\varepsilon_0$  is used internally to calculate Young's modulus  $E$  for the selected material used in this plasticity analysis. It is the user's responsibility to ensure this consistency. The number of bond families,  $n_f$ , is automatically set to the number of user-specified points on this flow curve. To maintain computational efficiency it is recommended to keep this number small, e.g., no greater than 30 points. Specifying only one point produces an elastic-perfectly plastic material response.

The response produced by the reactive plasticity material will agree nearly exactly with the user-specified points as long as the strain  $\varepsilon$  remains in the infinitesimal range. However, as the strain  $\varepsilon$  becomes finite, the reactive plasticity response may deviate non-negligibly from the user-specified flow curve, due to the nonlinearity of the hyperelastic relation describing the elastic response of the material. In practice, for isotropic elastic responses, using a “*natural neo-Hookean*” material (Section 4.1.4.19) will produce the least amount of deviation when using large strains, since its uniaxial stress-strain response is nearly linear with  $\varepsilon$ ,  $\Phi(\varepsilon) = E\varepsilon e^{-(1-2\nu)\varepsilon}$ , where  $E$  is Young's modulus and  $\nu$  is Poisson's ratio. As  $\nu$  approaches  $\frac{1}{2}$  this expression produces an exact linear response under uniaxial loading,  $\Phi(\varepsilon) = E\varepsilon$ , even under finite deformation. When using  $\nu < \frac{1}{2}$ , care must be taken not to exceed  $\varepsilon_{\max} = (1 - 2\nu)^{-1}$ , since the slope of  $\Phi(\varepsilon)$  becomes zero at that value and negative beyond it.

#### 4.8.3.2 Mathematical Expression for Flow Curve

Alternatively, a mathematical expression may be supplied to describe  $\Upsilon(\varepsilon)$ , using the plastic flow curve material type “*PFC math*”. The following parameters must also be defined:

|        |                                                                                   |     |
|--------|-----------------------------------------------------------------------------------|-----|
| <nf>   | Number of bond families $n_f$ ( $n_f > 0$ )                                       | [ ] |
| <e0>   | The value of $\varepsilon_0$ that produces $\Upsilon(\varepsilon_0) = \Upsilon_0$ | [ ] |
| <emax> | The highest value of $\varepsilon$ expected in this analysis                      | [ ] |

It is the user's responsibility to ensure that the supplied mathematical function produces  $\Upsilon_0/\varepsilon_0$  equal to Young's modulus for the elastic material model being used. The actual response produced by the reactive plasticity material will agree nearly exactly with the mathematical expression for  $\Upsilon(\varepsilon)$  as long as the strain  $\varepsilon$  remains in the infinitesimal range. However, as the strain  $\varepsilon$  becomes finite, the reactive plasticity response may deviate non-negligibly from the user-specified flow curve, due to the nonlinearity of the hyperelastic relation describing the elastic response of the material. In practice, for isotropic elastic responses, using a “*natural neo-Hookean*” material (Section 4.1.4.19) will produce the least amount of deviation when using large strains, since its uniaxial stress-strain response is nearly linear with  $\varepsilon$ ,  $\Phi(\varepsilon) = E\varepsilon e^{-(1-2\nu)\varepsilon}$ , where  $E$  is Young's modulus and  $\nu$  is Poisson's ratio. As  $\nu$  approaches  $\frac{1}{2}$  this expression produces an exact linear response under uniaxial loading,  $\Phi(\varepsilon) = E\varepsilon$ , even under finite deformation. When using  $\nu < \frac{1}{2}$ , care must be taken not to exceed  $\varepsilon_{\max} = (1 - 2\nu)^{-1}$ , since the slope of  $\Phi(\varepsilon)$  becomes zero at that value and negative beyond it.

#### 4.8.3.3 Custom Flow Curve

This custom method for specifying the flow curve was provided in the paper that introduced reactive plasticity [93], therefore it is called “*PFC paper*”. The following parameters must be defined:

|        |                                                                                                               |            |
|--------|---------------------------------------------------------------------------------------------------------------|------------|
| <nf>   | Number of bond families $n_f$ ( $n_f > 0$ )                                                                   | [ ]        |
| <Y0>   | Lowest yield threshold $\Upsilon_0$ (Figure 4.8.1a-b)                                                         | [ $\Phi$ ] |
| <Ymax> | Highest yield threshold $\Upsilon_{\max}$ (Figure 4.8.1a-b)                                                   | [ $\Phi$ ] |
| <w0>   | Fraction $w_0$ of bonds that yield at $\Upsilon_0$ ( $0 \leq w_0 \leq 1$ )                                    | [ ]        |
| <we>   | Fraction $w_e$ of bonds that never yield ( $0 \leq w_e \leq 1$ )                                              | [ ]        |
| <r>    | Bias factor $r$ for consecutive increments of $\Upsilon_\beta$ over the range $[\Upsilon_0, \Upsilon_{\max}]$ | [ ]        |

This method assumes that  $\Upsilon_\beta$  values are evenly distributed between the *initial yield threshold*  $\Upsilon_0$  (e.g., the yield stress) and a *final yield threshold*  $\Upsilon_{\max}$ , parameters which may be identified from a stress strain curve (Figure 4.8.1a-b). Beyond  $\Upsilon_{\max}$ , the material either behaves as if it is perfectly plastic (a scenario which may be valid around the ultimate strength, for example), or it transitions to a linear hardening regime.

The family mass fractions  $w_\beta$  govern the influence of each family on the overall material response. The simplest model for  $w_\beta$  involves specifying the mass fraction of the first yielding family  $w_0$ , which controls the slope of the initial post-yield response (Figure 4.8.1a), and then evenly weighting the rest of the bond families. In cases where the material transitions to a linear hardening regime, we can recover this behavior by adding one more bond family,  $\beta = n_f$ , that never yields, thus remaining elastic. The associated mass fraction  $w_\beta$  for  $\beta = n_f$  is called the *elastic mass fraction* and denoted  $w_e$ ; a non-zero value for this parameter may be specified whenever we wish to include linear hardening behavior (Figure 4.8.1b). The effect of the mass fraction parameters  $w_0$  and  $w_e$  is explored parametrically in Figure 4.8.1c and Figure 4.8.1d, respectively. In general, most ductile materials have  $w_0$  very close to unity, which provides hardening behavior over a finite strain range. As  $w_0 \rightarrow 1$  the stress-strain behavior approaches perfect plasticity. In contrast, when  $w_e = 0$ , the material response becomes perfectly plastic once the final yield threshold  $\Upsilon_{\max}$  has been exceeded. As  $w_e$  increases, a region of linear hardening is seen on a plot of



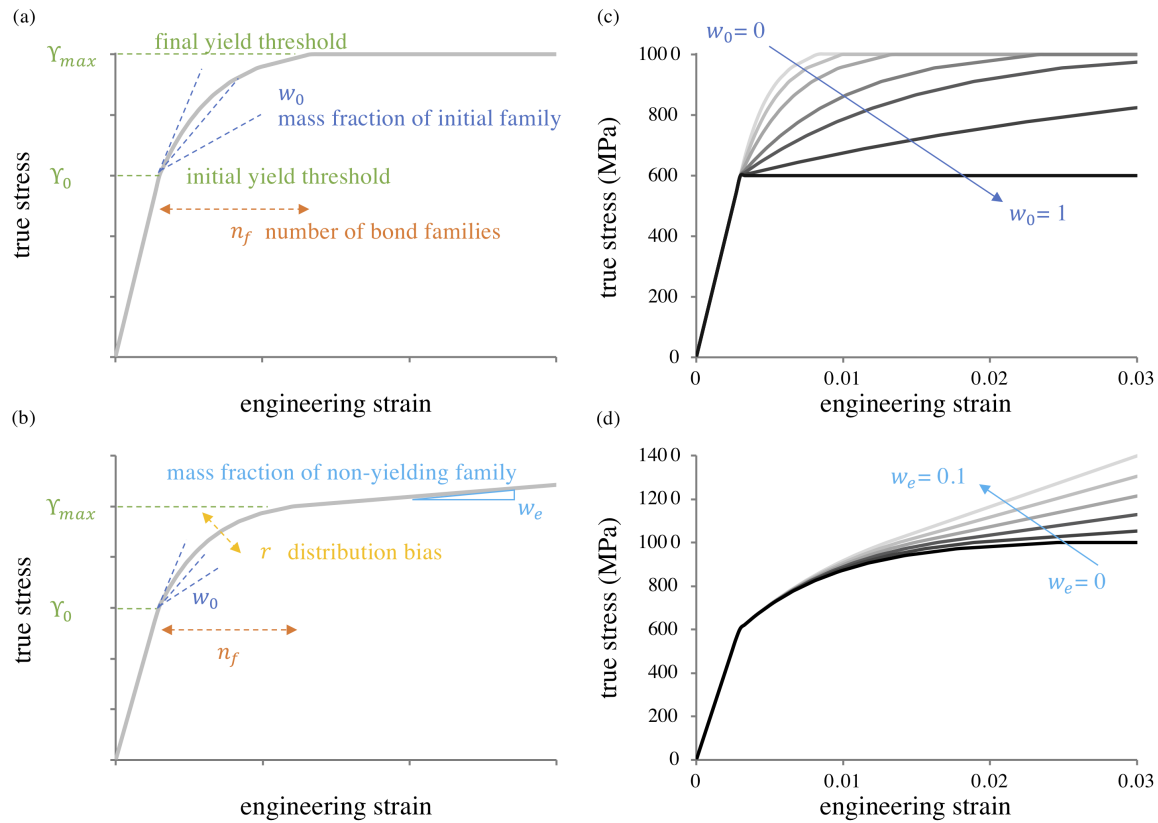


Figure 4.8.1: Modeling uniaxial stress-strain curves using the parameters given in Section 4.8.1. Identification of parameters on idealized stress-strain curves showing (a) a plateau in the stress, or (b) exhibiting a region of linear hardening. The yielding behavior is fully described by the set of parameters  $\{n_f, \Upsilon_0, \Upsilon_{max}, w_0, w_e, r\}$ . Parametric variations of (c)  $w_0$  and (d)  $w_e$  illustrate their influence on the stress-strain response; other parameters are held fixed. In (c-d)  $n_f = 10$ ,  $\Upsilon_0 = 600$  MPa,  $\Upsilon_{max} = 1000$  MPa, and  $r = 1$ . In (c),  $w_e = 0$  and in (d)  $w_0 = 0.75$ .

the true stress against strain. For most ductile materials,  $w_e$  is usually 0 or on the order of 0.001.

It is also possible to introduce a *bias factor*  $r$ , which allows a geometric progression rather than uniform spacing of the apparent yield thresholds and family mass fractions. The bias factor  $r$  has the effect of modifying the shape of the hardening region between  $\Upsilon_0$  and  $\Upsilon_{\max}$  (Figure 4.8.1b). The full set of material parameters is then given by  $\{n_f, \Upsilon_0, \Upsilon_{\max}, w_0, w_e, r\}$ . Setting  $r = 1$  recovers the uniform distribution. Figure 4.8.1a-b graphically describes the influence of each parameter on simplified stress-strain curves, showing how these parameters may be extracted from experimental data.

Default values in FEBio are  $n_f = 1$ ,  $w_0 = 1$ ,  $w_e = 0$ ,  $\Upsilon_0 = \Upsilon_{\max} = 0$ ,  $r = 0.9$ .

#### 4.8.4 Specification of Reactive Elasto-Plastic Solid

The material type of a reactive elasto-plastic solid with kinematic hardening is “*reactive plasticity*”. The following parameters must be defined:

|                   |                                                                                                                     |
|-------------------|---------------------------------------------------------------------------------------------------------------------|
| <isochoric>       | Flag (0 or 1) for enforcing isochoric plastic flow (1 implies $\det \mathbf{F}_\beta^{ys} = 1, \forall \beta$ ) [ ] |
| <elastic>         | Specification of the elastic material (Section 4.1.4)                                                               |
| <yield_criterion> | Specification of the yield criterion $\Phi$ (Section 4.6.3)                                                         |
| <flow_curve>      | Specification of the plastic flow curve (Section 4.8.3)                                                             |

The <elastic> tag encloses a description of the constitutive relation of the intact elastic material and associated material properties, as may be selected from the list provided in Section 4.1.4 for unconstrained materials. The <yield\_criterion> tag encloses a description of the yield criterion  $\Phi$ , as may be selected from the list presented in Section 4.6.3. The <flow\_curve> tag encloses a description of the relation between bond mass fractions  $w_\beta$  and apparent yield stresses  $\Upsilon_\beta$  for the bond family  $\beta$  as described in Section 4.8.3. The default value in FEBio is isochoric=1.

*Example: Idealized elastic-perfectly plastic response of steel (mm-N-s units)*

```
<material id="1" type="reactive plasticity">
  <elastic type="neo-Hookean">
    <density>8.05e-9</density>
    <E>200e3</E>
    <v>0.30</v>
  </elastic>
  <yield_criterion type="DC von Mises stress"/>
  <flow_curve type="PFC paper">
    <Y0>450</Y0>
  </flow_curve>
</material>
```

*Example: Annealed mild steel (mm-N-s units)*

```
<material id="1" type="reactive plasticity">
  <isochoric>1</isochoric>
  <elastic type="neo-Hookean">
    <density>7.85e-9</density>
    <E>205e3</E>
```

```

    <v>0.29</v>
  </elastic>
  <yield_criterion type="DC von Mises stress"/>
  <flow_curve type="PFC paper">
    <nf>15</nf>
    <Y0>220</Y0>
    <Ymax>490</Ymax>
    <w0>0.973</w0>
    <we>0</we>
    <r>1</r>
  </flow_curve>
</material>

```

*Example: Annealed copper (mm-N-s units)*

```

<material id="1" type="reactive plasticity">
  <isochoric>1</isochoric>
  <elastic type="neo-Hookean">
    <density>7.764e-9</density>
    <E>120e3</E>
    <v>0.34</v>
  </elastic>
  <yield_criterion type="DC von Mises stress"/>
  <flow_curve type="PFC paper">
    <nf>15</nf>
    <Y0>60</Y0>
    <Ymax>288</Ymax>
    <w0>0.988</w0>
    <we>0</we>
    <r>1</r>
  </flow_curve>
</material>

```

*Example: Unaged maraging steel (mm-N-s units)*

```

<material id="1" type="reactive plasticity">
  <isochoric>1</isochoric>
  <elastic type="neo-Hookean">
    <density>8.00e-9</density>
    <E>165e3</E>
    <v>0.33</v>
  </elastic>
  <yield_criterion type="DC von Mises stress"/>
  <flow_curve type="PFC paper">
    <nf>22</nf>
    <Y0>398</Y0>
    <Ymax>1010</Ymax>
    <w0>0</w0>
    <we>0</we>
  </flow_curve>
</material>

```

```

    <r>0.9</r>
  </flow_curve>
</material>

```

*Example: Annealed aluminum 1100 (mm-N-s units)*

```

<material id="1" type="reactive plasticity">
  <isochoric>1</isochoric>
  <elastic type="neo-Hookean">
    <density>2.71e-9</density>
    <E>68e3</E>
    <v>0.33</v>
  </elastic>
  <yield_criterion type="DC von Mises stress"/>
  <flow_curve type="PFC paper">
    <nf>18</nf>
    <Y0>63</Y0>
    <Ymax>112</Ymax>
    <w0>0.994</w0>
    <we>0</we>
    <r>0.6</r>
  </flow_curve>
</material>

```

*Example: Mild steel (mm-N-s units)*

```

<material id="1" type="reactive plasticity">
  <isochoric>1</isochoric>
  <elastic type="natural neo-Hookean">
    <density>2.71e-9</density>
    <E>206e3</E>
    <v>0.30</v>
  </elastic>
  <yield_criterion type="DC von Mises stress"/>
  <flow_curve type="PFC math">
    <nf>15</nf>
    <emin>0.0008403</emin>
    <emax>1.3</emax>
    <plastic_response>545.46*(0.011024+eps)^0.2589</plastic_response>
  </flow_curve>
</material>

```

*Example: Steel (in-lbf-s units)*

```

<material id="1" name="Steel" type="reactive plasticity">
  <density>1</density>
  <isochoric>1</isochoric>
  <elastic type="natural neo-Hookean">
    <E>29911000</E>

```

```

    <v>0.3</v>
  </elastic>
  <yield_criterion type="DC von Mises stress"/>
  <flow_curve type="PFC user">
    <plastic_response type="point">
      <interpolate>SMOOTH</interpolate>
      <points>
        <point> 0.002, 59822 </point>
        <point> 0.002841, 64450 </point>
        <point> 0.00469, 68500 </point>
        <point> 0.00953, 72000 </point>
        <point> 0.0193, 75000 </point>
      </points>
    </plastic_response>
  </flow_curve>
</material>

```

*Example: elastic-perfectly plastic response, using PFC math flow curve*

```

<material id="1" name="Elastoplastic Math" type="reactive plasticity">
  <density>1</density>
  <isochoric>1</isochoric>
  <elastic type="neo-Hookean">
    <E>200000</E>
    <v>0.3</v>
  </elastic>
  <yield_criterion type="DC von Mises stress"/>
  <flow_curve type="PFC math">
    <nf>1</nf>
    <e0>0.001</e0>
    <emax>1</emax>
    <plastic_response type="math">200</plastic_response>
  </flow_curve>
</material>

```

## 4.9 Reactive Elasto-Plastic Damage Mechanics

Plastic deformation (Section 4.8) is often coupled with damage, as the finite deformation and plastic flow of a loaded material typically induces some amount of failure. Within the constrained reactive mixture framework adopted in FEBio, damage is produced by bonds breaking permanently (Section 4.6), which reduces the generation mass fractions  $w_\beta^s$  or  $w_\beta^y$  [93]. In our treatment of elastoplastic damage we assume that both intact and yielded bonds may become damaged. Intact bonds belong to the  $s$ -generation which is present at  $t = -\infty$ . Once yielding occurs, all successive generations of that family are labeled as yielded bonds  $y$ . This distinction is necessary so we can then distinguish between damage to intact bonds (elastic damage) and damage to yielded bonds (plastic damage), since intact bonds which get damaged never have the ability to yield. Damage to intact bonds is represented by the reaction  $\mathcal{E}_\beta^s \rightarrow \mathcal{E}_\beta^b$ , where  $\mathcal{E}_\beta^b$  is the species associated with broken bonds in bond family  $\beta$ . It may represent some initial damage value for a material with defects, or damage due to intermolecular failure of bonds that never yielded; this damage alters the value of  $w_\beta^s$  and we refer to it as *elastic damage*. Damage to yielded bonds represents *plastic damage*; it is represented by the reaction  $\mathcal{E}_\beta^y \rightarrow \mathcal{E}_\beta^b$  and it alters the value of  $w_\beta^y$ . The mechanism of damage and the failure measure may be different for these two types of bonds, particularly since a stress- or energy-based failure measure may not be appropriate for plastic damage. It is important to note that the nature of the plastic deformation described in Section 4.8 remains unchanged. Damage modifies the material behavior by reducing the fraction of bonds in various generations, which scales the response accordingly.

In a reactive constrained mixture framework the insertion of damage into the reactive plasticity formulation is straightforward. Since bonds break permanently in a damage reaction, there is no need to define a function of state  $\mathbf{F}_\beta^{ys}$  to describe a (non-existing) reformed configuration. Furthermore, the specific free energy of broken bonds is zero. The scalar *elastic damage criterion*  $\Xi^e(\mathbf{F}^s)$ , which is taken to have the same functional form for all bond families  $\beta$ , is the analog to the yield criterion  $\Phi$  for plasticity. As shown in Section 4.6, the main contrast with reactive plasticity is that not all bonds in the family  $\beta$  break simultaneously at a single *elastic damage threshold*  $\Xi_{m\beta}^e$ . Instead, the fraction of broken bonds varies as a function of  $\Xi^e(\mathbf{F}^s)$ , denoted by  $F^e(\Xi_\beta^e)$ , such that  $0 \leq F^e(\Xi_\beta^e) \leq 1$ . Here,  $F^e(\Xi_\beta^e)$  is a function of state; it must be a monotonically increasing function of its argument to satisfy the Clausius-Duhem inequality [66]. We may view  $F^e$  as a cumulative distribution function (CDF), whose corresponding probability distribution function (PDF) represents the probability of damage at a particular value of  $\Xi_\beta^e$ .

### 4.9.1 Theoretical Formulation

Here, we sketch the structure of the elastoplastic damage theory in FEBio. Since each bond family in reactive plasticity yields all at once, we can easily split an elastoplastic damage theory into two parts to represent elastic and plastic damage regimes. Assume that the first yielding reaction for bond family  $\beta$  occurs at time  $t = u_\beta$ . Prior to this initial yielding, the damage behavior described in Section 4.6 applies, and the material composition is generally a mixture of intact ( $\mathcal{E}_\beta^s$ ) and broken ( $\mathcal{E}_\beta^b$ ) bonds satisfying the reaction  $\mathcal{E}_\beta^s \rightarrow \mathcal{E}_\beta^b$ . The corresponding bond mass fractions satisfy  $1 = w_\beta^s + w_\beta^b$  and  $w_\beta^y = 0$ , where  $w_\beta^b = F^e(\Xi_\beta^e)$  is the elastic damage in bond family  $\beta$ . At  $t = u_\beta$ , the remaining intact bonds  $w_\beta^s = 1 - w_\beta^b$  all yield, following the reaction  $\mathcal{E}_\beta^s \rightarrow \mathcal{E}_\beta^y$ . The family mass balance is then given as  $1 = w_\beta^y + w_\beta^b$ , since  $w_\beta^s = 0$  after yielding.

For time  $t > u_\beta$ , yielded bonds may continue to yield, but they may also sustain damage

according to the reaction  $\mathcal{E}_\beta^y \rightarrow \mathcal{E}_\beta^b$ , which reduces their mass fraction  $w_\beta^y$ . Damage to yielded bonds may occur based on a function of state (often described as a plastic strain, though it is not an observable kinematic variable), which is distinct from the measure of elastic damage. Therefore, we denote the *plastic damage measure* as  $\Xi^p(\mathbf{F}_\beta^{ys})$  and its cumulative distribution function by  $F^p(\Xi_\beta^p)$ , under the assumption that all bond families  $\beta$  share the same functional forms for  $\Xi^p$  and  $F^p$ . For each bond family  $\beta$ , only the remaining undamaged fraction  $1 - F^p(\Xi_\beta^p)$  of yielded bonds may break and reform as the next yielded generation.

It must be recognized that, just as  $\mathbf{F}_\beta^{ys}$  is a constitutively-prescribed function of state and does not carry the meaning of a plastic deformation gradient, the plastic damage measure  $\Xi^p(\mathbf{F}_\beta^{ys})$  is also a function of state. This quantity is called a plastic strain for convenience only.

#### 4.9.1.1 Stress and Damage

Recognizing that damaged (broken) bonds do not store free energy, the mixture stress is given by

$$\boldsymbol{\sigma} = \sum_{\beta} w_{\beta} \left( w_{\beta}^s \boldsymbol{\sigma}_0(\mathbf{F}^s) + w_{\beta}^y \boldsymbol{\sigma}_0(\mathbf{F}_\beta^y) \right), \quad (4.9.1)$$

where the stresses  $\boldsymbol{\sigma}_0$  are given by the standard hyperelasticity relation in Eq.(4.8.2). The reactive mixture equivalent of the damage variable  $D$  may be evaluated for elastoplastic damage as the fraction of all bonds that are broken,

$$D = w^b = \sum_{\beta} w_{\beta} w_{\beta}^b. \quad (4.9.2)$$

#### 4.9.1.2 Damage Measures

For elastic damage, we may use the same functional measure as proposed for plastic yielding (e.g., the von Mises stress); this implies that the functions  $\Xi^e$  and  $\Phi$  have the same form. For plastic damage of bonds in family  $\beta$ , we use the constitutively-determined mapping  $\mathbf{F}_\beta^{ys}$  to define plastic right Cauchy-Green and Lagrange strain tensors through

$$\begin{aligned} \mathbf{C}_\beta^{ys} &= \left( \mathbf{F}_\beta^{ys} \right)^T \cdot \mathbf{F}_\beta^{ys} \\ \mathbf{E}_\beta^{ys} &= \frac{1}{2} \left( \mathbf{C}_\beta^{ys} - \mathbf{I} \right). \end{aligned} \quad (4.9.3)$$

One possible constitutive relation for  $\Xi^p$ , which remains valid for general deformations, is to set it equal to the *effective plastic strain*  $e_\beta^p$  for the various bond families  $\beta$ ,

$$e_\beta^p = \sqrt{\frac{2}{3} \text{dev } \mathbf{E}_\beta^{ys} : \text{dev } \mathbf{E}_\beta^{ys}}. \quad (4.9.4)$$

In a numerical implementation, the effective plastic strain  $e_0^p$  of the first bond family to yield may be reported as the effective plastic strain in the entire material, for consistency with plastic strain measures in classical models of plasticity.

Quantities in this section do not represent plastic strains or plastic strain tensors, though we adopt the terminology due to similarities. Recall that the non-observable function of state  $\mathbf{F}_\beta^{ys} =$

$\mathbf{F}_\beta^{ys}(\mathbf{F}^s, \rho_{r\beta}^\alpha)$  is a time-invariant mapping providing the reference configuration of a yielded bond  $y$  with respect to the reference configuration of the master constituent  $s$ , for family  $\beta$ . The quantities  $\mathbf{C}_\beta^{ys}$  and  $\mathbf{E}_\beta^{ys}$  then also represent non-observable functions of state calculated as strain tensors. Consequently,  $e_\beta^p$  is a measure of the relative motion of the reference configuration of bond family  $\beta$ , expressed as a scalar “strain”. Physically, this amounts to the modeling assumption that once the breaking-and-reforming process takes a bond family out of a local neighborhood centered about its original position, the bond begins to degrade with further breaking-and-reforming processes. That each of these quantities exists for every bond family  $\beta$  emphasizes the lack of any true or unique plastic strain measure in this framework.

## 4.9.2 Specification of Reactive Elasto-Plastic Damage Solid

The material type of a reactive elasto-plastic damage solid with kinematic hardening is “*reactive plastic damage*”. The following parameters must be defined:

|                            |                                                                                                                     |
|----------------------------|---------------------------------------------------------------------------------------------------------------------|
| <isochoric>                | Flag (0 or 1) for enforcing isochoric plastic flow (1 implies $\det \mathbf{F}_\beta^{ys} = 1, \forall \beta$ ) [ ] |
| <elastic>                  | Specification of the elastic material (Section 4.1.4)                                                               |
| <yield_criterion>          | Specification of the yield criterion $\Phi$ (Section 4.6.3)                                                         |
| <flow_curve>               | Specification of the plastic flow curve (Section 4.8.3)                                                             |
| <plastic_damage>           | Specification of the plastic damage CDF $F^p(\Xi_\beta^p)$                                                          |
| <plastic_damage_criterion> | Specification of the plastic damage criterion $\Xi^p(\mathbf{F}_\beta^{ys})$                                        |
| <elastic_damage>           | Specification of the elastic damage CDF $F^e(\Xi_\beta^e)$                                                          |
| <elastic_damage_criterion> | Specification of the elastic damage criterion $\Xi^e(\mathbf{F}^s)$                                                 |

The <elastic> tag encloses a description of the constitutive relation of the intact elastic material and associated material properties, as may be selected from the list provided in Section 4.1.4 for unconstrained materials. The <yield\_criterion> tag encloses a description of the yield criterion  $\Phi$ , as may be selected from the list presented in Section 4.6.3. The <flow\_curve> tag encloses a description of the relation between bond mass fractions  $w_\beta$  and apparent yield stresses  $\Upsilon_\beta$  for the bond family  $\beta$  as described in Section 4.8.3. The plastic and elastic damage CDFs, in the <plastic\_damage> and <elastic\_damage> tags respectively, may be selected from Section 4.6.2, whereas the corresponding damage criteria <plastic\_damage\_criterion> and <elastic\_damage\_criterion> may be selected from Section 4.6.3. Default values are  $n_f = 1$ ,  $w_0 = 1$ ,  $w_e = 0$ ,  $\Upsilon_0 = \Upsilon_{\max} = 0$ ,  $r = 0.9$ , and isochoric=1. If only plastic damage needs to be modeled, the tags <elastic\_damage> and <elastic\_damage\_criterion> may be omitted. If <plastic\_damage> and <plastic\_damage\_criterion> are also omitted the material response becomes identical to the “*reactive plasticity*” material described in Section 4.8.4.

*Example: A-533 Grade B class 1 nuclear pressure vessel steel (mm-N-s units)*

```
<material id="1" type="reactive plasticity">
  <isochoric>1</isochoric>
  <elastic type="neo-Hookean">
    <density>8.00e-9</density>
    <E>206.9e3</E>
    <v>0.29</v>
```



```
</elastic>
<yield_criterion type="DC von Mises stress"/>
<flow_curve type="PFC paper">
  <nf>10</nf>
  <Y0>458</Y0>
  <Ymax>730</Ymax>
  <w0>0.985</w0>
  <we>0.00072</we>
  <r>1</r>
</flow_curve>
<plastic_damage type="CDF Weibull">
  <mu>3.3</mu>
  <alpha>3</alpha>
  <Dmax>1</Dmax>
</plastic_damage>
<plastic_damage_criterion type="DC octahedral shear strain"/>
</material>
```

## 4.10 Reactive Viscoplasticity

A reactive viscoelastic material (Section 4.5) consists of an “*elastic*” material, which imparts the equilibrium elastic response, and a “*bond*” material, which imparts the viscous response. The “*elastic*” response may be attributed to strong molecular bonds (e.g., covalent bonds) in a material region, that don’t break in the absence of damage, whereas the “*bond*” response is attributed to weak molecular bonds (e.g., hydrogen bonds) in the material region, that break under loading, and reform in a stress-free state.

To model viscoplasticity in a reactive framework, one may use a “*reactive plasticity*” material (Section 4.8) as the type of “*elastic*” material appearing in a “*reactive viscoelastic*” material. The “*bond*” material may be modeled as recommended previously (Section 4.5).

*Example: Reactive viscoelasticity with reactive plasticity*

```
<material id="2" name="Material2" type="reactive viscoelastic">
<density>1</density>
<wmin>0.001</wmin>
<kinetics>1</kinetics>
<trigger>0</trigger>
<emin>0</emin>
<elastic type="reactive plasticity">
<isochoric>1</isochoric>
<rtol>0.0001</rtol>
<secant_tangent>1</secant_tangent>
<elastic type="neo-Hookean">
<E>73000</E>
<v>0.3</v>
</elastic>
<yield_criterion type="DC von Mises stress"/>
<flow_curve type="PFC paper">
<Y0>330</Y0>
<Ymax>489</Ymax>
<w0>0.967</w0>
<we>0</we>
<nf>15</nf>
<r>1</r>
</flow_curve>
</elastic>
<bond type="neo-Hookean">
<density>1</density>
<E>7300</E>
<v>0.3</v>
</bond>
<relaxation type="relaxation-exponential">
<tau>1.5</tau>
</relaxation>
</material>
```

## 4.11 Reactive Viscoelasticity and Viscoplasticity with Damage

A reactive viscoelastic material may undergo damage as detailed in [15]. Reactive viscoelastic materials (Section 4.5) consist of an “*elastic*” material, which imparts the equilibrium elastic response, and a “*bond*” material, which imparts the viscous response. The “*elastic*” response may be attributed to strong molecular bonds (e.g., covalent bonds) in a material region, that don’t break in the absence of damage, whereas the “*bond*” response is attributed to weak molecular bonds (e.g., hydrogen bonds) in the material region, that break under loading, and reform in a stress-free state.

It follows from this definition that the “*elastic*” material of a “reactive viscoelastic” material may undergo damage, as described for example in Section 4.6. This damage is caused by the breaking of strong molecular bonds in the material region, and the associated damage variable  $D$  represents the mass fraction of the material region whose strong bonds have broken. However, when strong molecular bonds break, the gap between molecular constituents of that material region may increase, also preventing the formation of weak molecular bonds (e.g., since hydrogen bonds may only form when there is sufficient proximity between moieties associated with these bonds). Based on the work of Simo [79], we can adopt the constitutive assumption that the fraction of weak bonds that fails is the same as the fraction  $D$  of strong bonds that fail [15]. Therefore, the stress  $\sigma^e$  produced by strong bonds in a material region may be attenuated as  $(1 - D) \sigma^e$ , just like the stress  $\sigma^b$  produced by weak bonds may be reduced to  $(1 - D) \sigma^b$ .

The “reactive viscoelastic” material in FEBio automatically recognizes if its “*elastic*” component happens to be a “*elastic damage*” material, or a “*reactive plastic damage*” material. In either of these cases, it will prescribe the damage  $D$  calculated from the “*elastic*” response to the “*bond*” response. In other words, it is possible to model viscoelastic damage and viscoplastic damage in FEBio.

*Example: Reactive viscoelasticity with elastic damage*

```
<material id="1" name="Material1" type="reactive viscoelastic">
  <density>1</density>
  <wmin>0.001</wmin>
  <kinetics>1</kinetics>
  <trigger>0</trigger>
  <emin>0.0001</emin>
  <elastic type="elastic damage">
    <elastic type="neo-Hookean">
      <E>1000</E>
      <v>0.3</v>
    </elastic>
  <damage type="CDF Weibull">
    <Dmax>1</Dmax>
    <alpha>2.5</alpha>
    <mu>125</mu>
    <ploc>0</ploc>
  </damage>
  <criterion type="DC von Mises stress"/>
</elastic>
<bond type="neo-Hookean">
  <E>2000</E>
```

```

<v>0.3</v>
</bond>
<relaxation type="relaxation-exponential">
<tau>5</tau>
</relaxation>
</material>
Example: Reactive viscoelasticity with reactive plastic damage
<material id="2" name="Material2" type="reactive viscoelastic">
<density>1</density>
<wmin>0.001</wmin>
<kinetics>1</kinetics>
<trigger>0</trigger>
<emin>0</emin>
<elastic type="reactive plastic damage">
<isochoric>1</isochoric>
<rtol>0.0001</rtol>
<secant_tangent>1</secant_tangent>
<elastic type="neo-Hookean">
<E>73000</E>
<v>0.3</v>
</elastic>
<yield_criterion type="DC von Mises stress"/>
<flow_curve type="PFC paper">
<Y0>330</Y0>
<Ymax>489</Ymax>
<w0>0.967</w0>
<we>0</we>
<nf>15</nf>
<r>1</r>
</flow_curve>
<plastic_damage type="CDF Weibull">
<Dmax>1</Dmax>
<alpha>0.65</alpha>
<mu>7.95</mu>
<ploc>0</ploc>
</plastic_damage>
<plastic_damage_criterion type="DC octahedral shear strain"/>
</elastic>
<bond type="neo-Hookean">
<E>7300</E>
<v>0.3</v>
</bond>
<relaxation type="relaxation-exponential">
<tau>1.5</tau>
</relaxation>
</material>

```

## 4.12 Viscoelastic Damage

A viscoelastic material (Section 4.4) is fundamentally a continuum representation of the Standard Solid in classical viscoelasticity, valid under finite deformation in the context of quasilinear viscoelasticity. The standard solid consists of a spring  $e$  in parallel with a Maxwell element (a spring  $s$  and dashpot  $d$  in series). The spring stiffness in the Maxwell element is assumed to be equal to  $\gamma$  times the stiffness of the parallel spring. The elastic spring response may be a nonlinear function of the strain but the viscoelastic response must be linear (i.e., it must obey Boltzmann's superposition principle, equivalent to stating that the Maxwell element response is governed by a linear ordinary differential equation, or ODE). In general, a viscoelastic material of this type may undergo damage, using the framework provided by Simo [79]. In FEBio we model this framework by adopting additional constitutive assumptions not provided by Simo, but outlined in [15] and in the *FEBio Theory Manual*. The FEBio formulation for viscoelastic damage relies on a proper formulation of the strain energy density of the Standard Solid, which is then employed in the analysis of viscoelastic damage. The strain energy density  $\Psi_r(\mathbf{F})$  of the viscoelastic solid is given by

$$\Psi_r(\mathbf{F}) = \Psi_r^e(\mathbf{F}) + \gamma \Psi_r^e(\mathbf{F}_s)$$

where  $\mathbf{F}$  is the (observable) deformation gradient of the material (also of the elastic spring  $e$ ),  $\mathbf{F}_s$  is the deformation gradient associated with the Maxwell spring  $s$  (an internal variable which evolves based on a first-order ODE), and  $\Psi_r^e$  is the strain energy density of the elastic spring. Let  $\mathbf{S}^e$  denote the second Piola-Kirchhoff stress in the elastic spring, obtained from the standard hyperelasticity relation  $\mathbf{S}^e = \partial \Psi_r^e(\mathbf{F}) / \partial \mathbf{E}$ , where  $\mathbf{E}$  is the Lagrange strain tensor evaluated from  $\mathbf{F}$ . It follows from the above relation for the strain energy density that  $\mathbf{S}(\mathbf{F}) = \mathbf{S}^e(\mathbf{F}) + \gamma \mathbf{S}^e(\mathbf{F}_s)$ .

Damage  $D^e$  is produced in the elastic spring  $e$  according to the framework described in Section 4.6, based on a user-defined damage criterion  $\Xi^e(\mathbf{F})$  (such as von Mises stress or maximum principal normal stress), such that  $0 \leq D^e \leq 1$  represents the range from zero to complete damage. The resulting stress in the elastic spring is now given by the attenuated value  $(1 - D^e) \mathbf{S}^e(\mathbf{F})$ . Based on Simo [79], FEBio assumes that the damage in the Maxwell element is similarly given by  $(1 - D^e) \gamma \mathbf{S}^e(\mathbf{F}_s)$ . Thus, the stress in a damaged viscoelastic material in FEBio is evaluated from  $(1 - D^e) \mathbf{S}(\mathbf{F})$ .

The specification of a material with “*viscoelastic damage*” takes the same form as that of a viscoelastic material (Section 4.4), except that the “*elastic*” component should now consist of an “*elastic damage*” material type, whose specification is outlined in Section 4.6.

*Example: Viscoelastic damage*

```
<material id="1" name="Material1" type="viscoelastic damage">
  <density>1</density>
  <t1>5</t1>
  <t2>1</t2>
  <t3>1</t3>
  <t4>1</t4>
  <t5>1</t5>
  <t6>1</t6>
  <g0>1</g0>
  <g1>2</g1>
  <g2>0</g2>
  <g3>0</g3>
  <g4>0</g4>
```

```
<g5>0</g5>
<g6>0</g6>
<elastic type="elastic damage">
<elastic type="neo-Hookean">
<E>1000</E>
<v>0.3</v>
</elastic>
<damage type="CDF Weibull">
<Dmax>1</Dmax>
<alpha>2.5</alpha>
<mu>125</mu>
<ploc>0</ploc>
</damage>
<criterion type="DC von Mises stress"/>
</elastic>
</material>
```

## 4.13 Multigeneration Solids

This type of material [8] implements a mechanism for multigenerational interstitial growth of solids whereby each growth generation  $\gamma$  has a distinct reference configuration  $\mathbf{X}^\gamma$  determined at the time  $t^\gamma$  of its deposition. Therefore, the solid matrix of a growing material consists of a multiplicity of intermingled porous bodies, each representing a generation  $\gamma$ , all of which are constrained to move together in the current configuration  $\mathbf{x}$ . The deformation gradient of each generation is  $\mathbf{F}^\gamma = \partial \mathbf{x} / \partial \mathbf{X}^\gamma$ . The first generation ( $\gamma = 1$ ) is assumed to be present at time  $t^1 = 0$ , therefore its reference configuration is  $\mathbf{X}^1 \equiv \mathbf{X}$  and its deformation gradient  $\mathbf{F}^1 = \partial \mathbf{x} / \partial \mathbf{X}^1$  is equivalent to  $\mathbf{F} = \partial \mathbf{x} / \partial \mathbf{X}$ . Each generation's reference configuration  $\mathbf{X}^\gamma$  has a one-to-one mapping  $\mathbf{F}^{\gamma 1} = \partial \mathbf{X}^\gamma / \partial \mathbf{X}^1$  with the master reference configuration  $\mathbf{X}^1$ , which is that of the first generation. This mapping is postulated based on a constitutive assumption with regard to that generation's state of stress at the time of its deposition. In the current implementation, the newly deposited generation is assumed to be in a stress-free state, even though the underlying material is in a loaded configuration. Therefore, the mapping between generation  $\gamma$  and the first generation is simply  $\mathbf{F}^{\gamma 1} = \mathbf{F}^1(\mathbf{X}^1, t^\gamma) \equiv \mathbf{F}(\mathbf{X}, t^\gamma)$ . In other words, when generation  $\gamma$  first comes into existence, its reference configuration is the current configuration at time  $t^\gamma$ . Note that  $\mathbf{F}^{\gamma 1}$  is a time-invariant (though not necessarily homogeneous) quantity that is determined uniquely at the birth of a generation.

The state of stress in a multigeneration solid is given by

$$\boldsymbol{\sigma} = \sum_{\gamma} \frac{1}{J^{\gamma 1}} \boldsymbol{\sigma}^{\gamma}(\mathbf{F}^{\gamma})$$

where  $\boldsymbol{\sigma}^{\gamma}(\mathbf{F}^{\gamma})$  is the state of stress in the generation  $\gamma$ , as would be evaluated from a strain energy density function whose reference configuration is  $\mathbf{X}^\gamma$ . In the above equation,  $J^{\gamma 1} = \det \mathbf{F}^{\gamma 1}$  and the factor  $1/J^{\gamma 1}$  ensures that the strain energy density of each generation is properly normalized the volume of the material in the master reference configuration  $\mathbf{X}^1$ , when summing up the stresses in all the generations.

Multigeneration solids typically exhibit residual stresses when  $\mathbf{F}^{\gamma 1}$  is inhomogeneous.

### 4.13.1 General Specification of Multigeneration Solids

The material type for a multigeneration solid is “*multigeneration*”<sup>2</sup>. This material describes a mixture of elastic solids, each created in a specific generation. It is a container for any combination of the elastic materials described.

|              |                             |
|--------------|-----------------------------|
| <generation> | Definition of a generation. |
|--------------|-----------------------------|

The <generation> tag defines a new generation. It takes the following child elements.

|              |                                                             |     |
|--------------|-------------------------------------------------------------|-----|
| <start_time> | “birth”-time for this generation                            | [t] |
| <solid>      | Specification of the constitutive model for this generation |     |

The *solid* element defines the solid matrix constitutive relation and associated material properties.

*Example:*

<sup>2</sup>As of FEBio version 2.0, the format for defining multi-generation materials has changed. The previous format where the generations are defined in the *Globals* section is no longer supported.

```

<material id="1" name="Growing Solid" type="multigeneration">
  <generation id="1">
    <start_time>0.0</start_time>
    <solid type="Holmes-Mow">
      <density>1</density>
      <E>1</E>
      <v>0</v>
      <beta>0.1</beta>
    </solid>
  </generation>
  <generation id="2">
    <start_time>1.0</start_time>
    <solid type="Holmes-Mow">
      <density>1</density>
      <E>1</E>
      <v>0</v>
      <beta>0.1</beta>
    </solid>
  </generation>
</material>

```

The corresponding value of  $t^\gamma$  for each of the generations is provided in the <Globals> section.

*Example:*

```

<Globals>
  <Generations>
    <gen id="1">0.0</gen>
    <gen id="2">1.0</gen>
  </Generations>
</Globals>

```



## 4.14 Biphasic Materials

Biphasic materials may be used to model a porous medium consisting of a mixture of a porous-permeable solid matrix and an interstitial fluid. Each of these mixture constituents is assumed to be intrinsically incompressible. This means that the true densities of the solid and fluid are invariant in space and time; this assumption further implies that a biphasic mixture will undergo zero volume change when subjected to a hydrostatic fluid pressure. Yet the mixture is compressible because the pores of the solid matrix may gain or lose fluid under general loading conditions. The Cauchy stress  $\sigma$  in a biphasic material is given by

$$\sigma = -p\mathbf{I} + \sigma^e, \quad (4.14.1)$$

where  $p$  is the interstitial fluid pressure and  $\sigma^e$  is the stress resulting from the deformation of the porous solid matrix. Therefore, the constitutive relation of the solid matrix should be chosen from the list of unconstrained materials provided in Section 4.1.4. The user is referred to the [FEBio Theory Manual](#) for a general description of the biphasic theory.

In addition to selecting a constitutive relation for the solid matrix, a constitutive relation must also be selected for the hydraulic permeability of the interstitial fluid flowing within the porous deformable solid matrix. The hydraulic permeability relates the volumetric flux of the fluid relative to the solid,  $\mathbf{w}$ , to the interstitial fluid pressure gradient,  $\nabla p$ , according to

$$\mathbf{w} = -\mathbf{k} \cdot \text{grad } p, \quad (4.14.2)$$

where  $\mathbf{k}$  is the hydraulic permeability tensor. (Note that this expression does not account for the contribution of external body forces on the fluid.)

The governing equations for biphasic materials are the mixture momentum balance under quasi-static conditions, in the absence of external body force,

$$\text{div } \sigma = -\text{grad } p + \text{div } \sigma^e = \mathbf{0}, \quad (4.14.3)$$

and the mixture mass balance,

$$\text{div } (\mathbf{v}^s + \mathbf{w}) = 0, \quad (4.14.4)$$

where  $\mathbf{v}^s$  is the solid velocity.

#### 4.14.1 General Specification of Biphasic Materials

The material type for a biphasic material is “*biphasic*”. The following parameters must be defined:

|                  |                                                                                              |     |
|------------------|----------------------------------------------------------------------------------------------|-----|
| <solid>          | Specification of the solid matrix                                                            |     |
| <phi0>           | solid volume fraction $\varphi_r^s$ in the reference configuration ( $0 < \varphi_r^s < 1$ ) | [ ] |
| <fluid_density>  | Fluid density $\rho_T^w$                                                                     |     |
| <permeability>   | Specification of the hydraulic permeability                                                  |     |
| <solvent_supply> | Specification of the fluid supply $\hat{\varphi}^w$                                          |     |
| <tau>            | Stabilization parameter $\tau$                                                               | [t] |

The <solid> tag encloses a description of the solid matrix constitutive relation and associated material properties, as may be selected from the list provided in Section 4.1.4. The <permeability> tag encloses a description of the permeability constitutive relation and associated material properties, as may be selected from the list presented in Section 4.14.2. The parameter <phi0> must be greater than 0 (no solid) and less than 1 (no porosity). The volume fraction of fluid (also known as the porosity) in the reference configuration is given by  $1 - \varphi_r^s$ . The fluid density  $\rho_T^w$  specified in <fluid\_density> and the solid density  $\rho_T^s$  specified in <density> within the <solid> tag are needed only when body forces are prescribed.

*Example:*

```
<material id="1" name="Biphasic tissue" type="biphasic">
  <solid name="Elasticity" type="neo-Hookean">
    <E>1.0</E>
    <v>0.3</v>
  </solid>
  <phi0>0.2</phi0>
  <permeability name="Permeability" type="perm-const-iso">
    ... (description of permeability material)
  </permeability>
</material>
```

The stabilization parameter  $\tau$  can be used to reduced oscillations in the fluid pressure field produced under certain loading conditions, when the finite element mesh is relatively coarse in the vicinity of a free-draining boundary (a boundary on which the fluid pressure is set to zero). When this occurs it is best to refine the mesh (create thinner elements in the direction normal to the boundary) to overcome these oscillations. When mesh refinement is not feasible or practical, one may use a non-zero stabilization parameter  $\tau$ . FEBio uses an implementation based on the formulation of Aguilar et al. [1]. In our adaptation of this formulation the parameter  $\tau$  has units of time. Its value is best estimated by identifying the thickness  $h$  of elements on the free-draining boundary, the characteristic modulus  $E$  of the solid matrix and hydraulic permeability  $k$  in these elements. Then  $\tau$  may be set approximately to  $h^2 / (E \cdot k)$ .

#### **4.14.2 Permeability Materials**

Permeability materials provide a constitutive relation for the hydraulic permeability of a biphasic material.

#### 4.14.2.1 Constant Isotropic Permeability

The material type for constant isotropic permeability is “*perm-const-iso*”. The following material parameters must be defined:

|        |                            |                                            |
|--------|----------------------------|--------------------------------------------|
| <perm> | hydraulic permeability $k$ | $[\mathbf{L}^4/\mathbf{F}\cdot\mathbf{t}]$ |
|--------|----------------------------|--------------------------------------------|

This isotropic material model uses the biphasic theory for describing the time-dependent material behavior of materials that consist of both a solid and fluid phase [64].

When the permeability is isotropic,

$$\mathbf{k} = k \mathbf{I}$$

For this material model,  $k$  is constant. Generally, this assumption is only reasonable when strains are small.

*Example:*

```
<permeability name="Permeability" type="perm-const-iso">
  <perm>0.001</perm>
</permeability>
```

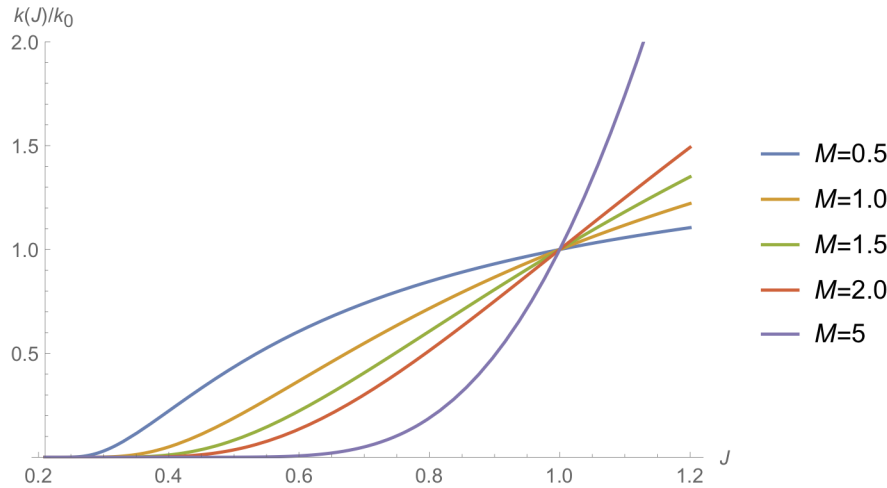


Figure 4.14.1: Exponential isotropic permeability, showing dependence of  $k(J)/k_0$  on material parameter  $M$ , when  $\varphi_0 = 0.2$  and  $\varphi_0 < J \leq 1.2$ .

#### 4.14.2.2 Exponential Isotropic Permeability

The material type for exponential isotropic permeability is “*perm-exp-iso*”. The following material parameters must be defined:

|        |                                                               |                                            |
|--------|---------------------------------------------------------------|--------------------------------------------|
| <perm> | isotropic hydraulic permeability $k_0$ in the reference state | $[\mathbf{L}^4/\mathbf{F}\cdot\mathbf{t}]$ |
| <M>    | exponential strain-dependence coefficient $M$ ( $M \geq 0$ )  | [ ]                                        |

This isotropic material model has the general form

$$\mathbf{k} = k(J) \mathbf{I}$$

where  $J = \det \mathbf{F}$  is the determinant of the deformation gradient. For this material model,

$$k(J) = k_0 \exp \left( M \frac{J - 1}{J - \varphi_0} \right),$$

where  $\varphi_0$  is the referential solid volume fraction of the porous solid matrix. Pore closure occurs as  $J$  approaches  $\varphi_0$  from above, in which case the permeability reduces to zero,

$$\lim_{J \rightarrow \varphi_0} k(J) = 0.$$

Representative variations of  $k(J)/k_0$  are shown in the figure.

*Example:*

```
<permeability name="Permeability" type="perm-exp-iso">
  <perm>0.001</perm>
  <M>1.5</M>
</permeability>
```

#### 4.14.2.3 Holmes-Mow

The material type for Holmes-Mow permeability is “*perm-Holmes-Mow*”. The following material parameters need to be defined:

|         |                                                               |                                              |
|---------|---------------------------------------------------------------|----------------------------------------------|
| <perm>  | isotropic hydraulic permeability $k_0$ in the reference state | [ $\mathbf{L}^4/\mathbf{F}\cdot\mathbf{t}$ ] |
| <M>     | exponential strain-dependence coefficient $M$ ( $M \geq 0$ )  | [ ]                                          |
| <alpha> | power-law exponent $\alpha$ ( $\alpha \geq 0$ )               | [ ]                                          |

This isotropic material is similar to the constant isotropic permeability material described above, except that it uses a strain-dependent permeability tensor [46]:

$$\mathbf{k} = k(J) \mathbf{I},$$

where,

$$k(J) = k_0 \left( \frac{J - \varphi_r^s}{1 - \varphi_0} \right)^\alpha e^{\frac{1}{2}M(J^2 - 1)},$$

and  $J$  is the Jacobian of the deformation, i.e.  $J = \det \mathbf{F}$  where  $\mathbf{F}$  is the deformation gradient. Here,  $\varphi_r^s$  is the referential solid volume fraction in the current configuration and  $\varphi_0 = \varphi_r^s(0)$  is its value in the reference configuration.

*Example:*

This example defines a permeability material of the Holmes-Mow type.

```
<permeability type="perm-Holmes-Mow">
  <perm>0.001</perm>
  <M>1.5</M>
  <alpha>2</alpha>
</permeability>
```

#### 4.14.2.4 Referentially Isotropic Permeability

The material type for a biphasic material with strain-dependent permeability which is isotropic in the reference configuration is “*perm-ref-iso*”. The following material parameters need to be defined:

|         |                                                              |                                            |
|---------|--------------------------------------------------------------|--------------------------------------------|
| <perm0> | hydraulic permeability $k_{0r}$                              | $[\mathbf{L}^4/\mathbf{F}\cdot\mathbf{t}]$ |
| <perm1> | hydraulic permeability $k_{1r}$                              | $[\mathbf{L}^4/\mathbf{F}\cdot\mathbf{t}]$ |
| <perm2> | hydraulic permeability $k_{2r}$                              | $[\mathbf{L}^4/\mathbf{F}\cdot\mathbf{t}]$ |
| <M>     | exponential strain-dependence coefficient $M$ ( $M \geq 0$ ) | [ ]                                        |
| <alpha> | power-law exponent $\alpha$ ( $\alpha \geq 0$ )              | [ ]                                        |

This material uses a strain-dependent permeability tensor that accommodates strain-induced anisotropy:

$$\mathbf{k} = \left( k_{0r} \mathbf{I} + \frac{k_{1r}}{J^2} \mathbf{b} + \frac{k_{2r}}{J^4} \mathbf{b}^2 \right) \left( \frac{J - \varphi_r^s}{1 - \varphi_0} \right)^\alpha e^{M(J^2 - 1)/2},$$

where  $J$  is the Jacobian of the deformation, i.e.  $J = \det \mathbf{F}$  where  $\mathbf{F}$  is the deformation gradient, and  $\mathbf{b} = \mathbf{F} \cdot \mathbf{F}^T$  is the left Cauchy-Green tensor. Here,  $\varphi_r^s$  is the referential solid volume fraction in the current configuration and  $\varphi_0 = \varphi_r^s(0)$  is its value in the reference configuration. Note that the permeability in the reference state ( $\mathbf{F} = \mathbf{I}$ ) is isotropic and given by  $\mathbf{k} = (k_{0r} + k_{1r} + k_{2r}) \mathbf{I}$ .

*Example:*

```
<permeability name="Permeability" type="perm-ref-iso">
  <perm0>0.001</perm0>
  <perm1>0.005</perm1>
  <perm2>0.002</perm2>
  <M>1.5</M>
  <alpha>2</alpha>
</permeability>
```

#### 4.14.2.5 Referentially Orthotropic Permeability

The material type for a poroelastic material with strain-dependent permeability which is orthotropic in the reference configuration is “*perm-ref-ortho*”. The following material parameters need to be defined:

|          |                                                                                              |                                              |
|----------|----------------------------------------------------------------------------------------------|----------------------------------------------|
| <perm0>  | isotropic hydraulic permeability $k_{0r}$                                                    | [ $\mathbf{L}^4/\mathbf{F}\cdot\mathbf{t}$ ] |
| <perm1>  | hydraulic permeabilities $k_{1r}^a$ along orthogonal directions ( $a = 1, 2, 3$ )            | [ $\mathbf{L}^4/\mathbf{F}\cdot\mathbf{t}$ ] |
| <perm2>  | hydraulic permeabilities $k_{2r}^a$ along orthogonal directions ( $a = 1, 2, 3$ )            | [ $\mathbf{L}^4/\mathbf{F}\cdot\mathbf{t}$ ] |
| <M0>     | isotropic exponential strain-dependence coefficient $M_0$ ( $M_0 \geq 0$ )                   | [ ]                                          |
| <M>      | orthotropic exponential strain-dependence coefficient $M_a$ ( $a = 1, 2, 3$ , $M_a \geq 0$ ) | [ ]                                          |
| <alpha0> | isotropic power-law exponent $\alpha_0$ ( $\alpha_0 \geq 0$ )                                | [ ]                                          |
| <alpha>  | power-law exponent $\alpha_a$ ( $a = 1, 2, 3$ , $\alpha_a \geq 0$ )                          | [ ]                                          |

This material uses a strain-dependent permeability tensor that accommodates strain-induced anisotropy:

$$\mathbf{k} = k_0 \mathbf{I} + \sum_{a=1}^3 k_1^a \mathbf{m}_a + k_2^a (\mathbf{m}_a \cdot \mathbf{b} + \mathbf{b} \cdot \mathbf{m}_a) ,$$

where,

$$\begin{aligned} k_0 &= k_{0r} \left( \frac{J - \varphi_r^s}{1 - \varphi_0} \right)^{\alpha_0} e^{M_0(J^2-1)/2} \\ k_1^a &= \frac{k_{1r}^a}{J^2} \left( \frac{J - \varphi_r^s}{1 - \varphi_0} \right)^{\alpha_a} e^{M_a(J^2-1)/2}, \quad a = 1, 2, 3. \\ k_2^a &= \frac{k_{2r}^a}{2J^4} \left( \frac{J - \varphi_r^s}{1 - \varphi_0} \right)^{\alpha_a} e^{M_a(J^2-1)/2} \end{aligned}$$

$J$  is the Jacobian of the deformation, i.e.  $J = \det \mathbf{F}$  where  $\mathbf{F}$  is the deformation gradient.  $\mathbf{m}_a$  are second order tensors representing the spatial structural tensors describing the orthogonal planes of symmetry, given by

$$\mathbf{m}_a = \mathbf{F} \cdot (\mathbf{V}_a \otimes \mathbf{V}_a) \cdot \mathbf{F}^T, \quad a = 1 - 3,$$

where  $\mathbf{V}_a$  are orthonormal vectors normal to the planes of symmetry (defined as described in Section 4.1.1.2). Here,  $\varphi_r^s$  is the referential solid volume fraction in the current configuration and  $\varphi_0 = \varphi_r^s(0)$  is its value in the reference configuration. Note that the permeability in the reference

state ( $\mathbf{F} = \mathbf{I}$ ) is given by  $\mathbf{k} = k_{0r} \mathbf{I} + \sum_{a=1}^3 (k_{1r}^a + k_{2r}^a) \mathbf{V}_a \otimes \mathbf{V}_a$ .

*Example:*

```
<permeability name="Permeability" type="perm-ref-ortho">
  <perm0>0.001</perm0>
  <perm1>0.01, 0.02, 0.03</perm1>
  <perm2>0.001, 0.002, 0.003</perm2>
  <M0>0.5</M0>
  <M>1.5, 2.0, 2.5</M>
```



<alpha>1.5</alpha>  
<alpha>2, 2.5, 3</alpha>  
</permeability>

#### 4.14.2.6 Referentially Transversely Isotropic Permeability

The material type for a biphasic material with strain-dependent permeability which is transversely isotropic in the reference configuration is “*perm-ref-trans-iso*”. The following material parameters need to be defined:

|          |                                                                             |                   |
|----------|-----------------------------------------------------------------------------|-------------------|
| <perm0>  | isotropic hydraulic permeability $k_{0r}$                                   | $[L^4/F \cdot t]$ |
| <perm1A> | axial hydraulic permeability $k_{1r}^A$                                     | $[L^4/F \cdot t]$ |
| <perm2A> | axial hydraulic permeability $k_{2r}^A$                                     | $[L^4/F \cdot t]$ |
| <perm1T> | transverse hydraulic permeability $k_{1r}^T$                                | $[L^4/F \cdot t]$ |
| <perm2T> | transverse hydraulic permeability $k_{2r}^T$                                | $[L^4/F \cdot t]$ |
| <M0>     | isotropic exponential strain-dependence coefficient $M_0$ ( $M_0 \geq 0$ )  | $[ ]$             |
| <MA>     | axial exponential strain-dependence coefficient $M_A$ ( $M_A \geq 0$ )      | $[ ]$             |
| <MT>     | transverse exponential strain-dependence coefficient $M_T$ ( $M_T \geq 0$ ) | $[ ]$             |
| <alpha0> | isotropic power-law exponent $\alpha_0$ ( $\alpha_0 \geq 0$ )               | $[ ]$             |
| <alphaA> | axial power-law exponent $\alpha_A$ ( $\alpha_A \geq 0$ )                   | $[ ]$             |
| <alphaT> | transverse power-law exponent $\alpha_T$ ( $\alpha_T \geq 0$ )              | $[ ]$             |

This material uses a strain-dependent permeability tensor that accommodates strain-induced anisotropy:

$$\begin{aligned} \mathbf{k} = & k_{0r} \left( \frac{J - \varphi_r^s}{1 - \varphi_0} \right)^{\alpha_0} e^{M_0(J^2-1)/2} \mathbf{I} \\ & + \left( \frac{k_{1r}^T}{J^2} (\mathbf{b} - \mathbf{m}) + \frac{k_{2r}^T}{2J^4} [2\mathbf{b}^2 - (\mathbf{m} \cdot \mathbf{b} + \mathbf{b} \cdot \mathbf{m})] \right) \left( \frac{J - \varphi_r^s}{1 - \varphi_0} \right)^{\alpha_T} e^{M_T(J^2-1)/2}, \\ & + \left( \frac{1}{J^2} k_{1r}^A \mathbf{m} + \frac{1}{2J^4} k_{2r}^A (\mathbf{m} \cdot \mathbf{b} + \mathbf{b} \cdot \mathbf{m}) \right) \left( \frac{J - \varphi_r^s}{1 - \varphi_0} \right)^{\alpha_A} e^{M_A(J^2-1)/2} \end{aligned}$$

where  $J$  is the Jacobian of the deformation, i.e.  $J = \det \mathbf{F}$  where  $\mathbf{F}$  is the deformation gradient, and  $\mathbf{b} = \mathbf{F} \cdot \mathbf{F}^T$  is the left Cauchy-Green tensor.  $\mathbf{m}$  is a second order tensor representing the spatial structural tensor describing the axial direction, given by

$$\mathbf{m} = \mathbf{F} \cdot (\mathbf{V} \otimes \mathbf{V}) \cdot \mathbf{F}^T,$$

where  $\mathbf{V}$  is a unit vector along the axial direction (defined as described in Section 4.1.1). Here,  $\varphi_r^s$  is the referential solid volume fraction in the current configuration and  $\varphi_0 = \varphi_r^s(0)$  is its value in the reference configuration. Note that the permeability in the reference state ( $\mathbf{F} = \mathbf{I}$ ) is given by,

$$\mathbf{k}_{|\mathbf{F}=\mathbf{I}} = (k_{0r} + k_{1r}^T + k_{2r}^T) \mathbf{I} + (k_{1r}^A - k_{1r}^T + k_{2r}^A - k_{2r}^T) (\mathbf{V} \otimes \mathbf{V})$$

*Example:*

```
<permeability name="Permeability" type="perm-ref-trans-iso">
  <perm0>0.002</perm0>
  <perm1A>0.01</perm1A>
  <perm2A>0.01</perm2A>
  <perm1T>0.001</perm1T>
  <perm2T>0.05</perm2T>
  <M0>1.0</M0>
```

```
<MA>0.5</MA>  
<MT>1.5</MT>  
<alpha0>1.0</alpha0>  
<alphaA>0.5</alphaA>  
<alphaT>2.0</alphaT>  
</permeability>
```

### 4.14.3 Fluid Supply Materials

Fluid supply materials may be used to simulate an external source of fluid, such as supply from microvasculature that is not modeled explicitly. The fluid supply term,  $\hat{\varphi}^w$ , appears in the mass balance relation for the mixture,

$$\operatorname{div}(\mathbf{v}^s + \mathbf{w}) = \hat{\varphi}^w.$$

$\hat{\varphi}^w$  has units of reciprocal time [ $\mathbf{t}^{-1}$ ]; it represents the rate at which the volume fraction of fluid changes with time.

#### 4.14.3.1 Starling Equation

The material type for Starling's equation for fluid supply is “*Starling*”. The following material parameters need to be defined:

|      |                                          |                                        |
|------|------------------------------------------|----------------------------------------|
| <kp> | hydraulic filtration coefficient, $k_p$  | [ <b>L</b> <sup>2</sup> / <b>F</b> ·t] |
| <pv> | fluid pressure in external source, $p_v$ | [ <b>P</b> ]                           |

The fluid supply is given by Starling's equation,

$$\hat{\varphi}^w = k_p (p_v - p) ,$$

where  $p$  is the fluid pressure in the biphasic material.

*Example:*

This example defines a fluid supply material of the Starling type.

```
<solvent_supply type="Starling">
  <kp>0.001</kp>
  <pv>0.1</pv>
</solvent_supply>
```

### 4.15 Biphasic-Solute Materials

Biphasic-solute materials may be used to model the transport of a solvent and a solute in a neutral porous solid matrix, under isothermal conditions. Each of these mixture constituents is assumed to be intrinsically incompressible. This means that their true densities are invariant in space and time; this assumption further implies that a biphasic-solute mixture will undergo zero volume change when subjected to a hydrostatic fluid pressure. Yet the mixture is compressible because the pores of the solid matrix may gain or lose fluid under general loading conditions. Therefore, the constitutive relation of the solid matrix should be chosen from the list of unconstrained materials provided in Section 4.1.4. The volume fraction of the solute is assumed to be negligible relative to the volume fractions of the solid or solvent. This means that the mixture will not change in volume as the solute concentration changes. As the solute transports through the mixture, it may experience frictional interactions with the solvent and the solid. This friction may act as a hindrance to the solute transport, or may help convect the solute through the mixture. The distinction between frictional exchanges with the solvent and solid is embodied in the specification of two diffusivities for the solute: The free diffusivity, which represents diffusivity in the absence of a solid matrix (frictional exchange only with the solvent) and the mixture diffusivity, which represents the combined frictional interactions with the solvent and solid matrix. The user is referred to the [FEBio Theory Manual](#) for a more detailed description of the biphasic-solute theory.

Due to steric volume exclusion and short-range electrostatic interactions, the solute may not have access to all of the pores of the solid matrix. In other words, only a fraction  $\kappa$  of the pores is able to accommodate a solute of a particular size ( $0 < \kappa \leq 1$ ). Furthermore, the activity  $\gamma$  of the solute (the extent by which the solute concentration influences its chemical potential) may be similarly altered by the surrounding porous solid matrix. Therefore, the combined effects of volume exclusion and attenuation of activity may be represented by the effective solubility  $\tilde{\kappa} = \kappa/\gamma$ , such that the chemical potential  $\mu$  of the solute is given by

$$\mu = \mu_0(\theta) + \frac{R\theta}{M} \ln \frac{c}{\tilde{\kappa}}. \quad (4.15.1)$$

In this expression,  $\mu_0$  is the solute chemical potential at some reference temperature  $\theta$ ;  $c$  is the solute concentration on a solution-volume basis (number of moles of solute per volume of interstitial fluid in the mixture);  $M$  is the solute molecular weight (an invariant quantity); and  $R$  is the universal gas constant. In a biphasic-solute material, a constitutive relation is needed for  $\tilde{\kappa}$ ; in general,  $\tilde{\kappa}$  may be a function of the solid matrix strain and the solute concentration. In FEBio, the dependence of the effective solubility on the solid matrix strain is currently constrained to a dependence on  $J = \det \mathbf{F}$ .

In a biphasic-solute material, the interstitial fluid pressure  $p$  is influenced by both mechanical and chemical environments. In other words, this pressure includes both mechanical and osmotic contributions, the latter arising from the presence of the solute. The solvent mechano-chemical potential  $\tilde{\mu}^w$  is given by

$$\tilde{\mu}^w = \mu_0^w(\theta) + \frac{1}{\rho_T^w} (p - R\theta\Phi c), \quad (4.15.2)$$

where  $\mu_0^w$  is the solvent chemical potential at some reference temperature  $\theta$ ;  $\rho_T^w$  is the true density of the solvent (an invariant property for an intrinsically incompressible fluid); and  $\Phi$  is the osmotic coefficient which represents the extent by which the solute concentration influences the solvent chemical potential. In a biphasic-solute material, a constitutive relation is needed for  $\Phi$ ; in general,  $\Phi$  may be a function of the solid matrix strain and the solute concentration. In FEBio, the depen-

dence of the osmotic coefficient on the solid matrix strain is currently constrained to a dependence on  $J = \det \mathbf{F}$ .

The solute mechano-chemical potential is nearly equal to its chemical potential because the solute volume fraction is assumed to be negligible. In general, momentum and energy balances evaluated across a boundary surface in a biphasic-solute mixture require that the mechano-chemical potentials of solvent and solute be continuous across that surface. These continuity requirements are enforced automatically in FEBio by defining the effective fluid pressure  $\tilde{p}$  and solute concentration  $\tilde{c}$  as

$$\begin{aligned}\tilde{p} &= p - R\theta\Phi c \\ \tilde{c} &= \frac{c}{\tilde{\kappa}}\end{aligned}\quad (4.15.3)$$

Therefore, nodal variables in FEBio consist of the solid matrix displacement  $\mathbf{u}$ , the effective fluid pressure  $\tilde{p}$ , and the effective solute concentration  $\tilde{c}$ . Essential boundary conditions must be imposed on these variables, and not on the actual pressure  $p$  or concentration  $c$ . (In a biphasic material however, since  $c = 0$ , the effective and actual fluid pressures are the same,  $p = \tilde{p}$ .)

The mixture stress in a biphasic-solute material is given by  $\boldsymbol{\sigma} = -p\mathbf{I} + \boldsymbol{\sigma}^e$ , where  $\boldsymbol{\sigma}^e$  is the stress arising from the solid matrix strain. The mixture traction on a surface with unit outward normal  $\mathbf{n}$  is  $\mathbf{t} = \boldsymbol{\sigma} \cdot \mathbf{n}$ . This traction is continuous across the boundary surface. Therefore, the corresponding natural boundary condition for a biphasic-solute mixture is  $\mathbf{t} = \mathbf{0}$ . (In other words, if no boundary condition is imposed on the solid matrix displacement or mixture traction, the natural boundary condition is in effect.)

The natural boundary conditions for the solvent and solute are similarly  $\mathbf{w} \cdot \mathbf{n} = 0$  and  $\mathbf{j} \cdot \mathbf{n} = 0$ , where  $\mathbf{w}$  is the volumetric flux of solvent relative to the solid and  $\mathbf{j}$  is the molar flux of solute relative to the solid. In general,  $\mathbf{w}$  and  $\mathbf{j}$  are given by

$$\begin{aligned}\mathbf{w} &= -\tilde{\mathbf{k}} \cdot \left( \nabla \tilde{p} + R\theta \frac{\tilde{\kappa}}{d_0} \mathbf{d} \cdot \nabla \tilde{c} \right) \\ \mathbf{j} &= \tilde{\kappa} \mathbf{d} \cdot \left( -\varphi^w \nabla \tilde{c} + \frac{\tilde{c}}{d_0} \mathbf{w} \right)\end{aligned}\quad (4.15.4)$$

where

$$\tilde{\mathbf{k}} = \left[ \mathbf{k}^{-1} + \frac{R\theta}{\varphi^w} \frac{\tilde{\kappa}}{d_0} \left( \mathbf{I} - \frac{\mathbf{d}}{d_0} \right) \right]^{-1} \quad (4.15.5)$$

is the effective hydraulic permeability of the interstitial fluid solution (solvent and solute) through the porous solid matrix;  $\mathbf{k}$  is the hydraulic permeability of the solvent through the porous solid matrix;  $\mathbf{d}$  is the solute diffusivity through the mixture (frictional interactions with solvent and solid); and  $d_0$  is the solute free diffusivity (frictional interactions with solvent only).  $\varphi^w \approx 1 - \varphi^s$  is the solid matrix porosity in the current configuration. The above expressions for the solvent and solute flux do not account for external body forces.

The governing equations for a biphasic-solute material are the momentum balance for the mixture, Eq.(4.14.3), the mass balance for the mixture, which reduces to Eq.(4.14.4) under the assumption of dilute solutions, and the mass balance for the solute,

$$\frac{\partial (\varphi^w c)}{\partial t} + \text{div} (\mathbf{j} + \varphi^w c \mathbf{v}^s) = 0. \quad (4.15.6)$$

### 4.15.1 Guidelines for Biphasic-Solute Analyses

#### 4.15.1.1 Prescribed Boundary Conditions

In most analyses, it may be assumed that the ambient fluid pressure in the external environment is zero, thus  $p_* = 0$ , where the subscripted asterisk is used to denote environmental conditions. The ambient solute concentration may be represented by  $c_*$ . It follows that the effective fluid pressure in the external environment is  $\tilde{p}_* = -R\theta\Phi_*c_*$  and the effective concentration is  $\tilde{c}_* = c_*/\tilde{K}_*$ . Therefore, in biphasic-solute analyses, whenever the external environment contains a solute at a concentration of  $c_*$ , the user must remember to prescribe non-zero boundary conditions for the effective solute concentration *and* the effective fluid pressure.

Letting  $p_* = 0$  also implies that prescribed mixture normal tractions (Section 3.13.2.6) represent only the traction above ambient conditions. Note that users are not obligated to assume that  $p_* = 0$ . However, if a non-zero value is assumed for the ambient pressure, then users must remember to incorporate this non-zero value whenever prescribing mixture normal tractions.

#### 4.15.1.2 Prescribed Initial Conditions

When a biphasic-solute material is initially exposed to a given external environment with effective pressure  $\tilde{p}_*$  and effective concentration  $\tilde{c}_*$ , the initial conditions inside the material should be set to  $\tilde{p} = \tilde{p}_*$  and  $\tilde{c} = \tilde{c}_*$  in order to produce the correct initial state. The values of  $\tilde{p}_*$  and  $\tilde{c}_*$  should be evaluated as described in Section 8.7.2.



### 4.15.2 General Specification of Biphasic-Solute Materials

The material type for a biphasic-solute material is “*biphasic-solute*”. Constitutive relations must be provided for the solid matrix, the hydraulic permeability  $k$ , the solute diffusivities  $d$  and  $d_0$ , the effective solubility  $\tilde{\kappa}$  and the osmotic coefficient  $\Phi$ . Therefore, the following parameters must be defined:

|                       |                                                                    |
|-----------------------|--------------------------------------------------------------------|
| <solid>               | specification of the solid matrix                                  |
| <phi0>                | solid volume fraction $\varphi_r^s$ in the reference configuration |
| <permeability>        | specification of the hydraulic permeability $k$                    |
| <osmotic_coefficient> | specification of the osmotic coefficient $\Phi$ $d d_0$            |
| <solute>              | specification of the solute properties                             |

The <solid> tag encloses a description of the solid matrix constitutive relation and associated material properties, as may be selected from the list provided in Section 4.1.4. The solid volume fraction in the reference configuration, <phi0>, must be greater than 0 (no solid) and less than 1 (only solid). The volume fraction of fluid (also known as the porosity) in the reference configuration is given by  $1 - \varphi_0$ . The <permeability> tag encloses a description of the permeability constitutive relation and associated material properties, as may be selected from the list presented in Section 4.14.2.

The <solute> tag provides a description of the solute in the biphasic-solute mixture. This tag includes the required *sol* attribute, which should reference a solute *id* from the <Solutes> description in the <Globals> section (Section 3.4.2). The following parameters must be defined in this description:

|               |                                                                   |
|---------------|-------------------------------------------------------------------|
| <diffusivity> | specification of the solute diffusivities $d$ and $d_0$           |
| <solubility>  | specification of the solute effective solubility $\tilde{\kappa}$ |

The <diffusivity> and <solubility> tags enclose descriptions of materials that may be selected from the lists presented in Sections 4.15.3 and 4.15.4, respectively. Each solute tag must include a “sol” attribute.

*Example:*

```
<material id="1" name="Biological tissue" type="biphasic-solute">
  <solid name="Elasticity" type="neo-Hookean">
    <E>1.0</E>
    <v>0.3</v>
  </solid>
  <phi0>0.2</phi0>
  <permeability name="Permeability" type="perm-const-iso">
    ... (description of permeability material)
  </permeability>
  <osmotic_coefficient name="Osmotic" type="osm-coef-const">
    ... (description of osmotic coefficient material)
  </osmotic_coefficient>
  <solute sol="1">
    <diffusivity name="Diffusivity" type="diff-const-iso">
      ... (description of diffusivity material)
```

```

    </diffusivity>
    <solubility name="Solubility" type="solub-const">
      ... (description of solubility material)
    </solubility>
  </solute>
</material>

```

When a biphasic-solute material is employed in an analysis, it is also necessary to specify the values of the universal gas constant  $R$  [**F·L/n·T**] and absolute temperature  $\theta$  [**T**] under <Constants> in the <Globals> section, using a self-consistent set of units. A solute must also be defined in the <Solutes> section, whose id should be associated with the “sol” attribute in the solute material description.

*Example:*

```

<Globals>
  <Constants>
    <R>8.314</R>
    <T>298</T>
  </Constants>
  <Solutes>
    <solute id="1" name="neutral">
      <charge_number>0</charge_number>
    </solute>
  </Solutes>
</Globals>

```

It is also possible to create models with biphasic-solute materials that use different solutes in different regions. In that case, introduce additional solute entries in the <Solutes> section and refer to those solute ids in the biphasic-solute material descriptions.

### 4.15.3 Diffusivity Materials

Diffusivity materials provide a constitutive relation for the solute diffusivity in a biphasic-solute material. In general, the diffusivity tensor  $\mathbf{d}$  may be a function of strain and solute concentration.

#### 4.15.3.1 Constant Isotropic Diffusivity

The material type for constant isotropic diffusivity materials is “*diff-const-iso*”. The following material parameters must be defined:

|             |                                                                         |                     |
|-------------|-------------------------------------------------------------------------|---------------------|
| <free_diff> | free diffusivity $d_0$ of solute (diffusivity in solution)              | [L <sup>2</sup> /t] |
| <diff>      | constant isotropic diffusivity $d$ of solute in biphasic-solute mixture | [L <sup>2</sup> /t] |

When the permeability is isotropic,

$$\mathbf{d} = d \mathbf{I}$$

For this material model,  $d$  is constant. This assumption is only true when strains are small. Note that the user must specify  $d \leq d_0$ , since a solute cannot diffuse through the biphasic-solute mixture faster than in free solution.

*Example:*

```
<diffusivity name="Permeability" type="diff-const-iso">
  <free_diff>1e-9</ free_diff >
  <diff>0.5e-9</diff>
</diffusivity>
```

#### 4.15.3.2 Constant Orthotropic Diffusivity

The material type for constant orthotropic diffusivity materials is “*diff-const-ortho*”. The following material parameters must be defined:

|             |                                                                   |                             |
|-------------|-------------------------------------------------------------------|-----------------------------|
| <free_diff> | free diffusivity $d_0$ of solute (diffusivity in solution)        | $[\mathbf{L}^2/\mathbf{t}]$ |
| <diff>      | diffusivities $d^a$ along orthogonal directions ( $a = 1, 2, 3$ ) | $[\mathbf{L}^2/\mathbf{t}]$ |

When the permeability is orthotropic,

$$\mathbf{d} = \sum_{a=1}^3 d^a \mathbf{V}_a \otimes \mathbf{V}_a$$

where  $\mathbf{V}_a$  are orthonormal vectors normal to the planes of symmetry (defined as described in Section 4.1.1). For this material model,  $d^a$ 's are constant. Therefore this model should be used only when strains are small. Note that the user must specify  $d^a \leq d_0$ , since a solute cannot diffuse through the biphasic-solute mixture faster than in free solution.

*Example:*

```
<diffusivity name="Permeability" type="diff-const-ortho">
  <free_diff>0.005</free_diff >
  <diff>0.002,0.001,0.003</diff>
</diffusivity>
```

### 4.15.3.3 Referentially Isotropic Diffusivity

The material type for a strain-dependent diffusivity tensor which is isotropic in the reference configuration is “diff-ref-iso”. The following material parameters need to be defined:

|             |                                                              |                     |
|-------------|--------------------------------------------------------------|---------------------|
| <free_diff> | free diffusivity $d_0$                                       | [L <sup>2</sup> /t] |
| <diff0>     | diffusivity $d_{0r}$                                         | [L <sup>2</sup> /t] |
| <diff1>     | diffusivity $d_{1r}$                                         | [L <sup>2</sup> /t] |
| <diff2>     | diffusivity $d_{2r}$                                         | [L <sup>2</sup> /t] |
| <M>         | exponential strain-dependence coefficient $M$ ( $M \geq 0$ ) | [ ]                 |
| <alpha>     | power-law exponent $\alpha$ ( $\alpha \geq 0$ )              | [ ]                 |

This material uses a strain-dependent diffusivity tensor that accommodates strain-induced anisotropy:

$$\mathbf{d} = \left( d_{0r} \mathbf{I} + \frac{d_{1r}}{J^2} \mathbf{b} + \frac{d_{2r}}{J^4} \mathbf{b}^2 \right) \left( \frac{J - \varphi_r^s}{1 - \varphi_r^s} \right) e^{M(J^2 - 1)/2},$$

where  $J$  is the jacobian of the deformation, i.e.  $J = \det \mathbf{F}$  where  $\mathbf{F}$  is the deformation gradient, and  $\mathbf{b} = \mathbf{F} \cdot \mathbf{F}^T$  is the left Cauchy-Green tensor. Note that the diffusivity in the reference state ( $\mathbf{F} = \mathbf{I}$ ) is isotropic and given by  $\mathbf{d} = (d_{0r} + d_{1r} + d_{2r}) \mathbf{I}$ .

*Example:*

```
<diffusivity name="Diffusivity" type="diff-ref-iso">
  <phi0>0.2</phi0>
  <free_diff>0.005</free_diff>
  <diff0>0.001</diff0>
  <diff1>0.005</diff1>
  <diff2>0.002</diff2>
  <M>1.5</M>
  <alpha>2</alpha>
</diffusivity>
```

#### 4.15.3.4 Referentially Orthotropic Diffusivity

The material type for a strain-dependent diffusivity which is orthotropic in the reference configuration is “diff-ref-ortho”. The following material parameters need to be defined:

|             |                                                                                           |                               |
|-------------|-------------------------------------------------------------------------------------------|-------------------------------|
| <free_diff> | free diffusivity $d_0$                                                                    | [ $\mathbf{L}^2/\mathbf{t}$ ] |
| <diff0>     | isotropic diffusivity $d_{0r}$                                                            | [ $\mathbf{L}^2/\mathbf{t}$ ] |
| <diff1>     | diffusivities $d_{1r}^a$ along orthogonal directions ( $a = 1, 2, 3$ )                    | [ $\mathbf{L}^2/\mathbf{t}$ ] |
| <diff2>     | diffusivities $d_{2r}^a$ along orthogonal directions ( $a = 1, 2, 3$ )                    | [ $\mathbf{L}^2/\mathbf{t}$ ] |
| <M0>        | isotropic exponential strain-dependence coefficient $M_0$ ( $M_0 \geq 0$ )                | [ ]                           |
| <M>         | orthotropic exponential strain-dependence coefficient $M_a$ ( $a = 1, 2, 3, M_a \geq 0$ ) | [ ]                           |
| <alpha0>    | isotropic power-law exponent $\alpha_0$ ( $\alpha_0 \geq 0$ )                             | [ ]                           |
| <alpha>     | power-law exponent $\alpha_a$ ( $a = 1, 2, 3, \alpha_a \geq 0$ )                          | [ ]                           |

This material uses a strain-dependent diffusivity tensor that accommodates strain-induced anisotropy:

$$\mathbf{d} = d_0 \mathbf{I} + \sum_{a=1}^3 d_1^a \mathbf{m}_a + d_2^a (\mathbf{m}_a \cdot \mathbf{b} + \mathbf{b} \cdot \mathbf{m}_a),$$

where,

$$\begin{aligned} d_0 &= d_{0r} \left( \frac{J - \varphi_r^s}{1 - \varphi_r^s} \right)^{\alpha_0} e^{M_0(J^2-1)/2} \\ d_1^a &= \frac{d_{1r}^a}{J^2} \left( \frac{J - \varphi_r^s}{1 - \varphi_r^s} \right)^{\alpha_a} e^{M_a(J^2-1)/2}, \quad a = 1, 2, 3 \\ d_2^a &= \frac{d_{2r}^a}{2J^4} \left( \frac{J - \varphi_r^s}{1 - \varphi_r^s} \right)^{\alpha_a} e^{M_a(J^2-1)/2} \end{aligned}$$

$J$  is the Jacobian of the deformation, i.e.  $J = \det \mathbf{F}$  where  $\mathbf{F}$  is the deformation gradient.  $\mathbf{m}_a$  are second order tensors representing the spatial structural tensors describing the orthogonal planes of symmetry, given by

$$\mathbf{m}_a = \mathbf{F} \cdot (\mathbf{V}_a \otimes \mathbf{V}_a) \cdot \mathbf{F}^T, \quad a = 1 - 3,$$

where  $\mathbf{V}_a$  are orthonormal vectors normal to the planes of symmetry (defined as described in Section 4.1.1). Note that the diffusivity in the reference state ( $\mathbf{F} = \mathbf{I}$ ) is given by  $\mathbf{d} = d_{0r} \mathbf{I} + \sum_{a=1}^3 (d_{1r}^a + d_{2r}^a) \mathbf{V}_a \otimes \mathbf{V}_a$ .

**Example:**

```
<diffusivity name="Diffusivity" type="diff-ref-ortho">
  <phi0>0.2</phi0>
  <free_diff>0.005</free_diff>
  <diff00>0.001</diff00>
  <diff1>0.01, 0.02, 0.03</diff1>
  <diff2>0.001, 0.002, 0.003</diff2>
  <M0>0.5</M0>
```

<M>1.5, 2.0, 2.5</M>  
<alpha0>1.5</alpha0>  
<alpha>2, 2.5, 3</alpha>  
</diffusivity>

#### 4.15.3.5 Albro Isotropic Diffusivity

The material type for a porosity and concentration-dependent diffusivity which is isotropic is “*diff-Albro-iso*”. The following material parameters need to be defined:

|             |                                                                                 |                     |
|-------------|---------------------------------------------------------------------------------|---------------------|
| <free_diff> | free diffusivity $d_0$                                                          | [L <sup>2</sup> /t] |
| <cdinv>     | inverse of characteristic concentration for concentration-dependence $c_D^{-1}$ | [L <sup>3</sup> /n] |
| <alphad>    | coefficient of porosity-dependence $\alpha_D$                                   | [ ]                 |

This material uses a porosity and concentration-dependent diffusivity tensor that remains isotropic with deformation:

$$\mathbf{d} = d_0 \exp \left( -\alpha_D \frac{1 - \varphi^w}{\varphi^w} - \frac{c}{c_D} \right) \mathbf{I},$$

where  $c_D^{-1} = 1/c_D$  and the porosity  $\varphi^w$  varies with deformation according to the kinematic constraint

$$\varphi^w = 1 - \frac{\varphi_r^s}{J}.$$

$J$  is the Jacobian of the deformation, i.e.  $J = \det \mathbf{F}$  where  $\mathbf{F}$  is the deformation gradient and  $\varphi_r^s$  is the referential solid volume fraction. Here,  $c$  represents the actual concentration of the solute whose diffusivity is given by  $\mathbf{d}$ . This constitutive relation is based on the experimental findings reported by Albro et al. [2].

*Example:*

```
<solute sol="1">
  <diffusivity type="diff-Albro-iso">
    <free_diff>123e-6</free_diff>
    <alphad>18</alphad>
    <cdinv>0.0625</cdinv>
  </diffusivity>
  <solubility type="solub-const">
    <solub>1</solub>
  </solubility>
</solute>
```



### 4.15.4 Solubility Materials

#### 4.15.4.1 Constant Solubility

The material type for constant solubility materials is “*solub-const*”. The following material parameters must be defined:

|         |                             |     |
|---------|-----------------------------|-----|
| <solub> | solubility $\tilde{\kappa}$ | [ ] |
|---------|-----------------------------|-----|

For this material model,  $\tilde{\kappa}$  is constant.

*Example:*

```
<solubility name="Solubility" type="solub-const">
  <solub>1</solub>
</solubility>
```

### 4.15.5 Osmotic Coefficient Materials

#### 4.15.5.1 Constant Osmotic Coefficient

The material type for constant osmotic coefficient materials is “*osm-coef-const*”. The following material parameters must be defined:

|                              |                            |     |
|------------------------------|----------------------------|-----|
| <code>&lt;osmcoef&gt;</code> | Osmotic coefficient $\Phi$ | [ ] |
|------------------------------|----------------------------|-----|

For this material model,  $\Phi$  is constant.

*Example:*

```
<osmotic_coefficient name="Osmotic coefficient" type="osm-coef-const">
  <osmcoef>1</osmcoef>
</osmotic_coefficient>
```

## 4.16 Triphasic and Multiphasic Materials

Triphasic materials may be used to model the transport of a solvent and a pair of monovalent salt counter-ions (two solutes with charge numbers  $+1$  and  $-1$ ) in a charged porous solid matrix, under isothermal conditions. Multiphasic materials may be used to model the transport of a solvent and any number of neutral or charged solutes; a triphasic mixture is a special case of a multiphasic mixture. Each of the mixture constituents is assumed to be intrinsically incompressible. This means that their true densities are invariant in space and time; this assumption further implies that a multiphasic mixture will undergo zero volume change when subjected to a hydrostatic fluid pressure. Yet the mixture is compressible because the pores of the solid matrix may gain or lose fluid under general loading conditions. Therefore, the constitutive relation of the solid matrix should be chosen from the list of unconstrained materials provided in Section 4.1.4. The volume fraction of the solutes is assumed to be negligible relative to the volume fractions of the solid or solvent. This means that the mixture will not change in volume as the solute concentrations change. As the solutes transport through the mixture, they may experience frictional interactions with the solvent and the solid. This friction may act as a hindrance to the solute transport, or may help convect the solutes through the mixture. The distinction between frictional exchanges with the solvent and solid is embodied in the specification of two diffusivities for each solute: The free diffusivity, which represents diffusivity in the absence of a solid matrix (frictional exchange only with the solvent) and the mixture diffusivity, which represents the combined frictional interactions with the solvent and solid matrix. The user is referred to the *FEBio Theory Manual* for a more detailed description of triphasic and multiphasic theory.

Due to steric volume exclusion and short-range electrostatic interactions, a solute  $\alpha$  may not have access to all of the pores of the solid matrix. In other words, only a fraction  $\kappa^\alpha$  of the pores is able to accommodate solute  $\alpha$  ( $0 < \kappa^\alpha \leq 1$ ). Furthermore, the activity  $\gamma^\alpha$  of solute  $\alpha$  (the extent by which the solute concentration influences its chemical potential) may be similarly altered by the surrounding porous solid matrix. Therefore, the combined effects of volume exclusion and attenuation of activity may be represented by the effective solubility  $\hat{\kappa}^\alpha = \kappa^\alpha / \gamma^\alpha$ , such that the chemical potential  $\mu$  of the solute is given by

$$\mu^\alpha = \mu_0^\alpha(\theta) + \frac{R\theta}{M^\alpha} \ln \frac{c^\alpha}{\hat{\kappa}^\alpha}. \quad (4.16.1)$$

In this expression,  $\mu_0^\alpha$  is the solute chemical potential at some reference temperature  $\theta$ ;  $c^\alpha$  is the solute concentration on a solution-volume basis (number of moles of solute per volume of interstitial fluid in the mixture);  $M^\alpha$  is the solute molecular weight (an invariant quantity); and  $R$  is the universal gas constant. In a triphasic material, a constitutive relation is needed for  $\hat{\kappa}^\alpha$ ; in general,  $\hat{\kappa}^\alpha$  may be a function of the solid matrix strain and the solute concentration. In FEBio, the dependence of the effective solubility on the solid matrix strain is currently constrained to a dependence on  $J = \det \mathbf{F}$ .

The solid matrix of a multiphasic material may be charged and its charge density is given by  $c^F$ . This charge density may be either negative or positive. The charge density varies with the deformation, increasing when the pore volume decreases. Based on the balance of mass for the solid,

$$c^F = \frac{1 - \varphi_r^s}{J - \varphi_r^s} c_r^F, \quad (4.16.2)$$

where  $\varphi_r^s$  is the solid volume fraction and  $c_r^F$  is the fixed charge density in the reference configuration.

In the multiphasic theory it is assumed that electroneutrality is satisfied at all times. In other words, the net charge of the mixture is always zero (neutral). This electroneutrality condition is represented by a constraint equation on the ion concentrations,

$$c^F + \sum_{\alpha} z^{\alpha} c^{\alpha} = 0, \quad (4.16.3)$$

where  $z^{\alpha}$  is the charge number of solute  $\alpha$   $z^{-} = -1$ . Since the concentrations of the cation and anion inside the triphasic material are not the same, an electrical potential difference is produced between the interstitial and external environments. The electric potential in the triphasic mixture is denoted by  $\psi$  and its effect combines with the chemical potential of each solute to produce the electrochemical potential  $\tilde{\mu}^{\alpha}$ , where

$$\tilde{\mu}^{\alpha} = \mu_0^{\alpha}(\theta) + \frac{R\theta}{M^{\alpha}} \ln \frac{c^{\alpha}}{\hat{\kappa}^{\alpha}} + \frac{z^{\alpha}}{M^{\alpha}} F_c \psi. \quad (4.16.4)$$

In this expression,  $F_c$  represents Faraday's constant. It is also possible to rearrange this expression as

$$\tilde{\mu}^{\alpha} = \mu_0^{\alpha}(\theta) + \frac{R\theta}{M^{\alpha}} \ln \left[ \exp \left( z^{\alpha} \frac{F_c \psi}{R\theta} \right) \frac{c^{\alpha}}{\hat{\kappa}^{\alpha}} \right]. \quad (4.16.5)$$

In a multiphasic material, the interstitial fluid pressure  $p$  is influenced by both mechanical and chemical environments. In other words, this pressure includes both mechanical and osmotic contributions, the latter arising from the presence of the solutes. The solvent mechano-chemical potential  $\tilde{\mu}^w$  is given by

$$\tilde{\mu}^w = \mu_0^w(\theta) + \frac{1}{\rho_T^w} \left( p - R\theta\Phi \sum_{\alpha} c^{\alpha} \right), \quad (4.16.6)$$

where  $\mu_0^w$  is the solvent chemical potential at some reference temperature  $\theta$ ;  $\rho_T^w$  is the true density of the solvent (an invariant property for an intrinsically incompressible fluid); and  $\Phi$  is the osmotic coefficient which represents the extent by which the solute concentrations influence the solvent chemical potential. In a multiphasic material, a constitutive relation is needed for  $\Phi$ ; in general,  $\Phi$  may be a function of the solid matrix strain and the solute concentrations. In FEBio, the dependence of the osmotic coefficient on the solid matrix strain is currently constrained to a dependence on  $J = \det \mathbf{F}$ .

The solute mechano-electrochemical potential is nearly equal to its electrochemical potential because the solute volume fraction is assumed to be negligible. The solvent mechano-electrochemical potential is the same as its mechano-chemical potential, since the solvent is neutral in a multiphasic mixture. In general, momentum and energy balances evaluated across a boundary surface in a multiphasic mixture require that the mechano-electrochemical potentials of solvent and solutes be continuous across that surface. These continuity requirements are enforced automatically in FEBio by defining the effective fluid pressure  $\tilde{p}$  and solute concentration  $\tilde{c}^{\alpha}$  as

$$\begin{aligned} \tilde{p} &= p - R\theta\Phi \sum_{\alpha} c^{\alpha}, \\ \tilde{c}^{\alpha} &= c^{\alpha} / \tilde{\kappa}^{\alpha} \end{aligned} \quad (4.16.7)$$

where

$$\tilde{\kappa}^{\alpha} = \hat{\kappa}^{\alpha} \exp \left( -z^{\alpha} \frac{F_c \psi}{R\theta} \right) \quad (4.16.8)$$

is the partition coefficient for solute  $\alpha$ . The partition coefficient incorporates the combined effects of solubility and long-range electrostatic interactions to determine the ratio of interstitial to external concentration for that solute. Therefore, the effective concentration represents a measure of the *activity* of the solute, as understood in chemistry. The effective fluid pressure represents that part of the pressure which is over and above osmotic effects.

In FEBio, nodal variables consist of the solid matrix displacement  $\mathbf{u}$ , the effective fluid pressure  $\tilde{p}$ , and the effective solute concentrations  $\tilde{c}^\alpha$ . Essential boundary conditions must be imposed on these variables, and not on the actual pressure  $p$  or concentrations  $c^\alpha$ . (In a biphasic material however, since  $c^\alpha = 0$ , the effective and actual fluid pressures are the same,  $p = \tilde{p}$ .)

The mixture stress in a triphasic material is given by  $\boldsymbol{\sigma} = -p\mathbf{I} + \boldsymbol{\sigma}^e$ , where  $\boldsymbol{\sigma}^e$  is the stress arising from the solid matrix strain. The mixture traction on a surface with unit outward normal  $\mathbf{n}$  is  $\mathbf{t} = \boldsymbol{\sigma} \cdot \mathbf{n}$ . This traction is continuous across the boundary surface. Therefore, the corresponding natural boundary condition for a multiphase mixture is  $\mathbf{t} = \mathbf{0}$ . (In other words, if no boundary condition is imposed on the solid matrix displacement or mixture traction, the natural boundary condition is in effect.)

The natural boundary conditions for the solvent and solutes are similarly  $\mathbf{w} \cdot \mathbf{n} = 0$  and  $\mathbf{j}^\alpha \cdot \mathbf{n} = 0$ , where  $\mathbf{w}$  is the volumetric flux of solvent relative to the solid and  $\mathbf{j}^\alpha$  is the molar flux of solute  $\alpha$  relative to the solid. In general,  $\mathbf{w}$  and  $\mathbf{j}^\alpha$  are given by

$$\begin{aligned}\mathbf{w} &= -\tilde{\mathbf{k}} \cdot \left( \nabla \tilde{p} + R\theta \sum_{\alpha} \frac{\tilde{\kappa}^\alpha}{d_0^\alpha} \mathbf{d}^\alpha \cdot \nabla \tilde{c}^\alpha \right), \\ \mathbf{j}^\alpha &= \tilde{\kappa}^\alpha \mathbf{d}^\alpha \cdot \left( -\varphi^w \nabla \tilde{c}^\alpha + \frac{\tilde{c}^\alpha}{d_0^\alpha} \mathbf{w} \right),\end{aligned}\tag{4.16.9}$$

where

$$\tilde{\mathbf{k}} = \left[ \mathbf{k}^{-1} + \frac{R\theta}{\varphi^w} \sum_{\alpha} \frac{\tilde{\kappa}^\alpha c^\alpha}{d_0^\alpha} \left( \mathbf{I} - \frac{\mathbf{d}^\alpha}{d_0^\alpha} \right) \right]^{-1}\tag{4.16.10}$$

is the effective hydraulic permeability of the interstitial fluid solution (solvent and solutes) through the porous solid matrix;  $\mathbf{k}$  is the hydraulic permeability of the solvent through the porous solid matrix;  $\mathbf{d}^\alpha$  is the diffusivity of solute  $\alpha$  through the mixture (frictional interactions with solvent and solid); and  $d_0^\alpha$  is its free diffusivity (frictional interactions with solvent only).  $\varphi^w \approx 1 - \varphi^s$  is the solid matrix porosity in the current configuration. The above expressions for the solvent and solute flux do not account for external body forces.

Also see Section 8.7 for additional guidelines for running multiphase materials.

### 4.16.1 Guidelines for Multiphasic Analyses

#### 4.16.1.1 Initial State of Swelling

Under traction-free conditions, a multiphasic material is usually in a state of swelling due to the osmotic pressure difference between the interstitial fluid and the external environment. This osmotic pressure arises from the difference in interstitial versus external concentrations of cations and anions, caused by the charged solid matrix and the requirement to satisfy electroneutrality. An osmotic pressure difference arising from such electrostatic interactions is known as the *Donnan osmotic pressure*. When the Donnan pressure is non-zero, traction-free conditions do not produce stress-free conditions for the solid matrix, since the matrix must expand until its stressed state resists the swelling pressure.

The Donnan pressure reduces to zero when the fixed charged density is zero, or when the external environment is infinitely hypertonic (having ion concentrations infinitely greater than the interstitial fixed charge density). Since these two conditions represent special cases, it is generally necessary to devise methods for achieving the desired initial state of swelling, in an analysis where loads or displacements need to be prescribed over and above this swollen state. Swelling occurs as a result of the influx of solvent into the porous solid matrix. This influx is a time-dependent process that could require extensive analysis time. Therefore, it is computationally efficacious to achieve the initial state of swelling by using a multi-step analysis (Chapter 6) where the first step is a steady-state analysis (Section 3.3.1). In this steady-state step, the fixed charge density may be ramped up from zero to the desired value using a load curve for that property or, alternatively, the external environmental conditions may be reduced from a very high hypertonic state down to the desired level. In the second step, prescribed displacement boundary conditions (when needed) may be specified to be of type *relative* (Section 3.11.1), so that they become superposed over and above the initial swelling state.

*Example:*

```
<Module type="multiphasic"/>
<Step>
  <Control>
    <analysis>STEADY-STATE</analysis>
    ...
  </Control>
</Step>
<Step>
  <Control>
    ...
  </Control>
  <Boundary>
    <bc type="prescribe" node_set="set1">
      <dof>z</dof>
      <scale>1</scale>
      <relative>1</relative>
    </prescribe>
  </Boundary>
</Step>
```

#### 4.16.1.2 Prescribed Boundary Conditions

In most analyses, it may be assumed that the ambient fluid pressure and electric potential in the external environment are zero, thus  $p_* = 0$  and  $\psi_* = 0$ , where the subscripted asterisk is used to denote environmental conditions. Since the external environment does not include a solid matrix, the fixed charge density there is zero. For example, in a triphasic analysis,  $c_*^+ = c_*^- \equiv c_*$ . It follows that the effective fluid pressure in the external environment is  $\tilde{p}_* = -R\theta\Phi_* \sum_{\alpha} c_*^{\alpha}$  and the effective concentrations are  $\tilde{c}_*^{\alpha} = c_*^{\alpha}/\tilde{\kappa}_*^{\alpha}$ . Therefore, in multiphasic analyses, whenever the external environment contains solutes with non-zero concentrations  $c_*^{\alpha}$ , the user must remember to prescribe non-zero boundary conditions for the effective solute concentrations *and* the effective fluid pressure.

Letting  $p_* = 0$  also implies that prescribed mixture normal tractions (Section 3.13.2.6) represent only the traction above ambient conditions. Note that users are not obligated to assume that  $p_* = 0$ . However, if a non-zero value is assumed for the ambient pressure, then users must remember to incorporate this non-zero value whenever prescribing mixture normal tractions. Similarly, users are not required to assume that  $\psi_* = 0$ ; when a non-zero value is assumed for the electric potential of the external environment, the prescribed boundary conditions for the effective concentrations should be evaluated using the corresponding partition coefficient,  $\tilde{c}_*^{\alpha} = c_*^{\alpha}/\tilde{\kappa}_*^{\alpha}$ .

#### 4.16.1.3 Prescribed Initial Conditions

When a multiphasic material is initially exposed to a given external environment with effective pressure  $\tilde{p}_*$  and effective concentrations  $\tilde{c}_*^{\alpha}$ , the initial conditions inside the material should be set to  $\tilde{p} = \tilde{p}_*$  and  $\tilde{c}^{\alpha} = \tilde{c}_*^{\alpha}$  in order to expedite the evaluation of the initial state of swelling. The values of  $\tilde{p}_*$  and  $\tilde{c}_*^{\alpha}$  should be evaluated as described in Section 8.7.2.

#### 4.16.1.4 Prescribed Effective Solute Flux

The finite element formulation for multiphasic materials in FEBio requires that the natural boundary condition for solute  $\alpha$  be prescribed as  $\tilde{j}_n^{\alpha} \equiv j_n^{\alpha} + \sum_{\beta} z^{\beta} j_n^{\beta}$ , where  $\tilde{j}_n^{\alpha}$  is the effective solute flux. For a mixture containing only neutral solutes ( $z^{\beta} = 0, \forall \beta$ ), it follows that  $\tilde{j}_n^{\alpha} = j_n^{\alpha}$ .

#### 4.16.1.5 Prescribed Electric Current Density

The electric current density in a mixture is a linear superposition of the ion fluxes,

$$\mathbf{I}_e = F_c \sum_{\alpha} z^{\alpha} \mathbf{j}^{\alpha}. \quad (4.16.11)$$

Since only the normal component  $j_n^{\alpha} = \mathbf{j}^{\alpha} \cdot \mathbf{n}$  of ion fluxes may be prescribed at a boundary, it follows that only the normal component  $I_n = \mathbf{I}_e \cdot \mathbf{n}$  of the current density may be prescribed. To prescribe  $I_n$ , it is necessary to know the nature of the ion species in the mixture, and how the current is being applied. For example, if the ions in the triphasic mixture consist of  $\text{Na}^+$  and  $\text{Cl}^-$ , and if the current is being applied using silver/silver chloride (Ag/AgCl) electrodes, a chemical reaction occurs at the anode where Ag combines with  $\text{Cl}^-$  to produce AgCl, donating an electron to the electrode to transport the current. At the cathode, the reverse process takes place. Therefore, in this system, there is only flux of  $\text{Cl}^-$  and no flux of  $\text{Na}^+$  ( $j_n^+ = 0$ ) at the electrode-mixture interface, so that the prescribed boundary condition should be  $j_n^- = -I_n/F_c$ .

Since  $z^+ = +1$  and  $z^- = -1$  in a triphasic mixture, the corresponding effective fluxes are given by  $\tilde{j}_n^+ = 2j_n^+ - j_n^- = I_n/F_c$  and  $\tilde{j}_n^- = j_n^+ = 0$ .

#### 4.16.1.6 Electrical Grounding

If a multiphasic mixture containing ions is completely surrounded by ion-impermeant boundaries it is necessary to ground the mixture to prevent unconstrained fluctuations of the electric potential. Grounding is performed by prescribing the effective concentration of one or more ions, at a location inside the mixture or on one of its boundaries corresponding to the location of the grounding electrode. The choice of ion(s) to constrain may be guided by the type of grounding electrode and the reference electrolyte bath in which it is inserted.

#### 4.16.1.7 Neglecting Osmotic Effects

Osmotic effects can be neglected in a multiphasic analysis by setting the osmotic coefficient to zero,  $\Phi = 0$ . In that case, the effective fluid pressure and the actual fluid pressure become equivalent measures,  $\tilde{p} = p$ , as for the case of a biphasic mixture.

#### 4.16.1.8 Solutes as 'Solid-Bound Molecules'

In multiphasic analyses, FEBio solves for the solid displacement  $\mathbf{u}$ , the effective fluid pressure  $\tilde{p}$ , and the effective solute concentrations  $\tilde{c}^\alpha$ , at all the nodes, representing all the degrees of freedom in an analysis. Once the solution has converged at each time point, FEBio solves for the referential mass densities  $\rho_r^\sigma$  of solid-bound molecules by solving the corresponding mass balance equation in eq.(4.17.7) at integration points, in a single numerical integration step. This approach reduces the number of degrees of freedom that need to be solved simultaneously by the FEBio solver. However, occasionally, the numerical solution for  $\rho_r^\sigma$  may produce numerical round-off errors that may get amplified as the analysis time progresses.

An alternative to using solid-bound molecules is to define solutes  $\sigma$  whose diffusivity  $d^\sigma$  in the mixture is set to 0 (however, the free diffusivity  $d_0^\sigma$  should not be set to zero, to prevent a division by zero; its exact value is not important, thus let  $d_0^\sigma = 1$ ). Then, FEBio treats these solutes as equivalent to solid-bound molecules: (1) Their concentration does not contribute to the osmolarity of the interstitial fluid; (2) their concentration contributes to the fixed charge density  $c^F$  in the current configuration; (3) these solutes do not contribute to the effective hydraulic permeability  $\tilde{k}$  of the porous multiphasic mixture; (4) these solutes can be involved in chemical reactions; (5) while their initial effective concentration may be prescribed, it must not contribute to the prescribed initial effective fluid pressure, and the user should not prescribe any boundary conditions on the effective concentration of these solutes. Using these 'solid-bound' solutes increases the number of degrees of freedom in an FEBio analysis; however, unlike solid-bound molecules, the solution for the effective concentration of these solutes remains as accurate as all other degrees of freedom in an analysis.

To evaluate the contribution of these solutes to the referential solid volume fraction  $\varphi_r^s$  and to chemical reactions, it should be recalled that the concentration  $c^\sigma$  is related to the referential mass density  $\rho_r^\sigma$  via eq.(4.17.8). Using the expression for  $\varphi_r^s$  in eq.(4.17.9), we find that

$$\varphi_r^s = \varphi_0^s + (J - \varphi_r^s) \sum_{\sigma} \frac{M^\sigma c^\sigma}{\rho_T^\sigma}$$



which can be solved for  $\varphi_r^s$  using the concentrations  $c^\sigma$ ,

$$\varphi_r^s = \frac{\varphi_0^s + J \sum_{\sigma} \frac{M^{\sigma} c^{\sigma}}{\rho_T^{\sigma}}}{1 + \sum_{\sigma} \frac{M^{\sigma} c^{\sigma}}{\rho_T^{\sigma}}}.$$

### 4.16.2 General Specification of Multiphasic Materials

The material type for a multiphasic material is “*multiphasic*”. Constitutive relations must be provided for the solid matrix, the mixture fixed charge density, the hydraulic permeability  $k$ , the osmotic coefficient  $\Phi$ , and the properties of each solute: the solute diffusivity in the mixture  $d^\alpha$ , the solute free diffusivity  $d_0^\alpha$ , and the solute effective solubility  $\hat{\kappa}^\alpha$ . Therefore, the following parameters must be defined:

|                        |                                                                    |                     |
|------------------------|--------------------------------------------------------------------|---------------------|
| <solid>                | specification of the solid matrix                                  |                     |
| <phi0>                 | solid volume fraction $\varphi_r^s$ in the reference configuration | [ ]                 |
| <fixed_charge_density> | fixed charge density $c_r^F$ in the reference configuration        | [n/L <sup>3</sup> ] |
| <permeability>         | specification of the hydraulic permeability $k$                    |                     |
| <solvent_supply>       | specification of the solvent supply $\hat{\varphi}^w$              |                     |
| <osmotic_coefficient>  | specification of the osmotic coefficient $\Phi$                    |                     |
| <solute>               | specification of the solute properties                             |                     |
| <solid_bound>          | specification of solid-bound molecule                              |                     |

The <solid> tag encloses a description of the solid matrix constitutive relation and associated material properties, as may be selected from the list provided in Section 4.1.4. The solid volume fraction in the reference configuration, <phi0>, must be greater than 0 (no solid) and less than 1 (only solid). The volume fraction of fluid (also known as the porosity) in the reference configuration is given by  $1 - \varphi_0$ . The <fixed\_charge\_density> may be negative, positive, or zero. The <permeability> and <osmotic\_coefficient> tags enclose descriptions of the permeability and osmotic coefficient constitutive relations and their associated material properties, as may be selected from the list presented in Sections 4.14.2 and 4.15.5.

The optional <solute> tag provides a description of each solute in the multiphasic mixture. Multiple solutes may be defined. Each tag includes the required *sol* attribute, which should reference a solute *id* from the <Solutes> description in the <Globals> section (Section 3.4.2). The following parameters must be defined in this description:

|               |                                                                        |
|---------------|------------------------------------------------------------------------|
| <diffusivity> | specification of the solute diffusivities $d^\alpha$ and $d_0^\alpha$  |
| <solubility>  | specification of the solute effective solubility $\hat{\kappa}^\alpha$ |

The <diffusivity> and <solubility> tags enclose descriptions of materials that may be selected from the lists presented in Sections 4.15.3 and 4.15.4, respectively. Each solute tag must include a “sol” attribute

The optional <solid\_bound> tag specifies which solid-bound molecule should be included in the multiphasic mixture. Multiple solid-bound molecules may be specified. Each tag should include the required *sbm* attribute, which references an *id* from the <SolidBoundMolecules> description in the <Globals> section (Section 3.4.3). The following parameter must be defined in this description:

|          |                                                                                               |                     |
|----------|-----------------------------------------------------------------------------------------------|---------------------|
| <rho0>   | initial value of the referential apparent density of the solid-bound molecule $\rho_r^\sigma$ | [M/L <sup>3</sup> ] |
| <rhomin> | optional minimum allowable value of $\rho_r^\sigma$ (zero by default)                         | [M/L <sup>3</sup> ] |
| <rhomax> | optional maximum allowable value of $\rho_r^\sigma$ (none by default)                         | [M/L <sup>3</sup> ] |

If a chemical reaction involves this solid-bound molecule its referential apparent density may evolve over time. The user may place lower and upper bounds on the allowable range of an evolving  $\rho_r^\sigma$ .

*Example:*

```
<material id="2" name="Media" type="multiphasic">
  <phi0>0.2</phi0>
  <fixed_charge_density>-40</fixed_charge_density>
  <solid name="Solid Matrix" type="Holmes-Mow">
    <density>1</density>
    <E>0.28</E>
    <v>0</v>
    <beta>0</beta>
  </solid>
  <permeability name="Permeability" type="perm-Holmes-Mow">
    <perm>1e-3</perm>
    <M>0</M>
    <alpha>0</alpha>
  </permeability>
  <osmotic_coefficient name="Ideal" type="osm-coef-const">
    <osmcoef>1.0</osmcoef>
  </osmotic_coefficient>
  <solute sol="1">
    <diffusivity name="Diffusivity" type="diff-const-iso">
      <free_diff>1e-3</free_diff>
      <diff>1e-3</diff>
    </diffusivity>
    <solubility name="Solubility" type="solub-const">
      <solub>1.0</solub>
    </solubility>
  </solute>
  <solute sol="2">
    <diffusivity name="Diffusivity" type="diff-const-iso">
      <free_diff>1e-3</free_diff>
      <diff>1e-3</diff>
    </diffusivity>
    <solubility name="Solubility" type="solub-const">
      <solub>1.0</solub>
    </solubility>
  </solute>
</material>
```

When a multiphasic material is employed in an analysis, it is also necessary to specify the values of the universal gas constant  $R$  [**F·L/n·T**], absolute temperature  $\theta$  [**T**], and Faraday's constant  $F_c$  [**Q/n**] in the <Globals> section, using a self-consistent set of units.

*Example:*

```
<Globals>
```

```

<Constants>
  <R>8.314e-6</R>
  <T>298</T>
  <Fc>96500e-9</Fc>
</Constants>
<Solutes>
  <solute id="1" name="Na">
    <charge_number>1</charge_number>
  </solute>
  <solute id="2" name="Cl">
    <charge_number>-1</charge_number>
  </solute>
</Solutes>
</Globals>

```

*Example:*

| Self-consistent units for a triphasic analysis |                                                |
|------------------------------------------------|------------------------------------------------|
| Primary Units                                  |                                                |
| time                                           | s                                              |
| length                                         | mm                                             |
| force                                          | N                                              |
| mole                                           | nmol                                           |
| charge                                         | C                                              |
| temperature                                    | K                                              |
| Derived Units                                  |                                                |
| stress                                         | N/mm <sup>2</sup> , MPa                        |
| permeability                                   | mm <sup>4</sup> /N·s, mm <sup>2</sup> /MPa · s |
| diffusivity                                    | mm <sup>2</sup> /s                             |
| concentration                                  | nmol/mm <sup>3</sup> , mM                      |
| charge density                                 | nEq/mm <sup>3</sup> , mEq/L                    |
| voltage                                        | mV                                             |
| current density                                | A/mm <sup>2</sup>                              |
| current                                        | A                                              |

It is also possible to create models with multiphasic materials that use different solutes in different regions. In that case, introduce additional solute entries in the <Solutes> section and refer to those solute ids in the multiphasic material descriptions. Generally, two adjoining multiphasic regions may share the same solute (e.g., Na in both regions), in which case that solute may transport freely across the interface separating these regions; or they may share no solute, in which case the interface is impermeable to all solutes.

### 4.16.3 Solvent Supply Materials

Solvent supply materials may be used to simulate an external source of solvent, such as supply from microvasculature that is not modeled explicitly. The solvent supply term,  $\hat{\varphi}^w$ , appears in the mass balance relation for the mixture,

$$\text{div}(\mathbf{v}^s + \mathbf{w}) = \hat{\varphi}^w. \quad (4.16.12)$$

$\hat{\varphi}^w$  has units of reciprocal time [ $\text{t}^{-1}$ ]; it represents the rate at which the volume fraction of solvent changes with time.

#### 4.16.3.1 Starling Equation

The material type for Starling's equation for fluid supply is “*Starling*”. The following material parameters need to be defined:

|                          |                                                                         |                       |
|--------------------------|-------------------------------------------------------------------------|-----------------------|
| <kp>                     | hydraulic filtration coefficient, $k_p$                                 | [L <sup>2</sup> /F.t] |
| <pv>                     | effective fluid pressure in external source, $\tilde{p}_v$              | [P]                   |
| <q <sub>c</sub> sol="n"> | osmotic filtration coefficient, $q_c^\alpha$                            | [L <sup>3</sup> /n.t] |
| <cv sol="n">             | effective solute concentration in external source, $\tilde{c}_v^\alpha$ | [n/L <sup>3</sup> ]   |

The fluid supply is given by Starling's equation,

$$\hat{\varphi}^w = k_p (\tilde{p}_v - \tilde{p}) + \sum_{\alpha} q_c^\alpha (\tilde{c}_v^\alpha - \tilde{c}^\alpha) ,$$

where  $\tilde{p}$  is the effective fluid pressure in the multiphasic material.

*Example:*

This example defines a solvent supply material of the Starling type for a multiphasic mixture containing one solute.

```
<solvent_supply type="Starling">
  <kp>0.001</kp>
  <pv>0.1</pv>
  <qc sol="1">1e-5</qc>
  <cv sol="1">150</cv>
</solvent_supply>
```

## 4.17 Chemical Reactions

The following sections describe FEBio's formulation for analyzing mechanochemical events in neutral or charged deformable porous media under finite deformation. The formulation allows full coupling of mechanical and chemical effects, providing a framework where material properties and response functions may depend on solid matrix strain as well as solute concentration.

If you use these capabilities in your published research, we request that you cite the following publication describing its development [10].

### 4.17.1 Guidelines for Chemical Reaction Analyses

Chemical reactions may be modeled within a multiphasic mixture. The reaction may involve solutes ( $\alpha = \iota$ ) and solid-bound molecules ( $\alpha = \sigma$ ) that move with the solid matrix ( $\mathbf{v}^\sigma = \mathbf{v}^s, \forall \sigma$ ). Consider a general chemical reaction,

$$\sum_{\alpha} \nu_R^\alpha \mathcal{E}^\alpha \rightarrow \sum_{\alpha} \nu_P^\alpha \mathcal{E}^\alpha, \quad (4.17.1)$$

where  $\mathcal{E}^\alpha$  is the chemical species representing constituent  $\alpha$  in the mixture;  $\nu_R^\alpha$  and  $\nu_P^\alpha$  represent stoichiometric coefficients of the reactants and products, respectively. To maintain consistency with classical chemical kinetics, the analysis of chemical reactions employs molar concentrations  $c^\alpha$  and molar supplies  $\hat{c}^\alpha$  on a solution-volume basis for all reactants and products, whether they are solutes or solid-bound molecular species.

Since the molar supply of reactants and products is constrained by stoichiometry, it follows that all molar supplies  $\hat{c}^\alpha$  in a specific chemical reaction may be related to a *molar production rate*  $\hat{\zeta}$  according to

$$\hat{c}^\alpha = \nu^\alpha \hat{\zeta}, \quad (4.17.2)$$

where  $\nu^\alpha$  represents the net stoichiometric coefficient for  $\mathcal{E}^\alpha$ ,

$$\nu^\alpha = \nu_P^\alpha - \nu_R^\alpha. \quad (4.17.3)$$

Thus, formulating constitutive relations for  $\hat{c}^\alpha$  is equivalent to providing a single relation for  $\hat{\zeta}(\theta, \mathbf{F}, c^\alpha)$ . When the chemical reaction is reversible,

$$\sum_{\alpha} \nu_R^\alpha \mathcal{E}^\alpha \rightleftharpoons \sum_{\alpha} \nu_P^\alpha \mathcal{E}^\alpha, \quad (4.17.4)$$

the relations of (4.17.2)-(4.17.3) still apply but the constitutive relation for  $\hat{\zeta}$  would be different.

*Example:*

Consider the dissociation of  $\text{CaCl}_2$  into ions  $\text{Ca}^{2+}$  and  $\text{Cl}^-$ ,



The mixture contains three constituents. The stoichiometric coefficients of the reactants are  $\nu_R^{\text{CaCl}_2} = 1$ ,  $\nu_R^{\text{Ca}^{2+}} = 0$ ,  $\nu_R^{\text{Cl}^-} = 0$ , and those of the products are  $\nu_P^{\text{CaCl}_2} = 0$ ,  $\nu_P^{\text{Ca}^{2+}} = 1$ ,  $\nu_P^{\text{Cl}^-} = 2$ .

The reaction production rate  $\hat{\zeta}$  enters into the governing equations of multiphasic mixtures via the mass balance relation for each solute,

$$\frac{1}{J} \frac{D^s [J(1 - \varphi^s) c^\iota]}{Dt} + \text{div } \mathbf{j}^\iota = (1 - \varphi^s) \nu^\iota \hat{\zeta}, \quad (4.17.5)$$

the mass balance for the mixture,

$$\text{div}(\mathbf{v}^s + \mathbf{w}) = (1 - \varphi^s) \hat{\zeta} \bar{\mathcal{V}}, \quad (4.17.6)$$

where  $\bar{\mathcal{V}} = \sum_{\alpha} \nu^{\alpha} \mathcal{V}^{\alpha}$  and  $\mathcal{V}^{\alpha} = M^{\alpha} / \rho_T^{\alpha}$  is the molar volume of  $\alpha$ , and the mass balance for solid-bound constituents,

$$D^s \rho_r^{\sigma} / Dt = \hat{\rho}_r^{\sigma}, \quad (4.17.7)$$

where  $\rho_r^{\sigma}$  is the referential apparent mass density (mass of  $\sigma$  per mixture volume in the reference configuration), and  $\hat{\rho}_r^{\sigma}$  is the referential apparent mass supply of solid constituent  $\sigma$ , related to molar concentrations and supplies via

$$c^{\sigma} = \frac{\rho_r^{\sigma}}{(J - \varphi_r^s) M^{\sigma}}, \quad \hat{c}^{\sigma} = \frac{\hat{\rho}_r^{\sigma}}{(J - \varphi_r^s) M^{\sigma}}. \quad (4.17.8)$$

Internally, the content of solid-bound species is stored in  $\rho_r^{\sigma}$  and (4.17.8) is used to evaluate  $c^{\sigma}$  when needed for the calculation of  $\hat{\zeta}$ . If a solid-bound molecule is involved in a chemical reaction, equation (4.17.7) is integrated to produce an updated value of  $\rho_r^{\sigma}$ , using  $\hat{\rho}_r^{\sigma} = (J - \varphi_r^s) M^{\sigma} \nu^{\sigma} \hat{\zeta}$  based on (4.17.2) and (4.17.8).

Evolving solid content due to chemical reactions implies that the referential solid volume fraction  $\varphi_r^s$  may not remain constant. This value is updated at every time point using

$$\varphi_r^s = \varphi_0^s + \sum_{\sigma} \frac{\rho_r^{\sigma}}{\rho_T^{\sigma}}, \quad (4.17.9)$$

where  $\varphi_0^s$  is the solid volume fraction specified by the multiphasic material parameter *phi0* (Section 4.16.2). Thus,  $\varphi_0^s$  may be used to account for the solid volume fraction not contributed explicitly by solid-bound molecules. Based on kinematics, the solid volume fraction in the current configuration is given by  $\varphi^s = \varphi_r^s / J$ . Therefore, since  $0 \leq \varphi^s \leq 1$  by definition, it follows that  $0 \leq \varphi_r^s \leq J$ , implying that the referential solid volume fraction may evolve to values greater than unity when growth leads to swelling of the multiphasic mixture.

Similarly, if solid-bound molecules are charged and their content evolves over time, the referential fixed charge density may also evolve with chemical reactions according to

$$c_r^F = c_0^F + \frac{1}{1 - \varphi_r^s} \sum_{\sigma} \frac{z^{\sigma} \rho_r^{\sigma}}{M^{\sigma}}, \quad (4.17.10)$$

where  $c_0^F$  is the referential fixed charge density specified by the multiphasic material parameter *fixed\_charge\_density* (Section 4.16.2). Thus,  $c_0^F$  may be used to account for the fixed charge density not contributed explicitly by solid-bound molecules.

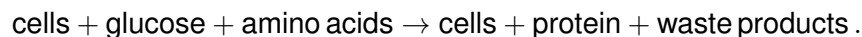
A chemical reaction is properly balanced when

$$\sum_{\alpha} \nu^{\alpha} M^{\alpha} = 0, \quad (4.17.11)$$

where  $M^{\alpha}$  is the molar mass of  $\alpha$ . This constraint implies that the net gain in mass of products must be the same as the net loss in mass of reactants. However, this constraint is not verified in the code, allowing users to model chemical reactions with implicit constituents (constituents that are neither explicitly modeled as solutes nor as solid-bound molecules, for which  $\nu^{\alpha}$  and  $M^{\alpha}$  are



not given). For example, a chemical reaction where cells consume glucose to form a protein from amino-acids building blocks may have the form



The user may opt to model only the glucose reactant and the protein product explicitly, while the presence of all other species in this reaction is implicit. In these types of analyses the user must beware of potential inconsistencies in the evolving mass of reactants and products since only some of those constituents are modeled explicitly. In particular, the evolution of  $\varphi_r^s$  as given in (4.17.9) can only account for the explicitly modeled solid-bound molecules. Furthermore, when some reactants and products are implicit, the value of the reaction molar volume  $\bar{V}$  calculated in the code becomes inaccurate and may produce unexpected results in the evaluation of the mixture mass balance relation in (4.17.6). Therefore, the user is given the option to override the value of  $\bar{V}$  calculated in the code. In particular, if the precise molar volumes of all the species in a reaction are not known, assuming that  $\bar{V} \approx 0$  is a reasonable choice equivalent to assuming that all the constituents have approximately the same density  $\rho_T^\alpha$ , as may be deduced from (4.17.11).

Since the electroneutrality condition is enforced in multiphasic mixtures in FEBio, it follows that chemical reactions must not violate this condition. Enforcing electroneutrality in a chemical reaction is equivalent to satisfying

$$\sum_{\alpha} z^{\alpha} \nu^{\alpha} = 0 . \quad (4.17.12)$$

This constraint is checked within the code and an error is generated when it is violated.

A constitutive relation must be provided for the molar production rate  $\hat{\zeta}(\theta, \mathbf{F}, c^{\alpha})$  of each chemical reaction.

### 4.17.2 General Specification for Chemical Reactions

The material type for a chemical reaction is “*reaction*”. The `<reaction>` tag must appear inside the definition of a multiphasic mixture and the reaction is defined only for that mixture. Multiple chemical reactions may be defined in a given mixture. Each `<reaction>` tag must include a *type* attribute that identifies the constitutive relation for  $\hat{\zeta}$ .

The stoichiometric coefficients  $\nu_R^\alpha$  of the reactants and  $\nu_P^\alpha$  for the products must be specified in every reaction. Optionally, the net reaction molar volume  $\bar{V}$  may be specified explicitly to override the internal value calculated in the code. Therefore, the following parameters need to be defined in a chemical reaction:

|                           |                                                    |                               |
|---------------------------|----------------------------------------------------|-------------------------------|
| <code>&lt;vR&gt;</code>   | reactant stoichiometric coefficient $\nu_R^\alpha$ | [ ]                           |
| <code>&lt;vP&gt;</code>   | product stoichiometric coefficient $\nu_P^\alpha$  | [ ]                           |
| <code>&lt;Vbar&gt;</code> | optional override value for $\bar{V}$              | [ $\mathbf{L}^3/\mathbf{n}$ ] |

Each `<vR>` and `<vP>` tag must include either the *sol* attribute, which should reference a solute *id* from the `<Solutes>` description in the `<Globals>` section (Section 3.4.2), or the *sbm* attribute, which should reference a solid-bound molecule *id* from the `<SolidBoundMolecules>` description in the `<Globals>` section (Section 3.4.3). Only solutes and solid-bound molecules that have been included in the parent multiphasic mixture may be specified as reactants or products of a chemical reaction.

Additional parameters may be needed in the definition of a chemical reaction, depending on the specific form of the constitutive relation for the production rate.

### 4.17.3 Chemical Reaction Materials

#### 4.17.3.1 Law of Mass Action for Forward Reactions

The material type for the Law of Mass Action for a forward reaction is *mass-action-forward*. The following parameters must be defined:

|                |                                    |
|----------------|------------------------------------|
| <forward_rate> | specific forward reaction rate $k$ |
|----------------|------------------------------------|

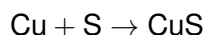
For this type of reaction the constitutive relation for the molar production rate is given by

$$\hat{\zeta} = k \prod_{\alpha} (c^{\alpha})^{\nu_R^{\alpha}}.$$

The constitutive form of the specific forward reaction rate must be selected from the list of materials given in Section 4.17.4. The units of  $\hat{\zeta}$  are  $[\mathbf{n/L}^3 \cdot \mathbf{t}]$  and those of  $c^{\alpha}$  are  $[\mathbf{n/L}^3]$ .

*Example:*

Consider the forward reaction that produces solid copper sulfide from solid copper and solid sulfur,



All three species are modeled explicitly in the mixture as solid-bound molecules, with id's 1 (Cu), 2 (for S) and 3 (for CuS). The chemical reaction material is given by:

```
<reaction name="copper sulfide production" type="mass-action-forward">
  <vR sbm="1">1</vR>
  <vR sbm="2">1</vR>
  <vP sbm="3">1</vP>
  <forward_rate type="constant reaction rate">
    <k>1e-3</k>
  </forward_rate>
</reaction>
```

#### 4.17.3.2 Law of Mass Action for Reversible Reactions

The material type for the Law of Mass Action for a reversible reaction is *mass-action-reversible*. The following parameters must be defined:

|                |                                      |
|----------------|--------------------------------------|
| <forward_rate> | specific forward reaction rate $k_F$ |
| <reverse_rate> | specific reverse reaction rate $k_R$ |

For this type of reaction the constitutive relation for the molar production rate is given by

$$\hat{\zeta} = \hat{\zeta}_F - \hat{\zeta}_R,$$

where

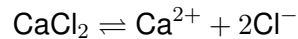
$$\hat{\zeta}_F = k_F \prod_{\alpha} (c^{\alpha})^{\nu_R^{\alpha}}$$

$$\hat{\zeta}_R = k_R \prod_{\alpha} (c^{\alpha})^{\nu_P^{\alpha}}.$$

The constitutive form of the specific forward and reverse reaction rates must be selected from the list of materials given in Section 4.17.4. The units of  $\hat{\zeta}_F$  and  $\hat{\zeta}_R$  are  $[\mathbf{n/L^3 \cdot t}]$  and those of  $c^{\alpha}$  are  $[\mathbf{n/L^3}]$ .

*Example:*

Consider the reversible dissociation of CaCl salt into  $\text{Ca}^{2+}$  and  $\text{Cl}^{-}$  in water,



All three species are modeled explicitly in the mixture as solutes, with id's 1 (for  $\text{CaCl}_2$ ), 2 (for  $\text{Ca}^{2+}$ ) and 3 (for  $\text{Cl}^{-}$ ). The chemical reaction material is given by:

```
<reaction name="CaCl2 dissociation" type="mass-action-reversible">
  <vR sol="1">1</vR>
  <vP sol="2">1</vP>
  <vP sol="3">2</vP>
  <forward_rate type="constant">
    <k>1.0</k>
  </forward_rate>
  <reverse_rate type="constant">
    <k>0.1</k>
  </reverse_rate>
</reaction>
```

### 4.17.3.3 Michaelis-Menten Reaction

The material type for the Michaelis-Menten reaction is *Michaelis-Menten*. The following parameters must be defined:

|                |                                                                  |                       |
|----------------|------------------------------------------------------------------|-----------------------|
| <forward_rate> | maximum rate at saturating substrate concentration $V_{\max}$    | [n/L <sup>3</sup> ·t] |
| <Km>           | substrate concentration when reaction rate is half of $V_{\max}$ | [n/L <sup>3</sup> ]   |
| <c0>           | optional minimum substrate concentration to trigger reaction     | [n/L <sup>3</sup> ]   |

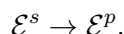
The Michaelis-Menten reaction may be used to model enzyme kinetics where the enzyme  $\mathcal{E}^e$  triggers the conversion of the substrate  $\mathcal{E}^s$  into the product  $\mathcal{E}^p$ . The product molar supply is given by

$$\hat{c}^p = \begin{cases} \frac{V_{\max} c^s}{K_m + c^s} & c^s \geq c_0 \\ 0 & c^s < c_0 \end{cases},$$

where  $c^s$  is the substrate concentration. The default value of  $c_0$  is 0. This relation may be derived, with some simplifying assumptions, by applying the law of mass action to the combination of two reactions,



Since the enzyme is not modeled explicitly in the Michaelis-Menten approximation to these two reactions, this reaction model is effectively equivalent to



Therefore, this reaction accepts only one reactant tag <vR> and one product tag <vP>. For consistency with the formulation of this reaction, the stoichiometric coefficients should be set to  $\nu_R^s = \nu_P^p = 1$ , so that  $\hat{c}^p = \hat{\zeta}$ .

The constitutive form of the specific forward reaction rate  $V_{\max}$  must be selected from the list of materials given in Section 4.17.4.

*Example:*

```
<reaction name="enzyme kinetics" type="Michaelis-Menten">
  <Vbar>0</Vbar>
  <vR sol="1">1</vR>
  <vP sol="2">1</vP>
  <forward_rate type="constant">
    <k>1.0</k>
  </forward_rate>
  <Km>5.0</Km>
</reaction>
```

#### **4.17.4 Specific Reaction Rate Materials**

The following sections define specific materials that can be used to define the reaction rate of a chemical reaction.

#### 4.17.4.1 Constant Reaction Rate

The material type for a constant specific reaction rate is *constant reaction rate*. The following parameter must be defined:

|     |                                     |              |
|-----|-------------------------------------|--------------|
| <k> | constant specific reaction rate $k$ | [units vary] |
|-----|-------------------------------------|--------------|

*Example:*

```
<reaction name="enzyme kinetics" type="Michaelis-Menten">
  <Vbar>0</Vbar>
  <vR sol="1">1</vR>
  <vP sol="2">1</vP>
  <forward_rate type="constant reaction rate">
    <k>1.0</k>
  </forward_rate>
</reaction>
```

#### 4.17.4.2 Huiskes Reaction Rate

The material type for the Huiskes reaction rate is *Huiskes reaction rate*. This material model employs the bone remodeling framework proposed by Weinans et al. [89] and extended by Mullender et al. [65]. The following parameters must be defined:

|        |                                                            |              |
|--------|------------------------------------------------------------|--------------|
| <B>    | reaction rate per specific strain energy $B$ (default=0)   | [units vary] |
| <psi0> | specific strain energy at homeostasis $\psi_0$ (default=0) | [F·L/M]      |
| <D>    | characteristic distance from sensors (default=0)           | [L]          |

The Huiskes specific reaction rate has the form

$$k = \frac{1}{(J - \varphi_r^s) M^s} \left( B \left( \frac{\Psi_r}{\rho_r^s} - \psi_0 \right) + \sum_{i=1}^N e^{-d_i/D} B_i \left( \frac{\Psi_{ri}}{\rho_{ri}^s} - \psi_0 \right) \right),$$

where  $\Psi_r$  is the strain energy density of the solid (strain energy in current configuration per mixture volume in the reference configuration) and  $\rho_r^s = \sum_{\sigma} \rho_r^{\sigma}$  is the referential apparent solid density (mass of solid in current configuration per mixture volume in reference configuration). The ratio  $\Psi_r/\rho_r^s$  is the specific strain energy (strain energy per mass of solid in the current configuration). The Huiskes specific reaction rate may assume positive and negative values; it reduces to zero at homeostasis, when  $\Psi_r/\rho_r^s = \psi_0$ .

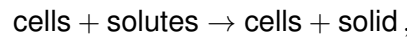
Per Mullender et al. [65], the remodeling rate  $k$  at the current material point may depend not only on the specific strain energy at the current point, but also on its value in the neighborhood of the current point (e.g., due to the presence of neighboring sensing cells that provide some form of feedback at the current location). Here,  $N$  represents the number of elements in the neighborhood of the current material point element, which fall within a characteristic sensing distance  $D$ , and  $d_i$  represents the distance between these neighboring points and the current material point.

When  $D = 0$  it follows that  $N = 0$  and the model uses the original formulation of Weinans et al. [89]. When  $D > 0$ , as per Mullender et al. [65], the finite element code first searches for all the neighboring elements  $N$  which fall within a distance of  $d_i \leq 4 \times D$  of the element at which  $k$  needs to be evaluated (here, using  $d_i \leq 4 \times D$  implies that the exponential term becomes negligible beyond this distance, since  $e^{-4} = 0.0183156$ ). This search is conducted once, at the start of the analysis, implying that it is not updated at subsequent time points when the deformation may alter distances. With increasing values of  $D$ , the number  $N$  of neighboring elements may increase considerably, especially in three-dimensional models, which will considerably slow down the computation of  $k$ , therefore users must be careful with choosing an appropriate value for  $D$ .

Since the referential solid volume fraction  $\varphi_r^s$  of a multiphasic mixture evolves with its composition as per eq.368, users must be careful not to allow  $J - \varphi_r^s \rightarrow 0$  as Huiskes remodeling takes place, or else the computation of  $k$  from the above formula becomes nearly singular. Typically this can be done by specifying the true density of the solid,  $\rho_T^s$  (<density> in Section 3.4.3), to be much greater than the maximum apparent density  $\rho_r^s$  allowed in the solid remodeling reaction (<rhomax> in Section 4.16.2). Then, the contribution of the evolving  $\rho_r^s$  to  $\varphi_r^s$  in eq.368 will remain negligible.

*Example:*

To reproduce the interstitial solid remodeling theory proposed by Weinans et al. [89], consider the forward reaction



where cells convert solutes (e.g., nutrients) into synthesized solid when the specific strain energy exceeds the homeostatic value. If the cells and solutes are implicit in this reaction (e.g., if it



is assumed that their concentrations change negligibly), the production rate of the solid may be given by  $\hat{\zeta} = k$  using the law of mass action for a forward reaction, where  $k$  has the form given above.

```
<reaction name="solid remodeling" type="mass-action-forward">
  <vP sbm="1">1</vP>
  <forward_rate type="Huiskes reaction rate">
    <B>1.0</B>
    <psi0>0.01</psi0>
    <D>0</D>
  </forward_rate>
</reaction>
```

## 4.18 Rigid Body

A rigid body can be defined using the rigid material model. The material type for a rigid body is “*rigid body*”. The following parameters are defined:

|                  |                                |                     |
|------------------|--------------------------------|---------------------|
| <density>        | Density of rigid body          | [M/L <sup>3</sup> ] |
| <center_of_mass> | Position of the center of mass | [L]                 |
| <E>              | Young’s modulus                | [P]                 |
| <v>              | Poisson’s ratio                | [ ]                 |

If the *center\_of\_mass* parameter is omitted, FEBio will calculate the center of mass automatically. In this case, a density *must* be specified. If the *center\_of\_mass* is defined the *density* parameter may be omitted. Omitting both will generate an error message.

The Young’s modulus *E* and Poisson ratio *v* currently have no effect on the results. The only place where they are used is in the calculation of a material stiffness for some auto-penalty contact formulation. If you are using contact it is advised to enter sensible values. Otherwise these parameters may be omitted.

The degrees of freedom of a rigid body are initially unconstrained<sup>3</sup>. This implies that a rigid body is free to move in all three directions and rotate about any of the coordinate axes. To constrain a rigid body degree of freedom you may use the *Constraints* sections (Section 3.12.1).

*Example:*

```
<material id="1" type="rigid body">
  <density>1.0</density>
  <center_of_mass>0,0,0</center_of_mass>
</material>
```

<sup>3</sup>This is different from previous versions of FEBio where rigid bodies were initially fully constrained.

## 4.19 Active Contraction

Active contraction materials may be used to impose a (typically) contractile stress within a solid material, by incorporating the contraction material within a solid mixture. The total stress in the solid mixture is simply  $\sigma = \sigma^s + \sigma^a$ , where  $\sigma^s$  is the solid stress (due to strain and strain history), and  $\sigma^a$  is the active contractile stress. The calculation of the contractile stress is the same, whether the solid mixture consists of uncoupled or unconstrained materials.

### 4.19.1 Contraction in Mixtures of Uncoupled Materials

When the solid mixture consists of uncoupled materials, the active contraction material should be selected from the list below.

#### 4.19.1.1 Uncoupled Prescribed Uniaxial Active Contraction

The material type for prescribed active contraction along a single direction (or fiber) in an uncoupled solid mixture is “*uncoupled prescribed uniaxial active contraction*”. This material must be combined with a stable uncoupled material that acts as a passive matrix, using a “*uncoupled solid mixture*” container as described in Section 4.1.2.16. The following material parameters need to be defined:

|      |                                            |     |
|------|--------------------------------------------|-----|
| <T0> | $T_0$ , representing the prescribed stress | [P] |
|------|--------------------------------------------|-----|

In the reference configuration, the fiber is oriented along the unit vector  $\mathbf{e}_1$ , where  $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$  are orthonormal basis vectors representing the local element coordinate system when specified (Section 4.1.1), or else the global Cartesian coordinate system. The active stress  $\boldsymbol{\sigma}^a$  for this material is given by

$$\boldsymbol{\sigma}^a = J^{-1} T_0 \mathbf{n} \otimes \mathbf{n},$$

where  $\mathbf{n} = \mathbf{F} \cdot \mathbf{n}_r$  is the stretched fiber orientation in the current (deformed) configuration.

*Example:*

Uniaxial contraction along  $\mathbf{e}_1$ , in a mixture containing a Mooney-Rivlin solid.

```
<material id="1" type="uncoupled solid mixture">
  <mat_axis type="local">0,0,0</mat_axis>
  <k>1000</k>
  <solid type="Mooney-Rivlin">
    <c1>1.0</c1>
    <c2>0</c2>
  </solid>
  <solid type="prescribed uniaxial active contraction uncoupled">
    <T0 lc="2">1</T0>
    <mat_axis type="angles">
      <theta>0</theta>
      <phi>90</phi>
    </mat_axis>
  </solid>
</material>
```

#### 4.19.1.2 Uncoupled Prescribed Transversely Isotropic Active Contraction

The material type for prescribed isotropic active contraction in a plane transverse to a given direction (or fiber), in an uncoupled solid mixture, is “*prescribed trans iso active contraction uncoupled*”. This material must be combined with a stable uncoupled material that acts as a passive matrix, using a “*uncoupled solid mixture*” container as described in Section 4.1.2.16. The following material parameters need to be defined:

|      |                                            |     |
|------|--------------------------------------------|-----|
| <T0> | $T_0$ , representing the prescribed stress | [P] |
|------|--------------------------------------------|-----|

In the reference configuration, the fiber is oriented along the unit vector  $\mathbf{e}_1$ , where  $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$  are orthonormal basis vectors representing the local element coordinate system when specified

(Section 4.1.1), or else the global Cartesian coordinate system. The active stress  $\sigma^a$  for this material is given by

$$\sigma^a = J^{-1}T_0 (\mathbf{B} - \mathbf{n} \otimes \mathbf{n}) ,$$

where  $\mathbf{n} = \mathbf{F} \cdot \mathbf{e}_1$  is the stretched fiber orientation in the current (deformed) configuration and  $\mathbf{B} = \mathbf{F} \cdot \mathbf{F}^T$  is the left Cauchy-Green tensor.

*Example:*

Isotropic contraction in plane transverse to  $\mathbf{e}_1$ , in a mixture containing a Mooney-Rivlin solid.

```
<material id="1" type="uncoupled solid mixture">
  <mat_axis type="local">0,0,0</mat_axis>
  <solid type="Mooney-Rivlin">
    <c1>1.0</c1>
    <c2>0</c2>
    <k>1000</k>
  </solid>
  <solid type="prescribed trans iso active contraction uncoupled">
    <T0 lc="2">1</T0>
    <mat_axis type="angles">
      <theta>0</theta>
      <phi>90</phi>
    </mat_axis>
  </solid>
</material>
```

#### 4.19.1.3 Uncoupled Prescribed Isotropic Active Contraction

The material type for prescribed isotropic active contraction, in an uncoupled solid mixture, is “*prescribed isotropic active contraction uncoupled*”. This material must be combined with a stable uncoupled material that acts as a passive matrix, using a “*uncoupled solid mixture*” container as described in Section 4.1.2.16. The following material parameters need to be defined:

|      |                                            |     |
|------|--------------------------------------------|-----|
| <T0> | $T_0$ , representing the prescribed stress | [P] |
|------|--------------------------------------------|-----|

The active stress  $\sigma^a$  for this material is given by

$$\sigma^a = J^{-1}T_0\mathbf{B} .$$

Note: If the solid material in the mixture is (nearly) incompressible, this isotropic contraction will cause no change in the deformation.

*Example:*

Isotropic contraction in a mixture containing a Mooney-Rivlin solid.

```
<material id="1" type="uncoupled solid mixture">
  <mat_axis type="local">0,0,0</mat_axis>
  <solid type="Mooney-Rivlin">
    <c1>1.0</c1>
    <c2>0</c2>
    <k>5.0</k>
```

```

</solid>
<solid type="prescribed isotropic active contraction uncoupled">
  <T0 lc="2">1</T0>
</solid>
</material>

```

#### 4.19.1.4 Prescribed Fiber Stress

An active fiber stress, based on a Hill formulation, can be added via the material “*uncoupled active fiber stress*”. This material must be combined with a stable compressible material that acts as a passive matrix, using a “*uncoupled solid mixture*” container as described in Section 4.1.2.16. The stress is given by,

$$\boldsymbol{\sigma}^a = J^{-1} T(\tilde{\lambda}) \operatorname{dev} \mathbf{A}.$$

Here,  $\mathbf{A} = \mathbf{a} \otimes \mathbf{a}$ , with  $\mathbf{a}$  the unit vector describing the fiber direction in the spatial frame,  $\tilde{\lambda}$  is the deviatoric fiber stretch,  $J$  is the jacobian of the deformation, and

$$T(\tilde{\lambda}) = s_{max} a(t) s_{TL}(\tilde{\lambda}) s_{TV}(\dot{\tilde{\lambda}})$$

The parameters are defined below.

|      |                                                                          |     |
|------|--------------------------------------------------------------------------|-----|
| smax | scale factor for defining maximum stress                                 | [P] |
| a    | activation level                                                         |     |
| stl  | Function defining the stress-stretch relationship for the material       |     |
| stv  | Function defining the stress-stretch rate relationship for the material. |     |

The parameters  $s_{TL}$  and  $s_{TV}$  are functions that need to be defined in place. There are currently two ways of defining these functions, either via a mathematical expression or a list of sample points. An example is given below. If these parameters are omitted, they are replaced by the constant 1 in the equation for the stress above.

*Example:*

An example defining the stl parameter via a mathematical expression.

```

<material id="1" type="uncoupled solid mixture">
  <k>100.0</k>
  <mat_axis type="local">0,0,0</mat_axis>
  <solid type="Mooney-Rivlin">
    <c1>1.0</c1>
    <c2>0</c2>
  </solid>
  <solid type="uncoupled active fiber stress">
    <smax>1.5</smax>
    <a lc="1">0.1</a>
    <stl type="math">
      <math>(1-l)^2</math>
    </stl>
  </solid>
</material>

```

An example defining the stl parameter via a point list.

```
<material id="1" type="uncoupled solid mixture">
  <k>100.0</k>
  <mat_axis type="local">0,0,0</mat_axis>
  <solid type="Mooney-Rivlin">
    <c1>1.0</c1>
    <c2>0</c2>
  </solid>
  <solid type="uncoupled active fiber stress">
    <smax>1.5</smax>
    <a lc="1">0.1</a>
    <stl type="point">
      <interpolate>linear</interpolate>
      <points>
        <pt>0,0</pt>
        <pt>1,0</pt>
        <pt>2,1</pt>
      </points>
    </stl>
  </solid>
</material>
```

### 4.19.2 Contraction in Mixtures of Unconstrained Materials

When the solid mixture consists of unconstrained materials, the active contraction material should be selected from the list below.

#### 4.19.2.1 Prescribed Uniaxial Active Contraction

The material type for prescribed active contraction along a single direction (or fiber) in a solid mixture of unconstrained materials is “*prescribed uniaxial active contraction*”. This material must be combined with a stable compressible material that acts as a passive matrix, using a “*solid mixture*” container as described in Section 4.1.4.27. The following material parameters need to be defined:

|      |                                            |     |
|------|--------------------------------------------|-----|
| <T0> | $T_0$ , representing the prescribed stress | [P] |
|------|--------------------------------------------|-----|

In the reference configuration, the fiber is oriented along the unit vector  $\mathbf{e}_1$ , where  $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$  are orthonormal basis vectors representing the local element coordinate system when specified (Section 4.1.1), or else the global Cartesian coordinate system. The active stress  $\boldsymbol{\sigma}^a$  for this material is given by

$$\boldsymbol{\sigma}^a = J^{-1} T_0 \mathbf{n} \otimes \mathbf{n},$$

where  $\mathbf{n} = \mathbf{F} \cdot \mathbf{e}_1$  is the stretched fiber orientation in the current (deformed) configuration.

*Example:*

Uniaxial contraction along  $\mathbf{e}_1$ , in a mixture containing a neo-Hookean solid.

```
<material id="1" type="solid mixture">
  <mat_axis type="local">0,0,0</mat_axis>
  <solid type="neo-Hookean">
    <E>1.0</E>
    <v>0.3</v>
  </solid>
  <solid type="prescribed uniaxial active contraction">
    <T0 lc="2">1</T0>
    <mat_axis type="angles">
      <theta>0</theta>
      <phi>90</phi>
    </mat_axis>
  </solid>
</material>
```

#### 4.19.2.2 Prescribed Transversely Isotropic Active Contraction

The material type for prescribed isotropic active contraction in a plane transverse to a given direction (or fiber), in a solid mixture of unconstrained materials, is “*prescribed trans iso active contraction*”. This material must be combined with a stable compressible material that acts as a passive matrix, using a “*solid mixture*” container as described in Section 4.1.4.27. The following material parameters need to be defined:

|      |                                            |     |
|------|--------------------------------------------|-----|
| <T0> | $T_0$ , representing the prescribed stress | [P] |
|------|--------------------------------------------|-----|



In the reference configuration, the fiber is oriented along the unit vector  $\mathbf{e}_1$ , where  $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$  are orthonormal basis vectors representing the local element coordinate system when specified (Section 4.1.1), or else the global Cartesian coordinate system. The active stress  $\boldsymbol{\sigma}^a$  for this material is given by

$$\boldsymbol{\sigma}^a = J^{-1}T_0(\mathbf{B} - \mathbf{n} \otimes \mathbf{n}),$$

where  $\mathbf{n} = \mathbf{F} \cdot \mathbf{e}_1$  is the stretched fiber orientation in the current (deformed) configuration and  $\mathbf{B} = \mathbf{F} \cdot \mathbf{F}^T$  is the left Cauchy-Green tensor.

*Example:*

Isotropic contraction in plane transverse to  $\mathbf{e}_1$ , in a mixture containing a neo-Hookean solid.

```
<material id="1" type="solid mixture">
  <mat_axis type="local">0,0,0</mat_axis>
  <solid type="neo-Hookean">
    <E>1.0</E>
    <v>0.3</v>
  </solid>
  <solid type="prescribed trans iso active contraction">
    <T0 lc="2">1</T0>
    <mat_axis type="angles">
      <theta>0</theta>
      <phi>90</phi>
    </mat_axis>
  </solid>
</material>
```

#### 4.19.2.3 Prescribed Isotropic Active Contraction

The material type for prescribed isotropic active contraction, in a solid mixture of unconstrained materials, is “*prescribed isotropic active contraction*”. This material must be combined with a stable compressible material that acts as a passive matrix, using a “*solid mixture*” container as described in Section 4.1.4.27. The following material parameters need to be defined:

|      |                                            |     |
|------|--------------------------------------------|-----|
| <T0> | $T_0$ , representing the prescribed stress | [P] |
|------|--------------------------------------------|-----|

The active stress  $\boldsymbol{\sigma}^a$  for this material is given by

$$\boldsymbol{\sigma}^a = J^{-1}T_0\mathbf{B}.$$

*Example:*

Isotropic contraction in a mixture containing a neo-Hookean solid.

```
<material id="1" type="solid mixture">
  <mat_axis type="local">0,0,0</mat_axis>
  <solid type="neo-Hookean">
    <E>1.0</E>
    <v>0</v>
  </solid>
  <solid type="prescribed isotropic active contraction">
    <T0 lc="2">1</T0>
  </solid>
</material>
```

#### 4.19.2.4 Prescribed Fiber Stress

An active fiber stress, based on a Hill formulation, can be added via the material “*active fiber stress*”. This material must be combined with a stable compressible material that acts as a passive matrix, using a “*solid mixture*” container as described in Section 4.1.4.27. The stress is given by,

$$\boldsymbol{\sigma}^a = J^{-1} T(\lambda) \mathbf{A}.$$

Here,  $\mathbf{A} = \mathbf{a} \otimes \mathbf{a}$ , with  $\mathbf{a}$  the unit vector describing the fiber direction in the spatial frame,  $\lambda$  is the fiber stretch,  $J$  is the jacobian of the deformation, and

$$T(\lambda) = s_{max} a(t) s_{TL}(\lambda) s_{TV}(\dot{\lambda})$$

The parameters are defined below.

|      |                                                                          |     |
|------|--------------------------------------------------------------------------|-----|
| smax | scale factor for defining maximum stress                                 | [P] |
| a    | activation level                                                         |     |
| stl  | Function defining the stress-stretch relationship for the material       |     |
| stv  | Function defining the stress-stretch rate relationship for the material. |     |

The parameters  $s_{TL}$  and  $s_{TV}$  are functions that need to be defined in place. There are currently two ways of defining these functions, either via a mathematical expression or a list of sample points. An example is given below. If these parameters are omitted, they are replaced by the constant 1 in the equation for the stress above.

*Example:*

An example defining the stl parameter via a mathematical expression.

```
<material id="1" type="solid mixture">
  <mat_axis type="local">0,0,0</mat_axis>
  <solid type="neo-Hookean">
    <E>1.0</E>
    <v>0</v>
  </solid>
  <solid type="active fiber stress">
    <smax>150</smax>
    <a lc="1">0.1</a>
    <stl type="math">
      <math>(1-1)^2</math>
    </stl>
  </solid>
</material>
```

An example defining the stl parameter via a point list.

```
<material id="1" type="solid mixture">
  <mat_axis type="local">0,0,0</mat_axis>
  <solid type="neo-Hookean">
    <E>1.0</E>
    <v>0</v>
```

```

</solid>
<solid type="active fiber stress">
  <smax>150</smax>
  <a lc="1">0.1</a>
  <stl type="point">
    <interpolate>linear</interpolate>
    <points>
      <pt>0,0</pt>
      <pt>1,0</pt>
      <pt>2,1</pt>
    </points>
  </stl>
</solid>
</material>

```

## 4.20 Viscous Fluids

Fluid mechanics analyses may be used to examine fluid flow over fixed domains. Fluid-structure interactions (FSI) may be used to examine fluid flow over deforming domains. It is also possible to perform solute (mass) transport analyses in fluid mixtures using the fluid-solutes module. FEBio's fluid, fluid-FSI and fluid-solutes modules treat the fluid as isothermal (constant and uniform temperature) and compressible; incompressible flow is simulated by prescribing a physically realistic bulk modulus  $K$  to model the fluid's elastic compressibility (for example,  $K = 2.2$  GPa for water at room temperature). The Cauchy stress  $\sigma^f$  in a fluid is given by

$$\sigma^f = -p\mathbf{I} + \tau, \quad (4.20.1)$$

where  $p$  is the fluid pressure arising from the elastic and osmotic responses and  $\tau$  is the viscous stress resulting from the fluid viscosity and its rate of deformation. FEBio's fluid solver uses the fluid dilatation  $e^f$ , instead of the *effective fluid pressure*  $\tilde{p}$ , as a nodal degree of freedom. The effective fluid pressure  $\tilde{p}$  may differ from the actual fluid pressure  $p$  if solutes are present in the fluid mixture, as per eq.(4.16.7). In fluid analyses the effective fluid pressure  $\tilde{p}$  is also the elastic contribution to the fluid pressure, resulting from changes in  $e^f$ . FEBio provides several alternative constitutive models for the elastic response  $\tilde{p}(e^f)$ , including

$$\begin{aligned}
 \tilde{p}(e^f) &= -Ke^f && \text{linear} \\
 \tilde{p}(e^f) &= K \left( \frac{1}{1+e^f} - 1 \right) && \text{nonlinear} \\
 \tilde{p}(e^f) &= -K \frac{\ln(1+e^f)}{1+e^f} && \text{log nonlinear}
 \end{aligned} \quad (4.20.2)$$

where  $K$  is the fluid's bulk modulus (a physical property that may be looked up or measured). The reason for selecting  $e^f$  instead of  $\tilde{p}$  is that the dilatation (volumetric strain) is a kinematic measure, just like the fluid velocity  $\mathbf{v}^f$  (the other nodal degree of freedom) and its associated rate of deformation tensor, whereas  $\tilde{p}$  is a function of state (it needs to be formulated as a function of the deformation, just like the viscous fluid stress  $\tau$ ). Here,  $\tilde{p}$  is called the *elastic pressure* because it represents the elastic response (equivalent to the stress-strain response in hyperelasticity for

solid mechanics) of the fluid under isothermal conditions. The *linear* model in eq.(4.20.2) is useful for nearly-incompressible flow, typically used for liquids. However, if one expects high volumetric strains in the flow (e.g., at a stagnation point in a high-velocity fluid flow problem), it may be better to employ one of the two *nonlinear* formulations in eq.(4.20.2). In particular, for an ideal gas, the gauge pressure  $\tilde{p}$  is given by

$$\tilde{p} = P_r \left( \frac{\theta}{J^f \theta_r} - 1 \right) \quad J^f = 1 + e^f \quad (4.20.3)$$

where  $\theta_r$  is the (arbitrarily selected) reference temperature and  $P_r$  is the corresponding referential absolute pressure (at  $\theta = \theta_r$  and  $J^f = 1$ ). When an ideal gas undergoes an isothermal process ( $\theta = \theta_r$ ), eq.(4.20.3) reduces to the *nonlinear* model in eq.(4.20.2), with bulk modulus  $K = P_r$ . Thus, the *nonlinear* model is consistent with ideal gas law under isothermal processes. Isothermal processes produce zero heat flux,  $\mathbf{q} = \mathbf{0}$ , however heat is generated by the fluid viscosity at the rate  $\tau : \mathbf{D}^f$  where  $\mathbf{D}^f$  is the rate of deformation tensor for the fluid (see below). Therefore, to maintain isothermal conditions, we must assume that this heat is implicitly radiated into or out of the fluid domain. For isentropic processes it can be shown that the temperature variation in an ideal gas obeys  $\theta/\theta_r = (J^f)^{1-\gamma}$ , where  $\gamma$  is the ratio of isobaric to isochoric specific heat capacities,  $\gamma = c_{p0}/c_{v0}$ . In that case, the pressure in an ideal gas evolves according to  $\tilde{p} = P_r \left( (1 + e^f)^{-\gamma} - 1 \right)$ , which represents a fourth alternative to the three models presented in eq.(4.20.2), with  $K = P_r$ . Contrary to the simplifying assumptions found in thermodynamic textbooks, an isentropic process is not automatically adiabatic (although the reverse is true). Therefore, when modeling ideal gases undergoing isentropic processes, it is also assumed that the equation of energy balance is implicitly satisfied by suitably radiating heat into or out of the fluid domain.

The fluid pressure  $\tilde{p}$  is evaluated from the dilatation  $e^f$  (the relative change in volume, or volumetric strain) using the user-specified model from eq.(4.20.2) (the default is *linear*), where  $e^f = J^f - 1$  and  $J^f$  is the fluid volume ratio. The fluid density  $\rho^f$  varies with  $J^f$  according to

$$\rho^f = \frac{\rho_r^f}{J^f}, \quad (4.20.4)$$

where  $\rho_r^f$  is the fluid density in the reference configuration (when the pressure  $p$  is zero). The dependence of  $\tau$  on the fluid rate of deformation tensor  $\mathbf{D}^f = \frac{1}{2} (\text{grad } \mathbf{v}^f + \text{grad}^T \mathbf{v}^f)$ , where  $\mathbf{v}^f$  = fluid velocity, is given by a constitutive model which may be chosen from the list provided in Section 4.20.2. Though Newtonian and non-Newtonian fluids may be analyzed in this framework, all fluids are purely viscous (no viscoelasticity is included in this formulation). Setting the viscosity to zero is allowable, for the purpose of analyzing inviscid flow. The user is referred to the [FEBio Theory Manual](#) for a general description of this isothermal compressible viscous flow framework.

For fluid analyses, which are performed over a fixed mesh, FEBio solves for the components of the fluid velocity  $\mathbf{v}^f$  and fluid dilatation  $e^f$  at each node. In contrast, fluid-FSI analyses are performed over deforming meshes and FEBio treats fluid-FSI domains as a mixture of a viscous fluid  $f$  and a massless, frictionless porous solid  $s$ . Thus, the fluid encounters zero resistance as it flows through the deforming mesh (porous solid). The solid constituent of a fluid-FSI domain regularizes the mesh deformation; it is a hyperelastic solid whose constitutive relation is selected by the user; the recommended choice is the unconstrained neo-Hookean solid defined in Section 4.1.4.20, with  $\nu = 0$  and  $E$  set to a very small (but non-zero) value. For fluid-FSI analyses, FEBio solves for the solid displacement  $\mathbf{u}$ , the fluid velocity  $\mathbf{w}$  relative to the mesh, and the fluid dilatation  $e^f$ . The fluid

velocity is then calculated as

$$\mathbf{v}^f = \mathbf{v}^s + \mathbf{w}, \quad (4.20.5)$$

where  $\mathbf{v}^s$  is the mesh velocity (the material time derivative of the solid displacement  $\mathbf{u}$ ). Since the mesh is fixed in a standard fluid analysis,  $\mathbf{v}^s = \mathbf{0}$  and  $\mathbf{v}^f = \mathbf{w}$  in that case. Therefore, we use  $\mathbf{w}$  to refer to the fluid velocity degrees of freedom for fluid and fluid-FSI analyses. For both types of analyses, the no-slip condition for viscous fluids flowing along a boundary is satisfied by setting  $\mathbf{w} = \mathbf{0}$  on that boundary.

On any fluid boundary, the outward surface normal may be denoted by  $\mathbf{n}$  and the traction vector on the fluid is given by  $\mathbf{t}^f = \boldsymbol{\sigma}^f \cdot \mathbf{n} = -p\mathbf{n} + \mathbf{t}^\tau$ , where  $\mathbf{t}^\tau = \boldsymbol{\tau} \cdot \mathbf{n}$  is the viscous traction. Essential (Dirichlet) boundary conditions may be prescribed on  $\mathbf{w}$  and  $e^f$ , while natural (Neumann) boundary conditions may be prescribed on  $\mathbf{t}^\tau$  and  $w_n$ . The appearance of velocity in both essential and natural boundary conditions may seem surprising at first. To better understand the nature of these boundary conditions, it is convenient to separate the velocity into its normal and tangential components,  $\mathbf{w} = w_n\mathbf{n} + \mathbf{w}_t$ , where  $\mathbf{w}_t = (\mathbf{I} - \mathbf{n} \otimes \mathbf{n}) \cdot \mathbf{w}$ . In particular, for inviscid flow, the viscous stress  $\boldsymbol{\tau}$  and its corresponding traction  $\mathbf{t}^\tau$  are both zero, leaving  $w_n$  as the sole natural boundary condition; similarly,  $e^f$  becomes the only essential boundary condition in such flows, since  $\mathbf{w}_t$  is unknown *a priori* on a frictionless boundary and must be obtained from the solution of the analysis.

In general, prescribing  $e^f$  is equivalent to prescribing the elastic fluid pressure, since  $p$  is only a function of  $e^f$  according to eq.(4.20.2). On a boundary where no conditions are prescribed explicitly, we conclude that  $w_n = 0$  and  $\mathbf{t}^\tau = \mathbf{0}$ , which represents a frictionless wall. Conversely, it is possible to prescribe  $w_n$  and  $\mathbf{t}^\tau$  on a boundary to produce a desired inflow or outflow while simultaneously stabilizing the flow conditions by prescribing a suitable viscous traction. Prescribing essential boundary conditions  $\mathbf{w}_t$  and  $e^f$  determines the tangential velocity on a boundary as well as the elastic fluid pressure  $p$ , leaving the option to also prescribe the normal component of the viscous traction,  $t_n^\tau = \mathbf{t}^\tau \cdot \mathbf{n}$ , to completely determine the normal traction  $t_n^f = \mathbf{t}^f \cdot \mathbf{n}$  (or else  $t_n^\tau$  naturally equals zero). Mixed boundary conditions represent common physical features: Prescribing  $w_n$  and  $\mathbf{w}_t$  completely determines the velocity  $\mathbf{w}$  on a boundary; prescribing  $\mathbf{t}^\tau$  and  $e^f$  completely determines the traction  $\mathbf{t}^f = \boldsymbol{\sigma}^f \cdot \mathbf{n}$  on a boundary. Note that  $w_n$  and  $e^f$  are mutually exclusive boundary conditions, and the same holds for  $\mathbf{w}_t$  and the tangential component of the viscous traction,  $\mathbf{t}_t^\tau = (\mathbf{I} - \mathbf{n} \otimes \mathbf{n}) \cdot \mathbf{t}^\tau$ .

For viscous fluids, the no-slip boundary condition at an impermeable wall is enforced by setting  $\mathbf{w}_t = \mathbf{0}$ , whereas the wall impermeability condition implies  $w_n = 0$ . Therefore, these conditions may be combined by prescribing  $w_x, w_y$  and  $w_z$  components of  $\mathbf{w}$  to zero. FEBio offers a range of options for conveniently prescribing natural and mixed conditions on boundary surfaces in fluid analyses (Section 3.13.2).

### 4.20.1 General Specification of Fluid Materials

The material type for a fluid material is “*fluid*”. The following parameters must be defined:

|           |                                           |                     |
|-----------|-------------------------------------------|---------------------|
| <density> | Fluid referential density $\rho_r$        | [M/L <sup>3</sup> ] |
| <k>       | Bulk modulus $K$                          | [P]                 |
| <viscous> | Specification of viscous model for $\tau$ |                     |

The <viscous> tag encloses a description of the viscous stress constitutive relation and associated material properties, as may be selected from the list provided in Section 4.20.2. The parameters <density> and <k> must be greater than 0. These parameters are always required.

*Example:*

```
<material id="1" name="Water" type="fluid">
  <density>1</density>
  <k>2.2e+09</k>
  <viscous type="Newtonian fluid">
    <mu>0.001</mu>
    <kappa>0</kappa>
  </viscous>
</material>
```

**4.20.2 Viscous Fluid Materials**

Viscous fluid materials provide a constitutive relation for the dependence of the viscous stress  $\tau$  on the rate of deformation tensor  $\mathbf{D}^f$  of a fluid.

#### 4.20.2.1 Newtonian Fluid

The material type for a Newtonian fluid is “*Newtonian fluid*”. The following material parameters must be defined:

|         |                         |                         |
|---------|-------------------------|-------------------------|
| <mu>    | shear viscosity $\mu$   | [ <b>P</b> · <b>t</b> ] |
| <kappa> | bulk viscosity $\kappa$ | [ <b>P</b> · <b>t</b> ] |

The viscous shear stress for this material model is

$$\boldsymbol{\tau} = \left( \kappa - \frac{2}{3}\mu \right) (\text{tr } \mathbf{D}) \mathbf{I} + 2\mu \mathbf{D}$$

Stokes’ condition is a constitutive assumption that sets  $\text{tr } \boldsymbol{\tau} = 0$ , implying that  $\kappa = 0$ . This assumption is only valid for some fluids (e.g., monoatomic gas [68]). Users may use or ignore this assumption by selecting an appropriate value for  $\kappa$ .

*Example:*

```
<viscous type="Newtonian fluid">
<mu>0.001</mu>
<kappa>0</kappa>
</viscous>
```



### 4.20.2.2 Bingham Fluid

The material type for a Bingham fluid [69] is “*Bingham*”. The following material parameters must be defined:

|        |                                                      |       |
|--------|------------------------------------------------------|-------|
| <mu>   | shear viscosity at infinite shear rate, $\mu_\infty$ | [P.t] |
| <tauy> | yield stress, $\tau_y$                               | [P]   |
| <n>    | exponent                                             |       |

The viscous shear stress for this material model is

$$\boldsymbol{\tau} = 2\mu\mathbf{D}$$

where

$$\mu = \mu_\infty + \frac{\tau_y}{\dot{\gamma}} (1 - e^{-n\dot{\gamma}})$$

Here,  $\dot{\gamma} = \sqrt{2\mathbf{D} : \mathbf{D}}$  is the engineering shear rate. In the limit as  $\dot{\gamma} \rightarrow 0$  the viscosity is given by  $\mu = \mu_\infty + n\tau_y$ . If we define the scalar shear stress  $\tau = \sqrt{\boldsymbol{\tau} : \boldsymbol{\tau}/2}$ , it follows that  $\tau = \mu_\infty\dot{\gamma} + \tau_y (1 - e^{-n\dot{\gamma}})$ . Effectively, this constitutive model represents a bilinear response for  $\tau$  versus  $\dot{\gamma}$ , with a slope of  $\mu_\infty + n\tau_y$  when  $\tau < \tau_y$  and  $\mu = \mu_\infty$  when  $\tau > \tau_y$ . The exponential function rounds the corner at the intersection of these two lines. For an ideal Bingham fluid one would need to let  $n \rightarrow \infty$ . In practice, values of  $n$  between 5 and 50 work well, with the lower end of this range producing faster convergence of the finite element solution.

*Example:*

```
<viscous type="Bingham">
  <mu>1</mu>
  <tauy>40</tauy>
  <n>40</n>
</viscous>
```

### 4.20.2.3 Carreau Model

The material type for a Carreau model [27] is “*Carreau*”. The following material parameters must be defined:

|          |                                                      |       |
|----------|------------------------------------------------------|-------|
| <mu0>    | shear viscosity at zero shear rate, $\mu_0$          | [P·t] |
| <mui>    | shear viscosity at infinite shear rate, $\mu_\infty$ | [P·t] |
| <lambda> | time constant                                        | [t]   |
| <n>      | power-law exponent                                   |       |

The viscous shear stress for this material model is

$$\tau = 2\mu\mathbf{D}$$

where

$$\mu = \mu_\infty + (\mu_0 - \mu_\infty) \left( 1 + (\lambda\dot{\gamma})^2 \right)^{(n-1)/2}$$

Here,  $\dot{\gamma} = \sqrt{2\mathbf{D} : \mathbf{D}}$  is the engineering shear rate.

*Example:*

```
<viscous type="Carreau">
<mu0>0.056</mu0>
<mui>0.0035</mui>
<lambda>3.3</lambda>
<n>0.36</n>
</viscous>
```

#### 4.20.2.4 Carreau-Yasuda Model

The material type for a Carreau-Yasuda model [27] is “*Carreau-Yasuda*”. The following material parameters must be defined:

|          |                                                      |       |
|----------|------------------------------------------------------|-------|
| <mu0>    | shear viscosity at zero shear rate, $\mu_0$          | [P·t] |
| <mui>    | shear viscosity at infinite shear rate, $\mu_\infty$ | [P·t] |
| <lambda> | time constant                                        | [t]   |
| <n>      | power-law exponent                                   |       |
| <a>      | power-law exponent divider                           |       |

The viscous shear stress for this material model is

$$\tau = 2\mu\mathbf{D}$$

where

$$\mu = \mu_\infty + (\mu_0 - \mu_\infty) (1 + (\lambda\dot{\gamma})^a)^{(n-1)/a}$$

Here,  $\dot{\gamma} = \sqrt{2\mathbf{D} : \mathbf{D}}$  is the engineering shear rate.

*Example:*

```
<viscous type="Carreau-Yasuda">
<mu0>0.056</mu0>
<mui>0.0035</mui>
<lambda>1.9</lambda>
<n>0.22</n>
<a>1.25</a>
</viscous>
```

#### 4.20.2.5 Powell-Eyring Model

The material type for a Powell-Eyring model [27] is “*Powell-Eyring*”. The following material parameters must be defined:

|          |                                                      |       |
|----------|------------------------------------------------------|-------|
| <mu0>    | shear viscosity at zero shear rate, $\mu_0$          | [P·t] |
| <mui>    | shear viscosity at infinite shear rate, $\mu_\infty$ | [P·t] |
| <lambda> | time constant                                        | [t]   |

The viscous shear stress for this material model is

$$\boldsymbol{\tau} = 2\mu\mathbf{D}$$

where

$$\mu = \mu_\infty + (\mu_0 - \mu_\infty) \frac{\sinh^{-1} \lambda \dot{\gamma}}{\lambda \dot{\gamma}}$$

Here,  $\dot{\gamma} = \sqrt{2\mathbf{D} : \mathbf{D}}$  is the engineering shear rate.

*Example:*

```
<viscous type="Powell-Eyring">
<mu0>0.056</mu0>
<mui>0.0035</mui>
<lambda>5.4</lambda>
</viscous>
```

**4.20.2.6 Cross Model**

The material type for a Cross model [27] is “Cross”. The following material parameters must be defined:

|          |                                                      |       |
|----------|------------------------------------------------------|-------|
| <mu0>    | shear viscosity at zero shear rate, $\mu_0$          | [P·t] |
| <mui>    | shear viscosity at infinite shear rate, $\mu_\infty$ | [P·t] |
| <lambda> | time constant                                        | [t]   |
| <m>      | exponent                                             |       |

The viscous shear stress for this material model is

$$\tau = 2\mu\mathbf{D}$$

where

$$\mu = \mu_\infty + (\mu_0 - \mu_\infty) \frac{1}{1 + (\lambda\dot{\gamma})^m}$$

Here,  $\dot{\gamma} = \sqrt{2\mathbf{D} : \mathbf{D}}$  is the engineering shear rate.

*Example:*

```
<viscous type="Cross">
<mu0>0.056</mu0>
<mui>0.0035</mui>
<lambda>1.0</lambda>
<m>1.0</m>
</viscous>
```

#### 4.20.2.7 Quemada Model

The material type for a Quemada model [72] is “*Quemada*”. The following material parameters must be defined:

|        |                                                                        |                    |
|--------|------------------------------------------------------------------------|--------------------|
| <mu0>  | shear viscosity at zero shear rate suspension volume fraction, $\mu_0$ | [P·t]              |
| <H>    | suspension volume fraction, $H$                                        | [ ]                |
| <k0>   | intrinsic relative viscosity at zero shear rate, $k_0$                 | [ ]                |
| <kinf> | intrinsic relative viscosity at infinite shear rate, $k_\infty$        | [ ]                |
| <gc>   | critical shear rate, $\dot{\gamma}_c$                                  | [t <sup>-1</sup> ] |

The viscous shear stress for this material model is

$$\tau = 2\mu\mathbf{D}$$

where

$$\mu = \frac{\mu_0}{\left(1 - k\frac{H}{2}\right)^2}$$

and

$$k = \frac{k_0 + k_\infty \sqrt{\dot{\gamma}_r}}{1 + \sqrt{\dot{\gamma}_r}}, \quad \dot{\gamma}_r = \frac{\dot{\gamma}}{\dot{\gamma}_c}$$

Here,  $\dot{\gamma} = \sqrt{2\mathbf{D} : \mathbf{D}}$  :  $\mathbf{D}$  is the engineering shear rate.

*Example:*

```
<viscous type="Quemada">
  <mu0>0.03</mu0>
  <H>0.5</H>
  <k0>3.691</k0>
  <kinf>1.778</kinf>
  <gc>2.30</gc>
</viscous>
```

### 4.20.3 General Specification of Fluid-FSI Materials

The material type for a fluid-FSI material is “*fluid-FSI*”. This type of material is used for fluid flow through a deforming mesh. The material is treated as a special case of fluid-solid mixture, with the solid material used to regularize the mesh deformation. The following parameters must be defined:

|         |                                                           |  |
|---------|-----------------------------------------------------------|--|
| <fluid> | Specification of fluid model                              |  |
| <solid> | Specification of elastic solid model for mesh deformation |  |

The <fluid> tag encloses a description of the fluid, as given in Section 4.20.1. The <solid> tag encloses a description of the solid, which should be modeled as an unconstrained elastic solid, as given in Section 4.1.4. The recommended choice is the unconstrained neo-Hookean solid defined in Section 4.1.4.20, with  $\nu = 0$  and  $E$  set to a very small (but non-zero) value.

*Example:*

```
<material id="1" name="Fluid-FSI domain" type="fluid-FSI">
  <fluid type="fluid">
    <density>1000</density>
    <k>2.2e9</k>
    <viscous type="Newtonian fluid">
      <mu>0.001</mu>
      <kappa>0</kappa>
    </viscous>
  </fluid>
  <solid type="neo-Hookean">
    <density>0</density>
    <E>1e-9</E>
    <v>0</v>
  </solid>
</material>
```

The value of <density> in the <solid> material is internally set to zero regardless of the user-defined value, since the deforming mesh should not be subjected to inertial forces.

### 4.20.4 General Specification of Biphasic-FSI Materials

The material type for a biphasic-FSI material is “*biphasic-FSI*”. This is a hybrid biphasic material used for modeling dynamic viscous fluid flow and frictional filtration through a porous-deformable solid. In contrast to the standard biphasic material described in Section 4.14, this material accommodates fluid viscosity and fluid dynamics; therefore it can be interfaced seamlessly with a fluid-FSI material to allow viscous fluid transport across that interface. The following parameters must be defined:

|                |                                                                                              |     |
|----------------|----------------------------------------------------------------------------------------------|-----|
| <phi0>         | solid volume fraction $\varphi_r^s$ in the reference configuration ( $0 < \varphi_r^s < 1$ ) | [ ] |
| <fluid>        | Specification of the fluid material                                                          |     |
| <solid>        | Specification of the porous solid material                                                   |     |
| <permeability> | Specification of the hydraulic permeability                                                  |     |

The parameter `<phi0>` must be greater than 0 (no solid) and less than 1 (no porosity). The `<fluid>` tag encloses a description of the fluid, as given in Section 4.20.1. The `<solid>` tag encloses a description of the solid matrix constitutive relation and associated material properties, such as those selected from the list provided in Sections 4.1-4.13. The `<permeability>` tag encloses a description of the permeability constitutive relation and associated material properties, as may be selected from the list presented in Section 4.14.2. To simulate the special case of zero friction between the fluid and solid (such as in the fluid-FSI material), it is permissible to set the permeability to zero.

*Example:*

```
<material id="2" name="Material2" type="biphasic-FSI">
  <phi0>0.25</phi0>
  <fluid type="fluid">
    <density>1e-9</density>
    <k>1e+09</k>
    <viscous type="Newtonian fluid">
      <mu>1</mu>
      <kappa>0</kappa>
    </viscous>
  </fluid>
  <solid type="neo-Hookean">
    <density>2e-9</density>
    <E>200</E>
    <v>0</v>
  </solid>
  <permeability type="perm-const-iso">
    <perm>0.005625</perm>
  </permeability>
</material>
```

The value of `<density>` in the `<fluid>` and `<solid>` materials represents the true mass density of each constituent.

#### 4.20.5 General Specification of Fluid-Solutes Materials

The material type for a fluid-solutes material is *“fluid-solutes”*. This type of material is used for fluid flow and solute mass transport through a fixed mesh. The following parameters must be defined:

|                                          |                                                                |  |
|------------------------------------------|----------------------------------------------------------------|--|
| <code>&lt;fluid&gt;</code>               | Specification of fluid model                                   |  |
| <code>&lt;osmotic_coefficient&gt;</code> | Specification of osmotic coefficient for fluid-solutes mixture |  |
| <code>&lt;solute&gt;</code>              | Specification of (any number of) solute(s)                     |  |

The `<fluid>` tag encloses a description of the fluid, as given in Section 4.20.1. The `<solid>` tag encloses a description of the solid, which should be modeled as an unconstrained elastic solid, as given in Section 4.1.4. The recommended choice is the unconstrained neo-Hookean solid defined in Section 4.1.4.20, with  $\nu = 0$  and  $E$  set to a very small (but non-zero) value.

*Example:*



```

<material id="1" name="Fluid-FSI domain" type="fluid-FSI">
  <fluid type="fluid">
    <density>1000</density>
    <k>2.2e9</k>
    <viscous type="Newtonian fluid">
      <mu>0.001</mu>
      <kappa>0</kappa>
    </viscous>
  </fluid>
  <solid type="neo-Hookean">
    <density>0</density>
    <E>1e-9</E>
    <v>0</v>
  </solid>
</material>

```

The value of `<density>` in the `<solid>` material is internally set to zero regardless of the user-defined value, since the deforming mesh should not be subjected to inertial forces.

## 4.21 Prestrain material

In FEBio a prestrain can be defined for most materials that allows the user to prestrain or prestress the model before loading is applied. This is useful for modeling biological tissues that exhibit residual strain, or whose initial state cannot assumed to be stress-free (e.g. arteries from in-vivo image data.)

If you use this feature in your published research, please cite the corresponding paper that details the theoretical framework and the implementation in FEBio [59].

### 4.21.1 Introduction

In the modeling of biological tissues there are various reasons why the model needs to start from a prestressed configuration. For example, the biological tissue may exhibit residual stress or in situ stress. Or the geometry is taken from in vivo data and is subjected to loads in the reference configuration. In these cases, the reference configuration cannot be considered stress-free and this so-called prestress must somehow be accounted for. In large strain analysis, stresses are usually not additive and in addition, it must somehow be ensured that these prestresses are in equilibrium in the reference configuration. In general, this makes prestressing the reference configuration challenging in large strain analyses.

Progress can be made by assuming that the material is hyperelastic and that a prestrain can be found that defines the prestress via the hyperelastic constitutive formulation. Usually this prestrain is characterized by a second-order tensor, termed the prestrain gradient and here denoted by  $F_p$ . The interpretation of this tensor is that its inverse, applied to a small neighborhood of a point in the reference configuration, would render this neighborhood stress-free. The total elastic response of the material is then defined via the elastic deformation gradient.

$$F_e = F \cdot F_p \quad (4.21.1)$$

Here,  $F$  is the deformation gradient that is the result of subsequent loading of the reference configuration. Finally, the Cauchy stress and spatial elasticity tangent are given by

$$\sigma = \mathcal{F}(F_e), c = \mathcal{G}(F_e) \quad (4.21.2)$$

Here,  $\mathcal{F}$  and  $\mathcal{G}$  are the same response functions from the “natural” material, i.e. the response of the material that starts from a stress-free configuration.

It must be noted that although  $F_p$  is termed the prestrain gradient, in general it will not be the derivative of a deformation map. Consequently, a global stress-free reference configuration may not exist and the mapping of a neighborhood in the reference configuration to a stress-free state can at best be defined only locally.

In addition, it must also be recognized that the prestrain gradient is not unique. Due to the requirements of objectivity in the presence of material symmetries, the prestrain gradients  $F_p$  and  $F'_p = F_p Q$ , where  $Q$  is any orthogonal tensor, must result in the same material response. This implies that the prestrain can only be dependent on the right stretch tensor  $V_p$ . However, in general even the right-stretch tensor cannot be defined uniquely. For instance, an isotropic material would only depend on the eigenvalues of this tensor and thus any tensor of the form  $RDR^T$ , where  $R$  is any orthogonal tensor and  $D$  is a diagonal tensor with the three eigenvalues of  $V_p$  on its diagonal, would render the same material response. In the presence of material symmetries, the situation is similar although  $R$  will only be part of the symmetry group.

Despite the fact that  $F_p$  is not unique, it is far from arbitrary. The most important restriction is that the resulting prestress field  $\sigma_p$  must be in equilibrium with the prestressed reference configuration. In other words it must hold that

$$\text{div} \sigma_p = 0 \quad (4.21.3)$$

in the interior of the reference domain (in the absence of body forces) and that on the free boundaries of the domain

$$\sigma_p \cdot n = 0 \quad (4.21.4)$$

In a finite element simulation, when these conditions are not satisfied, the mesh will distort and a new reference state is obtained as well as an altered prestrain field. We will denote the deformation gradient between the original (incompatible) reference configuration and prestressed reference configuration by  $F_c$ . When the mesh distorts the effectively applied prestrain field is also altered. In general, this is not desired and a correction to the prestrain field is necessary that either eliminates the distortion or finds a new reference geometry in which the desired prestrain field is supported.

The prestrain framework as currently implemented assumes that the total effective prestrain that is compatible with the possibly altered prestressed reference configuration is given by the following multiplicative decomposition.

$$F_p = F_c \cdot \hat{F}_p \quad (4.21.5)$$

Both the prestrain gradient and the distortion are in general unknown. The framework implements an iterative algorithm that updates  $F_c$  and  $\hat{F}_p$  until an effective prestrain gradient is found that is compatible with the possibly distorted reference configuration. Without loss of generality the framework assumes that the altered prestrain gradient  $\hat{F}_p$  is itself given by the multiplicative decomposition

$$\hat{\mathbf{F}}_p = \mathbf{G} \cdot \mathbf{F}_0 \quad (4.21.6)$$

where  $\mathbf{F}_0$  is an initial guess for the prestrain gradient and  $\mathbf{G}$  is a correction factor. The algorithm starts by initializing  $\mathbf{F}_0$  to a user-defined value and  $\mathbf{G}$  to the identity. This initial value can be specified in a variety of ways and the code that generates this initial guess is called the prestrain generator. The framework provides several prestrain generators and users can easily add new ones. Then, a forward analysis is executed keeping  $\hat{\mathbf{F}}_p$  fixed. Unless the initial guess was compatible with the reference configuration, the mesh will distort. This distortion will define the new value for  $\mathbf{G}$  which can be calculated directly from the nodal displacements in the usual manner. Next, the framework will update using a user-defined update rule. How  $\mathbf{G}$  is updated depends on the particular application and the framework expects the user to provide the desired update rule. The current implementation provides two particular update rules: one to eliminate the distortion and one to enforce a particular form of the prestrain gradient. See section 3.15.6 on how to use these update rules. In addition, users can easily implement new update rules. After this update, a new forward analysis is executed. This process is repeated until  $\mathbf{G}$  converges.

### 4.21.2 The Prestrain Material

Prestrain can be applied by defining the *prestrain* material. This material acts as a wrapper to the underlying elastic constitutive model. There are two flavors of this material, one for coupled materials and one for uncoupled. In either case, the material defines two properties. The *elastic* property defines the elastic constitutive model. Any of the materials in FEBio could be used, however, the prestrain material has only been validated for hyperelastic materials. The second property, called *prestrain*, defines the prestrain generator. This property defines how the initial prestrain gradient is calculated. Currently, it can be defined as a full second order matrix or as a fiber stretch. The prestrain data can be defined at the material level or at the element level.

The *prestrain* property defines the way that the initial prestrain gradient tensor is calculated. An algorithm that calculates the initial prestrain based on user input is here referred to as a prestrain generator. The type attribute is used to create a specific instance of a prestrain generator. Currently, the following generators are available.

#### 4.21.2.1 prestrain gradient

The *prestrain gradient* generator defines the full prestrain gradient matrix either for the entire material or for each element separately. It has the following parameters.

| parameter | description                                           |
|-----------|-------------------------------------------------------|
| ramp      | scale factor that ramps up the prestrain gradient (1) |
| F0        | 3x3 target prestrain gradient matrix                  |

*Comments:*

1. The prestrain gradient that is applied is calculated by the following equation.

$$\mathbf{F}_g = \mathbf{I}(1 - r) + \mathbf{F}_0 r$$

Here,  $\mathbf{F}_g$  is the applied prestrain gradient,  $\mathbf{I}$  is the 3x3 identity tensor,  $\mathbf{F}_0$  is the target prestrain gradient, defined by  $F_0$  parameter, and  $r$  is the ramp factor.

*Example:*

In this example, the prestrain gradient is ramped up to a desired gradient value via a load controller.

```
<material id="1" type="prestrain elastic">
  <elastic type="neo-Hookean">
    <E>1.0</E>
    <v>0.3</v>
  </elastic>
  <prestrain type="prestrain gradient">
    <ramp lc="1">1.0</ramp>
    <F0>2,0,0,0,2,0,0,0,2</F0>
  </prestrain>
</material>
```

#### 4.21.2.2 in-situ stretch

The *in-situ stretch* generator option calculates a prestrain gradient based on a fiber stretch and the fiber vector defined by the elastic material of the prestrain material. This implies that this elastic material must define a *fiber* property.

| parameter | description                                  | initial values |
|-----------|----------------------------------------------|----------------|
| stretch   | initial fiber stretch                        | 1.0            |
| isochoric | use the isochoric prestrain generator option | 1              |

This option generates one of the following prestrain gradients, depending on the isochoric option.

$$\hat{\mathbf{F}}_{p,iso} = \mathbf{Q} \begin{bmatrix} \lambda & & \\ & \lambda^{-1/2} & \\ & & \lambda^{-1/2} \end{bmatrix} \mathbf{Q}^T, \quad \hat{\mathbf{F}}_{p,uni} = \mathbf{Q} \begin{bmatrix} \lambda & & \\ & 1 & \\ & & 1 \end{bmatrix} \mathbf{Q}^T \quad (4.21.7)$$

If the *isochoric* option is set to 1, then  $\hat{\mathbf{F}}_{p,iso}$  is used. Otherwise,  $\hat{\mathbf{F}}_{p,uni}$  is used.

The rotation tensor is defined implicitly via the fiber vector  $\mathbf{a}$  and the prestrain gradient tensor is effectively determined via,

$$\hat{\mathbf{F}}_{p,iso} = \lambda \mathbf{A} + \mu (\mathbf{1} - \mathbf{A})$$

where  $\mathbf{A} = \mathbf{a} \otimes \mathbf{a}$ ,  $\lambda$  the prescribed fiber stretch, and  $\mu = \lambda^{-1/2}$  or  $\mu = 1$  depending on the *isochoric* option.

*Example:*

```

<material id="1" type="prestrain elastic">
  <elastic type="coupled trans-iso Mooney-Rivlin">
    <k>0.1</k>
    <density>1e-09</density>
    <c1>0.01</c1>
    <c2>0</c2>
    <c3>0.0139</c3>
    <c4>116.22</c4>
    <c5>535.09</c5>
    <lam_max>1.046</lam_max>
    <fiber type="vector">-0.0804,-0.508,-0.858</fiber>
  </elastic>
  <prestrain type="in-situ stretch">
    <stretch lc="1">1.05</stretch>
    <isochoric>1</isochoric>
  </prestrain>
</material>

```

## 4.22 Continuous-Discontinuous Damage

In this damage formulation, the damage variable  $D$  is embedded inside the strain-energy function. The implementation follows largely the formulation in Balzani [[17]] and is further detailed in [[3]].

The scalar damage variable  $D$  is embedded in the strain energy function of the fibers, its relationship is dependent on the Continuous Damage solid constituent chosen. The scalar damage variable  $D$  has continuous and discontinuous damage components defined by damage internal damage variables  $\beta$  and  $\gamma$ .

The following material damage parameters must be defined:

|             |                                                   |
|-------------|---------------------------------------------------|
| <Dmax>      | Parameter $D_\infty$ ( $0 \leq D_\infty \leq 1$ ) |
| <beta_s>    | Parameter $\beta_s$ ( $\beta_s > 0$ )             |
| <gamma_max> | Parameter $\gamma_\infty$ ( $\gamma_\infty > 0$ ) |
| <t0>        | Parameter $t_0$ ( $t_0 \geq 0$ )                  |

Note the parameter  $t_0$  is the time in the current loading situation where the damage model is initiated.

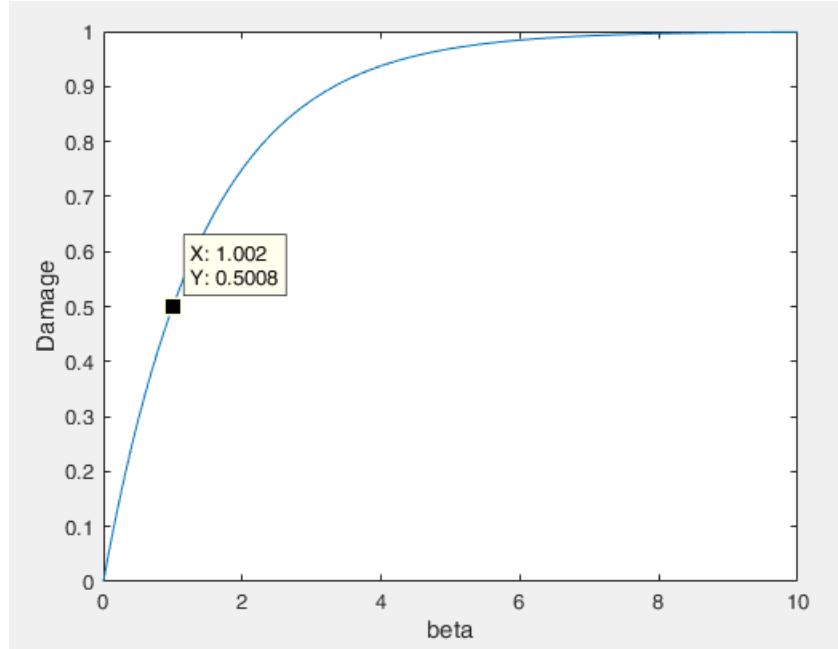
The scalar damage variable  $D$  is a function of  $\beta$ , the continuous damage variable, and is given by

$$D(\beta) = D_s \left[ 1 - \exp \left( \frac{\ln(1-r_s)}{\beta_s} \beta \right) \right],$$

where  $\beta$  is defined by

$$\beta = \langle \tilde{\beta} - \tilde{\beta}_{ini} \rangle \text{ with } \tilde{\beta} = \int_0^t \dot{\Psi}^0 ds,$$

and  $\beta_s$  is the value of the internal material variable  $\beta$  that has reached a certain fraction  $r_s$  ( $0 \leq r_s \leq 1$ ) of the maximal damage value  $D_s$  for a fixed maximum load level. At  $\beta = \beta_s$ , the overall damage  $D = r_s * D_s$ . For example, in Figure 1,  $D_s = 1$ ,  $r_s = 0.5$ , and  $\beta_s = 1$ , meaning the overall damage in this cycle has reached 50% of the maximal damage value  $D_s$ .



Overall Damage variable  $D$  as a function of Continuous Damage variable  $\beta$

To make sure damage evolution begins to accumulate only when damage is accrued,  $\beta_{ini}$  is the internal variable  $\tilde{\beta}$  at an initial damage state and  $\Psi^0$  is the undamaged effective strain energy density of the fibrils.

$D_s$  is a function increasing the maximally reachable damage value for increased maximum load levels. It is assumed that the maximum damage is itself a function of the internal variable  $\gamma$ , describing the discontinuous damage variable and is given by

$$D_s(\gamma) = D_\infty \left[ 1 - \exp \left( \frac{\ln(1-r_\infty)}{\gamma_\infty} \gamma \right) \right],$$

where  $\gamma$  is defined by

$$\gamma = \max_{s \in [0, t]} \langle \Psi^0 - \Psi_{ini}^0 \rangle,$$

and  $\gamma_\infty$  is the value of the internal material variable  $\gamma$  that has reached a certain fraction  $r_\infty$  ( $0 \leq r_\infty \leq 1$ ) of the maximal damage value  $D_\infty$  for a fixed maximum load level. At  $\gamma = \gamma_\infty$ , the overall damage  $D_s = r_\infty * D_\infty$  (the same relationship can be adapted from Figure 1 with  $\gamma \sim \beta, r_\infty \sim r_s, D_\infty \sim D_s, D_s \sim D$ ).  $\Psi_{ini}^0$  denotes the strain energy density at an initial damage state obtained at the limit of the physiological domain (e.g. in relation to arterial expansion). It is the initial damage strain energy threshold value for the fibers. The time associated with the loading history is denoted by  $s \in \mathbb{R}$  and the time associated with the current loading situation is represented by  $t \in \mathbb{R}^+$ .

The strain energy is assumed to be of the following form,

$$\Psi(C, D) = m(P(C, D)),$$

where the modified internal energy function  $P(C, D) = (1 - D)\bar{P} - c$  with the scalar damage variable  $D$  (defined above).  $m(P(C, D))$  is a convex and monotonically increasing function, whose first derivative is zero in the origin. Possible functions of  $\bar{P}$  are given by

$$\begin{aligned}\bar{P}_1 &:= J_4^{(a)}, \bar{P}_3 := K_1^{(a)}, \bar{P}_2 := K_2^{(a)}, \bar{P}_4 := K_3^{(a)}, \\ (\bar{P}_5 &:= I_1, \bar{P}_6 := I_2, \bar{P}_7 := I_3),\end{aligned}$$

where  $a$  denotes the direction of the fibers. In this formulation we take  $\bar{P} = \Psi^0$ . Note The fifth invariant is not polyconvex, therefore alternative polyconvex invariant functions are defined as

$$\begin{aligned}K_1^{(a)} &= \text{tr} [\text{Cof} C M_{(a)}] = J_5^{(a)} - I_1 J_4^{(a)} + I_2, \\ K_2^{(a)} &= \text{tr} [C(I - M_{(a)})] = I_1 - J_4^{(a)}, \\ K_3^{(a)} &= \text{tr} [\text{Cof} C(I - M_{(a)})] = I_1 J_4^{(a)} + J_5^{(a)}.\end{aligned}$$

The subtraction of  $c$  ensures the stress-free reference configuration for the undamaged case is satisfied,  $c$  is the value of the function  $\bar{P}$  in the natural state and is given by

$$c = \bar{P}_i(C = I) \text{ for } i = 1, \dots, 7).$$

#### 4.22.1 Damage Fiber Power

The material type for Damage Fiber Power is “*damage fiber power*”. The following material parameters must be defined:

|         |                                                 |
|---------|-------------------------------------------------|
| <a1>    | Parameter $\alpha_1$ ( $\alpha_1 > 0$ )         |
| <a2>    | Parameter $\alpha_2$ ( $\alpha_2 > 1$ )         |
| <kappa> | Parameter $\kappa$ ( $0 \leq \kappa \leq 2/3$ ) |

By setting,

$$\Psi^0 = \kappa I_1 + \left(1 - \frac{3}{2}\kappa\right) K_3$$

and  $m(P) = \alpha_1(P)^{\alpha_2}$ , the strain-energy form above can be made suitable for modeling damage,

$$\Psi(C, D) = \alpha_1 \left( (1 - D) \left[ \kappa I_1 + \left(1 - \frac{3}{2}\kappa\right) K_3 \right] - c \right)^{\alpha_2}.$$

The Cauchy stress takes on the following form,

$$\sigma = \frac{2}{J} (1 - D) \frac{dm}{dP} \left[ \kappa b + \left(1 - \frac{3}{2}\kappa\right) [bI_4 + I_1 I_4 m - I_4 a \odot ba] \right].$$

*Example:*

```
<solid type="damage fiber power">
  <a1>1400</a1>
  <a2>2.2</a2>
  <kappa>1e-08</kappa>
  <t0>0.98</t0>
  <Dmax>0.96</Dmax>
```

```

<beta_s>0.06</beta_s>
<gamma_max>17.98</gamma_max>
<fiber type="angles">
  <theta>-39.87</theta>
  <phi>90</phi>
</fiber>
</solid>

```

#### 4.22.2 Damage Fiber Exponential

The material type for Damage Fiber Exponential is “*damage fiber exponential*”. The following material parameters must be defined:

|         |                                                 |
|---------|-------------------------------------------------|
| <k1>    | Parameter $\kappa_1$ ( $\kappa_1 > 0$ )         |
| <k2>    | Parameter $\kappa_2$ ( $\kappa_2 > 0$ )         |
| <kappa> | Parameter $\kappa$ ( $0 \leq \kappa \leq 1/3$ ) |

The effective strain-energy function is given by,

$$\Psi^0 = \kappa I_1 + (1 - 3\kappa) I_4$$

and

$$m(P) = \frac{k_1}{2k_2} \left\{ \exp \left( k_2 \langle P \rangle^2 \right) - 1 \right\},$$

where  $a$  denotes the direction of the fibers.

So that,

$$\Psi = \frac{k_1}{2k_2} \left\{ \exp \left( k_2 \langle (1 - D_a) (\kappa I_1 - (1 - 3\kappa) I_4) - 1 \rangle^2 \right) - 1 \right\}.$$

The Cauchy stress then takes on the following form,

$$\sigma = \frac{2}{J} (1 - D) \frac{dm}{dP} [\kappa b + (1 - 3\kappa) I_4 m].$$

*Example:*

```

<solid type="damage fiber exponential">
  <k1>1288.97</k1>
  <k2>400</k2>
  <kappa>0.2</kappa>
  <t0>0.9</t0>
  <Dmax>0.99</Dmax>
  <beta_s>0.001</beta_s>
  <gamma_max>6.67</gamma_max>
  <fiber type="angles">
    <theta>-54.94</theta>
    <phi>90</phi>
  </fiber>
</solid>

```



### 4.22.3 Damage Fiber exp-linear

This continuous damage formulation uses a slightly modified damage formulation. The total damage is here defined as,

$$D = D_1 + D_2$$

Here,  $D_1$  defines the same damage term as defined above. The  $D_2$  is defined as follows,

$$D_2 = D_3 [a (\exp (b\beta) - 1) + c (\exp (d\beta) - 1)]$$

The material parameters  $a$  and  $c$  control the magnitude of the continuous damage of the two phenomena: 1) slow, constant increased in damage, 2) sharp jump in damage near failure. The other material parameters,  $b$  and  $d$ , control the rate of continuous damage accumulation for the two phenomena mentioned above, and is a function of the collagen discontinuous damage. Furthermore,

$$D_3 (\gamma) = \frac{D_{3\infty}}{1 + \exp(-(\gamma - \gamma_0)/r_\gamma)}$$

Here,  $D_{3\infty}$  is the maximum achievable discontinuous damage possible for the model,  $\gamma_0$  determines the shape of the curve, and  $r_\gamma$  controls the rate of discontinuous damage accumulation.

The additional damage parameters to define the  $D_2$  term, are as follows.

|        |                         |
|--------|-------------------------|
| D2_a   | parameter $a$           |
| D2_b   | parameter $b$           |
| D2_c   | parameter $c$           |
| D2_d   | parameter $d$           |
| D3_inf | parameter $D_{3\infty}$ |
| D3_g0  | parameter $\gamma_0$    |
| D3_rg  | parameter $r_\gamma$    |

The effective strain-energy function is given by,

$$\Psi^0 = \sqrt{I_4}$$

and

$$m(P) = \begin{cases} c_3 (\exp(-c_4) (Ei(c_4(P+1)) - Ei(c_4)) - \log(P+1)), & P \leq \lambda^* + 1 \\ c_5 P + c_6 \log(P+1) + d & P > \lambda^* + 1 \end{cases}$$

Here,  $d$  is determined by requiring continuity at  $\lambda^*$ .

The (elastic) material parameters for this material are as follows.

|        |                                                                                |
|--------|--------------------------------------------------------------------------------|
| c3     | parameter $c_3$                                                                |
| c4     | parameter $c_4$                                                                |
| c5     | parameter $c_5$                                                                |
| c6     | parameter $c_6$                                                                |
| lambda | parameter $\lambda^*$ , the transition between exponential and linear regions. |

## 4.23 First-Order Homogenization

This document describes how to use the first-order homogenization feature in FEBio. Homogenization refers to the process of evaluating the physical response of a macro model by explicitly solving a separate finite element model that represents the micro structure of the material. This micro-model is referred to as the Representative Volume Element (RVE). In the first-order approach it is assumed that the macro and micro scales are separated, which implies that the size of the RVE is negligible compared to the size of the macro model.

### 4.23.1 Macro model definition

For the most part, the definition of the macro-model is a standard finite element model with properly chosen boundary conditions. The use of first-order homogenization is controlled entirely by the material definition.

To use the first-order homogenization formulation use the “micro-material” material. This material has the following parameters.

| parameters | description                                                               | default    |
|------------|---------------------------------------------------------------------------|------------|
| RVE        | The file name of the RVE model (1)                                        | N/A        |
| rve_type   | The type of RVE model (2)                                                 | 0          |
| bc_set     | The name of the nodeset (in the RVE) that defines the outside surface (3) | (optional) |
| probe      | Defines a point in the macro model where the RVE will be tracked. (4)     | (none)     |

*Comments:*

1. The RVE parameter defines the file name of the RVE model. The RVE is a standard FE model, although somewhat slimmed down. See the rve\_type parameter definition and the section RVE model definition below.

2. This parameter determines the type of RVE (and its assumed boundary conditions) that will be used and must be one of the following values.

| rve_type | description                                                                                            |
|----------|--------------------------------------------------------------------------------------------------------|
| 0        | Prescribed displacement RVE: The position of all RVE boundary nodes is fully prescribed                |
| 1        | Periodic RVE: RVE with periodic boundary constraints. Constraints are enforced with linear constraints |

3. The bc\_set defines the name of the nodeset (defined in the RVE model), that defines the outside surface. In general, this parameter does not need to be defined. It is only needed for problems using rve\_type = 0, and for which the RVE geometry is not a closed cube.

4. By default, FEBio will only output the deformation of the macro model to the plot file. It will not output any results from the RVE models. (Given there could be thousands of RVE models, this would be too cumbersome to do.) However, users can request FEBio to output the deformation of the RVE model at specific points in the model using the probe option. The probe tag requires the following child tags.

| probe parameter | description                        |
|-----------------|------------------------------------|
| element_id      | the ID of the element to probe     |
| gausspt         | the index of the integration point |
| file            | output file name for RVE model     |

### 4.23.2 RVE model definition

The RVE model is a standard FE model, with a few caveats.

- The time stepping parameters will be ignored, since the macro model controls the time stepping of the RVE to make sure that both models advance in sync.
- The boundary conditions of the RVE depend on the `rve_type` parameter:
  - `rve_type = 0`: The RVE model should not have any boundary conditions applied. The RVE may define a nodeset that defines the outside surface of the RVE. This is generally not needed, but must be defined when the RVE geometry is not a closed cube.
  - `rve_type = 1`: The RVE should have periodic constraints applied for all opposing surfaces.
- The RVE model does not need an Output section, but may define the plotfile section. This will be used to define the contents of the probe's plotfiles.
- The size of the RVE is assumed to be much smaller than the macro model. It is allowed to use a different length unit for the coordinates of the RVE nodes. However, the unit of stress must be the same for both the macro model and the RVE model.



# Chapter 5

## Restart Input file

### 5.1 Introduction

Users can request FEBio to output a binary dump file. This dump file can be used to restart the analysis from the time point saved in the dump file. The user can restart the analysis either directly from this binary dump file or via a restart input file, which is described in this chapter. See [section 2.8.4](#) for more information on how to use the restart feature.

This chapter describes the format of the restart input file. This file is used to redefine some parameters when restarting a previously terminated run and can also be used to extend the analysis. The structure is very similar to the FEBio input file and also uses XML formatting. Since the file uses XML, the first line must be the XML header:

```
<?xml version="1.0" encoding="ISO-8859-1"?>
```

The next line contains the root element of the restart file, and has to be:

```
<febio_restart version="2.0">
```

The restart file is composed of the following sections. These sections are sub-elements of the *febio\_restart* root element.

**Archive**     define the binary dump file used for restarting.

**Control**     redefine some control parameters.

**LoadData**   redefine or add some loadcurves.

**Step**        add an analysis step.

All sections are optional except for the Archive section. In the following paragraphs we describe the different sections in more detail.

### 5.2 The Archive Section

The Archive section must be the first sub-element of the *febio\_restart* root element. This section defines the name of the binary dump file:

```
<Archive>archive.dmp</Archive>
```

This file will be created by FEBio when requested by the user and contains the solution of the analysis up to the last converged time step.

### 5.3 The Control Section

The following control parameters can be redefined:

| Parameter  | Description                                       |
|------------|---------------------------------------------------|
| dtol       | convergence tolerance for displacements           |
| etol       | convergence tolerance for energy                  |
| rtol       | convergence tolerance for residual                |
| lstol      | line search tolerance                             |
| max_refs   | maximum number of stiffness reformations          |
| max_ups    | maximum number of BFGS updates                    |
| restart    | restart file generation flag                      |
| plot_level | defines the frequency of the plot file generation |

### 5.4 The LoadData Section

In the LoadData section some or all of the load curves can be redefined, or new load curves can be added. The syntax is identical to the LoadData section of the FEBio input file:

```
<LoadData>
  <loadcurve id="1">
    <loadpoint>0, 0</loadpoint>
    ...
    <loadpoint>1, 0.54</loadpoint>
  </loadcurve>
</LoadData>
```

In this case, the loadcurve *id* is the loadcurve number of the loadcurve that the user wishes to redefine.

New loadcurves can be added as well. This is useful when in addition adding new Step sections, where new boundary conditions are defined. When adding new loadcurves, make sure to continue the numbering of the parent input file. That is, if the last loadcurve of the parent input file has an id of *n*, then the first new load curve defined here must have id of *n*+1.

### 5.5 The Step Section

The Step section can be used to add additional analysis steps to the model. These steps are executed after the initial model has completed. This section cannot be used to modify the steps defined in the parent file. See chapter 6 for more information on the Step section or multi-step analyses in general.

Using the Step section, additional boundary conditions, loads, contact definitions, etc. can be added, just like in a regular model input file. The syntax is the same as the Step section in the regular model input file. The only difference is that the type attribute is required to define the type of analysis. The type has the same role as the Module section in the regular FEBio input file.

## 5.6 Example

### 5.6.1 Example 1

The following example defines a restart input file. No parameters are redefined. Only the mandatory *Archive* element is defined. In this case the analysis will simply continue where it left off:

```
<?xml version="1.0" encoding="ISO-8859-1"?>
<febio_restart version="1.0">
  <Archive>out.dmp</Archive>
</febio_restart>
```

### 5.6.2 Example 2

In the following example an additional step is defined, which defines a new boundary condition and a new loadcurve.

```
<?xml version="1.0" encoding="utf-8"?>
<febio_restart version="2.0">
  <Archive>out.dmp</Archive>
  <Step type="solid">
    <Control>
      <time_steps>10</time_steps>
      <step_size>0.1</step_size>
    </Control>
    <Boundary>
      <prescribe bc="x" node_set="set1">
        <scale lc="2">1.0</scale>
      </prescribe>
    </Boundary>
  </Step>
  <LoadData>
    <loadcurve id="2" type="linear">
      <point>1, 0</point>
      <point>2, 1</point>
    </loadcurve>
  </LoadData>
</febio_restart>
```

Notice the “type” attribute in the Step section. This serves a similar purpose as the Module section in the FEBio input file and defines the solver that FEBio will use to solve this step. This attribute is required when defining steps in the restart file. (The Module section is not supported in the restart input file.)

The prescribed boundary condition references a node set named “set1”. This node set must be defined in the parent input file. No new node sets (or surfaces, etc.) can be defined in the restart input file.





## Chapter 6

# Multi-Step Analysis

A multi-step analysis is defined using multiple steps, where in each step the user can redefine control parameters, boundary conditions, loads, contact interfaces, and other model components. This is useful, for instance, for defining time-dependent boundary conditions or for switching between different analysis types during the simulation.

Multi-step models require a slightly different file organization compared to single-step models. More specifically, the *Control* section is not specified at the top of the input file. Instead, a **Steps** section is added at the bottom of the file, which contains a list of **step** sub-sections for each step. Each step then defines its own *Control* section, boundary conditions, loads, etc.

### 6.1 The Step Section

The multi-step analysis feature organizes the input file in a slightly different way: the **Control** section is no longer defined at the top of the file, and instead, a new **Steps** section is added to the bottom of the file, which will contain a list of analysis steps. Each analysis step requires its own **step** sub-section. In this step section, the user can redefine the control section, boundary section, loads, contact, and other model components. The following format is suggested when defining a multi-step analysis:

```
<febio_spec version="4.0">
  <Module type="solid"/>
  <Material>
    <!-- materials go here -->
  </Material>
  <Mesh>
    <!-- mesh definition goes here -->
  </Mesh>
  <MeshDomains>
    <!-- mesh domains goes here -->
  </MeshDomains>
  <Boundary>
    <!-- global boundary conditions -->
  </Boundary>
  <LoadData>
    <!-- load curve data goes here -->
```

```

</LoadData>
<Steps>
  <step>
    <Control>
      <!-- local control settings -->
    </Control>
    <Boundary>
      <!-- local boundary conditions -->
    </Boundary>
  </step>
</Steps>
</febio_spec>

```

The first part of the file looks similar to a normal input file, except that the control section is not specified. Also, the *Boundary*, *Loads*, *Constraints*, *Contact*, sections should only contain global boundary conditions, i.e. boundary conditions that will remain active in all analysis steps.

At the end of the file the user defines as many **step** sections as needed. In each **step** section, the user can now define the control parameters and model components.

The sub-sections that can be defined in a step are:

**Control** Defines the steps control parameter, such time stepping, solver settings, etc.

**Initial** Applies initial conditions that take effect at the start of the step.

**Boundary** Defines boundary conditions that are only applied in this step.

**Loads** Defines loads that are only applied in this step.

**Constraints** Defines constraints that are only applied in this step.

**Rigid** Defines rigid constraints, loads, and connectors in this step.

**MeshAdaptor** Defines mesh adaptors that will be applied in this step.

### 6.1.1 Boundary Conditions

In a multi-step analysis boundary conditions can be applied that are only active during the step. This applies to the *Boundary*, *Loads*, *Contact* and *Constraints* section of the FEBio input file. These sections can thus be placed inside the *Step* section to define a boundary condition that is only active during the step. If one of these sections is defined before the first *Step* section, the corresponding boundary conditions remain enforced during all the steps.

### 6.1.2 Relative Boundary Conditions

Some boundary conditions can be defined as relative boundary conditions. This means that the corresponding conditions will be applied to the final configuration of the previous step and not to the original reference configuration. For example, if a prescribed displacement is defined as relative the displacement will be taken with respect to the positions of the final configuration of the previous step. This makes it possible, for example, to switch between load and displacement controlled boundary conditions in multi-step analyses.

## 6.2 An Example

The following example illustrates the use of the multi-step feature of FEBio. This problem defines two steps. In the first step, a single element is stretched using a prescribed displacement boundary condition. In the second step, the boundary condition is removed and the analysis type is switched from quasi-static to a dynamic analysis. Note the presence of the global fixed boundary constraints, which will remain enforced during both steps:

```
<?xml version="1.0" encoding="ISO-8859-1"?>
<febio_spec version="4.0">
  <Module type="solid"/>
  <Material>
    <material id="1" name="Material 1" type="neo-Hookean">
      <density>1.0</density>
      <E>1</E>
      <v>0.45</v>
    </material>
  </Material>
  <Mesh>
    <Nodes>
      <node id="1">-2.0,-0.5, 0.0</node>
      <node id="2">-2.0,-0.5, 1.0</node>
      <node id="3">-2.0, 0.5, 0.0</node>
      <node id="4">-2.0, 0.5, 1.0</node>
      <node id="5"> 2.0,-0.5, 0.0</node>
      <node id="6"> 2.0,-0.5, 1.0</node>
      <node id="7"> 2.0, 0.5, 0.0</node>
      <node id="8"> 2.0, 0.5, 1.0</node>
    </Nodes>
    <Elements type="hex8" name="Part1">
      <elem id="1">1,5,7,3,2,6,8,4</elem>
    </Elements>
    <NodeSet name="set1">
      <node id="1"/>
      <node id="2"/>
      <node id="3"/>
      <node id="4"/>
    </NodeSet>
    <NodeSet name="set2">
      <node id="1"/>
      <node id="2"/>
      <node id="5"/>
      <node id="6"/>
    </NodeSet>
    <NodeSet name="set3">
      <node id="1"/>
      <node id="3"/>
      <node id="5"/>
```

```

<node id="7"/>
</NodeSet>
<NodeSet name="set4">
<node id="5"/>
<node id="6"/>
<node id="7"/>
<node id="8"/>
</NodeSet>
</Mesh>
<MeshDomains>
<SolidDomain name="Part1" mat="Material 1"/>
</MeshDomains>
<Boundary>
<bc type="zero displacement" node_set="set1">
<dofs>x,y,z</dofs>
</bc>
</Boundary>
<LoadData>
<load_controller id="1" type="loadcurve">
<points>
<point>0,0</point>
<point>1,0.1</point>
</points>
</load_controller>
</LoadData>
<Step>
<step id="1">
<Control>
<time_steps>10</time_steps>
<step_size>0.1</step_size>
</Control>
<Boundary>
<bc type="prescribed displacement" node_set="set4">
<dof>x</dof>
<scale lc="1">1.0</scale>
</bc>
</Boundary>
</step>
<step id="2">
<Control>
<time_steps>50</time_steps>
<step_size>0.5</step_size>
<analysis>DYNAMIC</analysis>
</Control>
</step>
</Step>
</febio_spec>

```

## Chapter 7

# Parameter Optimization

This chapter describes FEBio's parameter optimization module. This module tries to estimate model parameters by solving an inverse finite element problem, where the “solution” of the problem is known, and the problem's model parameters are sought. For instance, the solution can be an experimentally determined reaction force curve, and the unknown parameters are the material parameters that will recreate the reaction force curve by solving a forward FE problem. Another common application is determining the model parameters that achieve a desired load, displacement, or other goal.

In any case, the optimization module tries to minimize an objective function of the form,

$$\varphi(\mathbf{a}) = \sum_{i=1}^n [y_i - f(x_i; \mathbf{a})]^2.$$

Here, the  $(x_i, y_i)$  are user-defined data pairs and  $f(x; \mathbf{a})$  is the function that extracts the corresponding data from the model. The optimization module tries to find the model parameters  $\mathbf{a}$  that minimize the function  $\varphi$ . It does this by repeatedly evaluating the function  $f$ , which will usually call FEBio to solve a forward FE problem.

The optimization module requires a separate input file that describes all the information needed for solving an optimization problem. This input file is described next.

### 7.1 Optimization Input File

Like all FEBio input files, the parameter optimization input file is an XML-formatted text file. The following sections are defined:

**Task** optional section defining the task that is executed to solve the FE problem.

**Options** optional section defining the optimization control parameters.

**Objective** defines the objective function that will be minimized.

**Parameters** defines the model parameters that are to be determined.

**Constraints** defines linear constraints for the model parameters (requires a solution method that supports linear constraints, e.g. constrained levmar).

In the following paragraphs each section will be explained in detail.

### 7.1.1 Task Section

The optional Task section can be used to define the task that is executed by FEBio when solving the FE model. By default, FEBio will solve the standard forward problem, but if needed, a custom task can be defined.

### 7.1.2 Options Section

This section defines the control parameters for the optimization. The options that can be defined will depend on the chosen optimization method, which is set with an attribute of the Options section. The following table shows the currently supported optimization algorithms.

| type               | Description                                     |
|--------------------|-------------------------------------------------|
| levmar             | Levenberg-Marquardt                             |
| constrained levmar | Levenberg-Marquardt with linear constraints (1) |
| powell             | Powell's method                                 |
| scan               | Parameter domain scanning (2)                   |

*Comments:*

1. The “constrained levmar” uses the third-party levmar library <http://users.ics.forth.gr/~lourakis/levmar/>. For additional information please see the library's home page.
2. The “scan” method will iterate over the parameter range in increments defined by the fourth parameter value defined in the Parameter section. It returns the parameters that resulted in the lowest objective value of visited points, but it may not be the actual minimum.

Note that the Options section is optional. When omitted, the levmar method is used with default values for the control parameters.

The following parameters can be defined.

| Parameter    | Description                                                                                      | Default          |
|--------------|--------------------------------------------------------------------------------------------------|------------------|
| obj_tol      | Convergence tolerance for objective function. (1)                                                | 0.001            |
| f_diff_scale | Forward difference scale factor (2)                                                              | 0.001            |
| log_level    | Sets the amount of output that is generated on screen and in the logfile by the FEBio solver (3) | LOG_NEVER        |
| tau          | Step size scale factor (4)                                                                       | 1.0e-3           |
| print_level  | Sets the amount of output generated by the optimization module (5)                               | PRINT_ITERATIONS |

For example,

```
<Options type="constrained levmar">
  <obj_tol>0.001</obj_tol>
  <f_diff_scale>0.001</f_diff_scale>
  <print_level>PRINT_ITERATIONS</print_level>
</Options>
```

*Comments:*

1. The objective function that is to be minimized is a function of the form:

$$\varphi(\mathbf{a}) = \sum_{i=1}^n [y_i - f(x_i; \mathbf{a})]^2.$$

Here,  $f(x; \mathbf{a})$  is the function that describes the model,  $\mathbf{a}$  is a vector with the (unknown) model parameters and the  $(x_i, y_i)$  are the user-defined data that approximates the ideal model.

2. The optimization method currently implemented requires the calculation of the gradient of the model function  $f$  with respect to the model parameters  $\mathbf{a}$ . Since this gradient is not known, it will be approximated using forward differences. For example, the  $k$ -th component of the gradient is approximated as follows.

$$\frac{\partial f}{\partial a_k} \approx \frac{1}{\delta a_k} [f(a_1, \dots, a_k + \delta a_k, \dots, a_m) - f(a_1, \dots, a_k, \dots, a_m)]$$

The value for  $\delta a_k$  is determined from the following formula.

$$\delta a_k = \varepsilon (1 + a_k)$$

where,  $\varepsilon$  is the forward difference scale factor which can be set by the user with the *fdiff\_scale* option.

3. The *log\_level* allows the user to control exactly how much output from the time iterations of each optimization iteration is written to the screen and logfile. The following values are allowed:

| Value               | Description                                 |
|---------------------|---------------------------------------------|
| LOG_DEFAULT         | Use default settings (may be omitted)       |
| LOG_NEVER           | Don't generate any output                   |
| LOG_FILE_ONLY       | Save to the logfile only                    |
| LOG_SCREEN_ONLY     | Only print to the screen                    |
| LOG_FILE_AND_SCREEN | Save to the logfile and print to the screen |

4. The step size scale factor *tau* can be set as an input only for the Constrained Levenberg-Marquardt method. It scales the initial guess for the damping parameter.
5. The *print\_level* sets how much output is generated during the optimizations. In particular, it controls if the objective function values are printed or not. The following values are allowed:

| Value            | Description                                                 |
|------------------|-------------------------------------------------------------|
| PRINT_ITERATIONS | Print minimal iterations info                               |
| PRINT_VERBOSE    | Print a lot of information, incl. objective function values |

### 7.1.3 Parameters Section

This section defines the material parameters that are to be determined. Each parameter is defined using the *param* element:

```
<Parameters>
  <param name="[name]">[guess], [min], [max], [scale]</param>
</Parameters>
```

The *name* attribute gives the name of the parameter that is to be determined. The following section describes the format to reference a model parameter. Each parameter takes four values: [guess] is the initial guess for this parameter, [min] and [max] are the minimum and maximum values respectively for this parameter, and [scale] is a representative scale (magnitude) for this parameter (although the precise interpretation of this parameter depends on the particular solver). This value is used to normalize the optimization parameter and improve convergence. If not specified by the user, it defaults to the initial guess.

For example, to optimize the material parameters of a neo-Hookean solid, named *mat1*, use the following syntax.

```
<Parameters>
  <param name="fem.material('mat1').E">1.0, 0.0, 5.0, 1.0</param>
  <param name="fem.material('mat1').v">0.1, 0.0, 0.5, 1.0</param>
</Parameters>
```

This example defines two material parameters. The first component of the name (here *mat1*) is the name of the material as defined in the model input file. The second component (here *E* and *v*) are the names of the material parameters. See Appendix B for more information on how to reference model parameters.

### 7.1.4 Objective Section

This section defines the functional model that is used to evaluate the objective function (i.e. the function *f* above). The particular functional model is defined via the *type* attribute, and the following table lists the available options.

| Value        | Description                                                                     |
|--------------|---------------------------------------------------------------------------------|
| data-fit     | Fits data, usually specified as a function of time, to a model output parameter |
| target       | Optimizes until one or more output model parameters achieve a target value.     |
| element-data | The target data is defined as a tabulated list of element values.               |
| node-data    | The target data is defined as a tabulated list of nodal values.                 |

These options will be described separately in the following sections.

#### 7.1.4.1 The data-fit model

The *data-fit* model tries to fit predicted FE data to a user-defined data curve. The data curve usually represents some experimentally obtained result, e.g. reaction force as a function of time. The data-fit model needs a data source, i.e. a function for extracting the model data, and the data curve.



```

<Objective type="data-fit">
  <fnc type="[enter type]">
    ...
  </fnc>
  <data>
    <pt>0, 0</pt>
    ...
    <pt>1, 1</pt>
  </data>
</Objective>

```

The data source is defined by the type attribute of the fnc tag. The following table shows the supported values.

| type                 | Description                                                                                   |
|----------------------|-----------------------------------------------------------------------------------------------|
| parameter            | The function will track the evolution of a model parameter.                                   |
| filter_positive_only | The function tracks the evolution of a model parameter, but filters only the positive values. |
| filter_sum           | The function is defined by summing nodal values.                                              |

#### parameter

In the case of *parameter*, a child element defines the parameter that will act as the data source for the objective function. (See Appendix B on more information on referencing model parameters.)

```

<fnc type="parameter">
  <param name="fem.rigidbody('rigid').Fz"/>
</fnc>

```

This defines the data function as a function of time. You can also change the ordinate of the function to create other functional dependencies. For example, the following creates a load-displacement curve for a rigid body.

```

<fnc type="parameter">
  <ordinate name="fem.rigidbody('rigid').position.x"/>
  <param name="fem.rigidbody('rigid').Fz"/>
</fnc>

```

As another example, the following creates a stress-strain curve for a single solid element, using the left Hencky (logarithmic) strain component along the z-axis (denoted by *hz*), and the Cauchy normal stress component along z-axis (denoted by *sz*) in element number 1,

```

<fnc type="parameter">
  <ordinate name="fem.element_data('hz',1)"/>
  <param name="fem.element_data('sz',1)"/>
</fnc>

```

#### filter\_positive\_only

This option requires another data source that is needed to extract model data (e.g. parameter), but only positive values are returned. If the model parameter's value is negative, zero will be returned by the function.

```

<fnc type="filter_positive_only">
  <source type="parameter">
    <param name="fem.rigidbody('rigid').Fz"/>
  </source>
</fnc>

```

### filter\_sum

This option will sum up nodal values and use that in the objective function. The summed value is always assumed to be a function of time. The following example illustrates how this option can be used to define an objective function from the sum of nodal reaction forces.

```

<fnc type="filter_sum">
  <node_data data="Rx" node_set="top surface"/>
</fnc>

```

#### 7.1.4.2 The target model

The *target* model tries to optimize model parameters such that one or more output variables achieve a user-defined target value. This model requires a list of variables and target values for each variable.

```

<Objective type="target">
  <var name="[enter name]">[target value]</var>
  ...
</Objective>

```

The name refers to a model output parameter. The example below will optimize until the X-euler angle of the rigid body 'Material2' becomes 0.5.

```

<Objective type="target">
  <var name="fem.rigidbody('Material2').euler.y">0.5</var>
</Objective>

```

#### 7.1.4.3 The element-data model

The *element-data* model uses a tabulated list of element values as the target for the optimization. In addition, it requires an output variable that will be matched against the element data.

```

<Objective type="element-data">
  <var type="[enter variable name]" />
  <data>
    <elem id="1">0.123</elem>
    ...
  </data>
</Objective>

```

The variable can be any of the element output data listed in the logfile section (See [3.6.6](#)).

#### 7.1.4.4 The node-data model

The *node-data* model uses a tabulated list of nodal values as the target for the optimization. In addition, it requires an output variable that will be matched against the nodal data.

```
<Objective type="node-data">
  <var type="[enter variable name]"/>
  <data>
    <node id="1">0.123</elem>
    ...
  </data>
</Objective>
```

The variable can be any of the node output data listed in the logfile section (See [E.2.1](#)).

#### 7.1.5 Constraints Section

The Constrained Levenberg-Marquardt method allows linear constraints on the material parameters. If  $a$  is the material parameter vector, then the linear constraint is of the form:

$$c_1 a_1 + c_2 a_2 + \dots + c_n a_n + b = 0 \quad .$$

The coefficients  $c_1, c_2, \dots, c_n, b$  are the inputs of the constraint tag. For example, if the linear constraint is  $2a_1 - a_2 + 3 = 0$ , then the Constraints section would be:

```
<Constraints>
  <constraint>2, -1, 3</constraint>
</Constraints>
```

## 7.2 Running a Parameter Optimization

As explained above, a parameter optimization problem is described using two input files. First, a standard FEBio input file that defines the mesh, materials, boundary conditions, etc. The second input file describes the parameter optimization problem, such as the objective and which model parameters are to be optimized. The format of the second file is described above. A parameter optimization can only be initiated from the command line. For example:

```
>febio -i model.feb -s optim.feb
```

The output of a parameter optimization analysis is a log file that contains the screen output of the FEBio run as well as the optimized parameter values.

## 7.3 An Example Input File

Below follows a complete example of an optimization input file.

```

<?xml version="1.0"?>
<febio_optimize version="2.0">
  <Options>
    <obj_tol>0.001</obj_tol>
    <f_diff_scale>0.001</f_diff_scale>
  </Options>
  <Parameters>
    <param name="fem.material[0].E">1, 0, 5</param>
    <param name="fem.material[0].v">-0.5, 0, 0.5</param>
  </Parameters>
  <Objective type="data-fit">
    <fnc type="parameter">
      <param name="fem.rigidbody('rigid').Fz"/>
    </fnc>
    <data>
      <point>0.0, 0</point>
      <point>0.5, 1</point>
      <point>1.0, 2</point>
    </data>
  </Objective>
</febio_optimize>

```

*Comments:*

1. Notice that the xml root element is *febio\_optimize* for the optimization input file. The version number has to be 2.0 (the 1.0 version is no longer supported).
2. The FEBio input file that contains the actual FE model data has to be defined on the command line using the -i command option. This file is a standard FEBio input file that defines all geometry, materials, boundary conditions and more.
3. The initial values of the parameters are defined in the optimization input file. The values in the model input file are ignored.
4. The *Options* section is included here, but can be omitted. If omitted default values will be used for all control parameters.

## Chapter 8

# Troubleshooting

Running a nonlinear finite element analysis can be a very challenging task. The large deformations and complex constitutive models can make it very difficult to obtain a converged solution. There are many causes for nonconvergence, ranging from element inversions, material instability, failure to enforce contact constraints and many more. Sometimes it is possible that FEBio gives you a converged solution, but the solution is meaningless or at least not what was expected.

Fortunately, many of these issues can be prevented or solved with little effort. This chapter discusses some strategies to prevent common problems and to troubleshoot a problematic run. It offers some sanity checks before you run your model which can save you a lot of frustration down the road. And when things do go bad, we hope that the strategies suggested here may prove helpful.

However, keep in mind, that the finite element method cannot solve all problems. It is a very powerful numerical method, but with limitations. Understanding these limitations and how they affect your modeling work is crucial in becoming a good analyst.

### 8.1 Before You Run Your Model

In this section we'll discuss what a well-defined finite element model is and some things you may need to check before you run your model in FEBio.

A well-defined finite element model contains a finite element mesh, a valid material and properly defined boundary and contact conditions in order to define a unique solution to the problem under study. We will look at each of these requirements in more detail in the following sections.

#### 8.1.1 The Finite Element Mesh

A finite element mesh is required to solve a problem with FEBio. The mesh defines a discretization of the problem domain in nodes and connected elements. FEBio only supports certain elements and thus the mesh must be composed of elements from this set. See [Section 3.6.2](#) for a discussion of the supported elements. In addition, FEBio assumes a specific ordering of the nodes of an element. FEBio cannot discover if the nodes are in the correct order, but if they are not, FEBio will most likely have trouble converging or throw negative jacobians. If FEBio discovers negative jacobians before the first time step, it is likely that the nodes of the elements are not defined in the proper order.

For shell elements, the initial thickness of an element is also important. When elements are too thick (the thickness is of the same order as the element size), FEBio may complain about negative

jacobians. This may be particularly a problem in areas of high curvature. The only solution around this issue might be to remesh the problematic area.

### 8.1.2 Materials

A material in FEBio defines the constitutive response of the domain to which the material is assigned. See Chapter 4 for a detailed list of all the available materials in FEBio. It is important to understand that the module defines which materials you can use. For example, the biphasic material cannot be used in the *solid* module. Although some cases of invalid material use are caught, in many situations the resulting behavior is undefined. The following table shows a list of some of FEBio's special materials and the modules in which they can be used.

| Material                 | Solid | Biphasic | Solute | Multiphasic | Heat |
|--------------------------|-------|----------|--------|-------------|------|
| <i>biphasic</i>          |       | YES      | YES    | YES         |      |
| <i>biphasic-solute</i>   |       |          | YES    | YES         |      |
| <i>triphasic</i>         |       |          | YES    | YES         |      |
| <i>multiphasic</i>       |       |          |        | YES         |      |
| <i>isotropic Fourier</i> |       |          |        |             | YES  |

In addition to using the proper materials for a given module, it is also important to understand that most material parameters have a limited range in which they define valid constitutive behavior. For example, in a neo-Hookean material the Young's modulus must be positive and the Poisson's ratio must larger than -1 and less than 0.5. FEBio has most of these limits implemented in the code and will throw an error when a parameter value is defined that falls outside the valid range.

Since material parameters can be defined as functions of time through a loadcurve, FEBio will check all parameters at the beginning of each time step. When a parameter has become invalid the run will be aborted.

Some constitutive models are only valid for a limited amount of deformation. All materials in FEBio are designed for large deformation (in the sense that they are objective under arbitrary deformation), but that does not imply unlimited deformation. When the deformation becomes too large, the material response may become nonphysical and although FEBio gives an answer, the result may be meaningless (see Section 8.9 for a discussion on interpreting the result). Some materials also become unstable at extremely large deformations. A classical example is the Mooney-Rivlin material. For a certain range of values of the material parameters, and under certain loading conditions, the stress-strain response can have a zero slope at large deformations. In that case, FEBio will most likely not be able to converge since the slope of the stress-strain response is used to progress towards the solution.

### 8.1.3 Boundary Conditions

Boundary conditions are necessary to uniquely define the solution of the problem under study. Improperly defined boundary conditions can lead to underconstrained or overconstrained problems.

If the problem is underconstrained the solution is not uniquely defined and FEBio will not be able to find a solution. The most common example of an underconstrained problem is one where the rigid body modes are not constrained. A rigid body mode is a mode of deformation that does not alter the stress in the body. Affine translation and rotation of the entire mesh are two ways to introduce a rigid body mode. In a (quasi-)static finite element analysis all rigid body modes must

be properly constrained in order to find a unique solution. This can often very easily be achieved by constraining an edge or face of the model from moving at all.

If the problem is over-constrained, FEBio will usually find an answer but it is probably not the solution that was sought. The most common cause of an over-constrained model is one that has conflicting boundary conditions. For example, a force is applied to a node that is fixed.

## 8.2 Debugging a Model

When a model is not running as expected, the first thing to try is rerun the problem using FEBio's debug mode. Debug mode can be activated by adding a `-g` on the command line. For example,

```
>febio -i file.feb -g
```

When in debug mode, FEBio will do additional checks during the solution (e.g. check for NAN's, check for zeroes on the diagonal of the stiffness matrix, etc.) and will store all non-converged states to the plot file. Inspecting these non-converged states can often be useful to identify the cause of the problem. (E.g. when the model seems to disappear at some point might indicate a rigid body mode.)

Since storing all non-converged states may make the plot-file too large for post-processing, debug mode can also be turned on during the analysis. To do this, start the problem normally (without the debug flag `-g`). Then, right before the problem starts to fail, interrupt the run (using `ctrl+c`) and wait for the FEBio prompt to appear. Once it appears, enter *debug* [ENTER], and then *cont* [ENTER] to continue the run in debug mode. Alternatively, a break point can be specified on the command line or at the FEBio prompt, which will instruct FEBio to break at a particular point in the analysis. At that point, the febio prompt will appear, and the debug mode can be toggled. For more information on setting breakpoints, consult section [2.8.3](#).

Another option is to use the `-g2` flag, instead of the `-g` flag. This flag will do all of the debug checks, but only output the non-converged state that caused febio to fail to the plot file.

## 8.3 Common Issues

In this section we take a look at some of the most common problems you may run into when solving a finite element problem with FEBio. We'll discuss possible causes of and solutions to these issues.

### 8.3.1 Inverted elements

When an element inverts, the element becomes so distorted that in certain areas of the element the Jacobian becomes zero or negative. In order to obtain a physically realistic solution, the Jacobian of the deformation must be positive at all points of the domain. Thus, when a negative Jacobian is found in an element, FEBio cannot continue. Fortunately, FEBio's automatic time-stepping algorithm can usually circumvent this problem by cutting the time step back and retrying the step using a smaller time step. In addition, FEBio will recalculate the global stiffness matrix. The combination of a reduced time step and a refactored stiffness matrix will often be sufficient to overcome the negative Jacobian. However, when it is not, you may have run into a more serious problem. These are some of the common causes that may require additional user intervention.

By default, when FEBio encounters inverted elements, it will print an error message stating that “negative jacobians” were detected. You can request FEBio to print out a detailed list of the inverted elements, by setting the *output\_negative\_jacobians* configuration flag in the FEBio configuration file. (see [9](#))

#### 8.3.1.1 Material instability

At large deformations, some materials can become unstable. This is usually caused by a softening of the material at large deformations. When the material softens too much the global stiffness matrix can become ill-conditioned and FEBio will not be able to find the correct solution. Unfortunately, there is no easy solution around this issue and you may need to consider using a different constitutive model.

#### 8.3.1.2 Time step too large

Although the auto-time stepper will automatically adjust the time step in case of trouble, it is designed to work within user defined limits and sometimes they are not set properly to solve the problem. The most common issue is that the minimum time step is set too large. Cutting back the minimum in addition to the initial time step can often help in preventing negative Jacobians.

#### 8.3.1.3 Elements too distorted

When elements get too distorted they might not be able to deform any further without inverting the element. In this case, it may be necessary to adjust the mesh in the area where the elements are inverting.

#### 8.3.1.4 Shells are too thick

A shell is an element where it is assumed that one dimension is much smaller compared to the other two directions. When the mesh is really fine, this assumption may no longer be valid and it could happen that the shell becomes too thick. When a shell gets too thick it can create negative Jacobians in the out-of-plane integration points, especially in areas of high curvature. To overcome this, you may need to remesh the affected area, adjust the shell thickness or even replace the shells with solid elements.

#### 8.3.1.5 Rigid body modes

A rigid body mode is a mode of deformation that does not alter the stress in the body. Uniform translation or rotation of the entire model are two possible rigid body modes. In the presence of a rigid body mode, the solution is not uniquely defined and FEBio will most likely throw negative Jacobians. The solution is to identify the rigid body mode and to eliminate it by applying additional constraints to the model.

### 8.3.2 Failure to converge

FEBio uses an iterative method to solve the nonlinear finite element equations. This means that for each time step the solution for that step is obtained by a succession of iterations, where each iteration brings the model (hopefully) closer to the solution for that step. An iterative method



requires a convergence criterion to decide when to stop iterating and FEBio uses no less than three criteria. (In biphasic and multiphasic problems additional convergence criteria are used for the pressure and concentration degrees of freedom.) The convergence norm for each of these criteria is defined by the user in the control section of the input file.

Sometimes it happens that FEBio cannot converge on a time step. We'll take a look at some of the most common causes in the next following sections. Also take a look at possible causes of element inversions above. Sometimes, the issues that cause an element inversion may also manifest themselves in convergence problems before the element actually inverts.

### 8.3.2.1 No loads applied

Perhaps surprisingly, when no loads are applied, FEBio will often struggle to find a solution. The reason (and solution!) is simple. Although no loads are applied, FEBio still performs many calculations and due to round-off errors, the result of these calculations may render the convergence norms not exactly zero. FEBio then tries to apply the convergence criteria to these really small norms and again due to round-off error will not be able to satisfy the convergence criteria.

FEBio can actually detect this situation. It looks at the residual norm and when this is smaller than a user-defined limit it assumes that no loads are present and moves on to the next time step. However, this limit is actually problem dependent and it can happen that for your particular problem the limit is set too small. In that case, the solution is to increase the limit. This can be done by setting the *min\_residual* parameter in the *Control* section of the input file. For example,

```
<min_residual>1e-15</min_residual>
```

If the residual norm now drops below this value, FEBio will consider the time step converged and move on to the next time step.

### 8.3.2.2 Convergence Tolerance Too Tight

In some cases, the default convergence tolerances are too tight and FEBio will not be able to converge. In that case, adjusting the affected convergence norms is a possible solution. This is done by editing the corresponding norms in the Control section of the input file. However, this should only be done when no other cause can be identified for the convergence problems.

### 8.3.2.3 Forcing convergence

It is possible to force a time step to converge. This can be done by interrupting the run (using ctrl+c) and then enter the *conv* command at the FEBio prompt. This will force the time step to converge and FEBio will continue with the next time step. This is not a recommended practice as the solution may become unstable and in fact it is unlikely that FEBio will be able to converge any of the following time steps. However, it can be useful in some scenarios. For example, if there is no load applied in the initial time step and FEBio has trouble getting past this initial step (see Section 8.3.2.1) or when you want to store the current state to the plot file without having to rerun the problem in debug mode (although the *plot* command or *debug* command might be better alternatives).

### 8.3.2.4 Problems due to Contact

When contact interfaces are defined, convergence problems can often be traced back to problems with the contact interfaces. See Section 8.5 for more details on how to properly use contact interfaces in FEBio.

## 8.4 Guidelines for Dynamic Analyses

A dynamic analysis differs from a quasi-static analysis in that the mass inertia of the model is taken into account. This adds an explicit time-dependency to the model (through the nodal velocities and accelerations) and therefore the solver needs to integrate the equations of motion in time. There are several integration schemes available in FEBio.

### 8.4.1 Implicit Dynamics Solver

The implicit solid solver implements the generalized alpha integration rule, which can be controlled via several parameters. Although the default settings will usually suffice, there might be instances where the parameters should be modified. (For example, see 8.4.2).

The main parameter that controls the time integration of the solid solver is the `rhoi` parameter. The following rules apply:

1.  $\rho_{\infty} = -1$ , All other integration parameters are set to one.
2.  $0 \leq \rho_{\infty} \leq 1$ , Newmark integration parameters are set to defaults, based on the value of  $\rho_{\infty}$ .
3. Any other value for  $\rho_{\infty}$ , default/user-specified values for Newmark integration parameters are not modified.

### 8.4.2 False Transient Analysis

Quasi-static models may sometimes exhibit instabilities that may prevent the default Newton-based solver from converging. In such situations, a dynamic analysis can be tried instead, with the anticipation that the inertia of the model provides sufficient stability to solve the model successfully. Since in these situations, only the long-term quasi-static solution is sought, it is important to damp the oscillations introduced by the dynamics of the system. Such damping can be accomplished several ways.

The generalized-alpha time integration scheme of the implicit solver uses several parameters that control the overall energy-conservation of the model. By choosing these parameters such that the system loses energy over time, artificial damping can be introduced.

The `rhoi` control parameter controls the amount of artificial damping. This parameter takes a value between 0 (more numerical damping) and 1 (least numerical damping). Thus, by setting this value to 0, the most numerical damping can be introduced.

The `rhoi` control parameter sets the actual alpha, beta, gamma parameters of the generalized-alpha rule. Instead of using `rhoi`, users can set these actual parameters directly. To introduce the most numerical damping, set the values as follows.

```
<rhoi>-2</rhoi>
<gamma>1.5</gamma>
<beta>1</beta>
```

Note that `rhoi` needs to be set to -2 so that FEBio uses the actual alpha, beta, gamma parameters.

## 8.5 Guidelines for Contact Problems

FEBio offers many contact interfaces which allow the user to model complex boundary interactions. However, this capability comes at a price and the use of contact interfaces may create several problems in the solution process. It is hoped that the guidelines presented in this section may prevent many of the most common problems with contact.

Most of the contact algorithms implemented in FEBio offer two contact enforcement methods: a penalty method and an augmented Lagrangian method. Since the strategies for these two methods are slightly different, we will look at them separately. These algorithms are discussed in much more detail in the [FEBio Theory Manual](#).

### 8.5.1 The penalty method

The penalty method enforces contact by “penalizing” (in an energy sense) any deviation from the contact constraint. In other words, when the two contacting surfaces penetrate, a force is generated proportional to the distance of penetration, which has the effect that the surfaces will now repel each other. The strength of this repelling force is controlled by the user through the *penalty factor*.

The obvious downside of this method is that some penetration is necessary to generate the contact force. The larger the penalty factor, the smaller this penetration needs to be and thus the better the contact constraint is enforced. However, choosing the penalty factor too large may cause the problem to become unstable since the contact force (and corresponding stiffness) will dominate the other forces in the model. Choosing a good penalty factor can thus be often difficult and may require several attempts before getting it “right”. To help with choosing a penalty factor, FEBio offers the *auto\_penalty* option for many of its contact interfaces which will estimate a good penalty factor based on element size and initial material stiffness. However, even then it may still take a few tries before getting the penalty factor good enough.

### 8.5.2 Augmented Lagrangian Method

In theory, in the presence of constraints, the corresponding Lagrange multipliers must be calculated which, in the case of contact, correspond to the contact forces that enforce the contact constraint exactly. Unfortunately, the exact evaluation of these Lagrange multipliers is numerically very challenging and therefore in FEBio it was decided to evaluate these Lagrange multipliers using an iterative method, named the *Augmented Lagrangian* method. Using this method, FEBio will solve a time step several times (these iterations are termed *augmentations*), where each time the approximate Lagrange multipliers are updated (or *augmented*).

The obvious drawback of this method is that now each time step has to be solved several times. However, the advantage of avoiding the numerical problems of obtaining the exact Lagrange multipliers and the fact that in most contact problems very little extra work is needed to solve these augmentations, makes this method very attractive. In addition, it often gives better enforcement of the contact constraint compared to the penalty method.

So why not just use the augmented Lagrangian method? Well, often the penalty method will give good results and since the penalty method is much faster, it is often the preferred choice of many analysts.

In many cases, users are only interested in the final time step. For these users it may be of interest that it is possible to use the augmented Lagrange method only in the final time step. This can be done by defining a loadcurve for the *tolerance* contact parameter, which sets the convergence tolerance for the augmentations. (Note that you still need to set the enforcement method, i.e. the *laugon* parameter, to AUGLAG (=1)). Then, define the corresponding loadcurve as zero everywhere except for the final time step. FEBio will now only use the augmented Lagrangian method in the final time step (and the penalty method all other steps).

### 8.5.3 Initial Separation

FEBio often struggles with problems that have an initial separation. This is especially so when the problem is force-driven, meaning that the deformation of the model is driven by a force (opposed to displacement-driven problems). In general, when the contact interface is necessary to define a well-constrained problem it is best to avoid initial separation if possible. Another way around this issue is to first use a displacement to bring the surfaces in contact and then replace the displacement with a force.

## 8.6 Cautionary Note for Steady-State Biphasic and Multiphasic Analyses

Biphasic, biphasic-solute and multiphasic analyses are generally transient analyses, involving mass transport (solvent and solutes) through a porous permeable domain. The default setting for these analyses is the transient mode (see Section 3.2). When a steady-state analysis is requested (`<analysis type="steady-state"/>`), the code simplifies the governing equations by setting time derivatives of the solid displacement and solute concentrations to zero. It is important to understand that this simplification is only applicable to analyses where interface conditions between sub-domains, and boundary conditions with the external environment, allow mass exchanges for the solvent and all solutes. If a domain or sub-domain is completely sealed such that it prevents solvent or solute exchanges, a steady-state analysis should not be used, as it would not capture these sealed conditions and would lead to erroneous results. For those types of problems, always use a transient analysis to obtain the correct solution. In those cases, a steady-state response may be obtained efficiently by temporarily increasing the transport properties (hydraulic permeability and solute diffusivities) by orders of magnitude via load curves, and/or by increasing the size of time steps.

## 8.7 Guidelines for Multiphasic Analyses

### 8.7.1 Initial State of Swelling

Under traction-free conditions, a multiphasic material is usually in a state of swelling due to the osmotic pressure difference between the interstitial fluid and the external environment. This osmotic pressure arises from the difference in interstitial versus external concentrations of cations and anions, caused by the charged solid matrix and the requirement to satisfy electroneutrality. An osmotic pressure difference arising from such electrostatic interactions is known as the *Donnan osmotic pressure*. When the Donnan pressure is non-zero, traction-free conditions do not produce

stress-free conditions for the solid matrix, since the matrix must expand until its stressed state resists the swelling pressure.

The Donnan pressure reduces to zero when the fixed charged density is zero, or when the external environment is infinitely hypertonic (having ion concentrations infinitely greater than the interstitial fixed charge density). Since these two conditions represent special cases, it is generally necessary to devise methods for achieving the desired initial state of swelling, in an analysis where loads or displacements need to be prescribed over and above this swollen state. Swelling occurs as a result of the influx of solvent into the porous solid matrix. This influx is a time-dependent process that could require extensive analysis time. Therefore, it is computationally efficacious to achieve the initial state of swelling by using a multi-step analysis (Chapter 6) where the first step is a steady-state analysis (Section 3.3.1). In this steady-state step, the fixed charge density may be ramped up from zero to the desired value using a load curve for that property or, alternatively, the external environmental conditions may be reduced from a very high hypertonic state down to the desired level. In the second step, prescribed displacement boundary conditions (when needed) may be specified to be of type *relative* (Section 3.11.1), so that they become superposed over and above the initial swelling state.

*Example:*

```
<Module type="multiphasic"/>
<!-- rest of file -->
<Step>
  <Control>
    <analysis type="steady-state"/>
    ...
  </Control>
</Step>
<Step>
  <Control>
    ...
  </Control>
  <Boundary>
    <prescribe type="relative">
      <node id="22" bc="z" lc="4">1</node>
      ...
    </prescribe>
  </Boundary>
</Step>
```

## 8.7.2 Prescribed Boundary Conditions

In most analyses, it may be assumed that the ambient fluid pressure and electric potential in the external environment are zero, thus  $p_* = 0$  and  $\psi_* = 0$ , where the subscripted asterisk is used to denote environmental conditions. Since the external environment does not include a solid matrix, the fixed charge density there is zero. For example, for a triphasic analysis,  $c_*^+ = c_*^- \equiv c_*$ . It follows that the effective fluid pressure in the external environment is  $\tilde{p}_* = -R\theta\Phi_* \sum_{\alpha} c_*^{\alpha}$  and the effective concentrations are  $\tilde{c}_*^{\alpha} = c_*^{\alpha} / \hat{\kappa}_*^{\alpha} \tilde{c}_*^{\alpha}$ . Therefore, in multiphasic analyses, whenever the external environment contains solutes with concentrations  $c_*^{\alpha}$ , the user must remember to prescribe non-zero boundary conditions for the effective solute concentrations *and* the effective fluid pressure.

Letting  $p_* = 0$  also implies that prescribed mixture normal tractions (Section 3.13.2.6) represent only the traction above ambient conditions. Note that users are not obligated to assume that  $p_* = 0$ . However, if a non-zero value is assumed for the ambient pressure, then users must remember to incorporate this non-zero value whenever prescribing mixture normal tractions. Similarly, users are not required to assume that  $\psi_* = 0$ ; when a non-zero value is assumed for the electric potential of the external environment, the prescribed boundary conditions for the effective concentrations should be evaluated using the corresponding partition coefficient,  $\tilde{c}_*^\alpha = c_*^\alpha / \tilde{\kappa}_*^\alpha$ .

### 8.7.3 Prescribed Initial Conditions

When a multiphasic material is initially exposed to a given external environment with effective pressure  $\tilde{p}_*$  and effective concentrations  $\tilde{c}_*^\alpha$ , the initial conditions inside the material should be set to  $\tilde{p} = \tilde{p}_*$  and  $\tilde{c}^\alpha = \tilde{c}_*^\alpha$  in order to expedite the evaluation of the initial state of swelling. The values of  $\tilde{p}_*$  and  $\tilde{c}_*^\alpha$  should be evaluated as described in Section 8.7.2.

### 8.7.4 Prescribed Effective Solute Flux

The finite element formulation for multiphasic materials in FEBio requires that the natural boundary condition for solute  $\alpha$  be prescribed as  $\tilde{j}_n^\alpha \equiv j_n^\alpha + \sum_\beta z^\beta j_n^\beta$ , where  $\tilde{j}_n^\alpha$  is the effective solute flux. For a mixture containing only neutral solutes ( $z^\beta = 0, \forall \beta$ ), it follows that  $\tilde{j}_n^\alpha = j_n^\alpha$ .

### 8.7.5 Prescribed Electric Current Density

The electric current density in a mixture is a linear superposition of the ion fluxes,

$$\mathbf{I}_e = F_c \sum_\alpha z^\alpha \mathbf{j}^\alpha.$$

Since only the normal component  $j_n^\alpha = \mathbf{j}^\alpha \cdot \mathbf{n}$  of ion fluxes may be prescribed at a boundary, it follows that only the normal component  $I_n = \mathbf{I}_e \cdot \mathbf{n}$  of the current density may be prescribed. To prescribe  $I_n$ , it is necessary to know the nature of the ion species in the mixture, and how the current is being applied. For example, if the ions in the triphasic mixture consist of  $\text{Na}^+$  and  $\text{Cl}^-$ , and if the current is being applied using silver/silver chloride (Ag/AgCl) electrodes, a chemical reaction occurs at the anode where Ag combines with  $\text{Cl}^-$  to produce AgCl, donating an electron to the electrode to transport the current. At the cathode, the reverse process takes place. Therefore, in this system, there is only flux of  $\text{Cl}^-$  and no flux of  $\text{Na}^+$  ( $j_n^+ = 0$ ) at the electrode-mixture interface, so that the prescribed boundary condition should be  $j_n^- = -I_n/F_c$ . Since  $z^+ = +1$  and  $z^- = -1$  in a triphasic mixture, the corresponding effective fluxes are given by  $\tilde{j}_n^+ = 2j_n^+ - j_n^- = I_n/F_c$  and  $\tilde{j}_n^- = j_n^+ = 0$ .

### 8.7.6 Electrical Grounding

If a multiphasic mixture containing ions is completely surrounded by ion-impermeant boundaries it is necessary to ground the mixture to prevent unconstrained fluctuations of the electric potential. Grounding is performed by prescribing the effective concentration of one or more ions, at a location inside the mixture or on one of its boundaries corresponding to the location of the grounding electrode. The choice of ion(s) to constrain may be guided by the type of grounding electrode and the reference electrolyte bath in which it is inserted.



## 8.8 Guidelines for Fluid Analyses

Currently, the fluid solvers of FEBio can be used to solve for isothermal fluid mechanics problems (“fluid” solver), isothermal fluid-structure interaction problems (“fluid-FSI” solver, Section 8.8.6) and isothermal fluid mechanics with solute (mass) transport (“fluid-solutes” solver). When solutes are included in an analysis, their treatment is very similar to the “multiphasic” solver described in Section 4.16. The fluid pressure  $p$  and solute concentrations  $c^\iota$  in a fluid-solutes domain are thus related to the effective fluid pressure  $\tilde{p}$  and effective solute concentrations  $\tilde{c}^\iota$  as per eq.(4.16.7), to account for the possibility that the solutes are electrically charged. The theoretical treatment of solutes in the “fluid-solutes” solver follows that of the “multiphasic” solver, with the following simplifications: (a) The effective solubility  $\hat{\kappa}^\iota$  appearing in the expression for the partition coefficient  $\tilde{\kappa}^\iota$  in eq.(4.16.7) only embodies non-ideal physico-chemical behavior (whereas  $\hat{\kappa}^\iota$  in a multiphasic mixture with a porous solid matrix can also embody the steric volume exclusion of solutes from the pore space); therefore, one should always set  $\hat{\kappa}^\iota = 1$  when modeling ideal behavior in a “fluid-solutes” analysis. (b) Since there is no porous solid matrix that can hinder solute transport, the diffusivity tensor  $\mathbf{d}^\iota$  of a solute in a “fluid-solutes” mixture is always equal to  $d_0^\iota \mathbf{I}$ , where  $d_0^\iota$  is the solute diffusivity in free solution. Thus, even though FEBio requires users to specify a value for  $\mathbf{d}^\iota$ , that value is essentially ignored in the “fluid-solutes” solver code, where it is replaced by  $d_0^\iota \mathbf{I}$ . If users would like to neglect osmotic pressure in a “fluid-solutes” analysis, the osmotic coefficient  $\Phi$  should be set to zero, producing  $\tilde{p} = p$ . Moreover, in the absence of solutes ( $c^\iota = 0$  for all  $\iota$ , as in the “fluid” and “fluid-FSI” solvers) this relation holds automatically. Therefore, when dealing with the “fluid” and “fluid-FSI” solvers, we do not need to add the “effective” modifier when mentioning the fluid pressure.

There are two broad categories of materials available for “fluid-FSI” analyses: (1) *fluid-FSI* (Section 4.20.3), and (2) *biphasic-FSI* (Section 4.20.4). A *fluid-FSI* material is used to describe a fluid domain whose mesh is deformable; though a solid material (typically *neo-Hookean*) must be assigned to the mesh of a *fluid-FSI* material, its mass density is ignored and it is the user’s responsibility to ascribe a negligible elasticity to this solid domain, whose sole purpose is to regularize the mesh deformation. A *biphasic-FSI* material considers that the deformable domain is a biphasic material (see Section 4.14), with a porous-permeable solid domain that has non-negligible mass density, solid volume fraction, solid matrix stiffness, and hydraulic permeability. The interstitial fluid of such a domain may also be ascribed a viscosity (Newtonian or non-Newtonian).

### 8.8.1 Degrees of Freedom and Boundary Conditions

The nodal degrees of freedom for “fluid” analyses are the nodal fluid velocity  $\mathbf{v}^f$  and nodal fluid dilatation  $e^f$ . For a “fluid-FSI” analysis the nodal fluid velocity  $\mathbf{v}^f$  is replaced with the relative fluid flux  $\mathbf{w} = \varphi^f (\mathbf{v}^f - \mathbf{v}^s)$  and supplemented with the nodal solid displacement,  $\mathbf{u}$ , whose material time derivative equals the solid velocity  $\mathbf{v}^s$ . Here,  $\varphi^f$  represents the fluid volume fraction (porosity) of the fluid-FSI domain, which is prescribed by the user in a *biphasic-FSI* material (via the complementary solid volume fraction) or automatically set to  $\varphi^f = 1$  in a *fluid-FSI* material. Thus, the degrees of freedom of a fluid-FSI analysis reduce to those of a fluid (or fluid-solutes) analysis in the limit when  $\varphi^f \rightarrow 1$  and  $\mathbf{v}^s \rightarrow \mathbf{0}$ . For “fluid-solutes” analyses, the nodal degrees of freedom include the effective solute concentrations  $\tilde{c}^\iota$ ; as for “multiphasic” analyses, it is assumed that solutes occupy a negligible volume fraction in the mixture (thus  $\varphi^f = 1$  for “fluid-solutes” analyses).

The fluid Cauchy stress is given by  $\boldsymbol{\sigma}^f = -p\mathbf{I} + \boldsymbol{\tau}$ , where  $p$  is the actual (not the effective) fluid pressure, as per eq.(4.16.7). The viscous stress  $\boldsymbol{\tau}$  depends on the fluid rate of deformation tensor  $\mathbf{D}^f$  according to a user-selected constitutive relation, such as *Newtonian fluid*.

In any of the fluid analysis types, the user may prescribe any of the components of  $\mathbf{w}$  as an essential boundary condition, or the corresponding component of the viscous traction  $\mathbf{t}^\tau = \boldsymbol{\tau} \cdot \mathbf{n}$  as a natural boundary condition ( $\mathbf{n}$  being the normal to the boundary surface). When neither a component of  $\mathbf{w}$  nor the corresponding component of  $\mathbf{t}^\tau$  is prescribed explicitly, the latter is naturally assumed to be zero. Similarly, the user may prescribe  $e^f$  (or  $\tilde{p}(e^f)$ , see Section 3.13.2.14) as an essential boundary condition, or  $w_n = \mathbf{w} \cdot \mathbf{n}$  as a natural boundary condition. When neither is prescribed explicitly, it is naturally assumed that  $w_n = 0$ . Natural boundary conditions are useful for creating symmetry planes in fluid analyses, where the normal fluid velocity and the viscous traction are zero: On those symmetry planes, no boundary conditions should be specified.

It is noteworthy that the fluid formulation in FEBio allows the prescription of the nodal value of  $\mathbf{w}$  as an essential boundary condition, and the surface value of  $w_n$  as a natural boundary condition. On boundaries where the fluid velocity is completely prescribed, i.e., when normal and tangential components are known (and not zero), prescribing  $\mathbf{w}$  and  $w_n$  instead of just  $\mathbf{w}$  produces the best computational outcome. In those cases, the user should use the *fluid velocity* surface load, which combines both boundary conditions (Section 3.13.2.21). If the user only wants to prescribe a known normal velocity  $w_n$  and leave the tangential velocity unspecified, use the *fluid normal velocity* surface load (Section 3.13.2.20); this surface load also provides the option of setting the tangential fluid velocity to zero. Finally, if the user wants the velocity to remain normal to the selected surface (e.g., an inlet or outlet surface on which  $e^f$  is prescribed or fixed), but does not know the velocity magnitude *a priori*, use the *normal fluid velocity* constraint (Section 3.15.4) to enforce this condition.

A viscous fluid satisfies the no-slip condition on a physical boundary surface. This means that  $\mathbf{w} = \mathbf{0}$  on no-slip boundaries. This boundary condition can be enforced simply by fixing the three components of  $\mathbf{w}$  (Section 3.11.6) and not specifying any value for  $e^f$  on that surface (which naturally enforces  $w_n = 0$ ).

Computational fluid dynamics analyses are often performed over a mesh that truncates the real fluid domain. Thus, fluid may enter the finite element domain across some upstream inlet boundary and leave the domain across some downstream outlet boundary. The exact flow conditions on these boundaries may not be known, thus they have to be approximated using best guesses. Arguably, the viscous traction  $\mathbf{t}^\tau$  is the least intuitive boundary condition to guess. On an outlet boundary, it is necessary to prescribe or fix the dilatation  $e^f$  to overcome the natural boundary condition  $w_n = 0$ . However, in most cases the tangential components of  $\mathbf{w}$  on that boundary are not known, therefore they cannot be fixed or prescribed, leading to the natural condition  $\mathbf{t}^\tau = \mathbf{0}$ . This natural condition may work fine in some cases, but it will often fail numerically when vortices being shed inside the finite element domain try to cross the outlet boundary. In those cases, consider using the *fluid backflow stabilization* (Section 3.13.2.16) and *fluid tangential stabilization* (Section 3.13.2.17) surface loads, which prescribe a velocity-dependent viscous traction  $\mathbf{t}^\tau$ .

Similarly, on an inlet boundary that shares an edge with a no-slip surface, prescribing the fluid velocity  $\mathbf{w}$  on the inlet boundary may not accurately reproduce the velocity profile in the boundary layer that would be produced on the adjoining no-slip surface. This potential mismatch could result in numerical instabilities; in such cases, prescribing a dilatation  $e^f$  on the shared edge may mitigate this issue.

### 8.8.2 Biased Meshes for Boundary Layers

In fluid analyses, boundary layers will form in the vicinity of no-slip boundaries, where the velocity magnitude varies rapidly with the distance from a no-slip surface. The thickness of boundary layers decreases with increasing Reynolds number. To capture these boundary layers accurately



in a fluid finite element analysis, it is necessary to refine the mesh to produce thinner elements closer to the no-slip boundaries. This is done most conveniently by using mesh biasing tools, available in various meshing software programs, or post-hoc boundary-layer meshing tool that modify an existing mesh. Both of these options are available in FEBioStudio. Boundary layer meshing should always be used in fluid analyses in FEBio. Whereas other finite element codes employ numerical stabilization techniques that partially alleviate the need for biased meshes, FEBio does not employ such techniques. Therefore, using uniform finite element meshes may fail to produce numerical convergence, even in some seemingly simple problems. Users should keep in mind that biased meshes don't necessarily require more nodes and elements, therefore there is no automatic computational cost for using biased meshes.

### 8.8.3 Computational Efficiency: Broyden's Method

Fluid analyses produce a non-symmetric stiffness matrix and the resulting system of equations may be efficiently solved using Broyden's quasi-Newton method, where an approximation to the matrix inverse is produced at each iteration after the first full-Newton iteration of a time step. In fluid analyses, it is often possible to achieve convergence at each time step without having to perform additional full-Newton updates. Therefore, it is recommended to set *max\_updates* (Section 3.3) to a large number (e.g., 50), along with setting *diverge\_reform* to 0 (false), which leads to a more efficient solution scheme. A further advantage arises in fluid analyses since the mesh remains invariant over time: it is often possible to continue using Broyden updates even across time steps, without performing a full-Newton iteration at the start of that step, by setting *reform\_each\_time\_step* to 0 (false). In that case, the solver will continue using Broyden updates up to the value of *max\_updates*, before performing a full Newton update.

### 8.8.4 Dynamic versus Steady-State Analyses

FEBio runs fluid analyses in dynamic mode by default, because many fluid flow problems do not have a steady-state solution, since the governing equations are nonlinear. For example, many flows may exhibit vortex shedding as fluid flows across an obstacle, such as a flow constriction or a solid body. Even if some form of periodicity emerges under specific flow conditions, as in the von Karman vortex street, the corresponding solution is not in steady state. However, for some low Reynolds number laminar flows, a steady-state response may exist and FEBio will attempt to find that solution when using analysis type "steady-state" (Section 3.3). The user should be aware that, under steady-state analyses, the solution may not necessarily converge as efficiently to the final solution as would a dynamic analysis that allows the solution to reach steady state after a sufficiently long time. The best choice of analysis type for steady-state problems may need to be determined by trial and error.

### 8.8.5 Isothermal Compressible Flow versus Acoustics

The governing equations for fluid flow in FEBio represent isothermal conditions, thus eliminating temperature as a variable in the numerical analysis. Since the fluid solver accommodates some measure of fluid compressibility (via the fluid dilatation variable  $e^f$ ), it is possible to solve for pressure wave propagation in fluids using this formulation. However, from a theoretical perspective, it should be recognized that the speed of sound (i.e., the wave speed) in isothermal (constant temperature) flow is different from the speed of sound in isentropic (constant entropy) flow. The field of acoustics typically examines pressure wave propagations under isentropic conditions, which are

found to be more consistent with experimental measurements of the speed of sound in air. Therefore, the FEBio fluid solver may not produce realistic wave speeds when simulating acoustics. A fluid solver for isentropic flow may be developed for FEBio in the future.

### 8.8.6 Fluid-Structure Interactions

Fluid-structure interactions (FSI) may be modeled in FEBio using the *fluid-FSI* module (Section 3.2). In an FSI analysis the finite element mesh defining the fluid domain is deformable, so that the shape of the fluid domain does not have to remain fixed over time. The fluid flows through the deforming mesh, just as it does in standard computational fluid dynamics (CFD) analyses with fixed meshes. The deformation of the fluid domain may be controlled by prescribing suitable boundary conditions (so-called *moving boundary problems*) and by interactions with solid domains. For example, fluid may flow inside a flexible tube and the fluid pressure may cause the tube to expand or contract; the tube itself may be pinched at some location, reducing the cross-section through which fluid flows. The tube wall is a deformable solid domain which may be modeled in FEBio using any of the solid material models described in Sections 4.1 through 4.13, or it may be modeled as a porous deformable domain containing a viscous fluid, in which case one should use the *biphasic-FSI* material for the tube wall (Section 4.20.4). The inner tube (the fluid domain) should be modeled as a *fluid-FSI* material (Section 4.20.3).

In FEBio, a *fluid-FSI* material is modeled as a specialized fluid-solid mixture where the solid matrix density is zero and its stiffness should be set to a negligible value. In contrast, a *biphasic-FSI* material is a full-fledged fluid-solid mixture, where the solid may have mass and exhibit any desired nonlinear elastic or inelastic response. Both FSI materials require the user to define a *fluid* material from the same list as those available for CFD analyses (Section 4.20). It also requires the user to define a *solid* material to regularize the mesh deformation, which may be selected from Sections 4.1 through 4.13 for a *biphasic-FSI* material. In contrast, the *solid* material of a *fluid-FSI* material is massless (the *density* value entered by the user is ignored and reset internally to zero). Among all the choices available in Section 4.1, it is recommended to use the *neo-Hookean* elastic solid (Section 4.1.4.20), setting Poisson's ratio  $\nu$  to zero and Young's modulus  $E$  to a very low value (a few orders of magnitude smaller than the moduli of surrounding solid domains).

The degrees of freedom at each node of a FSI domain consist of the components of the solid displacement  $\mathbf{u}$  (which describe the mesh deformation), the fluid velocity  $\mathbf{w}$  relative to the solid, and the fluid dilatation  $e^f$ . Essential boundary conditions may be prescribed on any of these degrees of freedom. For example, in a moving boundary problem, the motion of a boundary may be prescribed by setting the relevant components of  $\mathbf{u}$ . A no-slip boundary between an FSI material and a solid domain may be defined by simply setting  $\mathbf{w} = \mathbf{0}$ , regardless of the motion of that boundary. In general, all boundary conditions described for CFD analyses in Section 8.8.1 may be applied identically to FSI analyses.

Currently in FEBio, the mesh of a *fluid/biphasic-FSI* material domain must be continuous with the mesh of surrounding solid domains. The surface between domains, which consists of the shared faces of adjoining finite elements from each domain, is called the FSI interface. Because the mesh is continuous, the solid displacement and the solid component of the traction are automatically continuous across FSI interfaces; when fluid is present on both sides of the interface, the relative fluid flux is also automatically continuous. In other words, when two FSI domains are adjoining (e.g., fluid-FSI with fluid-FSI, or fluid-FSI with biphasic-FSI), all boundary conditions are automatically satisfied across those interfaces. However, when an FSI domain interfaces with a solid domain, the fluid component of the traction in the *fluid-FSI* and *biphasic-FSI* domains must be explicitly transferred to the solid domain across that interface. This is done by prescribing a

*fluid-FSI traction* or a *biphasic-FSI traction* on that interface (Section 3.13.2.23). This surface load does not require any user-defined parameters; its sole purpose is to identify those FSI interfaces where the fluid traction must be properly transferred to the surrounding solid domain. Omitting this boundary condition means that the fluid traction (i.e., the fluid pressure and the normal and tangential viscous components of the fluid stress) have no effect on this interface; the interface may still deform in response to the motion of the solid domain. For example, when a fluid interacts with a rigid solid domain whose motion is entirely prescribed, it is not necessary to apply a *fluid-FSI traction* at the interface between the fluid and rigid solid domains.

A *fluid-FSI traction* is also required on free fluid surfaces, such as the surface of a fluid in open channel flow, in order to properly capture the motion of that surface (such as waves). This condition is required since the net traction  $\mathbf{t}$  on a free surface is zero. As the fluid-FSI domain is modeled as a specialized solid-fluid mixture, the net traction consists of the sum of solid and fluid tractions,  $\mathbf{t} = \mathbf{t}^s + \mathbf{t}^f$ . Setting  $\mathbf{t}$  to zero produces a traction  $\mathbf{t}^s = -\mathbf{t}^f$  on the solid mesh, allowing it to deform to the proper shape.

Here are a few cautionary notes to keep in mind when running FSI analyses: (1) Depending on the problem being analyzed, fluid-structure interactions may cause significant mesh deformations that lead to mesh distortion. Currently, adaptive meshing is not yet available to relieve this mesh distortion, therefore analyses and meshes need to be designed to minimize this effect or to terminate before the solution becomes too corrupted. (2) It is possible to pinch a solid domain that surrounds a fluid domain until the flow cross-section has reduced considerably; however it is not possible to completely shut off the flow, as this would require reducing the volume of some finite elements to zero. (3) While it is possible to model the deformation of free fluid surfaces, such as the formation of surface waves in open channel flow or the deformation of fluid blobs floating in a gravity-free environment, it is not possible to model the separation of such fluid domains into sub-domains such as spraying fluid droplets; this means that analyses where such phenomena are expected to happen will likely fail. (4) Whereas standard CFD analyses may be jump-started by prescribing a relatively large fluid velocity at the very first time step of an analysis, these types of boundary conditions may lead to dynamic oscillations and potential instabilities in an FSI analysis; therefore, users must be more considerate when prescribing boundary conditions for FSI analyses.

### 8.8.7 Fluid-Solutes Analyses

Fluid-solutes analyses introduce two types of molar fluxes into an analysis: (1) The diffusive molar flux  $\mathbf{j}_d^\iota = -\tilde{\kappa}^\iota d_0^\iota \text{grad } \tilde{c}^\iota$  and (2) the sedimentation flux  $\mathbf{j}_b^\iota = s^\iota c^\iota \mathbf{b}^\iota$ , where  $s^\iota \equiv d_0^\iota \frac{M^\iota}{R\theta}$  is the sedimentation coefficient and  $\mathbf{b}^\iota$  is the body force acting on solute  $\iota$ . These molar fluxes represent the flux of the solute relative to the solvent. The natural boundary condition for solutes is  $\tilde{j}_n^\iota = 0$ , where

$$\tilde{j}_n^\iota = \left( \mathbf{j}^\iota + \sum_{\gamma} z^\gamma \mathbf{j}^\gamma \right) \cdot \mathbf{n} \quad (8.8.1)$$

is the normal component of the effective molar flux  $\mathbf{j}^\iota + \sum_{\gamma} z^\gamma \mathbf{j}^\gamma$  on a boundary, with  $\mathbf{j}^\iota = \mathbf{j}_d^\iota + \mathbf{j}_b^\iota$  representing the sum of diffusive and sedimentation fluxes. The effective molar flux is constructed in this manner to enforce the electroneutrality condition,

$$\sum_{\iota} z^\iota c^\iota = 0 \quad (8.8.2)$$

as also done in “multiphasic” analyses (Section 4.16.1.4).

Zero natural flux  $\tilde{j}_n^t$  implies that the solute moves at the velocity of the solvent (the fluid). Therefore, if the fluid satisfies  $v_n^f = 0$  on that boundary, the solute will not transport across it. But if the fluid is allowed to transport across the boundary (e.g., using a prescribed fluid pressure), the solute can also transport across the boundary via convection. Therefore, for open flows or flows through conduits, where the fluid pressure is prescribed (e.g., set to zero) at an outlet boundary, no specific boundary condition needs to be prescribed on the solute in order for the latter to convect freely with the solvent at that outlet boundary. However, if the solvent exhibits back flow on that boundary in a manner that requires the application of the *fluid backflow stabilization* (Section 3.13.2.16) and *fluid tangential stabilization* (Section 3.13.2.17) surfaces loads, then consider also using the *solute backflow stabilization* surface load (Section 3.13.2.26) on that boundary, to produce more physically-realistic solute flows on such truncated domains, and improve numerical convergence.

The fluid pressure  $p$  in a fluid-solutes analysis is given by eq.(4.16.7). To prescribe the fluid pressure  $p$  on the boundary of a fluid-solutes domain, use the fluid pressure boundary condition specified in Section 3.11.18.

By default, the fluid-solutes solver employs the *staggered equation\_scheme* format for the stiffness matrix, and solves for the nodal fluid velocity  $\mathbf{v}^f$ , nodal dilatation  $e^f$  and nodal effective solute concentrations  $\tilde{c}^t$  using a *coupled solve\_strategy*. This default settings works well when the Peclet number (the ratio of solute convective velocity to solute diffusive velocity) is relatively low (e.g., less than 100). For these problems, using the default value of  $ctol=0.01$  is generally appropriate. However, if users notice significant convergence difficulties on the solute concentration as the Peclet number approaches 100 or greater, consider setting  $ctol=0$ . When solving strongly convective solute transport problems with high Peclet numbers (e.g., greater than 1000), diffusion will be negligible compared to convection. In that case, users should switch to *equation\_scheme=block* and *solve\_strategy=sequential* and set  $ctol=0$  to increase the chances of achieving a converged numerical solution. The rationale for these choices is explained in the FEBio journal article associated with this fluid-solutes solver [78].

As done in the multiphasic solver, the solute volume fraction is considered negligible in fluid mixtures modeled in the fluid-solutes solver. Therefore, the volume of a fluid mixture will not change with increasing solute concentrations. When a solute is convected by the solvent, it means that the solvent exerts a frictional force on the solute, whose magnitude is inversely proportional to the solute diffusivity  $d_0^t$ . In the fluid-solutes solver, the acceleration of solutes is considered negligible, implying that the velocity of the solute in the solvent reaches its terminal velocity instantaneously (thus, inertial forces are negligible compared to the drag force between solute and solvent).

According to Newton's third law stating that every action force has an equal and opposite reaction force, the drag force exerted by the solvent on the solute is balanced by an equal and opposite reaction force exerted by the solute on the solvent. However, by default, this reaction force on the solvent is turned off in the fluid-solutes solver. In other words, the motion of the solvent affects that of the solute, but the motion of the solute does not affect the solvent. This default setting is adopted because it represent the common form of diffusion-reaction equations modeled in most of the mass transport literature. Therefore, when modeling solute diffusion problems in a stationary solvent (e.g., in the absence of pressure or velocity-driven solvent flows), the diffusive flux of a solute will not drag the solvent, keeping it in a stationary state. When the solvent motion is driven by solute concentration gradients, we call it *osmosis*. There is no common term describing solvent motion caused by solute sedimentation (motion of solute under the action of a body force), so we use the term *osmosis* to describe this phenomenon as well. Users can override the default setting of the fluid-solutes solver by checking the *include osmosis* box in the fluid-solutes material definition. When this box is checked, the solvent will have a force exerted on it by the solute (either

due to diffusion or sedimentation, or both), causing it to move.

High-Peclet number flows are notoriously difficult to solve numerically. In FEBio we have reported valid solutions for a Peclet number as high as  $10^{11}$  [78], thanks to the specific finite element formulation employed in the fluid-solutes solver, and the use of the sequential solver when needed. Nevertheless, the fluid-solutes solver is not fool-proof. In some problems, spurious increases in solute concentration can occur, seemingly randomly in the mesh, even when using a highly regular mesh. In many of these cases, this type of numerical artifact may go away by simply refining the mesh. However, users should be aware that this artifact may emerge 'silently', meaning that it does not necessarily produce slower numerical convergence that could hint at this problem.

The explanation for this behavior is that the numerical solutions to the effective solute concentrations at large Peclet numbers are effectively solutions to the partial differential equation  $D^f(J^f c^e)/Dt = 0$ , as reviewed in the Theory Manual. This integration is performed based on the current solution for  $J^f = 1 + e^f$  and  $\mathbf{v}^f$  as well as the structure of the mesh. Any ill-conditioning in the mesh (such as a mesh which is too coarse) may introduce a numerical round-off error in the calculation of  $\tilde{c}^e$ , and there is no other sufficiently dominant term in the solute mass balance equation that can dampen or correct for such numerical round-offs. Thus, they may grow over time. Mesh refinements are required when encountering these circumstances.

Other boundary conditions for the solute (such as the essential boundary condition  $\tilde{c}^e = 0$ ) may also improve predictions when using relatively coarse meshes, preventing the occurrence of spurious rises which often initiate on boundaries. Unfortunately, in some problems, this option forces users to adopt a non-physical boundary condition which may produce unacceptable results. This workaround may not be needed when using a sufficiently refined mesh, though finer meshes typically require longer solution times. Also note that highly irregular meshes near no-slip boundaries may cause spurious rises in solute concentration in high Peclet number flows. Therefore, it is recommended to use suitable meshing techniques for generating boundary layer meshes, such as stacking pentahedral elements atop tetrahedral elements when using unstructured tetrahedral meshes. For diffusion-dominated or reaction-dominated flows however, these numerical concerns may not arise. Nevertheless, a thorough understanding of the underlying physics of these problems is essential for properly prescribing boundary conditions, and the number and size of time increments in a finite element analysis.

The default settings for the fluid-solutes solver force it to reform the stiffness matrix at the start of each time step. This is a standard requirement in most of the FEBio solvers (including solid and structural mechanics, and standard biphasic and multiphasic domains), though it had previously been demonstrated that the fluid solver could avoid reforming the stiffness matrix even after achieving convergence at a given time step, producing a very efficient computational scheme. Unfortunately, this increased efficiency is not always an option with the fluid-solutes solver, because the stiffness matrix contains terms that depend significantly on the solute concentrations. As these concentrations evolve over time and space, it becomes essential to update the stiffness matrix to reflect those changes. Therefore, consider using the direct sparse solver from the Intel MKL library (*linear\_solver=Mkl dss*) to improve computational efficiency during matrix factorization, in comparison to the Pardiso solver from the same library. Moreover, in applications where the initial solute concentration is non-zero, and non-negligible compared to the prescribed change in concentration on an inlet boundary, it may be possible to solve a fluid-solutes problem more efficiently without reforming the stiffness matrix at each time step.

## 8.9 Understanding the Solution

Okay, FEBio found a solution. Great, but how do you know it is the right one? Well, it turns out this is in fact a harder question than it may seem. In any finite element model there are numerous assumptions and approximations and trying to discuss all of these is outside the scope of this section. Instead, we'll look at some of the most typical problems that FEBio users have run into.

### 8.9.1 Mesh convergence

The solution calculated by FEBio only applies to the mesh for which it was solved. However, the “exact” solution must be independent of the mesh and therefore a mesh convergence study is almost always necessary to make sure the FEBio solution is close to this exact solution. In practice this means that a model must be run several times with an increasingly finer mesh to find out at which mesh density the FEBio solution no longer changes.

### 8.9.2 Constraint enforcement

FEBio uses many iterative algorithms for enforcing constraints. For each of these constraints the user must verify that the solution indeed satisfies these constraints sufficiently. The two most common constraints used in FEBio are incompressibility and contact.

For uncoupled materials, incompressibility (for hexahedral elements) is handled in FEBio using a three-field formulation in addition to the enforcement of the incompressibility constraint. This constraint is usually enforced using a penalty formulation (although an augmented Lagrangian method is also available). To inspect whether the incompressibility constraint is satisfied sufficiently the user can look at the volume ratio which has to be close to one. If it deviates from one too much the user needs to increase the “bulk modulus” of the material.

For coupled materials, incompressibility is not treated explicitly and care must be taken when using coupled materials in a near-incompressible regime (e.g. setting the Poisson's ratio too close to 0.5 for a neo-Hookean material). In that case, the solution will most likely “lock” which manifests itself in displacements that are too small. Inspection of the volume ratio may not be sufficient to identify locking. However, a mesh convergence study will often help in identifying this problem.

When a contact constraint is not sufficiently enforced, the contacting surfaces will have penetrated. This problem is usually identified easily by looking at the deformed model in a post-processing software (e.g. FEBioStudio). Increasing the penalty factor and/or decreasing the augmented Lagrangian tolerance should solve this problem.

## 8.10 Guidelines for Using Prestrain

There are a variety of ways in which prestrain can be added to an FEBio model, but the recommended approach – a two-step approach – is outlined here.

In the first step, the prestrain is enforced on the model. This is accomplished by making sure the part that will be prestrained uses the prestrain elastic material (or uncoupled prestrain elastic for uncoupled hyperelastic materials). A suitable prestrain generator must be defined as well. In addition, one of the update rules can be specified as a nonlinear constraint in order to modify the effective prestrain field. The boundary conditions in this step should be minimal but sufficient to ensure a well-defined finite element problem.



After the first step converges, the model is successfully prestrained. However, depending on the application, the reference geometry could have changed and this may complicate the application of further loads (most of FEBio will still use the original geometry as the reference geometry). To eliminate this potential source of confusion, it is recommended to apply the prestrain initial condition (see chapter 4) which essentially fixes the prestrain gradient and resets the reference geometry to the current geometry. FEBio will then apply any subsequent loads to this new reference geometry. One caveat of this approach is that the plotfile still stores the original reference geometry. To prevent inconsistencies in the plotfile it is recommended suppressing the creation of the plot file in the first step. This is done by setting the following control flag in the Control section of the first step.

```
<plot_level>PLOT_NEVER</plot_level>
```

Then, in the Control section of the second step, restore the plot level value.

```
<plot_level>PLOT_DEFAULT</plot_level>
```

Now, the new reference configuration resulting after the prestrain analysis step, will be the reference configuration that is used in the plot file.

## 8.11 Guidelines for using the CG-solid solver

The nonlinear conjugate gradient solver, CG-solid, is an alternative solution algorithm for static solid mechanics problems. It does not currently support dynamic problems, fluids, etc. This solver has advantages in several situations:

- It uses much less memory and therefore allows bigger problems to be solved on a given computer than the standard Newton-based solvers
- Its performance scales better for large problems and it may be faster for very large problems (but see the limitations below);
- For nonlinear problems with buckling or other instabilities it can converge much more reliably than the standard solver;
- For many problems it will converge in a single timestep.

However, it also has some disadvantages:

- For small models it is typically slower than the BFGS solver;
- Although it usually converges in a single timestep, it requires a much larger number of iterations to do so. It is therefore inefficient if many timesteps are needed to produce intermediate results or follow the progression of a problem;
- The number of iterations it requires to converge is proportional to the number of elements across the longest dimension of the mesh. It therefore performs best for compact problems with similar numbers of elements in each direction. For example a cube with equal numbers of elements in each direction would solve much faster than a flat sheet or a long, thin beam with the same total number of elements;

- It does not currently work well for shell elements or freely moving rigid bodies. The way it works is quite different from the Newton-based solver and the problem may need to be defined differently for optimum results:
- The convergence criteria in FEBio measure the change in the solution on each iteration as a fraction of the total change for the timestep. Because the CG solver converges in a much larger number of smaller steps, the convergence criteria must be set smaller than for the BFGS solver. The default values are  $\text{dtol}=1\text{e-}6$  and  $\text{etol}=0.001$ , but it may be necessary to reduce these further to ensure full convergence. Check the solution carefully to make sure that it is correct and reduce  $\text{dtol}$  further if necessary.
- The rate of convergence is quite slow near to the solution so if the convergence tolerance is set too small the solution may take much longer.
- It is usually best to use a single timestep if possible, rather than many smaller steps as is normal for the Newton solver. More timesteps will greatly increase the solution time.
- Unlike the Newton-based solvers, prescribed displacements are applied only to the nodes they affect at first and take many iterations to propagate through the adjoining mesh. This means that they can often cause excessive deformation of the adjacent elements, for example a compressive prescribed displacement can push the surface layer of nodes right through the adjoining elements causing a negative Jacobian error. Unlike the BFGS solver, where prescribed displacements often give more stable convergence, it is better to apply forces where possible and to avoid prescribed displacements.
- If prescribed displacements are essential, it may be necessary to use smaller timesteps to avoid excessive distortion of the adjoining elements. In this case it is possible to slacken the convergence criteria for the intermediate steps and then apply a tighter tolerance for the final step to generate a correct solution.
- The preconditioner improves convergence when the stiffness of the nodes varies, for example because of different element sizes, different materials or features such as quadratic elements with midside nodes. For problems where all the elements are the same it offers no great advantage and may even cause slower convergence in some cases. It is therefore best to use it for irregular meshes, multiple materials, quadratic elements and contact problems, but for problems with regular meshes where all the elements are a similar size it may be better to turn it off.
- For problems with large deformation and displacements the solver may take a very large number of iterations to converge. The solver proceeds by many small, cautious steps, unlike the Newton-based solvers, which attempt to jump straight to the final answer on each iteration. It is therefore much more reliable but may need thousands or tens of thousands of iterations to reach the solution.

## 8.12 Limitations of FEBio

If you run into a problem that you can't seem to solve, maybe you ran into a limitation of FEBio. This section discusses some important limitations of the software and how they may affect the outcome of your analysis.



### 8.12.1 Geometrical instabilities

A geometrical instability is a point in the solution at which several paths for continuing the solution are possible. Typical examples are buckling of a column. FEBio's Newton based solvers cannot handle buckling since usually at these points, the material's elasticity tangent is not uniquely defined. Although in some cases FEBio will be able to pass a buckling point, in most cases, FEBio will not be able to converge to a solution.

### 8.12.2 Material instabilities

A material instability is a point in the stress-strain response where the slope of the stress-strain curve effectively becomes zero. Since FEBio's Newton based solvers use this slope to advance the solution, a zero slope would mean an infinite step and FEBio will not be able to continue.

The slope (or more accurately the elasticity tangent tensor) does not have to be zero to cause problems. For materials that soften at large deformations, this slope may become so small that it renders the global stiffness matrix ill-conditioned. It is unlikely that FEBio will be able to continue and even if it finds a solution, it may not be useful.

### 8.12.3 Remeshing

FEBio is designed to solve problems that can undergo large deformations. However, when the deformations become too large it can happen that the finite element mesh cannot be distorted any further without inverting elements. In that case, FEBio will not be able to continue the solution and will most likely terminate with a negative jacobian error message.

One possible solution to this problem is to remesh the model at the point at which the deformation become too large, but currently FEBio does not have any remeshing capabilities. The only remedy at this point is to plan ahead and design your mesh in such a way that the elements will not invert in the range of deformation that you are interested in (e.g. place a butterfly mesh in sharp corners or make the mesh finer in areas of larger deformation).

### 8.12.4 Force-driven Problems

When the primary method of deforming the model is a force, the problem is said to be force-driven. (Opposed to a displacement-driven model where the deformation is caused by displacement boundary conditions.) Force-driven forces are inherently unstable and FEBio has limited capabilities of handling such problems<sup>1</sup>. The cause of this instability can easily be explained using a simple example. Imagine a box that is constrained in all directions except one. In the unconstrained direction, apply two equal but opposite forces on opposite faces. In order to prevent the box from flying away, the box must generate stresses which balance the applied forces exactly. Unfortunately, the numerical solution will only be approximate due to numerical round-off errors inherent in the calculation. Even the slightest deviation from balancing the forces exactly will create a net-force which in turn causes the model to fly off in the unconstrained direction (in which a rigid body mode now exists).

Usually, these problems are circumvented by adjusting the boundary conditions so that rigid body modes cannot exist. In the previous example this can be done by only modeling half of the box and enforcing a symmetry boundary condition. However, in some cases this is not possible.

---

<sup>1</sup>As of FEBio 2.0, some features are available that may help with this issue, but they are still experimental.

This is often the case in force-driven contact problems, where the contact interface is necessary to define a well-constrained model. When there is an initial separation (or even when the surfaces have no initial overlap) the initial contact force is zero, which is equivalent to having no contact enforcement, and thus the model is under-constrained. Therefore, for force-driven contact problems it is important to have some initial contact before starting the analysis.

### 8.12.5 Solutions obtained on Multi-processor Machines

Although technically not a limitation of FEBio it is important that users are aware that in some cases you may get a different convergence history or sometimes even a slightly different answer when you run the same model twice on a multi-processor machine. This is caused by the fact that on multi-processor machines the same calculations can be executed in a different order and due to the accumulation of numerical round-off errors may result in slightly different answers. The type of problems that are mostly affected by this are problems that are close to being ill-conditioned. Also, problems that have many time steps or require many iterations for each time step may be affected by this.

If you experience different answers for the same problem, try running the model on just one processor (if possible). See Section 2.6 for more information on how to run FEBio on multi-processor machines.

## 8.13 Where to Get More Help

When you get here, you may be ready to pull all your hair out, but fret not, all is not lost. In fact, some people may have run into a similar issue and found a solution. The first place to find these people is on the FEBio user's forum (<https://forums.febio.org/>). This forum contains hundreds of posts by FEBio users and is monitored by the FEBio developers who will always be more than happy to help you with your problems. In addition, this forum can be used to report bugs or to request a new feature. It is the ideal portal to find help with your problems so don't hesitate to use it!

## Chapter 9

# Configuration File

### 9.1 Overview

FEBio requires a configuration file to run. The purpose of this file is to store platform-specific settings such as the default linear solver. See Section [2.5](#) for more information on how to use this file. This section details the format of this file.

The configuration file uses an xml format. The root element must be *febio\_config*. The required attribute *version* specifies the version number of the format. Currently this value must be set to “3.0”. The following elements are defined.

| Parameter                 | Description                                                                             |
|---------------------------|-----------------------------------------------------------------------------------------|
| default_linear_solver     | Set the default linear solver for the platform (1)                                      |
| import                    | Load a plugin file (2)                                                                  |
| set                       | define an alias (3)                                                                     |
| if_debug                  | Process child tags only for debug versions of FEBio (4)                                 |
| if_release                | Process child tags only for release versions of FEBio (4)                               |
| output_negative_jacobians | Output a detailed list of inverted elements when negative jacobians are encountered (5) |
| print_model_params        | Print the values of load controlled parameters at the start of each time step (6)       |
| show_warnings_and_errors  | Print warnings and errors to the screen (7)                                             |

*Comments:*

1. FEBio supports several linear solvers, such as Pardiso and Skyline. Not all solvers are available for all platforms. Only the Skyline solver is available for all platforms. As of version 1.2, the Pardiso solver is available on all platforms for which FEBio is distributed. See section [9.2](#) on how to configure the linear solvers.
2. As of FEBio 2.0 the user can create and use plugins designed for FEBio. These plugins extend the standard capabilities without the need to recompile the FEBio code. See Chapter [10](#) for more information on using plugins in FEBio.
3. As of FEBio 2.5 the configuration file supports aliases which can be used to define plugin paths. The *set* tag defines an alias. The name of the alias is specified via the *name* attribute. Aliases are accessed using the  $\$(name)$  syntax. See below for an example.

4. The *if\_debug/if\_release* tags allow users to customize settings based on whether a debug or release version of FEBio is processing the configuration file. (Note that the debug version refers to how FEBio was compiled. It does not refer to the -g command line option.) The versions of FEBio distributed through the website are release versions. A debug version of FEBio can only be obtained by building FEBio with the `_DEBUG` preprocessor command.
5. When elements become inverted, FEBio prints an error message that “negative jacobians” are encountered. If you wish to see a detailed list of the inverted elements, you can set the *output\_negative\_jacobians* flag to 1.
6. The *print\_model\_params* parameter can be used to print values of load-controlled parameters to the screen and log file. If disabled, the values are not printed to the screen or log file. For models that use many load-controlled parameters this can be useful to reduce the amount of output generated by FEBio. By default, this parameter is on.
7. The *show\_warnings\_and\_errors* parameter can be used to decide whether or not to print warning and error messages. The default value is on.

An example configuration file that sets the default linear solver to pardiso, defines an alias, and loads a plugin.

```
<?xml version="1.0" encoding="ISO-8859-1"?>
<febio_config version="3.0">
  <default_linear_solver type="pardiso"/>
  <set name="PluginsDir">C:\path\to\plugins</set>
  <import>$(PluginsDir)\myplugin.dll</import>
</febio_config>
```

## 9.2 Configuring Linear Solvers

FEBio offers several direct and iterative linear solvers that can be used to solve the linear system of equations of the FE model. Users can configure these linear solvers in the configuration file with the *default\_linear\_solver* tag. The particular linear solver is selected via the *type* attribute. The following table lists the possible values, the solver type, and the supported matrix format.

| <b>solver</b>           | <b>description</b>                                       | <b>type</b> | <b>matrix format</b> |
|-------------------------|----------------------------------------------------------|-------------|----------------------|
| <code>pardiso</code>    | A sparse direct solver, from the MKL library. (default)  | D           | S/U                  |
| <code>skyline</code>    | A sparse direct solver, but not as efficient as pardiso. | D           | S                    |
| <code>mkl_dss</code>    | The DSS solver from the MKL library.                     | D           | S/U                  |
| <code>fgmres</code>     | An implementation of the GMRES algorithm, from MKL .     | It          | U                    |
| <code>cg</code>         | An implementation of the CG algorithm, from MKL .        | It          | S                    |
| <code>boomeramg</code>  | An algebraic multigrid solver, from the HYPRE library.   | It          | S/U                  |
| <code>schur</code>      | A block based iterative solver.                          | It          | B                    |
| <code>accelerate</code> | Apple-specific sparse solver                             | D/It        | S/U                  |

D=direct solver, It=iterative solver, S = symmetric format, U = unsymmetric format, B = block format.

Iterative solvers can be configured with preconditioners, which attempt to reduce the condition number of the matrix. Preconditioning is highly recommended and often necessary for reducing the number of iterations needed by the iterative solver to converge the solution. Any of the linear solvers listed above can be used as a preconditioner. In addition, the following specialized preconditioners are also available.

| preconditioner | description                                                  | matrix format |
|----------------|--------------------------------------------------------------|---------------|
| ilu0           | Incomplete LU decomposition with no fill-in                  | unsymmetric   |
| ilut           | Incomplete LU decomposition with custom fill-in              | unsymmetric   |
| ichol          | Incompletely Cholesky decomposition with no fill-in          | symmetric     |
| diagonal       | Diagonal preconditioner made from the diagonal of the matrix | any           |

The following sections describe each of these solvers and preconditioners in more detail.

### 9.2.1 Pardiso

The Pardiso solver is an efficient sparse direct linear solver and is the default linear solver. FEBio uses the implementation from the MKL library. It does not require any configuration parameters. It can take symmetric, unsymmetric, and structurally symmetric sparse matrix formats.

*Example:*

```
<default_linear_solver type="pardiso"/>
```

### 9.2.2 Skyline

The Skyline solver was initially added for users who build their own copy of FEBio and did not have access to third party linear solvers. It is a sparse direct solver, but not as efficient as Pardiso and not recommended for large problems. When using this solver, it is highly recommended to set the `optimize_bw` flag in the model file's Control section, which attempts to minimize the matrix bandwidth. It does not require any configuration parameters and only works with symmetric matrices.

### 9.2.3 MKL\_DSS

The MKL library also offers another direct solver, called DSS (Direct-Sparse-Solver). This solver is very promising and appears to perform better for many problems than the pardiso solver. Currently, in FEBio4, pardiso remains the default mostly for backward compatibility reasons, but it is often worthwhile to see if the `mkl_dss` solver performs better.

### 9.2.4 FGMRES

This iterative linear solver, from the MKL library, implements the GMRES algorithm, which is an efficient Krylov-based iterative solution strategy. It requires several configuration parameters, listed below, and only works with unsymmetric matrices. In the table below, **N** refers to the number of equations.

| parameter      | description                                                  | default    |
|----------------|--------------------------------------------------------------|------------|
| max_iter       | The maximum number of iterations                             | min(N,150) |
| tol            | relative residual tolerance                                  | 1e-6       |
| abs_tol        | absolute residual tolerance                                  | 1e-12      |
| print_level    | The amount of information printed (0,1, or 2)                | 0          |
| fail_max_iters | Indicates whether reaching max iters is a failure or not (1) | 1 (true)   |
| pc_left        | The left preconditioner                                      | (none)     |
| pc_right       | The right preconditioner                                     | (none)     |

*Comments:*

1. When using FGMRES as a preconditioner it is recommended to set the fail\_max\_iters flag to zero.

**Example 1.** This example shows how to set up the FGMRES solver with an ILU0 preconditioner.

```
<default_linear_solver type="fgmres">
  <max_iter>100</max_iter>
  <tol>1e-5</tol>
  <pc_left type="ilu0"/>
</default_linear_solver>
```

**Example 2.** This example sets up an FGMRES solver with a Schur preconditioner. For the Schur system we use pardiso for solving the A-block outside of the Schur complement and approximate the A-block inside the Schur complement with its diagonal. This solver requires a block-structured matrix. In order to generate the block structure you need to set the equation\_scheme control parameter to 1 in the model input file.

```
<default_linear_solver type="fgmres">
  <print_level>2</print_level>
  <max_iter>100</max_iter>
  <tol>1e-5</tol>
  <abs_tol>1e-9</abs_tol>
  <pc_left type="schur">
    <print_level>0</print_level>
    <do_jacobi>0</do_jacobi>
    <A_solver type="pardiso"/>
    <schur_pc>0</schur_pc>
    <schur_A_solver type="diagonal"/>
    <schur_solver type="fgmres">
      <print_level>0</print_level>
      <max_iter>100</max_iter>
      <tol>0.05</tol>
      <fail_max_iters>0</fail_max_iters>
    </schur_solver>
  </pc_left>
</default_linear_solver>
```

### 9.2.5 CG

This iterative linear solver, from the MKL library, implements the Conjugate Gradient algorithm, which is an efficient Krylov-based iterative solution strategy for symmetric systems. It requires several configuration parameters, listed below, and only works with symmetric matrices. In the table below, **N** refers to the number of equations.

| parameter   | description                                   | default    |
|-------------|-----------------------------------------------|------------|
| max_iter    | The maximum number of iterations              | min(N,150) |
| tol         | relative residual tolerance                   | 1e-6       |
| print_level | The amount of information printed (0,1, or 2) | 0          |
| pc_left     | The left preconditioner                       | (none)     |

### 9.2.6 BoomerAMG

This solver, from the HYPRE library (<https://hypre.readthedocs.io/en/latest/index.html>), is an adaptive multigrid solver. It often works better as a preconditioner to an iterative solver than as a stand-alone solver.

| parameter        | description                                           | default           |
|------------------|-------------------------------------------------------|-------------------|
| max_iter         | The maximum number of iterations                      | 20                |
| tol              | relative residual tolerance                           | 1e-7              |
| print_level      | The amount of information printed [0,1, or 2]         | 0                 |
| max_levels       | The max number of multigrid levels                    | 25                |
| coarsen_type     | Method of grid coarsening (1)                         | (library default) |
| use_num_funcs    | Use dimensionality of solution variable in coarsening | 0 (false)         |
| relax_type       | relaxation type (2)                                   | 3                 |
| interp_type      | interpolation type (3)                                | 6                 |
| strong_threshold | strong threshold                                      | 0.5               |
| p_max_elmts      | P max elements                                        | 4                 |
| num_sweeps       | Number of sweeps                                      | 1                 |
| agg_num_levels   | Aggressive number of levels                           | 0                 |
| nodal            | Nodal option                                          | 0 (false)         |
| do_jacobi        | Do jacobi preconditioning                             | 0 (false)         |
| fail_max_iters   | Fail on max iterations (4)                            | 1 (true)          |

*Comments:*

1. *coarsen\_type* selects the algorithm for the coarsening phase. The following values can be used.

| value | description                                                                 |
|-------|-----------------------------------------------------------------------------|
| 0     | CLJP-coarsening (a parallel coarsening algorithm using independent sets)    |
| 1     | classical Ruge-Stueben coarsening, no boundary treatment (not recommended!) |
| 3     | classical Ruge-Stueben coarsening, with boundary treatment                  |
| 6     | Falgout coarsening                                                          |
| 7     | CLJP-coarsening (using fix random vector, for debugging purposes only)      |
| 8     | PMIS-coarsening (might lead to slower convergence)                          |
| 9     | PMIS-coarsening (using a fixed random vector ,for debugging purposes only)  |
| 10    | HMIS-coarsening                                                             |
| 11    | one-pass Ruge-Stueben coarsening (not recommended!)                         |
| 21    | CGC coarsening                                                              |
| 22    | CGC-E coarsening                                                            |

2. *relax\_type* selects the algorithm for the relaxation phase. The following values can be used.

| value | description                                                            |
|-------|------------------------------------------------------------------------|
| 0     | Jacobi                                                                 |
| 1     | Gauss-Seidel, sequential (very slow!)                                  |
| 2     | Gauss-Seidel, interior points in parallel, boundary sequential (slow!) |
| 3     | hybrid Gauss-Seidel or SOR, forward solve                              |
| 4     | hybrid Gauss-Seidel or SOR, backward solve                             |
| 5     | hybrid chaotic Gauss-Seidel (works only with OpenMP)                   |
| 6     | hybrid symmetric Gauss-Seidel or SSOR                                  |
| 8     | l1-scaled hybrid symmetric Gauss-Seidel                                |
| 9     | Gaussian elimination (only on coarsest level)                          |
| 15    | CG (warning - not a fixed smoother - may require RGMRES)               |
| 16    | Chebyshev                                                              |
| 17    | FCF-Jacobi                                                             |
| 18    | l1-scaled Jacobi                                                       |

3. *interp\_type* selects the algorithm for the interpolation phase. The following values can be used.

| value | description                                                             |
|-------|-------------------------------------------------------------------------|
| 0     | classical modified interpolation                                        |
| 1     | LS interpolation (for use with GSMG)                                    |
| 2     | classical modified interpolation for hyperbolic PDEs                    |
| 3     | direct interpolation (with separation of weights)                       |
| 4     | multipass interpolation                                                 |
| 5     | multipass interpolation (with separation of weights)                    |
| 6     | extended+i interpolation                                                |
| 7     | extended+i (if no common C neighbor) interpolation                      |
| 8     | standard interpolation                                                  |
| 9     | standard interpolation (with separation of weights)                     |
| 10    | classical block interpolation (for use with nodal systems version only) |
| 11    | like 10, with diagonalized blocks                                       |
| 12    | FF interpolation                                                        |
| 13    | FF1 interpolation                                                       |
| 14    | extended interpolation                                                  |



4. When using this solver as a preconditioner, it is recommended to set the fail\_max\_iters flag to 0.

*Example:*

```
<default_linear_solver type="boomeramg">
  <max_iter>100</max_iter>
  <print_level>3</print_level>
  <tol>1e-05</tol>
  <max_levels>25</max_levels>
  <coarsen_type>-1</coarsen_type>
  <use_num_funcs>1</use_num_funcs>
  <relax_type>5</relax_type>
  <interp_type>6</interp_type>
  <strong_threshold>0.9</strong_threshold>
  <p_max_elmts>4</p_max_elmts>
  <num_sweeps>1</num_sweeps>
  <agg_interp_type>0</agg_interp_type>
  <agg_num_levels>0</agg_num_levels>
  <nodal>0</nodal>
  <do_jacobi>0</do_jacobi>
  <fail_max_iters>0</fail_max_iters>
</default_linear_solver>
```

### 9.2.7 Schur

The Schur solver is a linear solver that takes a 2x2 block structured matrix and solves the linear system by decomposing it in its Schur-complement form. Thus, instead of solving the following system directly,

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} a \\ b \end{bmatrix}$$

it first solves for  $v$ ,

$$Sv = b - CA^{-1}a$$

where  $S = D - CA^{-1}B$  is the Schur complement of  $A$ . Then the Schur solver solves for  $u$ ,

$$u = A^{-1}(a - Bv)$$

In order to use this solver, users need to select how they wish to solve the A-block, and how to solve the Schur complement. For the A-block users can choose any of the linear solvers listed above, either direct or iterative. For the Schur complement users can only choose a Krylov-based iterative solver (FGMRES or CG) because FEBio does not form the Schur complement explicitly.

This solver is often more effective as a preconditioner for FGMRES. In this case, the A-block and Schur complement only need to be solved approximately. See the examples section below for some example configurations. This solver requires a block-structured matrix. In order to generate the block structure you need to set the equation\_scheme control parameter to 1 in the model input file.

| parameter      | description                                               | default            |
|----------------|-----------------------------------------------------------|--------------------|
| print_level    | The amount of information printed (0,1, or 2)             | 0                  |
| schur_block    | Take the Schur complement of A (=0) or D (=1)             | 0                  |
| do_jacobi      | Do Jacobi preconditioning                                 | 0                  |
| zero_D_block   | Is the D-block zero?                                      | 0 (false)          |
| A_solver       | The solver for the A-block (outside the Schur complement) | user specified     |
| schur_solver   | The iterative solver for solving Schur complement         | user specified     |
| schur_A_solver | The solver for the A block in the Schur complement        | (same as A_solver) |

### 9.2.8 Accelerate

The Accelerate sparse solver is only available on Apple computers ([https://developer.apple.com/documentation/accelerate/sparse\\_solvers](https://developer.apple.com/documentation/accelerate/sparse_solvers)). It uses the native Accelerate framework of Mac OS. It may be used as a direct solver or an iterative solver. Users should consult the Apple documentation to determine optimal settings for this solver, based on their application.

| parameter              | description                       | default             |
|------------------------|-----------------------------------|---------------------|
| print_condition_number | Print the matrix condition number | 0 (false)           |
| iterative              | Use iterative solver              | 0 (false)           |
| factorization          | Factorization method (0 to 6)     | U:5=QR, S:4=DLTPP   |
| order_method           | Ordering algorithm (0 to 2)       | U:2=COLAMD, S:0=AMD |
| iterative_method       | Iterative method (0 to 4)         | 4=LSMR              |

The factorization methods include: 0=Cholesky, 1=LDLT, 2=LDLTUnpivoted, 3=LDLTSBK, 4=DLTPP, 5=QR, 6=CholeskyAtA ([https://developer.apple.com/documentation/accelerate/sparsefactorization\\_t](https://developer.apple.com/documentation/accelerate/sparsefactorization_t)). The ordering algorithms include: 0=AMD, 1=Metis, 2=COLAMD ([https://developer.apple.com/documentation/accelerate/sparseorder\\_t](https://developer.apple.com/documentation/accelerate/sparseorder_t)). The iterative methods include: 0=Conjugate-Gradient, 1=GMRES, 2=DQGMRES, 3=FGMRES, 4=LSMR ([https://developer.apple.com/documentation/accelerate/sparse\\_solvers/sparse\\_iterative\\_methods](https://developer.apple.com/documentation/accelerate/sparse_solvers/sparse_iterative_methods)).

### 9.2.9 Pardiso\_64

The pardiso\_64 solver is variant of the pardiso solver that can be used for very large problems since it uses 64 bit integers. Other than that, it is identical to the standard pardiso solver.

*Example:*

```
<default_linear_solver type="pardiso_64"/>
```

# Chapter 10

## FEBio Plugins

As of version 2.0 FEBio supports plugins. Plugins are dynamic libraries that extend the capabilities of FEBio at runtime without the need to recompile the entire source code. This offers the user a powerful mechanism for extending the default feature set of FEBio with little effort. In addition, the development of the plugin is independent of the FEBio source code, which implies that upgrading to a newer version of FEBio will not require a recompilation of the plugin<sup>1</sup>. Using plugins, users can create custom materials, boundary conditions, loads, output variables, etc., and even couple FEBio to other codes.

The *FEBio Developer's Manual* explains in detail how to create these plugins. In this section we outline how to use the plugins with FEBio.

### 10.1 Using Plugins

A plugin is compiled into a dynamic library (dll or dylib) or shared object (so) and contains the compiled code of the library. In order to use this plugin, you must add the path to the plugin file in the FEBio configuration file (see Chapter 9 for a discussion of the configuration file). For each plugin, an *import* item has to be added in this configuration file. For example,

```
<import>myplugin.dll</import>
```

You may need to add the full path to the plugin if FEBio has problems locating the plugin.

```
<import>C:\path\to\my\plugins\myplugin.dll</import>
```

When FEBio starts, it will first load all the plugins that are defined in the configuration file before it reads the input file. This way, sections in the input file that require the plugin's capabilities will already be available for use.

An alternative way for loading a plugin is using the **-import** command line option.

```
>febio -i input.feb -import myplugin.dll
```

In this example, FEBio will look for the file *myplugin.dll* in the current working directory. If the plugin file is located in a different folder than the current working directory, you must prepend the relative or absolute path to the file. Note that if the path contains spaces, you need to wrap it in quotes.

---

<sup>1</sup>When upgrading to a newer version of FEBio, it may be necessary to recompile the plugin if the plugin interface was changed. As long as the plugin remains compatible with the new interface, no code changes are required.

```
>febio -i input.feb -import "C:\path\to\my custom plugin\myplugin.dll"
```

## 10.2 Error Messages

If FEBio cannot find a plugin it will print an error message. There are several reasons why a plugin fails to load. Below is a list of possible error messages and some actions that can be taken to prevent the error.

- *“Failed to load the file”*: Most likely FEBio cannot find the plugin file specified in the configuration file. Make sure that the file name including the path is correct and that there are no spaces at the start or end of the import tag’s value string.
- *“Required plugin function PluginNumClasses not found”*: The plugin is missing a required function that FEBio needs to communicate with the plugin. Notify the developer of the plugin to fix this.
- *“Required plugin function PluginGetFactory not found”*: The plugin is missing a required function that FEBio needs to communicate with the plugin. Notify the developer of the plugin to fix this.
- *“Invalid number of classes returned by PluginNumClasses”*: There was a problem with the plugin’s initialization. Notify the developer of the plugin to fix this.
- *“Required plugin function GetSDKVersion not found”*: The plugin is missing a required function that FEBio needs to communicate with the plugin. Notify the developer of the plugin to fix this.
- *“Invalid SDK version”*: The plugin was compiled with a version of the SDK that is not compatible with the current FEBio version. This most likely means that the plugin must be rebuilt with a newer version of the SDK.

## Appendix A

# Heterogeneous model parameters

Many model parameters can be made dependent on the reference coordinates. This is useful, for instance, for defining inhomogeneous materials, where some material parameters depend on the spatial position.

There are two mechanisms for creating heterogeneous parameters: a mathematical expression, or a map. The type of the parameter is set via the *type* attribute, which can take on two values.

| type  | description                                           |
|-------|-------------------------------------------------------|
| math  | Define a math parameter via a mathematical expression |
| map   | Define a mapped parameter                             |
| const | Define a constant parameter (default)                 |

The two types of parameters are described in more detail in the sections below.

The *type* attribute can be omitted. In that case, it is assumed to be a const parameter, unless the tag's value is not a number. In that case, it is assumed to be a math parameter.

FEBio performs parameter validation at the start of an analysis where the values of parameters are checked against the valid ranges as defined in the code. However, it is important to keep in mind that this validation is only performed on constant parameters. For math and map parameters, it is up to the user to ensure that all values over the relevant mesh partition are valid.

### A.1 Math parameters

For math parameters, add the `type="math"` attribute to the parameter's definition. The value of the parameter tag can then be any mathematical expression. The upper case letters X, Y, Z, are used to denote material coordinates. The letter *t* denotes time. The algebraic operators +, -, \*, /, ^ are supported, as well as the list of functions listed in Appendix C.

```
<material id="1" name="my_material" type="neo-Hookean">
  <E type="math">X+Y+Z</E>
  <v>0.3</v>
</material>
```

You can also use the component's other parameters inside mathematical expressions.

```
<material id="1" name="my_material" type="neo-Hookean">
  <density>1.0</density>
  <E type="math">2*density+0.1</E>
  <v>0.3</v>
</material>
```

The names of maps are also allowed in mathematical expressions. For instance, assume that a map, named “E\_map” is defined in the MeshData section. Then the following is a valid mathematical expression.

```
<E type="math">2*E_map+X</E>
```

## A.2 Mapped parameters

For mapped parameters, use the attribute type=“map”. In this case, the value of the parameter is the name of the map that is defined in the MeshData section.

```
<material id="1" name="my_material" type="neo-Hookean">
  <E type="map">map_E</E>
  <v>0.3</v>
</material>
```

The map is defined in the MeshData section.

```
<ElementData name="map_E" elem_set="Part1">
  <elem lid="1">1.23</elem>
  <elem lid="2">4.56</elem>
</ElementData>
```

Note that it is also possible to use maps in mathematical expressions.

```
<material id="1" name="my_material" type="neo-Hookean">
  <E type="math">5.0*map_E</E>
  <v>0.3</v>
</material>
```

## Appendix B

# Referencing Parameters

This appendix details how that model parameters can be referenced in the FEBio input files. This is used to

- write model parameters to the output file.
- reference model parameters in the parameter optimization input file.

The general format follows the following rule:

```
fem.property.property...property.parameter.component
```

Properties and parameters are referenced by tags separated by a periods. The first tag must be the *fem* tag, which serves as the name of the model. (In future version this name can be user defined, but for now is fixed). Next, the name of a model property must follow. The following properties are currently defined:

- **material**: references a material, defined in the *Material* section.
- **bc\_fixed**: references a fixed bc, defined in the *Boundary* section.
- **bc\_prescribed**: references a prescribed bc, defined in the *Boundary* section.
- **nodal\_load**: references a nodal load, defined in the *Loads* section.
- **surface\_load**: references a surface load, defined in the *Loads* section.
- **body\_load**: references a body load, define in the *Loads* section.
- **rigid\_body**: references a rigid body. Rigid bodies are defined implicitly via rigid materials.

All of these properties define arrays. A specific property can be referenced via square brackets. For instance, to reference the first material in the model, do this

```
fem.material[0]
```

Alternatively, a property can also be reference by name or by ID. In either case, use round brackets. For instance, to reference by ID, do

```
fem.material(1)
```

The ID refers to the “id” attribute that can be specified for most model components. To reference by name,

```
fem.material('MyMaterial')
```

Note the single quotes around the material name. The name is defined via the “name” attribute.

Note that rigid bodies can only be referenced via their material name.

```
fem.rigidbody('MyRigidMaterial')
```

Each property can have a list of parameters and sub-properties. To reference a parameter, just end the reference string with the name of the parameter. For example,

```
fem.material[0].E
```

Sub-properties are referenced similarly. For example, assume that the first material is a biphasic material that defines the solid property.

```
fem.material[0].solid.E
```

If the parameter itself is an array, a specific item in the array is referenced using square brackets. For example, for an EFD-neo Hookean material,

```
fem.material[0].ksi[0]
```

In cases where the model parameter does not define a scalar it is possible to reference the components of the parameter. For instance, a vec3 parameter defines an x, y, and z component.

```
fem.body_load[0].force.x
```

Nodal positions can also be referenced using the *mesh* property.

```
fem.mesh.node[0].position.x
```

In certain applications (e.g. optimization), it is also possible to reference output data, i.e. data that is normally used in the logfile section of the input file. This data is considered read-only. For example, the following references the 'sx' (xx-stress) of element with id 1.

```
fem.element_data('sx', 1)
```

Another example evaluates the enclosed volume of a surface (assuming the surface is closed).

```
fem.surface_data('volume', 'SomeSurface1')
```



## Appendix C

# Math Expression

Many model parameters can be defined via mathematical expression. This appendix documents the syntax of this capability.

### C.1 Functions

The following functions are currently supported.

**sin(x)** The sine function.

**cos(x)** The cosine function

**tan(x)** The tangent function

**csc(x)** The cosecans function

**sec(x)** The secans function

**cot(x)** The cotangent function

**acos(x)** The arccosine function

**asin(x)** The arcsine function

**cosh(x)** The hyperbolic cosine function

**sinh(x)** The hyperbolic sine function

**tanh(x)** The hyperbolic tangent function

**ln(x)** The natural logarithm

**log(x)** The logarithm with base 10

**exp(x)** The exponential function

**sqrt(x)** The square root function

**abs(x)** The absolute value function

**sgn(x)** The sign function

**H(x)** The Heaviside function ( $x = 0.5$  at zero)

**step(x)** The right-continuous unit step function

**J0(x)** Bessel function of the first kind, order 0

**J1(x)** Bessel function of the first kind, order 1

**Jn(x,n)** Bessel function of the first kind, order  $n$

**Y0(x)** Bessel function of the second kind, order 0

**Y1(x)** Bessel function of the second kind, order 1

**Yn(x,n)** Bessel function of the second kind, order  $n$

**sinc(x)** The *sinc* function (i.e.  $\sin(x)/x$ )

**fac(n)** The factorial of  $n$

**erf(x)** The *error* function

**erfc(x)** The complementary error function ( $\text{erfc}(x) = 1 - \text{erf}(x)$ )

**tgamma(x)** The *Gamma* function

**atan2(x,y)** The two-parameter arctangent function

**pow(x,y)** The  $y$ -th power of  $x$

**Tn(n,x)** The Chebyshev polynomial of order  $n$

**binomial(n,m)** The binomial of  $(n,m)$

**max(x1,...,xn)** Returns the maximum value of the values listed

**min(x1,...,xn)** Returns the minimum value of the values listed

**avg(x1,...,xn)** Returns the average value of the values listed

## C.2 Constants

The following predefined constants are supported:

**pi** The value of  $\pi$  ( $= 3.1415926535897932385$ )

**e** The value of  $\exp(1)$  ( $= 2.7182818284590452354$ )

**gamma** The Euler-Mascheroni constant ( $= 0.57721566490153286061$ )

**\_inf\_** A very large number ( $= 1\text{e}308$ )

## Appendix D

# FEBio Binary Database Specification

### D.1 Introduction

This appendix describes the structure of the FEBio binary database (more commonly referred to as the FEBio plot file), which stores the results of a FEBio analysis. The FEBio binary database format is both self-describing and extendable. A user can understand the contents of the file by simply parsing the block structure of the file. The file format is made extensible by providing an abstract layer between the data and the file system. This layer defines the file structure as a hierarchy of data blocks. The result is that each file is now structured similarly as a file folder. Each block can be viewed as a file in the folder and each branch in the hierarchy as a sub-folder.

The database consists of four parts:

1. the **header** contains some general info that may be useful for parsing the rest of the file.
2. the **dictionary** presents a textual description for each data field in the file. This can be used to identify the contents of each data field.
3. the **mesh section** defines the mesh of the model.
4. the **state sections** contain the actual data or results for the field variables.

The following sections describe the details of the database format. In the next section, the block-structure of the file is explained. The sections thereafter describe the different parts of the database.

#### D.1.1 Changes in this version

This appendix describes version 3.0 of the binary database format. This version of the FEBio Binary Database Specification differs in a few important aspects from the previous version. It addresses the following issues:

- Reduce plot file size by describing rigid bodies via rigid body mechanics.
- Reduce plot file size by identifying non-mutable data, which is data that does not change in different states.
- Add support for more generic “objects” that may not be represented via the mesh. In FE simulations, additional structures can be present that interact with the FE parts, but are themselves not part of the FE mesh. Examples are rigid bodies and the various mechanisms to connect rigid bodies (e.g. springs, joints, etc.).

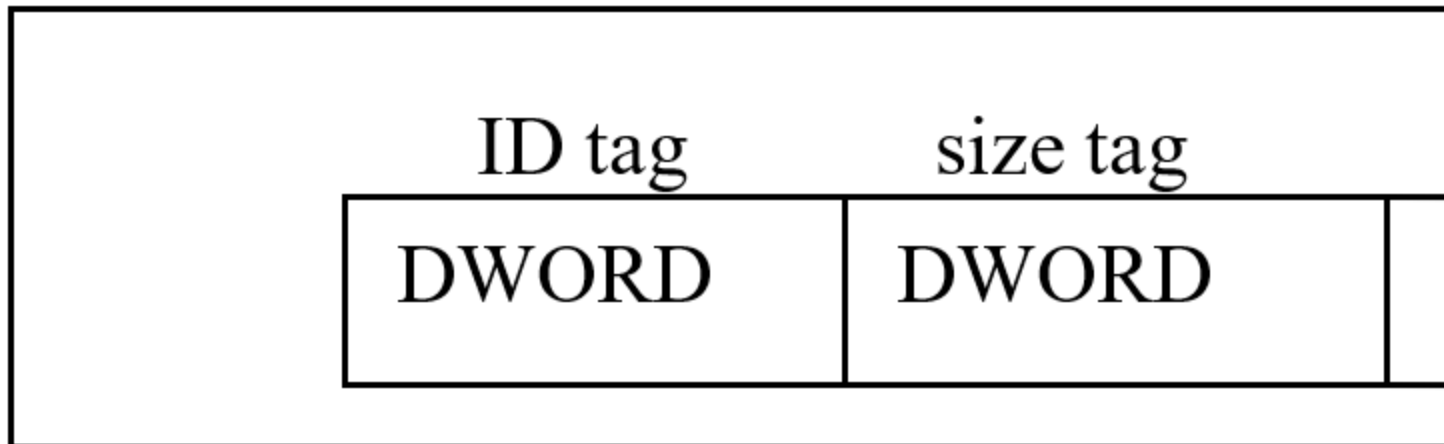


Figure D.2.1: Each block in the plot file consists of three fields: a DWORD with an identifier, a DWORD containing the size of the block and finally the data of the block, which can be child blocks.

- The State section was modified so that the position of non-rigid nodes is redefined in each section.

## D.2 Block Structure

### D.2.1 Overview

The FEBio binary database uses an abstract layer to communicate with the file system. This achieves two goals. The first is that the content of a file becomes independent of the file system that wrote the file. Big endian systems can read files created with small endian systems and vice versa. The second goal is that the data is now stored in a hierarchical structure which is easy to search and modify. This means that future additions and changes can be made fairly easily without losing backward compatibility. In addition, the self-describing feature of the format even allows for some forward compatibility.

As mentioned above, the file is structured as a hierarchy of blocks. Each block consists of three fields: a DWORD identifier, followed by a DWORD containing the size of the data chunk in bytes and then the actual data. (A DWORD is a “double word” meaning an unsigned integer of 4 bytes.)

This data may be either numeric data (such as the data of a field variable) or child blocks. If the block has children, it is referred to as a *branch*. If the block only contains numeric data, it is referred to as a *leaf*.

### D.2.2 Parsing the FEBio plot file

Although the FEBio plot file in essence is a hierarchy of blocks as described above, there are a few more caveats that are important for parsing the FEBio file.

The first DWORD of the file is a tag that identifies the FEBio plot file.

Parsers should read this number and use it to identify whether the file is indeed a proper FEBio plot file. If the value differs, then that means that the file is either not a valid plot file, or that the

| Tag   | ID         | description          |
|-------|------------|----------------------|
| FEBio | 0x00464542 | FEBio identifier tag |

Table D.1: The first tag of the plot file is an identifier that can be used to see if the file is indeed an febio plot file.

| TAG   | ID         | description                                          |
|-------|------------|------------------------------------------------------|
| ROOT  | 0x01000000 | Root block of FEBio plot file                        |
| MESH  | 0x01040000 | Mesh block containing the mesh definition            |
| STATE | 0x02000000 | State block containing results of a single time step |

Table D.2: The overall structure of the plot file consists of a root section, a mesh section, and multiple state sections.

endianness of the system that wrote the file is different than that of the system that is reading the file. In the latter case, a byte swap will be necessary when reading data from the file.

**NOTE: Note that the FEBio tag is not followed by a size tag. The reason is that when FEBio is writing the file it does not know the length of the final file yet.**

After the FEBio tag, the content of the file follows, organized in the hierarchical block structure described above. At the highest level, the plot file has a single root block, followed by a mesh block and a series of state blocks.

After the root block, the mesh section follows which defines the mesh of the model. Then, at least one state blocks follow. There will be one state block for each time step in the FEBio plot file. Note that these blocks are not child blocks of the root block. In the following figure, the high-level structure of the FEBio plot file is depicted for a file that has two state blocks.

As explained above, the first DWORD will be the FEBio identifier. The first block in the file will always be the ROOT block, which will contain the header and dictionary. Then, the mesh section follows, which defines the mesh. After that, the state blocks follow, which will contain the actual results. Each block has three fields. The first field is a DWORD, identifying the section. For the first block, this will be ROOT, for the second block this will be MESH, and for the other blocks this will be STATE. The next DWORD is the size of the entire block. Finally, the actual data will follow, which for the ROOT, MESH and STATE blocks will contain child blocks.

## D.3 Root Section

The ROOT section is the first block in the file. This block contains the *header* and *dictionary* sections as child blocks.

These sections will be detailed below.

| Tag        | ID         | description                     |
|------------|------------|---------------------------------|
| HEADER     | 0x01010000 | contains the header section     |
| DICTIONARY | 0x01020000 | contains the dictionary section |

Table D.3: The Root section has two child sections, the *header* and the *dictionary* sections.

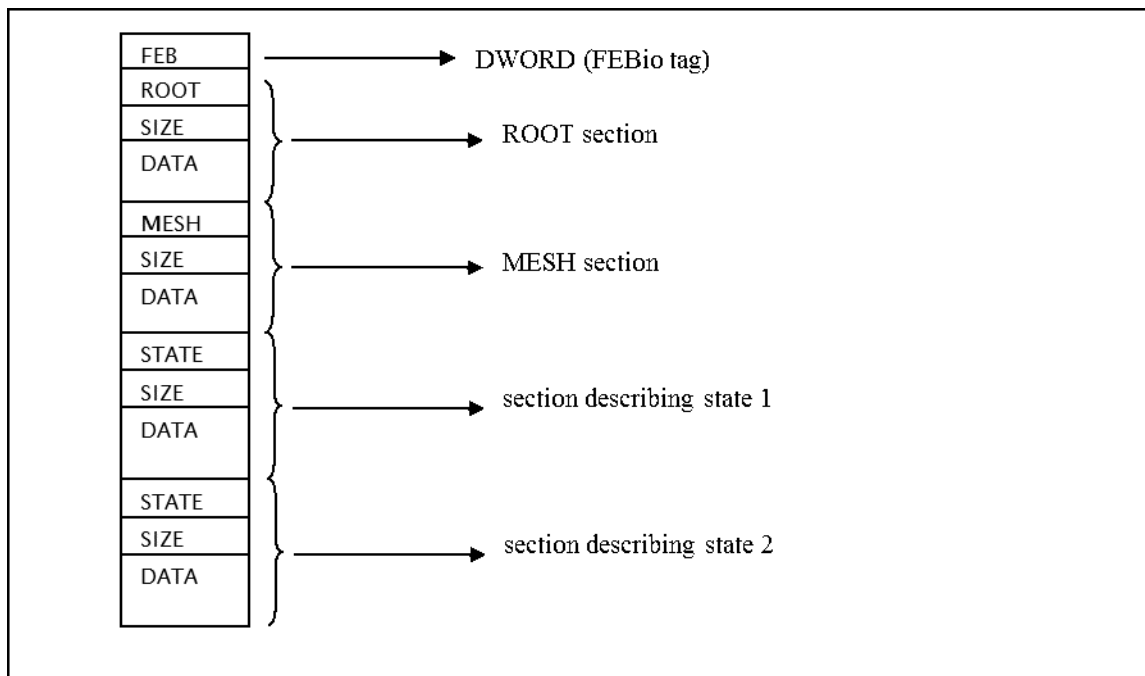


Figure D.2.2: Example structure for a plot file containing a root section and two data sections. Each section is composed of three fields. A DWORD containing the section ID, followed by a DWORD containing the size of the section block and finally the actual data (which may be child blocks).

| Tag         | ID         | description                       | size   |
|-------------|------------|-----------------------------------|--------|
| VERSION     | 0x01010001 | version of the file format        | DWORD  |
| COMPRESSION | 0x01010004 | Compression flag                  | DWORD  |
| AUTHOR      | 0x01010005 | Author name                       | CHAR64 |
| SOFTWARE    | 0x01010006 | Software that generated this file | CHAR64 |

Table D.4: The Header Section

### D.3.1 Header Section

The first section of the root block is the header section. It stores the following data:

Currently, the VERSION will always be 0x0008. The COMPRESSION flag indicates whether the mesh and state sections are compressed or not. Data compression is an optional feature for plot files, and will not be discussed in this document. The AUTHOR contains the name of the person who created the file. The SOFTWARE tag contains the name of the software that generated the file. (Note that not all header data might be present in a plot file. For instance, as of FEBio3, the AUTHOR tag is not written to the plot file.)

### D.3.2 Dictionary Section

When running a FEBio analysis, FEBio will store the values of certain user-selected variables (e.g. stress, temperature, fluid pressure, etc.) to the plot file. In order for a parser to know which variables were written, it needs to read the dictionary section. This section stores a list of

| Tag          | ID         | description                              |
|--------------|------------|------------------------------------------|
| GLOBAL_DATA  | 0x01021000 | lists global data in model               |
| NODESET_DATA | 0x01023000 | lists data associated with the node sets |
| DOMAIN_DATA  | 0x01024000 | lists data associated with domains       |
| SURFACE_DATA | 0x01025000 | lists data associated with surfaces      |

Table D.5: The Dictionary Section

| Tag             | ID         | description                  |
|-----------------|------------|------------------------------|
| DICTIONARY_ITEM | 0x01020001 | beginning of dictionary item |

Table D.6: The dictionary consists of *dictionary items*.

variables, including their names, which are stored in the plot file. Specifically, three attributes are stored for each data variable: a name that provides a description of the data, the data type (scalar, vector, tensor) and the storage format. In addition, the variables are grouped by category. Data variables can be defined for node sets, domains (element sets) and surfaces. In addition, global variables (which are not associated with any part of the model) can also be defined. Each of these categories has their own sub-section in the dictionary.

Note that all of these sections are optional. For example, if a model does not define global data, that section will not be part of the plot file.

Each of these sub-sections defines a list of dictionary items defined by the following tag.

Each dictionary item contains three fields.

The ITEM\_ARRAY\_SIZE and ITEM\_ARRAY\_NAME are only used for array variables. The ITEM\_ARRAY\_SIZE defines the number of data items in the array. For each data item, the ITEM\_ARRAY\_NAME defines an optional name for that item.

The type of the data can be any of the following values.

As explained below, data will be stored for different regions of the mesh, where a region can be a node set, a surface, or a domain (element set). A region is composed of items where an item is a single shape in the region. For node sets, an item will refer to a node, for surfaces this will be a facet and for domains an item will refer to an element. The storage format defines how many values are written for each region and how the values relate to the geometry of the region or items. The following values are currently defined.

The easiest format to understand is the ITEM format which simply stores one value for each item of the region. Thus, one value for each node or for each facet or for each element, depending on the region type. The MULT format defines a value for each node of each item. For example, for a surface of quads, the data will contain four values for each facet, one for each of the four nodes. The NODE format stores a single value for each node of the region. Each type of region (node

| Tag             | ID         | description                 | size   |
|-----------------|------------|-----------------------------|--------|
| ITEM_TYPE       | 0x01020002 | type of data                | DWORD  |
| ITEM_FORMAT     | 0x01020003 | storage format of data      | DWORD  |
| ITEM_NAME       | 0x01020004 | textual description of data | CHAR64 |
| ITEM_ARRAY_SIZE | 0x01020005 | Size of array variables     | DWORD  |
| ITEM_ARRAY_NAME | 0x01020006 | Name of array variable      | CHAR64 |

Table D.7: Structure of dictionary item.

| ITEM_TYPE   | VALUE | description                                                                        |
|-------------|-------|------------------------------------------------------------------------------------|
| FLOAT       | 0     | single precision (s.p.) floating point                                             |
| VEC3F       | 1     | 3D vector of s.p. floats (stored in x, y, z order)                                 |
| MAT3FS      | 2     | symmetric 2nd order tensor of s.p. floats (stored in xx, yy, zz, xy, yz, xz order) |
| MAT3FD      | 3     | diagonal matrix of s.p. floats (stored in xx, yy, zz order).                       |
| TENS4FS     | 4     | symmetric fourth-order tensor of s.p. floats.                                      |
| MAT3F       | 5     | 3x3 matrix of floats (stored in row order).                                        |
| ARRAY_FLOAT | 6     | Array of floats.                                                                   |
| ARRAY_VEC3F | 7     | Array of vec3f.                                                                    |

Table D.8: The supported data types.

| ITEM_FORMAT | VALUE | description                           |
|-------------|-------|---------------------------------------|
| NODE        | 0     | one value for each node of the region |
| ITEM        | 1     | one value for each item               |
| MULT        | 2     | one value for each node for each item |

Table D.9: The different storage format identifiers.

set, surface, or domain) implicitly also defines a set of nodes, namely all the nodes that are part of that region. With the NODE format, the user defines a single value for each of the nodes in this implicit node set. This will be explained in more detail below in the state section.

Note that the storage formats are only important for surfaces and domains. For other categories (e.g. node sets), all formats are essentially equivalent, and can safely be ignored.

## D.4 Mesh Section

The Mesh section defines the mesh of the model and its decomposition into separate regions. A region can be a node set, a surface (i.e. facet set), or a domain (element set). The mesh section is defined by the following sub-sections.

### D.4.1 Node Section

The NODE section defines a set of nodes of the mesh. Each node section starts with a NODE\_HEADER followed by a NODE\_COORDS section.

The header contains the following information.

| Tag             | ID         | description                   |
|-----------------|------------|-------------------------------|
| NODE_SECTION    | 0x01041000 | beginning of node section     |
| DOMAIN_SECTION  | 0x01042000 | beginning of domain section   |
| SURFACE_SECTION | 0x01043000 | beginning of surface section  |
| NODESET_SECTION | 0x01044000 | beginning of node set section |
| PARTS_SECTION   | 0x01045000 | beginning of parts section    |

Table D.10: The child sections of the *Mesh* section.



| Tag         | ID         | description    |
|-------------|------------|----------------|
| NODE_HEADER | 0x01041100 | node header    |
| NODE_COORDS | 0x01041200 | node data list |

Table D.11: The Node section consists of a header and a list of coordinates.

| Tag   | ID         | description                                |
|-------|------------|--------------------------------------------|
| NODES | 0x01041101 | number of nodes (N) in this NODE section   |
| DIM   | 0x01041102 | dimension of node set (i.e. nr. of coords) |
| NAME  | 0x01041103 | name of node set.                          |

Table D.12: The header of the Node section.

The NODES parameter defines the number of nodes defined in this section. The DIM parameter sets the number of coordinates that will be defined for each node (e.g. 3 for 3D problems). Each NODE section also implicitly defines a node set. The NAME attribute defines the name of this node set.

Then the NODE\_COORDS section follows, which defines the nodal data for each node. This data field stores the nodal IDs and coordinates for all the nodes in the mesh. The size of the field is defined by  $N \cdot \text{DWORD} + \text{DIM} \cdot N \cdot \text{FLOAT}$  where  $N$  is the number of nodes in the mesh (as defined in the header section), DIM is the dimension, and FLOAT is the size of a single precision floating point number (4 bytes). The order of the data is (e.g. if DIM equals 3),

|     |    |    |    |     |     |    |    |    |
|-----|----|----|----|-----|-----|----|----|----|
| ID1 | x1 | y1 | z1 | ... | IDn | xn | yn | zn |
|-----|----|----|----|-----|-----|----|----|----|

Here,  $x[i]$  is the x-coordinate of node  $i$  and similarly for  $y$  and  $z$ .

### D.4.2 Domain Section

The Domain section lists all domains (i.e. element sets) in the mesh. Each domain is identified by a DOMAIN section.

Each domain is defined by two sub-sections, a *domain header* and an *element list*.

The *domain header* contains the following data fields.

The ELEM\_TYPE defines the type of elements stored in the domain. It can have one of the following values.

The PART\_ID corresponds to the ID of one of the parts defined in the Parts section.

The ELEMENTS field is the number of elements in the domain.

After the domain header the element list follows. For each element it defines an ELEMENT data field.

If NE is the number of nodes per element, then this data field stores  $NE+1$  DWORDS. The first DWORD is the element ID, a unique number that identifies the element. The following NE DWORD's define the element connectivity.

| Tag    | ID         | description                   |
|--------|------------|-------------------------------|
| DOMAIN | 0x01042100 | beginning of a domain section |

Table D.13: The Domain section consists of a list of domains.

| Tag           | ID         | description                  |
|---------------|------------|------------------------------|
| DOMAIN_HEADER | 0x01042101 | beginning of domain header   |
| ELEMENT_LIST  | 0x01042200 | list of element connectivity |

Table D.14: Each domain section consists of a header section and an element list.

| Tag       | ID         | description                      | size   |
|-----------|------------|----------------------------------|--------|
| ELEM_TYPE | 0x01042102 | element type                     | DWORD  |
| PART_ID   | 0x01042103 | part ID                          | DWORD  |
| ELEMENTS  | 0x01032104 | number of elements               | DWORD  |
| NAME      | 0x01032105 | an optional name for this domain | CHAR64 |

Table D.15: The domain header section.

| ELEM_TYPE | VALUE | description                           |
|-----------|-------|---------------------------------------|
| HEX8      | 0     | 8-node hexahedron solid element       |
| PENTA6    | 1     | 6-node pentahedron solid element      |
| TET4      | 2     | 4-node tetrahedron solid element      |
| QUAD4     | 3     | 4-node quadrilateral shell element    |
| TRI3      | 4     | 3-node triangular shell element       |
| TRUSS2    | 5     | 2-node linear truss element           |
| HEX20     | 6     | 20-node quadratic hexahedral element  |
| TET10     | 7     | 10-node quadratic tetrahedral element |
| TET15     | 8     | 15-node quadratic tetrahedral element |
| HEX27     | 9     | 27-node quadratic hexahedral element  |

Table D.16: The supported element types.

| Tag     | ID         | description                 | size         |
|---------|------------|-----------------------------|--------------|
| ELEMENT | 0x01042201 | Element ID and connectivity | DWORD*(NE+1) |

Table D.17: Each element is defined by an ID and a list of nodes that defines the connectivity.

| Tag     | ID         | description                    |
|---------|------------|--------------------------------|
| SURFACE | 0x01043100 | beginning of a surface section |

Table D.18: The Surface section consists of a list of surface definitions.

| Tag            | ID         | description                 |
|----------------|------------|-----------------------------|
| SURFACE_HEADER | 0x01043101 | beginning of surface header |
| FACET_LIST     | 0x01043200 | facet connectivity list     |

Table D.19: Each surface is defined by a header section and a facet list.

### D.4.3 Surface Section

The surface section defines the surfaces of the mesh for which data is stored in the plot file. Each surface begins with a SURFACE section.

The surface section follows a similar structure as the domain section, namely a *surface header* followed by a *facet list*.

The surface header contains the following data fields.

The SURFACE\_ID is a unique identifier and FACETS is the number of facets in the surface.

The FACET\_LIST follows the header and contains a FACET data field for each facet in the surface.

Here,  $NF = MAX\_FACET\_NODES + 2$ . The first DWORD is a unique identifier (which currently should be ignored). The second DWORD is the number of nodes for this facet. This also defines the facet type. (For instance, 3 nodes define a triangle, 4 nodes define a quadrilateral). Next, the node ID's follow.

### D.4.4 Node Set Section

The Node Set section defines all the node sets where a node set is a named collection of nodes. Each nodeset begins with the NODESET section.

Next, a header section and a node list section follow.

The header is composed of the following chunks.

The node list is a list of all the nodes that belong to this node set. All node indices are zero-based.

### D.4.5 Parts Section

The Parts section lists the parts defined in the plot file. For each part, a PART section is written.

Then, for each part the following fields are written.

| Tag             | ID         | description             | size   |
|-----------------|------------|-------------------------|--------|
| SURFACE_ID      | 0x01043102 | surface ID              | DWORD  |
| FACETS          | 0x01043103 | Number of facets        | DWORD  |
| NAME            | 0x01043104 | A name for this surface | CHAR64 |
| MAX_FACET_NODES | 0x01043105 | max nodes per facet     | DWORD  |

Table D.20: The surface header section.

| Tag   | ID         | description      | size     |
|-------|------------|------------------|----------|
| FACET | 0x01043201 | facet definition | DWORD*NF |

Table D.21: A facet definition.

| Tag     | ID         | description                    |
|---------|------------|--------------------------------|
| NODESET | 0x01044100 | beginning of a nodeset section |

Table D.22: The Node Set section consists of a list of NODESET sections.

| Tag            | ID         | description                 |
|----------------|------------|-----------------------------|
| NODESET_HEADER | 0x01044101 | beginning of nodeset header |
| NODE_LIST      | 0x01044200 | node list                   |

Table D.23: Each nodeset consists of a header and a list of node numbers.

| Tag        | ID         | description                       | size   |
|------------|------------|-----------------------------------|--------|
| NODESET_ID | 0x01044102 | nodeset ID                        | DWORD  |
| NODES      | 0x01044104 | Number of nodes                   | DWORD  |
| NAME       | 0x01044103 | An optional name for this nodeset | CHAR64 |

Table D.24: The node set header.

| Tag  | ID         | description                    |
|------|------------|--------------------------------|
| PART | 0x01045100 | beginning of a part definition |

Table D.25: The Part section consists of a list of parts.

| Tag       | ID         | description  | size   |
|-----------|------------|--------------|--------|
| PART_ID   | 0x01045101 | ID of part   | DWORD  |
| PART_NAME | 0x01045102 | Name of part | CHAR64 |

Table D.26: A part definition.

| Tag          | ID         | description          |
|--------------|------------|----------------------|
| STATE_HEADER | 0x02010000 | state header section |
| STATE_DATA   | 0x02020000 | state data section   |

Table D.27: The State section consists of a header and a data section.

| Tag        | ID         | description         | size  |
|------------|------------|---------------------|-------|
| STATE_TIME | 0x02010002 | time stamp of state | FLOAT |

Table D.28: The state header section.

The ID is a unique number that will be used in the domain definitions to refer to this part. The NAME is a textual description that can be used by the post-processor to present the part to the user.

## D.5 State Section

The *state* section is where all the actual data is stored. FEBio will store one state section for each time step in the analysis. Each state defines two sub-sections, namely the *state header* and the *state data*.

### D.5.1 State Header

The STATE\_HEADER contains the following data field.

### D.5.2 State Data Section

The STATE\_DATA section stores the data for this state. It is composed of several sub-sections where each section corresponds to a data category as defined in the dictionary. The following sub-sections can thus be defined.

**Note that FEBio currently doesn't write global data sections.**

Each of these sub-sections follows a similar structure. For each of the variables defined in the dictionary corresponding to the data category, a STATE\_DATA section is defined.

Each state data section has two fields.

The REGION\_ID refers to the ID of the region for which this data variable is defined.

It is important to note that for node sets, a value of zero for the REGION\_ID refers to the "master" node set which is the set containing all the nodes in the model. This node set is defined implicitly and will not be part of the list of node sets.

| Tag          | ID         | description                 |
|--------------|------------|-----------------------------|
| GLOBAL_DATA  | 0x02020100 | define global data section  |
| NODE_DATA    | 0x02020300 | define nodal data section   |
| DOMAIN_DATA  | 0x02020400 | define domain data section  |
| SURFACE_DATA | 0x02020500 | define surface data section |

Table D.29: Each data section is composed of several child sections.

| Tag        | ID         | description            |
|------------|------------|------------------------|
| STATE_DATA | 0x02020001 | defines a data section |

Table D.30: State Data section.

| Tag       | ID         | description  | size  |
|-----------|------------|--------------|-------|
| REGION_ID | 0x02020002 | ID of region | DWORD |
| DATA      | 0x02020003 | actual data  | ?     |

Table D.31: State data sub-sections.

The DATA field contains the actual data. The size of this field is variable and is determined by the data type and storage format as defined in the dictionary as well as the number of items in the corresponding region. For example, for a domain that has NE elements, the size of the DATA field, using a type of FLOAT and a storage format of FMT\_ITEM will be NE\*FLOAT.

When a surface or domain stores its data in the FMT\_NODE format, then the parser needs to figure out how many nodes are implicitly defined by the region and what the node order is. An implicit nodeset can be constructed by simple enumeration: each item of the region lists the nodes its visits and no nodes can be visited more than once. So for example, consider the surface of three faces shown in the figure below.

Each facet has four nodes. The first four nodes of the implicit node set are simply the four nodes of the first element. The second element skips its first node, since it is already visited, add its second node (which becomes node 5) and its third (node 6) and skips its fourth node. Similarly, the third element adds its second (node 7) and third (node 8) and skips its third and fourth. Thus, this surface defines an implicit node set containing 6 nodes in the order as shown in figure 3. Consequently, if this surface stores its data in FMT\_NODE format, then the corresponding data section will contain six values, one for each of the nodes in the implicit node set.

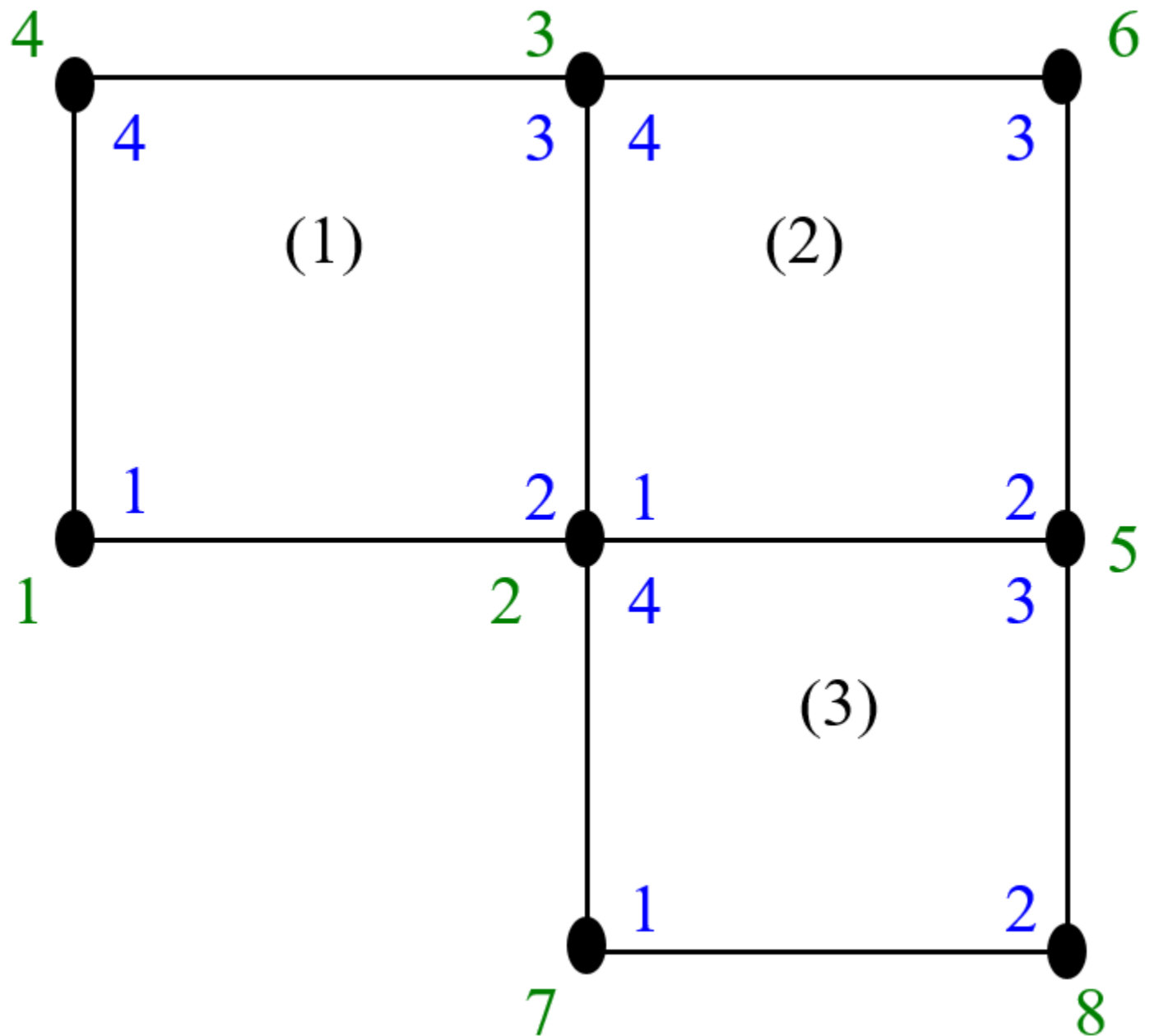


Figure D.5.1: Example illustrating enumeration algorithm that defines the node ordering for the implicit node set defined by a surface.





# Appendix E

## FEBio Output

FEBio supports two main data output mechanisms: the plot file and the log file. The plot file is a binary format that stores the data in format that makes it easy to process with FEBio Studio. The log file uses a text format that is more convenient for processing data further via scripting. See [3.19](#) to learn more about these features. This appendix list the variables that are supported by these two output formats.

### E.1 Plotfile Variables

There are four categories of plot file variables:

- **Nodal:** These typically correspond to data that is directly available at nodes. For instance, the primary solution variables are usually stored as nodal variables. (E.g. “displacement”)
- **Element:** These quantities correspond to data that is typically stored on the elements (e.g. stress). In FEBio, element data is usually stored at the integration points during the solution phase. Since this is often not a convenient format for post-processing, element data is processed before export and will be written to the plot file as either a constant, element-averaged value, or projected to the nodes of the element.
- **Surface:** These quantities correspond to data that is only available at the surface of the mesh (e.g. contact traction). If the surface data is defined at the nodes of the surface, it will be written directly to the plot file. However, if the data is associated with the surface facets - or more accurately, with the integration points of the surface facet - then it will be processed before export, and a constant, facet-averaged value, will be written to the plot file.
- **Rigid body:** These are quantities that correspond to data stored on rigid bodies.

The following table lists the possible values for the *type* attribute of the plot file variable tag.

An asterisk (\*) in the Category column denotes that the attribute is required and must be included in the definition of the plot variable.

The data type column specifies the type of the data. The following table lists the available data types.

| Data Type | Description                                                 |
|-----------|-------------------------------------------------------------|
| SCALAR    | a single scalar value                                       |
| VEC3      | a three-component vector                                    |
| MAT3      | a 3x3 matrix                                                |
| MAT3D     | a diagonal 3x3 matrix (only diagonal components are stored) |
| MAT3S     | a symmetric 3x3 matrix (only 6 components are stored)       |
| TENS4S    | a 4th order tensor with both major and minor symmetries.    |

The following table lists all the available plot variables.

| Type                             | Category | Data Type | Description                                                                                                                                                                                         |
|----------------------------------|----------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| acceleration                     | Element  | VEC3      | Average element accelerations                                                                                                                                                                       |
| Almansi strain                   | Element  | MAT3S     | Almansi strain tensor                                                                                                                                                                               |
| concentration gap                | Surface  | SCALAR    | The concentration gap across a biphasic contact interface.                                                                                                                                          |
| contact area                     | Surface  | SCALAR    | Net contact area across contact interface. Evaluated by integrating the surface area at integration points where the contact pressure is not zero.                                                  |
| contact force                    | Surface  | VEC3      | Net contact force across contact interface. Evaluated by integrating the contact traction over the contact surface.                                                                                 |
| contact gap                      | Surface  | SCALAR    | Contact gap distance, evaluated at contact surface faces                                                                                                                                            |
| contact penalty                  | Surface  | SCALAR    | Contact penalty value                                                                                                                                                                               |
| contact pressure                 | Surface  | SCALAR    | Contact pressure, evaluated at contact surface faces                                                                                                                                                |
| contact status                   | Surface  | SCALAR    | The nr. of integration points of the surface element that are considered in contact.                                                                                                                |
| contact stick                    | Surface  | SCALAR    | Stick status in stick-slip frictional contact (1=stick, 0=slip). Values between 0 and 1 imply that a fraction of that face is in stick mode. Currently works only with sliding-tension-compression. |
| contact traction                 | Surface  | VEC3      | Contact traction vectors, evaluated at contact surface faces                                                                                                                                        |
| current density                  | Element  | SCALAR    | Electric current density, Eq.(4.16.11)                                                                                                                                                              |
| current element angular momentum | Element  | VEC3      | The element's total angular momentum at the current configuration.                                                                                                                                  |
| current element center of mass   | Element  | VEC3      | The element's center of mass at the current configuration.                                                                                                                                          |
| current element kinetic energy   | Element  | SCALAR    | The element's total kinetic energy at the current configuration.                                                                                                                                    |

|                                      |         |        |                                                                                                                                                          |
|--------------------------------------|---------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| current element linear momentum      | Element | VEC3   | The element's total linear momentum at the current configuration                                                                                         |
| current element strain energy        |         | SCALAR | The element's total strain energy at the current configuration.                                                                                          |
| damage                               | Element | SCALAR | Damage variable $D$ (Section 4.6)                                                                                                                        |
| density                              | Element | SCALAR | The current density                                                                                                                                      |
| deviatoric elasticity                | Element | TENS4S | The deviatoric elasticity tensor of an uncoupled material                                                                                                |
| deviatoric strain energy density     | Element | SCALAR | Deviatoric strain energy density $\tilde{\Psi}(\tilde{\mathbf{C}})$ , Eq.(4.1.1)                                                                         |
| deviatoric strong bond SED           | Element | SCALAR | Strain energy density of strong bonds in reactive viscoelastic material                                                                                  |
| deviatoric weak bond SED             | Element | SCALAR | Strain energy density of weak bonds in reactive viscoelastic material                                                                                    |
| dilatation gradient                  | Element | VEC3   | gradient of $J$ in biphasic-FSI                                                                                                                          |
| discrete element direction           | Element | VEC3   | Unit vector along direction of discrete element                                                                                                          |
| discrete element percent elongation  | Element | SCALAR | ratio of change in length over original length                                                                                                           |
| discrete element signed force        | Element | SCALAR | The dot product of the force and the element's direction vector.                                                                                         |
| discrete element strain energy       | Element | SCALAR | strain energy of the discrete element's material                                                                                                         |
| displacement                         | Node    | VEC3   | Nodal displacements                                                                                                                                      |
| effective elasticity                 | Element | TENS4S | The elasticity of the elastic material in a biphasic/triphasic/multiphasic material.                                                                     |
| effective fluid pressure             | Node    | SCALAR | fluid pressure for biphasic, biphasic-solute, triphasic, and multiphasic materials.                                                                      |
| effective shell fluid pressure       | Node    | SCALAR | fluid pressure for biphasic, biphasic-solute, triphasic, and multiphasic materials of shell domains.                                                     |
| effective shell solute concentration | Node    | SCALAR | Effective solute concentration for biphasic-solute, triphasic, and multiphasic materials of shell domains.                                               |
| effective solute concentration       | Node    | SCALAR | Effective solute concentration for biphasic-solute, triphasic, and multiphasic materials.                                                                |
| effective fluid pressure             | Node    | SCALAR | Fluid pressure $p$ for biphasic materials, Eq.(4.14.1), or $\tilde{p}$ for biphasic-solute, Eq.(4.15.3) and triphasic/multiphasic materials, Eq.(4.16.7) |
| effective friction coefficient       | Surface | SCALAR | Effective friction coefficient at a biphasic contact interface                                                                                           |

|                                |            |        |                                                                                                                                         |
|--------------------------------|------------|--------|-----------------------------------------------------------------------------------------------------------------------------------------|
| effective solute concentration | Node       | SCALAR | Effective solute concentration $\tilde{c}$ for biphasic-solute materials, Eq.(4.15.3), and triphasic/multiphasic materials, Eq.(4.16.7) |
| elasticity                     | Element    | TENS4S | Spatial elasticity tensor components                                                                                                    |
| electric potential             | Element    | SCALAR | Electric potential $\psi$ in triphasic/multiphasic materials, Section 4.16                                                              |
| element angular momentum       | Element    | VEC3   | Element total angular momentum                                                                                                          |
| element center of mass         | Element    | VEC3   | Element center of mass                                                                                                                  |
| element kinetic energy         | Element    | SCALAR | Element total kinetic energy                                                                                                            |
| element linear momentum        | Element    | VEC3   | Element total linear momentum                                                                                                           |
| element strain energy          | Element    | SCALAR | Element total strain energy                                                                                                             |
| element stress power           | Element    | SCALAR | Element total stress power                                                                                                              |
| enclosed volume                | Surface*   | SCALAR | Volume enclosed by named surface                                                                                                        |
| enclosed volume change         | Surface*   | SCALAR | Change in volume enclosed by named surface                                                                                              |
| Euler angle                    | Rigid body | VEC3   | Rigid body Euler angles                                                                                                                 |
| facet area                     | Surface    | SCALAR | The area of the surface element in the deformed configuration.                                                                          |
| fatigue bond fraction          | Element    | SCALAR | fatigue bond fraction in a reactive-viscoelastic material                                                                               |
| fiber stretch                  | Element    | SCALAR | Element's average fiber stretch                                                                                                         |
| fiber vector                   | Element    | VEC3   | Material fiber vector                                                                                                                   |
| field                          | Node       |        | Output the nodal values of a field variable.                                                                                            |
| fixed charge density           | Element    | SCALAR | Fixed charge density $c^F$ in current configuration, Eq.(4.16.2)                                                                        |
| fluid acceleration             | Element    | VEC3   | Fluid acceleration (material time derivative of fluid velocity $\mathbf{v}^f$ , Section (4.20))                                         |
| fluid body force               | Element    | VEC3   | The element's average body force applied to the element.                                                                                |
| fluid bulk modulus             | Element    | SCALAR | The element's average bulk modulus, calculate from the pressure.                                                                        |
| fluid density                  | Element    | SCALAR | Fluid density $\rho^f$ in fluid and fluid-FSI materials, Eq.(4.20.4)                                                                    |
| fluid density rate             | Element    | SCALAR | The element's average fluid density rate.                                                                                               |
| fluid dilatation               | Node       | SCALAR | Fluid dilatation $e$ , Section (4.20))                                                                                                  |
| fluid element angular momentum | Element    | VEC3   | Total angular momentum of fluid element                                                                                                 |
| fluid element center of mass   | Element    | VEC3   | Center of mass of fluid element                                                                                                         |
| fluid element kinetic energy   | Element    | SCALAR | Total kinetic energy of fluid element                                                                                                   |
| fluid element linear momentum  | Element    | VEC3   | Total linear momentum of fluid element                                                                                                  |
| fluid element strain energy    | Element    | SCALAR | Total strain energy of fluid element                                                                                                    |
| fluid energy density           | Element    | SCALAR | Average energy density of fluid element                                                                                                 |

|                              |          |        |                                                                                                                                                                                                                                  |
|------------------------------|----------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| fluid flow rate              | Surface* | SCALAR | Volumetric fluid flow rate $Q = \int_A \mathbf{w} \cdot \mathbf{n} dA$ across a named surface (biphasic and multiphasic analyses)                                                                                                |
| fluid flux                   | Element  | VEC3   | Fluid flux $\mathbf{w}$ in biphasic, Eq.(4.14.2), biphasic-solute, Eq.(4.15.4), triphasic/multiphasic, Eq.(4.16.9), fluid-FSI and biphasic-FSI materials.                                                                        |
| fluid force                  | Surface  | VEC3   | Net fluid force across biphasic (sliding-biphasic), biphasic-solute (sliding-biphasic-solute) and multiphasic (sliding-multiphasic) contact interface. Evaluated by integrating the fluid pressure $p$ over the contact surface. |
| fluid force2                 | Surface  | VEC3   | Alternative calculation of the net fluid force                                                                                                                                                                                   |
| fluid heat supply density    | Element  | SCALAR | Average heat supply density of a fluid element                                                                                                                                                                                   |
| fluid kinetic energy density | Element  | SCALAR | Average kinetic energy density of a fluid element                                                                                                                                                                                |
| fluid load support           | Surface  | SCALAR | The total fluid load support across a named surface                                                                                                                                                                              |
| fluid mass flow rate         | Surface* | SCALAR | Mass flow rate across a named surface, $\dot{m} = \int_A \rho \mathbf{v} \cdot \mathbf{n} dA$ , Section (4.20)                                                                                                                   |
| fluid pressure               | Element  | SCALAR | Fluid pressure $p$ in biphasic, Eq.(4.14.1), biphasic-solute, Eq.(4.15.3), triphasic/multiphasic materials, Eq.(4.16.7), and fluid materials, Eq.(4.20.2)                                                                        |
| fluid rate of deformation    | Element  | MAT3S  | Fluid rate of deformation tensor $\mathbf{D}$ , Section (4.20)                                                                                                                                                                   |
| fluid shear viscosity        | Element  | SCALAR | Average shear viscosity of a fluid element                                                                                                                                                                                       |
| fluid strain energy density  | Element  | SCALAR | Average shear energy density of fluid element                                                                                                                                                                                    |
| fluid stress                 | Element  | MAT3S  | Fluid stress $\sigma$ , Eq.(4.20.1)                                                                                                                                                                                              |
| fluid stress power density   | Element  | SCALAR | Fluid stress power $\sigma : \mathbf{D}$ , Section (4.20)                                                                                                                                                                        |
| fluid surface energy flux    | Surface  | VEC3   | The total energy flux across a named surface for a fluid analysis                                                                                                                                                                |
| fluid surface force          | Surface* | VEC3   | Net force exerted by a fluid on a named surface, $\mathbf{f} = - \int_A \mathbf{t} dA$ , Section (4.20)                                                                                                                          |
| fluid surface pressure       | Surface* | SCALAR | Average fluid pressure on each face of a surface, evaluated from the nodal values of the fluid dilatation on that face                                                                                                           |
| fluid surface traction power | Surface  | SCALAR | Net traction power across a named surface of a fluid analysis                                                                                                                                                                    |
| fluid velocity               | Element  | VEC3   | Element fluid velocity $\mathbf{v}^f$ in fluid and fluid-FSI materials, Section (4.20)                                                                                                                                           |
| fluid volume ratio           | Element  | SCALAR | Fluid volume ratio $J$ , Section (4.20)                                                                                                                                                                                          |
| fluid vorticity              | Element  | VEC3   | Fluid vorticity, Section (4.20)                                                                                                                                                                                                  |

|                                          |            |        |                                                                                                                   |
|------------------------------------------|------------|--------|-------------------------------------------------------------------------------------------------------------------|
| growth infinitesimal strain              | Element    | MAT3S  | growth infinitesimal strain                                                                                       |
| growth Lagrange strain                   | Element    | MAT3S  | growth Lagrange strain                                                                                            |
| growth left Hencky                       | Element    | MAT3S  | growth left Hencky                                                                                                |
| growth left stretch                      | Element    | MAT3S  | growth left stretch                                                                                               |
| growth relative volume                   | Element    | MAT3S  | growth relative volume                                                                                            |
| growth right Hencky                      | Element    | MAT3S  | growth right Hencky                                                                                               |
| growth right stretch                     | Element    | MAT3S  | growth right stretch                                                                                              |
| heat flux                                | Element    | VEC3   | The average heat flux inside an element of a heat-transfer analysis                                               |
| ideal gas pressure                       | Surface    | SCALAR | current pressure value of the “ideal gas pressure” load                                                           |
| incremental displacement                 | Node       | VEC3   | The displacement increment from the previously written plot state.                                                |
| intact bond fraction                     | Element    | SCALAR | fraction of intact bonds in a reactive-viscoelastic material                                                      |
| kinetic energy density                   | Element    | SCALAR | The average kinetic energy density in an element                                                                  |
| Lagrange strain                          | Element    | MAT3S  | The average Lagrange strain in an element                                                                         |
| left Hencky                              | Element    | MAT3S  | Left Hencky strain $\mathbf{h} = \frac{1}{2} \ln \mathbf{b}$ , where $\mathbf{b} = \mathbf{F} \cdot \mathbf{F}^T$ |
| left stretch                             | Element    | MAT3S  | Left stretch tensor $\mathbf{V} = \mathbf{b}^{1/2}$                                                               |
| local fluid load support                 | Surface    | SCALAR | Local fluid load support on biphasic contact surfaces                                                             |
| mesh_data                                | (multiple) |        | Stores the value of a mesh_data field to the plotfile. The output class depends on the mesh_data field.           |
| mixture deviatoric strain energy density | Element    | SCALAR | Deviatoric strain energy density in mixture components                                                            |
| mixture specific strain energy           | Element    | SCALAR | Specific strain energy in mixture components                                                                      |
| mixture strain energy density            | Element    | SCALAR | Strain energy density in mixture components                                                                       |
| nested damage                            | Element    | SCALAR | The average damage in an elastic mixture                                                                          |
| nodal acceleration                       | Node       | VEC3   | The acceleration at the nodes                                                                                     |
| nodal contact gap                        | Surface    | SCALAR | Contact gap distance, evaluated at contact surface nodes                                                          |
| nodal contact pressure                   | Surface    | SCALAR | Contact pressure, evaluated at contact surface nodes                                                              |
| nodal contact traction                   | Surface    | VEC3   | Contact traction vectors, evaluated at contact surface nodes                                                      |
| nodal fluid flux                         | Node       | VEC3   | Fluid flux in biphasic, biphasic-solute, triphasic, multiphasic, fluid-FSI and biphasic-FSI materials             |
| nodal fluid velocity                     | Node       | VEC3   | Fluid velocity $\mathbf{v}^f$ in fluid and fluid-FSI materials                                                    |
| nodal shell director                     | Node       | VEC3   | The “director” of a shell node.                                                                                   |
| nodal stress                             | Element    | MAT3S  | The Cauchy stress, projected to the nodes.                                                                        |
| nodal surface traction                   | Surface    | VEC3   | Contact nodal tractions                                                                                           |

|                                  |            |        |                                                                                                                |
|----------------------------------|------------|--------|----------------------------------------------------------------------------------------------------------------|
| nodal vector gap                 | Surface    | SCALAR | Contact gap vector at surface nodes                                                                            |
| nodal velocity                   | Node       | VEC3   | The nodal velocities                                                                                           |
| octahedral plastic strain        | Element    | SCALAR | octahedral plastic strain                                                                                      |
| osmolarity                       | Element    | SCALAR | Element's average osmolarity in biphasic/triphasic/multiphasic analysis                                        |
| parameter                        | Element    |        | Stores heterogeneous material parameter data                                                                   |
| pressure gap                     | Surface    | SCALAR | Pressure gap between opposing surfaces in a biphasic contact analysis                                          |
| reaction forces                  | Node       | VEC3   | The nodal reaction forces.                                                                                     |
| receptor-ligand concentration    | Element    | SCALAR | Receptor-ligand concentration in a biphasic-solute analysis.                                                   |
| referential fixed charge density | Element    | SCALAR | Referential fixed charge density $c_r^F$ , Eq.(4.16.2), which may evolve with chemical reactions, Eq.(4.17.10) |
|                                  | Element    |        | $\varphi_r^s$ , which may evolve with chemical reactions, Eq.(4.17.9)                                          |
| relative fluid velocity          | Element    | VEC3   | Average element fluid velocity relative to mesh $w$ in fluid and fluid-FSI materials                           |
| relative volume                  | Element    | SCALAR | Relative volume $J = \det \mathbf{F}$                                                                          |
| right Hencky                     | Element    | MAT3S  | Right Hencky strain $\mathbf{H} = \frac{1}{2} \ln \mathbf{C}$                                                  |
| right stretch                    | Element    | MAT3S  | Right stretch tensor $\mathbf{U} = \mathbf{C}^{1/2}$                                                           |
| rigid acceleration               | Rigid body | VEC3   | Rigid body center of mass acceleration                                                                         |
| rigid angular acceleration       | Rigid body | VEC3   | Rigid body angular acceleration                                                                                |
| rigid angular momentum           | Rigid body | VEC3   | Rigid body angular momentum                                                                                    |
| rigid angular position           | Rigid body | VEC3   | Rigid body rotation pseudo-vector                                                                              |
| rigid angular velocity           | Rigid body | VEC3   | Rigid body angular velocity                                                                                    |
| rigid force                      | Rigid body | VEC3   | Total force applied to the rigid body                                                                          |
| rigid kinetic energy             | Rigid body | SCALAR | Rigid body kinetic energy                                                                                      |
| rigid linear momentum            | Rigid body | VEC3   | Rigid body linear momentum                                                                                     |
| rigid position                   | Rigid body | VEC3   | Rigid body center of mass position                                                                             |
| rigid rotation vector            | Rigid body | VEC3   | Rigid body rotation pseudo-vector                                                                              |
| rigid torque                     | Rigid body | VEC3   | Rigid body moment                                                                                              |
| rigid velocity                   | Rigid body | VEC3   | Rigid body center of mass velocity                                                                             |
| RVE generations                  | Element    | SCALAR | Number of generations in reactive viscoelastic material                                                        |
| RVE reforming bonds              | Element    | SCALAR | Fraction of weak bonds available to break and reform in reactive viscoelastic material                         |
| RVE strain                       | Element    | SCALAR | Strain measure used in strain-dependent reactive viscoelastic relaxation and weak bond recruitment             |
| sbm concentrationMAT3D           | Element    | SCALAR | Average element solid-bound molecule concentration                                                             |
| sbm referential apparent density | Element    | SCALAR | Average element solid-bound molecule referential apparent density                                              |

|                           |         |        |                                                                                                                           |
|---------------------------|---------|--------|---------------------------------------------------------------------------------------------------------------------------|
| shell bottom nodal strain | Node    | MAT3S  | Shell Lagrange strain extrapolated to bottom nodes                                                                        |
| shell bottom nodal stress | Node    | MAT3S  | Shell Cauchy stress extrapolated to bottom nodes                                                                          |
| shell bottom strain       | Element | MAT3S  | Average of shell Lagrange strain extrapolated to bottom surface                                                           |
| shell bottom stress       | Element | MAT3S  | Average of shell Cauchy stress extrapolated to bottom surface                                                             |
| shell director            | Element | VEC3   | The shell director                                                                                                        |
| shell relative volume     | Element | SCALAR | Relative volume of shell element                                                                                          |
| shell strain              | Element | MAT3S  | Average strain in shell element                                                                                           |
| shell thickness           | Node    | SCALAR | Shell thickness                                                                                                           |
| shell top nodal strain    | Node    | MAT3S  | Shell Lagrange strain extrapolated to top nodes                                                                           |
| shell top nodal stress    | Node    | MAT3S  | Shell Cauchy stress extrapolated to top nodes                                                                             |
| shell top strain          | Element | MAT3S  | Average of shell Lagrange strain extrapolated to top surface                                                              |
| shell top stress          | Element | MAT3S  | Average of shell Cauchy stress extrapolated to top surface                                                                |
| solute concentration      | Element | SCALAR | Solute concentration $c$ in biphasic-solute materials, or $c^\alpha$ in triphasic/multiphasic materials                   |
| solute flux               | Element | VEC3   | Solute flux $j$ in biphasic-solute materials, Eq.(4.15.4), or $j^\alpha$ in triphasic/multiphasic materials, Eq.(4.16.9). |
| specific strain energy    | Element | SCALAR | Average element specific strain energy                                                                                    |
| SPR principal stress      | Element | MAT3D  | Principal stress values obtained via SPR projection                                                                       |
| SPR stress                | Element | MAT3S  | Cauchy stress obtained via SPR projection                                                                                 |
| SPR-P1 stress             | Element | MAT3S  | Cauchy stress obtained via SPR projection (first-order projection)                                                        |
| strain energy density     | Element | SCALAR | Strain energy density $\Psi$ (C)                                                                                          |
| stress                    | Element | MAT3S  | Cauchy stress                                                                                                             |
| surface traction          | Surface | VEC3   | Nodal surface tractions of nodes on a contact surface                                                                     |
| uncoupled pressure        | Element | SCALAR | The pressure of an uncoupled material                                                                                     |
| ut4 nodal stress          | Element | MAT3S  | Stresses at the nodes of the UT4 element                                                                                  |
| vector gap                | Element | VEC3   | The vector gap at nodes of a contact surface                                                                              |
| velocity                  | Node    | VEC3   | Nodal velocities (fluid nodal velocities in fluid analyses)                                                               |
| volume fraction           | Element | SCALAR | Average element volume fraction of a solid mixture                                                                        |
| in-situ target stretch    | Element | SCALAR | the fiber stretch induced by the initial prestrain gradient                                                               |



|                          |         |        |                                                                                  |
|--------------------------|---------|--------|----------------------------------------------------------------------------------|
| PK1 norm                 | Element | SCALAR | The norm of the PK1 stress of a micro material                                   |
| PK1 stress               | Element | MAT3   | The PK1 stress of the element.                                                   |
| prestrain stretch        | Element | SCALAR | the stretch induced by the effective prestrain gradient                          |
| prestrain correction     | Element | SCALAR | the 3x3 prestrain correction tensor                                              |
| SPR infinitesimal strain | Element | MAT3S  | The infinitesimal strain tensor recovered using SPR (1)                          |
| SPR Lagrange strain      | Element | MAT3S  | The Lagrange strain recovered using SPR (1)                                      |
| SPR prestrain correction | Element | MAT3   | The 3x3 prestrain correction tensor (recovered with SPR) (1)                     |
| SPR relative volume      | Element | SCALAR | The relative volume (i.e. $\det \mathbf{F}$ ) recovered using the SPR method (1) |

*Comments:*

1. FEBio calculates element variables at the integration points. However, this is not convenient for output, so by default the integration point values are averaged over the element and a single value per element is stored. In FEBio Studio, this data is further processed and in order to get a unique nodal value, the values of adjacent elements are averaged. For linear elements and a sufficiently fine mesh, this typically produces an accurate and smooth representation of the datafield. However, for higher-order elements, this often produces artifacts, such as “bubbles” that degrade the accuracy. For these types of meshes, it may be better to use one of the other available recovery methods. At this time, FEBio offers two alternative recovery methods that often produce better results with higher-order meshes:
  - (a) SPR method: The SPR (Superconvergent Patch Recovery) method fits at each node a quadratic function to the surrounding integration point values. The value of this quadratic function at the node will be stored to the plotfile.
  - (b) Nodal projection: The nodal projection method extrapolates the integration point values to the nearest nodes and stores those values to the plot file.

## E.2 Logfile Variables

As discussed in [3.19.1](#), there are different category or classes of log variables. The following sections lists the available variables.

### E.2.1 Node\_Data Class

The *node\_data* class defines a set of nodal variables. The data is stored for each node that is listed in the item list of the *node\_data* element or for all nodes if no list is defined. The following nodal variables are defined.

| Node variables | Description                            |
|----------------|----------------------------------------|
| x              | x-coordinate of current nodal position |
| y              | y-coordinate of current nodal position |
| z              | z-coordinate of current nodal position |
| ux             | x-coordinate of nodal displacement     |
| uy             | y-coordinate of nodal displacement     |
| uz             | z-coordinate of nodal displacement     |
| vx             | x-coordinate of nodal velocity         |
| vy             | y-coordinate of nodal velocity         |
| vz             | z-coordinate of nodal velocity         |
| ax             | x-coordinate of nodal acceleration     |
| ay             | y-coordinate of nodal acceleration     |
| az             | z-coordinate of nodal acceleration     |
| Rx             | x-coordinate of nodal reaction force   |
| Ry             | y-coordinate of nodal reaction force   |
| Rz             | z-coordinate of nodal reaction force   |

For analyses using biphasic, biphasic-solute, and triphasic materials, the following additional variables can be defined.

| Node variables | Description                                                                   |
|----------------|-------------------------------------------------------------------------------|
| p              | effective fluid pressure                                                      |
| vx             | x-component of solid velocity                                                 |
| vy             | y-component of solid velocity                                                 |
| vz             | z-component of solid velocity                                                 |
| ce<n>          | effective concentration of solute n, with n from 1 to 8 (e.g. ce1. ce2. etc.) |

For heat transfer problems the following nodal variables are available.

| Node variables | Description       |
|----------------|-------------------|
| T              | Nodal temperature |

For fluid analyses, the following variables can be defined.

| Node variables | Description                   |
|----------------|-------------------------------|
| nfvx           | x-component of fluid velocity |
| nfvy           | y-component of fluid velocity |
| nf vz          | z-component of fluid velocity |

For example, to store the current nodal positions of all nodes, use the following *node\_data* element:

```
<node_data data="x;y;z"></node_data>
```

You can store the total nodal displacement for nodes 1 through 100, and all even numbered nodes 200 through 400 as follows:

```
<node_data data="ux;uy;uz">1:100:1,200:400:2</node_data>
```

### E.2.2 Face\_Data Class

The *face\_data* class defines a set of surface variables. This item requires the *surface* attribute, which defines the surface that will be used to extract the data from. The surface must be defined in the *Mesh* section. The data is stored for each surface facet that is listed in the item list, or for all facets of the surface if no list is defined.

The following surface variables are defined. Note that the actual value is the average over the facet's integration points values (if applicable).

| Element variables  | Description                                                                  |
|--------------------|------------------------------------------------------------------------------|
| contact gap        | The gap (separation) of the contact interface this surface belongs to        |
| contact pressure   | The normal contact pressure of the contact interface this surface belongs to |
| contact_traction.x | The x-component of the contact traction on the face                          |
| contact_traction.y | The y-component of the contact traction on the face                          |
| contact_traction.z | The z-component of the contact traction on the face                          |

#### Example:

This example stores the contact gap and contact pressure of all the facets of surface "Face-tOnFacetSliding1\_primary". The surface is defined in the Mesh section.

```
<logfile>
  <face_data data="contact gap;contact pressure" surface="FacetOnFacetSliding1_primary"/>
</logfile>
```

### E.2.3 Element\_Data Class

The *element\_data* class defines a set of element variables. The data is stored for each element that is listed in the item list of the *element\_data* element or for all nodes if no list is defined. The following element variables are defined. Note that the actual value is the average over the element's integration points values (if applicable).

| Element variables | Description                                       |
|-------------------|---------------------------------------------------|
| x                 | x-coordinate of current element centroid position |
| y                 | y-coordinate of current element centroid position |
| z                 | z-coordinate of current element centroid position |
| sx                | xx-component of the Cauchy stress                 |
| sy                | yy-component of the Cauchy stress                 |
| sz                | zz-component of the Cauchy stress                 |
| sxy               | xy-component of the Cauchy stress                 |
| syz               | yz-component of the Cauchy stress                 |
| sxz               | xz-component of the Cauchy stress                 |
| s1                | first eigenvalue of Cauchy stress tensor          |
| s2                | second eigenvalue of Cauchy stress tensor         |

|     |                                                   |
|-----|---------------------------------------------------|
| s3  | third eigenvalue of Cauchy stress tensor          |
| Ex  | xx-component of the Green-Lagrange strain         |
| Ey  | yy-component of the Green-Lagrange strain         |
| Ez  | zz-component of the Green-Lagrange strain         |
| Exy | xy-component of the Green-Lagrange strain         |
| Eyz | yz-component of the Green-Lagrange strain         |
| Exz | xz-component of the Green-Lagrange strain         |
| E1  | first eigenvalue of Green-Lagrange strain tensor  |
| E2  | second eigenvalue of Green-Lagrange strain tensor |
| E3  | third eigenvalue of Green-Lagrange strain tensor  |
| ex  | xx-component of the infinitesimal strain          |
| ey  | yy-component of the infinitesimal strain          |
| ez  | zz-component of the infinitesimal strain          |
| exy | xy-component of the infinitesimal strain          |
| eyz | yz-component of the infinitesimal strain          |
| exz | xz-component of the infinitesimal strain          |
| Ux  | xx-component of the right stretch tensor          |
| Uy  | yy-component of the right stretch tensor          |
| Uz  | zz-component of the right stretch tensor          |
| Uxy | xy-component of the right stretch tensor          |
| Uyz | yz-component of the right stretch tensor          |
| Uxz | xz-component of the right stretch tensor          |
| U1  | first eigenvalue of right stretch tensor          |
| U2  | second eigenvalue of right stretch tensor         |
| U3  | third eigenvalue of right stretch tensor          |
| Vx  | xx-component of the left stretch tensor           |
| Vy  | yy-component of the left stretch tensor           |
| Vz  | zz-component of the left stretch tensor           |
| Vxy | xy-component of the left stretch tensor           |
| Vyz | yz-component of the left stretch tensor           |
| Vxz | xz-component of the left stretch tensor           |
| V1  | first eigenvalue of left stretch tensor           |
| V2  | second eigenvalue of left stretch tensor          |
| V3  | third eigenvalue of left stretch tensor           |
| Hx  | xx-component of the right Hencky strain           |
| Hy  | yy-component of the right Hencky strain           |
| Hz  | zz-component of the right Hencky strain           |
| Hxy | xy-component of the right Hencky strain           |
| Hyz | yz-component of the right Hencky strain           |
| Hxz | xz-component of the right Hencky strain           |
| H1  | first eigenvalue of right Hencky strain           |
| H2  | second eigenvalue of right Hencky strain          |
| H3  | third eigenvalue of right Hencky strain           |
| hx  | xx-component of the left Hencky strain            |
| hy  | yy-component of the left Hencky strain            |

|                         |                                                                |
|-------------------------|----------------------------------------------------------------|
| hz                      | zz-component of the left Hencky strain                         |
| hxy                     | xy-component of the left Hencky strain                         |
| hyz                     | yz-component of the left Hencky strain                         |
| hxz                     | xz-component of the left Hencky strain                         |
| h1                      | first eigenvalue of left Hencky strain                         |
| h2                      | second eigenvalue of left Hencky strain                        |
| h3                      | third eigenvalue of left Hencky strain                         |
| Fxx                     | xx-component of the deformation gradient                       |
| Fyy                     | yy-component of the deformation gradient                       |
| Fzz                     | zz-component of the deformation gradient                       |
| Fxy                     | xy-component of the deformation gradient                       |
| Fyz                     | yz-component of the deformation gradient                       |
| Fxz                     | xz-component of the deformation gradient                       |
| Fyx                     | yx-component of the deformation gradient                       |
| Fzy                     | zy-component of the deformation gradient                       |
| Fzx                     | xz-component of the deformation gradient                       |
| J                       | relative volume (determinant of deformation gradient)          |
| cxxxx                   | xxxx component of spatial elasticity tensor (a.k.a. c11)       |
| cxyxy                   | xyxy component of spatial elasticity tensor (a.k.a. c12)       |
| cyyyy                   | yyyy component of spatial elasticity tensor (a.k.a. c22)       |
| cxxzz                   | xxzz component of spatial elasticity tensor (a.k.a. c13)       |
| cyyzz                   | yyzz component of spatial elasticity tensor (a.k.a. c23)       |
| czzzz                   | zzzz component of spatial elasticity tensor (a.k.a. c33)       |
| cxxxy                   | xxxy component of spatial elasticity tensor (a.k.a. c14)       |
| cyyxy                   | yyxy component of spatial elasticity tensor (a.k.a. c24)       |
| czzxy                   | zzxy component of spatial elasticity tensor (a.k.a. c34)       |
| cxyxy                   | xyxy component of spatial elasticity tensor (a.k.a. c44)       |
| cxyyz                   | xyyz component of spatial elasticity tensor (a.k.a. c15)       |
| cyyyz                   | yyyz component of spatial elasticity tensor (a.k.a. c25)       |
| czzyz                   | zzyz component of spatial elasticity tensor (a.k.a. c35)       |
| cxyyz                   | xyyz component of spatial elasticity tensor (a.k.a. c45)       |
| cyzyz                   | yzyz component of spatial elasticity tensor (a.k.a. c55)       |
| cxxxz                   | xxxz component of spatial elasticity tensor (a.k.a. c16)       |
| cyyxz                   | yyxz component of spatial elasticity tensor (a.k.a. c26)       |
| czzxz                   | zzxz component of spatial elasticity tensor (a.k.a. c36)       |
| cxyxz                   | xyxz component of spatial elasticity tensor (a.k.a. c46)       |
| cyzyz                   | yzyz component of spatial elasticity tensor (a.k.a. c56)       |
| cxzxz                   | xzxz component of spatial elasticity tensor (a.k.a. c66)       |
| sed                     | strain energy density                                          |
| devsed                  | deviatoric strain energy density                               |
| D                       | damage variable (or sum of damage variables in solid mixtures) |
| effective left Hencky   | The effective (or von-Mises) norm of the left Hencky tensor    |
| effective right Hencky  | The effective (or von-Mises) norm of the right Hencky tensor   |
| effective left stretch  | The effective (or von-Mises) norm of the left stretch tensor   |
| effective right stretch | The effective (or von-Mises) norm of the right stretch tensor  |

|                        |                                                                                      |
|------------------------|--------------------------------------------------------------------------------------|
| fiber_stretch          | The element's fiber stretch, averaged over integration points.                       |
| fiber_x                | The x-component of the element's fiber vector                                        |
| fiber_y                | The y-component of the element's fiber vector                                        |
| fiber_z                | The z-component of the element's fiber vector                                        |
| Ft_xx                  | The xx component of the total deformation gradient of a prestrain material           |
| Ft_xy                  | The xy component of the total deformation gradient of a prestrain material           |
| Ft_xz                  | The xz component of the total deformation gradient of a prestrain material           |
| Ft_yx                  | The yx component of the total deformation gradient of a prestrain material           |
| Ft_yy                  | The yy component of the total deformation gradient of a prestrain material           |
| Ft_yz                  | The yz component of the total deformation gradient of a prestrain material           |
| Ft_zx                  | The zx component of the total deformation gradient of a prestrain material           |
| Ft_zy                  | The zy component of the total deformation gradient of a prestrain material           |
| Ft_zz                  | The zz component of the total deformation gradient of a prestrain material           |
| mixture_stress[<n>].xx | The xx-component of the stress of the mixture's solid component <n> (e.g. mixture_st |
| mixture_stress[<n>].xy | The xy-component of the stress of the mixture's solid component <n> (e.g. mixture_st |
| mixture_stress[<n>].xz | The xz-component of the stress of the mixture's solid component <n> (e.g. mixture_st |
| mixture_stress[<n>].yy | The yy-component of the stress of the mixture's solid component <n> (e.g. mixture_st |
| mixture_stress[<n>].yz | The yz-component of the stress of the mixture's solid component <n> (e.g. mixture_st |
| mixture_stress[<n>].zz | The zz-component of the stress of the mixture's solid component <n> (e.g. mixture_st |
| Pxx                    | The xx component of the PK1 stress                                                   |
| Pxy                    | The xy component of the PK1 stress                                                   |
| Pxz                    | The xz component of the PK1 stress                                                   |
| Pyx                    | The yx component of the PK1 stress                                                   |
| Pyy                    | The yy component of the PK1 stress                                                   |
| Pyz                    | The yz component of the PK1 stress                                                   |
| Pxz                    | The xz component of the PK1 stress                                                   |
| Pzy                    | The zy component of the PK1 stress                                                   |
| Pzz                    | The zz component of the PK1 stress                                                   |
| s1x                    | The x-component of the 1st Eigen vector of the Cauchy stress                         |
| s1y                    | The y-component of the 1st Eigen vector of the Cauchy stress                         |
| s1z                    | The z-component of the 1st Eigen vector of the Cauchy stress                         |
| s2x                    | The x-component of the 2nd Eigen vector of the Cauchy stress                         |
| s2y                    | The y-component of the 2nd Eigen vector of the Cauchy stress                         |
| s2z                    | The z-component of the 2nd Eigen vector of the Cauchy stress                         |
| s3x                    | The x-component of the 3rd Eigen vector of the Cauchy stress                         |
| s3y                    | The y-component of the 3rd Eigen vector of the Cauchy stress                         |
| s3z                    | The z-component of the 3rd Eigen vector of the Cauchy stress                         |
| Sx                     | The xx-component of the PK2 stress                                                   |
| Sy                     | The yy-component of the PK2 stress                                                   |
| Sz                     | The zz-component of the PK2 stress                                                   |
| Sxy                    | The xy-component of the PK2 stress                                                   |
| Sxz                    | The xz-component of the PK2 stress                                                   |
| Syz                    | The yz-component of the PK2 stress                                                   |
| wy                     | fraction of yielded bonds                                                            |
| wf                     | fraction of fatigued bonds                                                           |

For example, to store the (average) Cauchy stress for all elements, define the following data element:

```
<element_data data="sx;sy;sz;sxy;syx;sxz" name="element stresses"> </element_data>
```

For analyses using biphasic, biphasic-solute, and triphasic materials, the following additional variables can be defined:

x

| Element variables                   | Description                                                              |
|-------------------------------------|--------------------------------------------------------------------------|
| p                                   | actual fluid pressure                                                    |
| $c_n$                               | actual concentration of solute $n$                                       |
| wx                                  | x-component of fluid flux                                                |
| wy                                  | y-component of fluid flux                                                |
| wz                                  | z-component of fluid flux                                                |
| $j_n^x$                             | x-component of flux of solute $n$                                        |
| $j_n^y$                             | y-component of flux of solute $n$                                        |
| $j_n^z$                             | z-component of flux of solute $n$                                        |
| jx                                  | x-component of flux of solute in biphasic-solute material                |
| jy                                  | y-component of flux of solute in biphasic-solute material                |
| jz                                  | z-component of flux of solute in biphasic-solute material                |
| Kp <sub>xx</sub>                    | xx-component of the permeability tensor.                                 |
| Kp <sub>xy</sub>                    | xy-component of the permeability tensor.                                 |
| Kp <sub>xz</sub>                    | xz-component of the permeability tensor.                                 |
| Kp <sub>yy</sub>                    | yy-component of the permeability tensor.                                 |
| Kp <sub>yz</sub>                    | yz-component of the permeability tensor.                                 |
| Kp <sub>zz</sub>                    | zz-component of the permeability tensor.                                 |
| lex                                 | The x-component of the element current density in a multiphasic material |
| ley                                 | The y-component of the element current density in a multiphasic material |
| lez                                 | The z-component of the element current density in a multiphasic material |
| es <sub>xx</sub>                    | The xx component of the solid stress of a biphasic material              |
| es <sub>xy</sub>                    | The xy component of the solid stress of a biphasic material              |
| es <sub>xz</sub>                    | The xz component of the solid stress of a biphasic material              |
| es <sub>yy</sub>                    | The yy component of the solid stress of a biphasic material              |
| es <sub>yz</sub>                    | The yz component of the solid stress of a biphasic material              |
| es <sub>zz</sub>                    | The zz component of the solid stress of a biphasic material              |
| sbm<n>                              | Concentration of solute-bound molecule <n> (e.g. sbm1)                   |
| sbm<n>_referential_apparent_density | Apparent density in reference configuration of solute-bound molecule <n> |

For fluid analyses, the following additional variables can be defined:

| Element variables | Description                                |
|-------------------|--------------------------------------------|
| fx                | x-coordinate of element centroid position  |
| fy                | y-coordinate of element centroid position  |
| fz                | z--coordinate of element centroid position |
| fp                | fluid pressure                             |
| fvx               | x-component of fluid velocity              |
| fvy               | y-component of fluid velocity              |
| fvz               | z-component of fluid velocity              |
| fJ                | fluid volume ratio                         |
| fd                | fluid density                              |
| fsp               | fluid stress power                         |
| fax               | x-component of fluid acceleration          |
| fay               | y-component of fluid acceleration          |
| faz               | z-component of fluid acceleration          |
| fwx               | x-component of fluid vorticity             |
| fwy               | y-component of fluid vorticity             |
| fwz               | z-component of fluid vorticity             |
| fsxx              | xx-component of fluid Cauchy stress        |
| fsyy              | yy-component of fluid Cauchy stress        |
| fszz              | zz-component of fluid Cauchy stress        |
| fsxy              | xy-component of fluid Cauchy stress        |
| fsyz              | yz-component of fluid Cauchy stress        |
| fsxz              | xz-component of fluid Cauchy stress        |
| fmxs              | maximum shear of fluid Cauchy stress       |
| fs1               | 1st principal fluid Cauchy stress          |
| fs2               | 2nd principal fluid Cauchy stress          |
| fs3               | 3rd principal fluid Cauchy stress          |
| fdxx              | xx-component of fluid rate of deformation  |
| fdyy              | yy-component of fluid rate of deformation  |
| fdzz              | zz-component of fluid rate of deformation  |
| fdxy              | xy-component of fluid rate of deformation  |
| fdyz              | yz-component of fluid rate of deformation  |
| fdxz              | xz-component of fluid rate of deformation  |

For discrete elements, the following log variables are defined.

| Discrete element variables  | Description                                                                       |
|-----------------------------|-----------------------------------------------------------------------------------|
| discrete element elongation | The discrete element's elongation (i.e. current length minus initial length)      |
| discrete element force      | The scalar force along the element's axis.                                        |
| discrete element stretch    | The discrete element's stretch (i.e. ratio of current length over initial length) |
| Fde.x                       | The x-component of the discrete element's force vector.                           |
| Fde.y                       | The y-component of the discrete element's force vector.                           |



|       |                                                         |
|-------|---------------------------------------------------------|
| Fde.z | The z-component of the discrete element's force vector. |
|-------|---------------------------------------------------------|

### E.2.4 Domain Data Class

The *domain\_data\_class* defines variables that are calculated on an entire domain. The variables are often quantities integrated over the entire domain. In the table below, the notation <n> is often used to represent a number that ranges from 1 to 8, unless specified otherwise. So for instance, the variable “*c<n>\_integral*” means “*c1\_integral*”, or “*c2\_integral*”, and so forth.

| domain variables           | Description                                                                       |
|----------------------------|-----------------------------------------------------------------------------------|
| c<n>_integral              | Integral of concentration of solute <n>                                           |
| sbm<n>_integral            | Integral of concentration of sbm <n>                                              |
| normalized internal energy | The integral over space and time of the internal energy, normalized by the volume |
| avg                        | Average of another log variable over domain                                       |
| pct                        | Percentile of another log variable over domain                                    |

### E.2.5 Surface Data Class

The *surface\_data* class defines variables that are calculated over an entire surface.

| surface variables | Description                                                      |
|-------------------|------------------------------------------------------------------|
| max contact gap   | The maximum contact gap value evaluated over the entire surface. |

### E.2.6 Rigid\_Body\_Data Class

The *rigid\_body\_data* class defines a set of variables for each rigid body. The data is stored for each rigid body that is listed in the item list of the *rigid\_body\_data* element or for all rigid bodies if no list is defined. The following variables are defined. Note that the item referenced in the item list is the material number of the rigid body.

| Rigid body variables | Description                             |
|----------------------|-----------------------------------------|
| x                    | x-coordinate of center of mass position |
| y                    | y-coordinate of center of mass position |
| z                    | z-coordinate of center of mass position |
| vx                   | x-component of center of mass velocity  |
| vy                   | y-component of center of mass velocity  |

|        |                                                                          |
|--------|--------------------------------------------------------------------------|
| vz     | z-component of center of mass velocity                                   |
| ax     | x-component of center of mass acceleration                               |
| ay     | y-component of center of mass acceleration                               |
| az     | z-component of center of mass acceleration                               |
| thx    | x-component of rotation pseudo-vector                                    |
| thy    | y-component of rotation pseudo-vector                                    |
| thz    | z-component of rotation pseudo-vector                                    |
| omx    | x-component of angular velocity                                          |
| omy    | y-component of angular velocity                                          |
| omz    | z-component of angular velocity                                          |
| alx    | x-component of angular acceleration                                      |
| aly    | y-component of angular acceleration                                      |
| alz    | z-component of angular acceleration                                      |
| qx     | x-component of rotation quaternion                                       |
| qy     | y-component of rotation quaternion                                       |
| qz     | z-component of rotation quaternion                                       |
| qw     | w-component of rotation quaternion                                       |
| Fx     | x-component of the rigid body reaction force                             |
| Fy     | y-component of the rigid body reaction force                             |
| Fz     | z-component of the rigid body reaction force                             |
| Mx     | x-component of the rigid body reaction torque                            |
| My     | y-component of the rigid body reaction torque                            |
| Mz     | z-component of the rigid body reaction torque                            |
| KE     | kinetic energy                                                           |
| EulerX | X-Euler angle                                                            |
| EulerY | Y-Euler angle                                                            |
| EulerZ | Z-Euler angle                                                            |
| FHAsx  | x-component of the finite helical axis                                   |
| FHAsy  | y-component of the finite helical axis                                   |
| FHAsz  | z-component of the finite helical axis                                   |
| FHAwx  | x-component of angular rotation vector around finite helical axis        |
| FHAwy  | y-component of angular rotation vector around finite helical axis        |
| FHAwz  | z-component of angular rotation vector around finite helical axis        |
| IHAsx  | x-component of the instantaneous helical axis                            |
| IHAsy  | y-component of the instantaneous helical axis                            |
| IHAsz  | z-component of the instantaneous helical axis                            |
| IHAwx  | x-component of angular rotation vector around instantaneous helical axis |
| IHAwy  | y-component of angular rotation vector around instantaneous helical axis |
| IHAwz  | z-component of angular rotation vector around instantaneous helical axis |
| R11    | 11-component of the rigid body's rotation matrix                         |
| R12    | 12-component of the rigid body's rotation matrix                         |
| R13    | 13-component of the rigid body's rotation matrix                         |
| R21    | 21-component of the rigid body's rotation matrix                         |
| R22    | 22-component of the rigid body's rotation matrix                         |
| R23    | 23-component of the rigid body's rotation matrix                         |

|     |                                                  |
|-----|--------------------------------------------------|
| R31 | 31-component of the rigid body's rotation matrix |
| R32 | 32-component of the rigid body's rotation matrix |
| R33 | 33-component of the rigid body's rotation matrix |

For example, to store the rigid body reaction force of rigid body 2 and 4 add the following data element. Note that the 2 and 4 refer to the rigid body material number as defined in the *Material* section of the input file:

```
<rigid_body_data data="Fx;Fy;Fz">2,4</rigid_body_data>
```

### E.2.7 Rigid\_Connector\_Data Class

The *rigid\_connector\_data* class defines a set of variables for each rigid joint or rigid connector. The data is stored for each rigid joint or rigid connector that is listed in the item list of the *rigid\_connector\_data* element or for all rigid connectors if no list is defined. The following variables are defined. Note that the item referenced in the item list is the rigid connector number in the order in which rigid connectors appear in the input file.

| Rigid body variables | Description                                |
|----------------------|--------------------------------------------|
| RCFx                 | x-component of rigid connector force       |
| RCFy                 | y-component of rigid connector force       |
| RCFz                 | z-component of rigid connector force       |
| RCMx                 | x-component of rigid connector moment      |
| RCMy                 | y-component of rigid connector moment      |
| RCMz                 | z-component of rigid connector moment      |
| RCx                  | x-component of rigid connector translation |
| RCy                  | y-component of rigid connector translation |
| RCz                  | z-component of rigid connector translation |
| RCthx                | x-component of rigid connector rotation    |
| RCthy                | y-component of rigid connector rotation    |
| RCthz                | z-component of rigid connector rotation    |

For example, to store the reaction forces and moments at rigid joints 2 and 4 add the following data element:

```
<rigid_connector_data data="RCFx;RCFy;RCFz;RCMx;RCMy;RCMz">2,4</rigid_connector_data>
```

The rigid connector translation and rotation variables return relative motions of the rigid bodies connected by rigid joints and other rigid connectors (Section 3.12.4). The rotation components represent the components of a vector whose direction is along the axis of rotation, and whose magnitude is the (counter-clockwise) angle of rotation. These relative motions are calculated as

the motion of *body\_b* relative to *body\_a*, expressed in the local Cartesian basis of *body\_a*, whose initial origin is at *joint\_origin*. This Cartesian basis (which generally moves with *body\_a*) has its first axis along *rotation\_axis* (for revolute and planar joints) or *translation\_axis* (for prismatic joints) or *joint\_axis* (for cylindrical joints); the second axis is along the *transverse\_axis* (for revolute, prismatic and cylindrical joints) or *translation\_axis\_1* (for planar joints). For spherical joints, springs, dampers, angular dampers and contractile forces, the axes are always aligned with the global Cartesian basis.

For springs, dampers and contractile forces, the returned translation components represent the relative position vector between insertion points on *body\_b* and *body\_a*. The magnitude of this vector represents the distance between those points.

### E.2.8 Model Data Class

This category encompasses variables that are not necessarily tied to specific components of a model.

| Model variables | Description                        |
|-----------------|------------------------------------|
| solution_norm   | The L2 norm of the solution vector |

# Bibliography

- [1] G Aguilar, F Gaspar, F Lisbona, and C24558811158 Rodrigo. Numerical stabilization of biot's consolidation model by a perturbation on the flow equation. *International journal for numerical methods in engineering*, 75(11):1282–1300, 2008.
- [2] Michael B Albro, Vikram Rajan, Roland Li, Clark T Hung, and Gerard A Ateshian. Characterization of the concentration-dependence of solute diffusivity and partitioning in a model dextran-agarose transport system. *Cell Mol Bioeng*, 2(3):295–305, Sep 2009.
- [3] Alexandra N. Allan, Jared L. Zitnay, Steve A. Maas, and Jeffrey A. Weiss. Development of a continuum damage model to predict accumulation of sub-failure damage in tendons. *Journal of the Mechanical Behavior of Biomedical Materials*, 135:105342, 2022.
- [4] E.M. Arruda and M.C. Boyce. A three-dimensional constitutive model for the large stretch behavior of rubber elastic materials. *J. Mech. Phys. Solids*, 41(2):389–412, 1993.
- [5] G. A. Ateshian. Anisotropy of fibrous tissues in relation to the distribution of tensed and buckled fibers. *J Biomech Eng*, 129(2):240–9, 2007.
- [6] G. A. Ateshian and K. D. Costa. A frame-invariant formulation of fung elasticity. *J Biomech*, 42(6):781–5, 2009.
- [7] G. A. Ateshian, V. Rajan, N. O. Chahine, C. E. Canal, and C. T. Hung. Modeling the matrix of articular cartilage using a continuous fiber angular distribution predicts many observed phenomena. *J Biomech Eng*, 131(6):061003, 2009.
- [8] G. A. Ateshian and T. Ricken. Multigenerational interstitial growth of biological tissues. *Biomech Model Mechanobiol*, 9(6):689–702, 2010.
- [9] G. A. Ateshian, W. H. Warden, J. J. Kim, R. P. Grelsamer, and V. C. Mow. Finite deformation biphasic material properties of bovine articular cartilage from confined compression experiments. *J Biomech*, 30(11-12):1157–64, 1997.
- [10] G.A. Ateshian, M. B. Albro, S.A. Maas, and J.A. Weiss. Finite element implementation of mechanochemical phenomena in neutral deformable porous media under finite deformation. *Journal of Biomechanical Engineering*, 133(8):1005–1017, 2011.
- [11] G.A Ateshian, Morrison B., J.W. Holmes, and C. T. Hung. Mechanics of cell growth. *Mechanics Research Communications*, 42:118–125, 2012.
- [12] GA Ateshian, SA Maas, and J.A. Weiss. Finite element algorithm for frictionless contact of porous permeable media under finite deformation and sliding. *J. Biomech. Engn.*, 132(6):1006–1019, 2010.

- [13] Gerard A Ateshian. Viscoelasticity using reactive constrained solid mixtures. *J Biomech*, 48(6):941–7, Apr 2015.
- [14] Gerard A Ateshian, Benjamin J Ellis, and Jeffrey A Weiss. Equivalence between short-time biphasic and incompressible elastic material responses. *J Biomech Eng*, 129(3):405–12, Jun 2007.
- [15] Gerard A Ateshian, Kimberly R Kroupa, Courtney A Petersen, Brandon K Zimmerman, Steve A Maas, and Jeffrey A Weiss. Damage mechanics of biological tissues in relation to viscoelasticity. *J Biomech Eng*, 145(4), Apr 2023.
- [16] Gerard A Ateshian, Steve Maas, and Jeffrey A Weiss. Solute transport across a contact interface in deformable porous media. *J Biomech*, 45(6):1023–7, Apr 2012.
- [17] Daniel Balzani, Sarah Brinkhues, and Gerhard A. Holzapfel. Constitutive framework for the modeling of damage in collagenous soft tissues with application to arterial walls. *Computer Methods in Applied Mechanics and Engineering*, 213-216:139–151, 2012.
- [18] Klaus-Jürgen Bathe and Eduardo N Dvorkin. A formulation of general shell elements—the use of mixed interpolation of tensorial components. *International journal for numerical methods in engineering*, 22(3):697–722, 1986.
- [19] P Betsch and E Stein. An assumed strain approach avoiding artificial thickness straining for a non-linear 4-node shell element. *Communications in Numerical Methods in Engineering*, 11(11):899–909, 1995.
- [20] Anna M. Birzle, Christian Martin, Stefan Uhlig, and Wolfgang A. Wall. A coupled approach for identification of nonlinear and compressible material models for soft tissue based on different experimental setups - exemplified and detailed for lung parenchyma. *Journal of the Mechanical Behavior of Biomedical Materials*, 94:126–143, 2019.
- [21] M Bischoff and E Ramm. Shear deformable shell elements for large strains and rotations. *International Journal for Numerical Methods in Engineering*, 40(23):4427–4449, 1997.
- [22] Manfred Bischoff, E Ramm, and J Irslinger. Models and finite elements for thin-walled structures. *Encyclopedia of Computational Mechanics Second Edition*, pages 1–86, 2018.
- [23] SS Blemker. *3D Modeling of Complex Muscle Architecture and Geometry*. Thesis, 2004.
- [24] Javier Bonet and Richard D. Wood. *Nonlinear continuum mechanics for finite element analysis*. Cambridge University Press, 1997.
- [25] D R Carter and W C Hayes. Bone compressive strength: the influence of density and strain rate. *Science*, 194(4270):1174–6, Dec 1976.
- [26] D R Carter and W C Hayes. The compressive behavior of bone as a two-phase porous structure. *J Bone Joint Surg Am*, 59(7):954–62, Oct 1977.
- [27] Y I Cho and K R Kenney. Effects of the non-Newtonian viscosity of blood on flows in a diseased arterial vessel. Part 1: Steady flows. *Biorheology*, 28(3-4):241–62, 1991.
- [28] JC Criscione, SA Douglas, and WC Hunter. Physically based strain invariant set for materials exhibiting transversely isotropic behavior. *J. Mech. Phys. Solids*, 49:871–897, 2001.

- [29] John C Criscione, Jay D Humphrey, Andrew S Douglas, and William C Hunter. An invariant basis for natural strain which yields orthogonal stress response terms in isotropic hyperelasticity. *J. Mech. Phys. Solids*, 48(12):2445–2465, 2000.
- [30] A. Curnier, He Qi-Chang, and P. Zysset. Conewise linear elastic materials. *J Elasticity*, 37(1):1–38, 1994.
- [31] VS Deshpande and NA Fleck. Multi-axial yield behaviour of polymer foams. *Acta materialia*, 49(10):1859–1866, 2001.
- [32] Daniel Charles Drucker. Relation of experiments to mathematical theories of plasticity. *Journal of Applied Mechanics*, 1949.
- [33] Daniel Charles Drucker and William Prager. Soil mechanics and plastic analysis or limit design. *Quarterly of applied mathematics*, 10(2):157–165, 1952.
- [34] Ana Cristina Estrada, Kyoko Yoshida, Samantha A Clarke, and Jeffrey W Holmes. Longitudinal reinforcement of acute myocardial infarcts improves function by transmurally redistributing stretch and stress. *J Biomech Eng*, 142(2), 02 2020.
- [35] Y. C Fung. *Biomechanics: mechanical properties of living tissues*. Springer-Verlag, New York, 1981.
- [36] Y. C. Fung. *Biomechanics : mechanical properties of living tissues*. Springer-Verlag, New York, 2nd edition, 1993.
- [37] Y. C. Fung, K. Fronek, and P. Patitucci. Pseudoelasticity of arteries and the choice of its mathematical expression. *Am J Physiol*, 237(5):H620–31, 1979.
- [38] Y. C Fung, Nicholas Perrone, and M Anliker. *Biomechanics, its foundations and objectives*. Prentice-Hall, Englewood Cliffs, N.J., 1972.
- [39] T Christian Gasser, Ray W Ogden, and Gerhard A Holzapfel. Hyperelastic modelling of arterial layers with distributed collagen fibre orientations. *J R Soc Interface*, 3(6):15–35, Feb 2006.
- [40] M.W. Gee, C.R. Dohrmann, S.W. Key, and W.A. Wall. A uniform nodal strain tetrahedron with isochoric stabilization. *Int. J. Numer. Meth. Engng*, (78):429–443, 2009.
- [41] M.J.A. Girard, J.C. Downs, and C. F. Burgoyne. Peripapillary and posterior scleral mechanics - part i: Development of an anisotropic hyperelastic constitutive model. *J Biomech Eng*, 131(5):051011, 2009.
- [42] C.L.M. Gouget, M.J.A. Girard, and C.R. Ethier. A constrained von mises distribution to describe fiber organization in thin soft tissues. *Biomechanics And Modeling in Mechanobiology*, 11(3-4):475–482, 2012.
- [43] J.M. Guccione and A.D. McCulloch. Mechanics of active contraction in cardiac muscle: part i - constitutive relations for fiber stress that describe deactivation. *J. Biomechanical Engineering*, vol. 115(no. 1):72–83, 1993.
- [44] Osman Gültekin, Hüsnü Dal, and Gerhard A Holzapfel. On the quasi-incompressible finite element analysis of anisotropic hyperelastic materials. *Computational mechanics*, 63(3):443–453, 2019.

- [45] J Helfenstein, M Jabareen, Edoardo Mazza, and S Govindjee. On non-physical response in models for fiber-reinforced hyperelastic materials. *International Journal of Solids and Structures*, 47(16):2056–2061, 2010.
- [46] M. H. Holmes and V. C. Mow. The nonlinear characteristics of soft gels and hydrated connective tissues in ultrafiltration. *J Biomech*, 23(11):1145–56, 1990.
- [47] C Hou and G.A. Ateshian. A gauss-kronrod-trapezoidal integration scheme for modeling biological tissues with continuous fiber distributions. *Computer Methods in Biomechanics and Biomedical Engineering*, 19(8):883–893, 2016.
- [48] Jay C Hou, Steve A Maas, Jeffrey A Weiss, and Gerard A Ateshian. Finite element formulation of multiphasic shell elements for cell mechanics analyses in febio. *Journal of Biomechanical Engineering*, 140(12), 2018.
- [49] P J Hunter, A D McCulloch, and H E ter Keurs. Modelling the mechanical properties of cardiac muscle. *Prog Biophys Mol Biol*, 69(2-3):289–331, 1998.
- [50] J. C. Iatridis, L. A. Setton, R. J. Foster, B. A. Rawlins, M. Weidenbaum, and V. C. Mow. Degeneration affects the anisotropic and nonlinear behaviors of human anulus fibrosus in compression. *J Biomech*, 31(6):535–44, 1998.
- [51] David Kamensky, Fei Xu, Chung-Hao Lee, Jinhui Yan, Yuri Bazilevs, and Ming-Chen Hsu. A contact formulation based on a volumetric potential: Application to isogeometric simulations of atrioventricular valves. *Computer Methods in Applied Mechanics and Engineering*, 330:522–546, 2018.
- [52] DE Kiouisis, TC Gasser, and GA Holzapfel. Smooth contact strategies with emphasis on the modeling of balloon angioplasty with stenting. *International journal for numerical methods in engineering*, 75(7):826–855, 2008.
- [53] Dimitrios E Kiouisis, Alexander R Wulff, and Gerhard A Holzapfel. Experimental studies and numerical analysis of the inflation and interaction of vascular balloon catheter-stent systems. *Annals of Biomedical Engineering*, 37(2):315–330, 2009.
- [54] S Klinkel, F Gruttmann, and W Wagner. A continuum based three-dimensional shell element for laminated structures. *Computers & Structures*, 71(1):43–62, 1999.
- [55] W M Lai, J S Hou, and V C Mow. A triphasic theory for the swelling and deformation behaviors of articular cartilage. *J Biomech Eng*, 113(3):245–58, Aug 1991.
- [56] Y. Lanir. Constitutive equations for fibrous connective tissues. *J Biomech*, 16(1):1–12, 1983.
- [57] T. A. Laursen and B. N. Maker. Augmented lagrangian quasi-newton solver for constrained nonlinear finite element applications. *International Journal for Numerical Methods in Engineering*, 38(21):3571–3590, 1995.
- [58] T. A. Laursen and J. C. Simo. Continuum-based finite element formulation for the implicit solution of multibody, large deformation frictional contact problems. *International Journal for Numerical Methods in Engineering*, 36(20):3451–3485, 1993.
- [59] S.A. Maas, A. Erdemir, J.P. Halloran, and J.A. Weiss. A general framework for applications of prestrain to computational models of biological materials. *Journal of Biomechanical Behavior of Biomedical Materials*, 61:499–510, 2016.



- [60] Richard H MacNeal. A simple quadrilateral shell element. *Computers & Structures*, 8(2):175–183, 1978.
- [61] B. N. Maker. Rigid bodies for metal forming analysis with nike3d. *University of California, Lawrence Livermore Lab Rept*, UCRL-JC-119862:1–8, 1995.
- [62] A Ya Malkin. Continuous relaxation spectrum-its advantages and methods of calculation. *Applied Mechanics and Engineering*, 11(2):235, 2006.
- [63] Andreas Menzel and Ellen Kuhl. Frontiers in growth and remodeling. *Mech Res Commun*, 42:1–14, Jun 2012.
- [64] V.C. Mow, S.C. Kuei, W.M. Lai, and C.G. Armstrong. Biphasic creep and stress relaxation of articular cartilage in compression: Theory and experiments. *J. Biomech. Eng.*, 102:73–84, 1980.
- [65] M G Mullender, R Huiskes, and H Weinans. A physiological approach to the simulation of bone remodeling as a self-organizational control process. *J Biomech*, 27(11):1389–94, Nov 1994.
- [66] Robert J Nims, Krista M Durney, Alexander D Cigan, Antoine Dusséaux, Clark T Hung, and Gerard A Ateshian. Continuum theory of fibrous tissue damage mechanics using bond kinetics: application to cartilage tissue engineering. *Interface Focus*, 6(1):20150063, Feb 2016.
- [67] J T Overbeek. The donnan equilibrium. *Prog Biophys Biophys Chem*, 6:57–84, 1956.
- [68] Ronald L Panton. *Incompressible flow*. John Wiley & Sons, 2006.
- [69] Tasos C Papanastasiou. Flows of materials with yield. *Journal of Rheology*, 31(5):385–404, 1987.
- [70] M. A. Puso and J. A. Weiss. Finite element implementation of anisotropic quasi-linear viscoelasticity using a discrete spectrum approximation. *J Biomech Eng*, 120(1):62–70, 1998.
- [71] K. M. Quapp and J. A. Weiss. Material characterization of human medial collateral ligament. *J Biomech Eng*, 120(6):757–63, 1998.
- [72] DJRA Quemada. Rheology of concentrated disperse systems ii. a model for non-newtonian shear viscosity in steady flows. *Rheologica Acta*, 17(6):632–642, 1978.
- [73] Adam Rauff, Michael R. Herron, Steve A. Maas, and Jeffrey A. Weiss. An algorithmic and software framework to incorporate orientation distribution functions in finite element simulations for biomechanics and biophysics. *Acta Biomaterialia*, November 2024.
- [74] Adam Rauff, Lucas H. Timmins, Ross T. Whitaker, and Jeffrey A. Weiss. A Nonparametric Approach for Estimating Three-Dimensional Fiber Orientation Distribution Functions (ODFs) in Fibrous Materials. *IEEE Transactions on Medical Imaging*, 41(2):446–455, February 2022. Conference Name: IEEE Transactions on Medical Imaging.
- [75] E K Rodriguez, A Hoger, and A D McCulloch. Stress-dependent finite growth in soft elastic tissues. *J Biomech*, 27(4):455–67, Apr 1994.
- [76] Carlo Sansour. On the physical assumptions underlying the volumetric-isochoric split and the case of anisotropy. *European Journal of Mechanics-A/Solids*, 27(1):28–39, 2008.

- [77] Lei Shi, Lingfeng Hu, Nicole Lee, Shuyang Fang, and Kristin Myers. Three-dimensional anisotropic hyperelastic constitutive model describing the mechanical response of human and mouse cervix. *Acta Biomaterialia*, 150:277–294, 2022.
- [78] Jay J Shim, Steve A Maas, Jeffrey A Weiss, and Gerard A Ateshian. Finite element implementation of computational fluid dynamics with reactive neutral and charged solute transport in febio. *J Biomech Eng*, pages 1–26, May 2023.
- [79] JC Simo. On a fully three-dimensional finite-strain viscoelastic damage model: formulation and computational aspects. *Computer methods in applied mechanics and engineering*, 60(2):153–173, 1987.
- [80] JC Simo, F Armero, and RL Taylor. Improved versions of assumed enhanced strain tri-linear elements for 3d finite deformation problems. *Computer methods in applied mechanics and engineering*, 110(3-4):359–386, 1993.
- [81] J.C. Simo and R.L. Taylor. Quasi-incompressible finite elasticity in principal stretches: Continuum basis and numerical algorithms. *Computer Methods in Applied Mechanics and Engineering*, 85:273–310, 1991.
- [82] Juan C Simo and MS10587420724 Rifai. A class of mixed assumed strain methods and the method of incompatible modes. *International journal for numerical methods in engineering*, 29(8):1595–1638, 1990.
- [83] RP Skelton, HJ Maier, and H-J Christ. The baushinger effect, masing model and the ramberg–osgood relation for cyclic deformation in metals. *Materials Science and Engineering: A*, 238(2):377–390, 1997.
- [84] Anthony James Merril Spencer. *Continuum Theory of the Mechanics of Fibre-Reinforced Composites*. Springer-Verlag, New York, 1984.
- [85] A.M. Swedberg, S.P. Reese, S.A. Maas, B. J. Ellis, and J.A. Weiss. Continuum description of the poisson’s ratio of ligament and tendon under finite deformation. *Journal of Biomechanics*, 47(12):3201–3209, 2014.
- [86] D.R. Veronda and R.A. Westmann. Mechanical characterization of skin - finite deformations. *J. Biomechanics*, Vol. 3:111–124, 1970.
- [87] Irene E Vignon-Clementel, C Alberto Figueroa, Kenneth E Jansen, and Charles A Taylor. Outflow boundary conditions for three-dimensional finite element modeling of blood flow and pressure in arteries. *Comput. Methods Appl. Mech. Engrg.*, 195(29):3776–3796, 2006.
- [88] L Vu-Quoc and XG Tan. Optimal solid shells for non-linear analyses of multilayer composites. i. statics. *Computer methods in applied mechanics and engineering*, 192(9-10):975–1016, 2003.
- [89] H Weinans, R Huiskes, and H J Grootenboer. The behavior of adaptive bone-remodeling simulation models. *J Biomech*, 25(12):1425–41, Dec 1992.
- [90] J.A. Weiss, J.C. Gardiner, and Bonifasi-Lista C. Ligament material behavior is nonlinear, viscoelastic and rate-independent under shear loading. *Journal of Biomechanics*, 35(7):943–950, 2002.

- [91] J.A. Weiss, B.N. Maker, and S. Govindjee. Finite element implementation of incompressible, transversely isotropic hyperelasticity. *Computer Methods in Applications of Mechanics and Engineering*, 135:107–128, 1996.
- [92] H Xiao and LS Chen. Hencky's elasticity model and linear stress-strain relations in isotropic finite hyperelasticity. *Acta Mechanica*, 157(1):51–60, 2002.
- [93] Brandon K. Zimmerman, David Jiang, Jeffrey A. Weiss, Lucas H. Timmins, and Gerard A. Ateshian. On the use of constrained reactive mixtures of solids to model finite deformation isothermal elastoplasticity and elastoplastic damage mechanics. *Journal of the Mechanics and Physics of Solids*, page 104534, 2021.
- [94] Brandon K Zimmerman, Steve A Maas, Jeffrey A Weiss, and Gerard A Ateshian. A finite element algorithm for large deformation biphasic frictional contact between porous-permeable hydrated soft tissues. *J Biomech Eng*, 144(2), 02 2022.